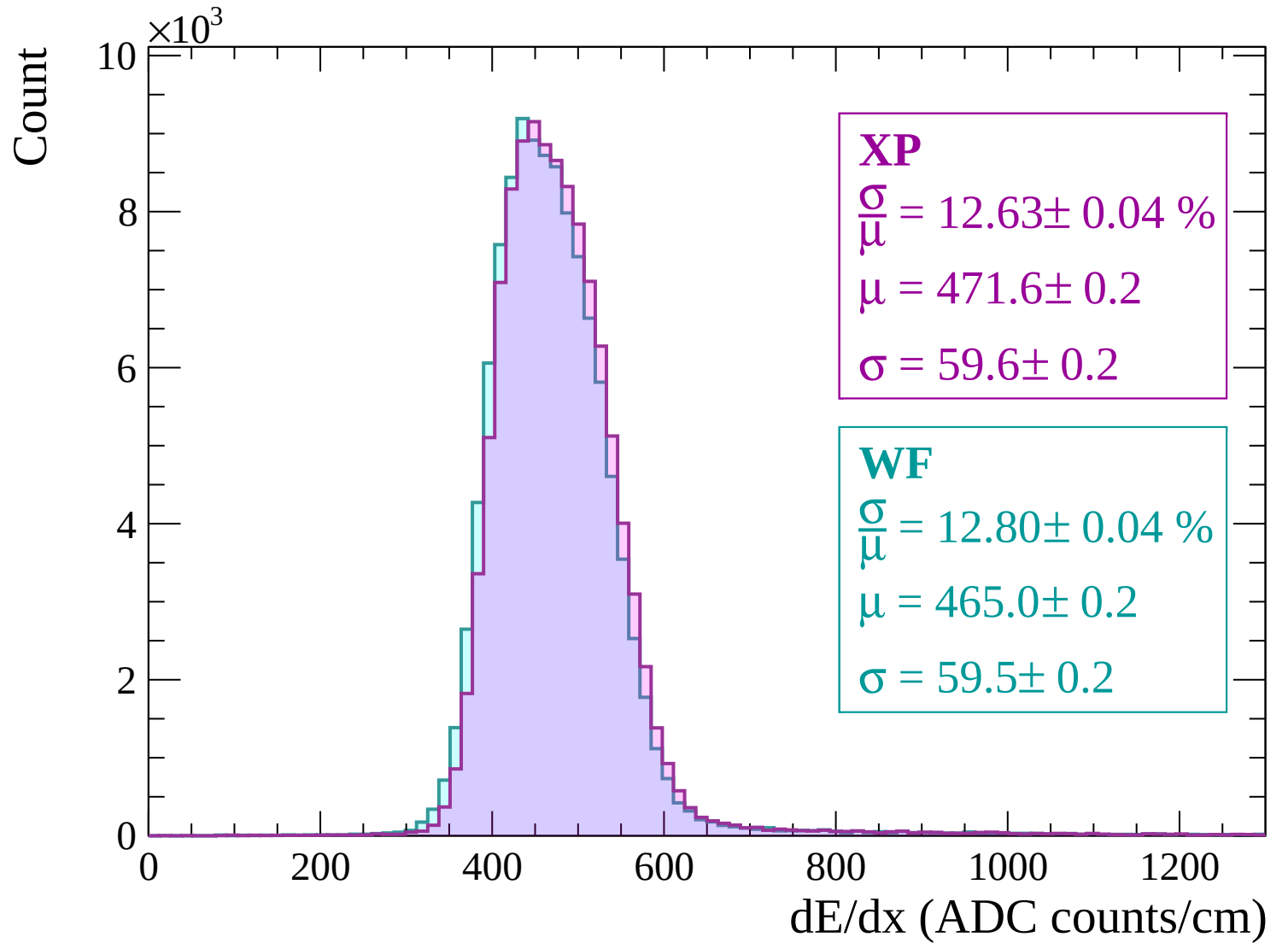
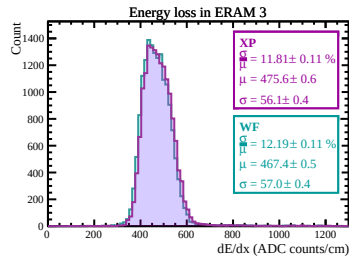
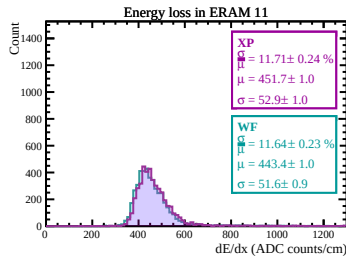
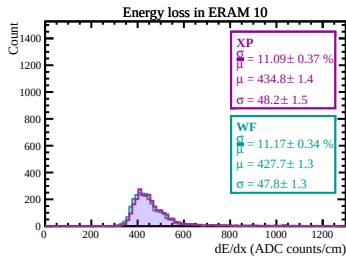
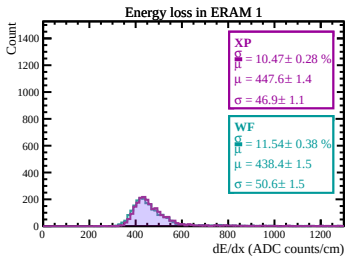
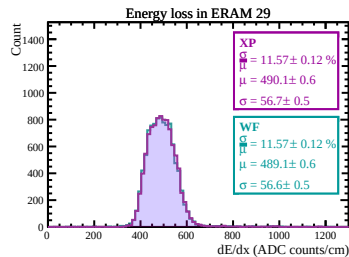
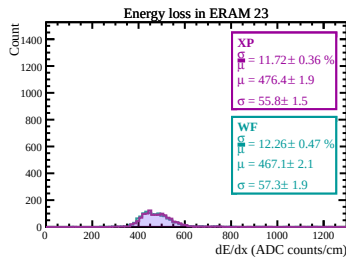
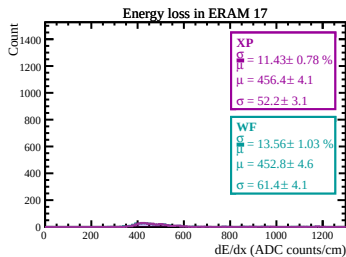
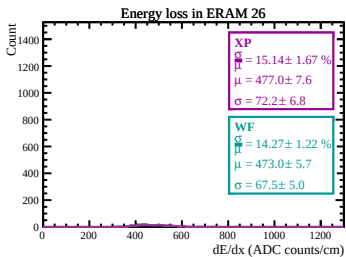
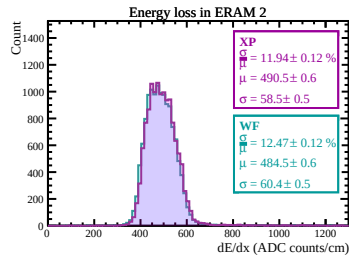
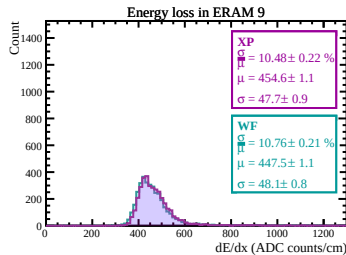
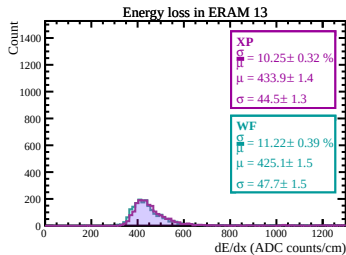
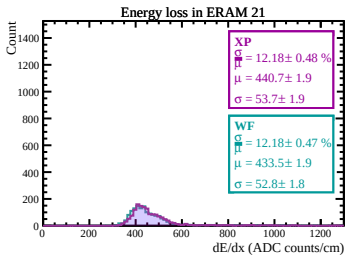
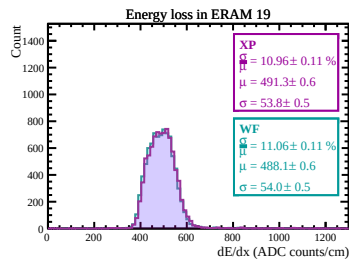
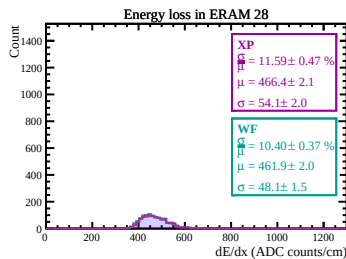
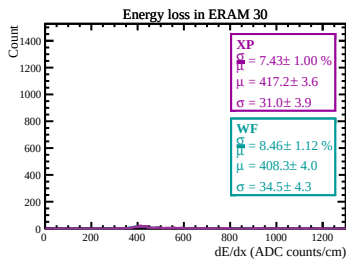
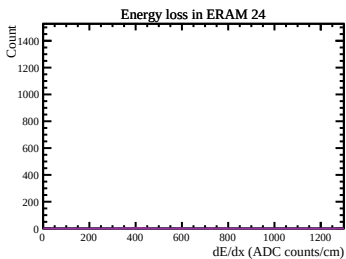
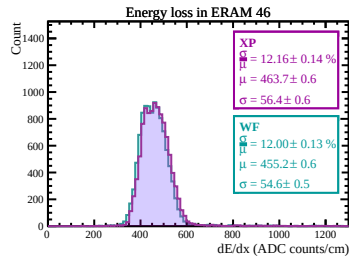
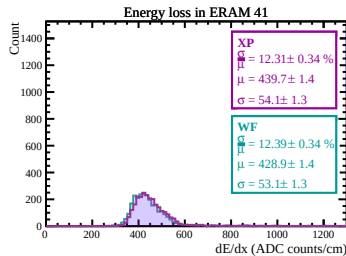
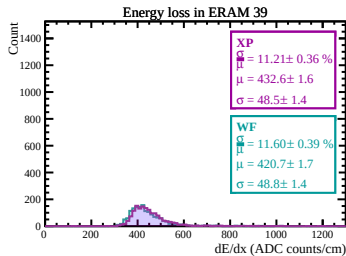
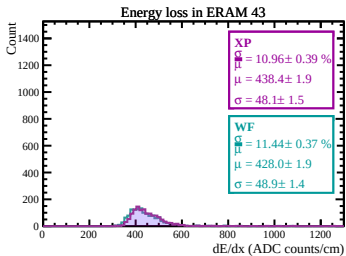
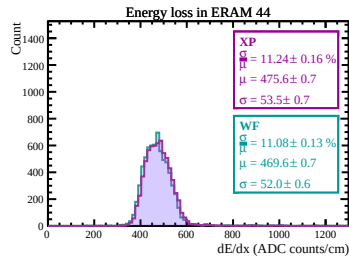
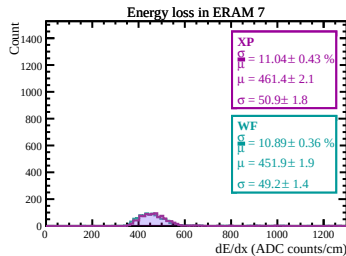
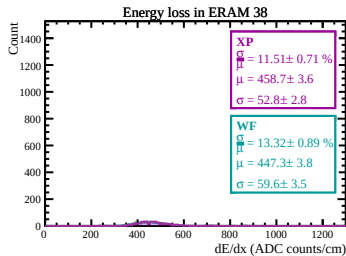
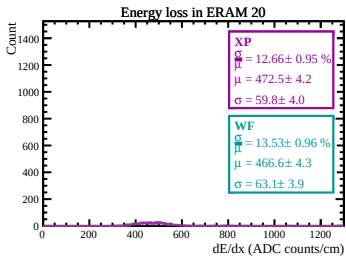
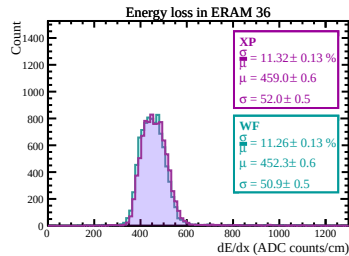
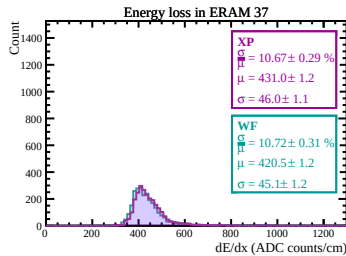
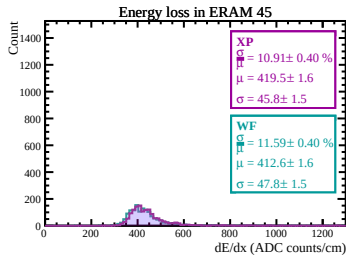
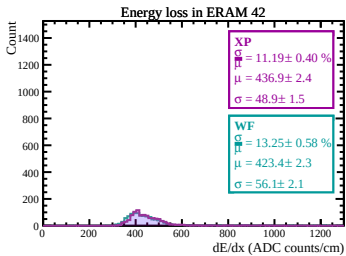
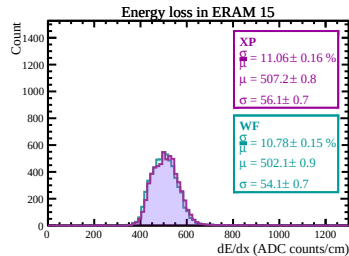
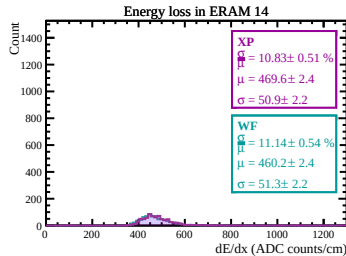
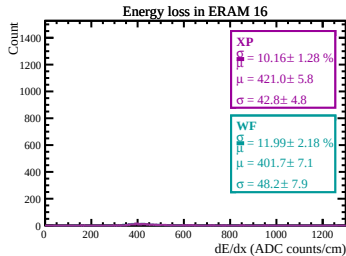
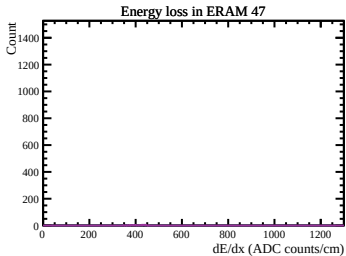
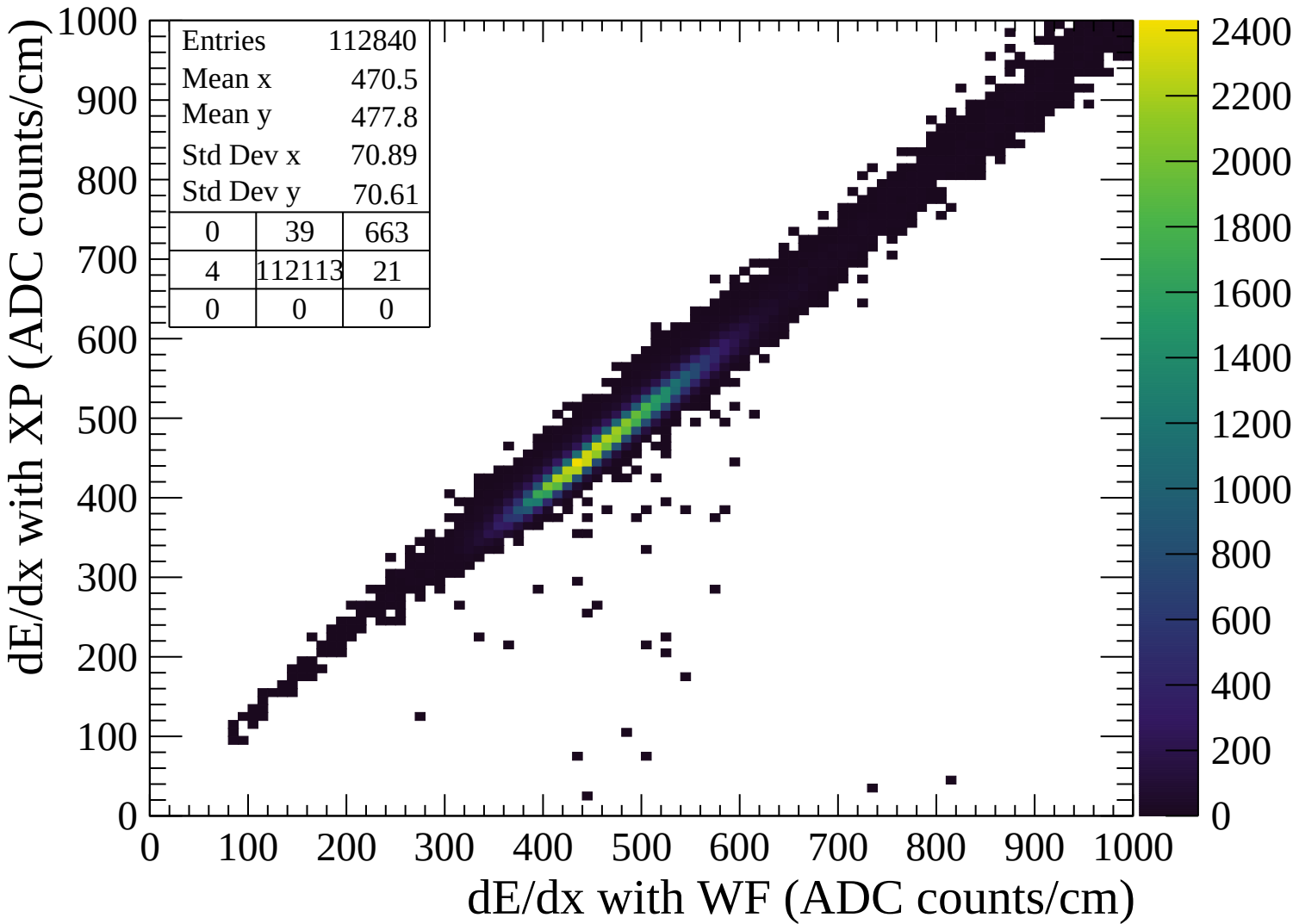
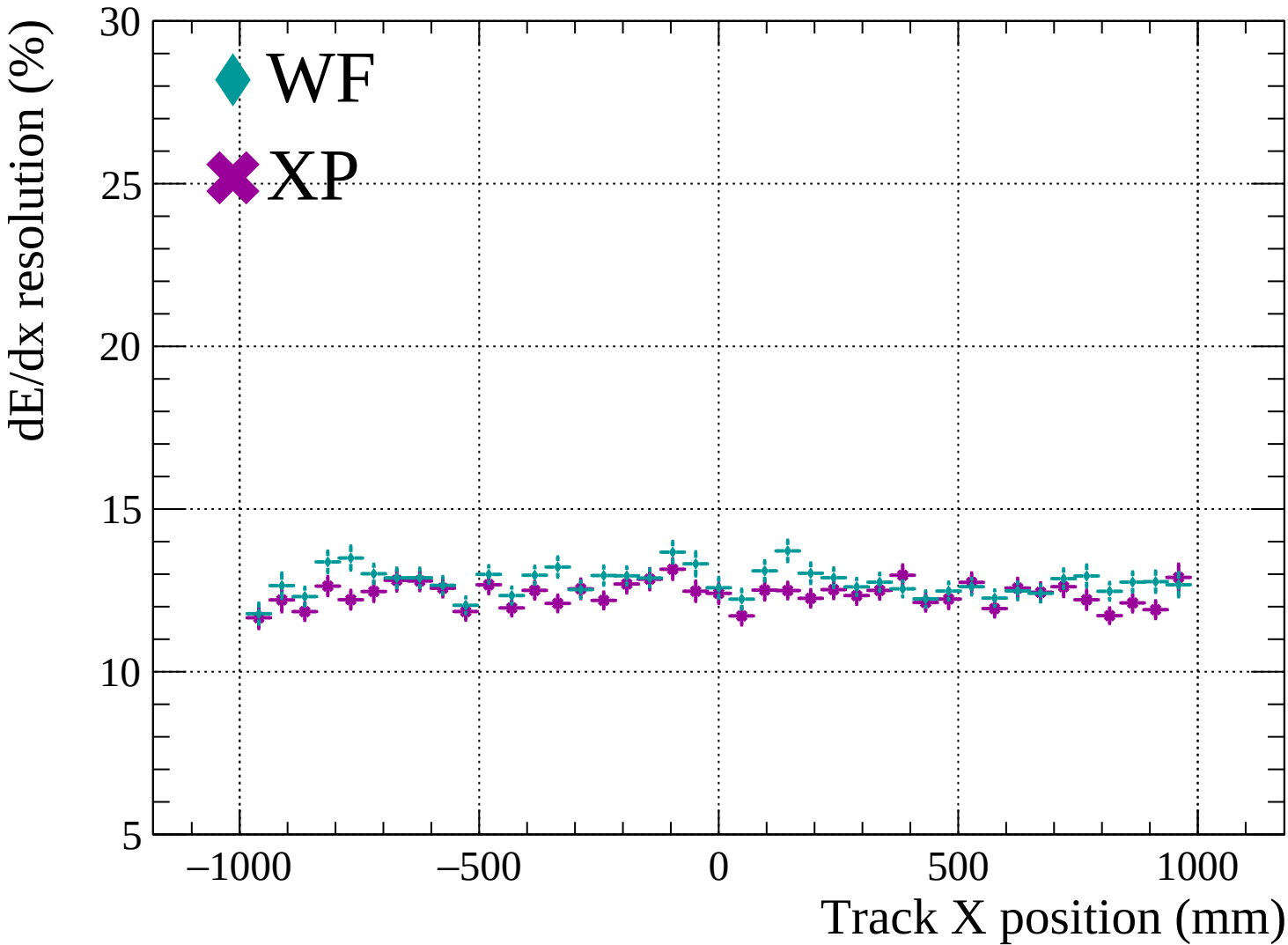


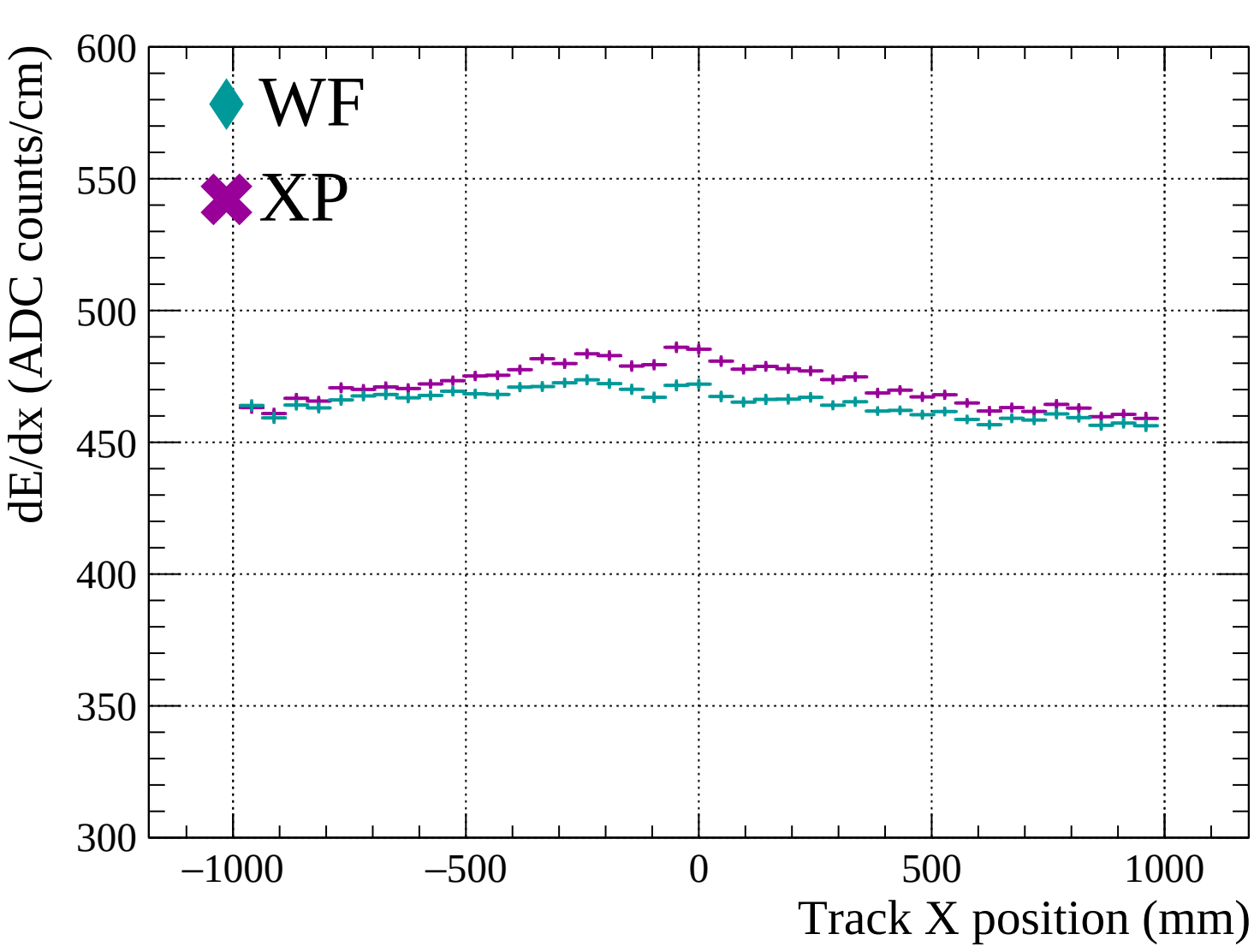


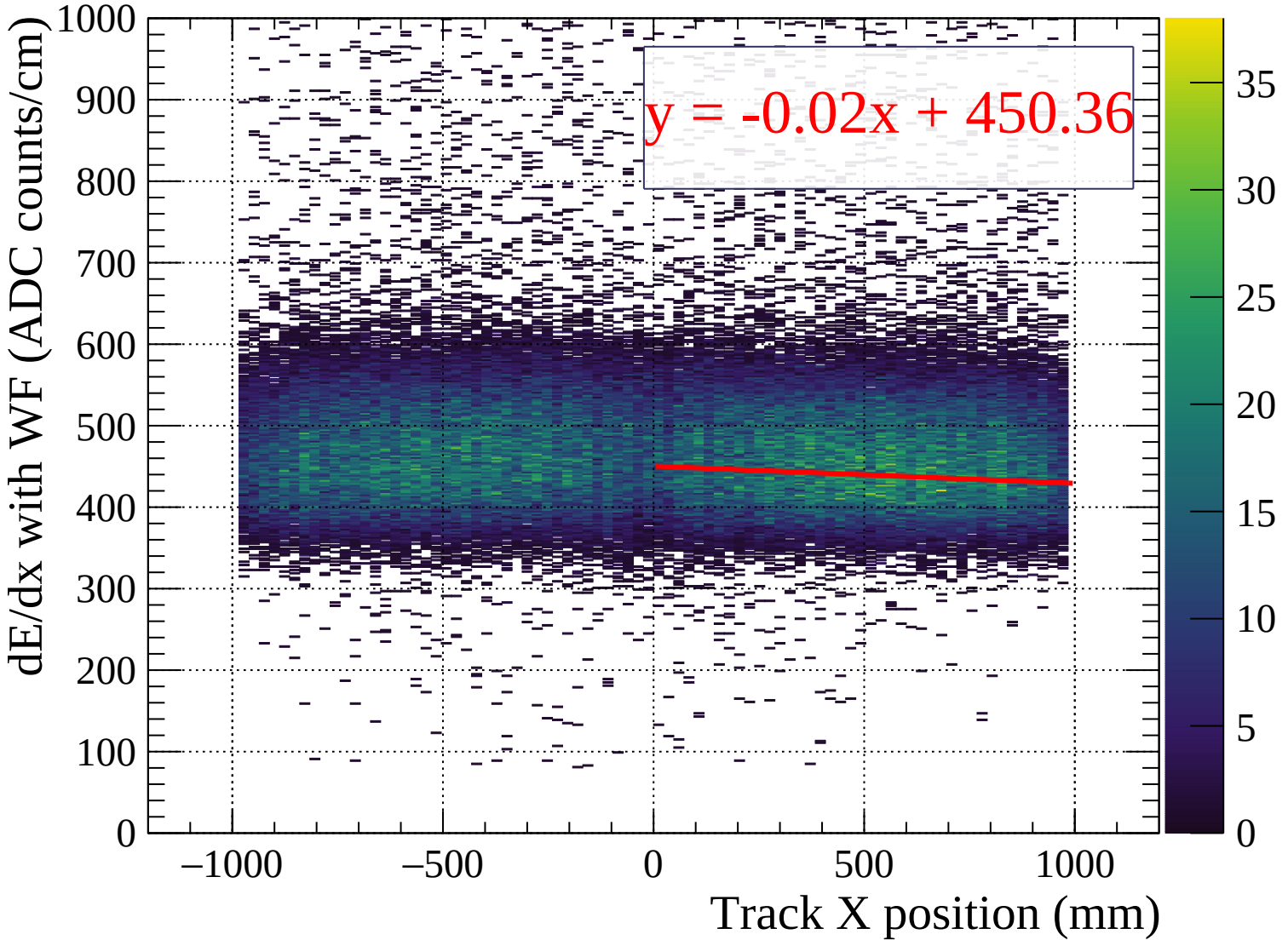


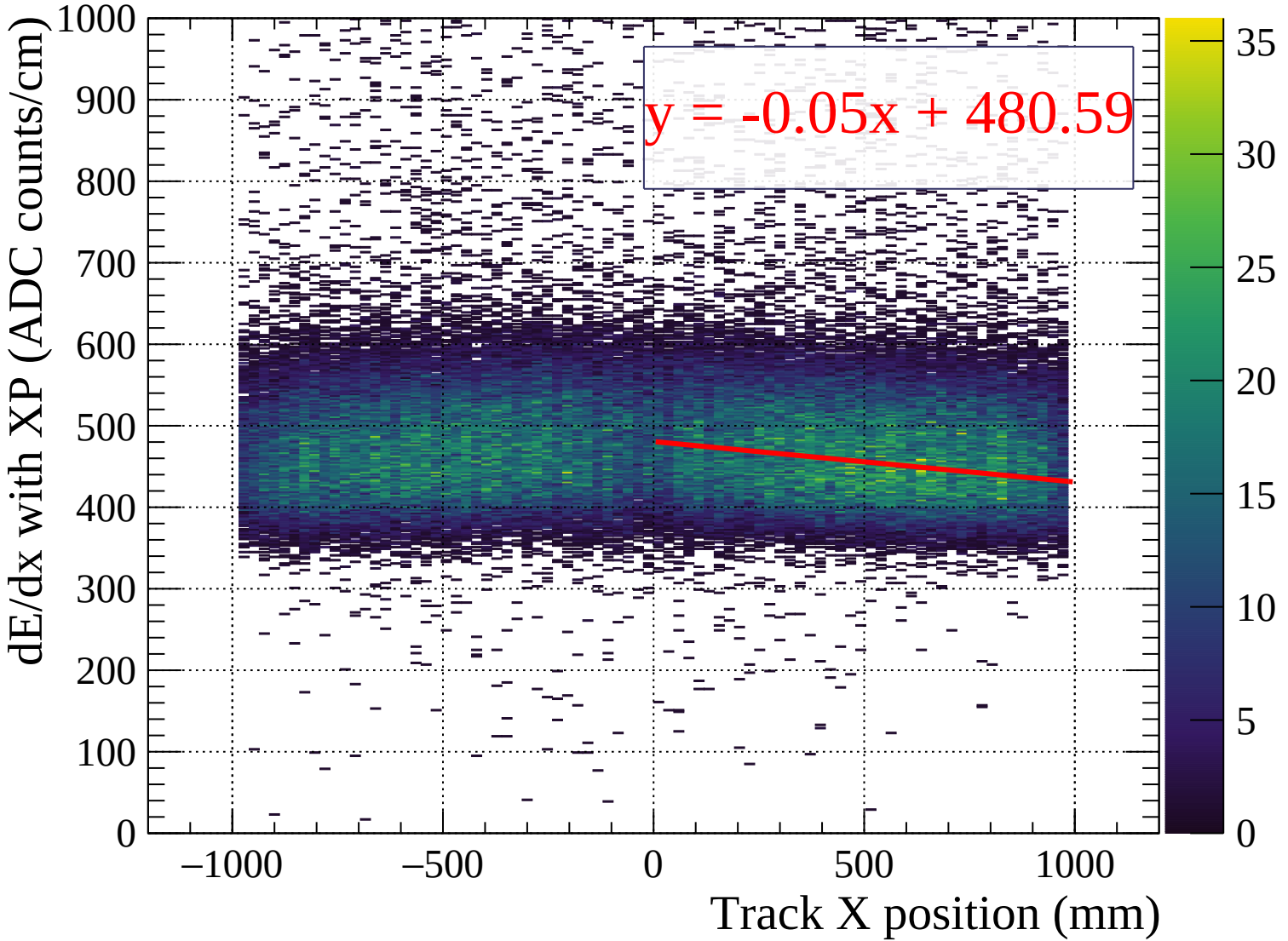


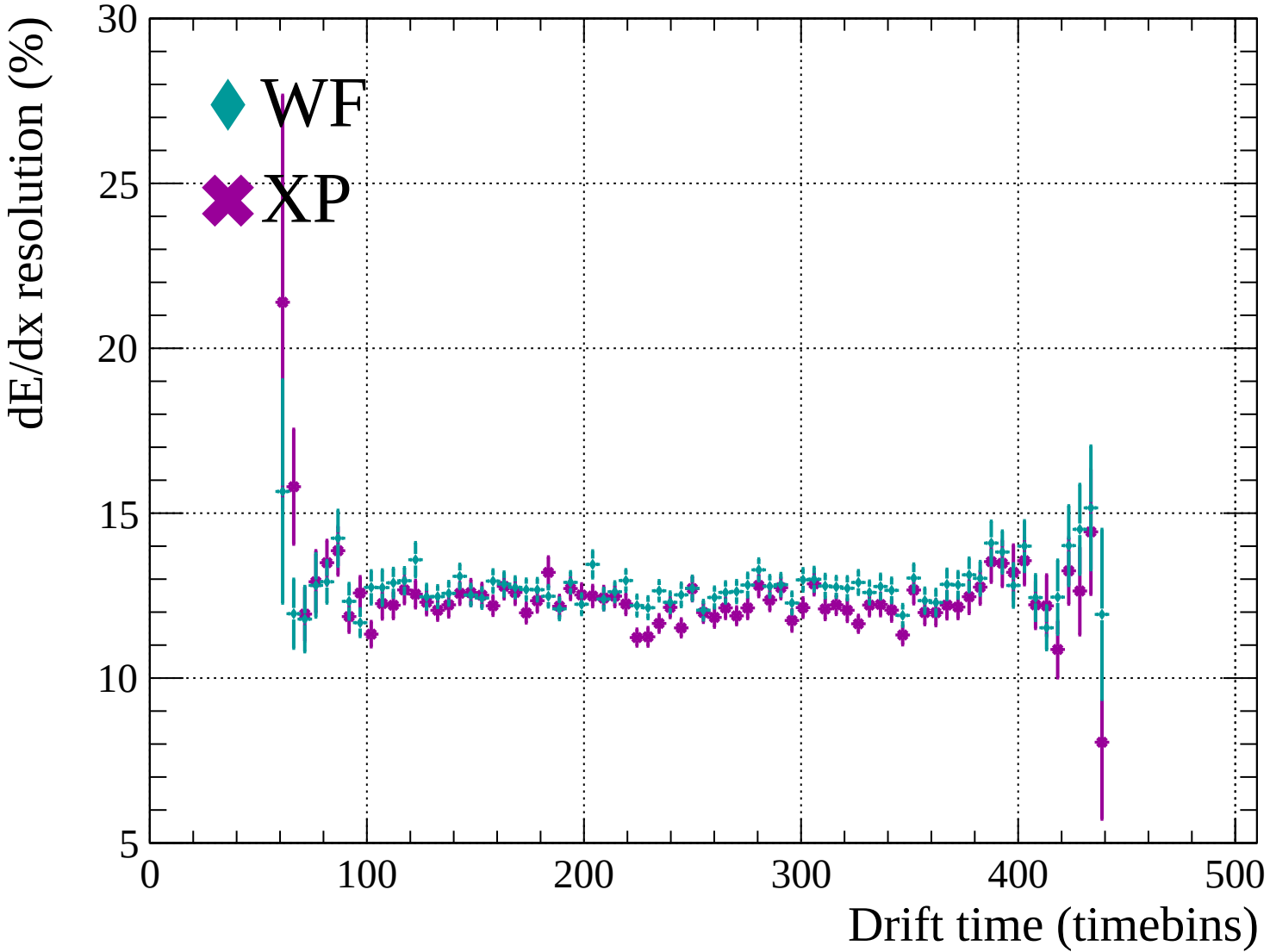


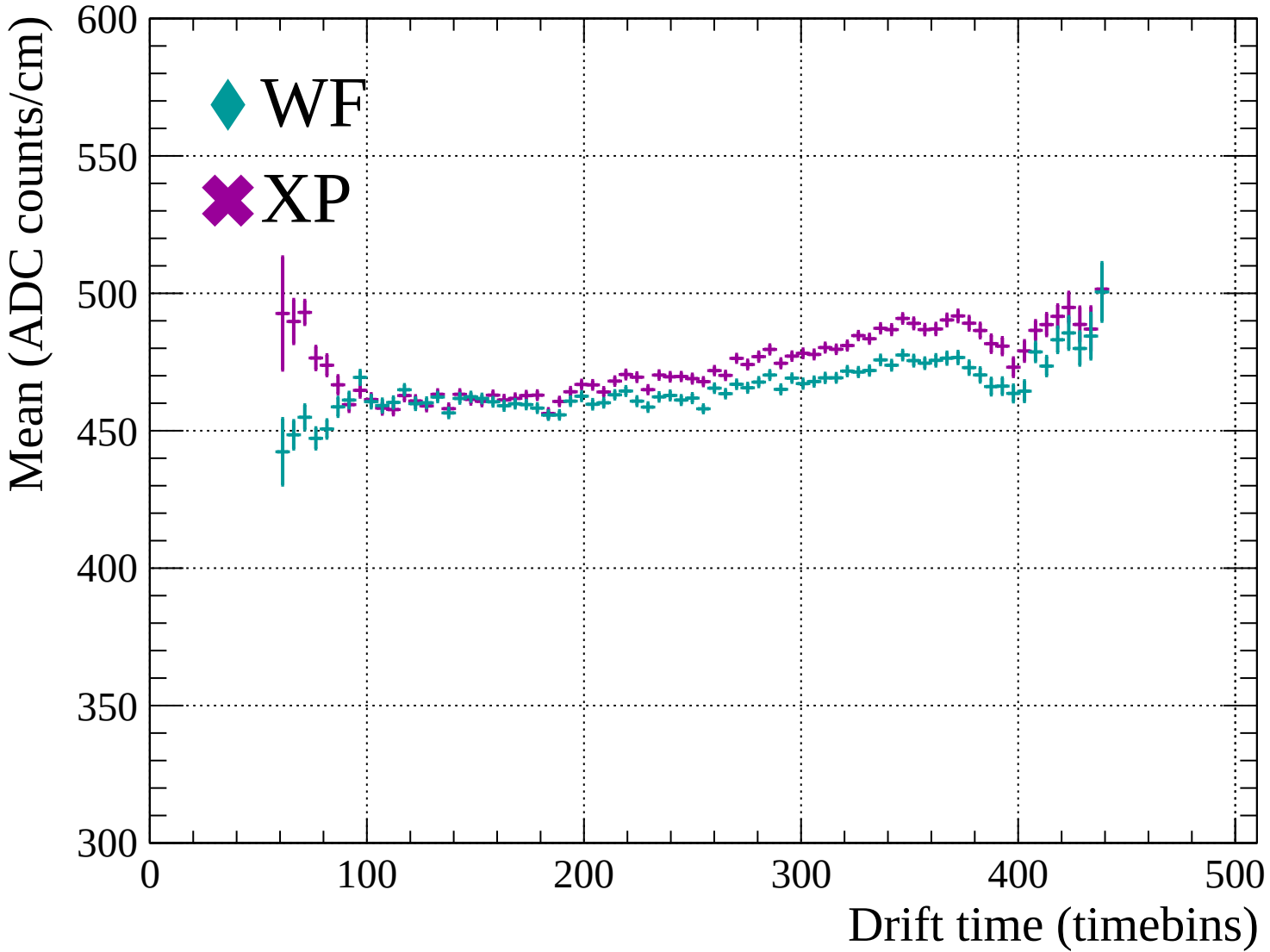


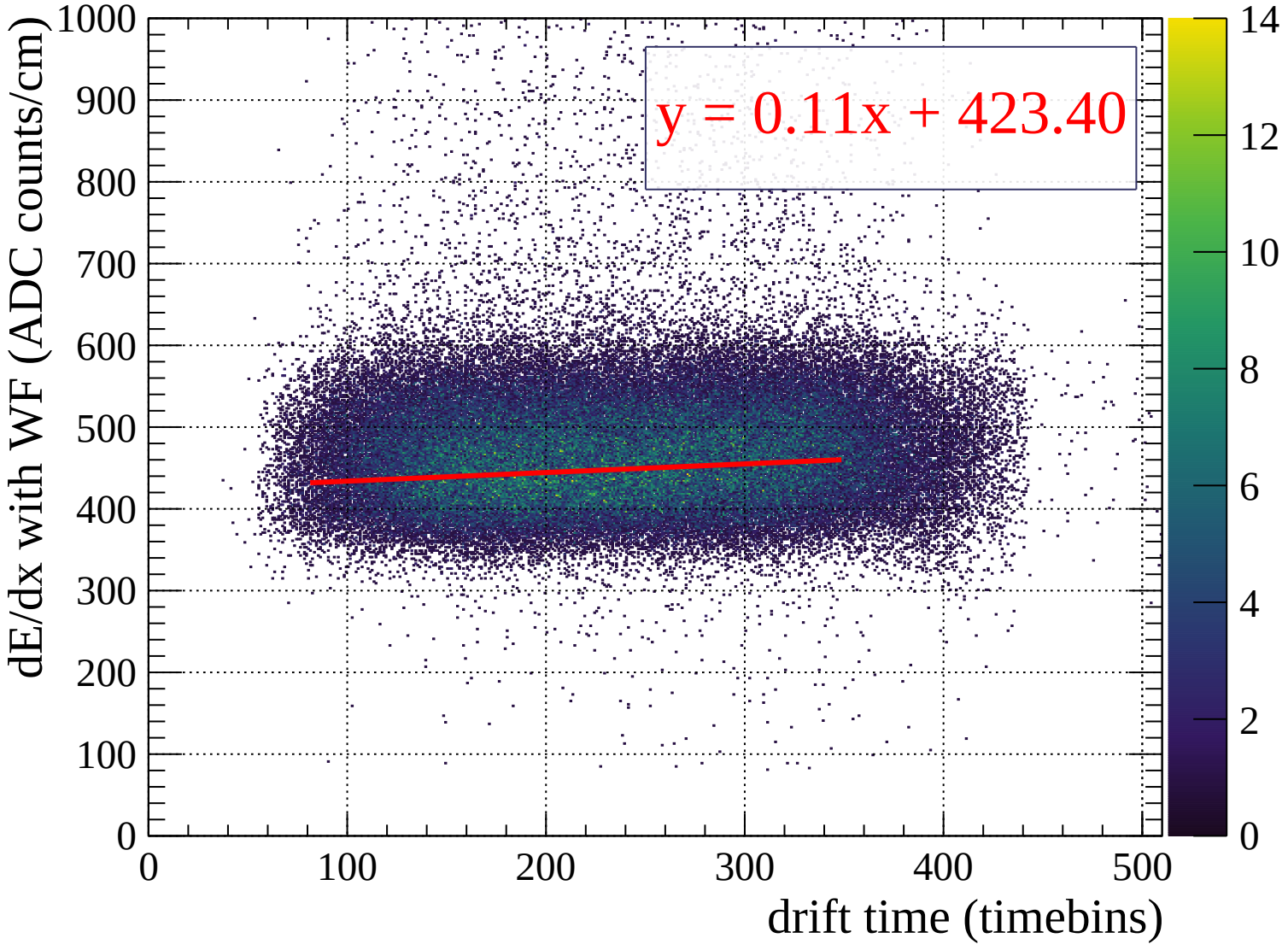


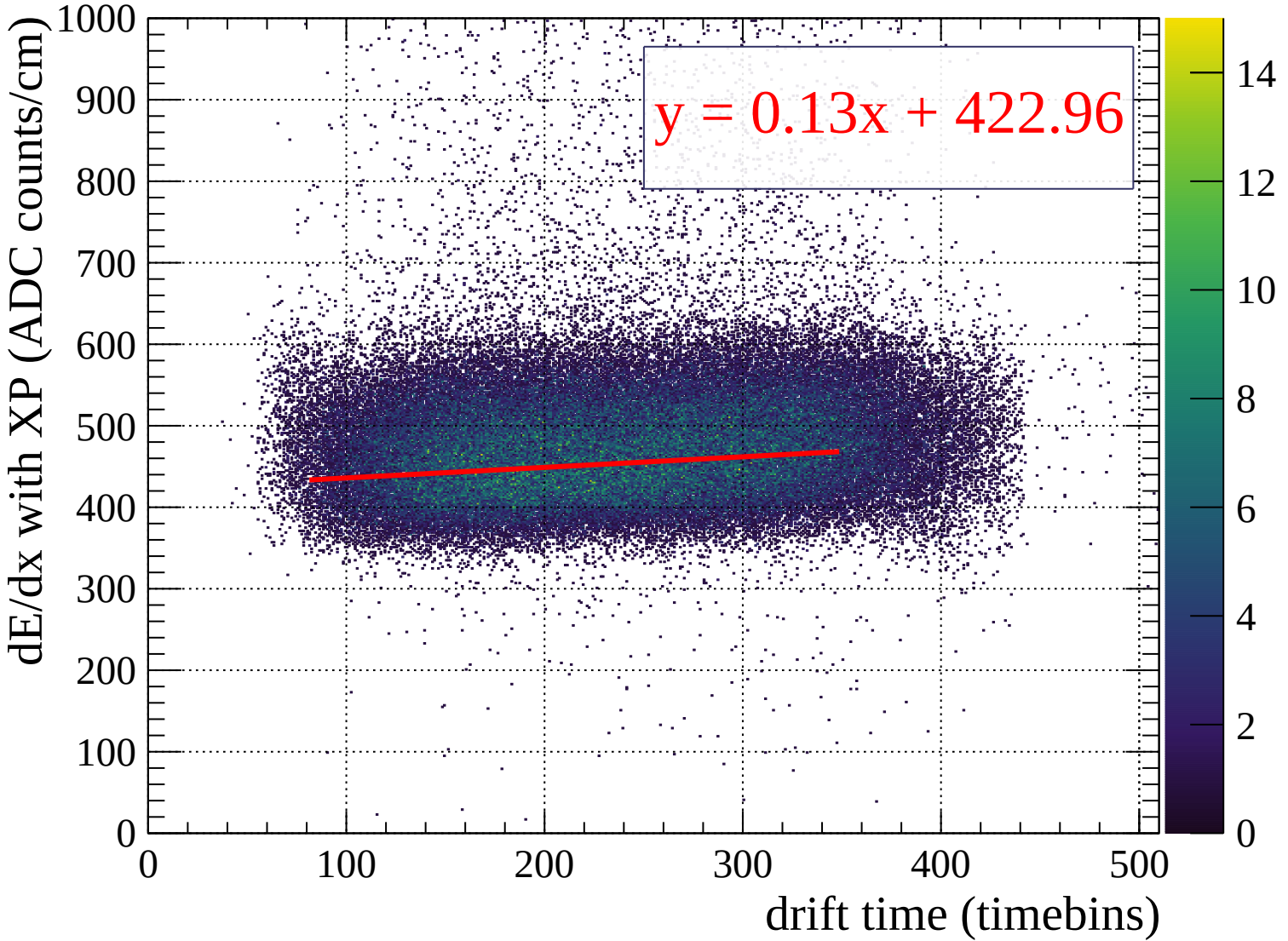


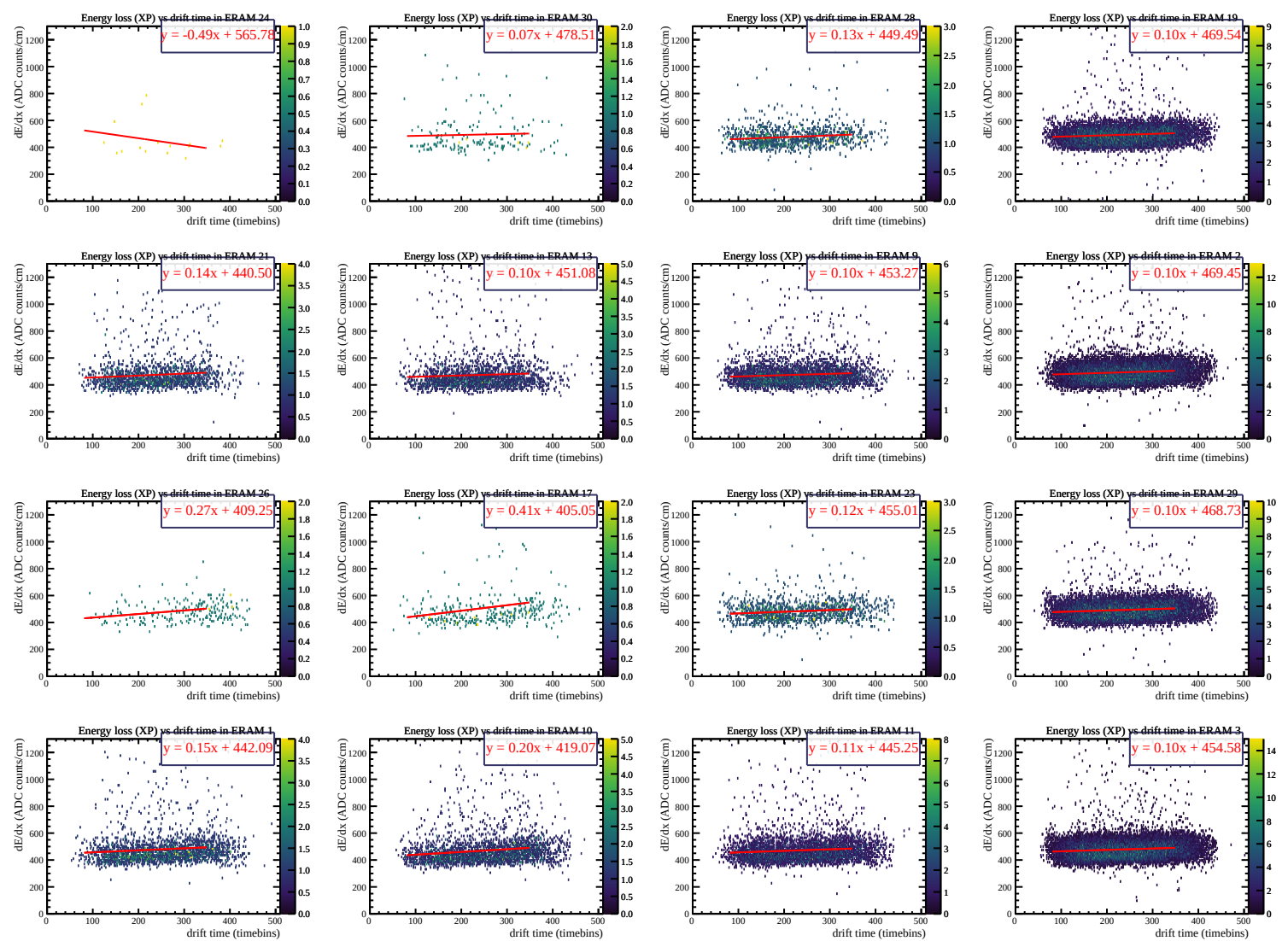


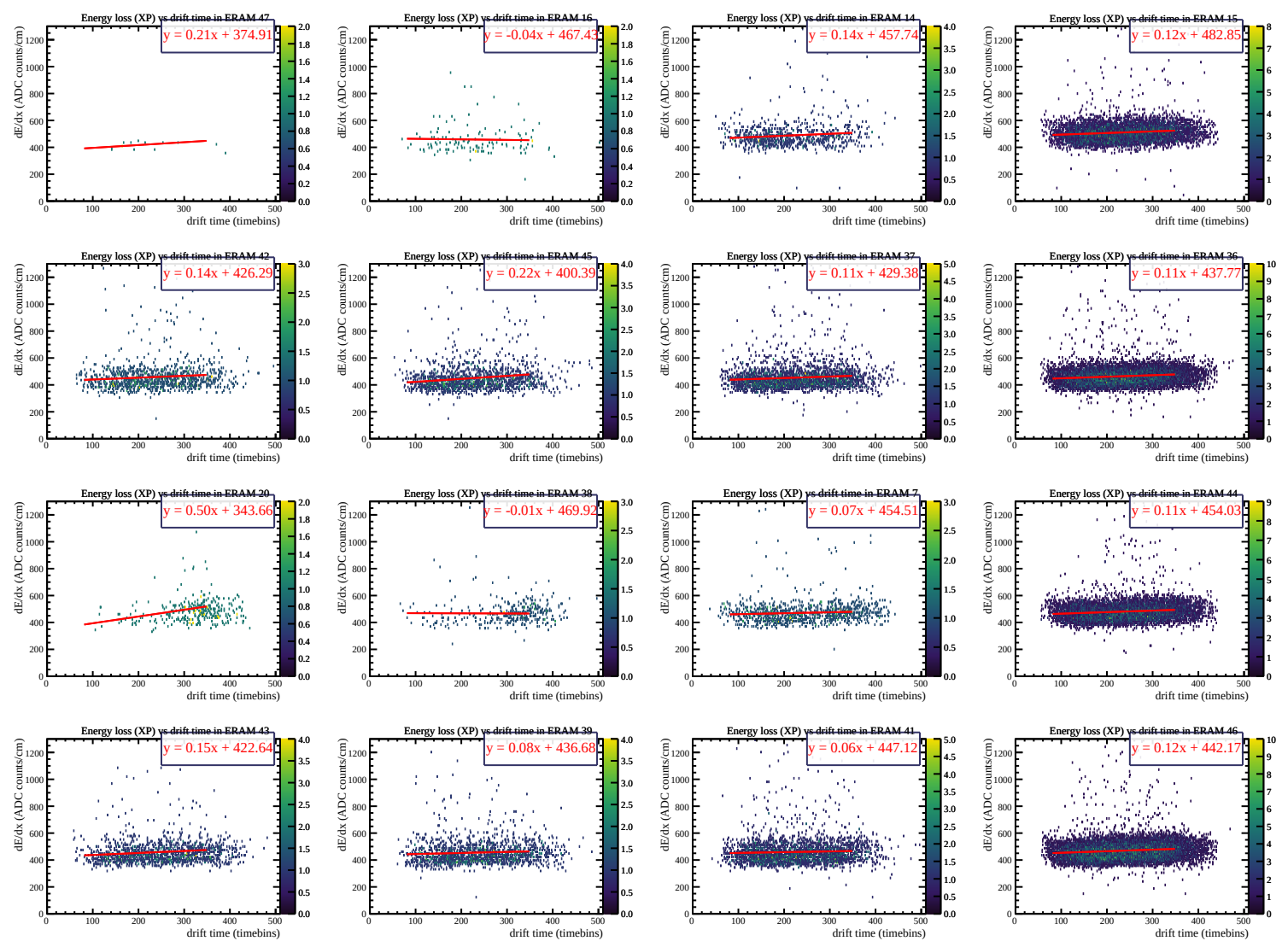


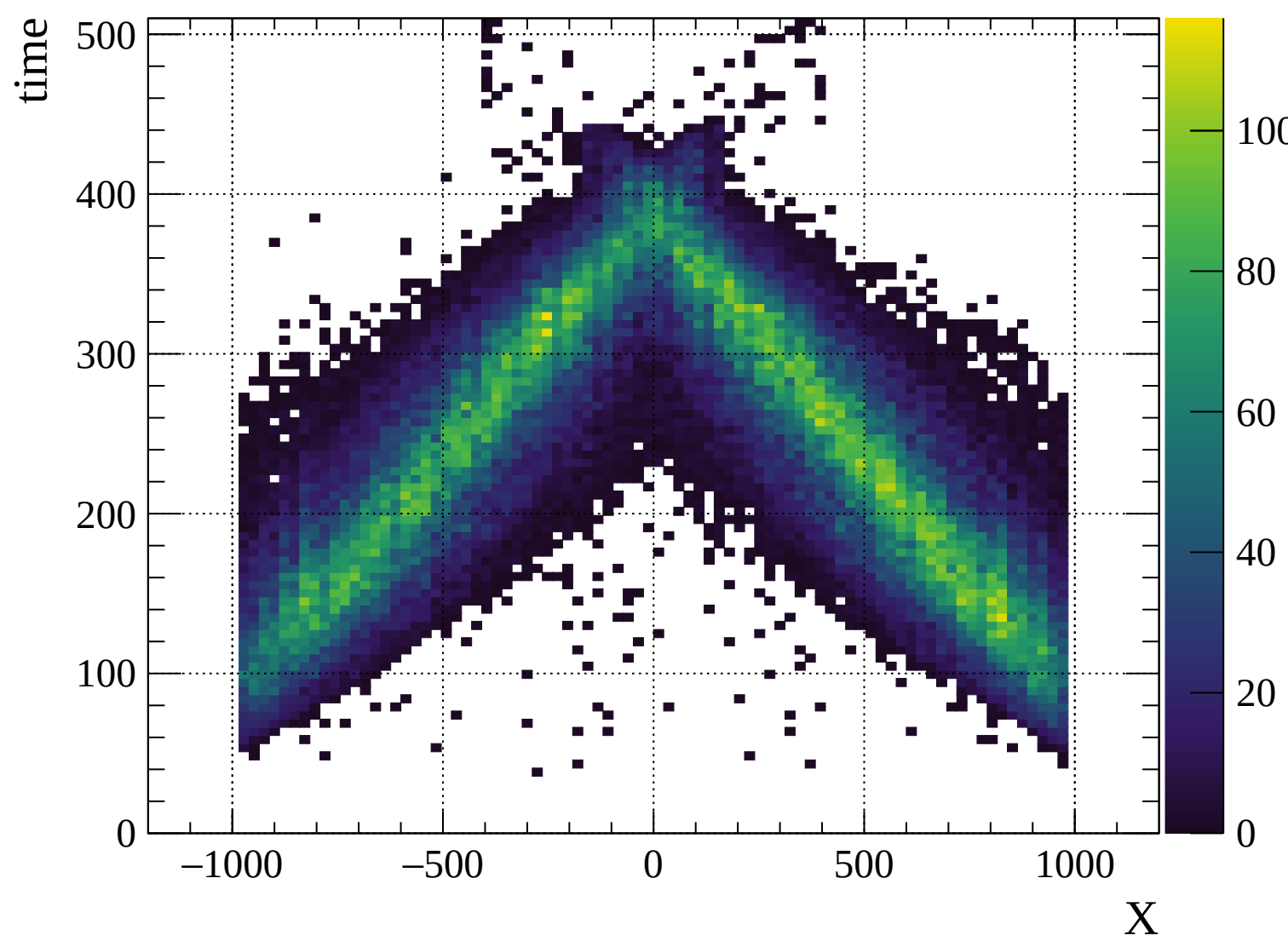


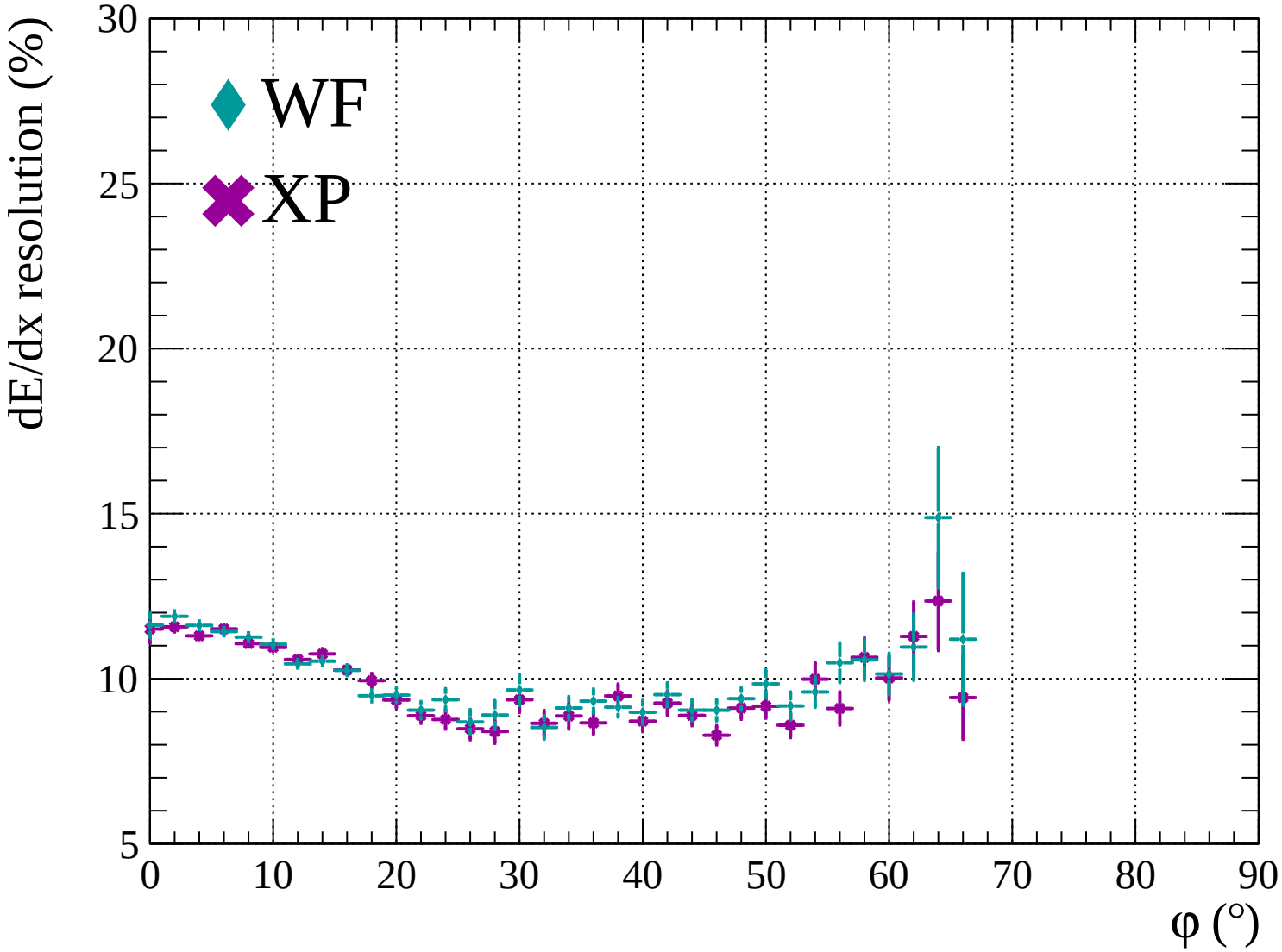


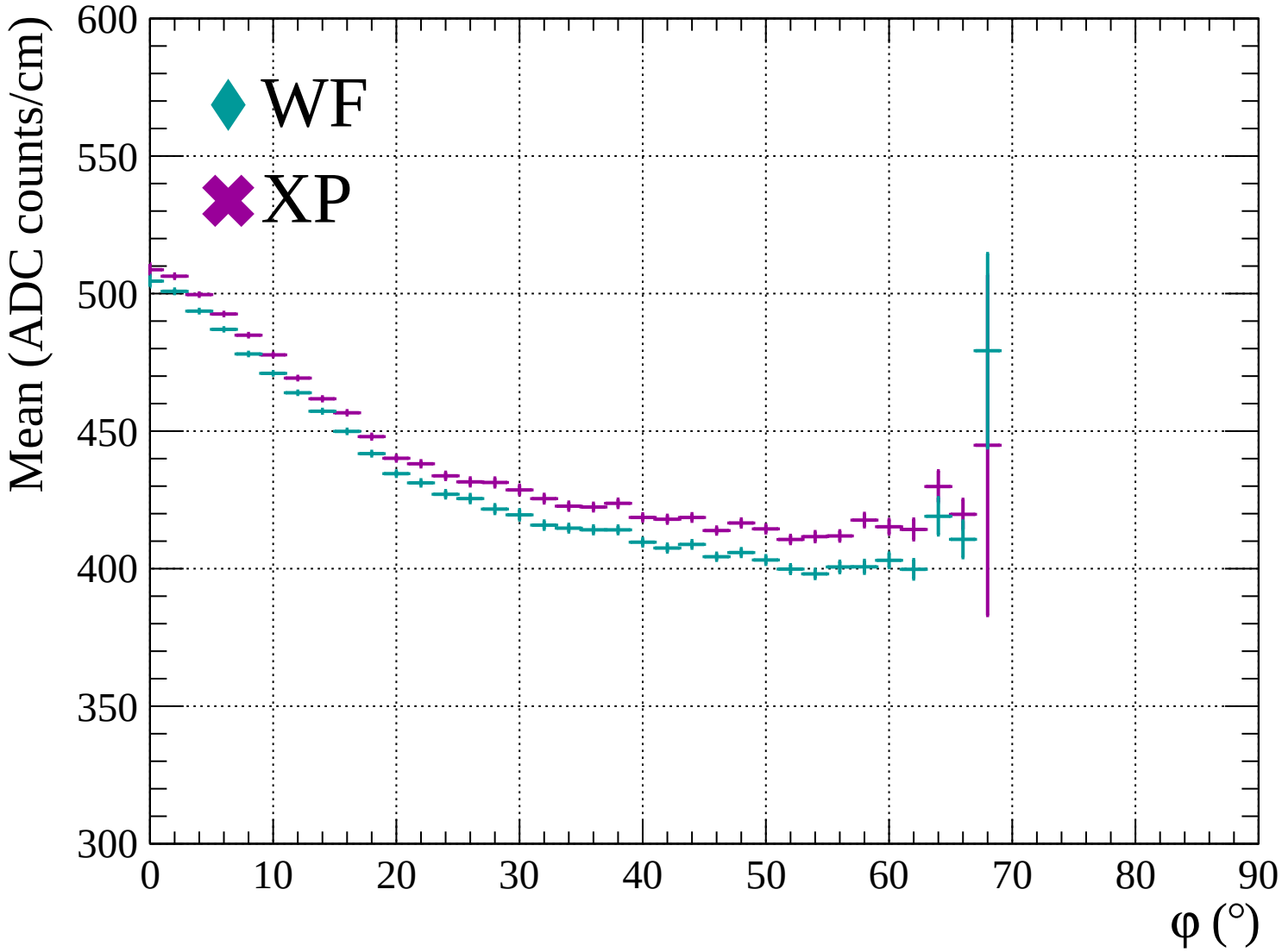


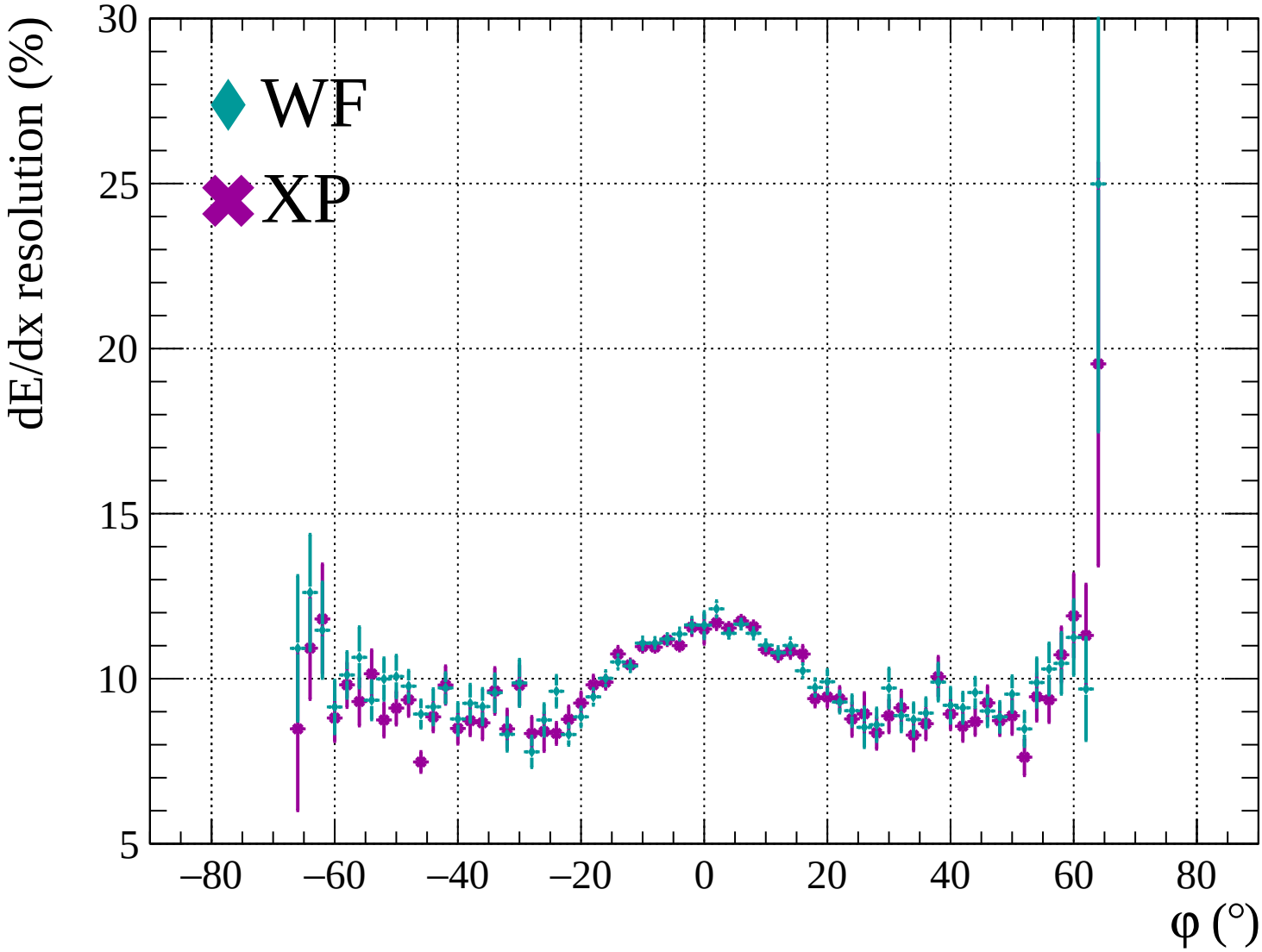


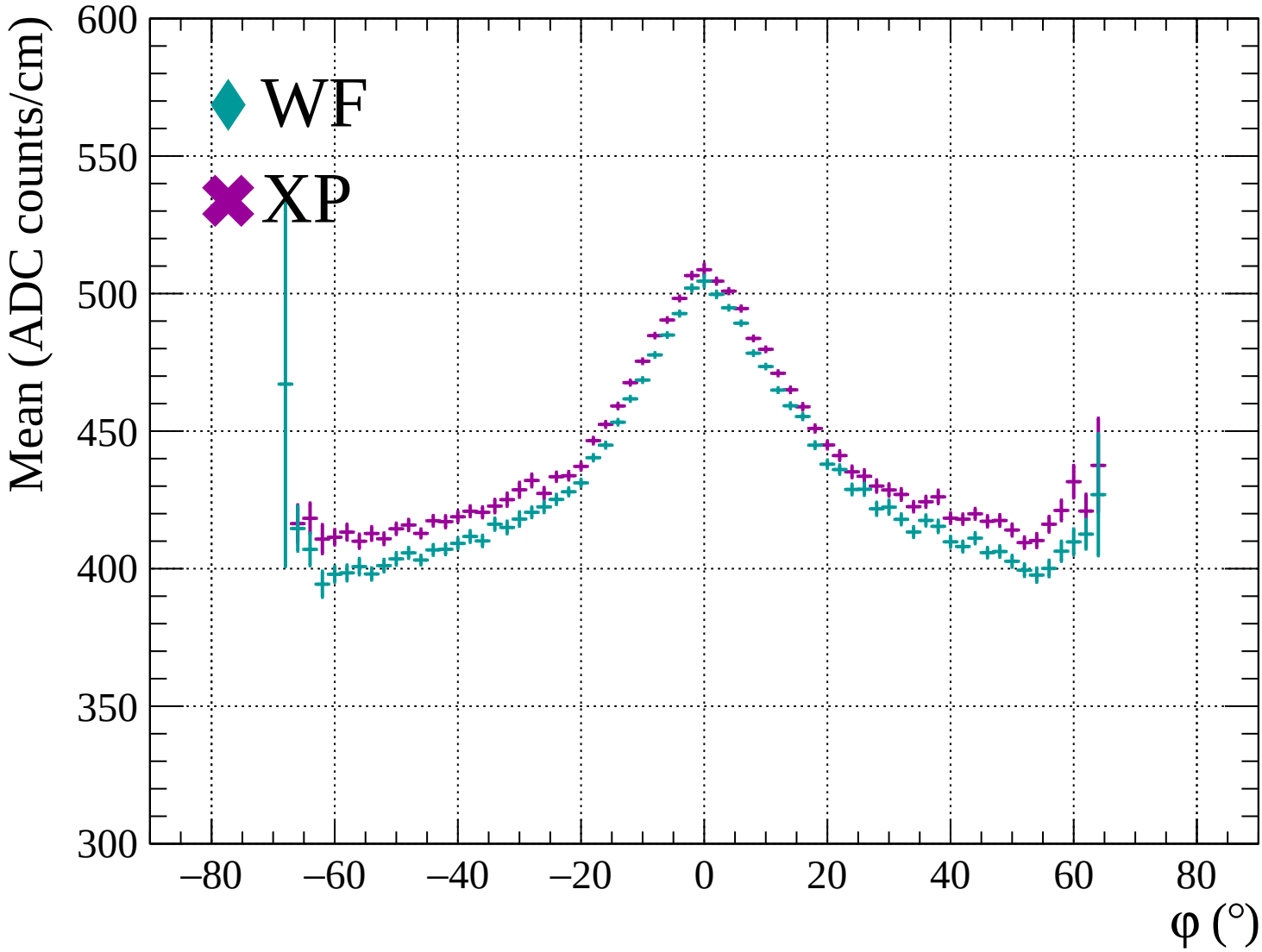


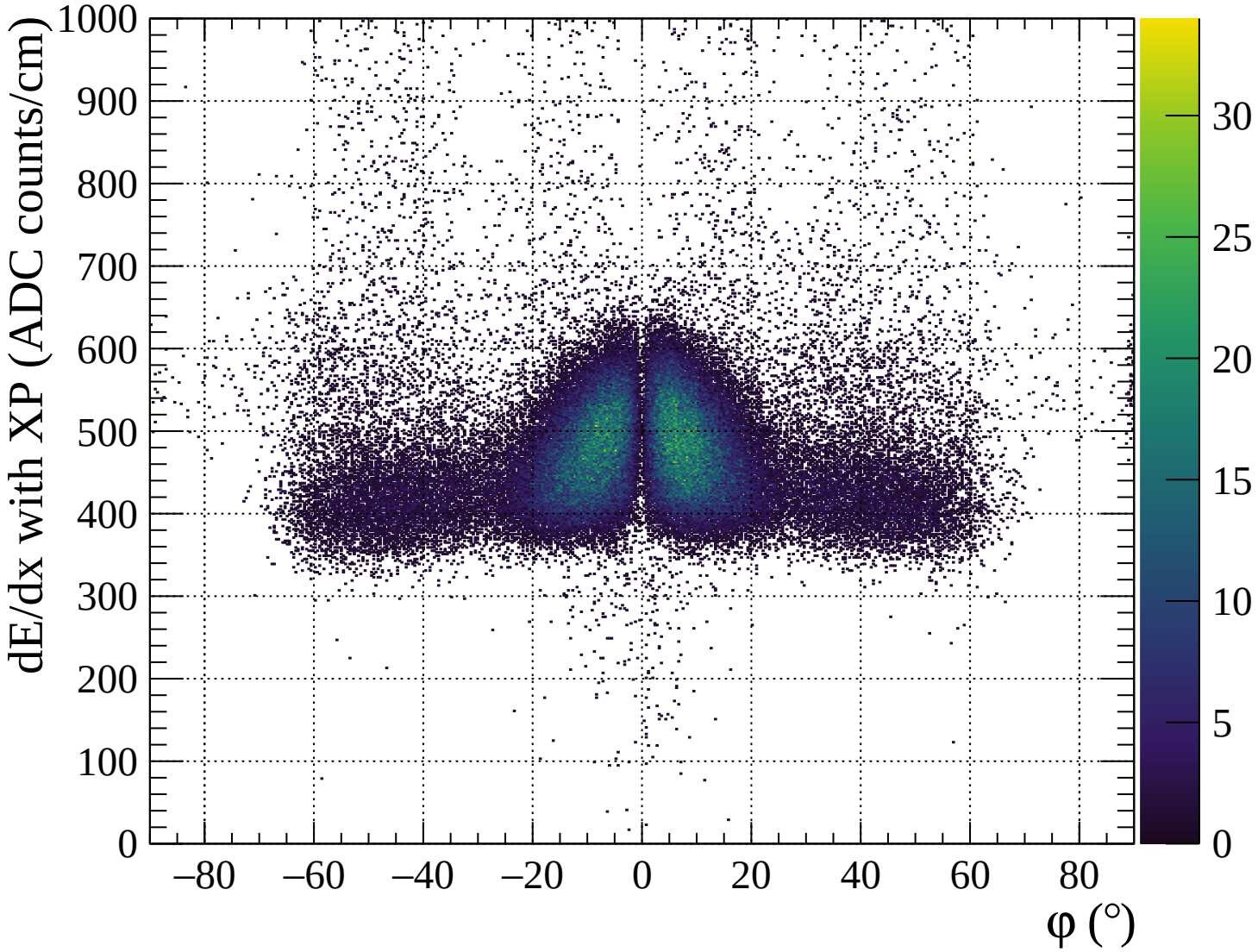


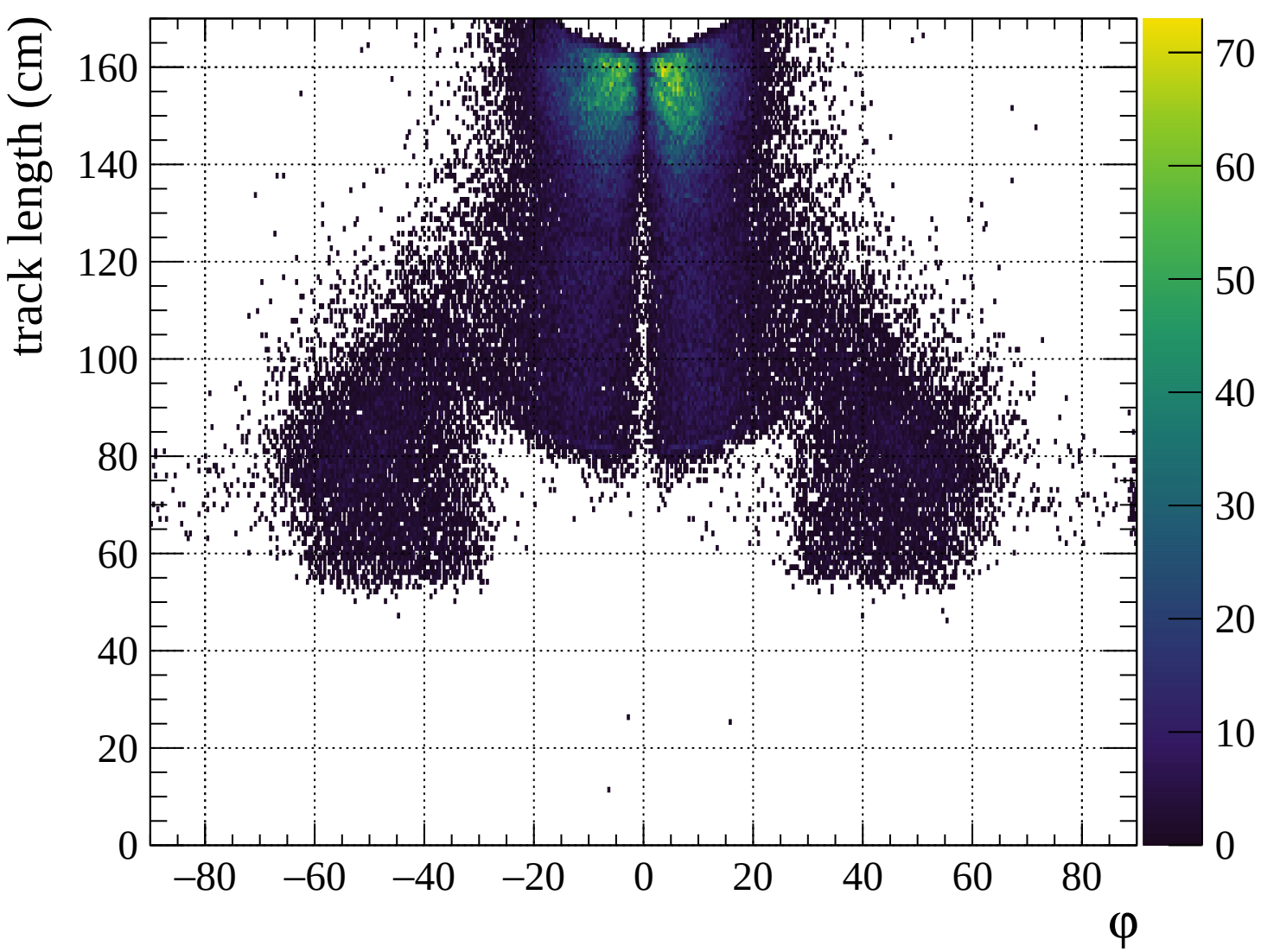


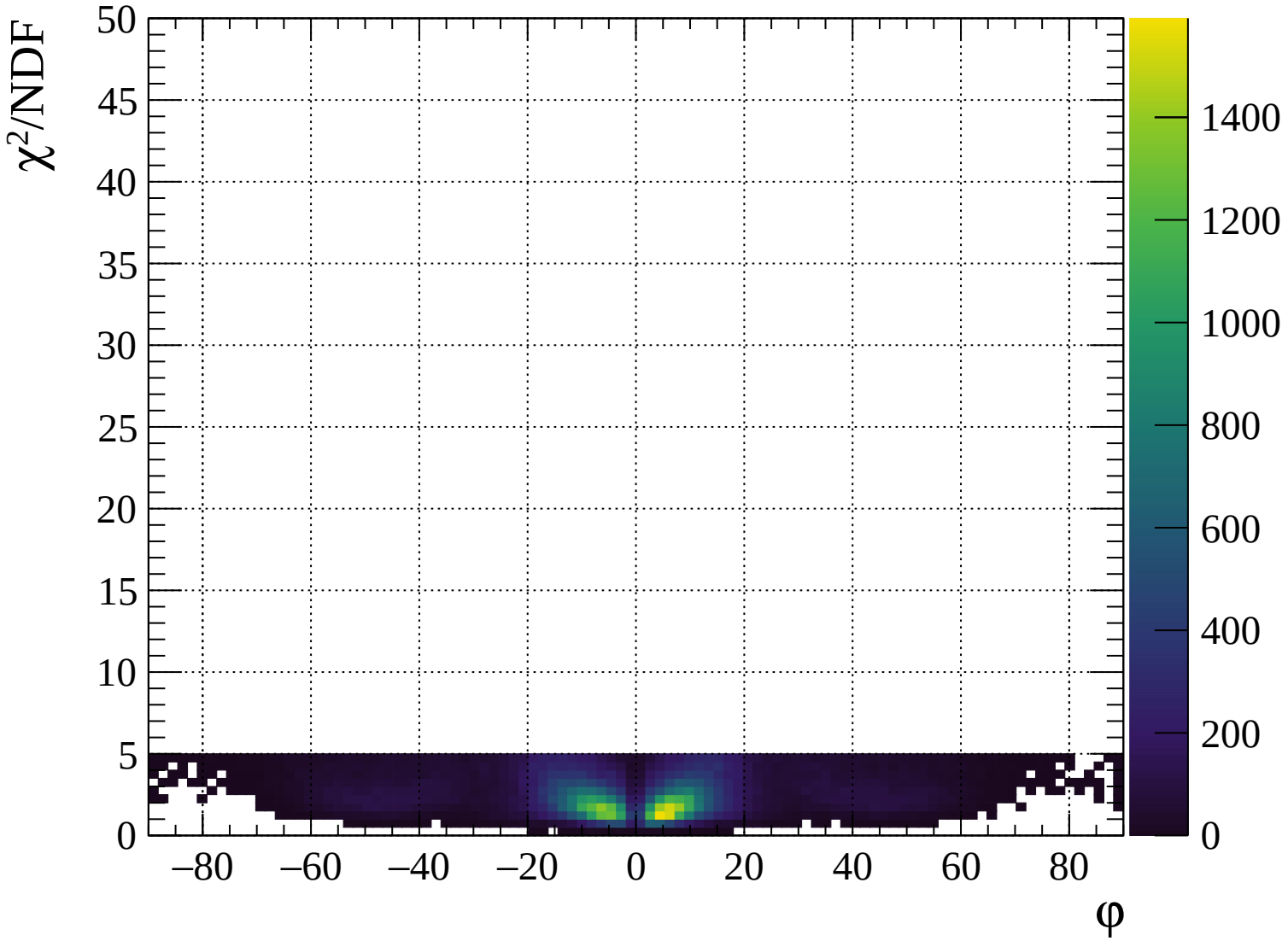


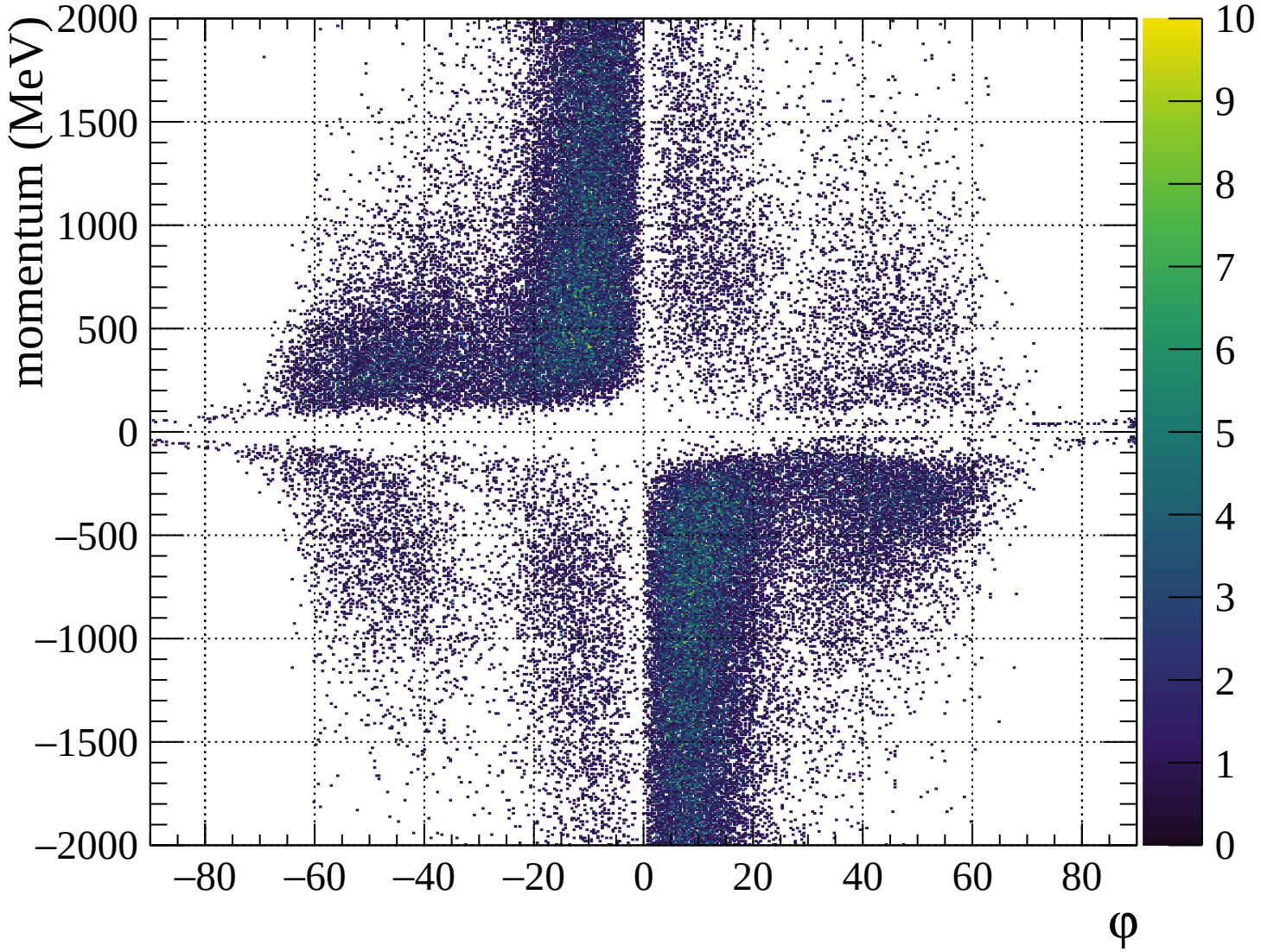


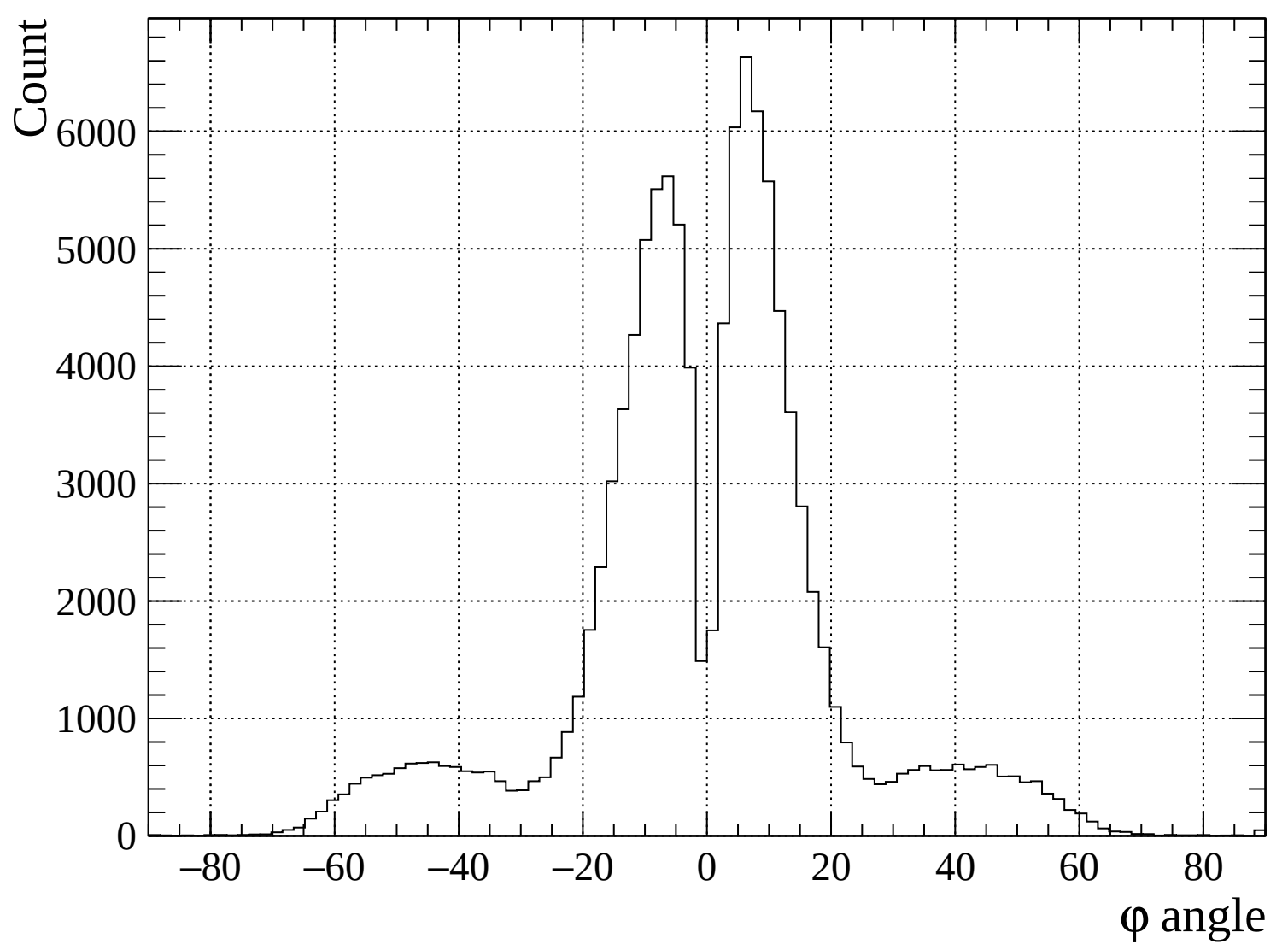


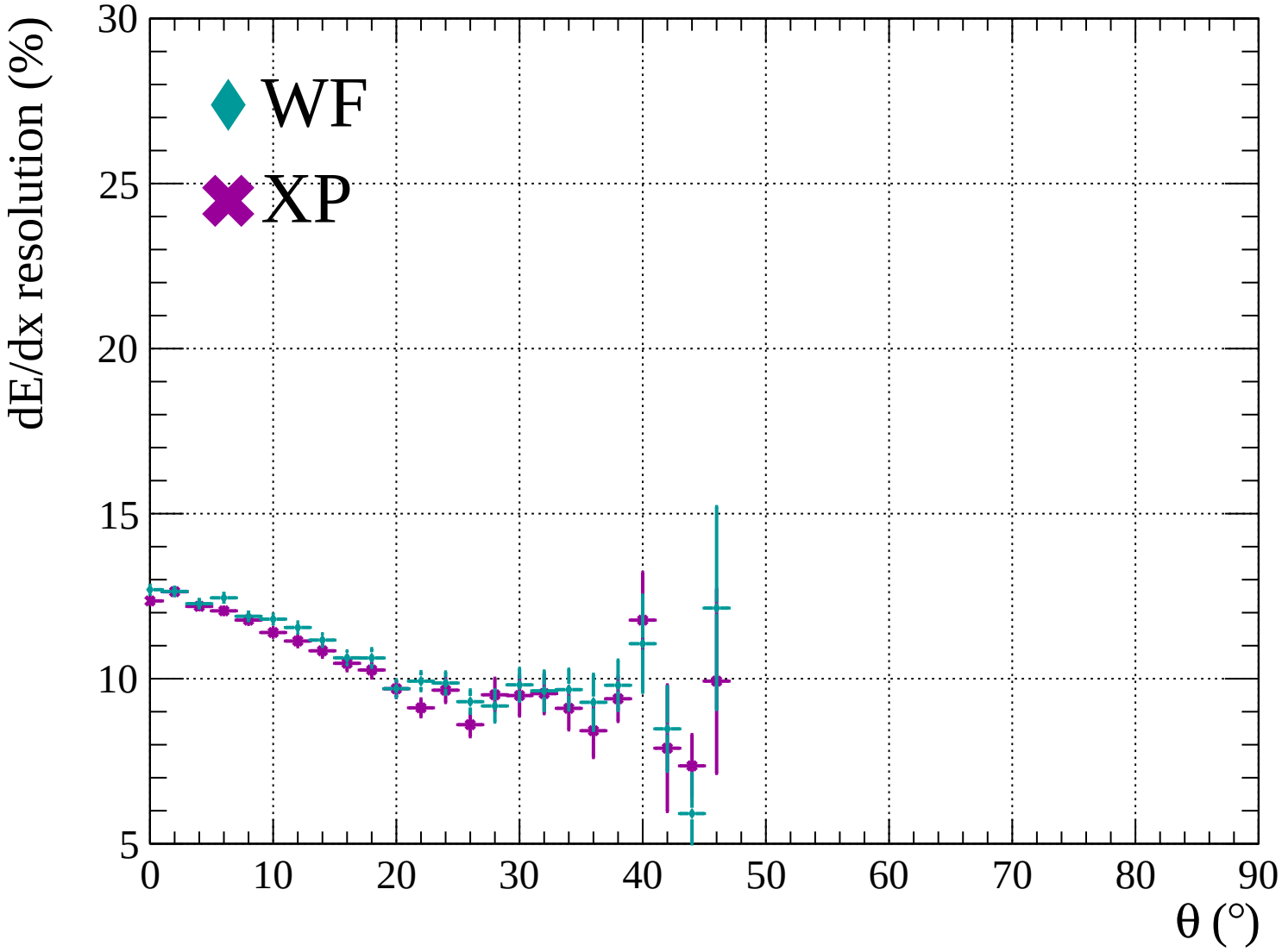


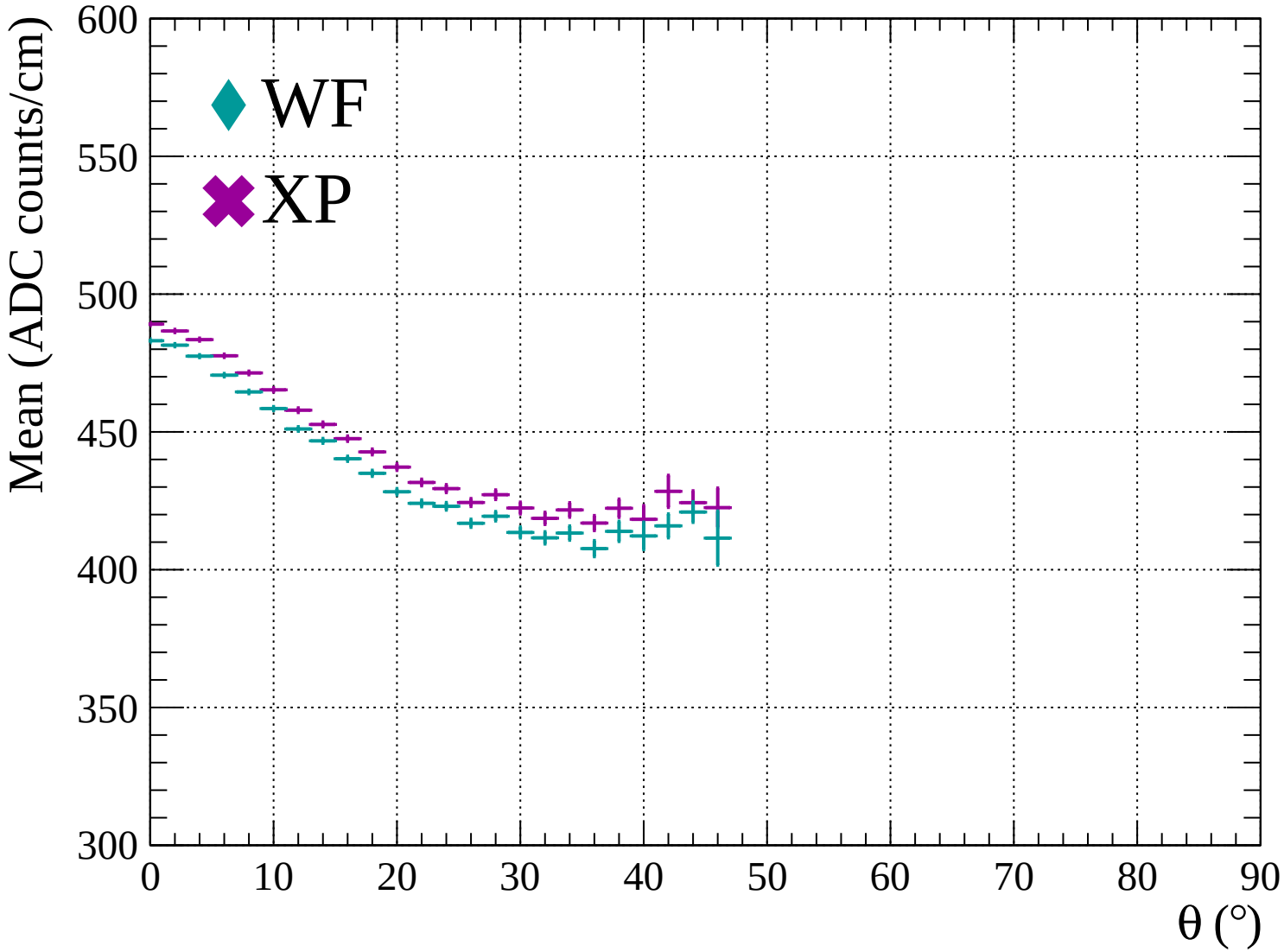


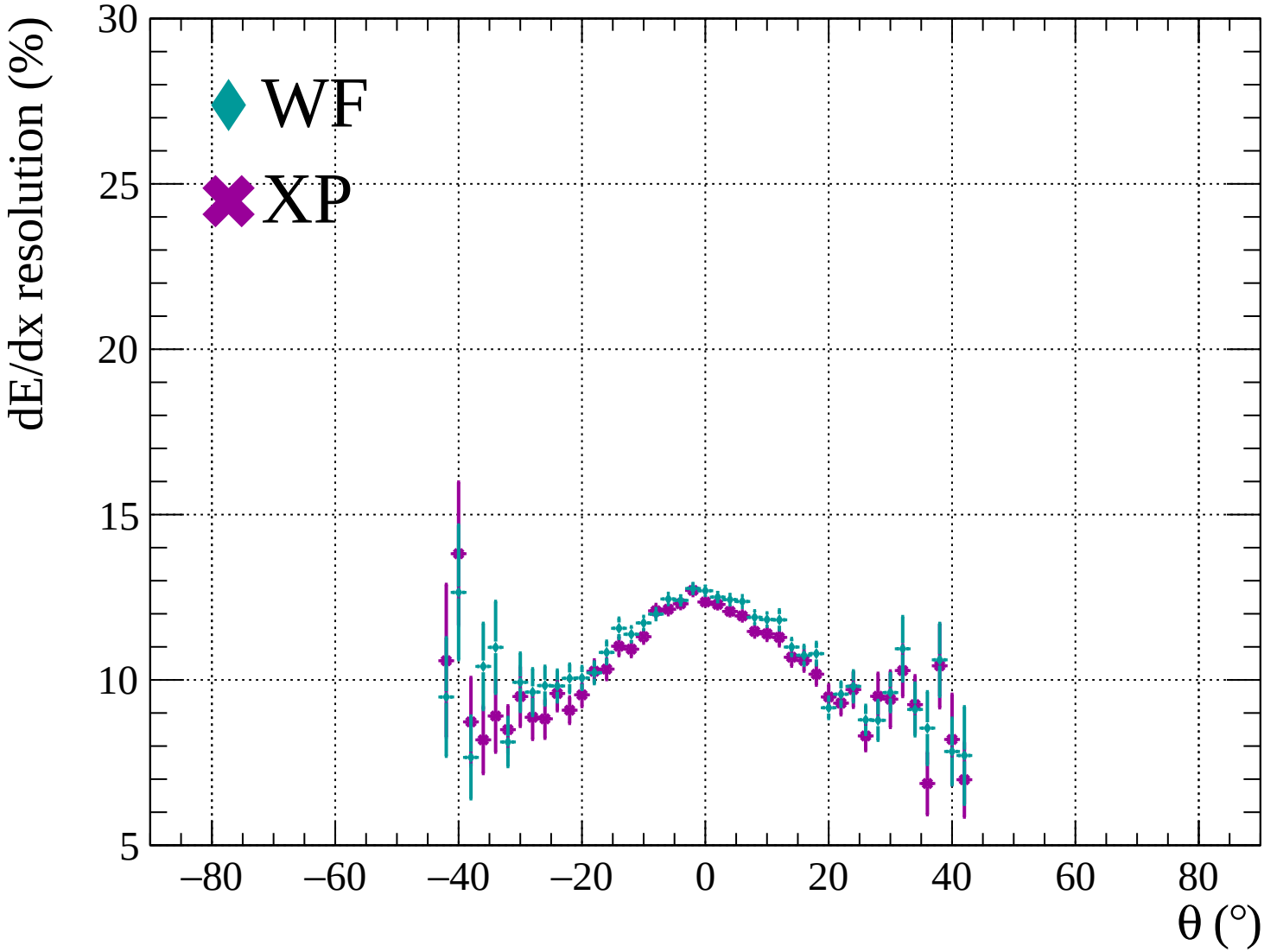


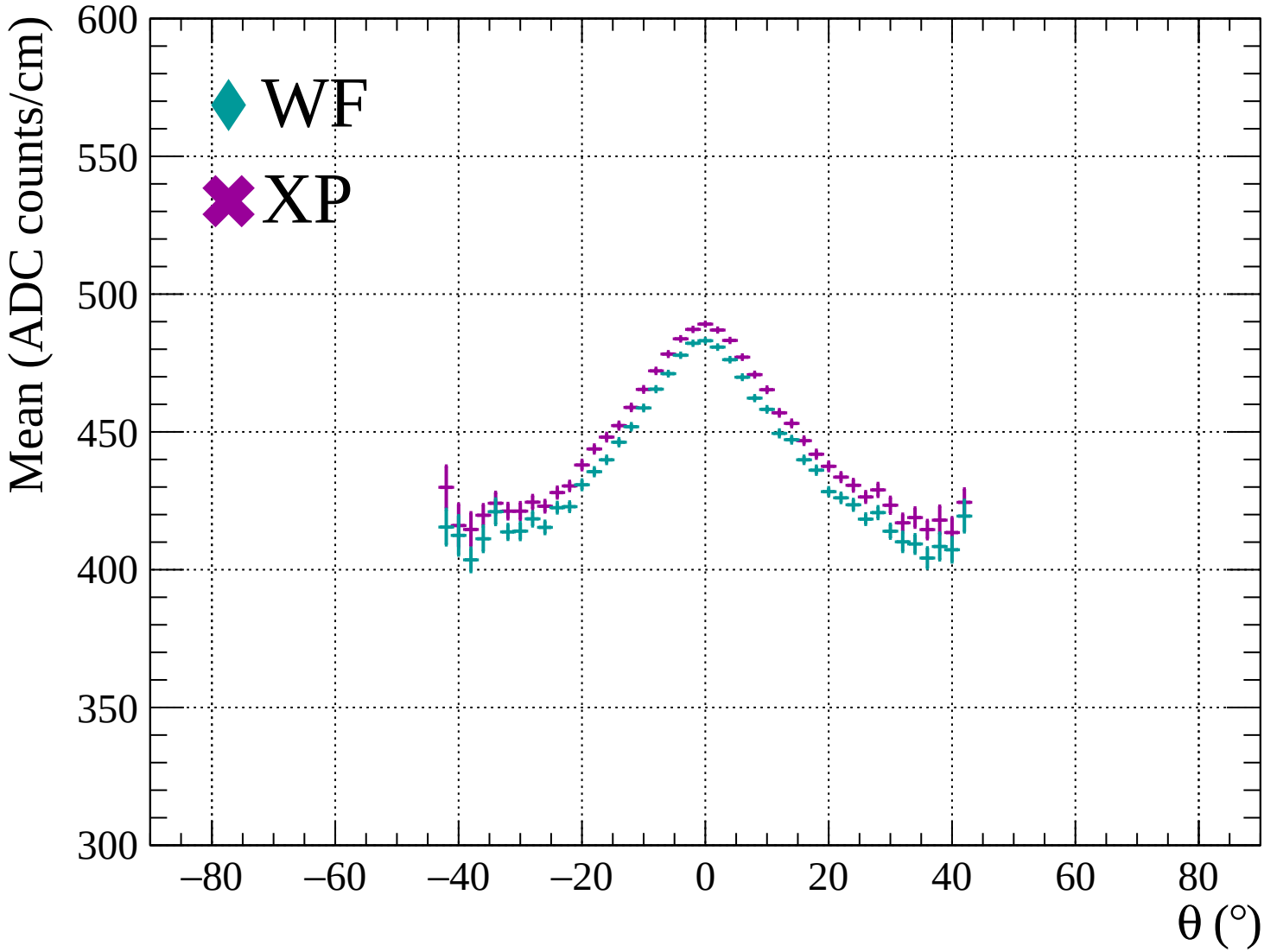


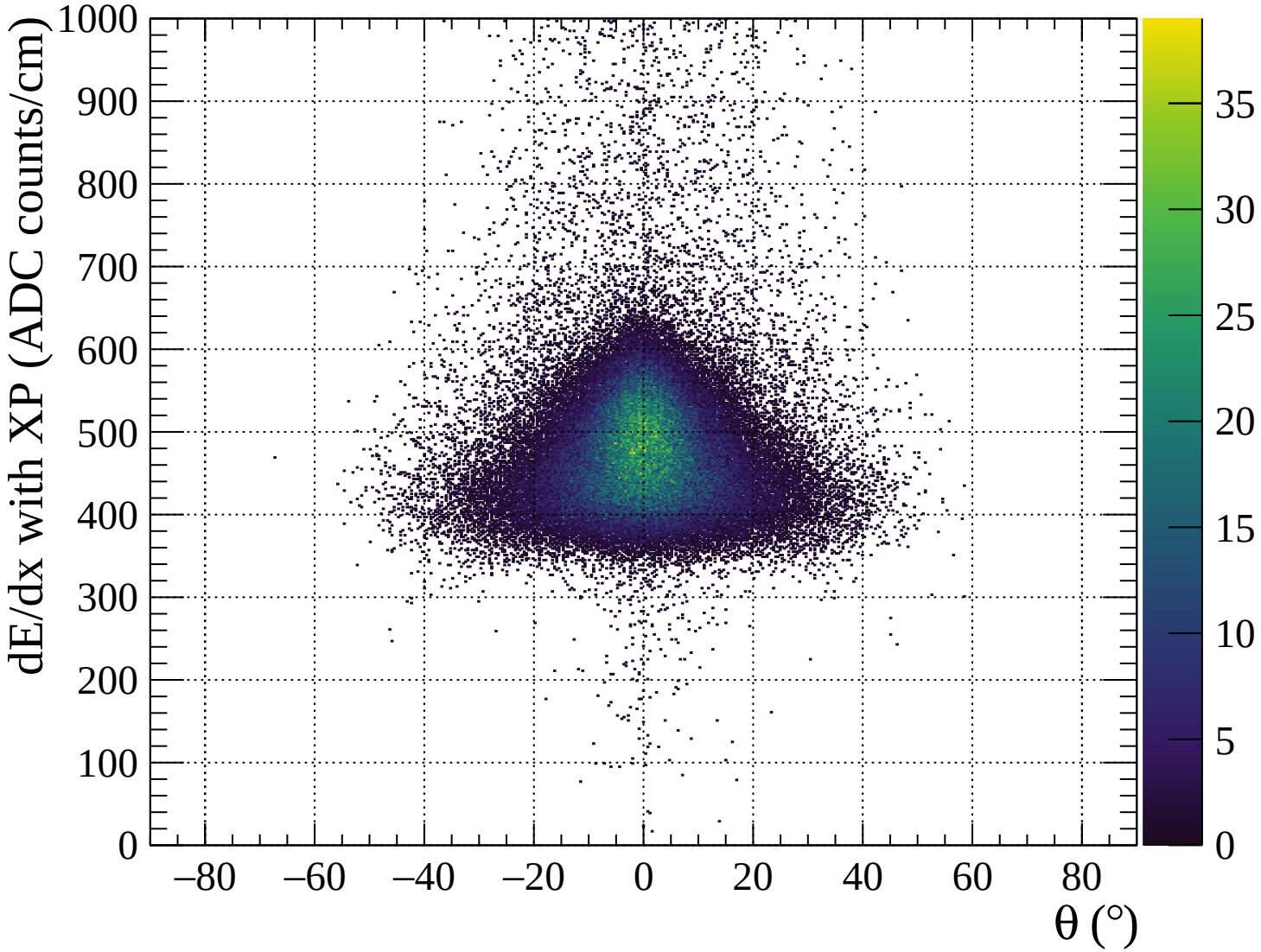


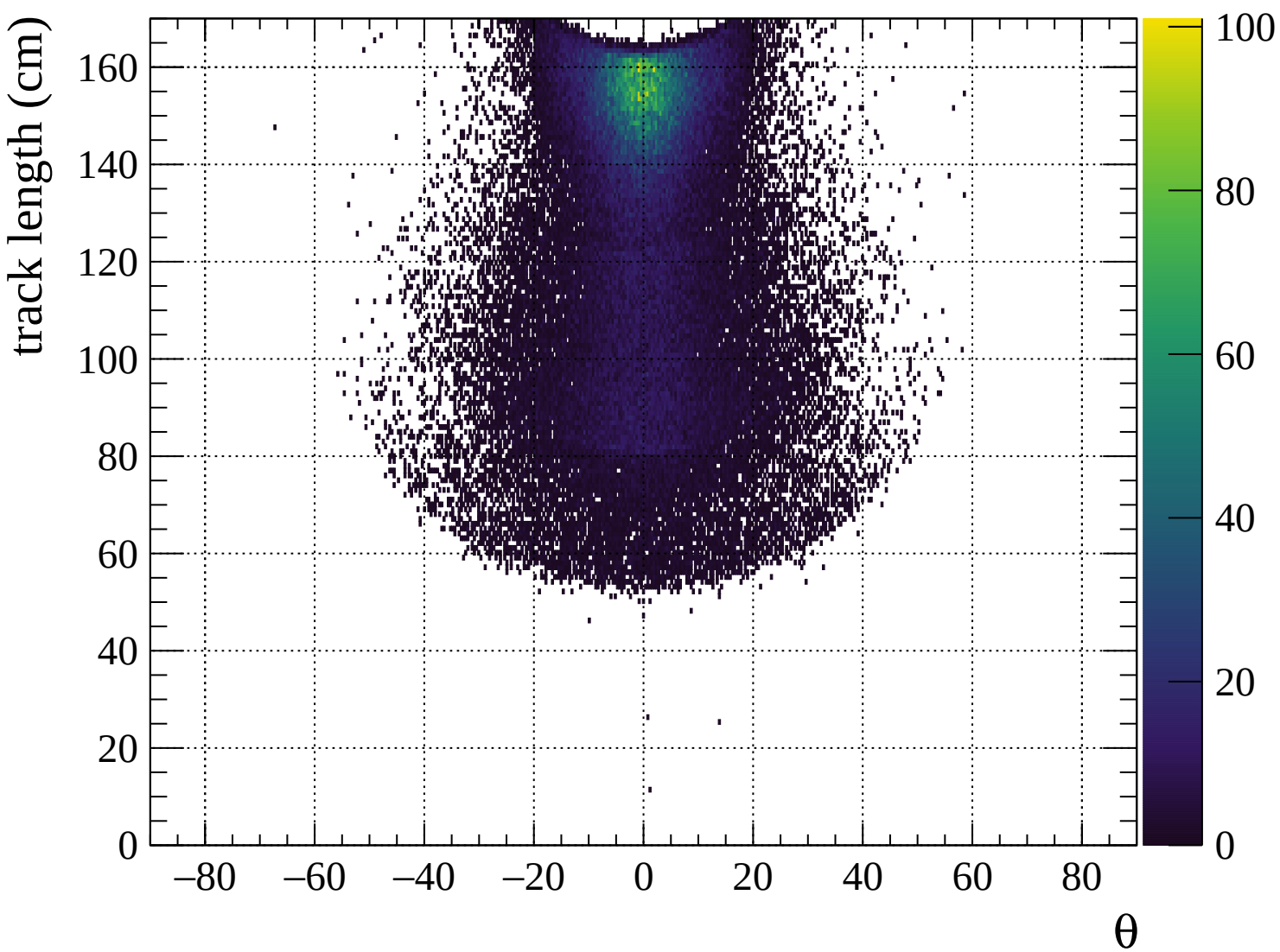


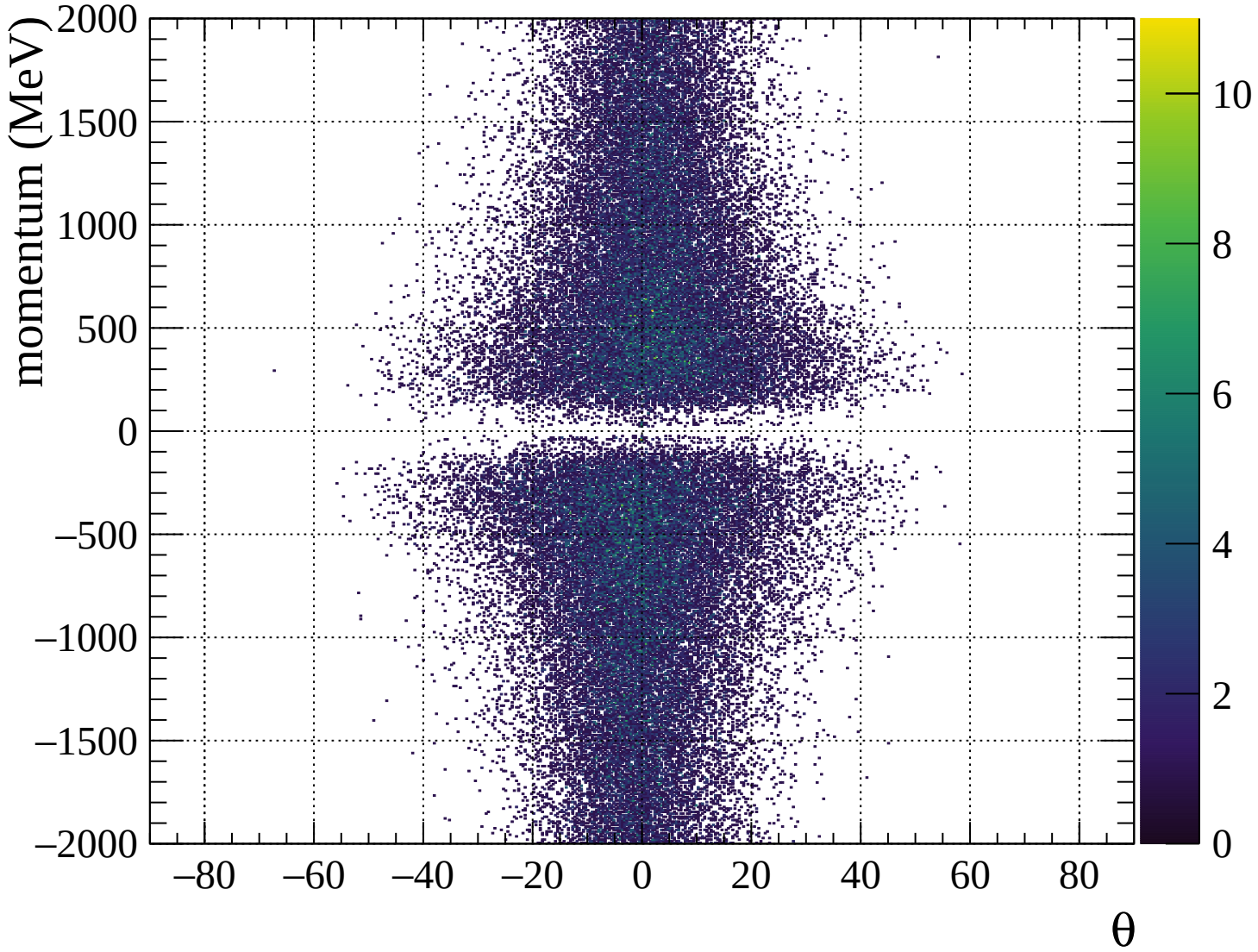


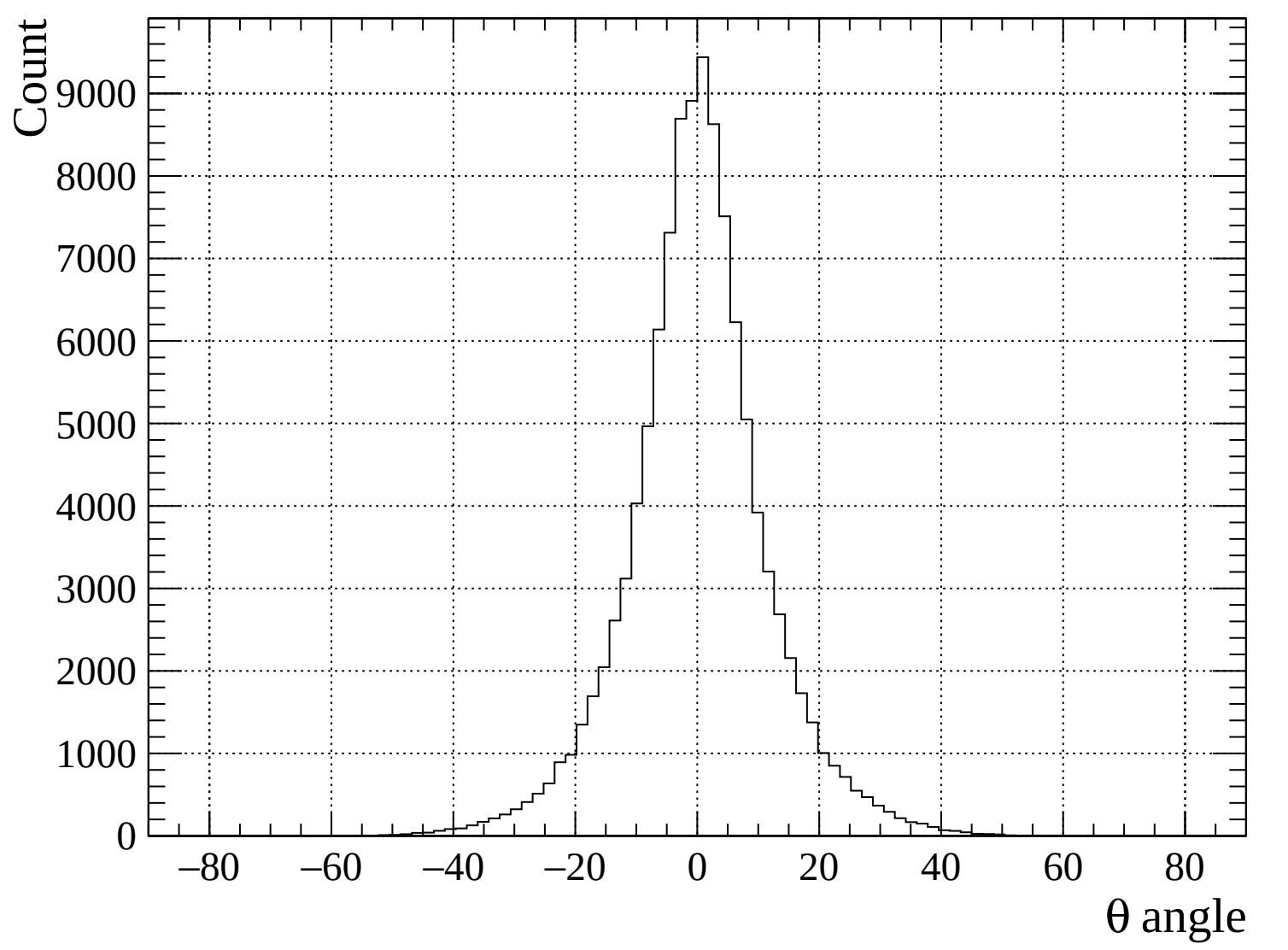


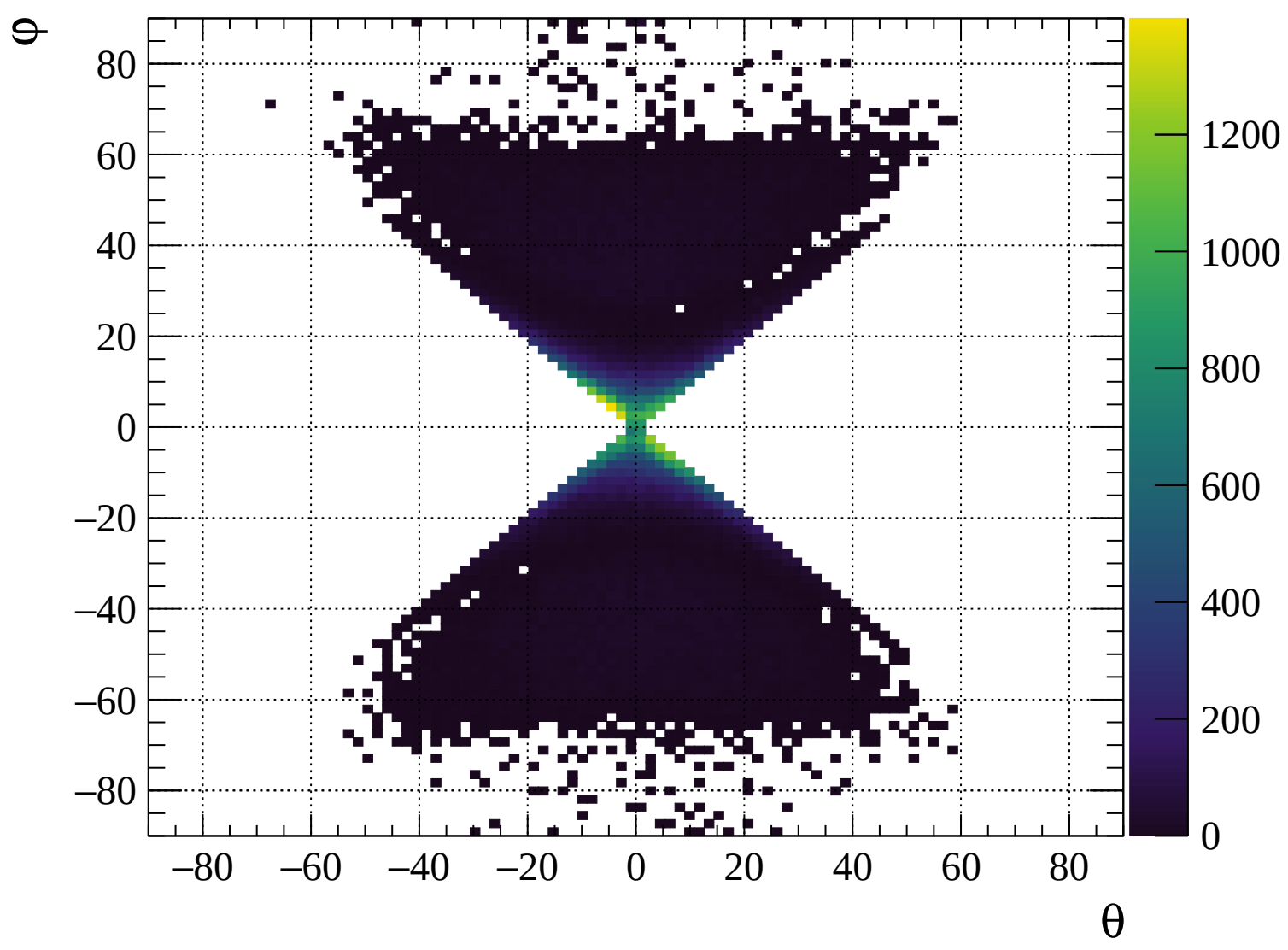


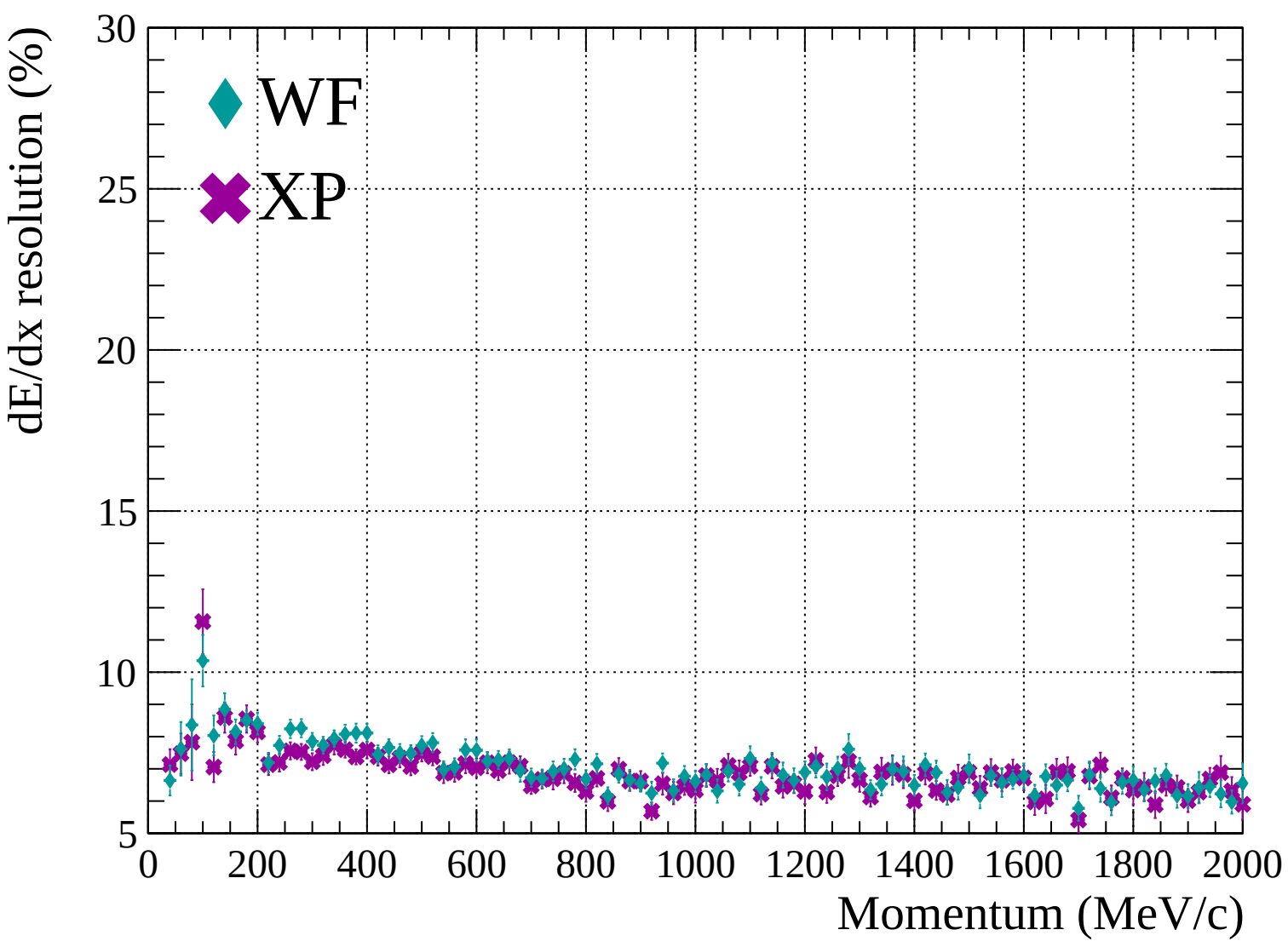


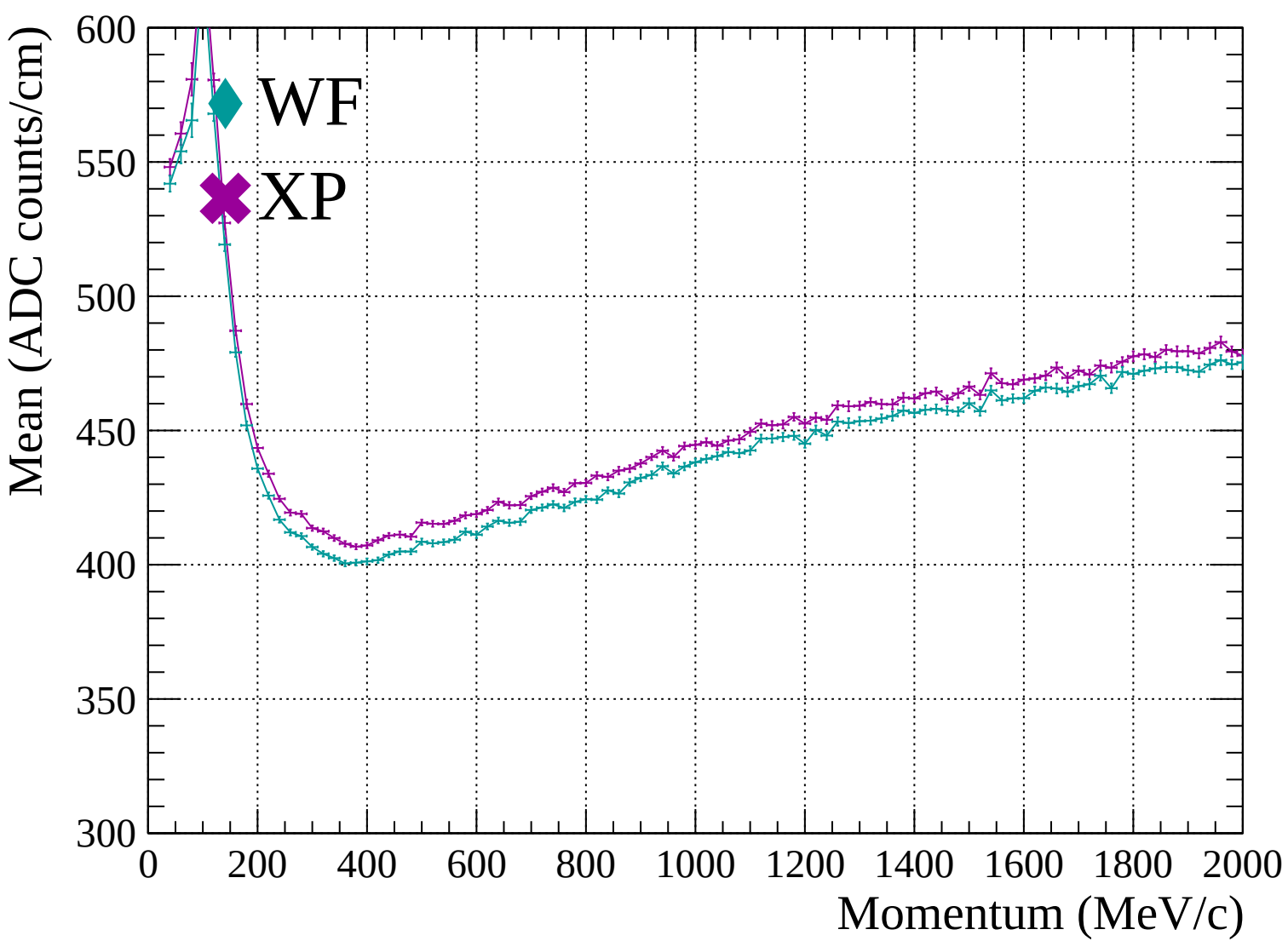


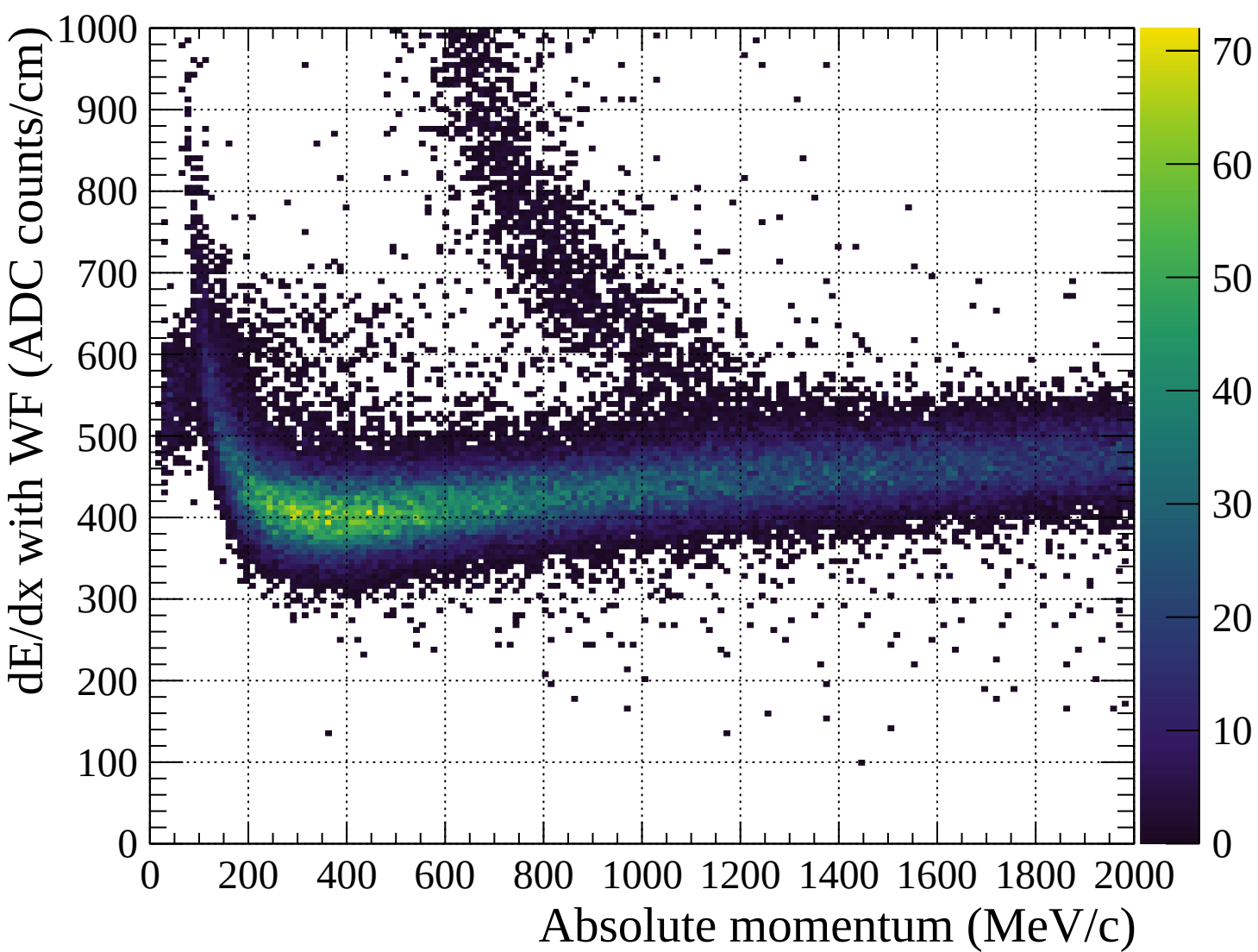


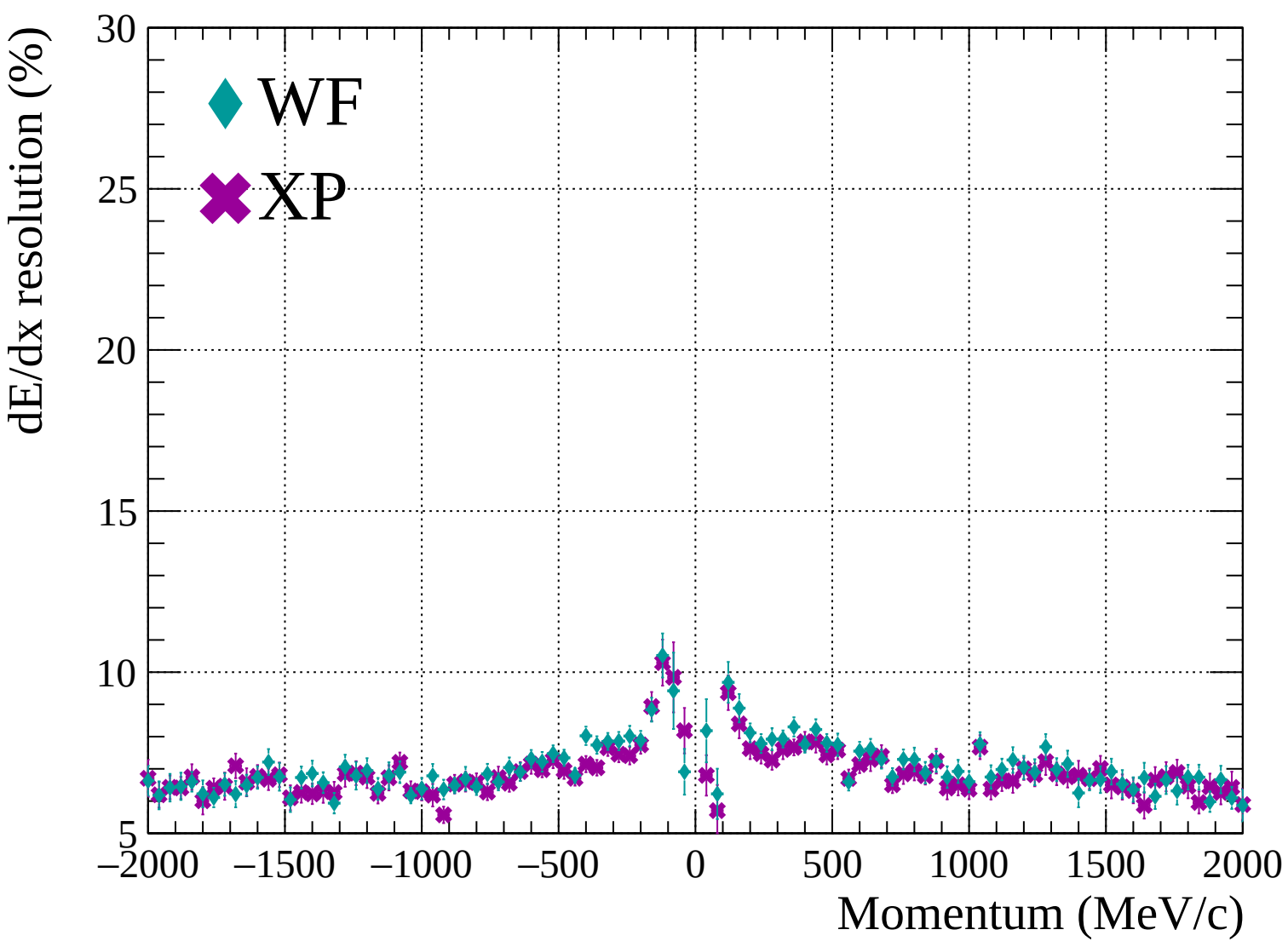


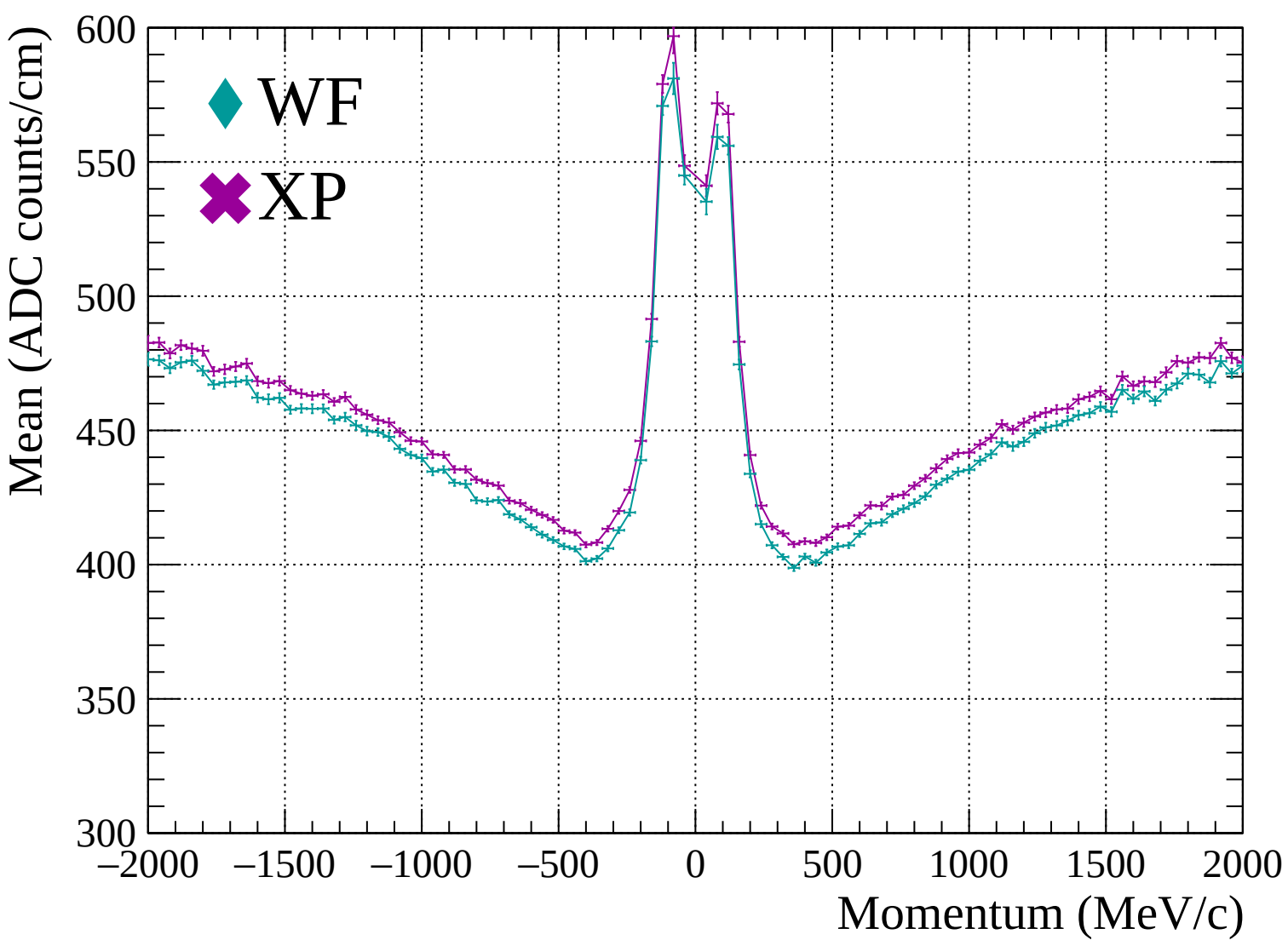


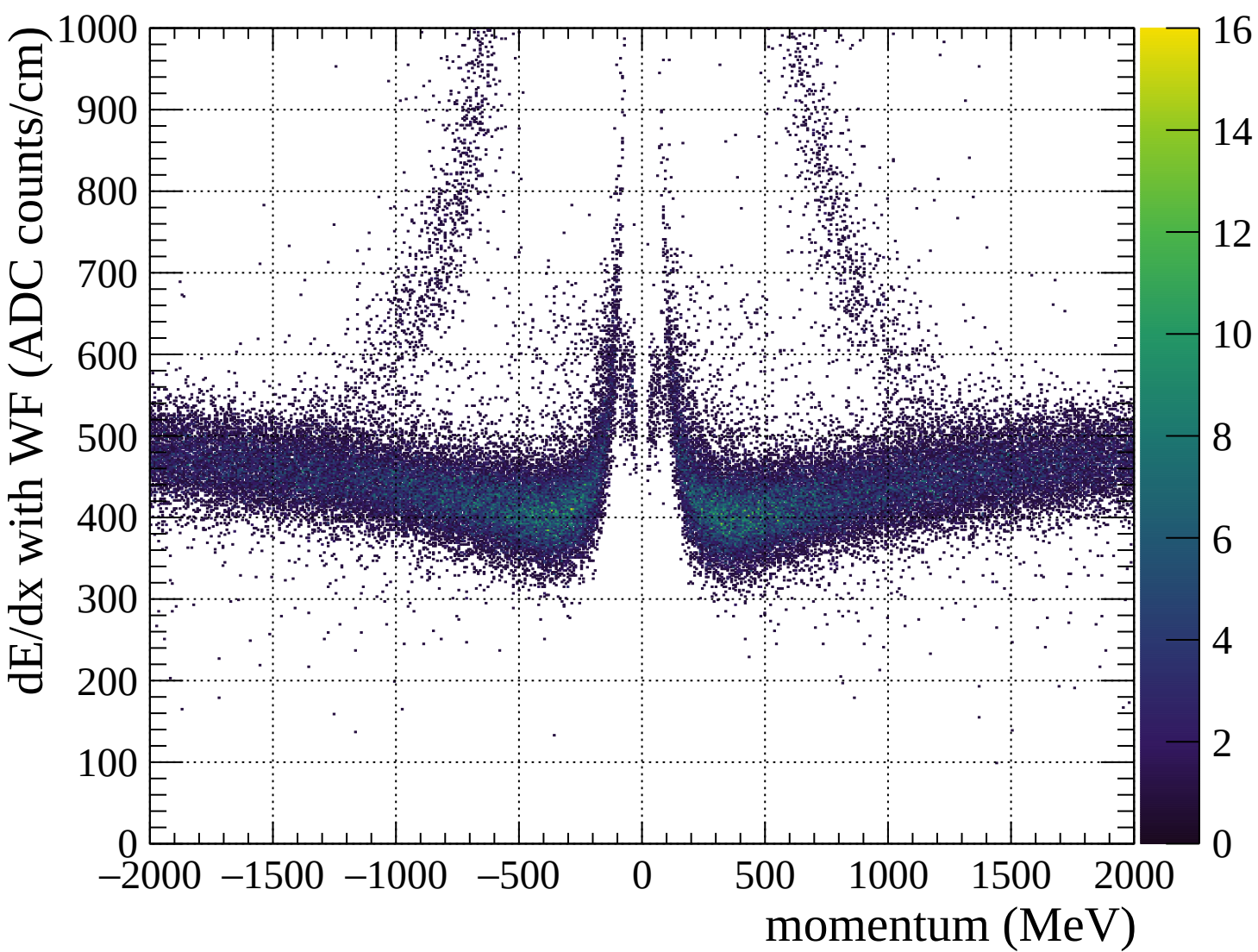


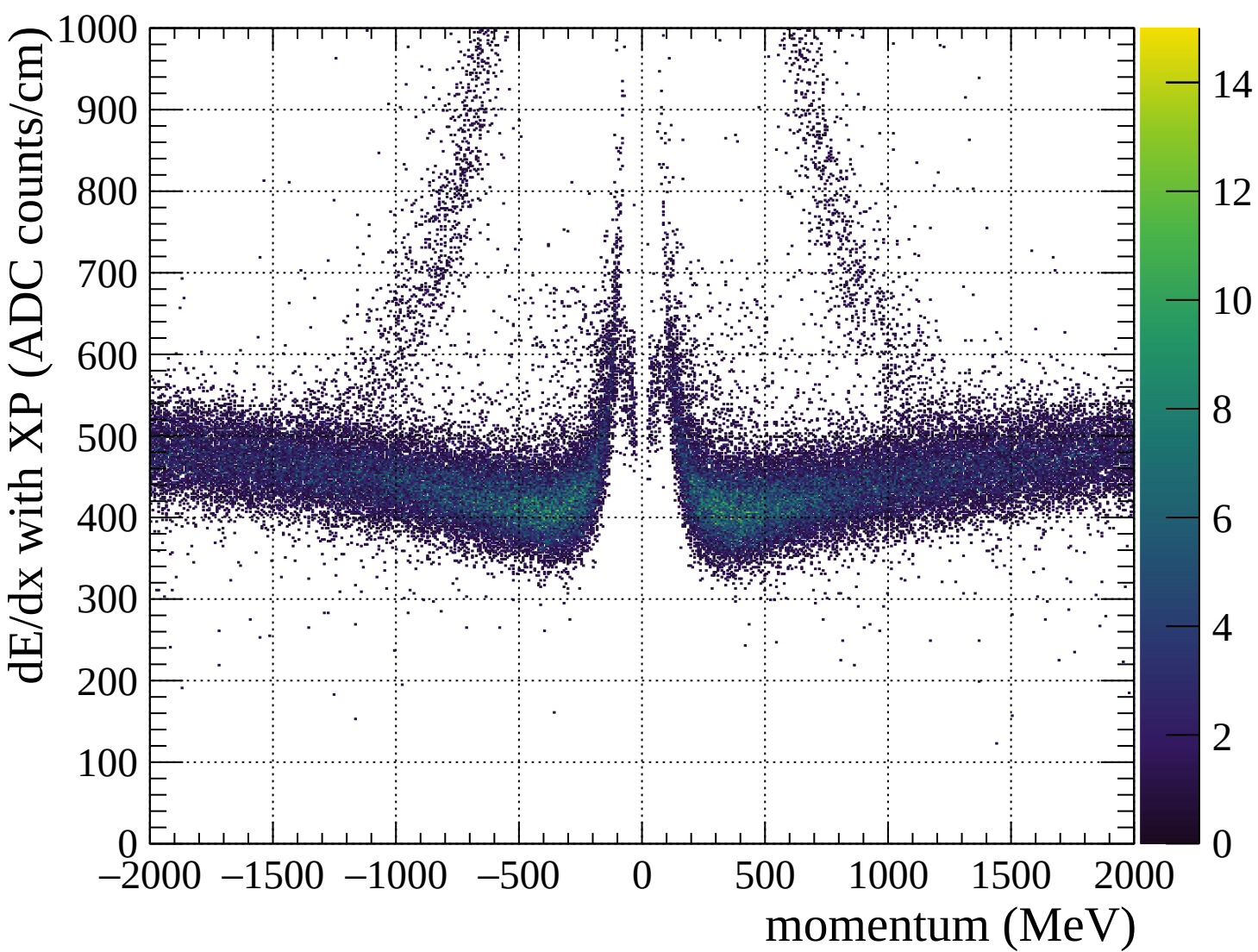


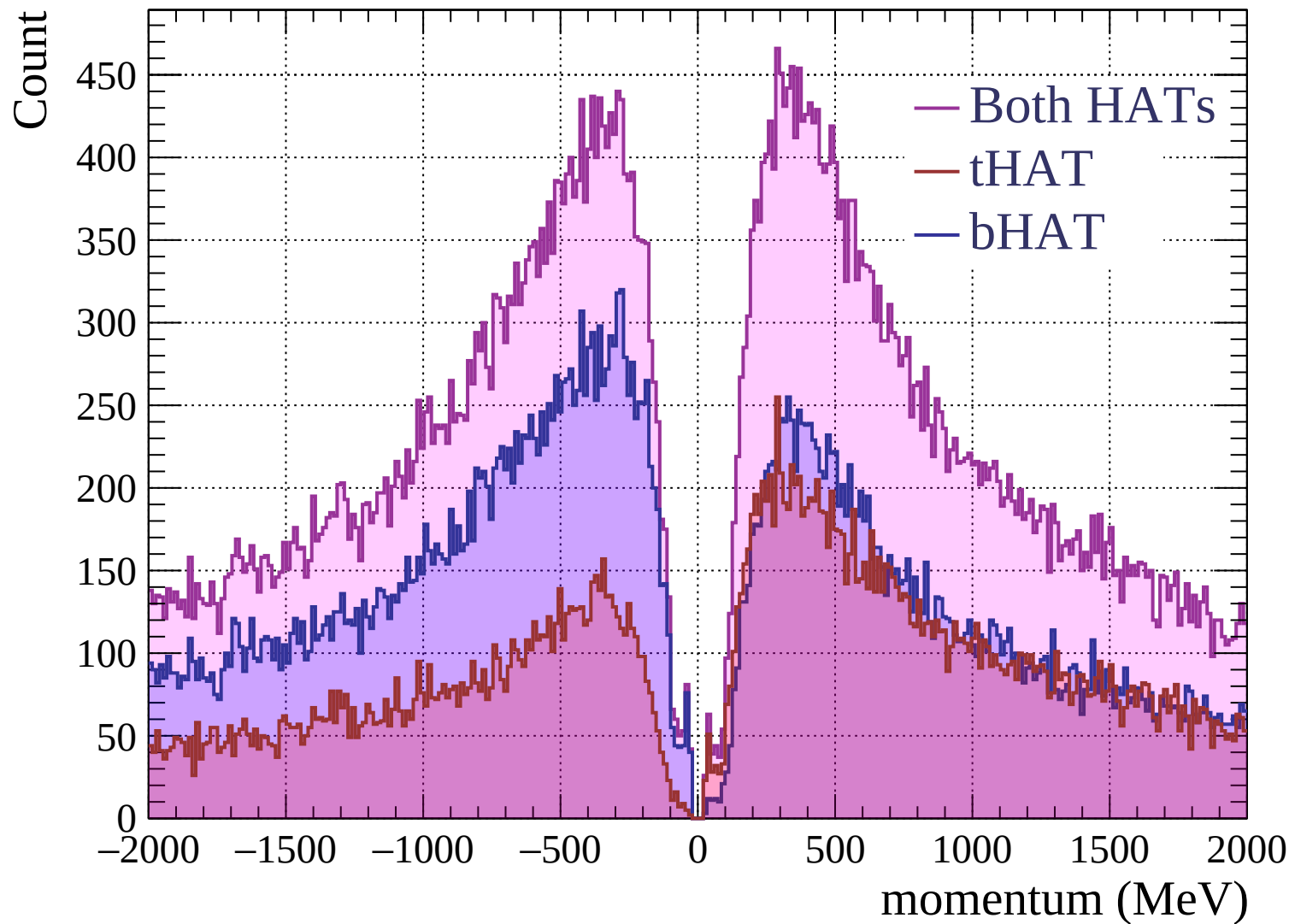


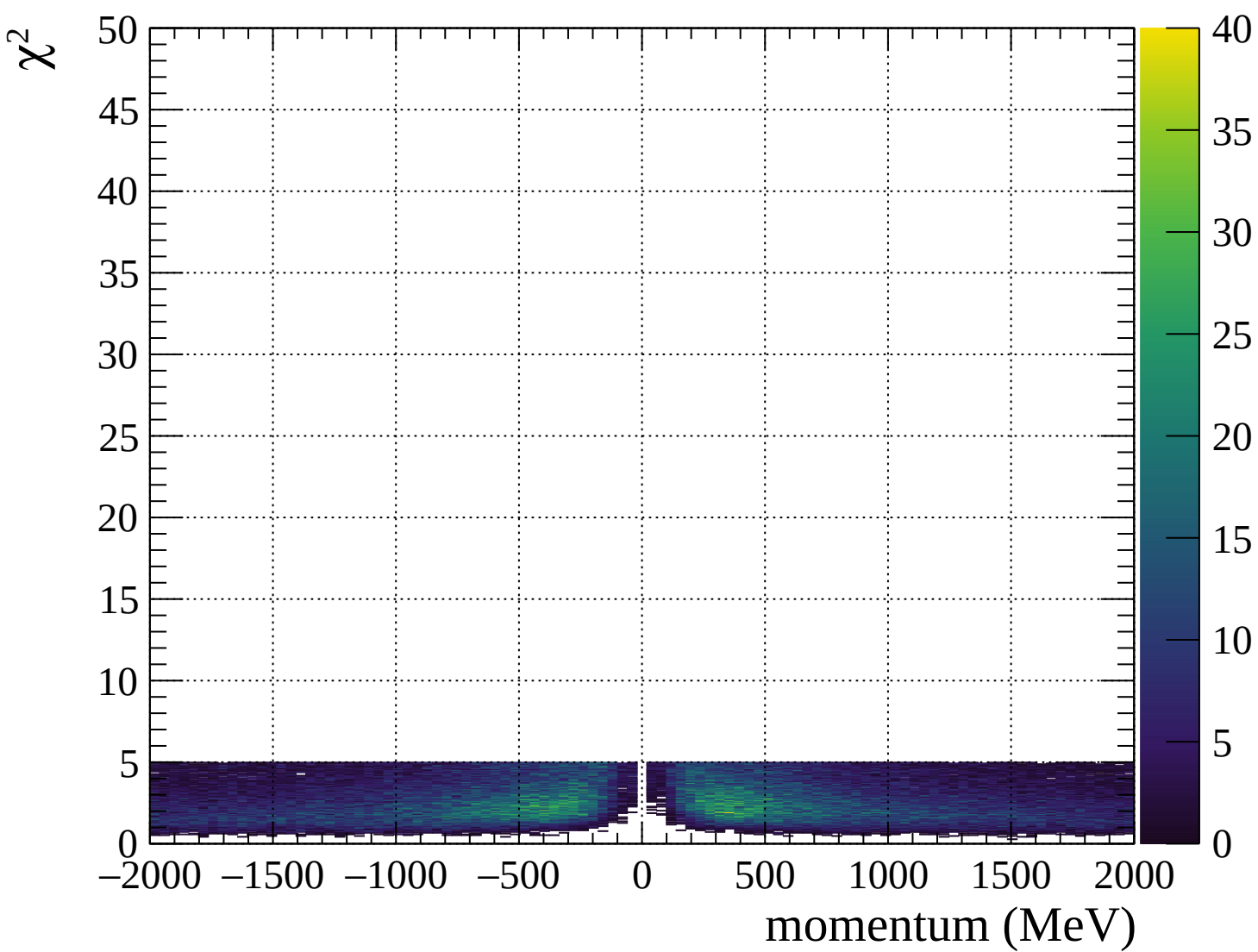












Pulls standard deviation

1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0

-80

-60

-40

-20

0

20

40

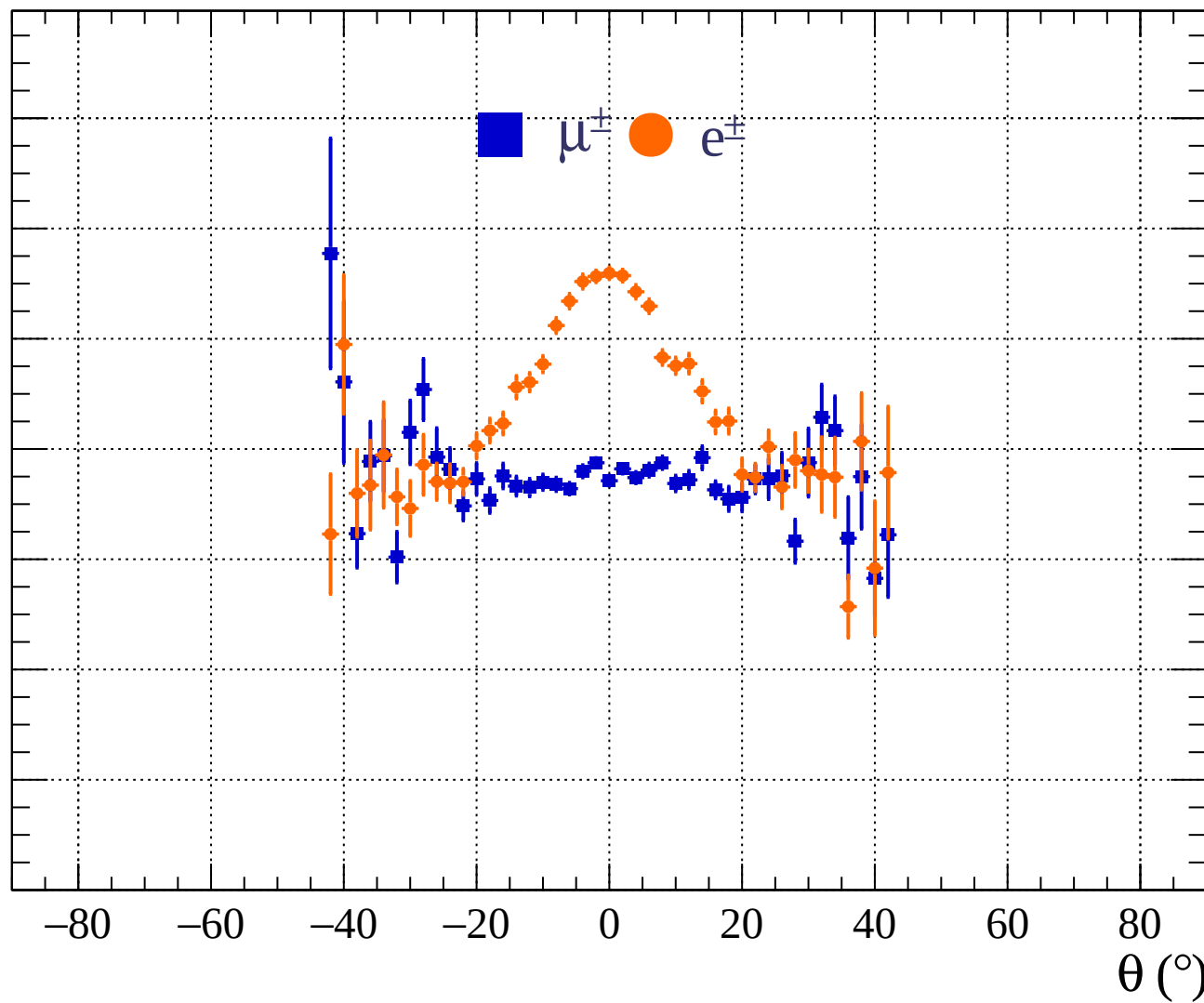
60

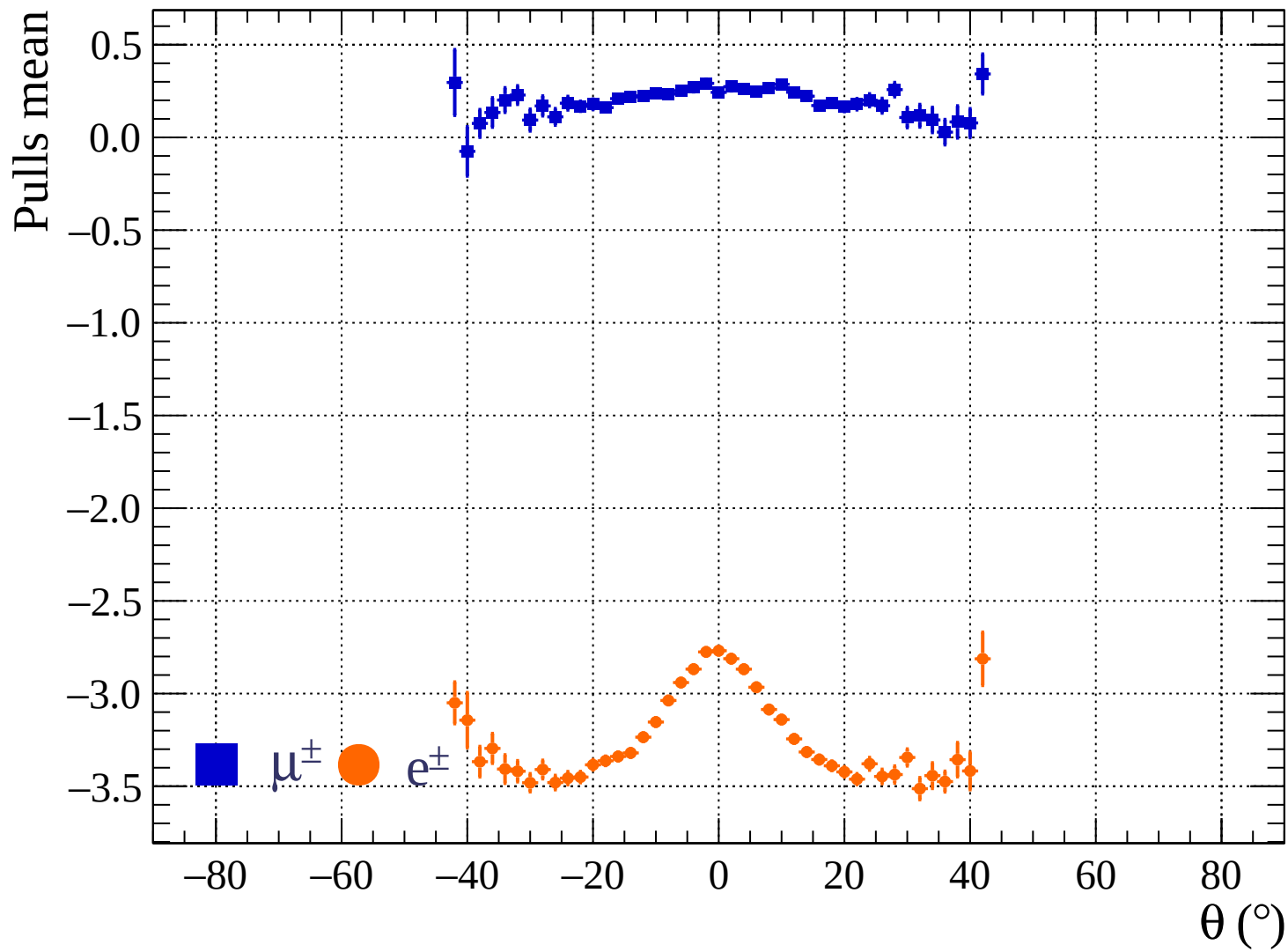
80

$\theta (^{\circ})$

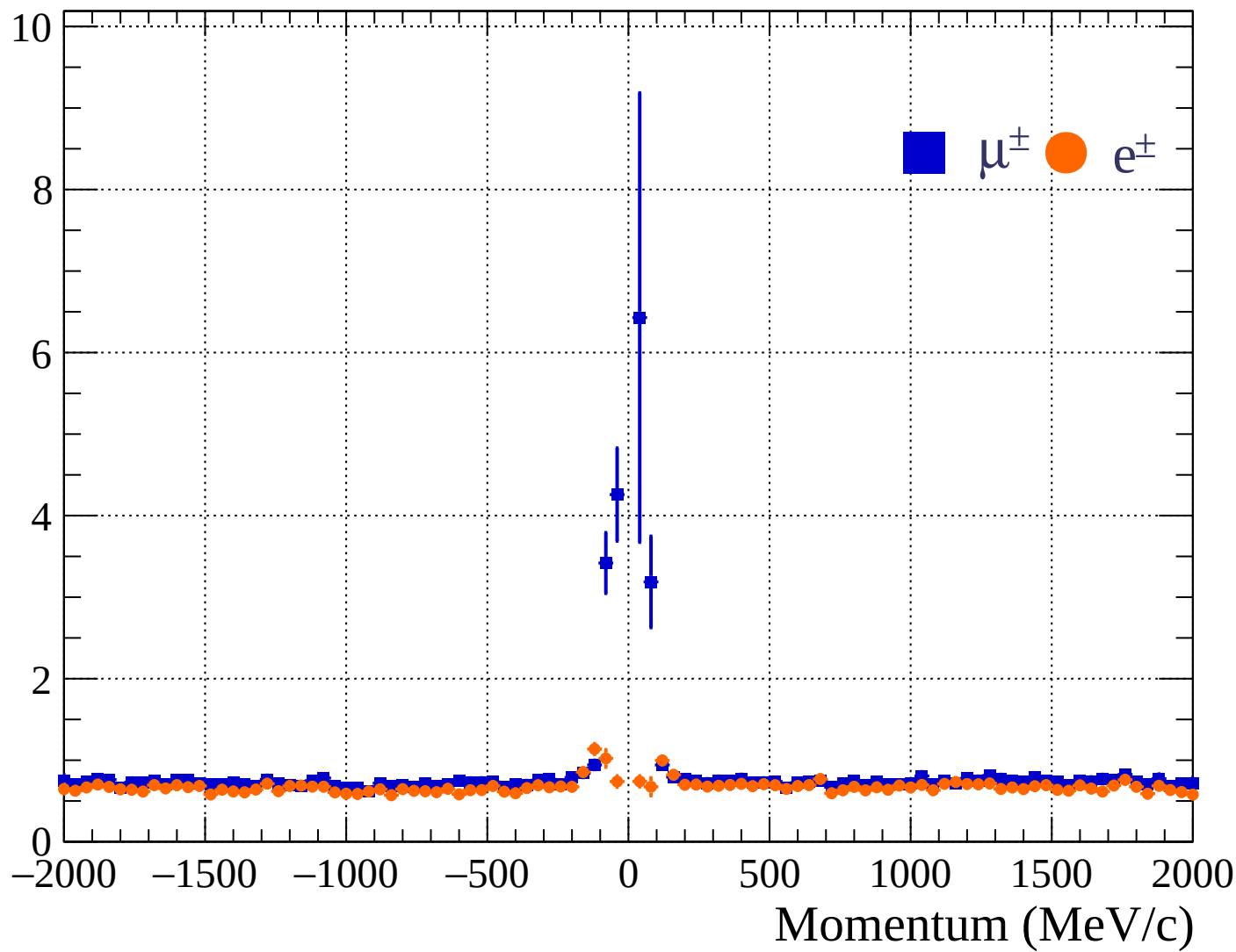
$\mu^{\pm}$

$e^{\pm}$

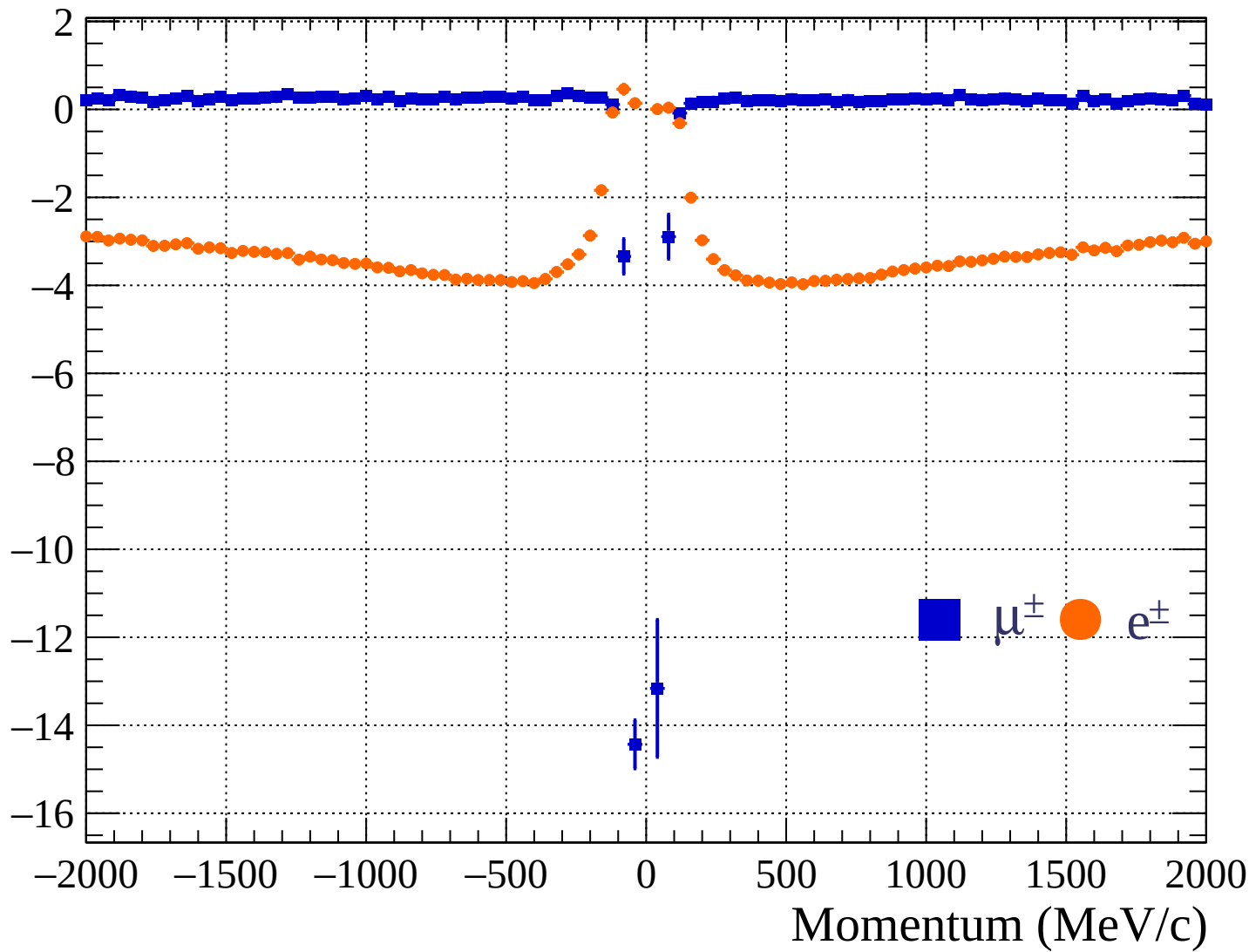


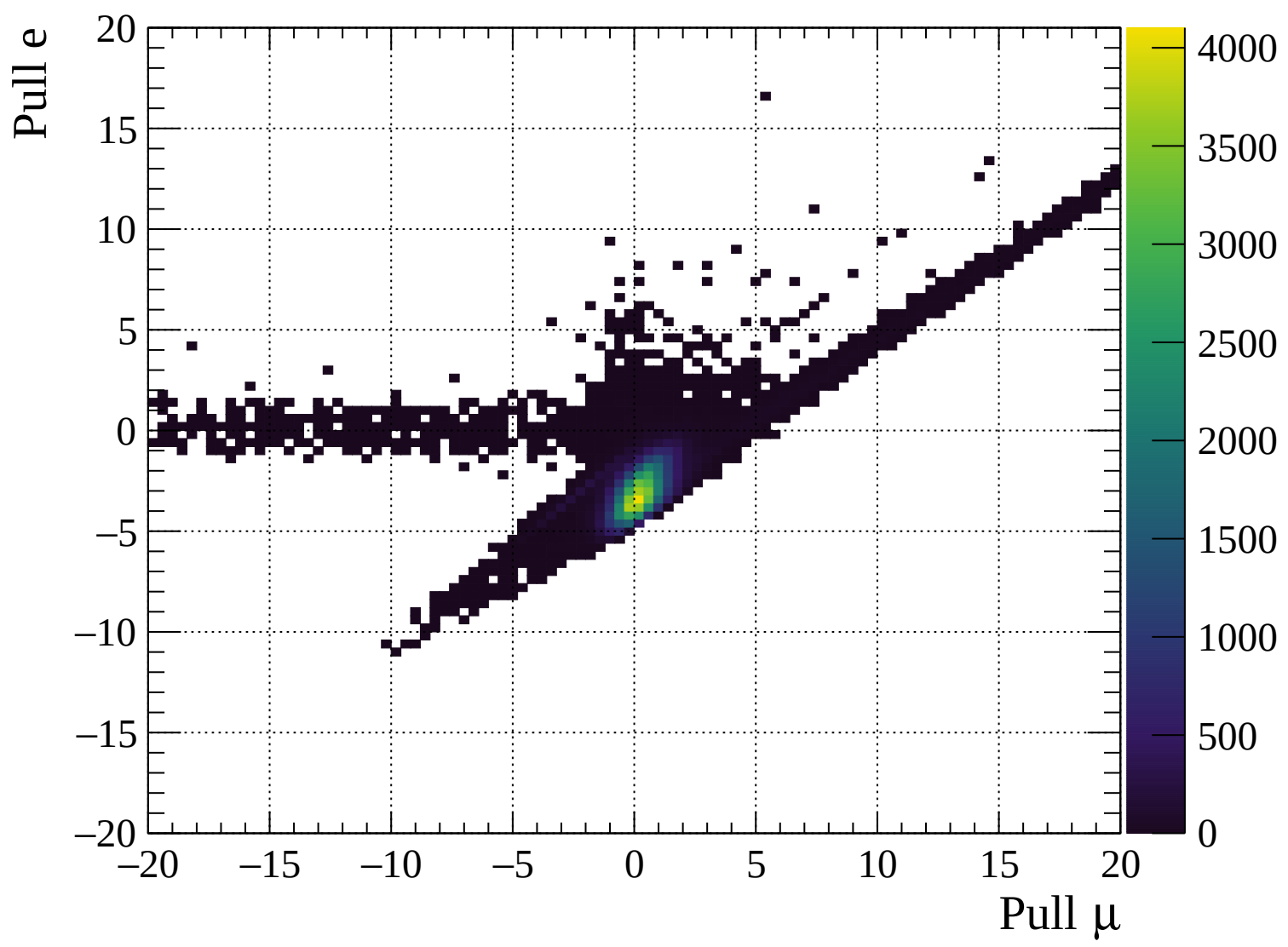


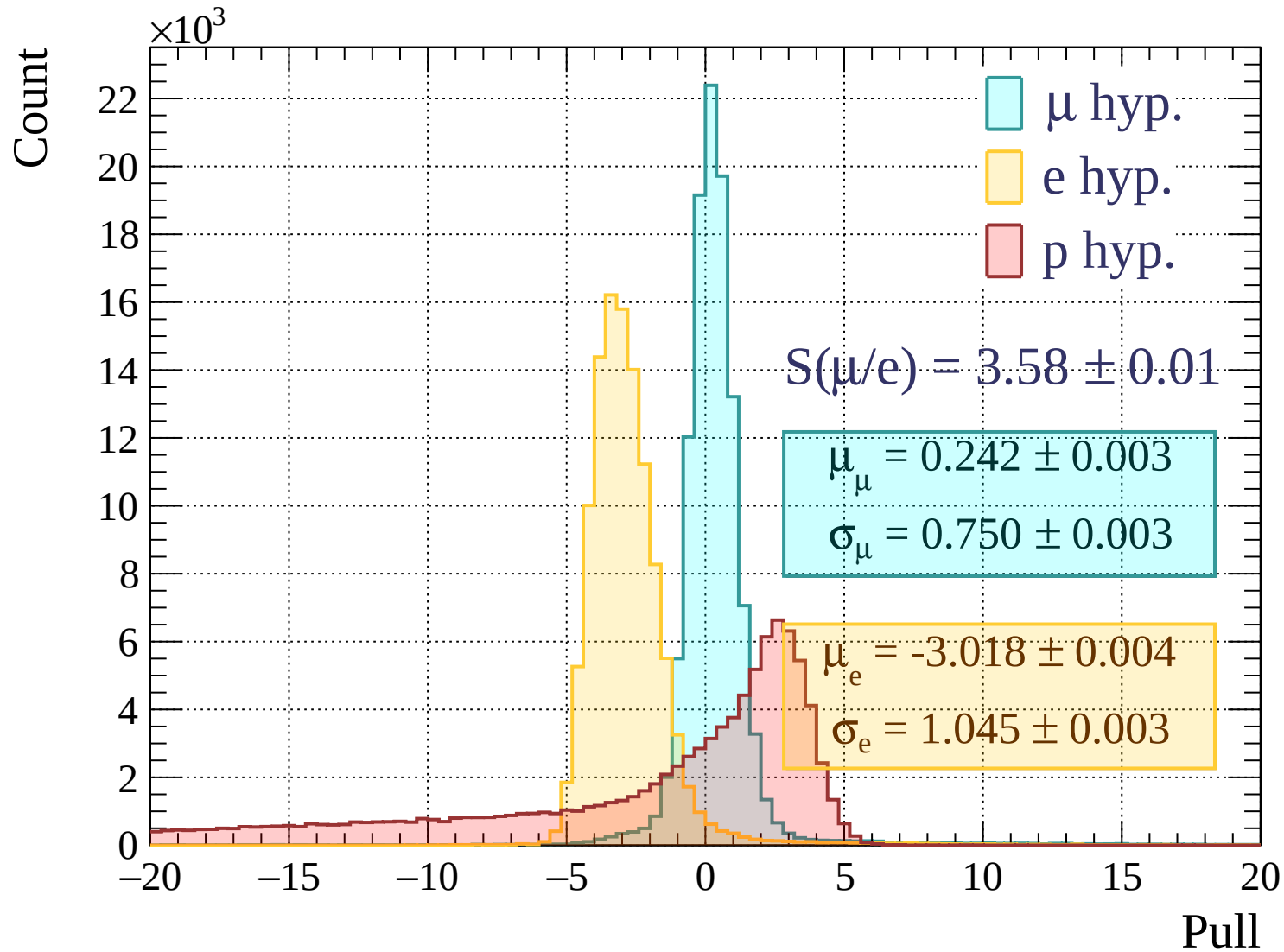
Pulls standard deviation

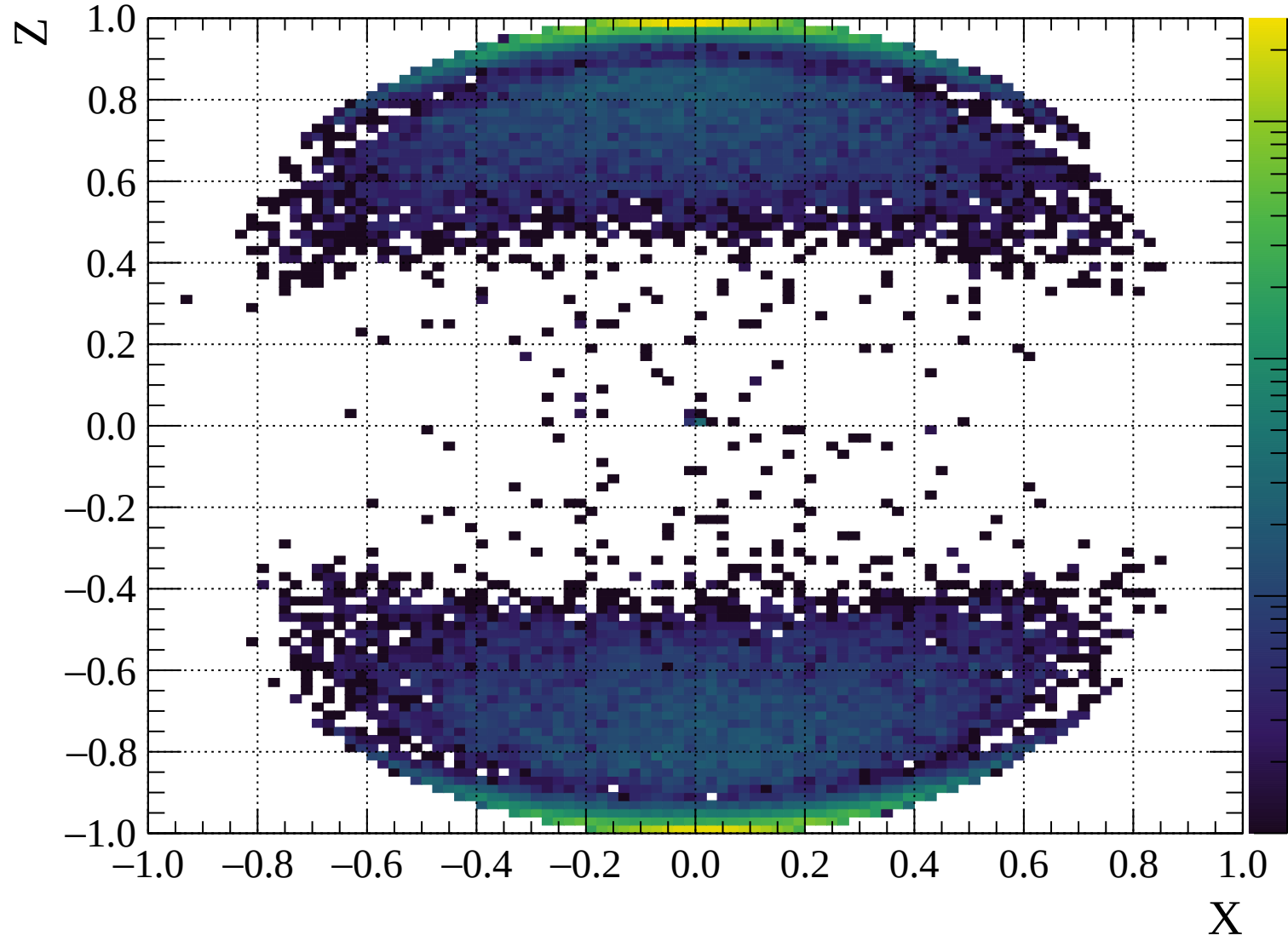


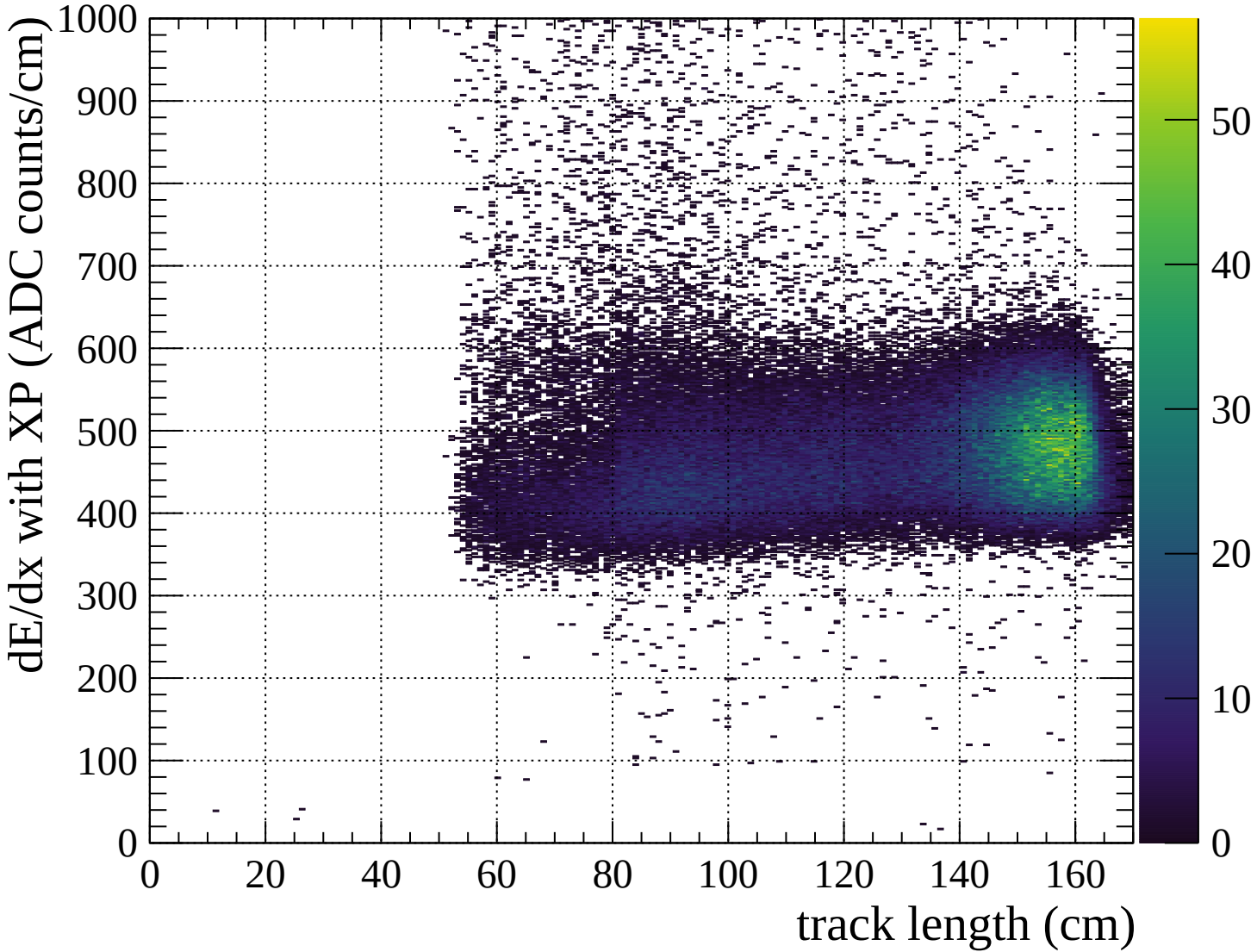
Pulls mean

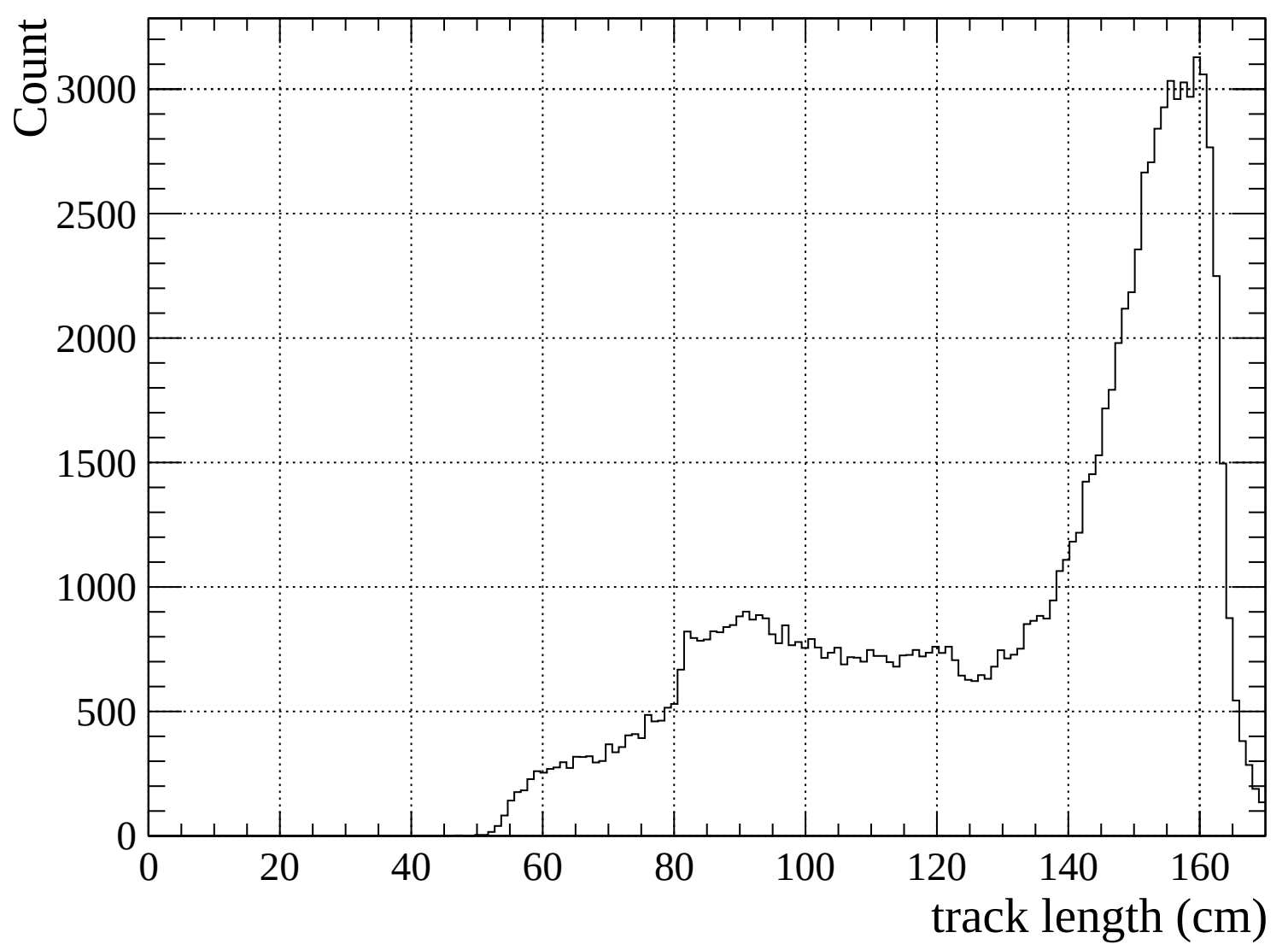


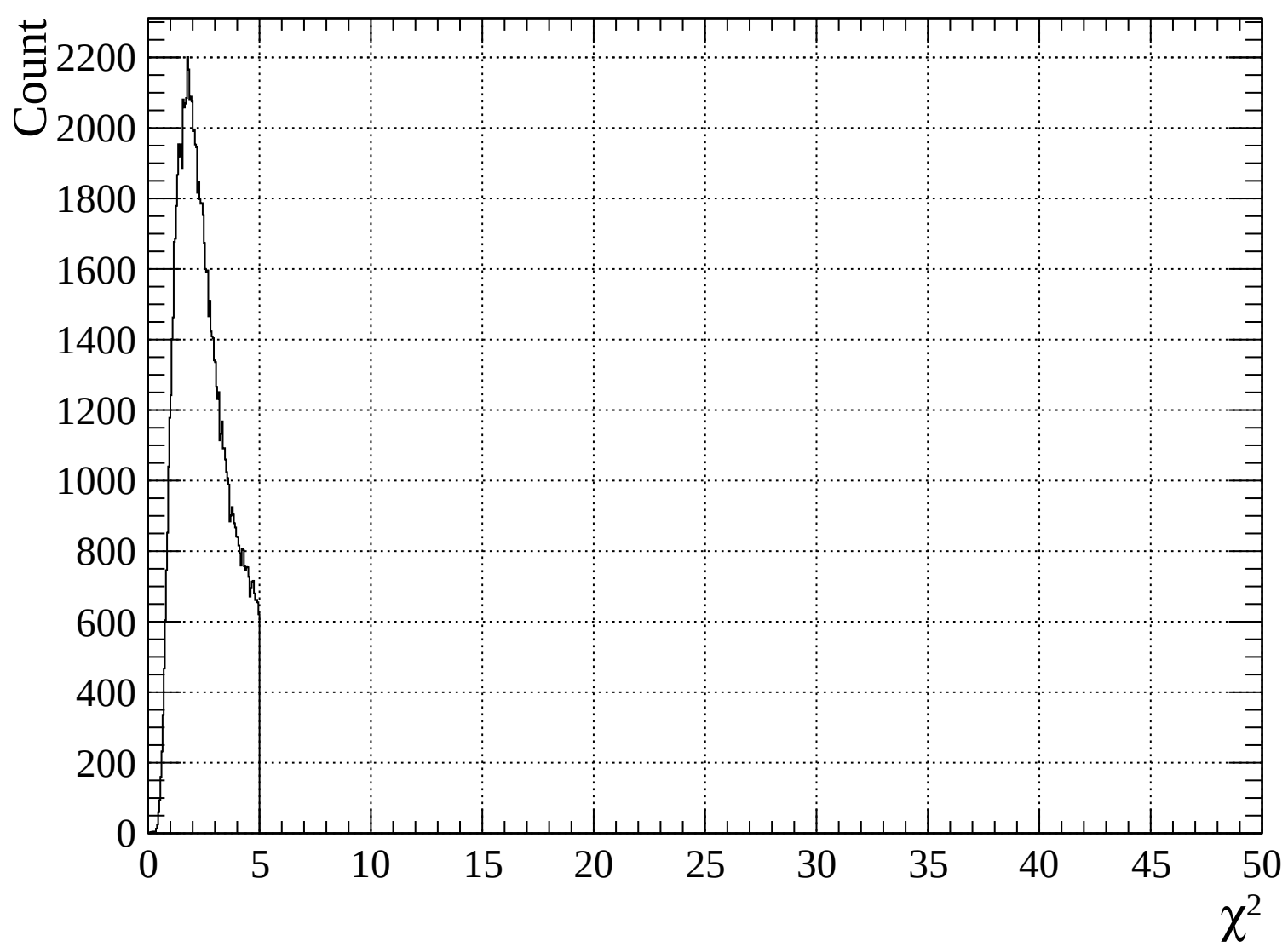


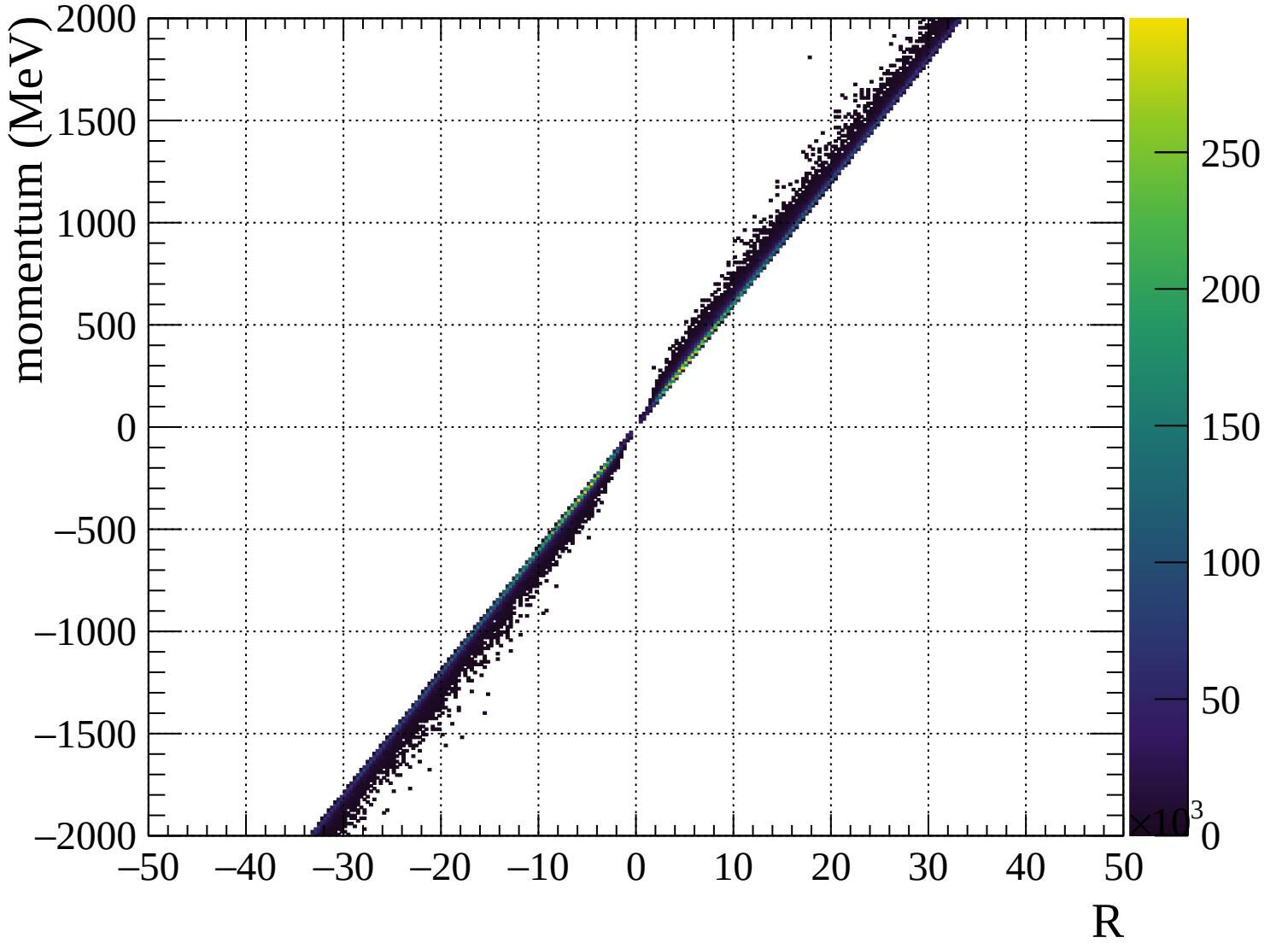


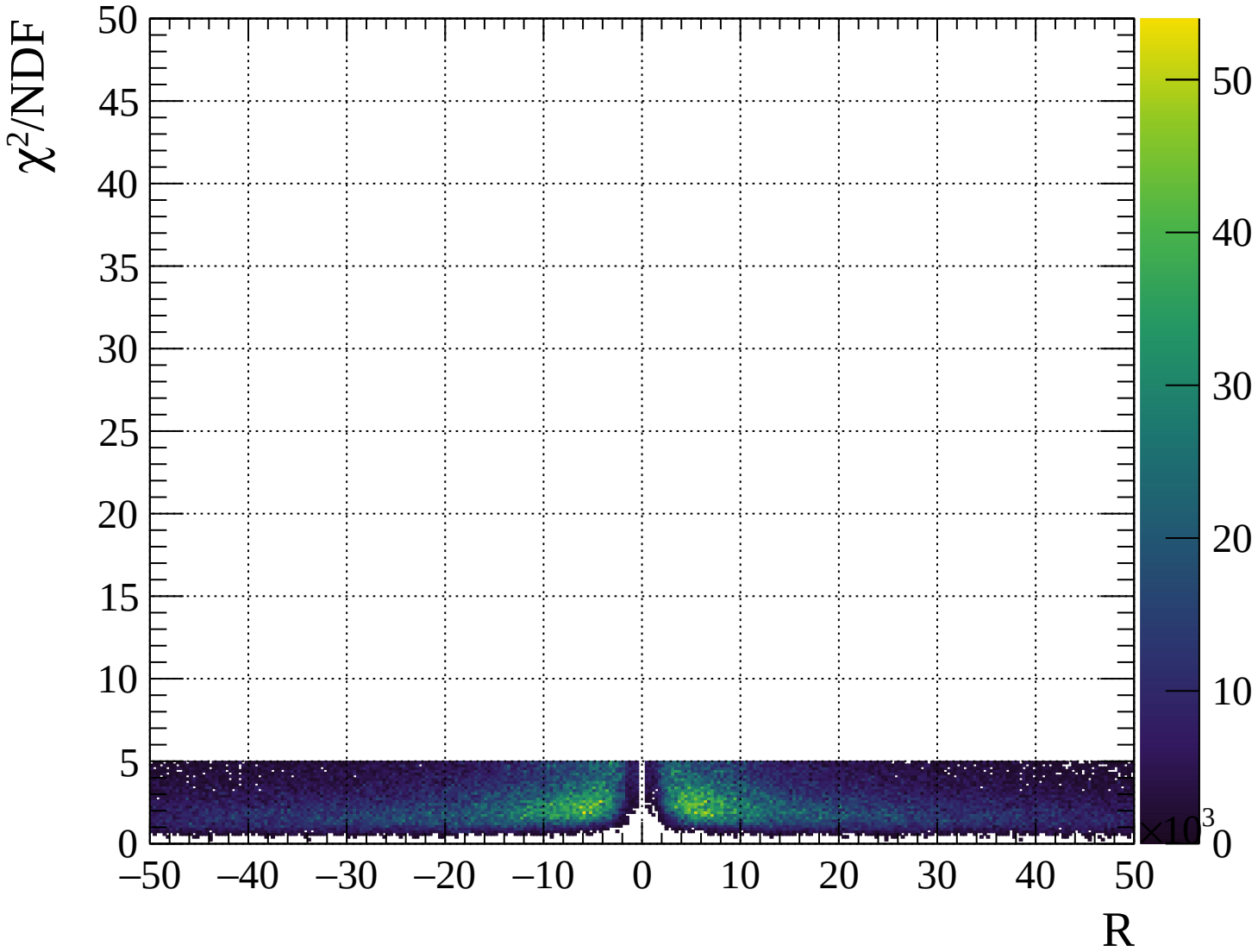






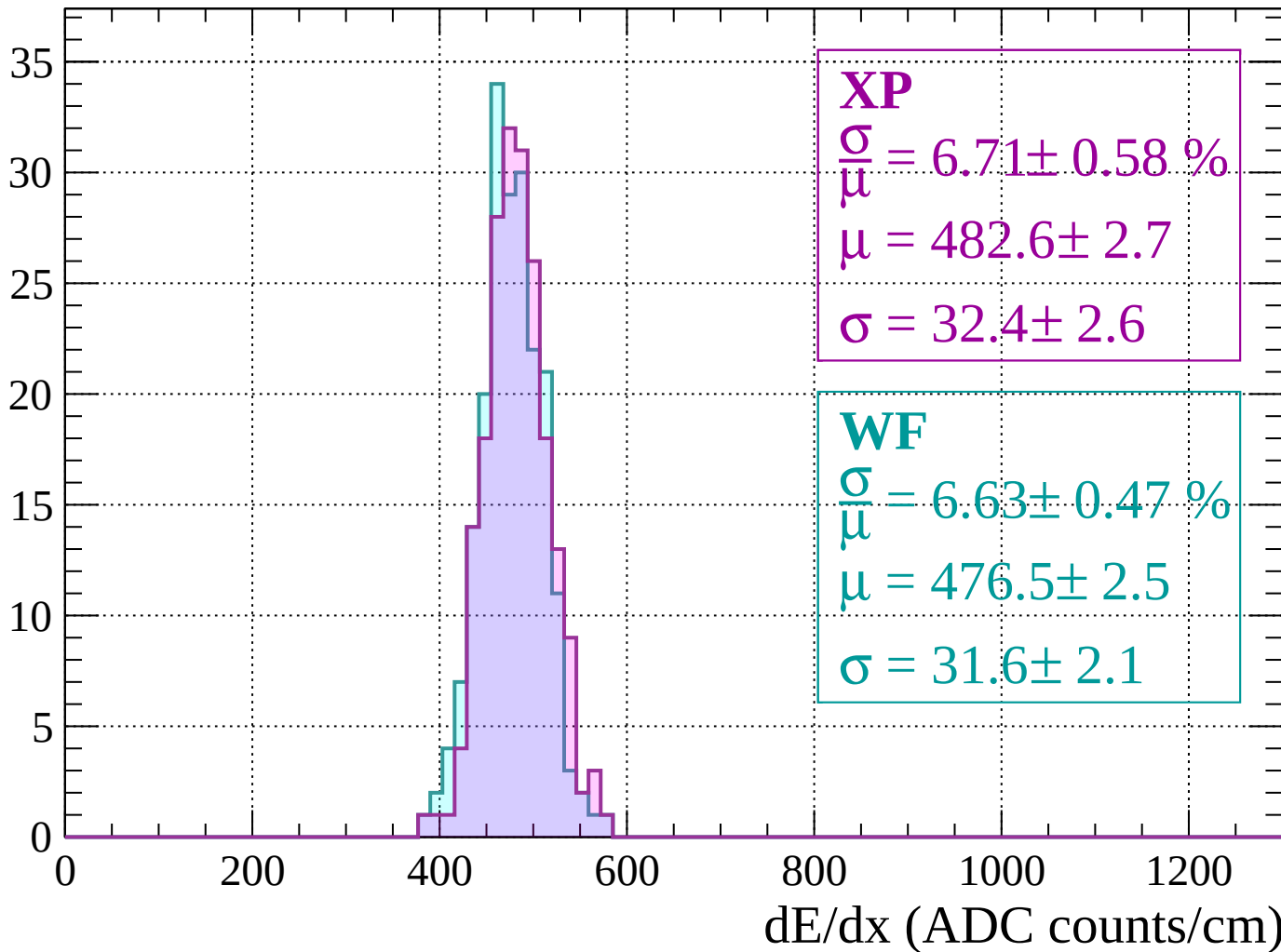






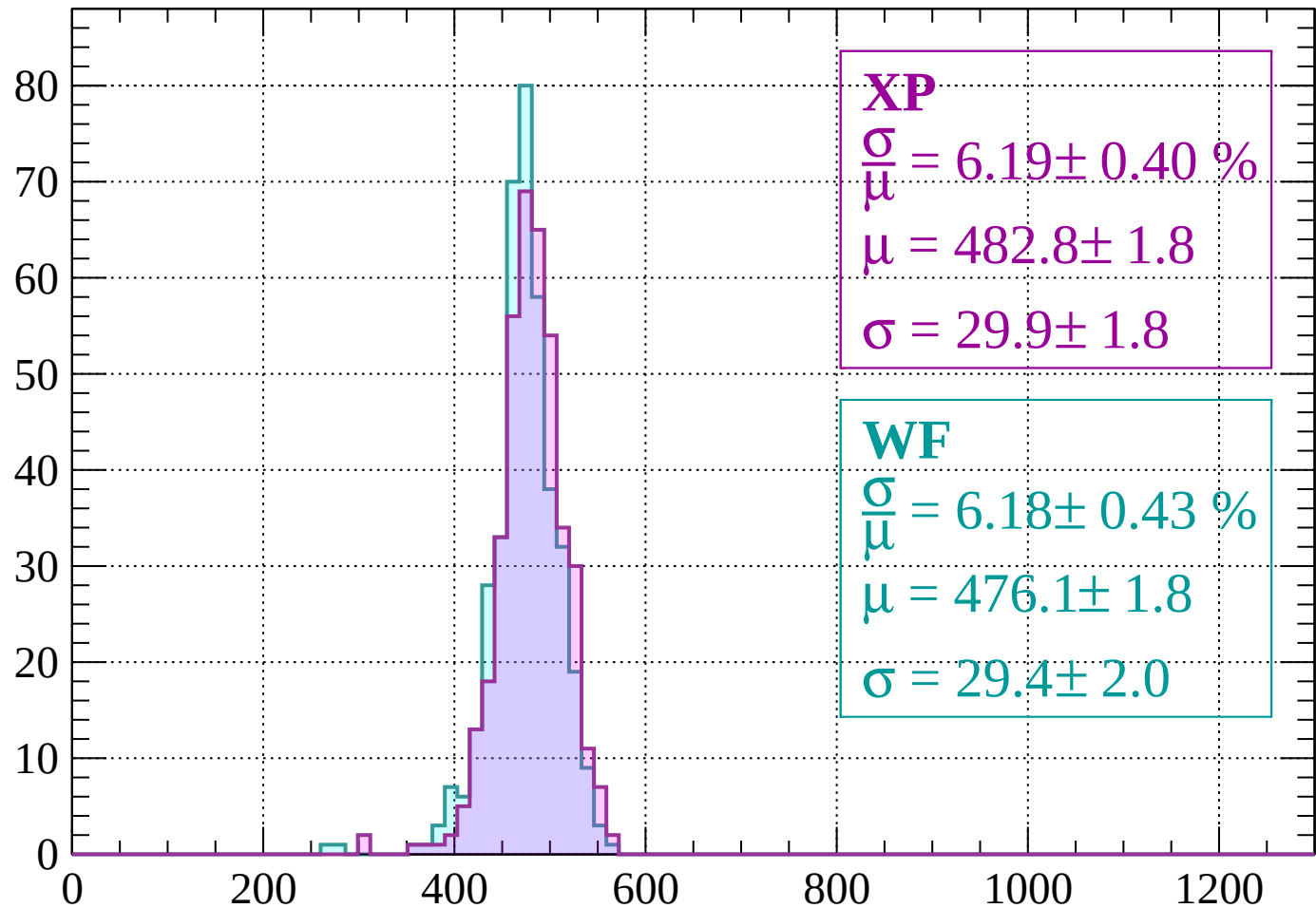
Energy loss | $-2000 < p < -1960$

Count



Energy loss | $-1960 < p < -1920$

Count



XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.40 \%$$

$$\mu = 482.8 \pm 1.8$$

$$\sigma = 29.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 6.18 \pm 0.43 \%$$

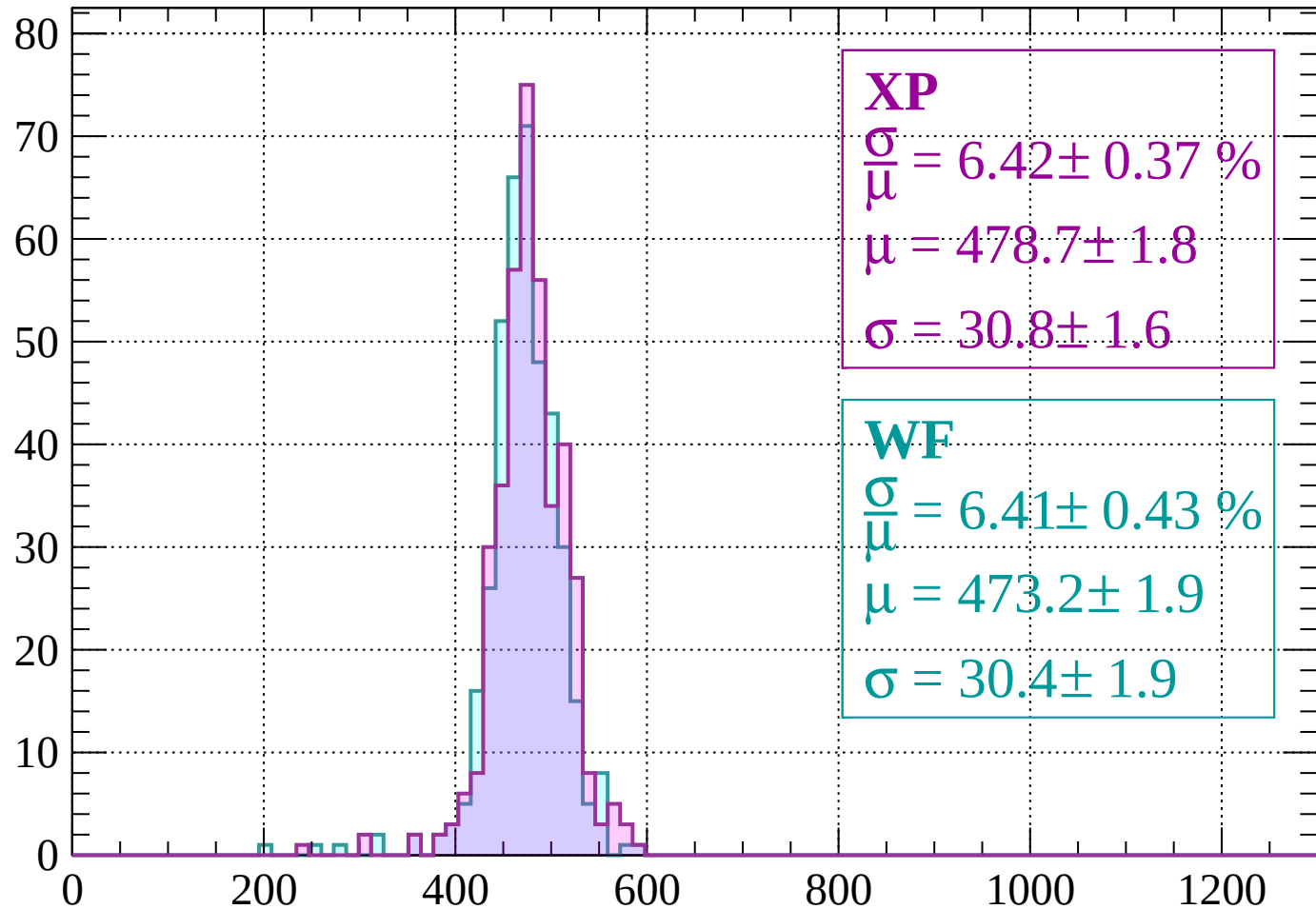
$$\mu = 476.1 \pm 1.8$$

$$\sigma = 29.4 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

Count



XP

$$\frac{\sigma}{\mu} = 6.42 \pm 0.37 \%$$

$$\mu = 478.7 \pm 1.8$$

$$\sigma = 30.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 0.43 \%$$

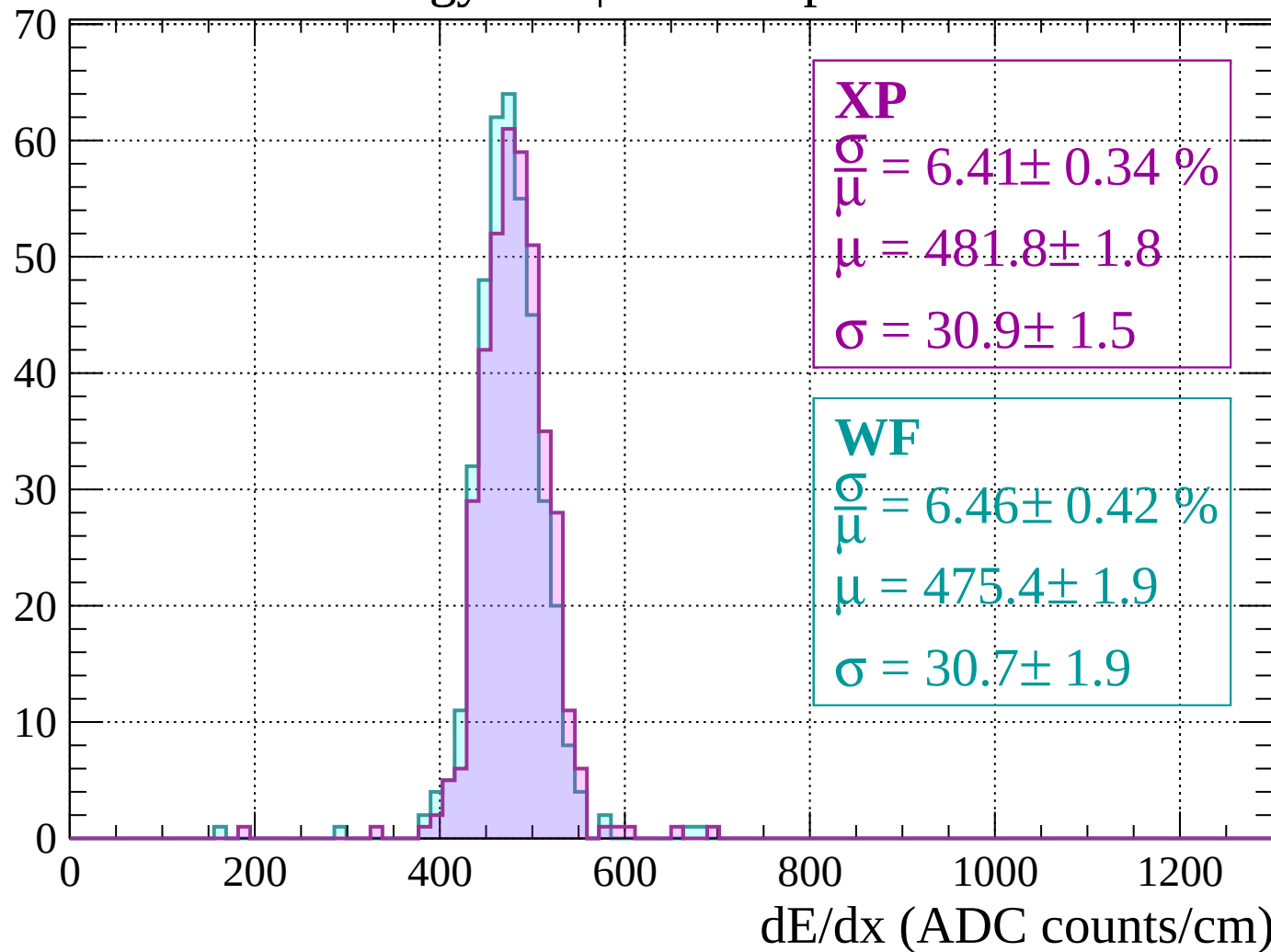
$$\mu = 473.2 \pm 1.9$$

$$\sigma = 30.4 \pm 1.9$$

dE/dx (ADC counts/cm)

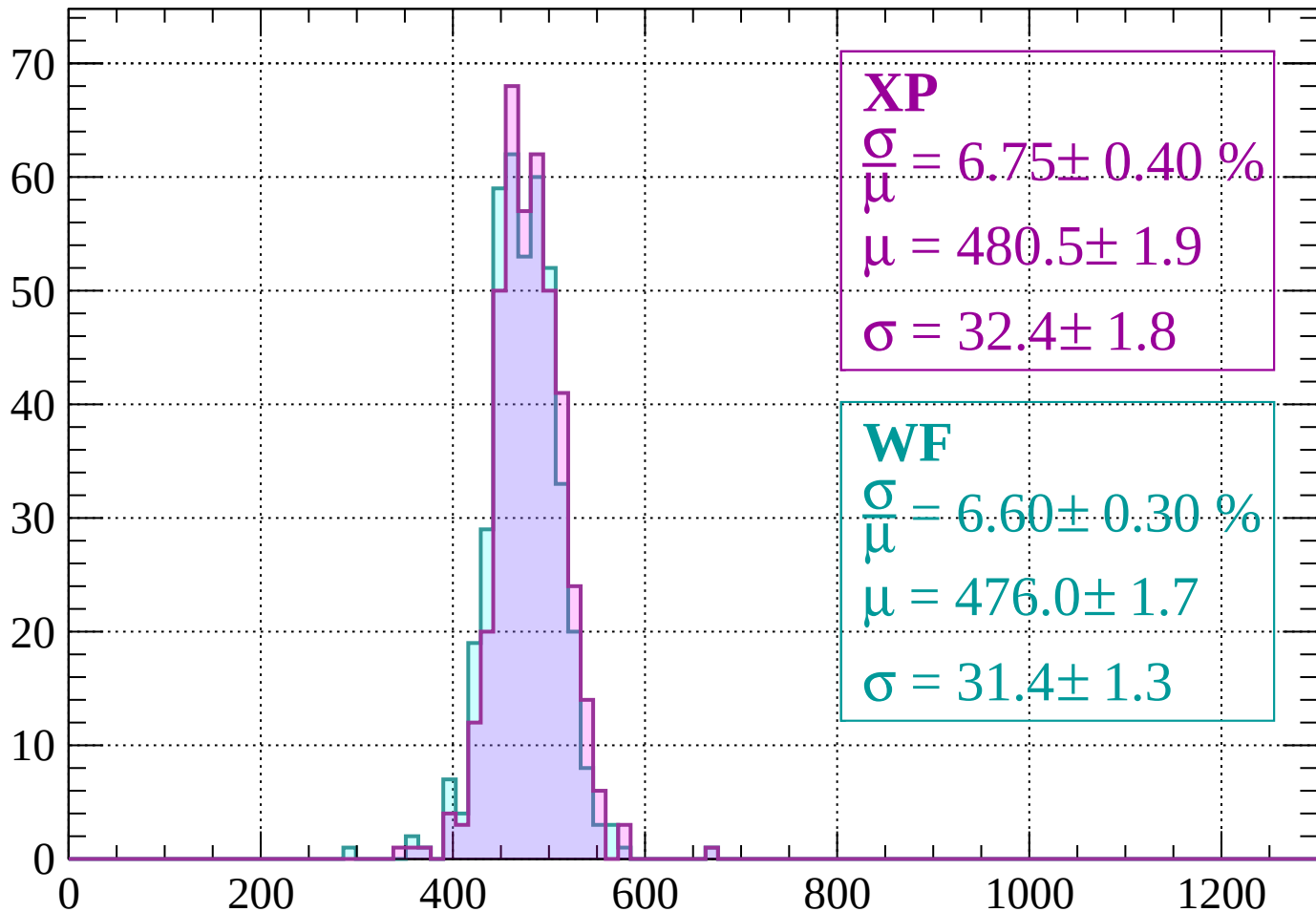
Energy loss | $-1880 < p < -1840$

Count



Energy loss | -1840 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.40 \%$$

$$\mu = 480.5 \pm 1.9$$

$$\sigma = 32.4 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 6.60 \pm 0.30 \%$$

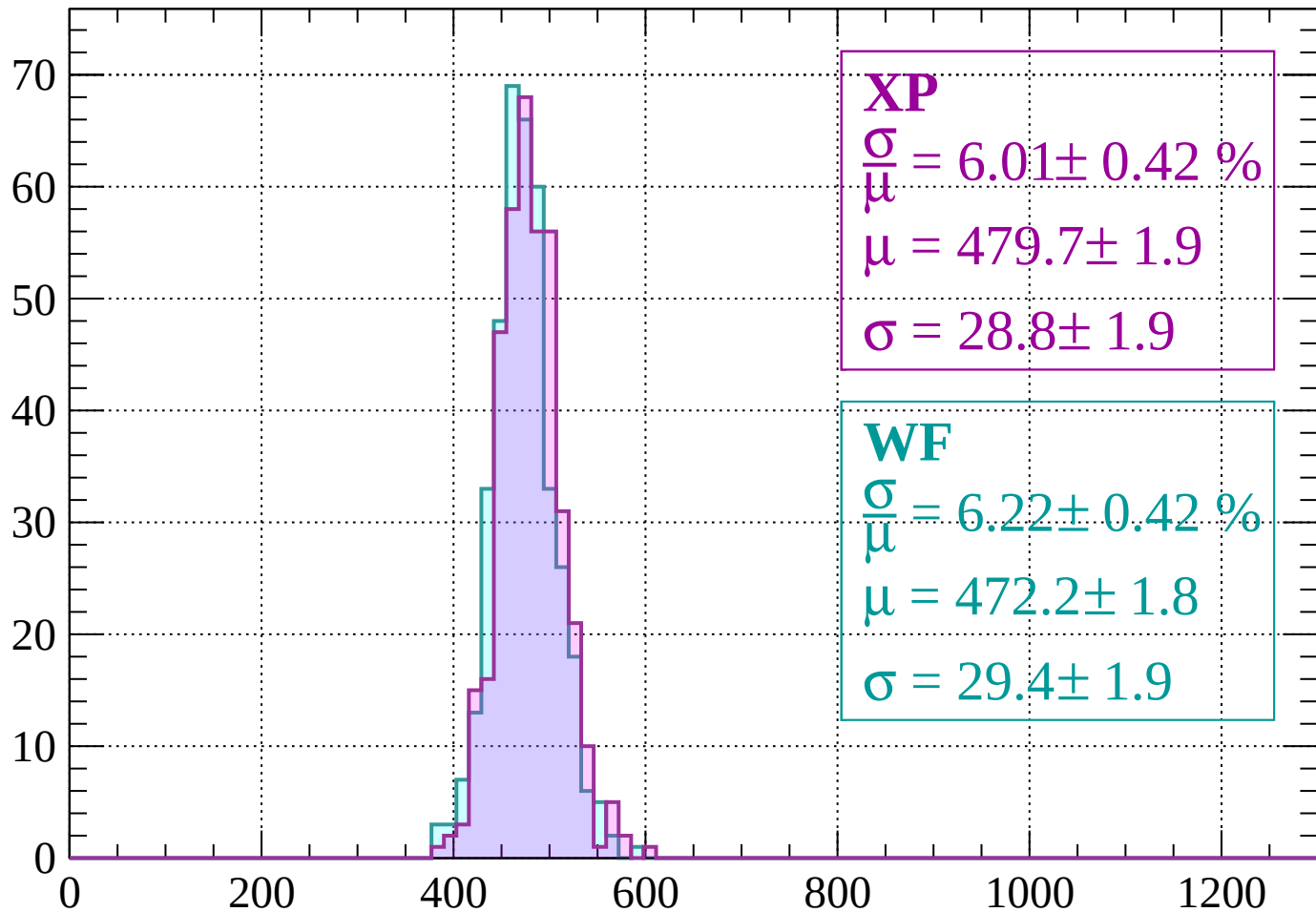
$$\mu = 476.0 \pm 1.7$$

$$\sigma = 31.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.01 \pm 0.42 \%$$

$$\mu = 479.7 \pm 1.9$$

$$\sigma = 28.8 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.42 \%$$

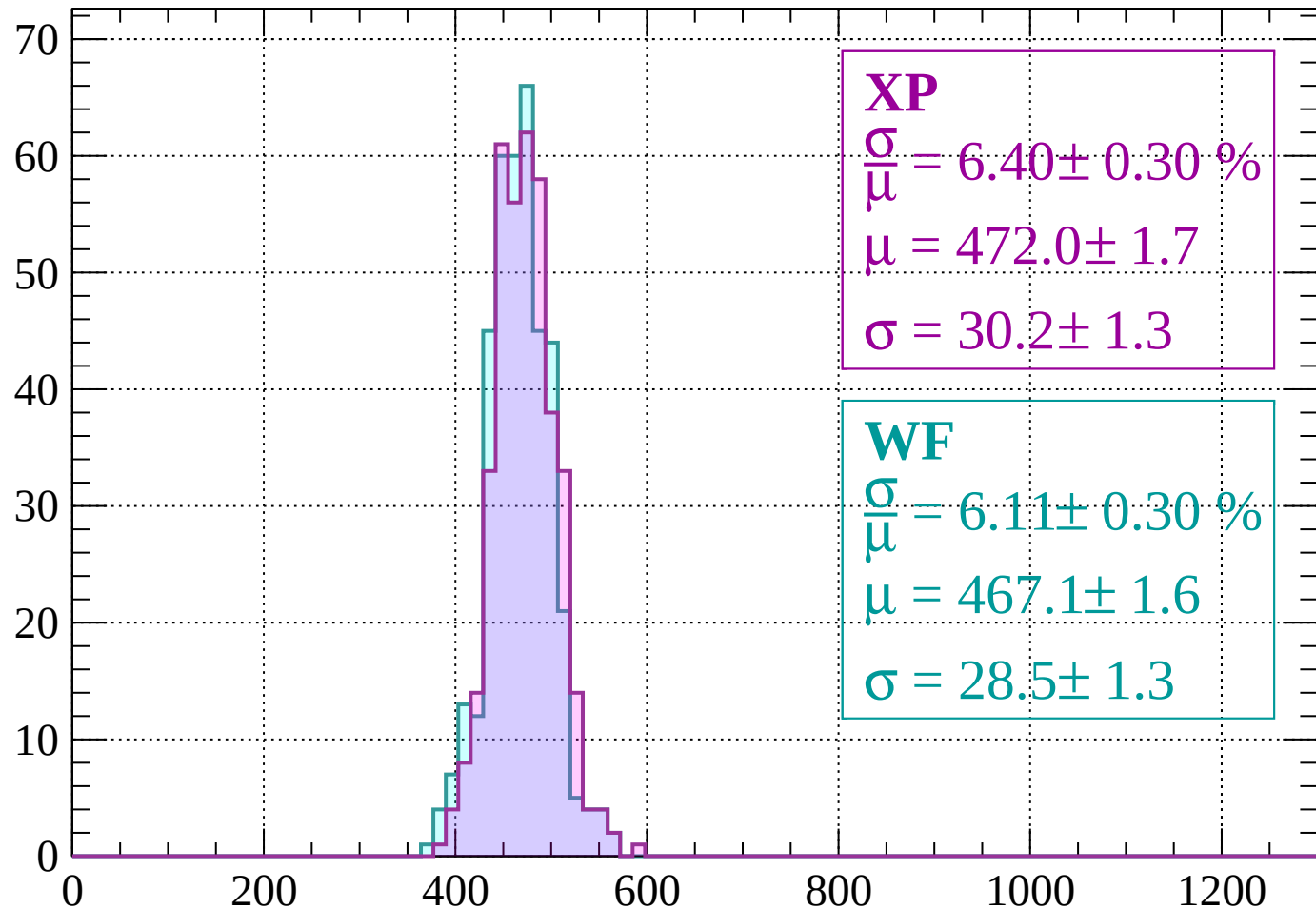
$$\mu = 472.2 \pm 1.8$$

$$\sigma = 29.4 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 6.40 \pm 0.30 \%$$

$$\mu = 472.0 \pm 1.7$$

$$\sigma = 30.2 \pm 1.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 6.11 \pm 0.30 \%$$

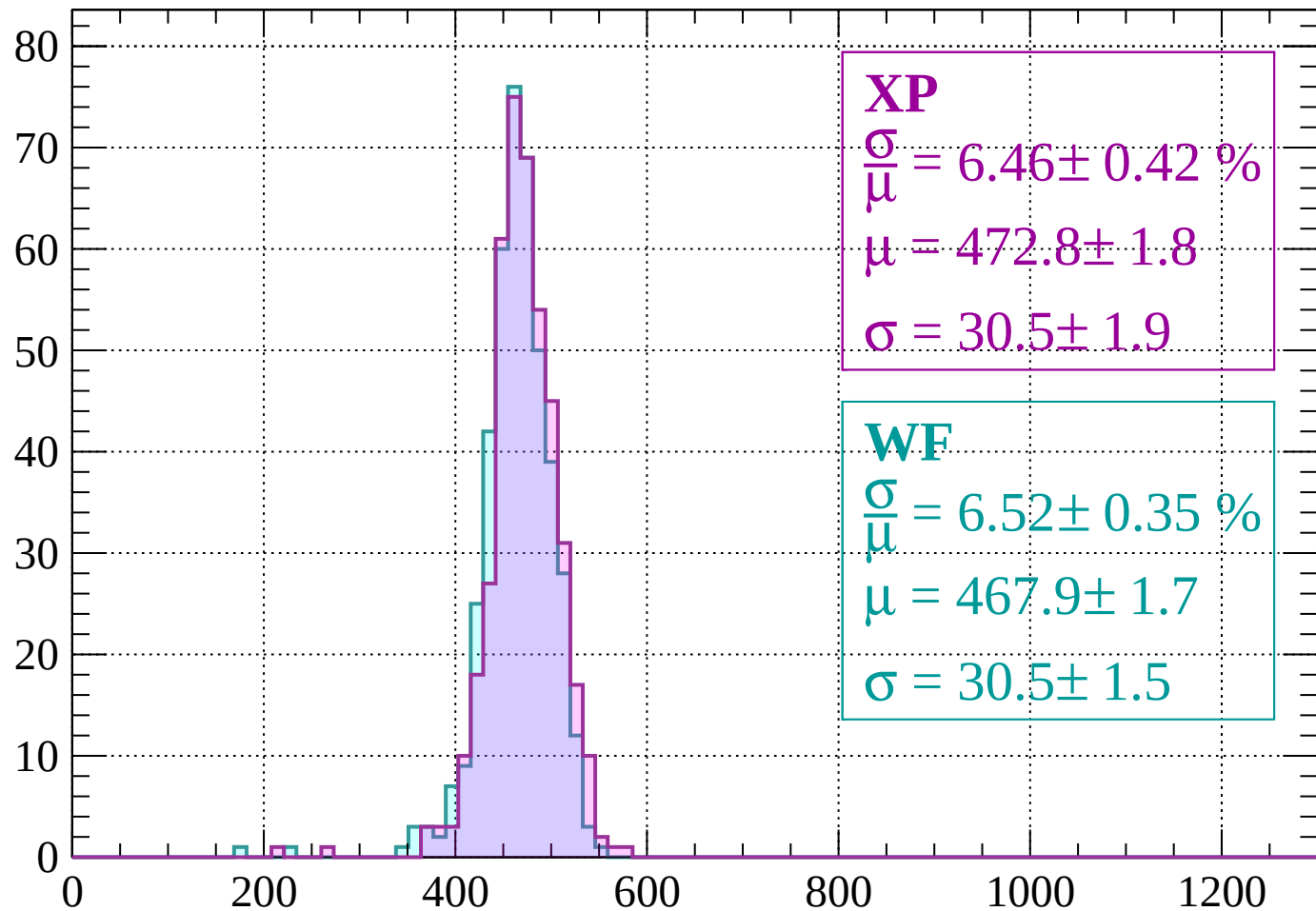
$$\mu = 467.1 \pm 1.6$$

$$\sigma = 28.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.46 \pm 0.42 \%$$

$$\mu = 472.8 \pm 1.8$$

$$\sigma = 30.5 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 6.52 \pm 0.35 \%$$

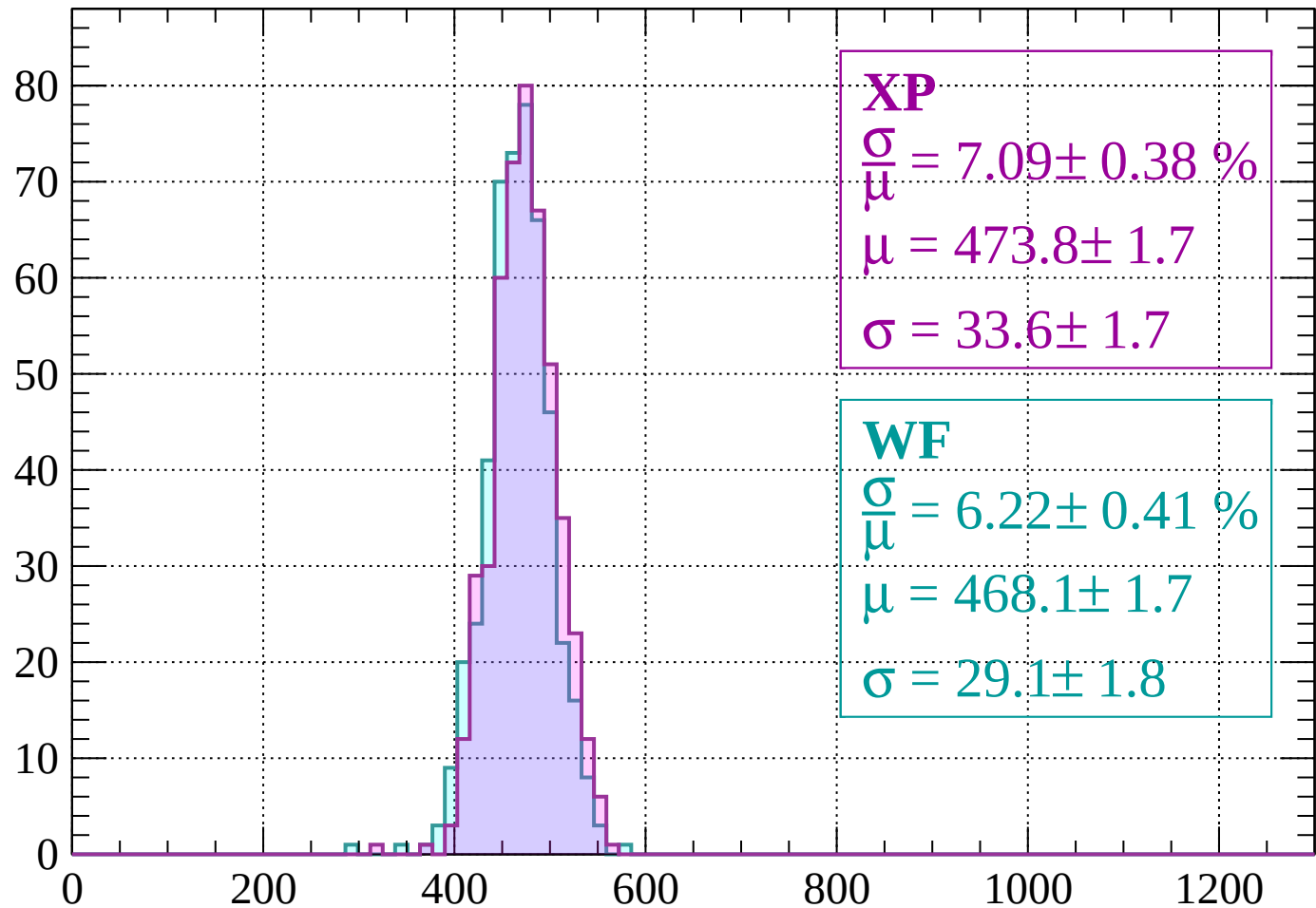
$$\mu = 467.9 \pm 1.7$$

$$\sigma = 30.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 7.09 \pm 0.38 \%$$

$$\mu = 473.8 \pm 1.7$$

$$\sigma = 33.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.41 \%$$

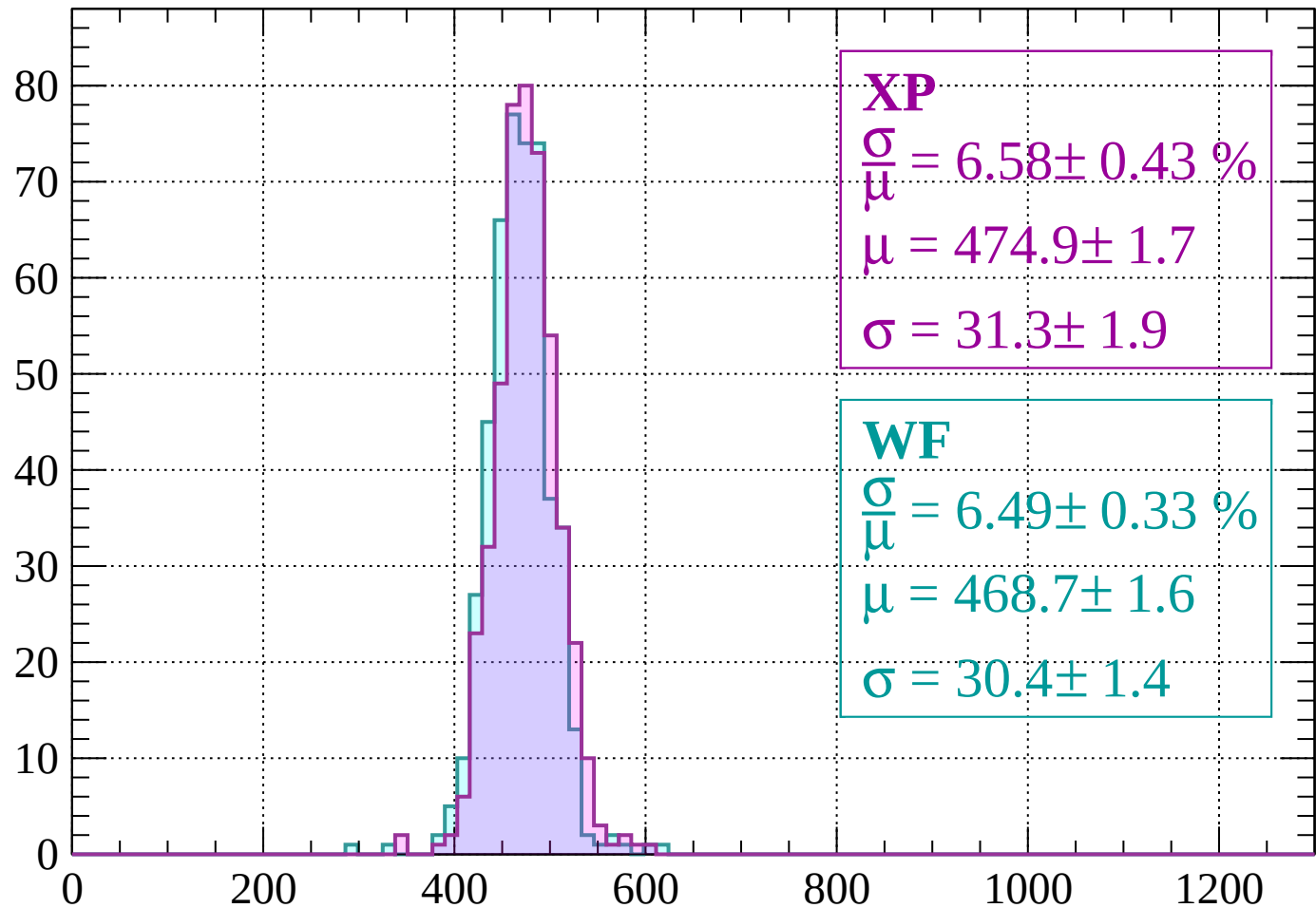
$$\mu = 468.1 \pm 1.7$$

$$\sigma = 29.1 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 6.58 \pm 0.43 \%$$

$$\mu = 474.9 \pm 1.7$$

$$\sigma = 31.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 6.49 \pm 0.33 \%$$

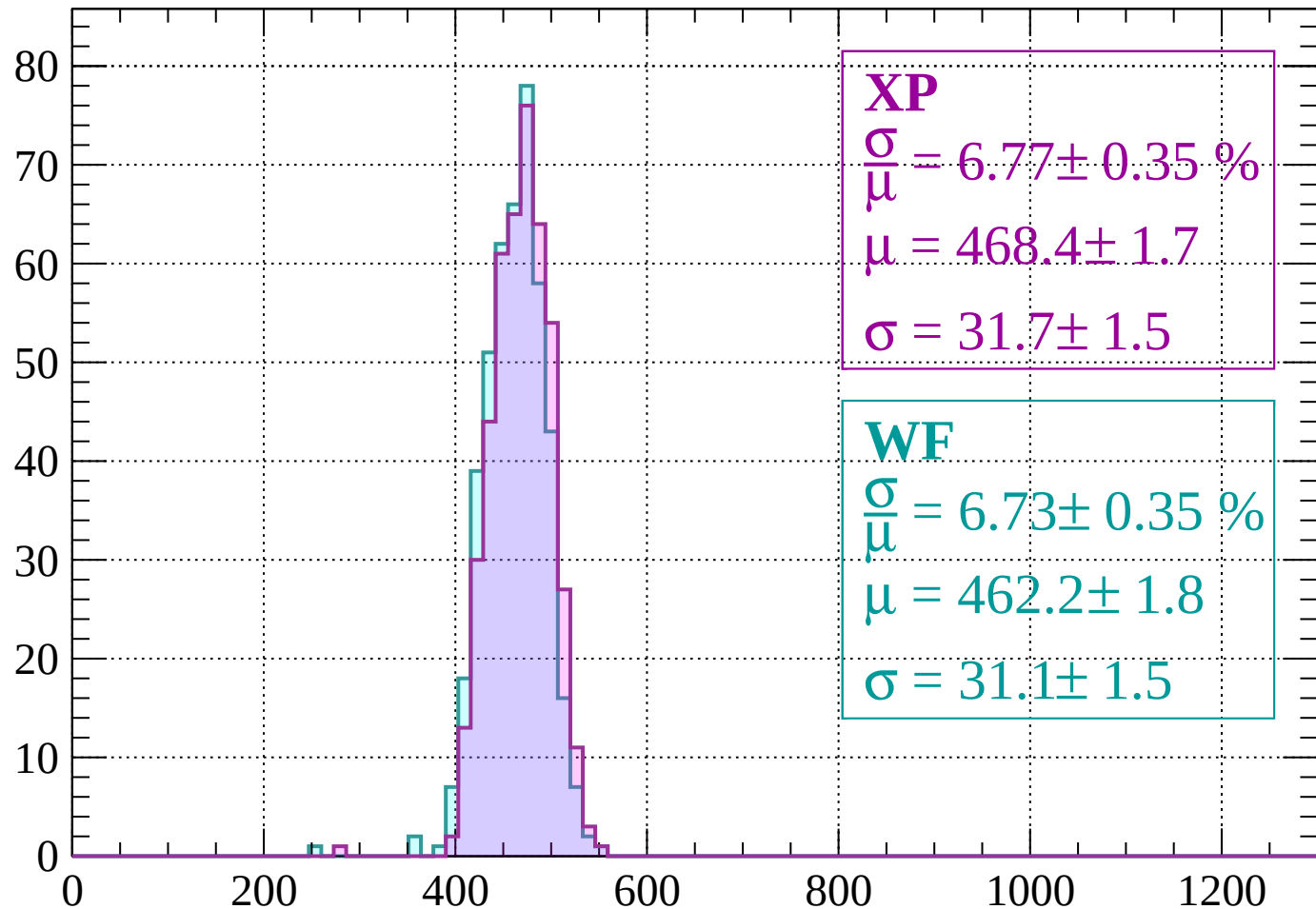
$$\mu = 468.7 \pm 1.6$$

$$\sigma = 30.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 6.77 \pm 0.35 \%$$

$$\mu = 468.4 \pm 1.7$$

$$\sigma = 31.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 6.73 \pm 0.35 \%$$

$$\mu = 462.2 \pm 1.8$$

$$\sigma = 31.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

Count

XP

$$\frac{\sigma}{\mu} = 6.67 \pm 0.33 \%$$

$$\mu = 467.6 \pm 1.7$$

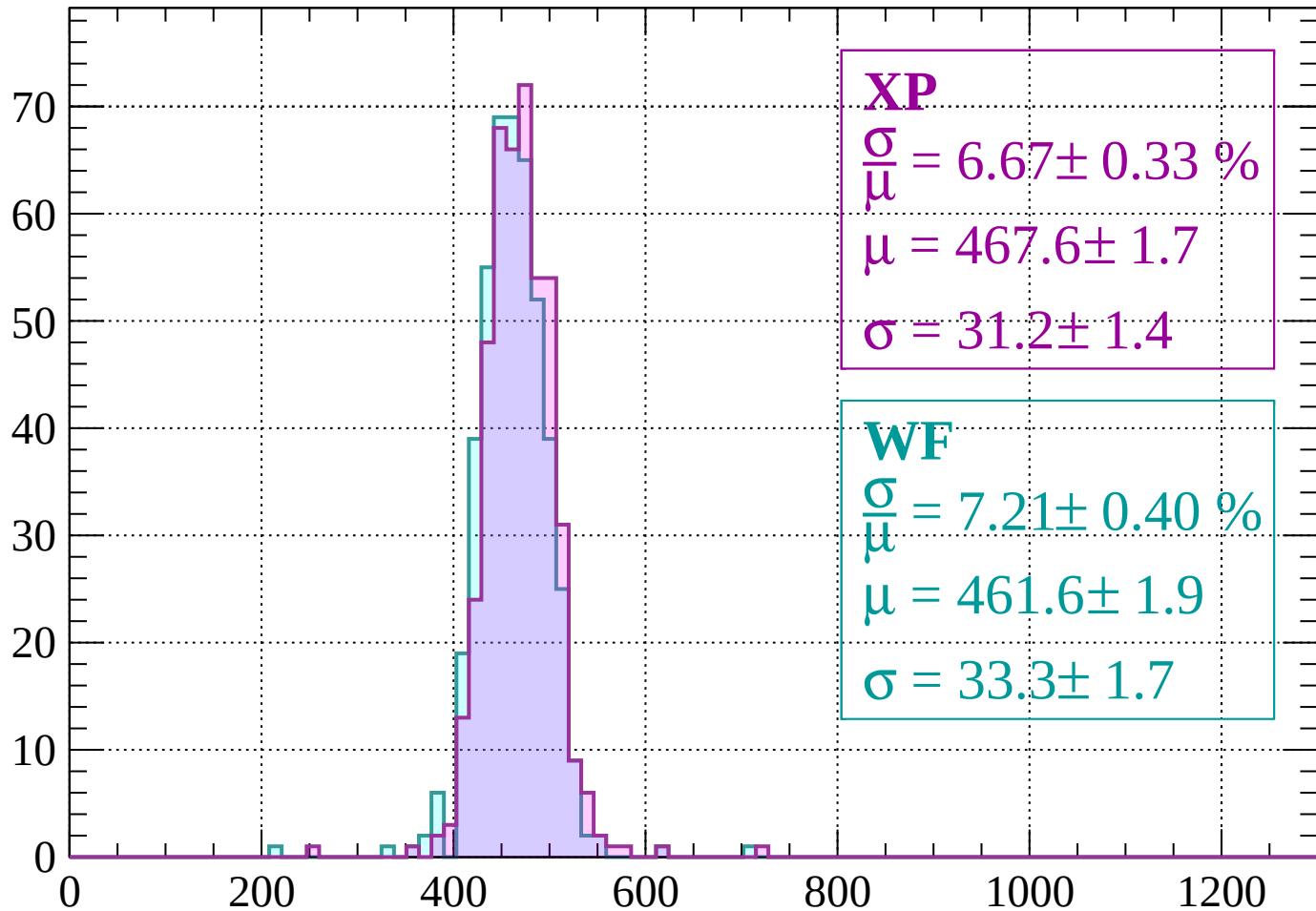
$$\sigma = 31.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.21 \pm 0.40 \%$$

$$\mu = 461.6 \pm 1.9$$

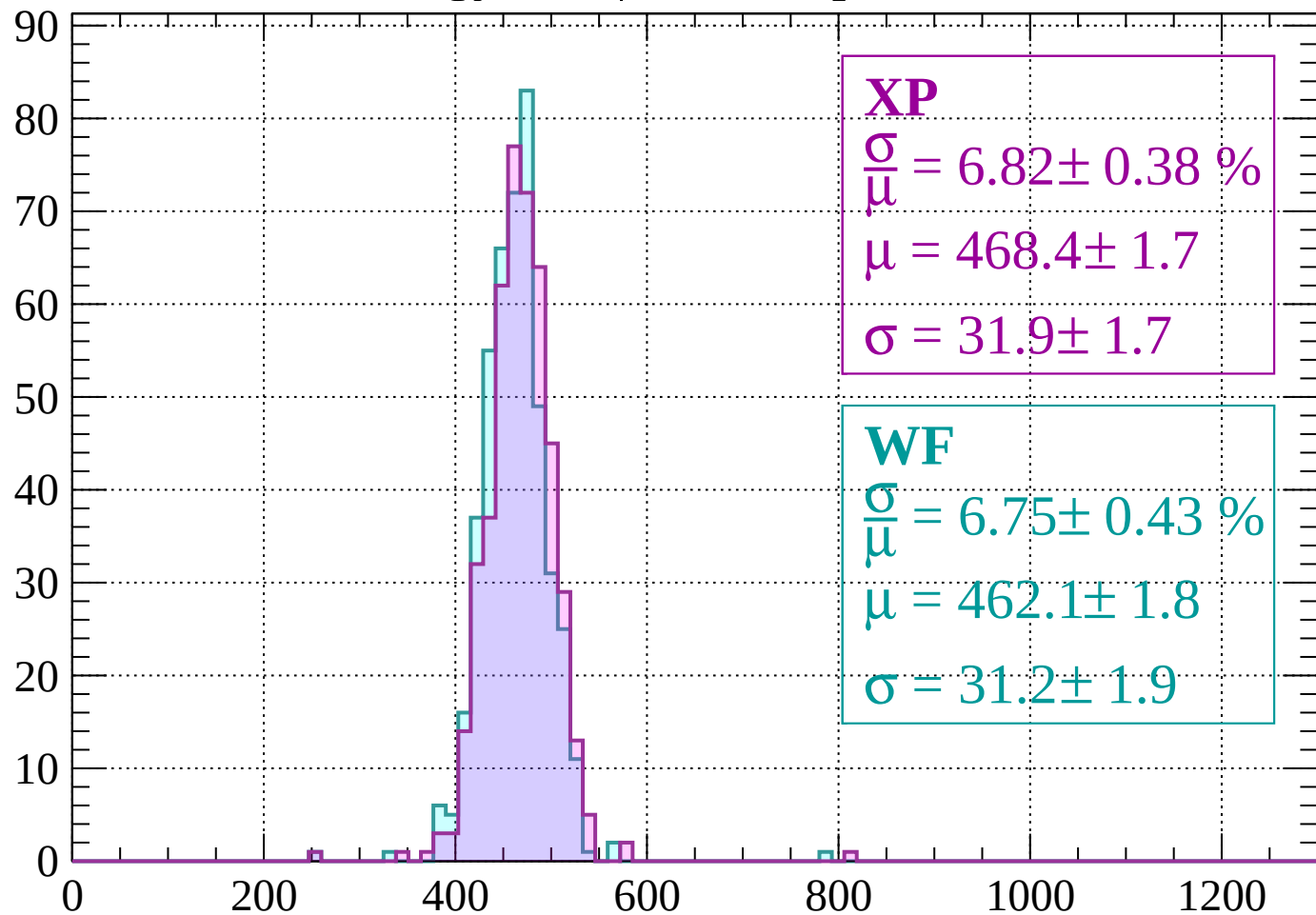
$$\sigma = 33.3 \pm 1.7$$



dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 6.82 \pm 0.38 \%$$

$$\mu = 468.4 \pm 1.7$$

$$\sigma = 31.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.75 \pm 0.43 \%$$

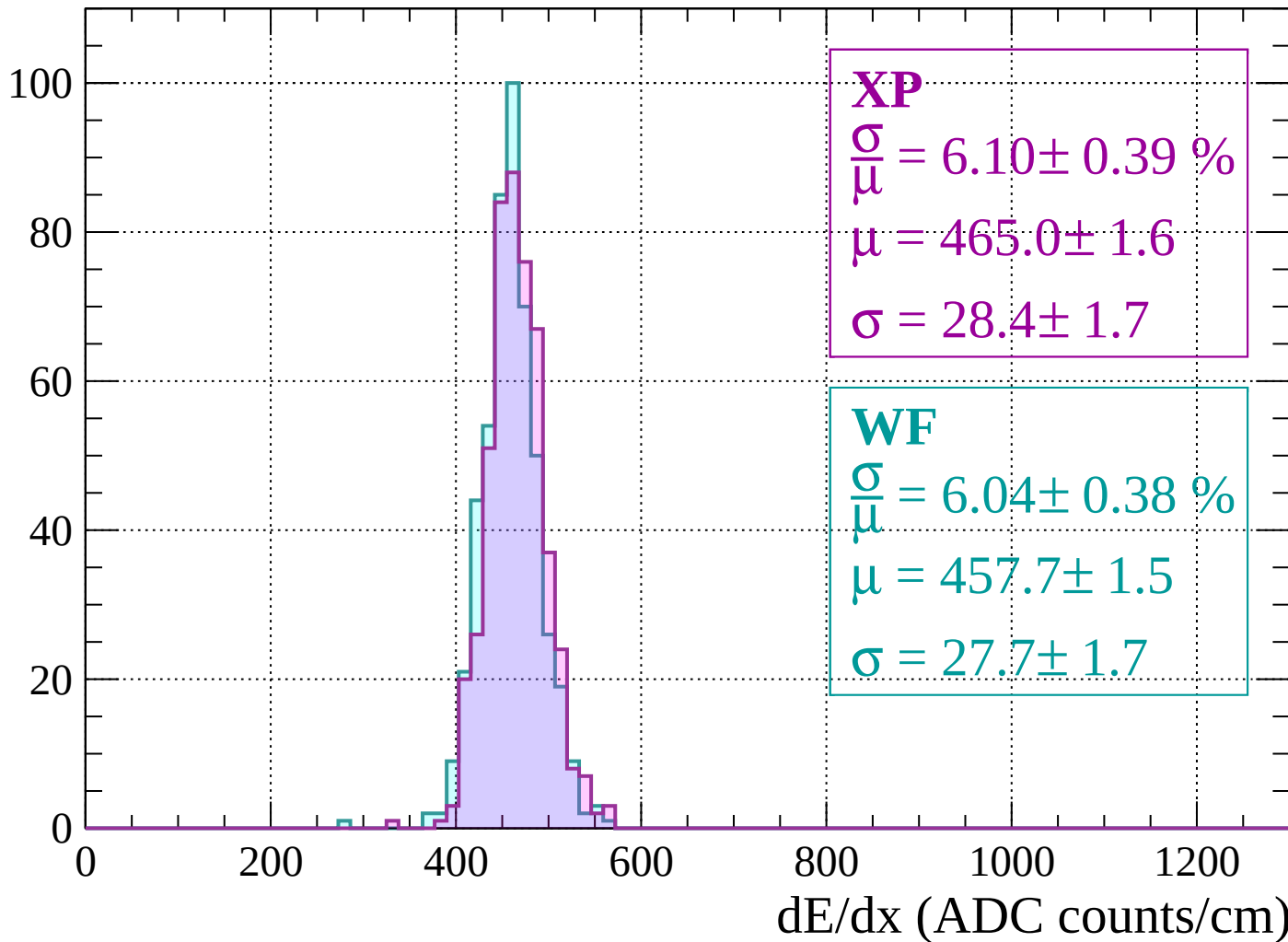
$$\mu = 462.1 \pm 1.8$$

$$\sigma = 31.2 \pm 1.9$$

dE/dx (ADC counts/cm)

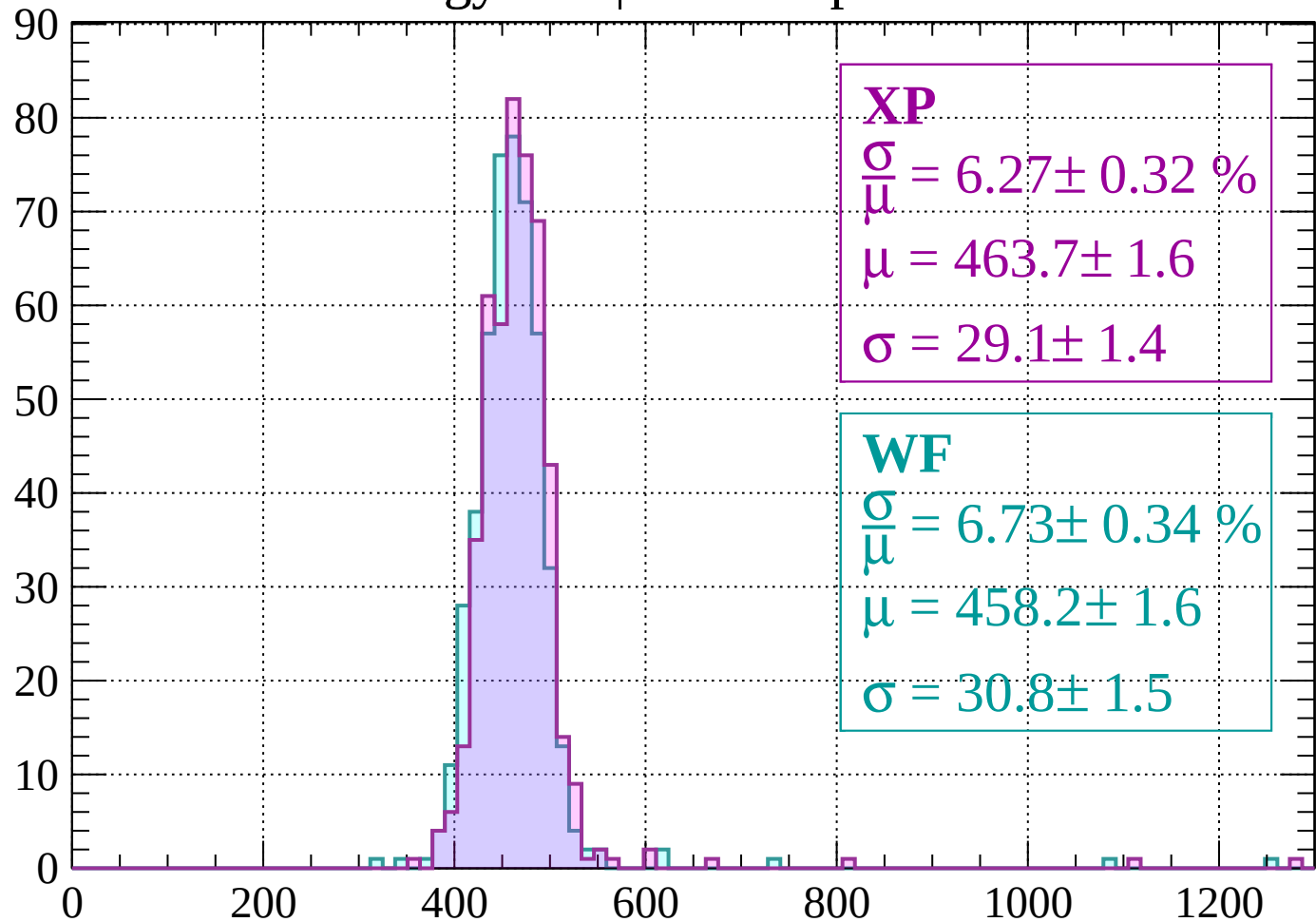
Energy loss | $-1480 < p < -1440$

Count



Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 6.27 \pm 0.32 \%$$

$$\mu = 463.7 \pm 1.6$$

$$\sigma = 29.1 \pm 1.4$$

WF

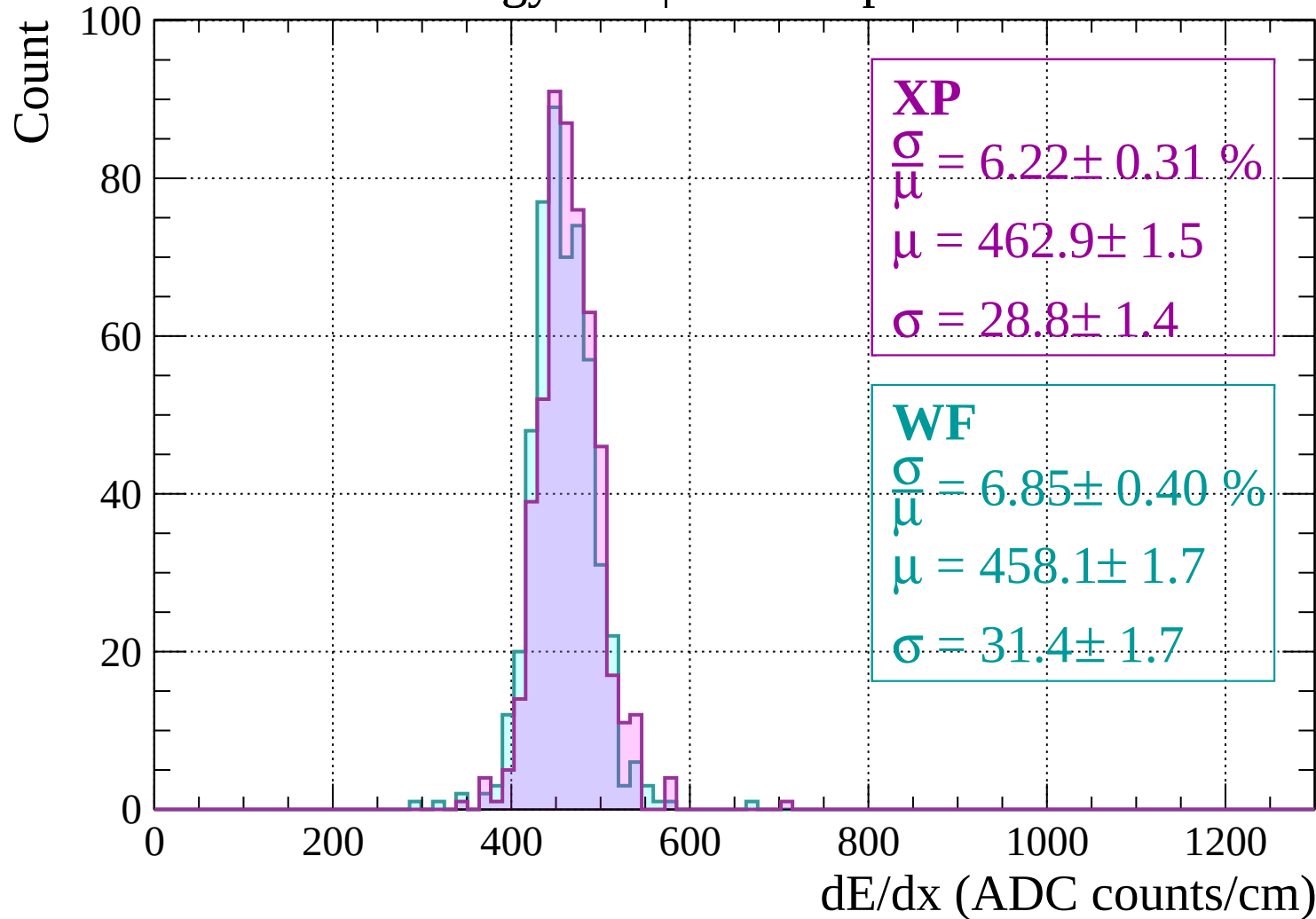
$$\frac{\sigma}{\mu} = 6.73 \pm 0.34 \%$$

$$\mu = 458.2 \pm 1.6$$

$$\sigma = 30.8 \pm 1.5$$

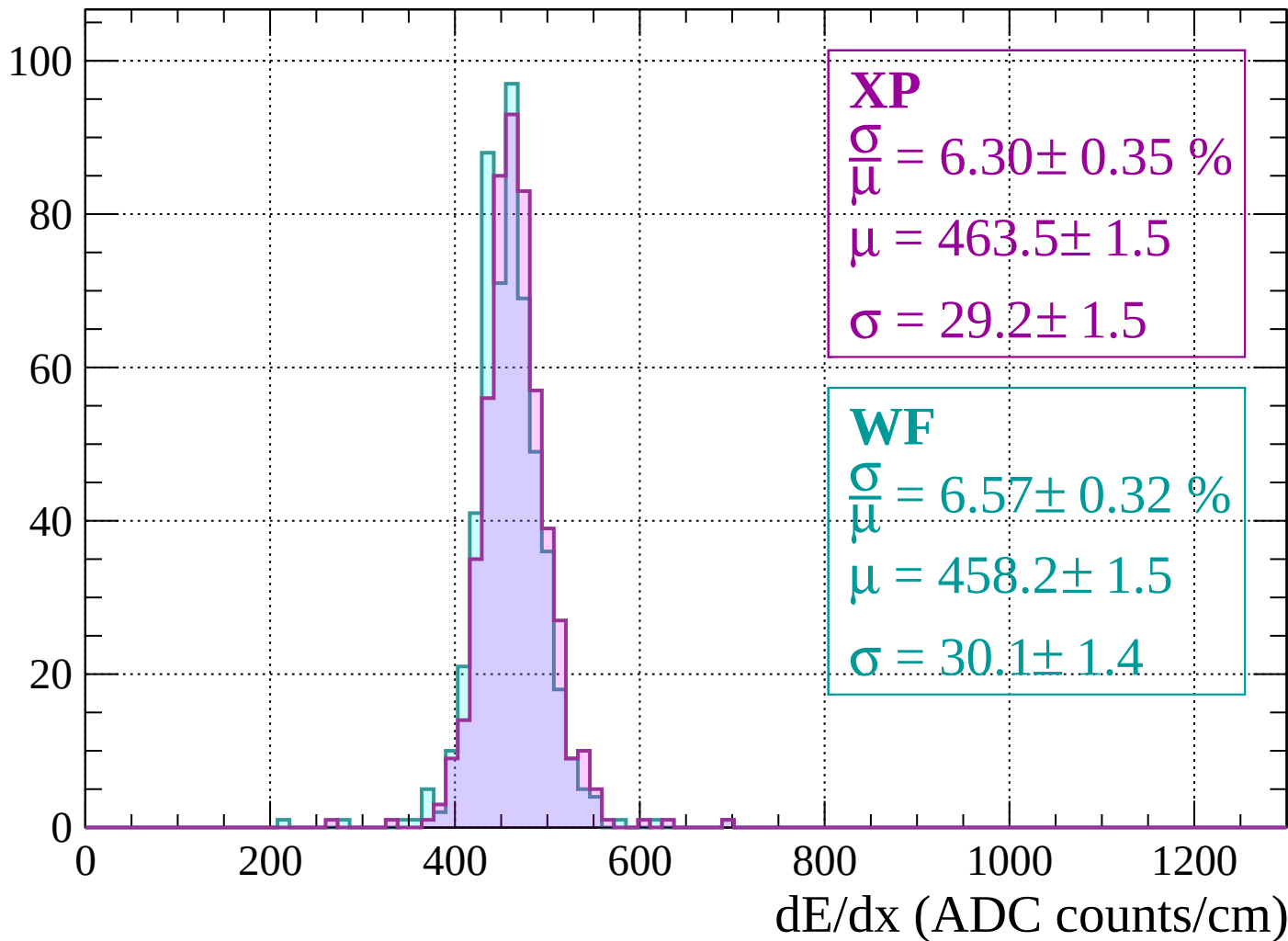
dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$



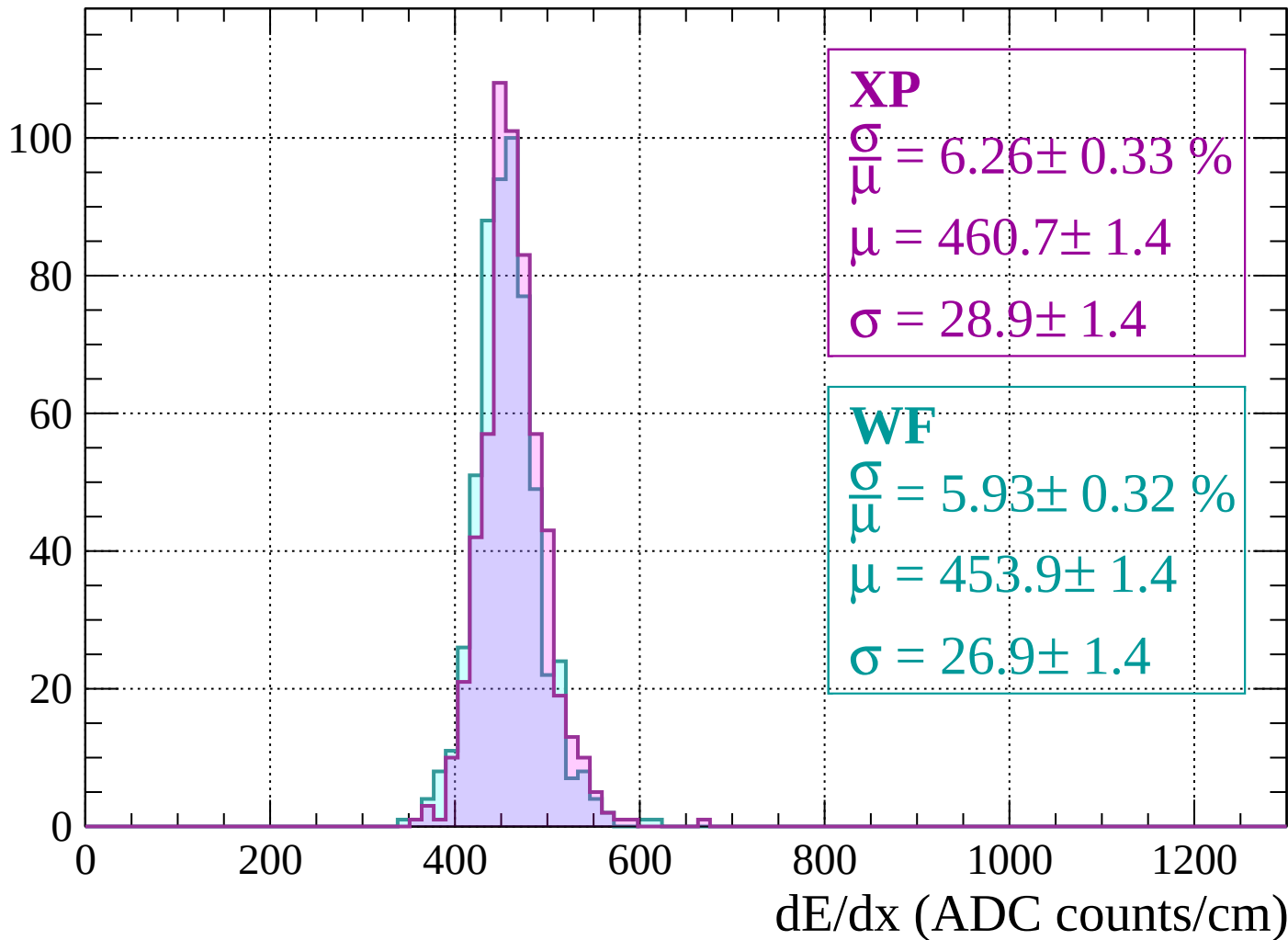
Energy loss | $-1360 < p < -1320$

Count

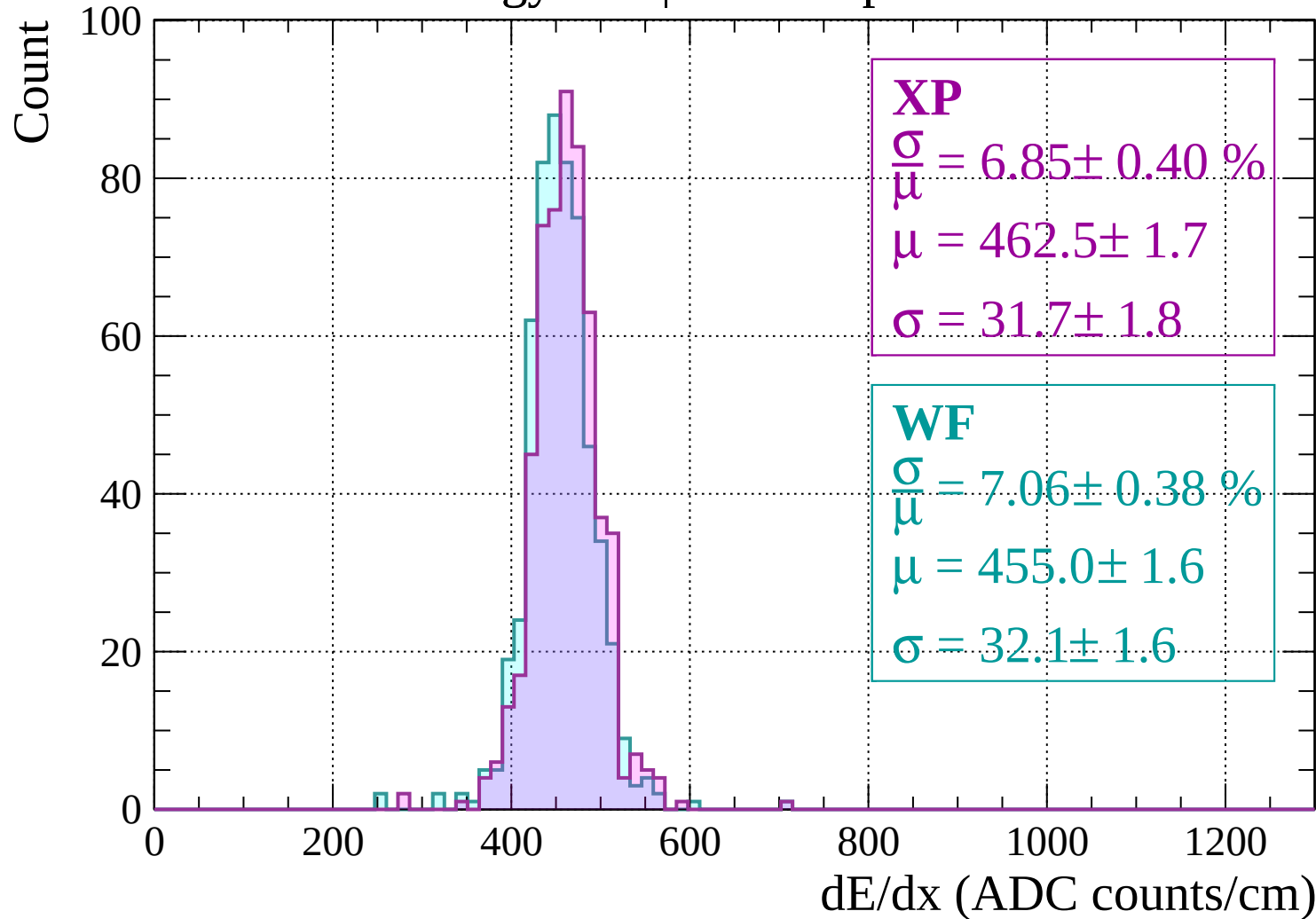


Energy loss | $-1320 < p < -1280$

Count

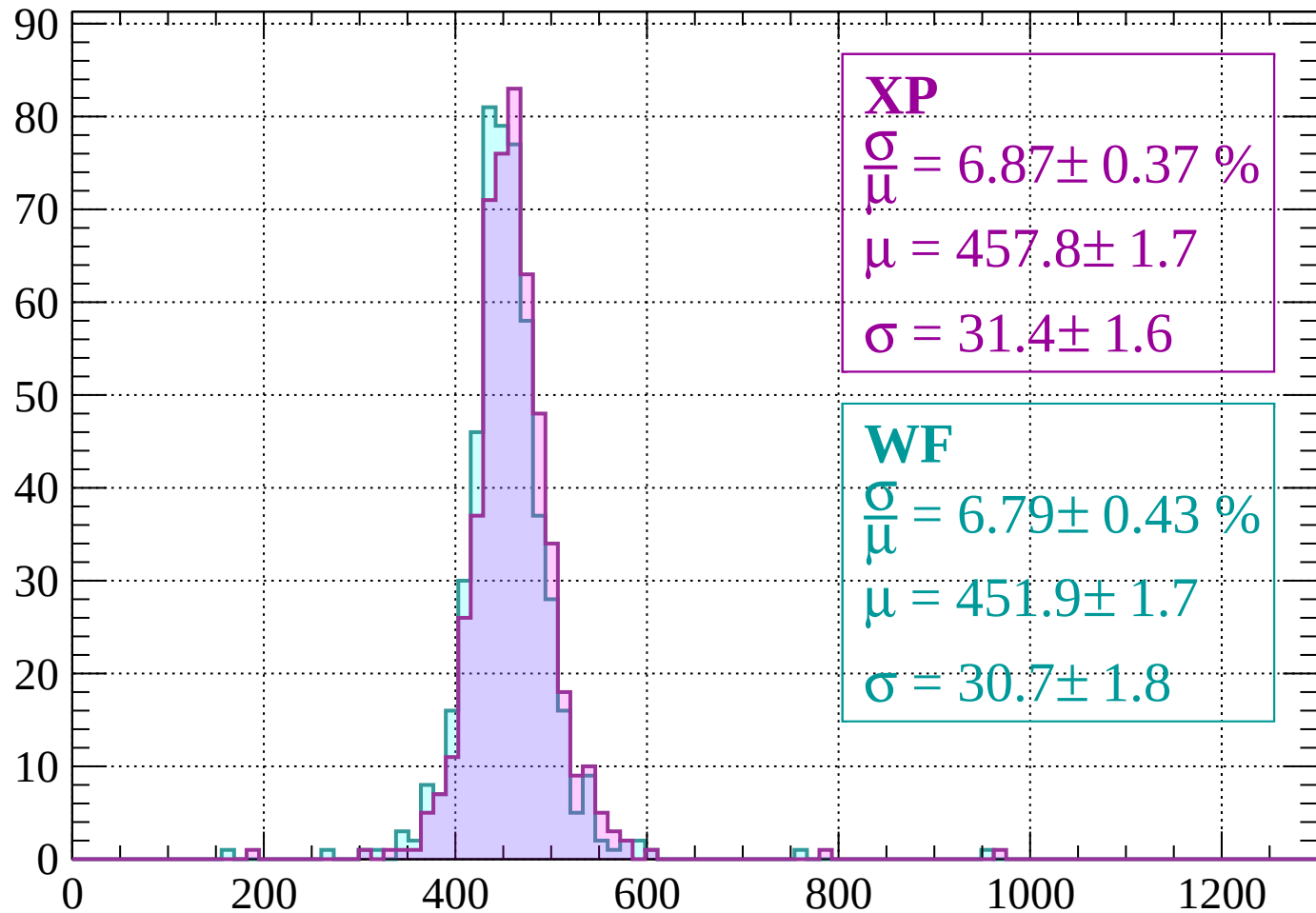


Energy loss | $-1280 < p < -1240$



Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 6.87 \pm 0.37 \%$$

$$\mu = 457.8 \pm 1.7$$

$$\sigma = 31.4 \pm 1.6$$

WF

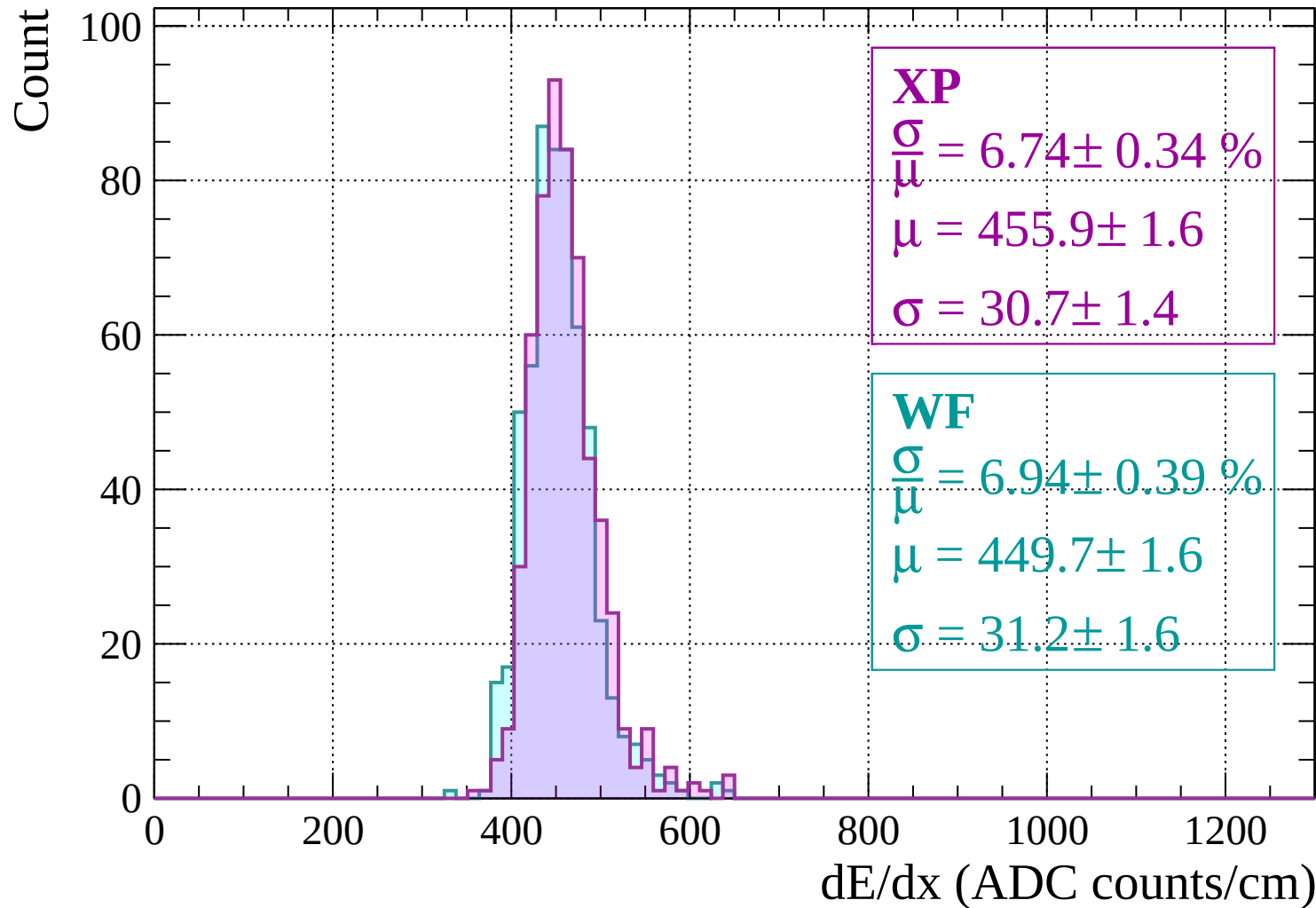
$$\frac{\sigma}{\mu} = 6.79 \pm 0.43 \%$$

$$\mu = 451.9 \pm 1.7$$

$$\sigma = 30.7 \pm 1.8$$

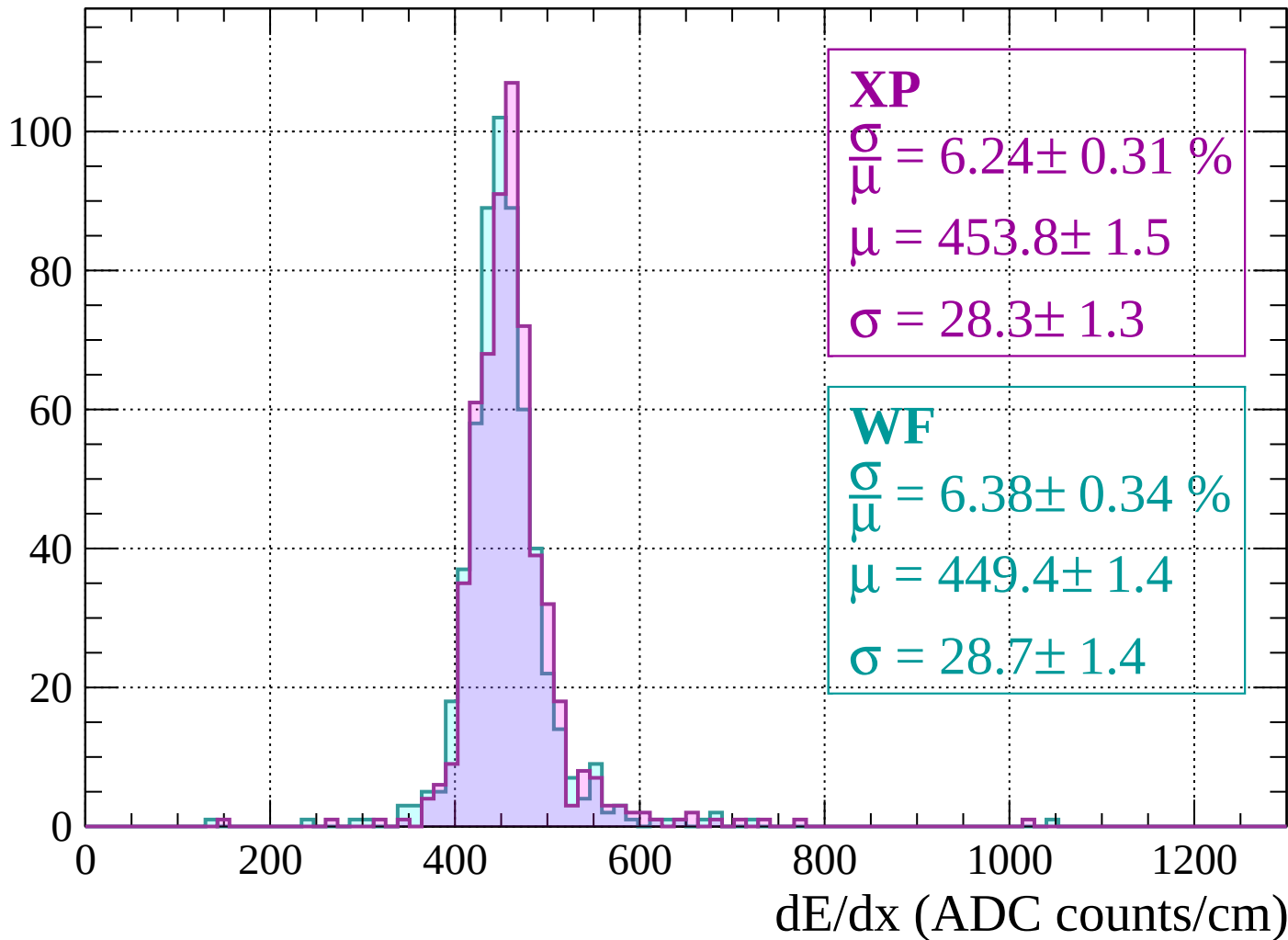
dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160



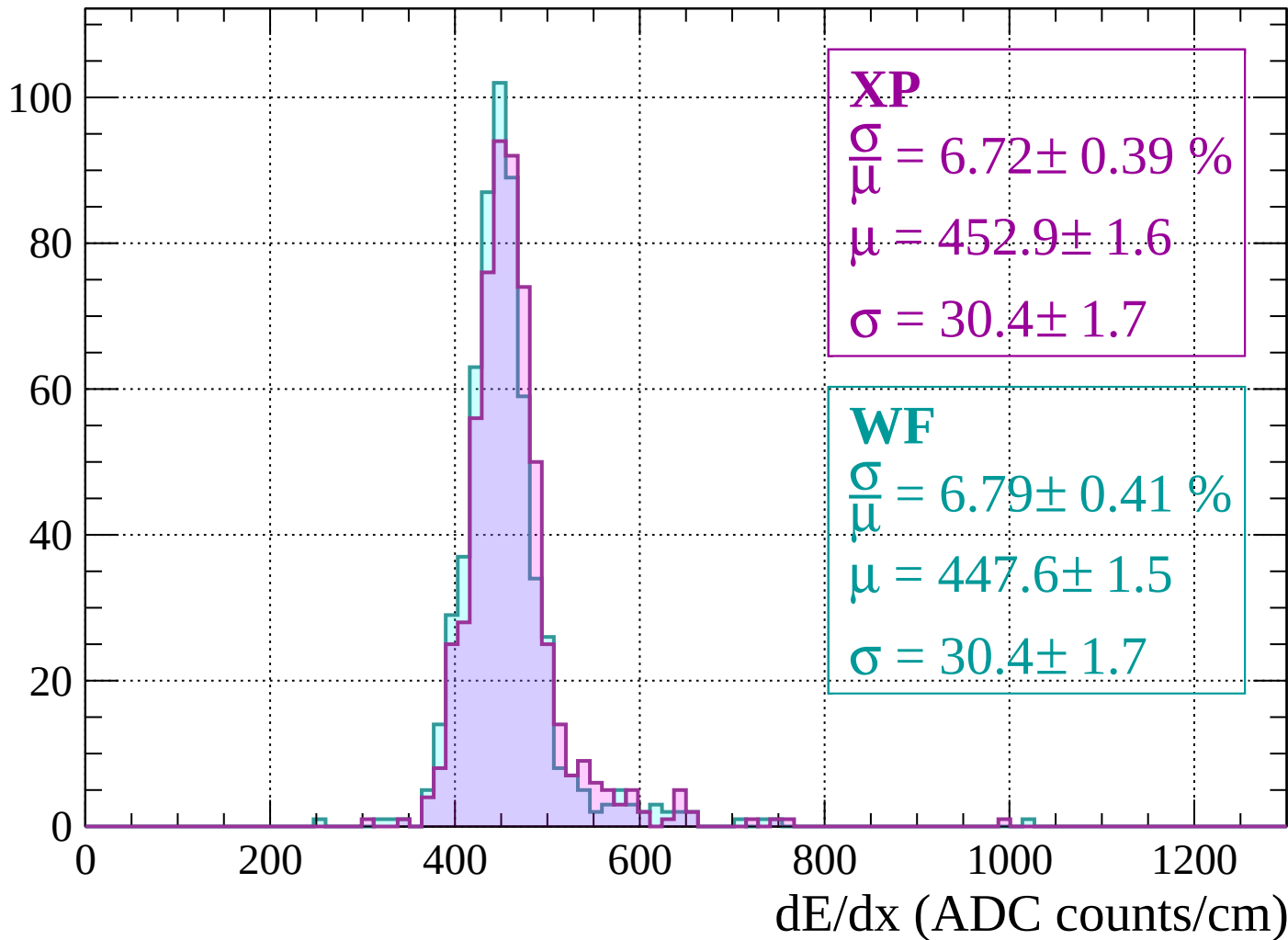
Energy loss | $-1160 < p < -1120$

Count



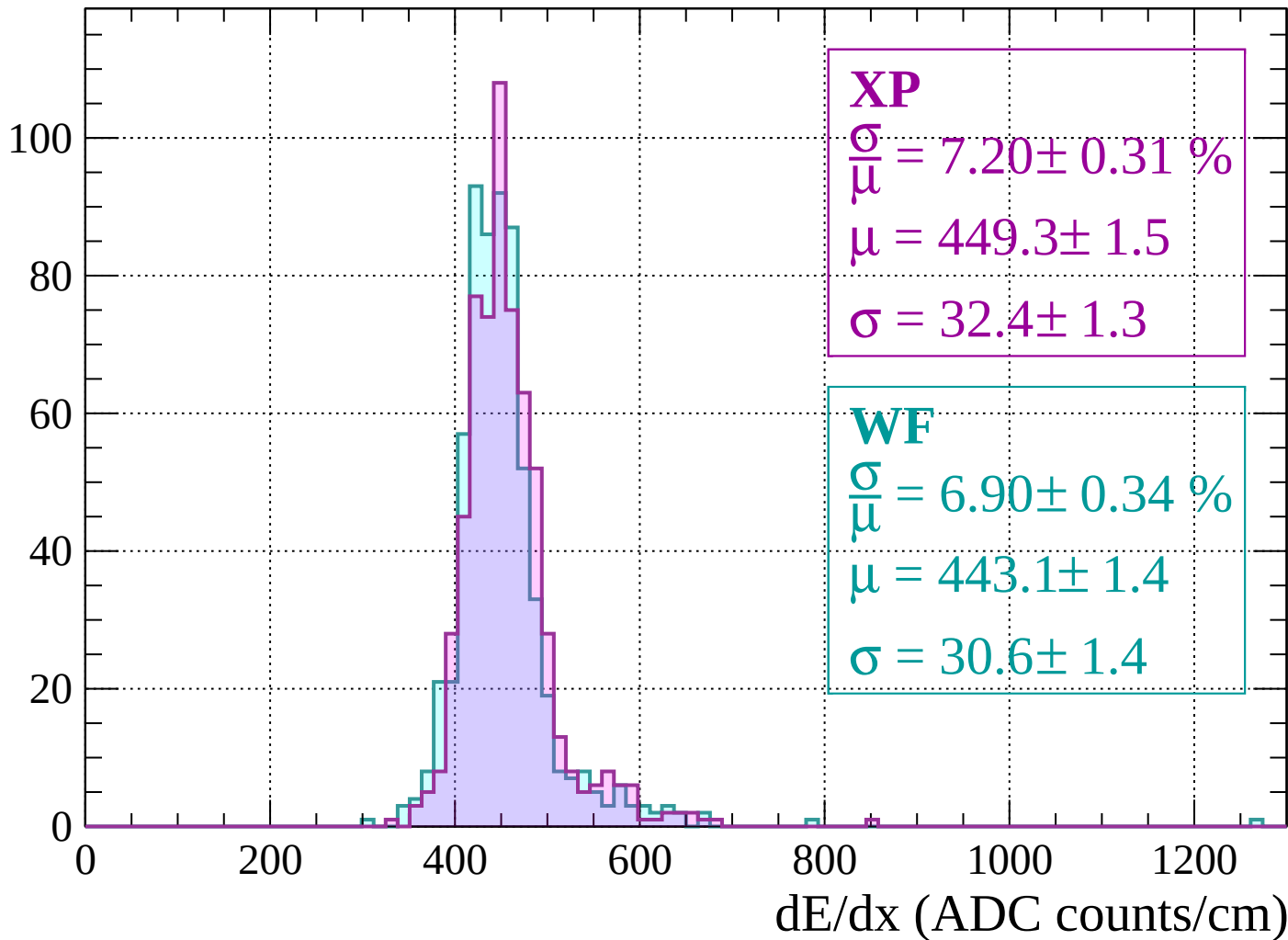
Energy loss | $-1120 < p < -1080$

Count



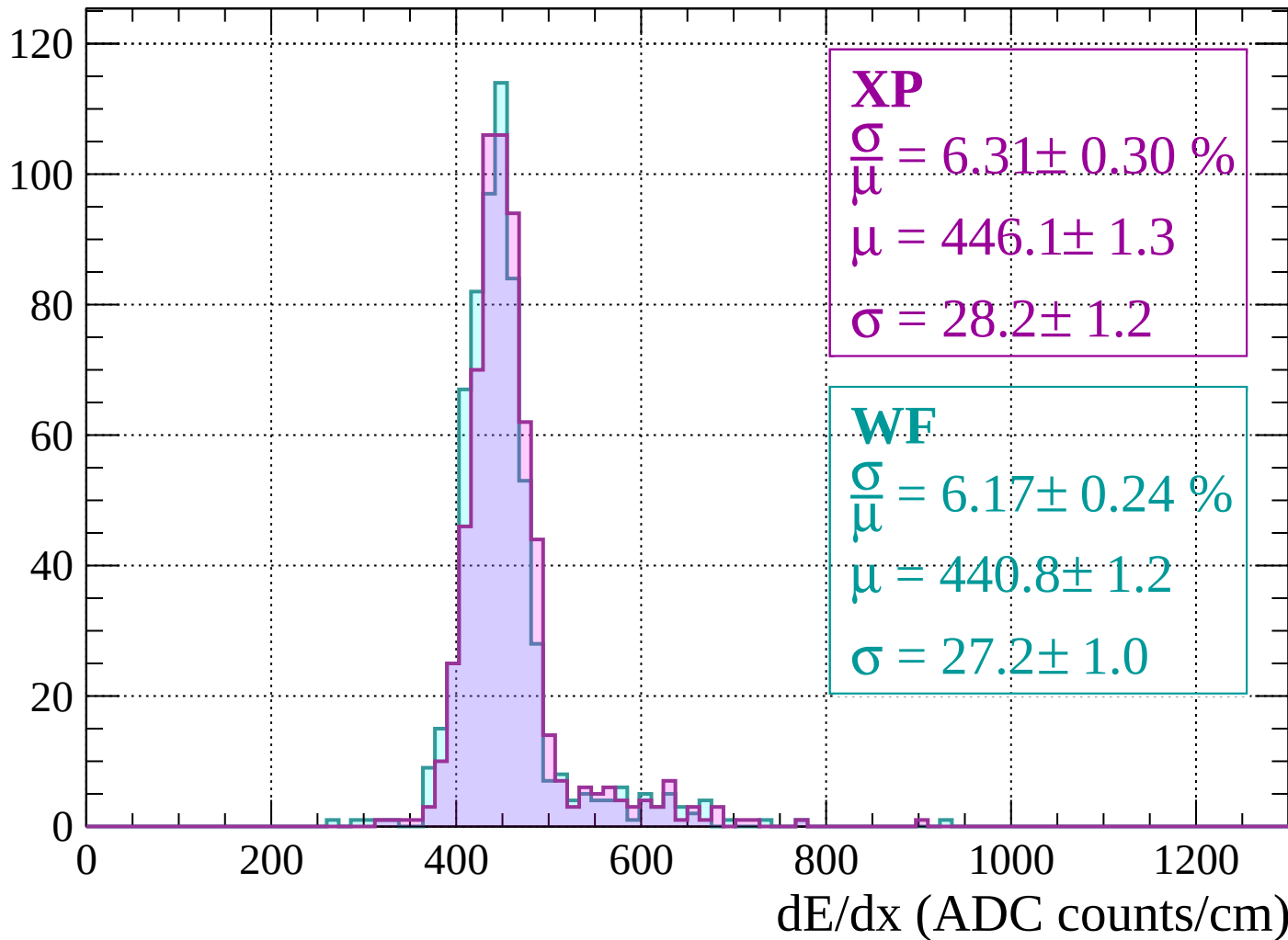
Energy loss | $-1080 < p < -1040$

Count



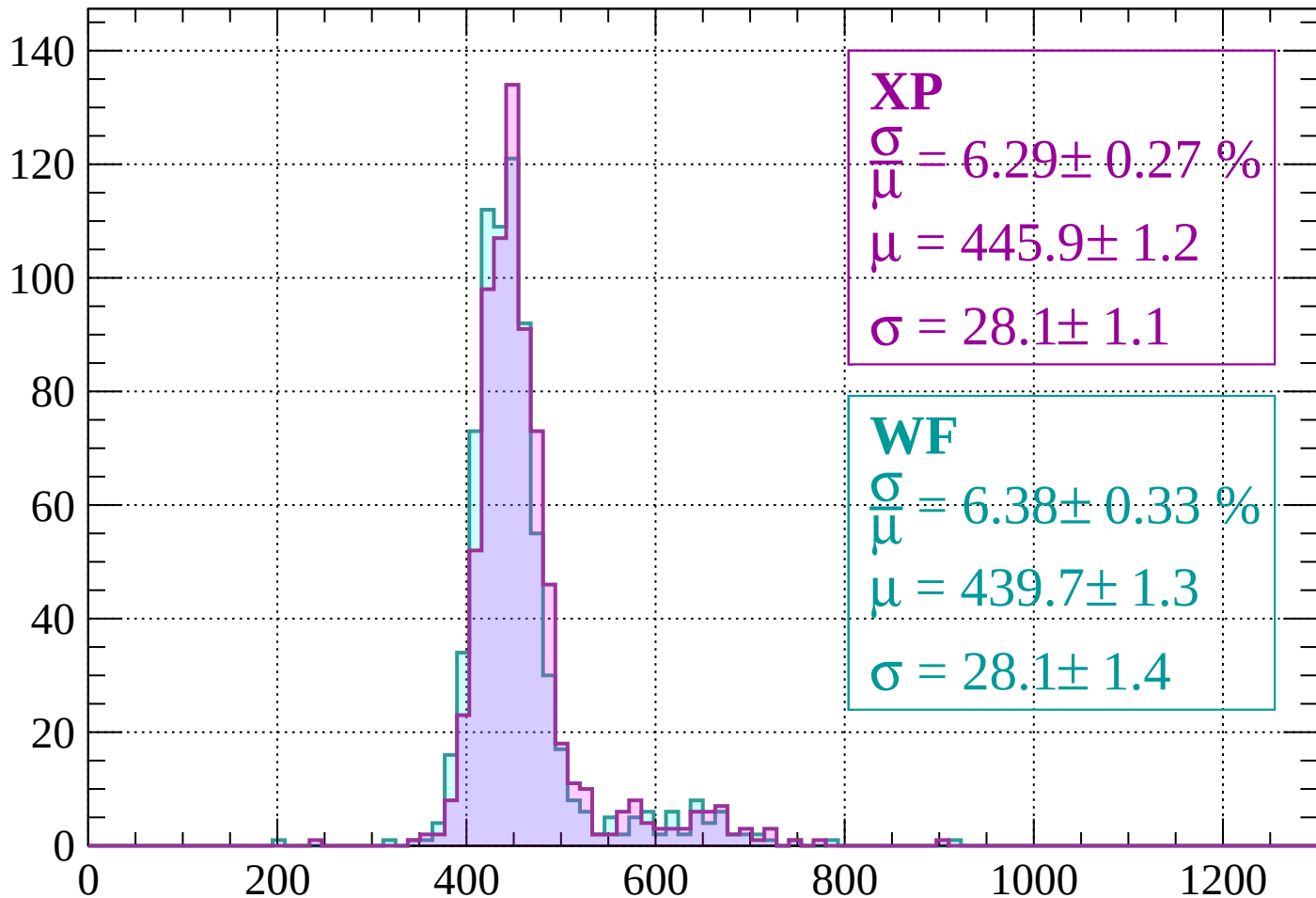
Energy loss | -1040 < p < -1000

Count



Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 6.29 \pm 0.27 \%$$

$$\mu = 445.9 \pm 1.2$$

$$\sigma = 28.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.38 \pm 0.33 \%$$

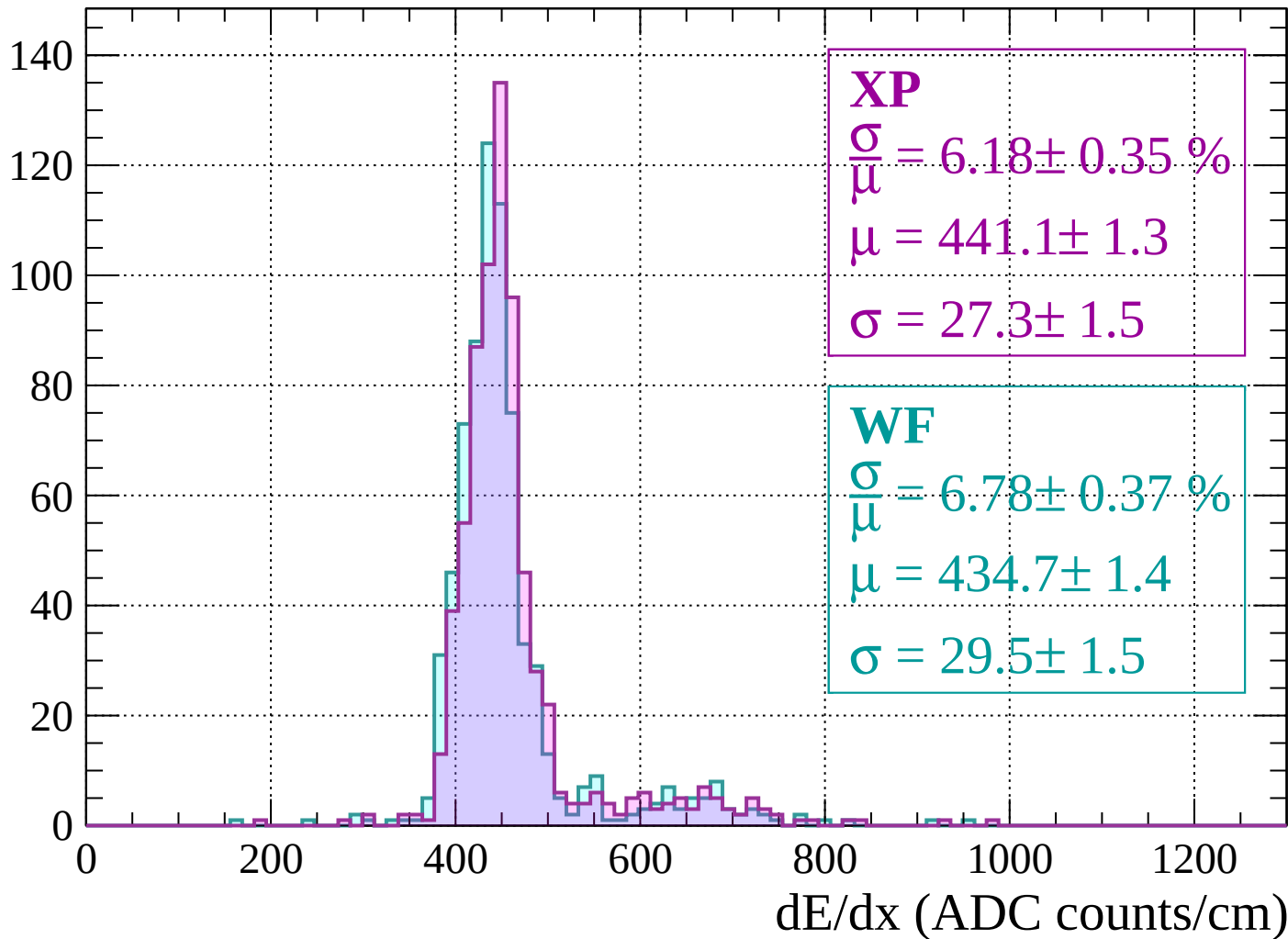
$$\mu = 439.7 \pm 1.3$$

$$\sigma = 28.1 \pm 1.4$$

dE/dx (ADC counts/cm)

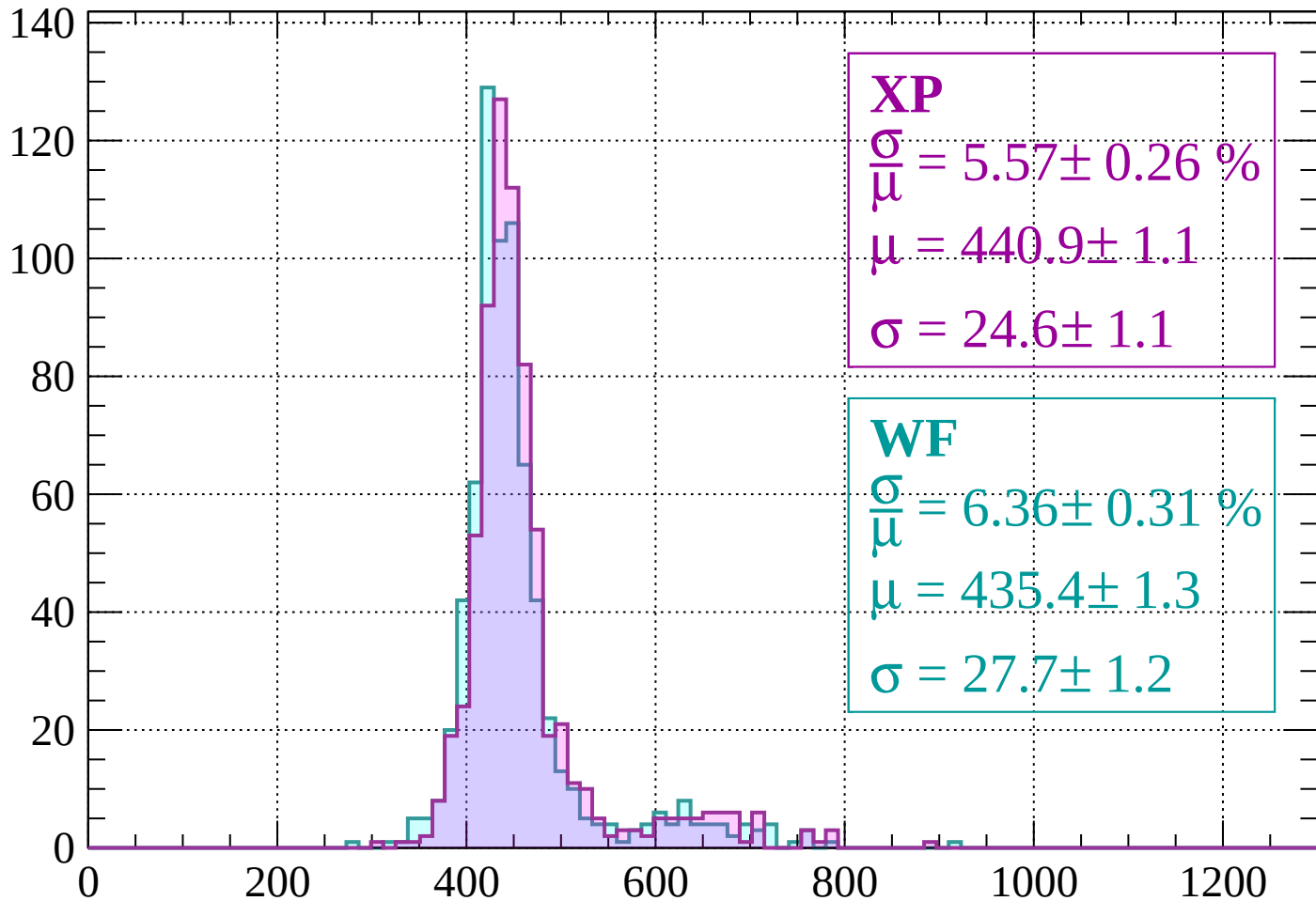
Energy loss | $-960 < p < -920$

Count



Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 5.57 \pm 0.26 \%$$

$$\mu = 440.9 \pm 1.1$$

$$\sigma = 24.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.36 \pm 0.31 \%$$

$$\mu = 435.4 \pm 1.3$$

$$\sigma = 27.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 6.53 \pm 0.29 \%$$

$$\mu = 435.5 \pm 1.3$$

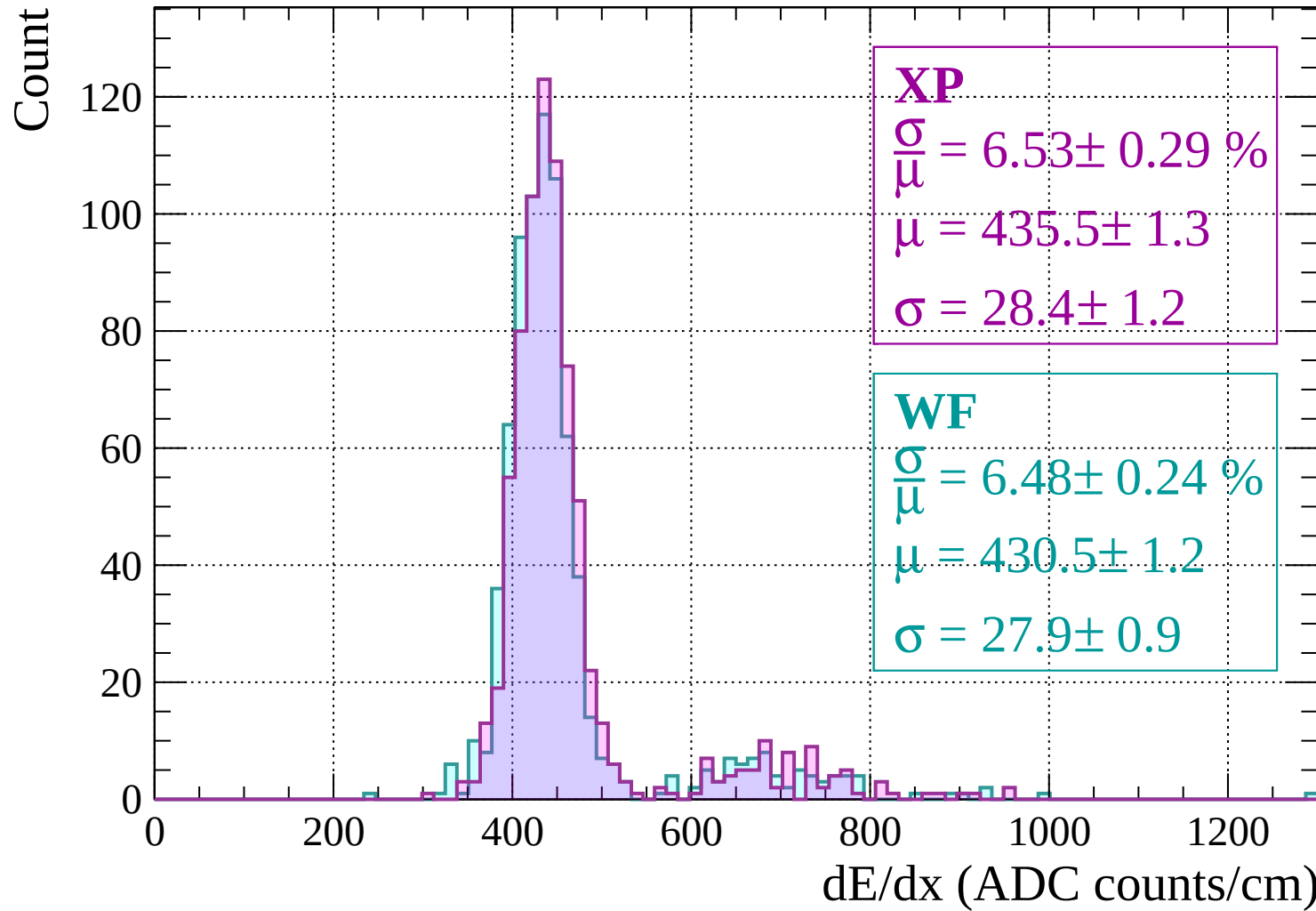
$$\sigma = 28.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.24 \%$$

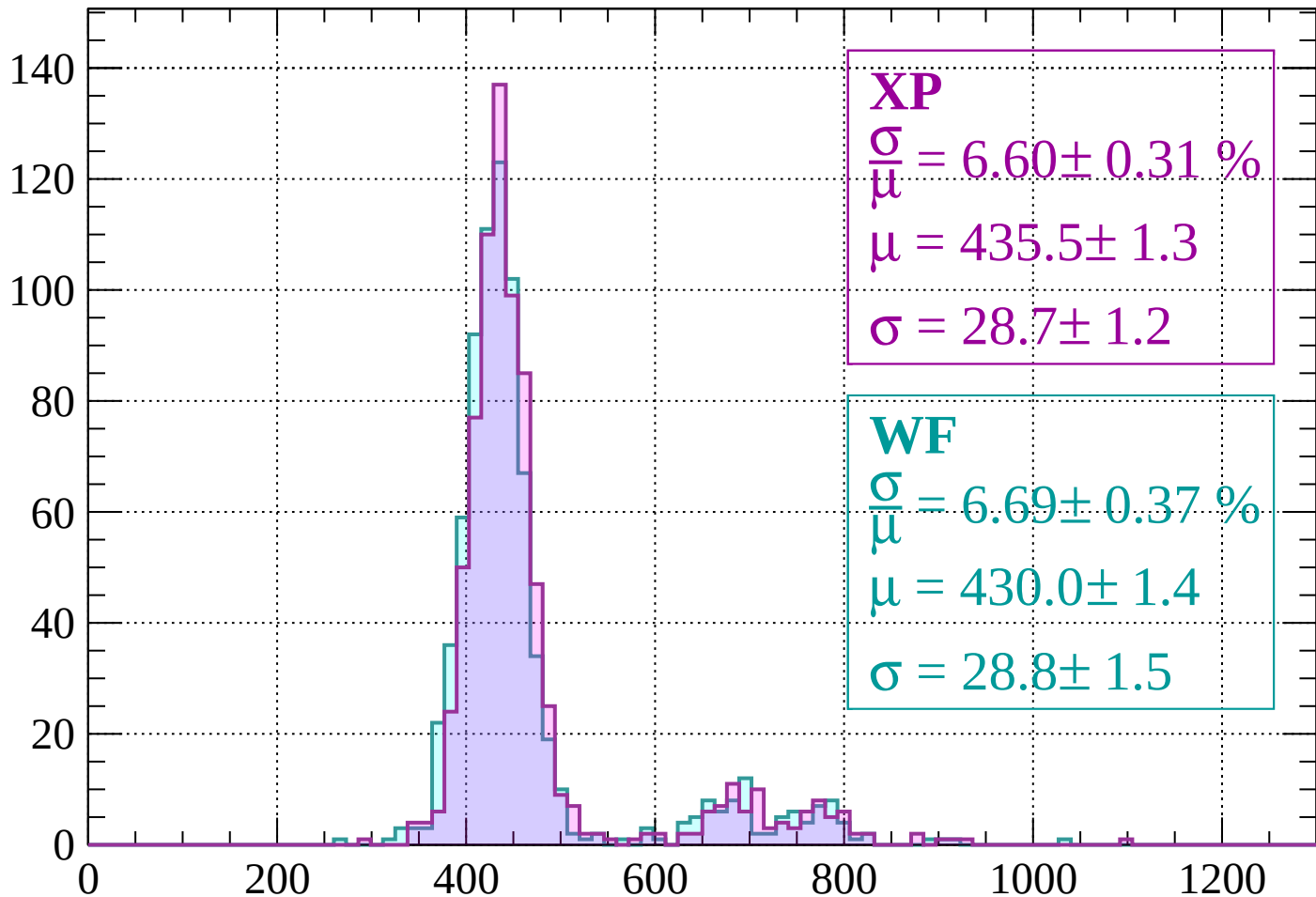
$$\mu = 430.5 \pm 1.2$$

$$\sigma = 27.9 \pm 0.9$$



Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 6.60 \pm 0.31 \%$$

$$\mu = 435.5 \pm 1.3$$

$$\sigma = 28.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.69 \pm 0.37 \%$$

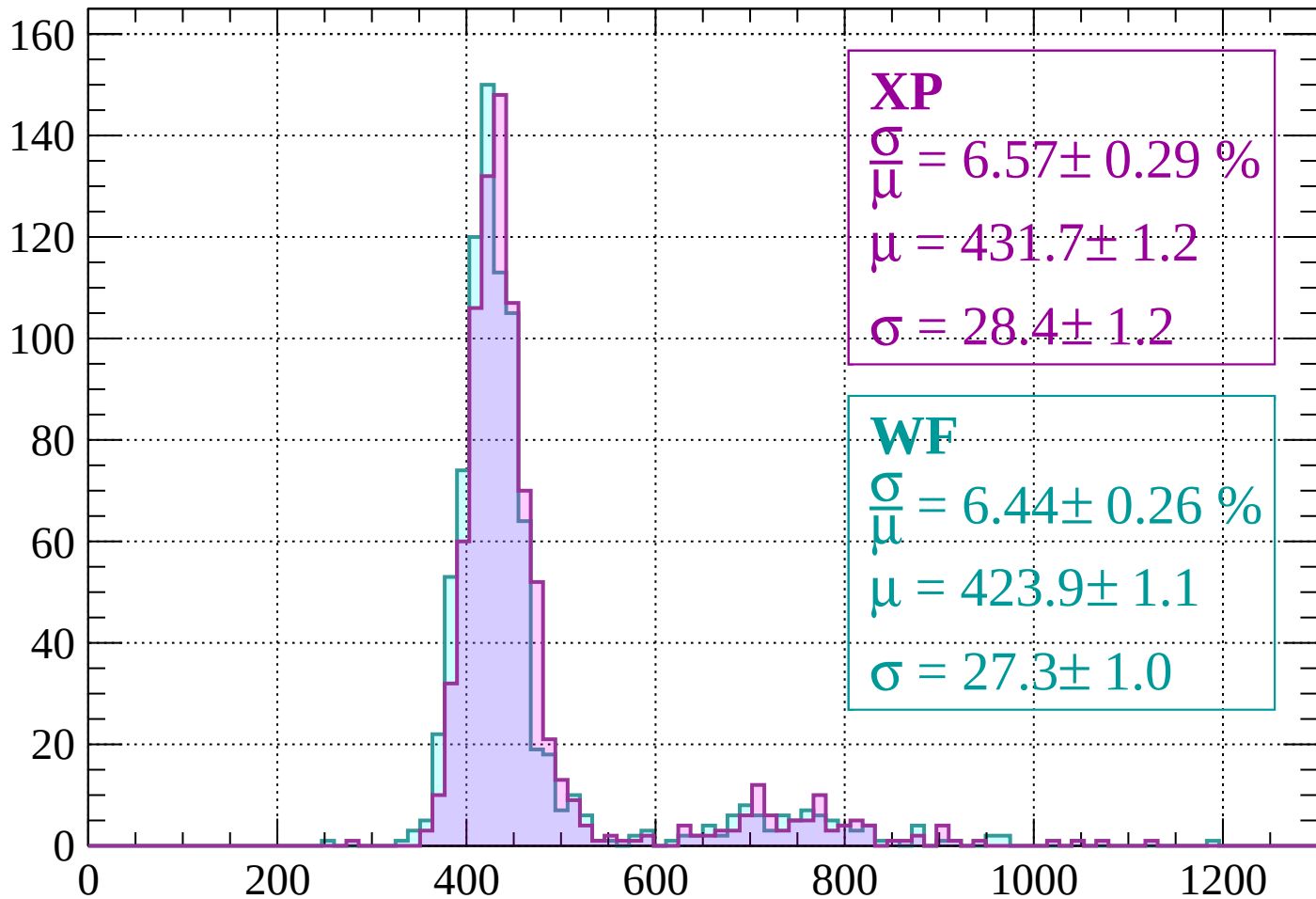
$$\mu = 430.0 \pm 1.4$$

$$\sigma = 28.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 6.57 \pm 0.29 \%$$

$$\mu = 431.7 \pm 1.2$$

$$\sigma = 28.4 \pm 1.2$$

WF

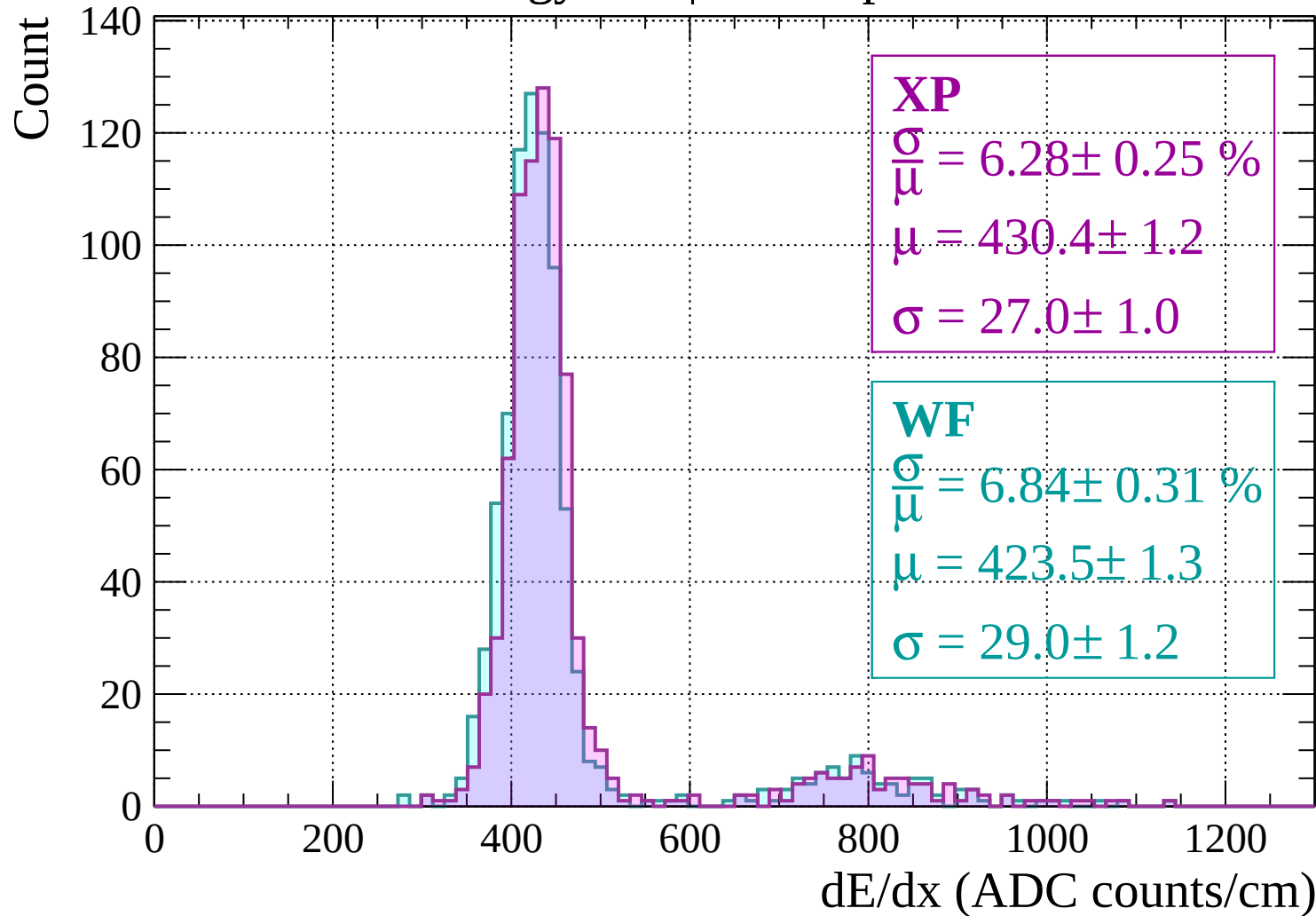
$$\frac{\sigma}{\mu} = 6.44 \pm 0.26 \%$$

$$\mu = 423.9 \pm 1.1$$

$$\sigma = 27.3 \pm 1.0$$

dE/dx (ADC counts/cm)

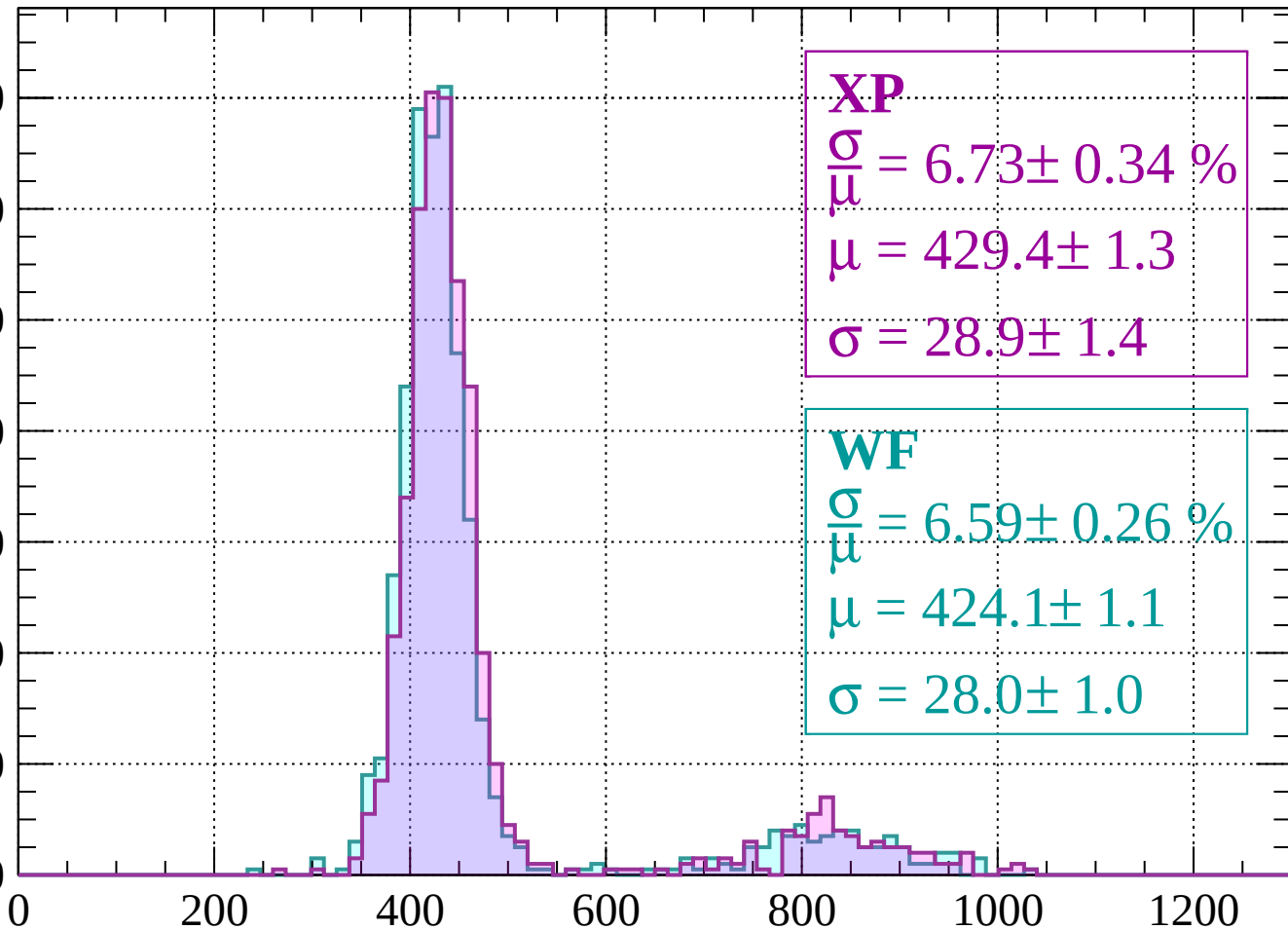
Energy loss | $-760 < p < -720$



Energy loss | $-720 < p < -680$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.73 \pm 0.34 \%$$

$$\mu = 429.4 \pm 1.3$$

$$\sigma = 28.9 \pm 1.4$$

WF

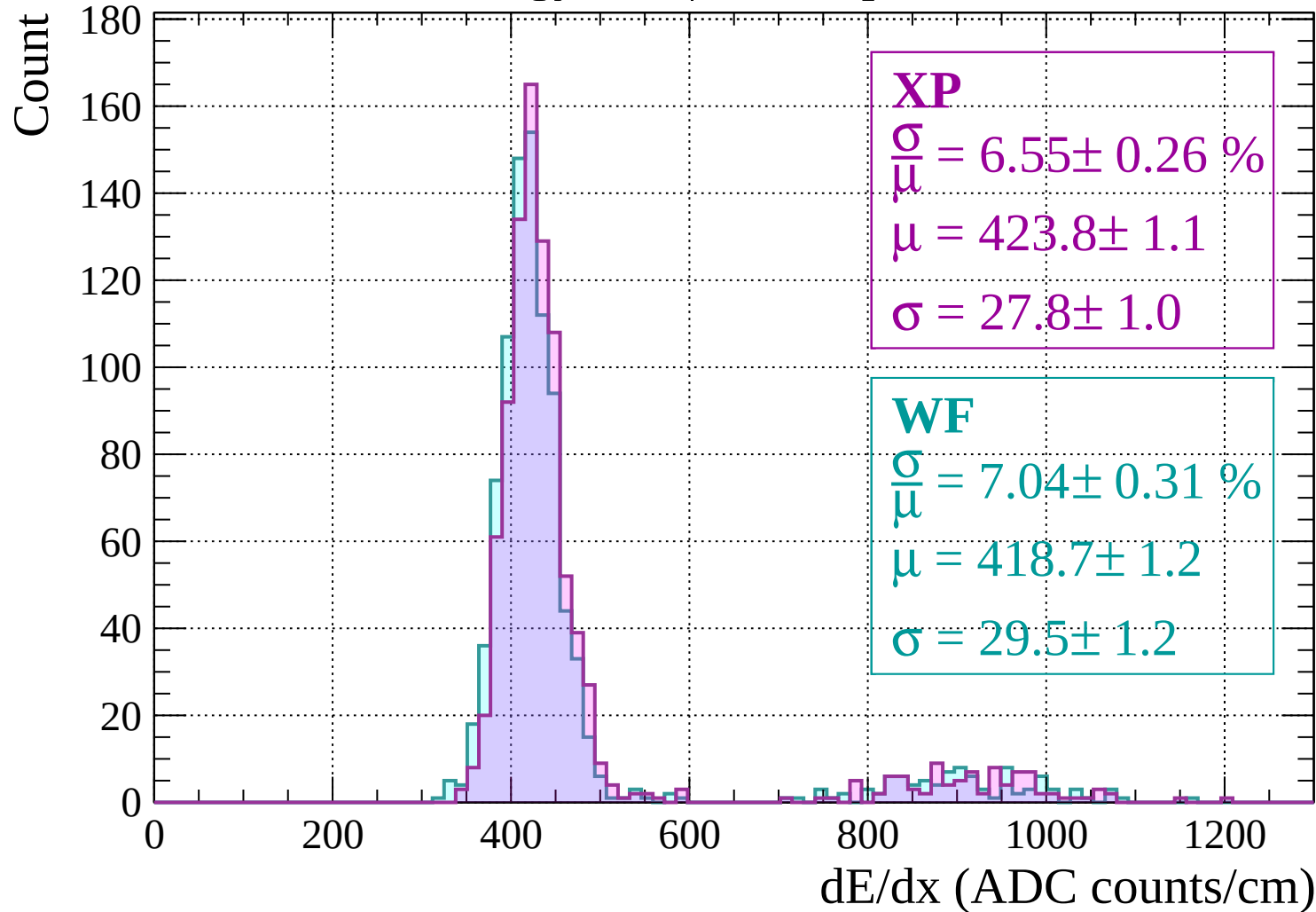
$$\frac{\sigma}{\mu} = 6.59 \pm 0.26 \%$$

$$\mu = 424.1 \pm 1.1$$

$$\sigma = 28.0 \pm 1.0$$

dE/dx (ADC counts/cm)

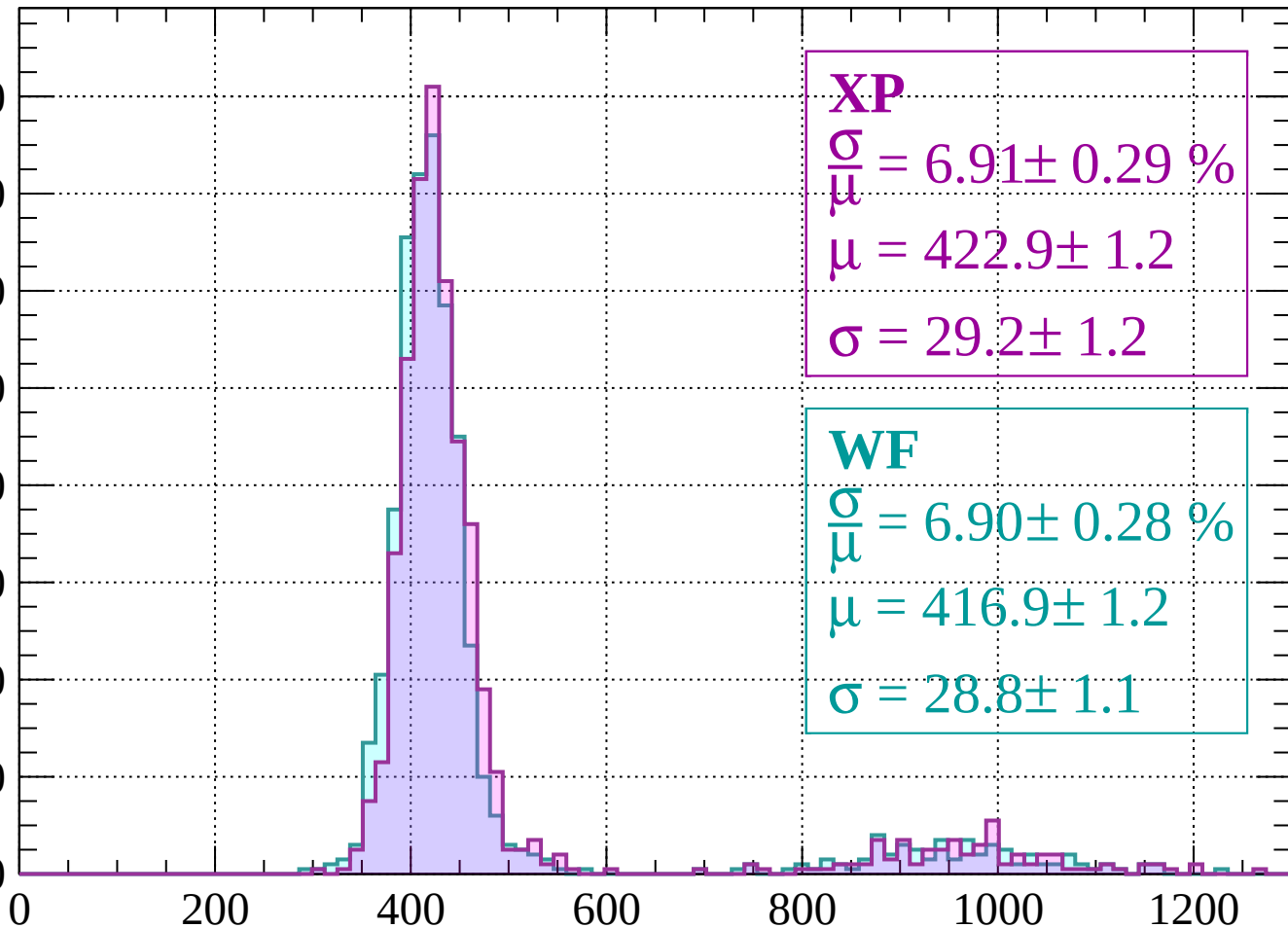
Energy loss | $-680 < p < -640$



Energy loss | $-640 < p < -600$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.91 \pm 0.29 \%$$

$$\mu = 422.9 \pm 1.2$$

$$\sigma = 29.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.28 \%$$

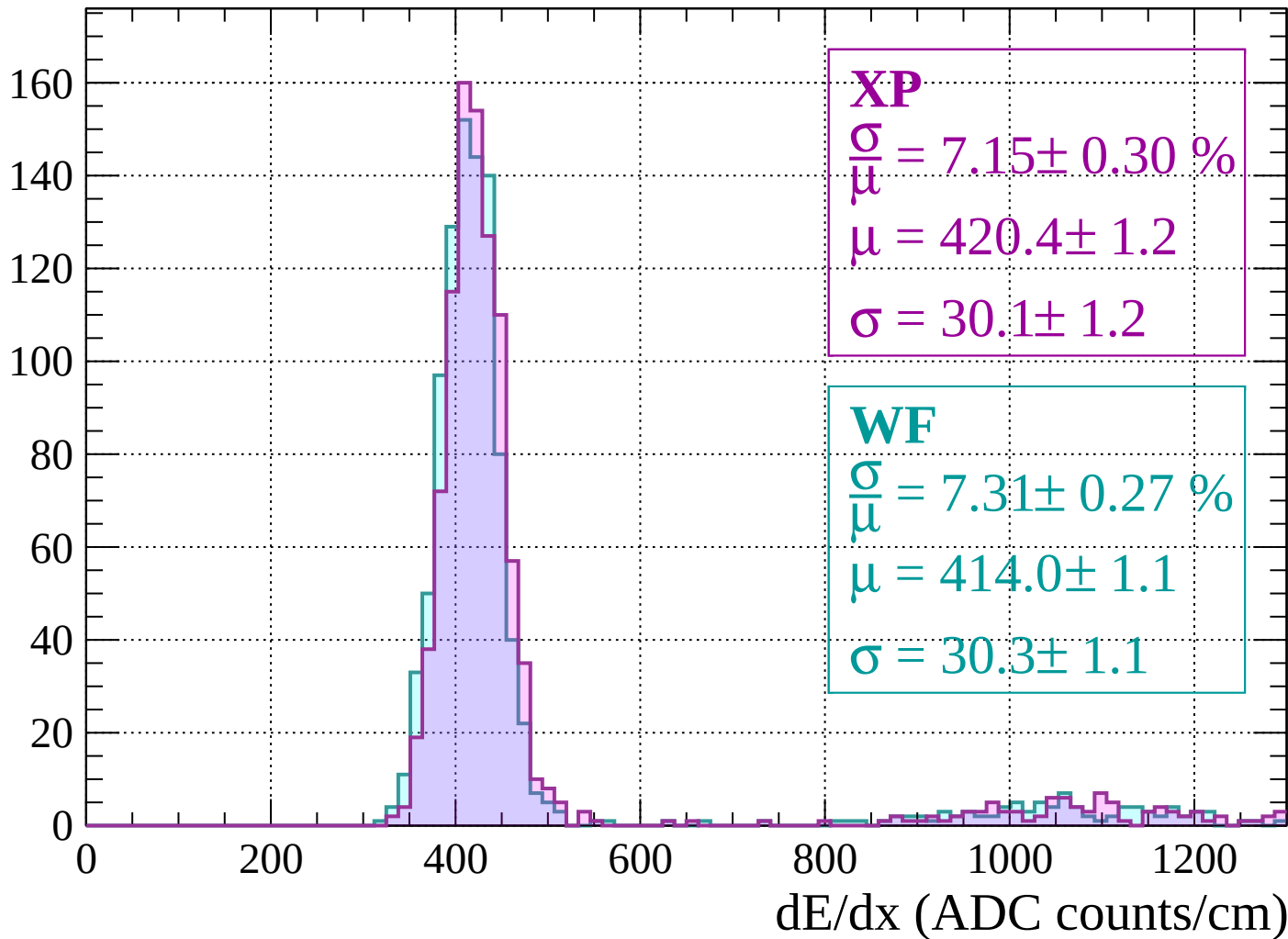
$$\mu = 416.9 \pm 1.2$$

$$\sigma = 28.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

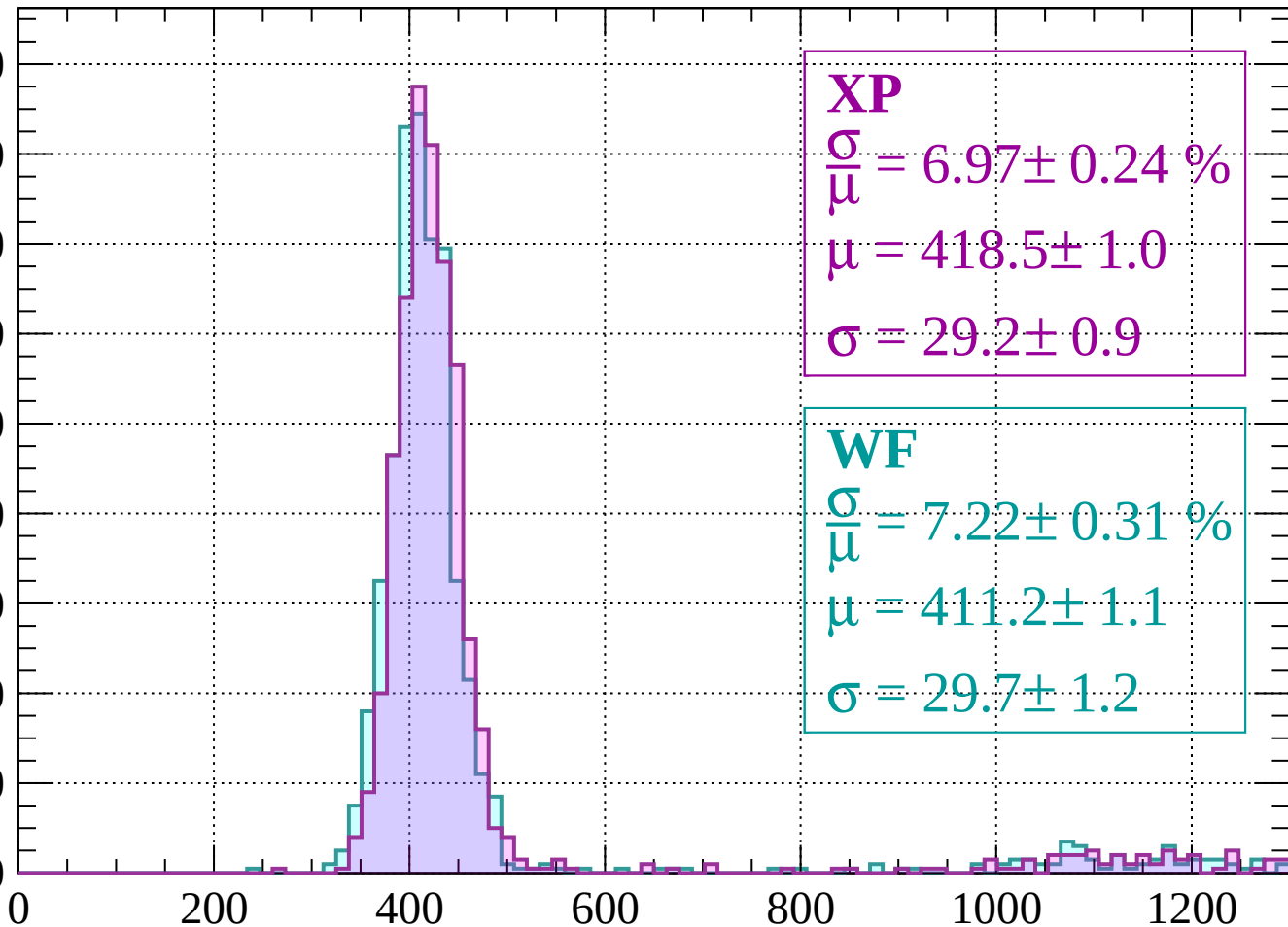
Count



Energy loss | $-560 < p < -520$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.97 \pm 0.24 \%$$

$$\mu = 418.5 \pm 1.0$$

$$\sigma = 29.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.22 \pm 0.31 \%$$

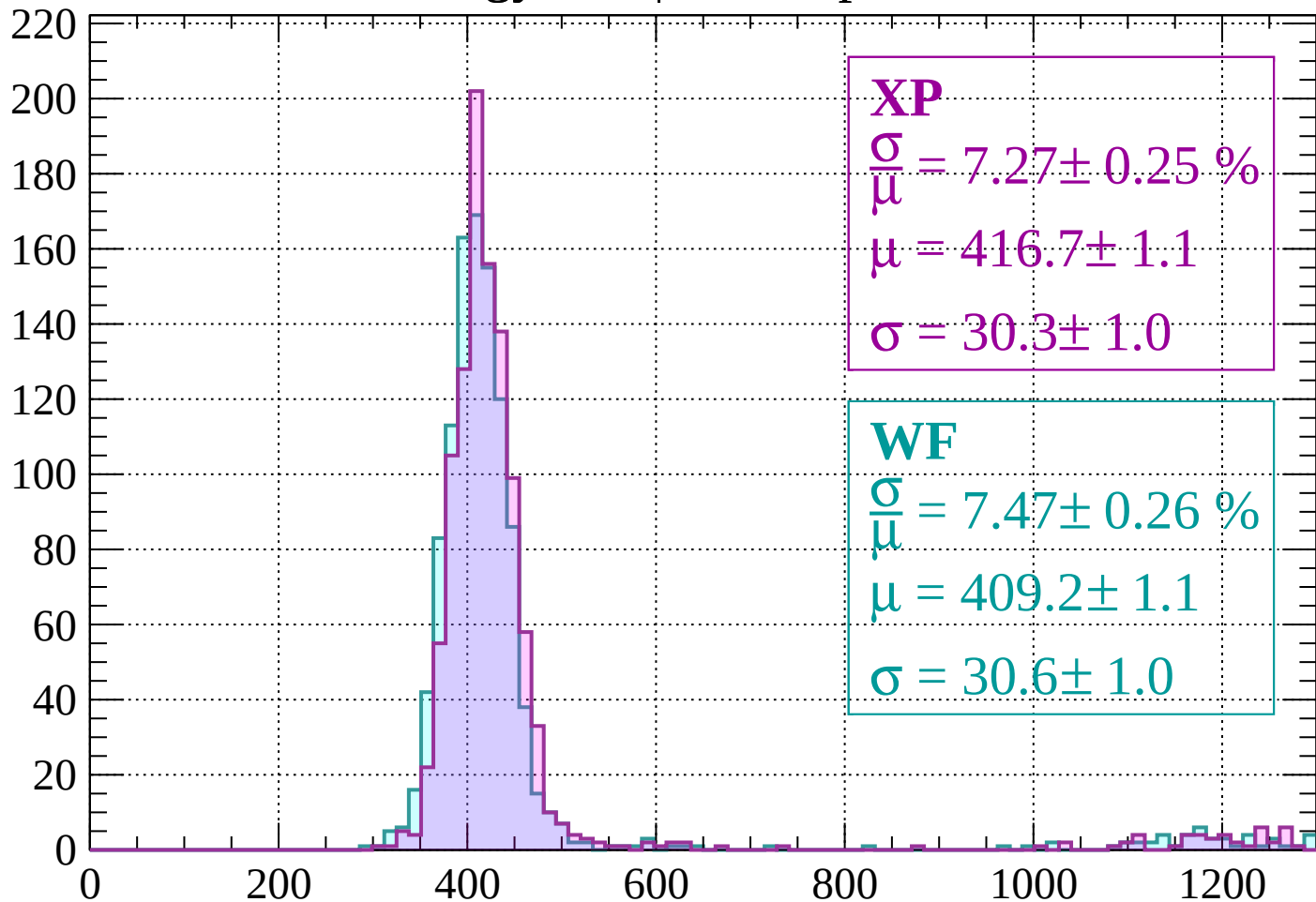
$$\mu = 411.2 \pm 1.1$$

$$\sigma = 29.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.25 \%$$

$$\mu = 416.7 \pm 1.1$$

$$\sigma = 30.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.26 \%$$

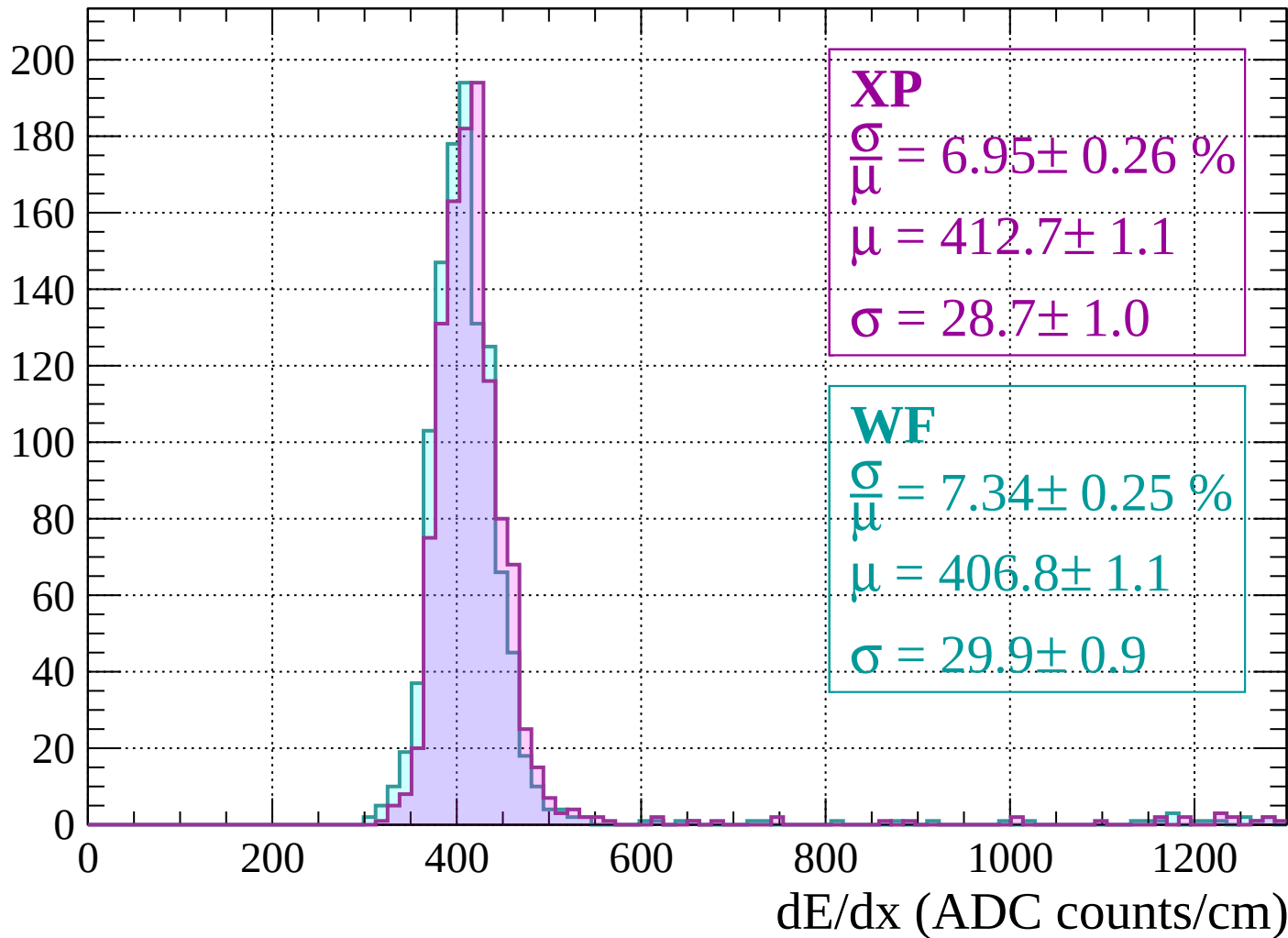
$$\mu = 409.2 \pm 1.1$$

$$\sigma = 30.6 \pm 1.0$$

dE/dx (ADC counts/cm)

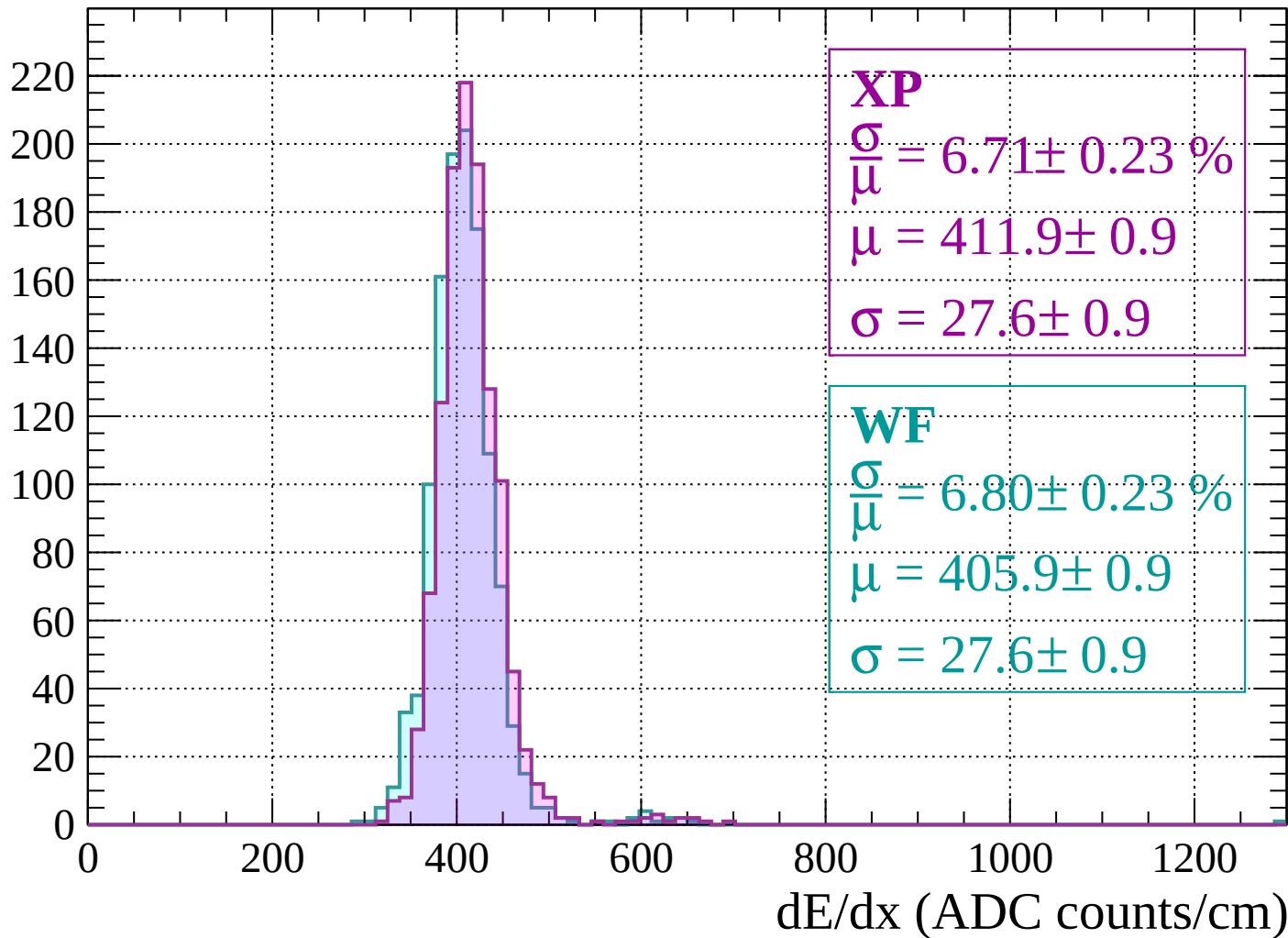
Energy loss | $-480 < p < -440$

Count



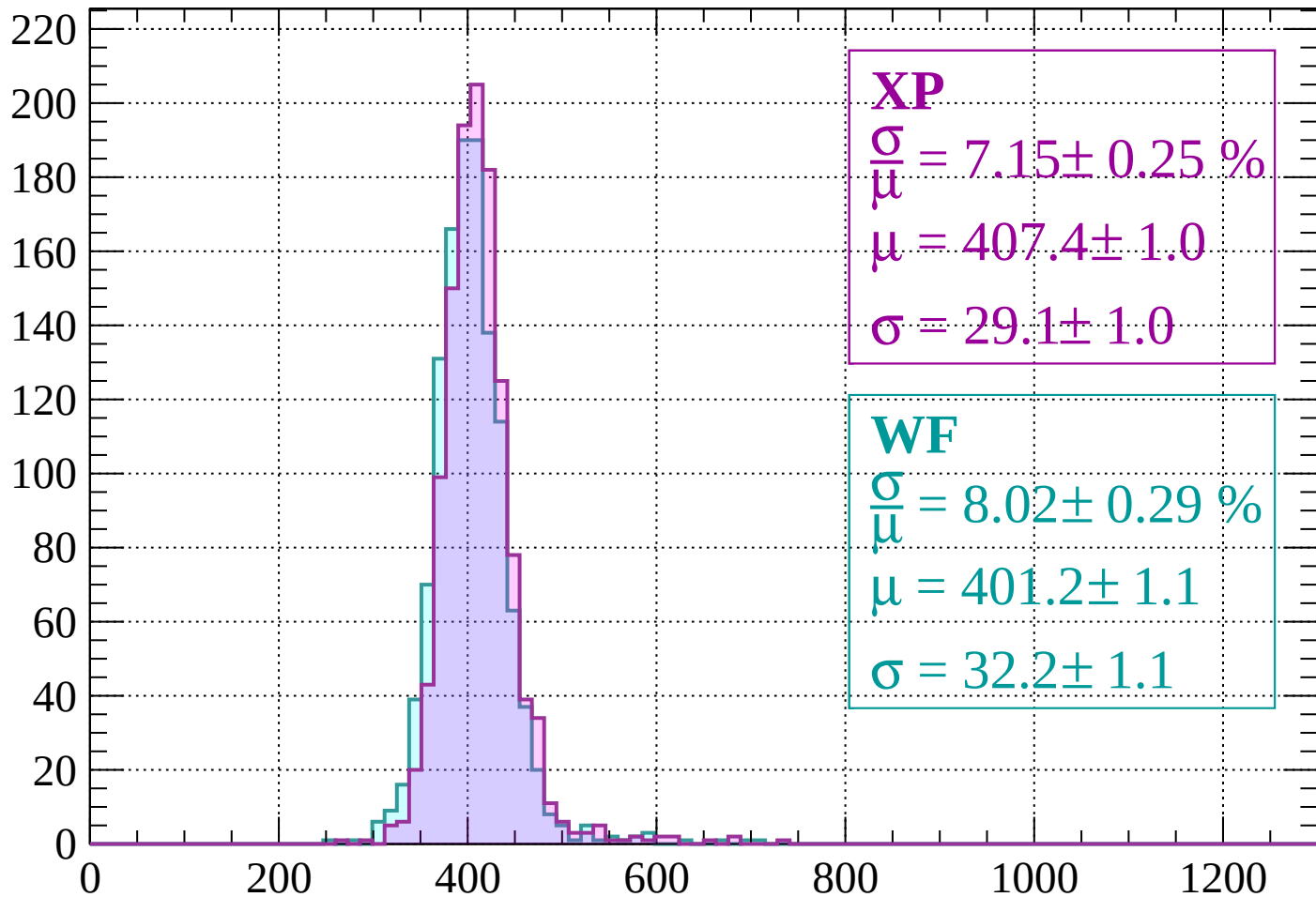
Energy loss | $-440 < p < -400$

Count



Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.25 \%$$

$$\mu = 407.4 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 0.29 \%$$

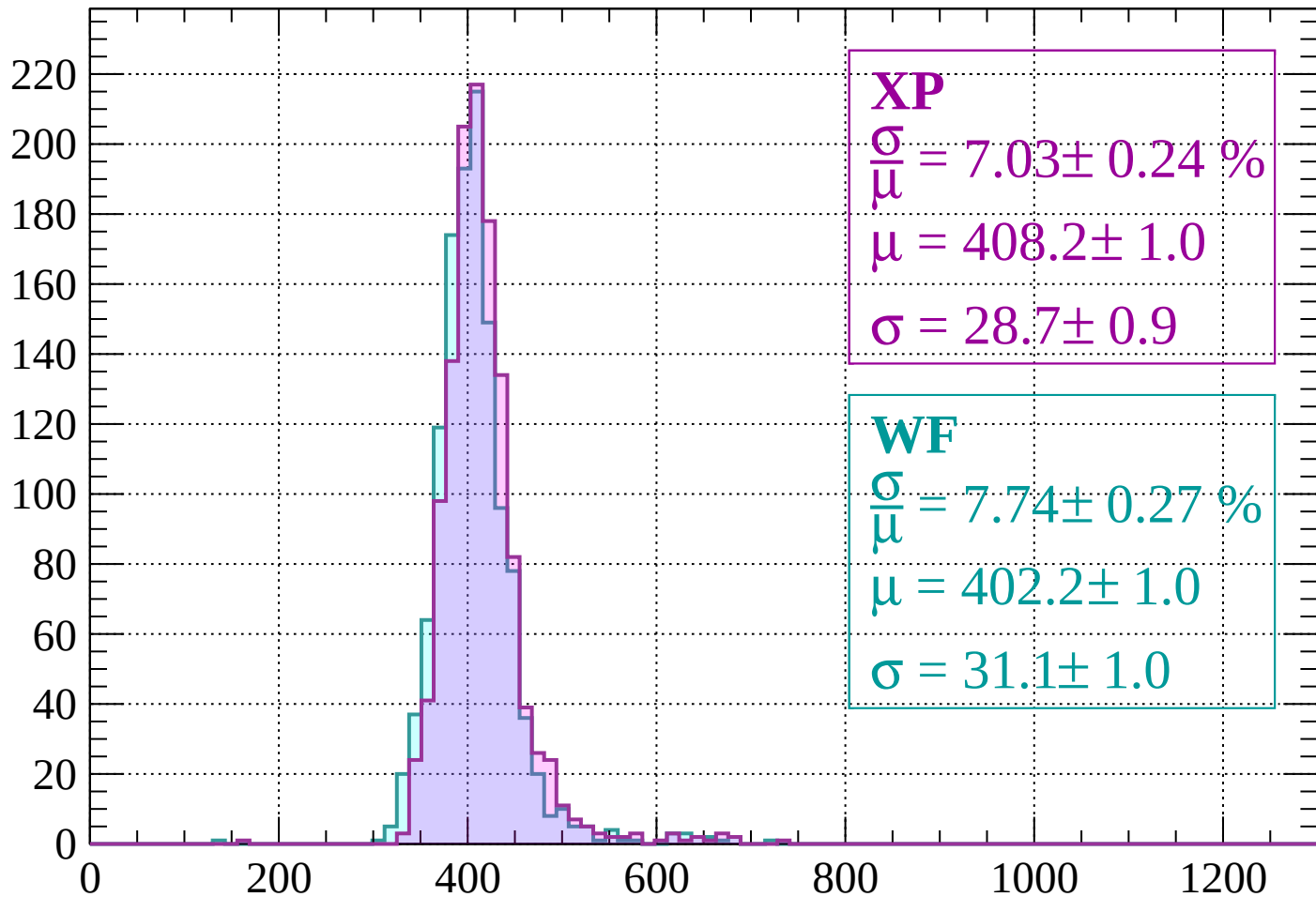
$$\mu = 401.2 \pm 1.1$$

$$\sigma = 32.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.24 \%$$

$$\mu = 408.2 \pm 1.0$$

$$\sigma = 28.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.27 \%$$

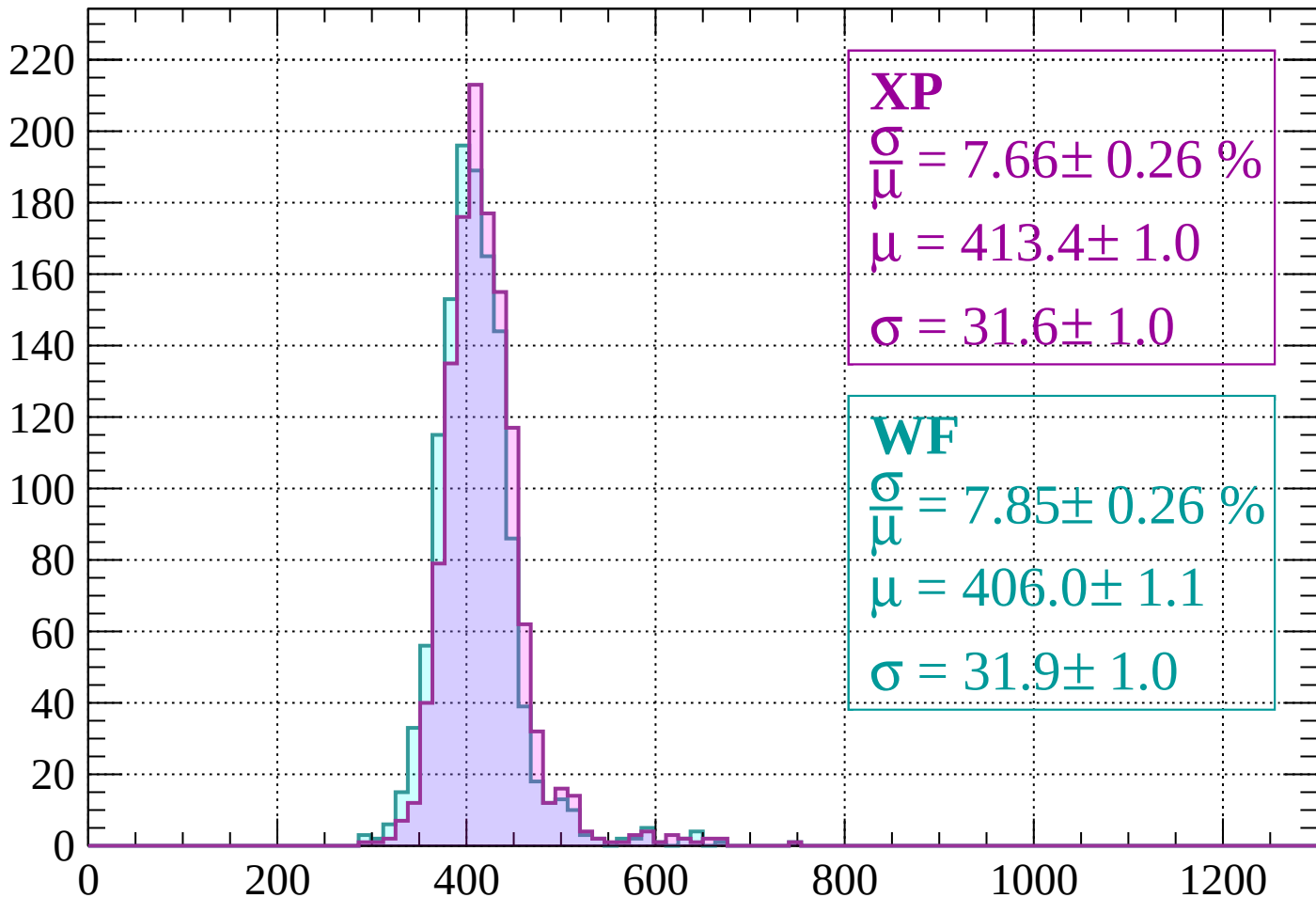
$$\mu = 402.2 \pm 1.0$$

$$\sigma = 31.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.26 \%$$

$$\mu = 413.4 \pm 1.0$$

$$\sigma = 31.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.26 \%$$

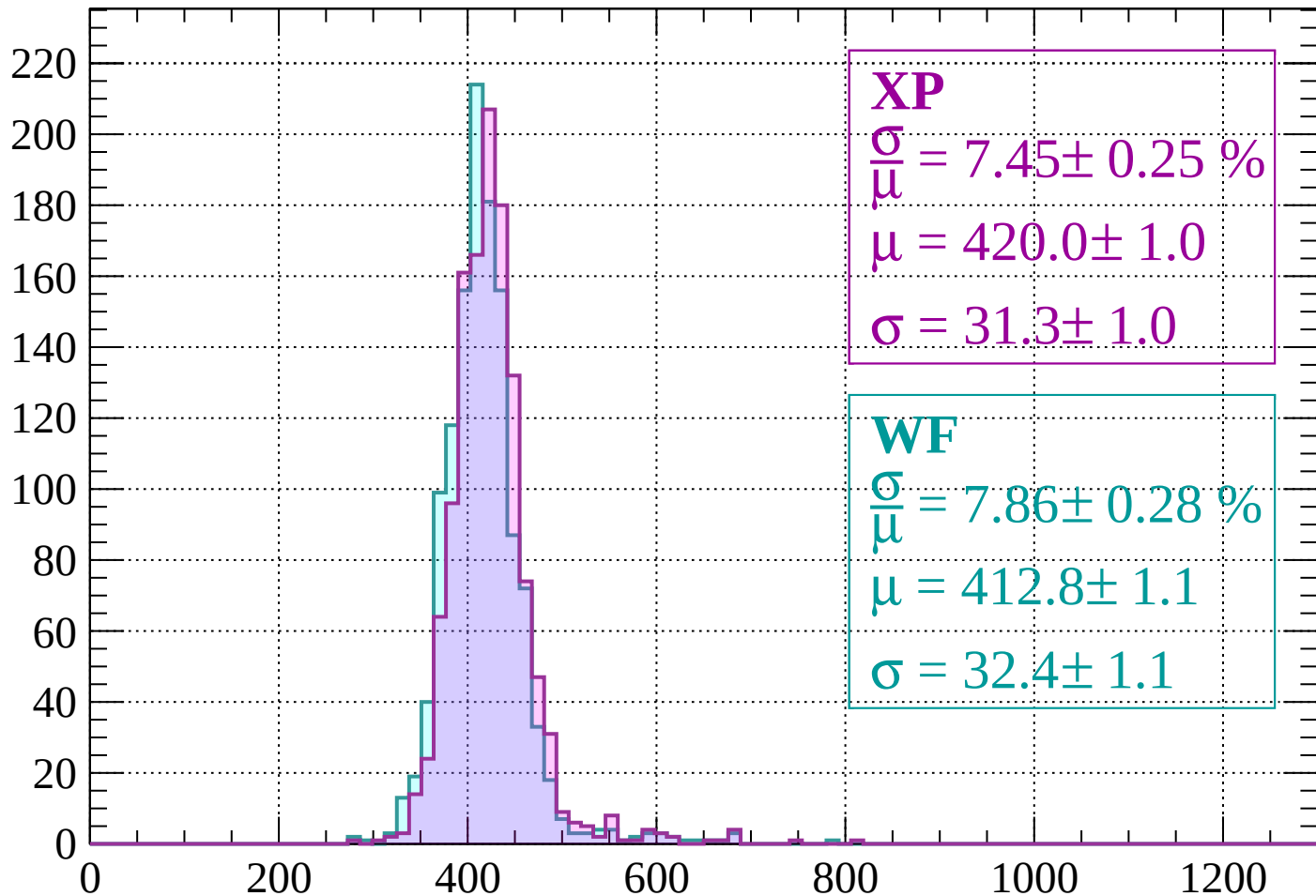
$$\mu = 406.0 \pm 1.1$$

$$\sigma = 31.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 7.45 \pm 0.25 \%$$

$$\mu = 420.0 \pm 1.0$$

$$\sigma = 31.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.86 \pm 0.28 \%$$

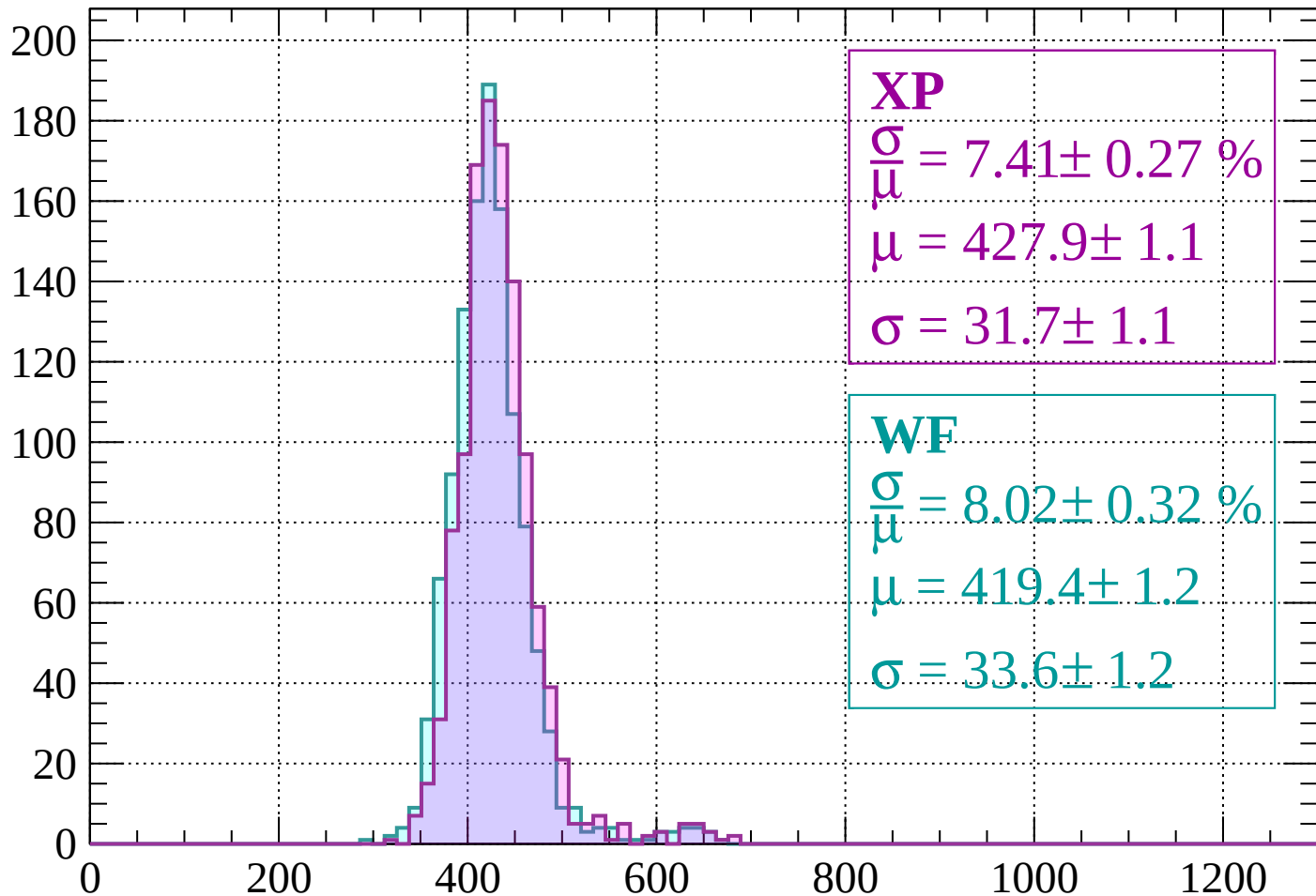
$$\mu = 412.8 \pm 1.1$$

$$\sigma = 32.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 7.41 \pm 0.27 \%$$

$$\mu = 427.9 \pm 1.1$$

$$\sigma = 31.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 0.32 \%$$

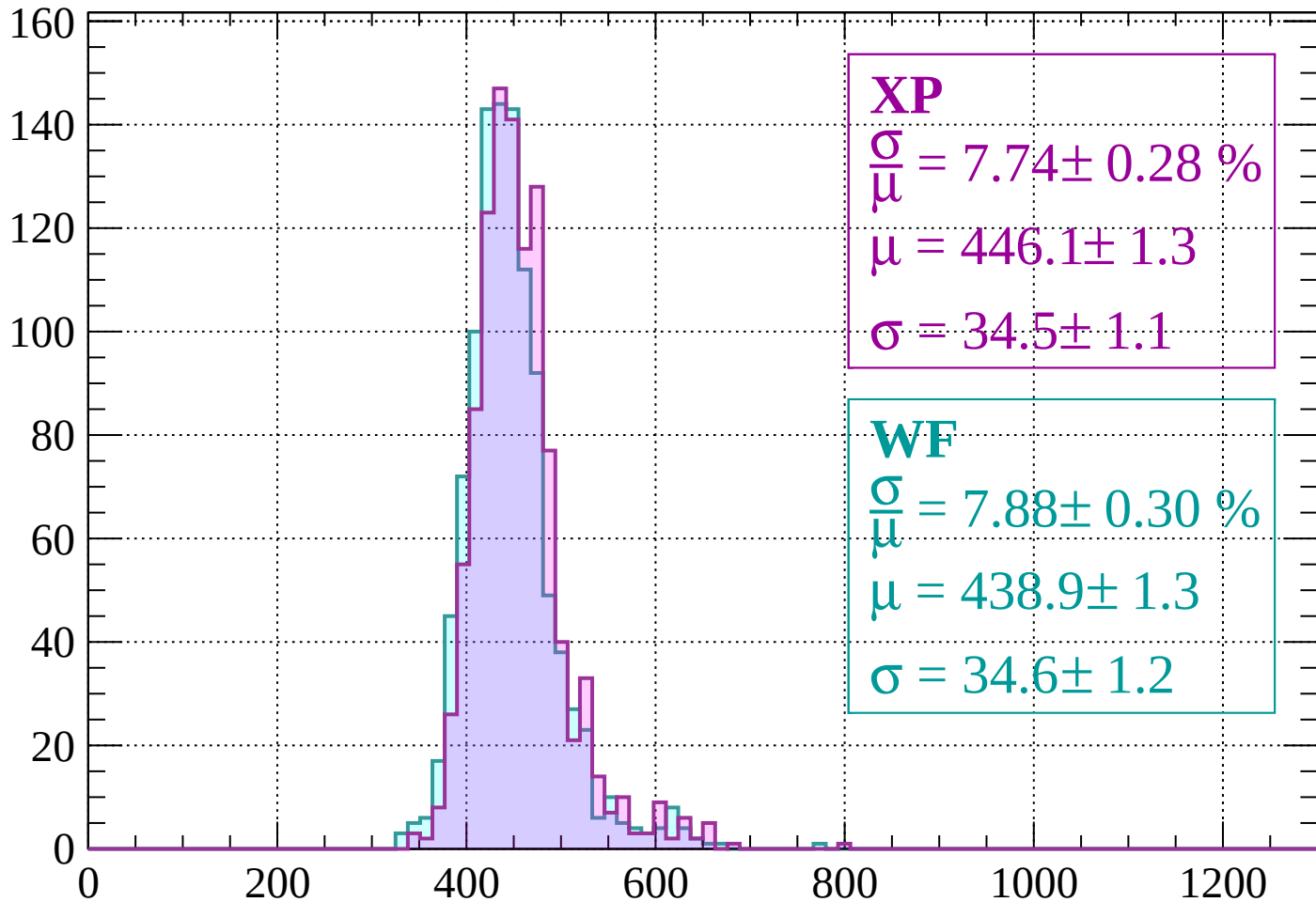
$$\mu = 419.4 \pm 1.2$$

$$\sigma = 33.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 7.74 \pm 0.28 \%$$

$$\mu = 446.1 \pm 1.3$$

$$\sigma = 34.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.30 \%$$

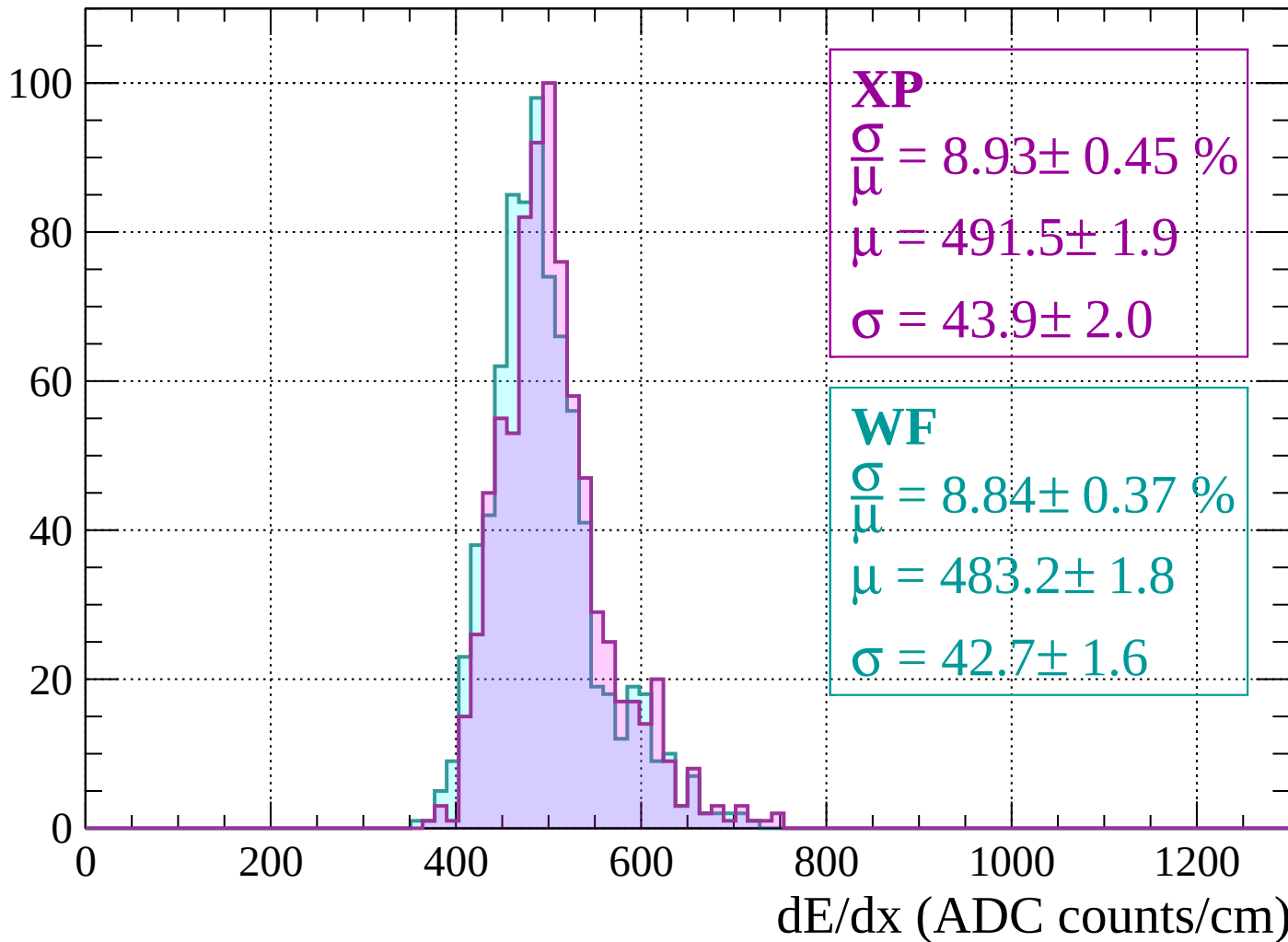
$$\mu = 438.9 \pm 1.3$$

$$\sigma = 34.6 \pm 1.2$$

dE/dx (ADC counts/cm)

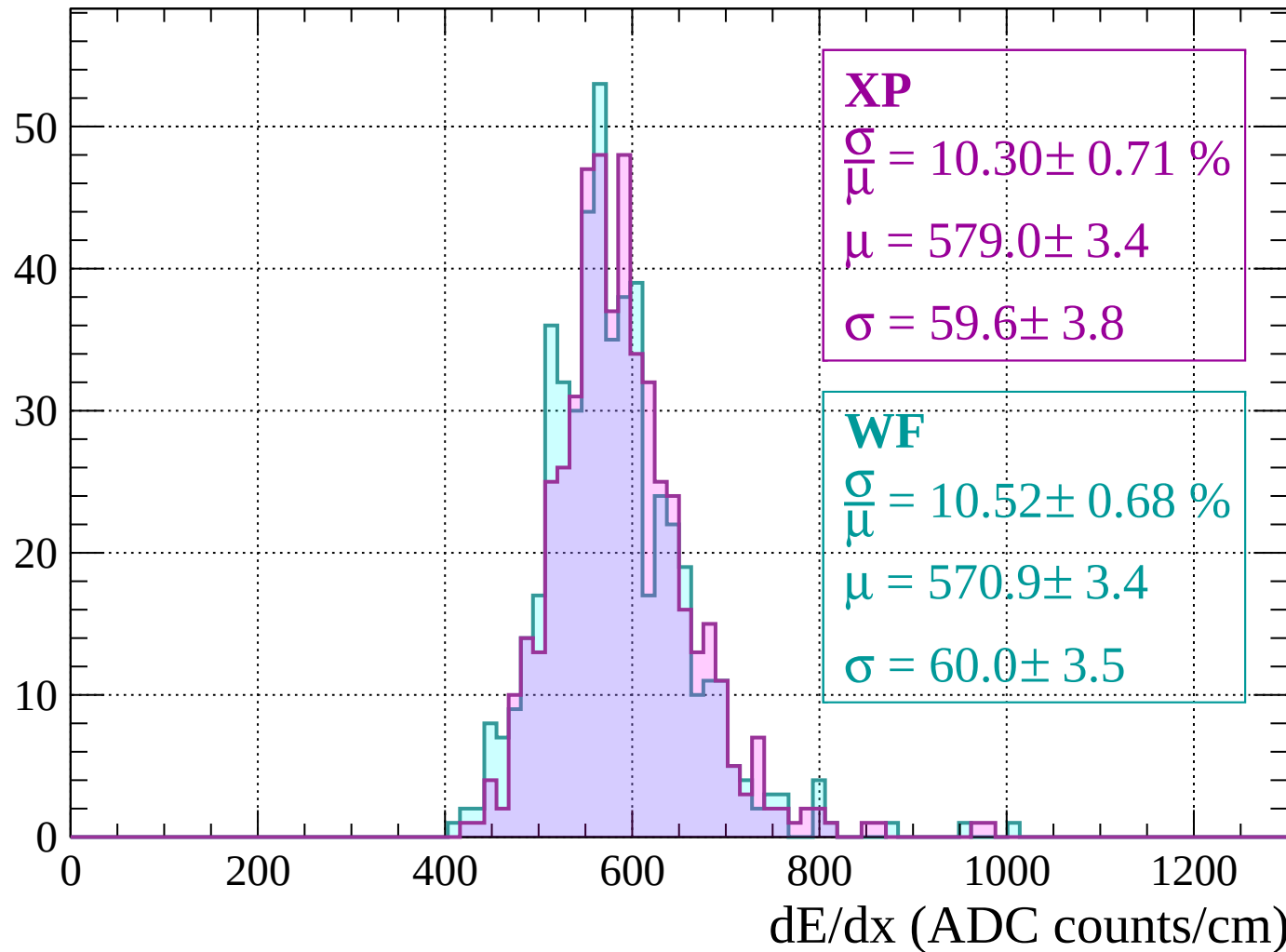
Energy loss | $-160 < p < -120$

Count



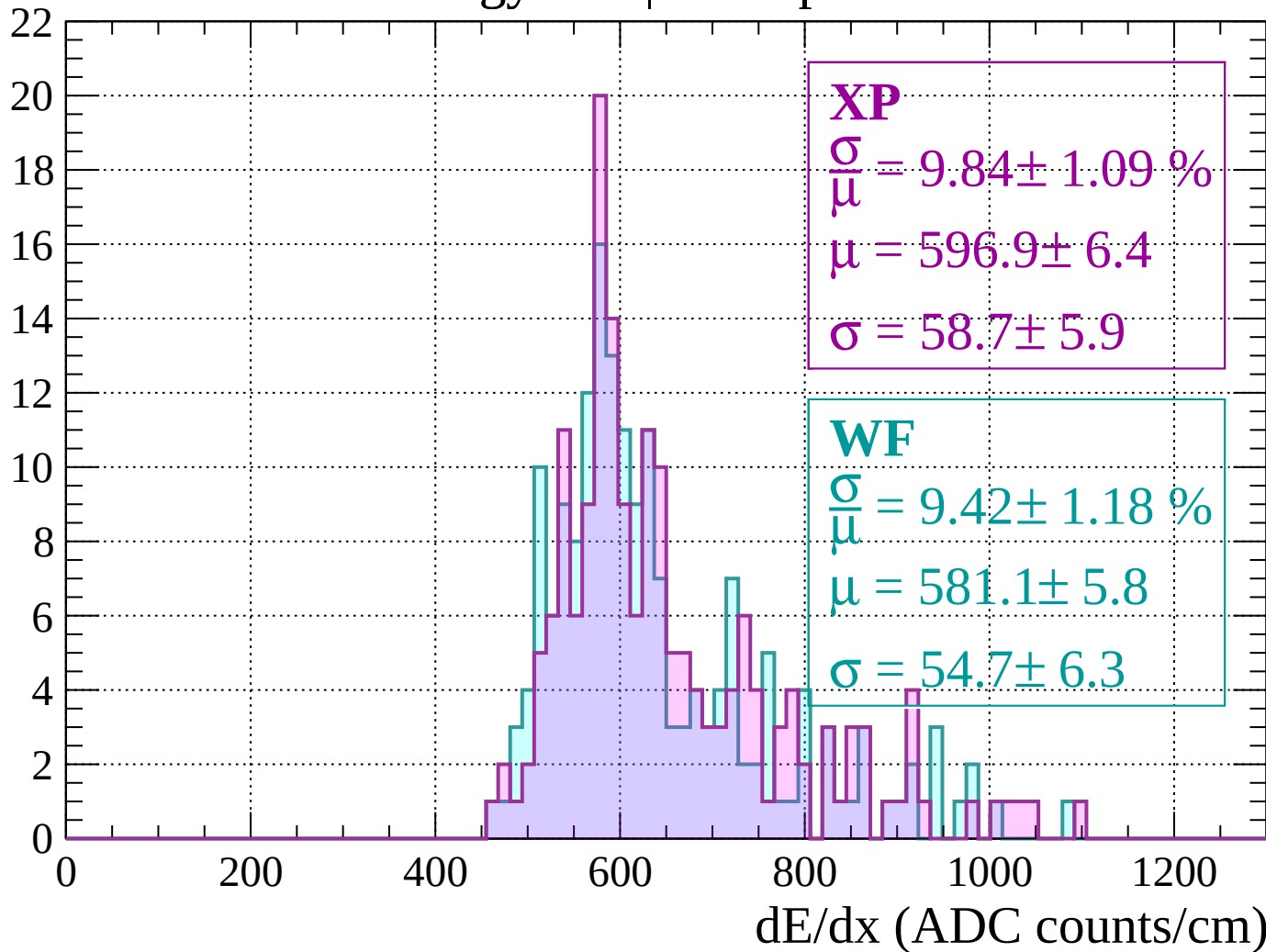
Energy loss | $-120 < p < -80$

Count



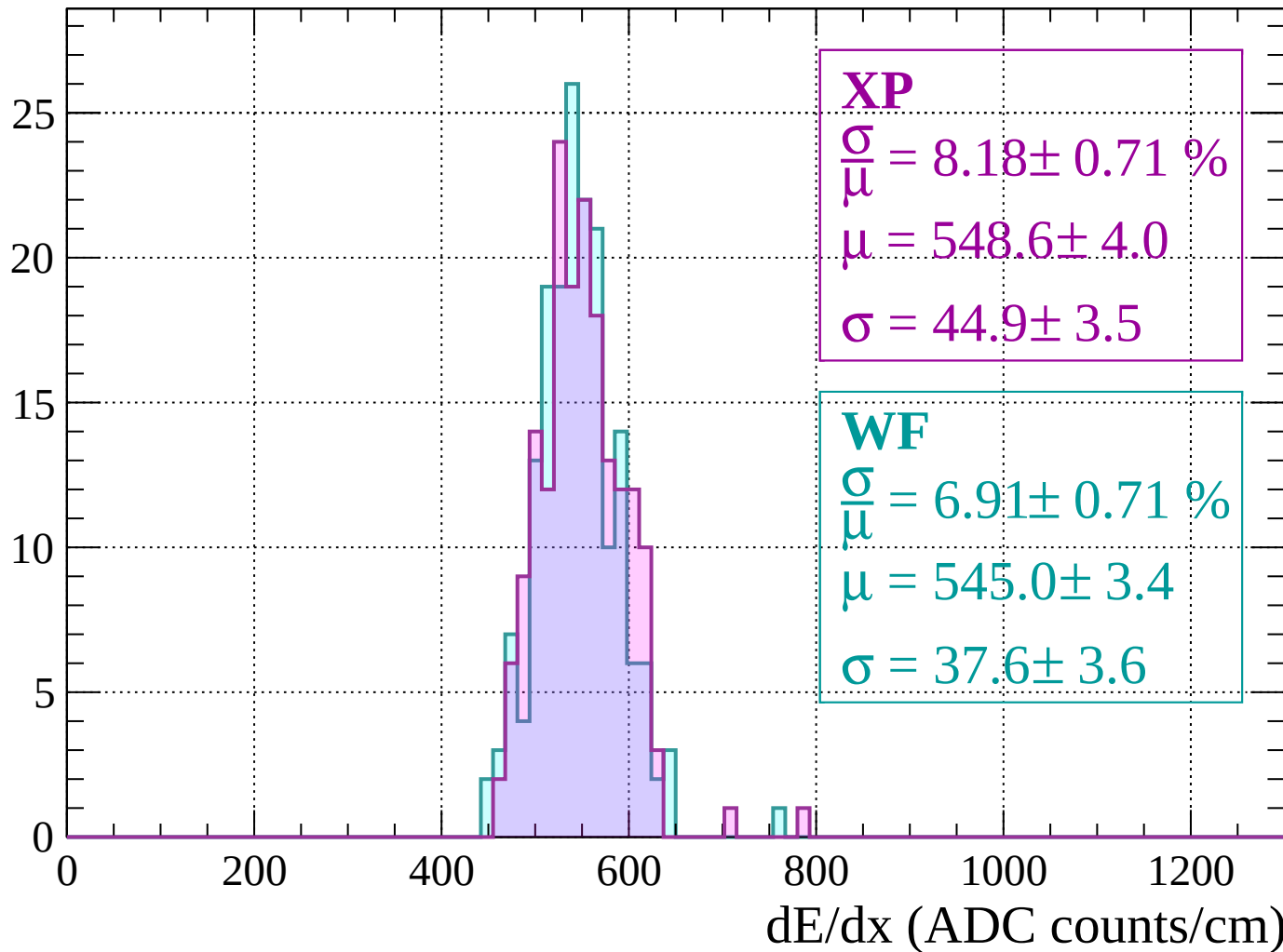
Energy loss | $-80 < p < -40$

Count



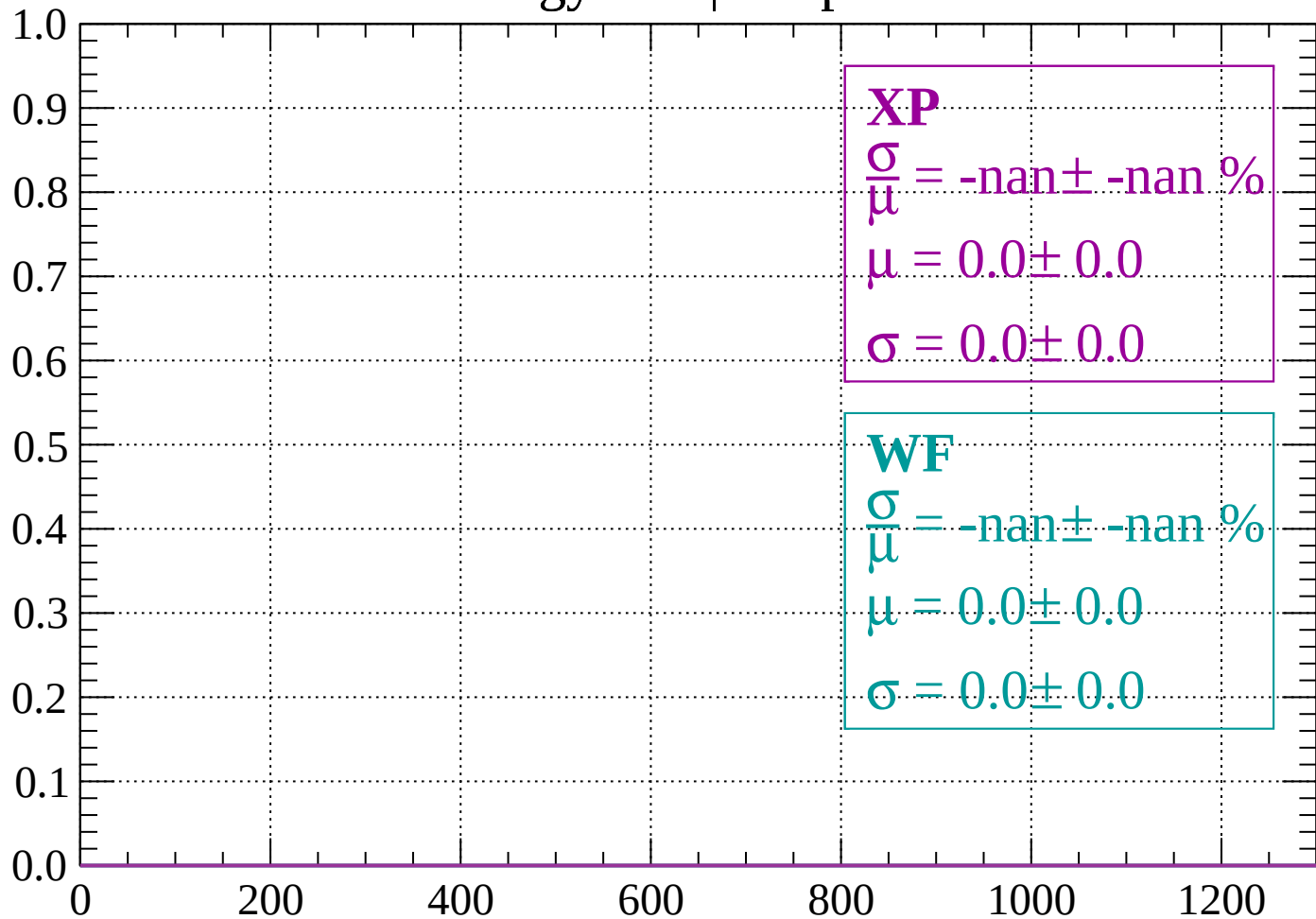
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

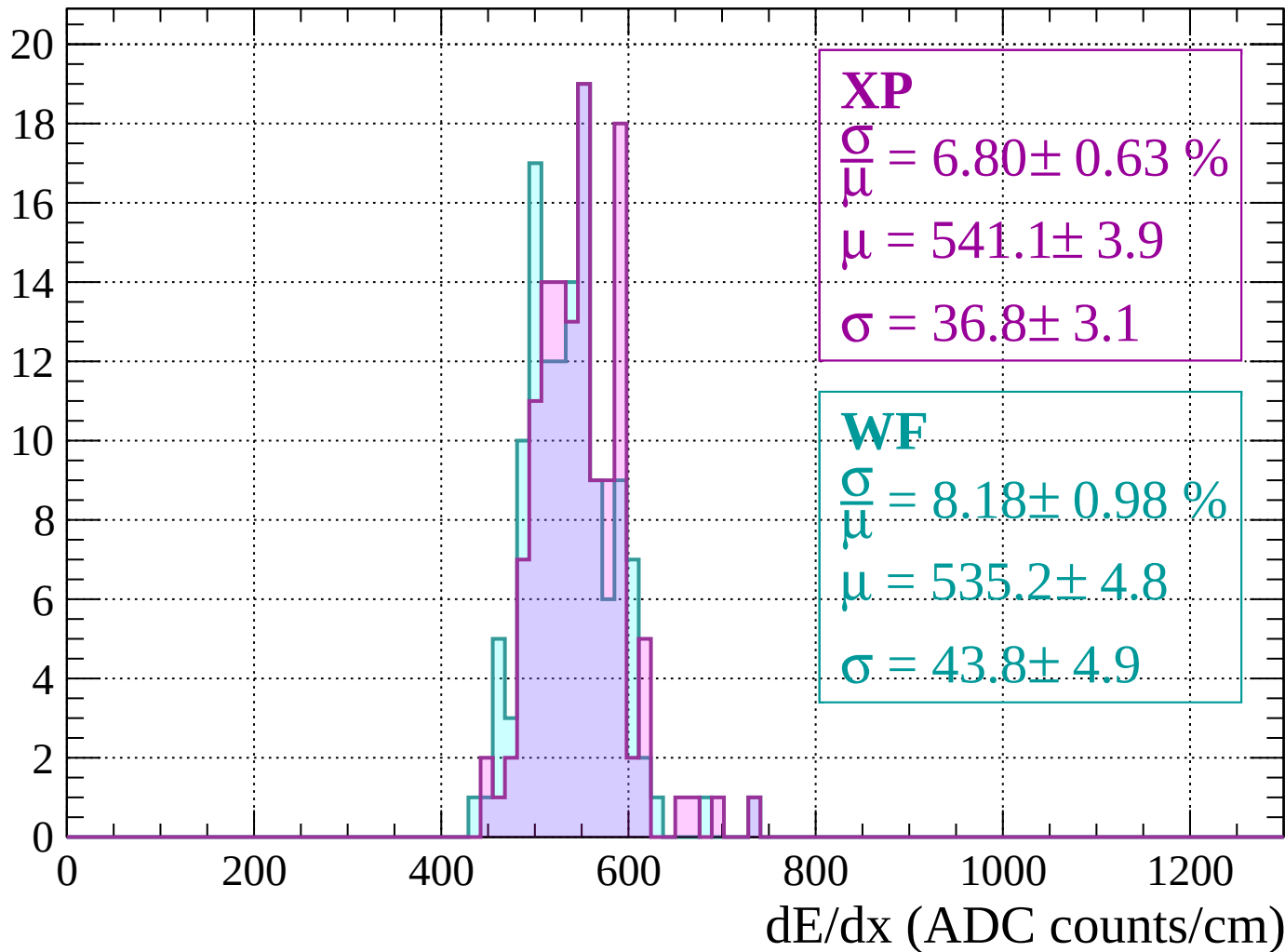
Count



dE/dx (ADC counts/cm)

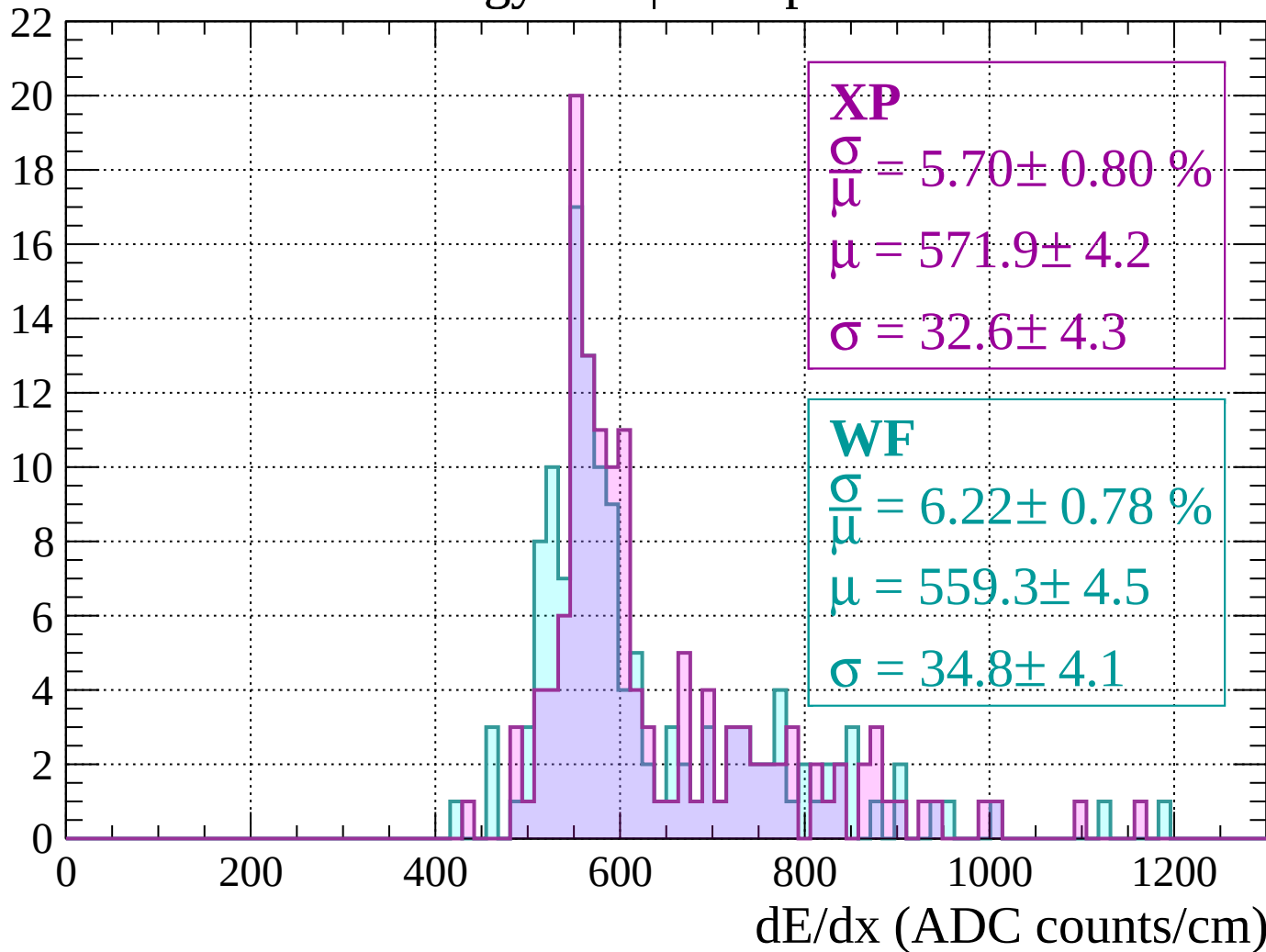
Energy loss | $40 < p < 80$

Count



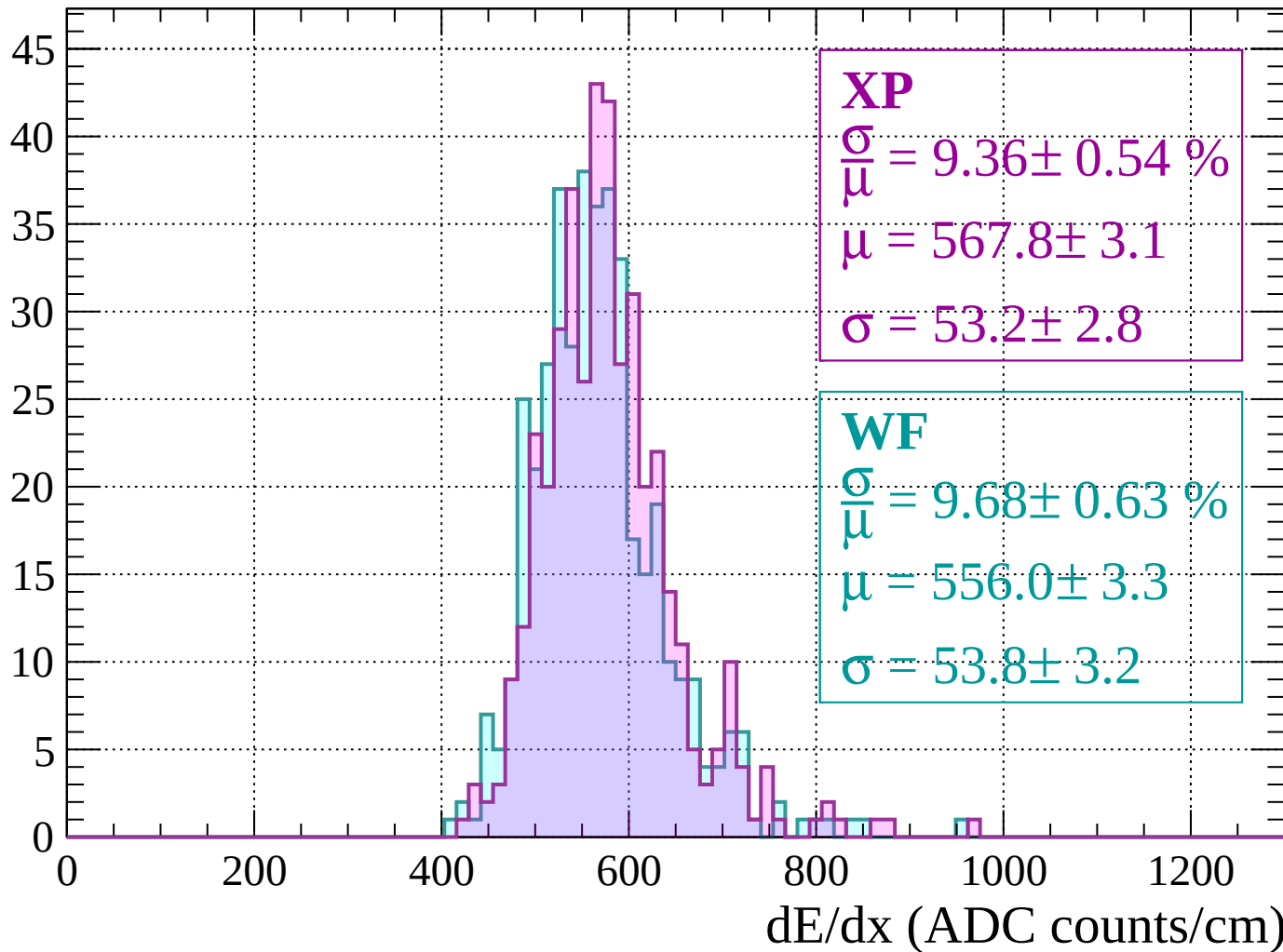
Energy loss | $80 < p < 120$

Count



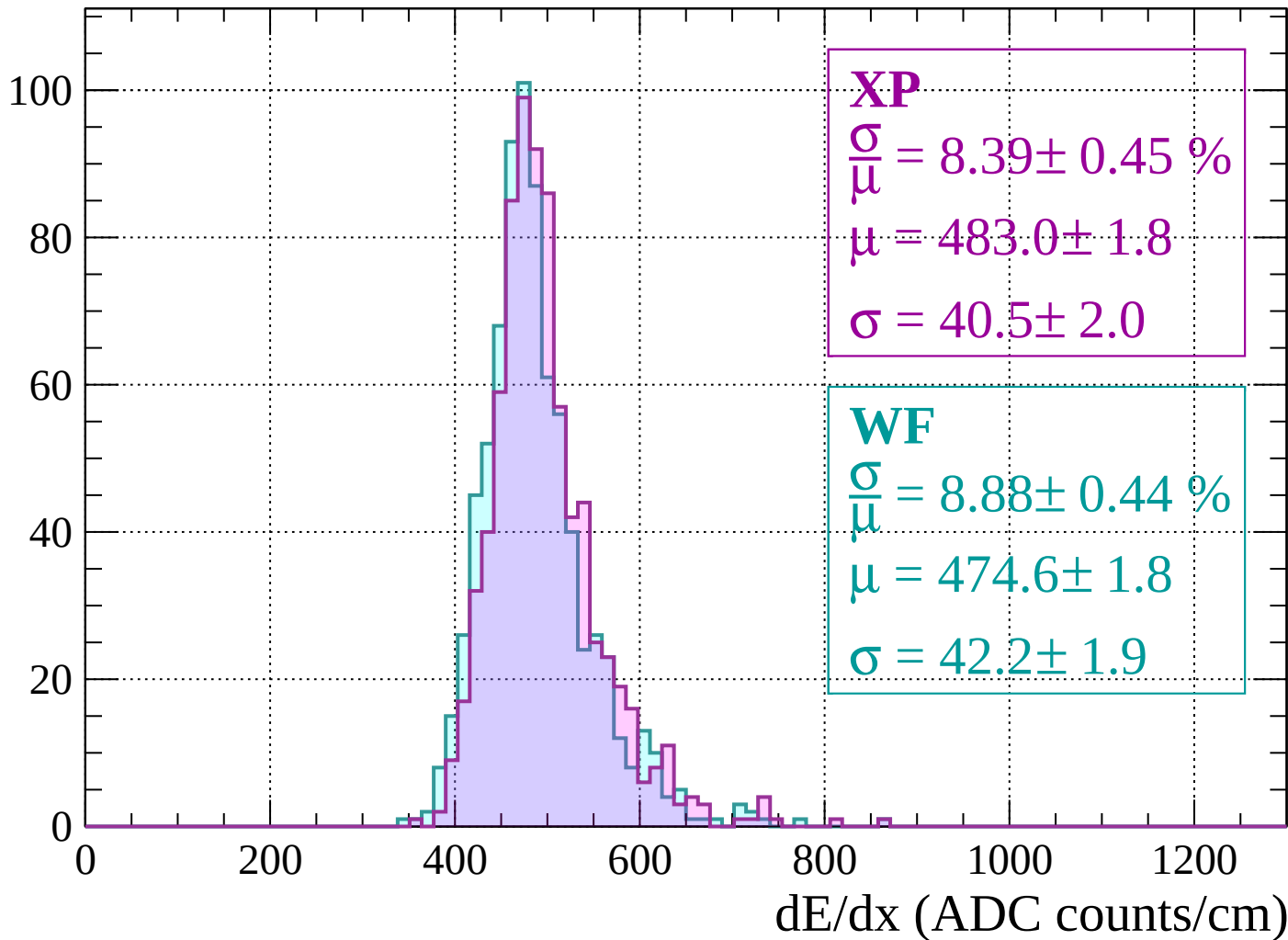
Energy loss | $120 < p < 160$

Count



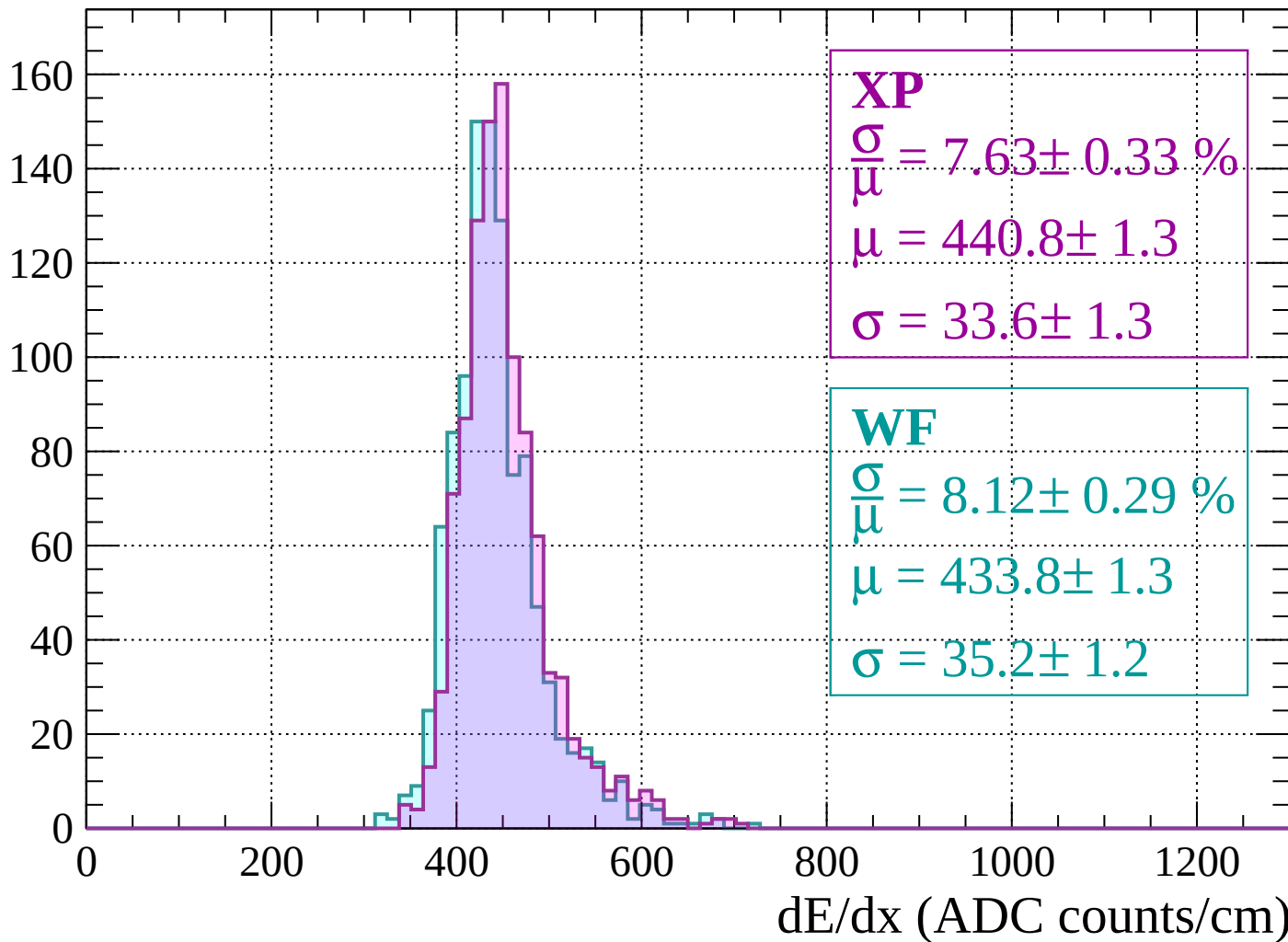
Energy loss | $160 < p < 200$

Count



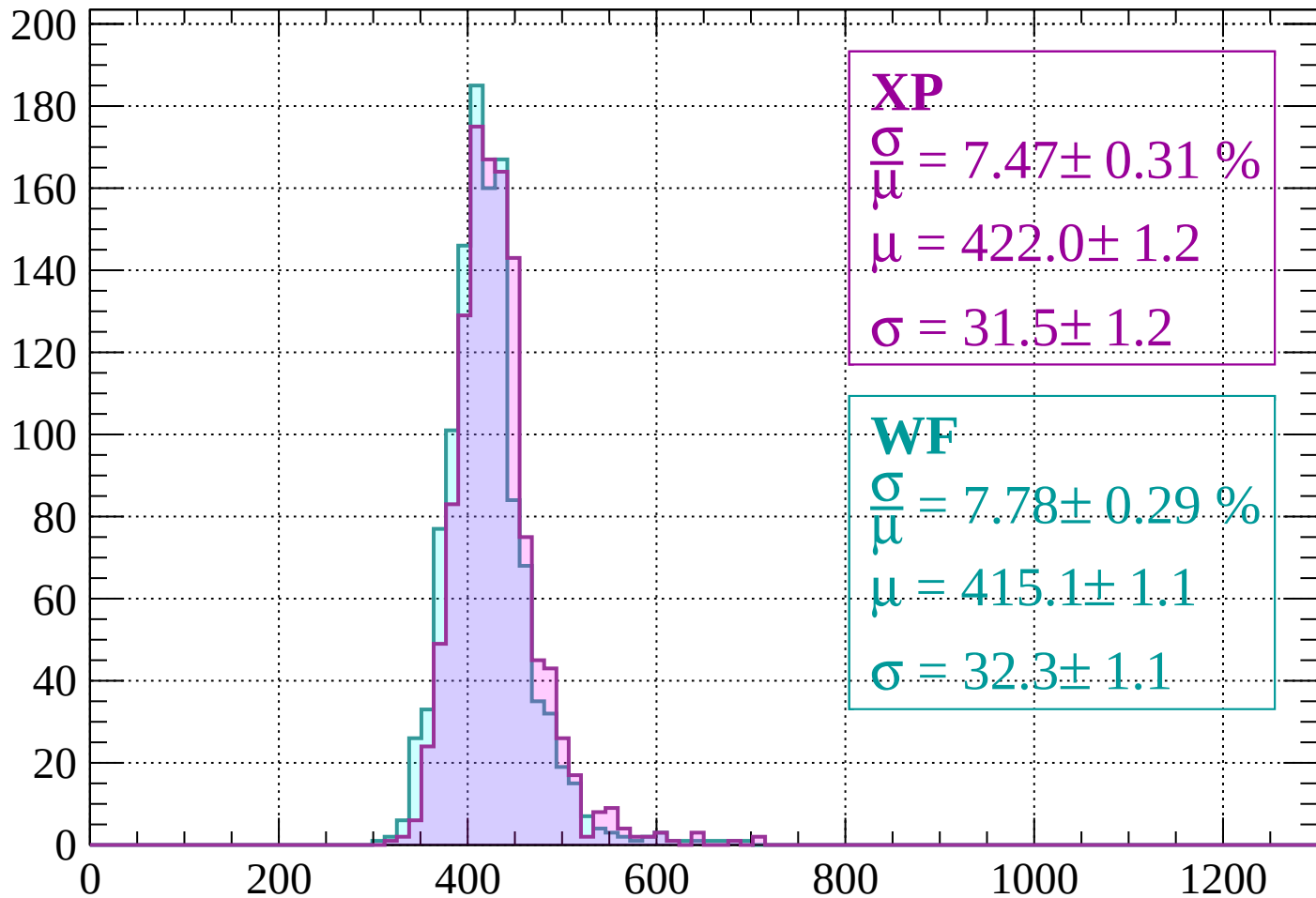
Energy loss | $200 < p < 240$

Count



Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.31 \%$$

$$\mu = 422.0 \pm 1.2$$

$$\sigma = 31.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.29 \%$$

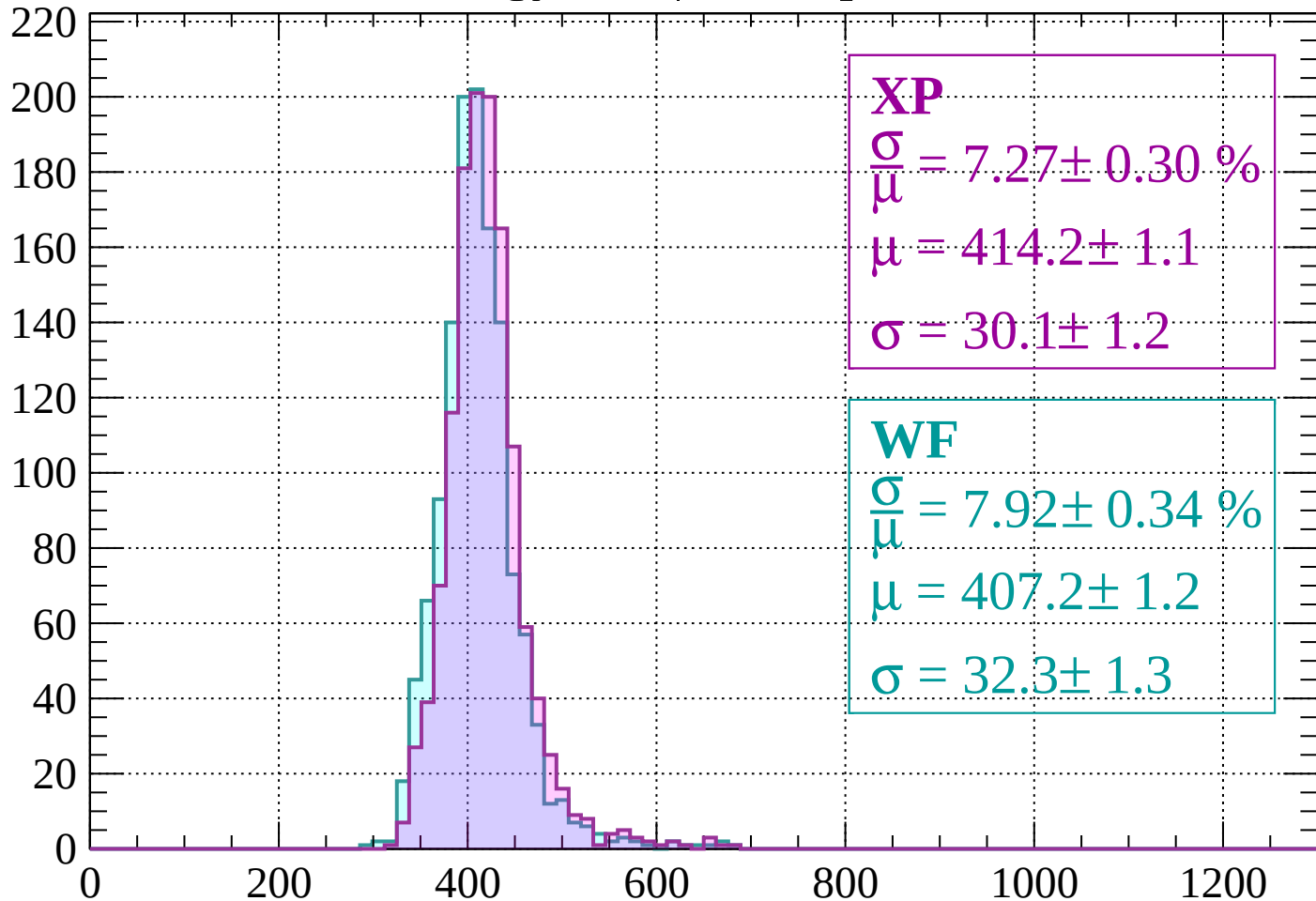
$$\mu = 415.1 \pm 1.1$$

$$\sigma = 32.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.30 \%$$

$$\mu = 414.2 \pm 1.1$$

$$\sigma = 30.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.92 \pm 0.34 \%$$

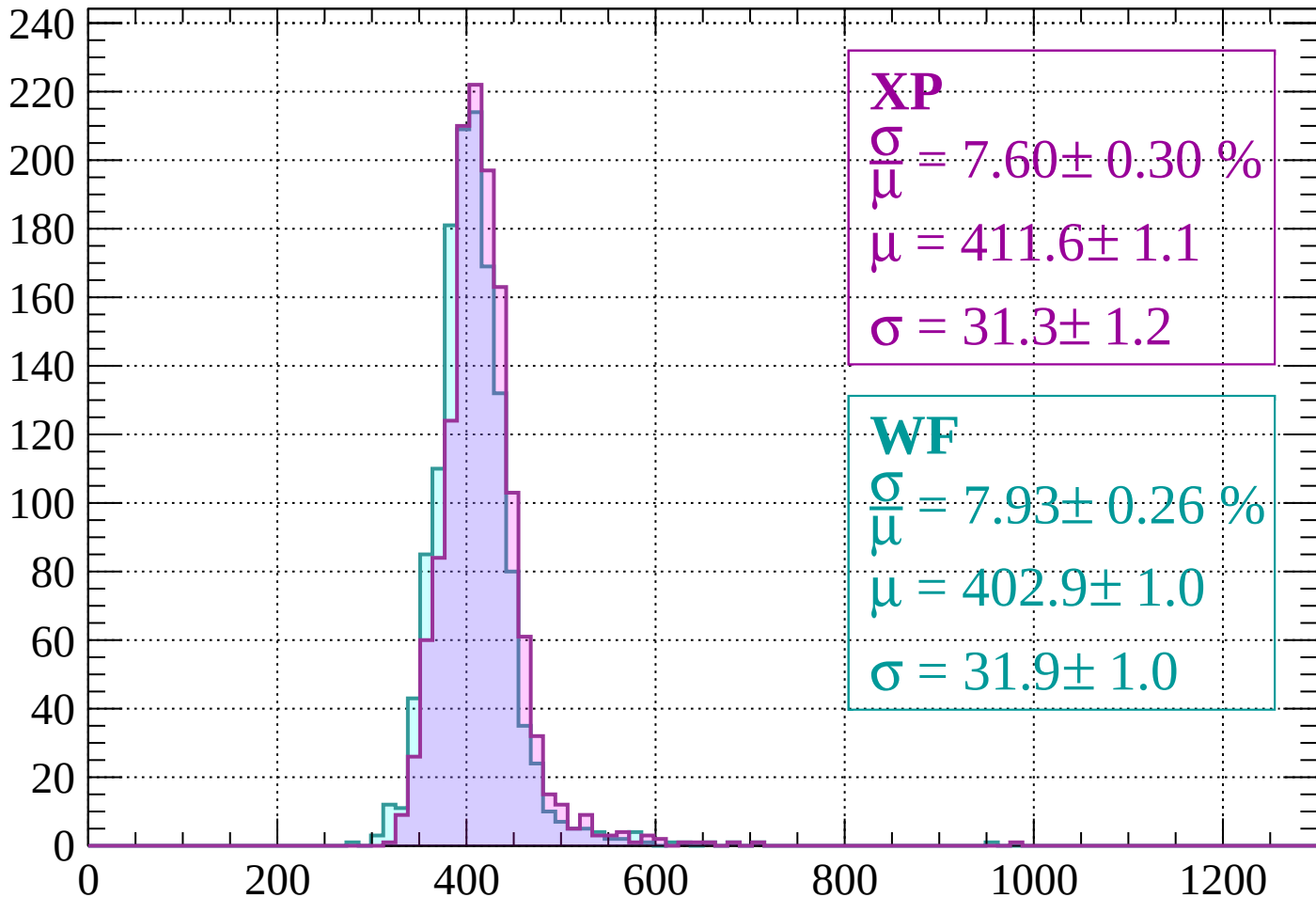
$$\mu = 407.2 \pm 1.2$$

$$\sigma = 32.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.30 \%$$

$$\mu = 411.6 \pm 1.1$$

$$\sigma = 31.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.93 \pm 0.26 \%$$

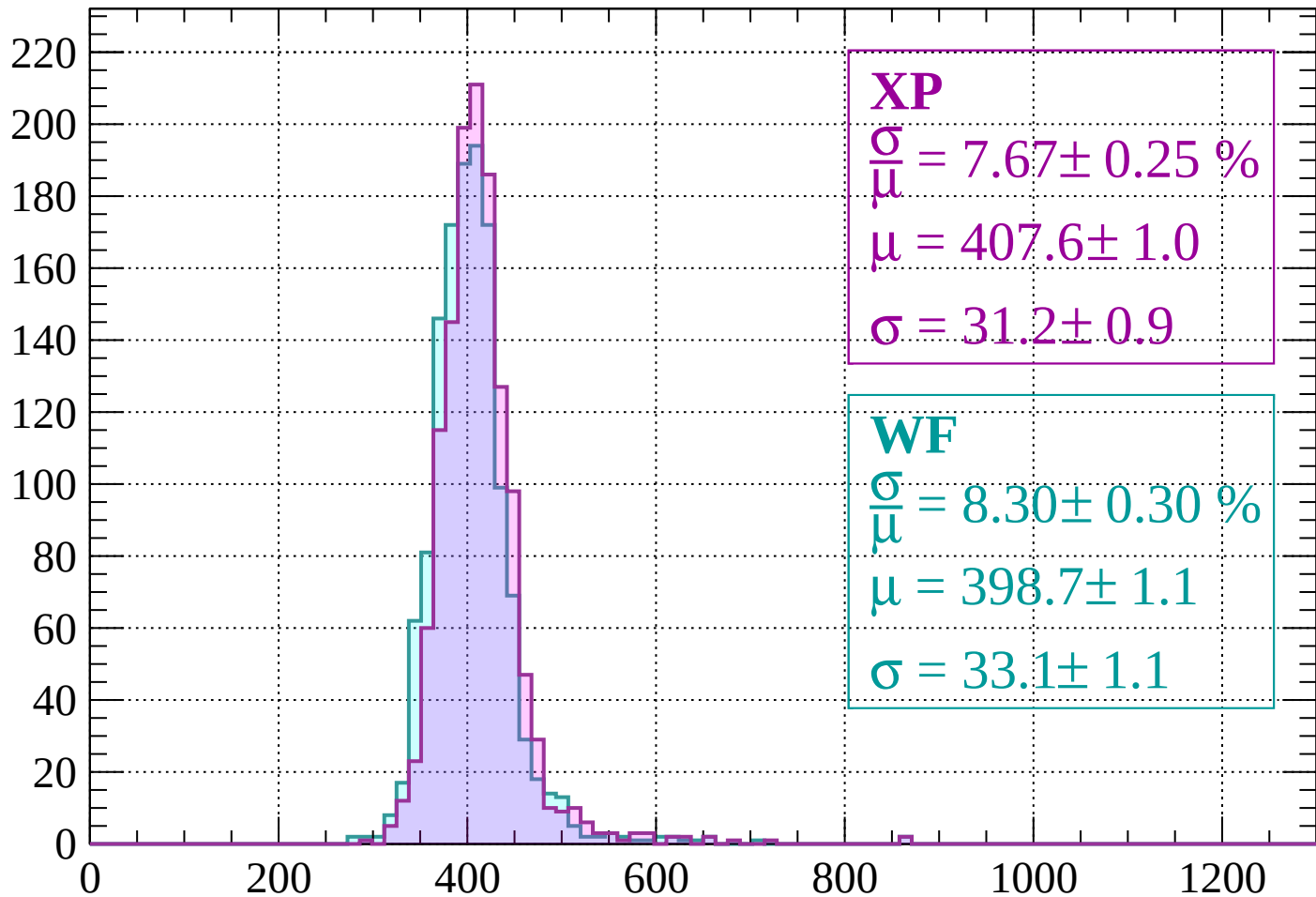
$$\mu = 402.9 \pm 1.0$$

$$\sigma = 31.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.25 \%$$

$$\mu = 407.6 \pm 1.0$$

$$\sigma = 31.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.30 \%$$

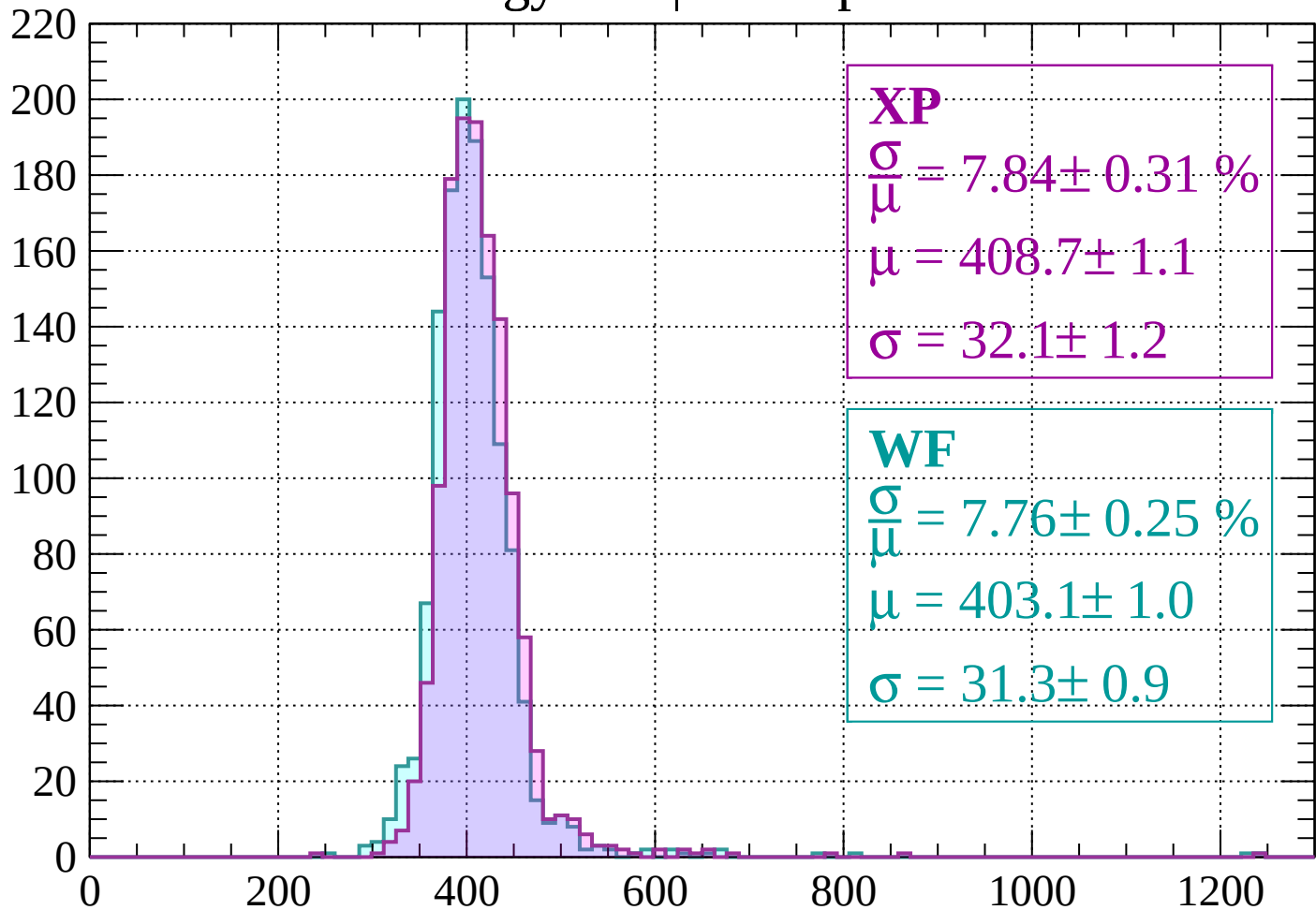
$$\mu = 398.7 \pm 1.1$$

$$\sigma = 33.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.31 \%$$

$$\mu = 408.7 \pm 1.1$$

$$\sigma = 32.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.76 \pm 0.25 \%$$

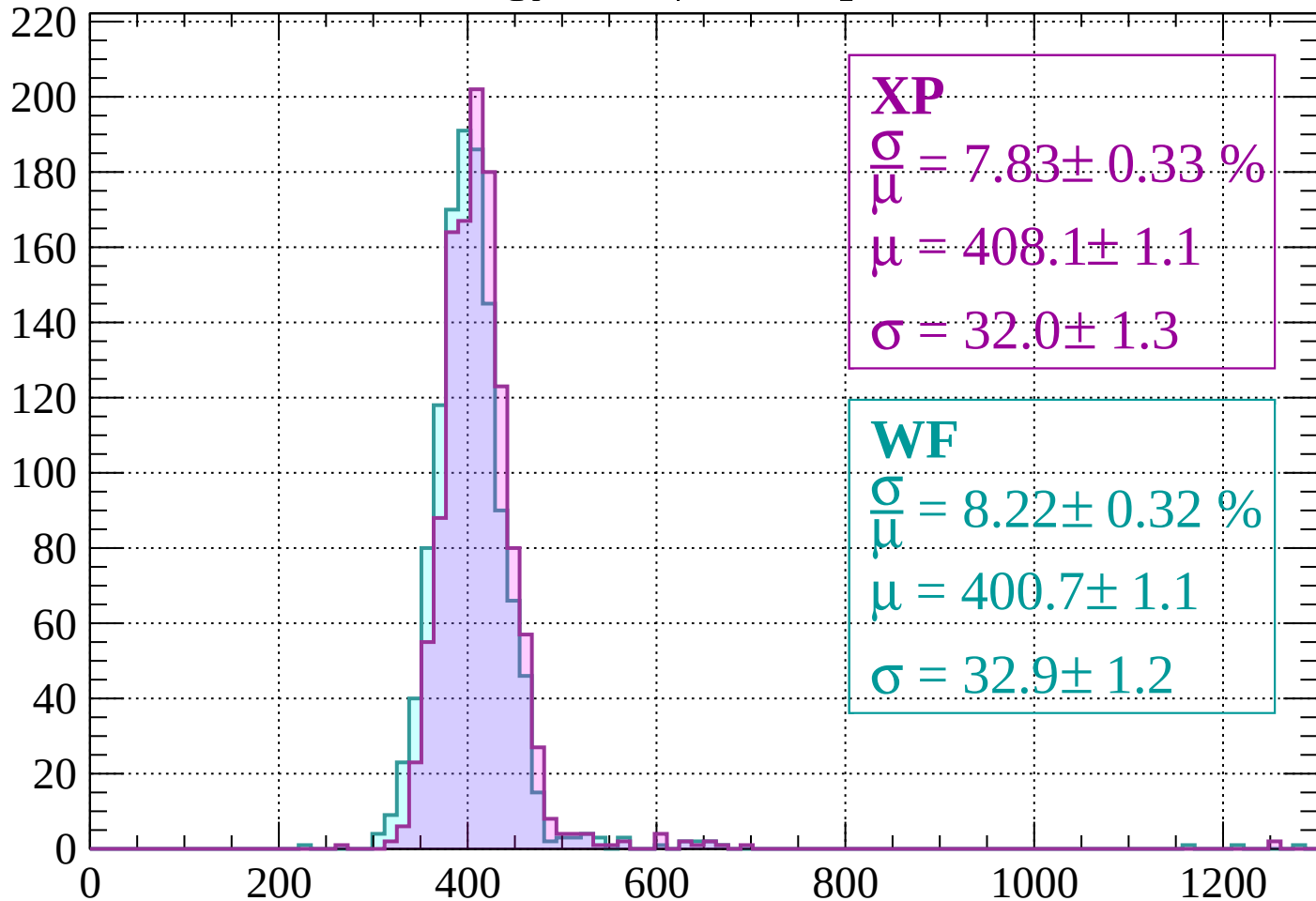
$$\mu = 403.1 \pm 1.0$$

$$\sigma = 31.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 7.83 \pm 0.33 \%$$

$$\mu = 408.1 \pm 1.1$$

$$\sigma = 32.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.32 \%$$

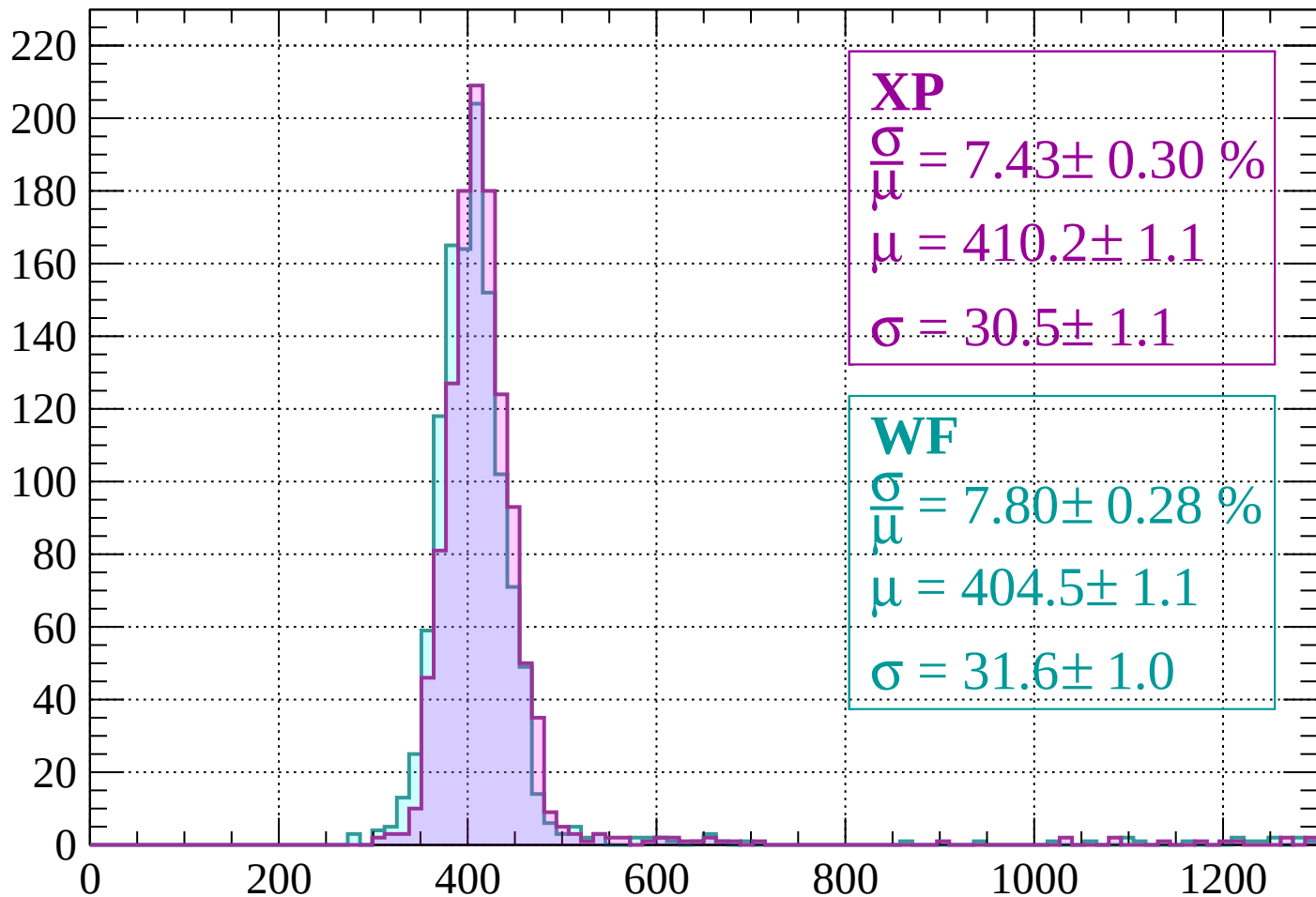
$$\mu = 400.7 \pm 1.1$$

$$\sigma = 32.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.30 \%$$

$$\mu = 410.2 \pm 1.1$$

$$\sigma = 30.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.28 \%$$

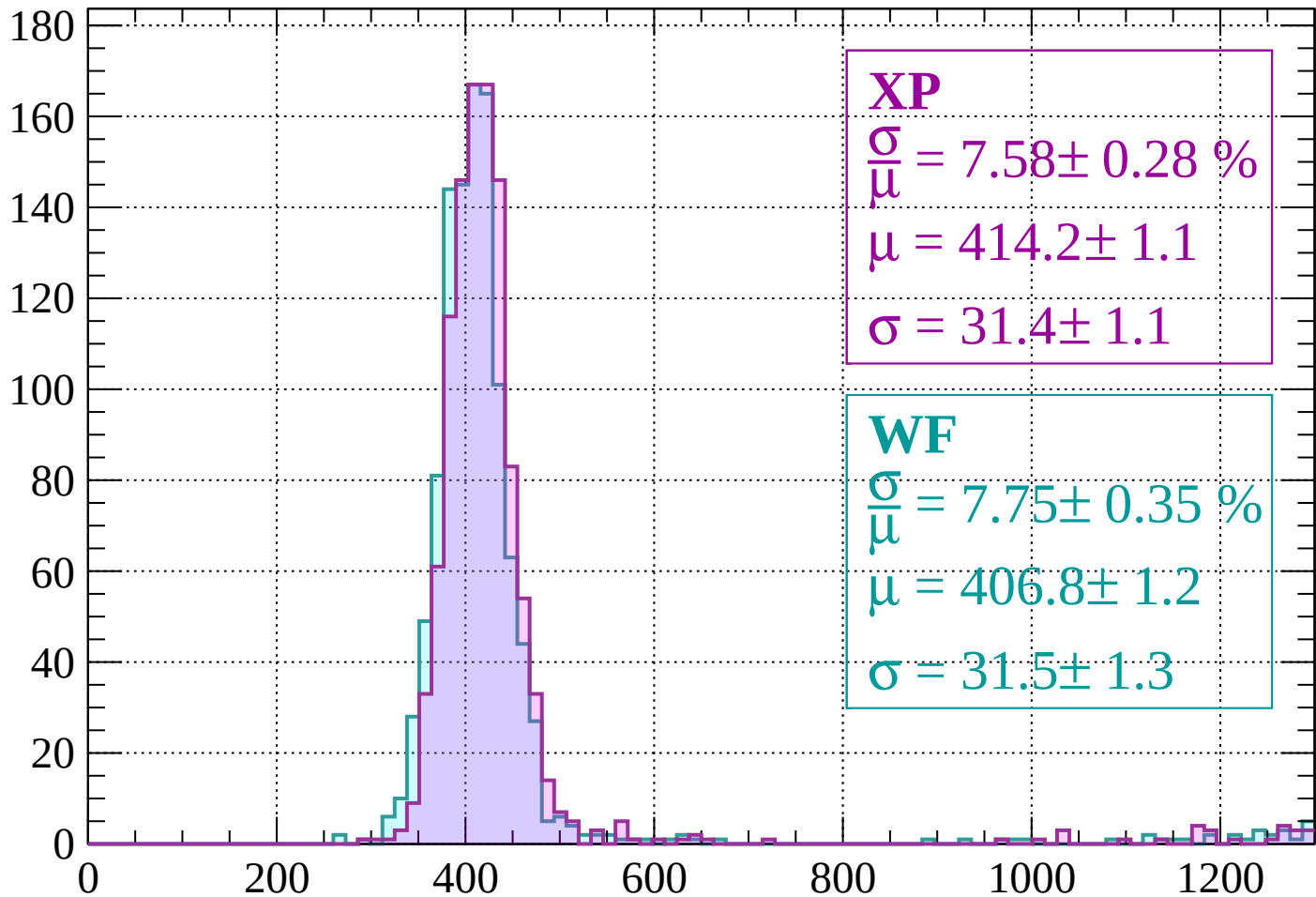
$$\mu = 404.5 \pm 1.1$$

$$\sigma = 31.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.28 \%$$

$$\mu = 414.2 \pm 1.1$$

$$\sigma = 31.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.35 \%$$

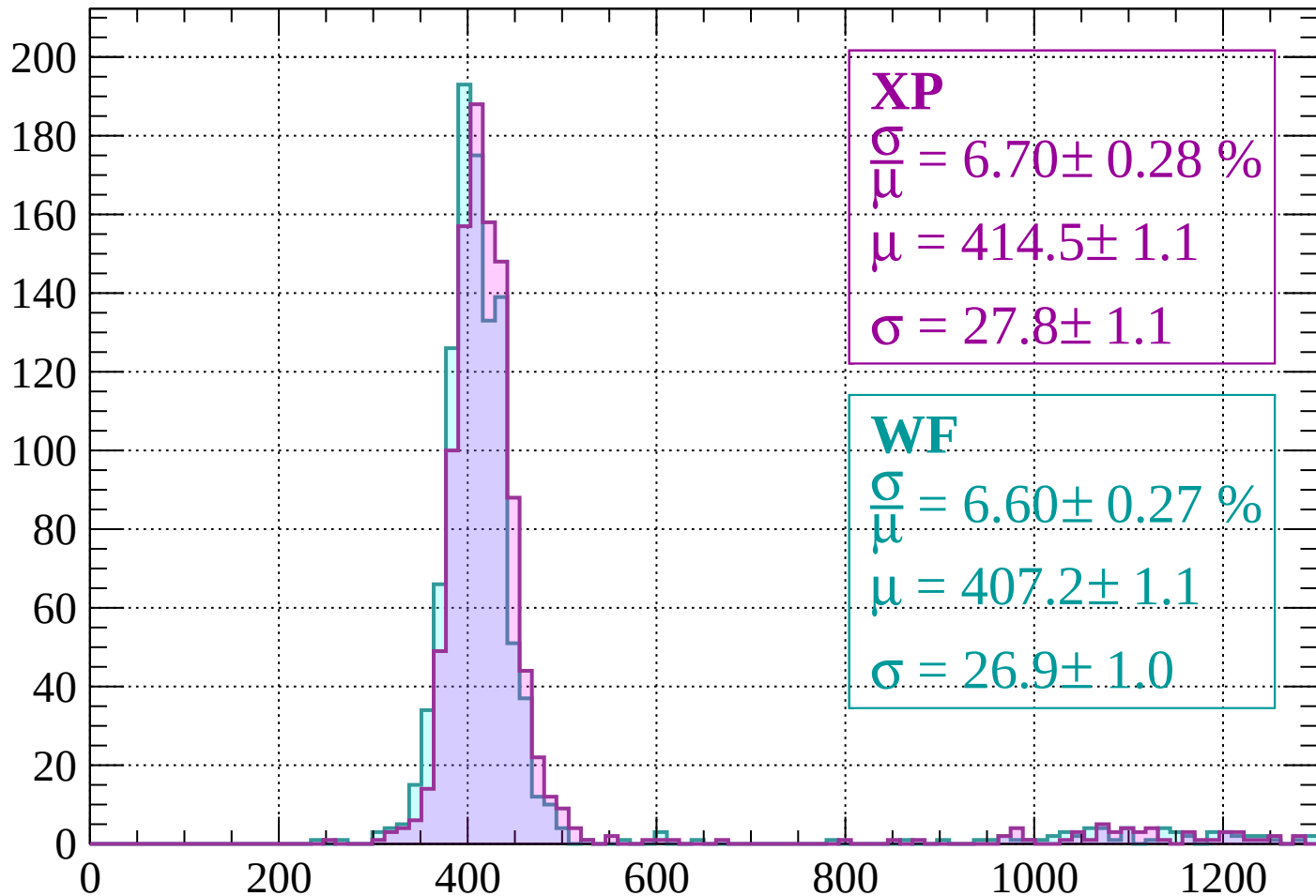
$$\mu = 406.8 \pm 1.2$$

$$\sigma = 31.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 6.70 \pm 0.28 \%$$

$$\mu = 414.5 \pm 1.1$$

$$\sigma = 27.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.60 \pm 0.27 \%$$

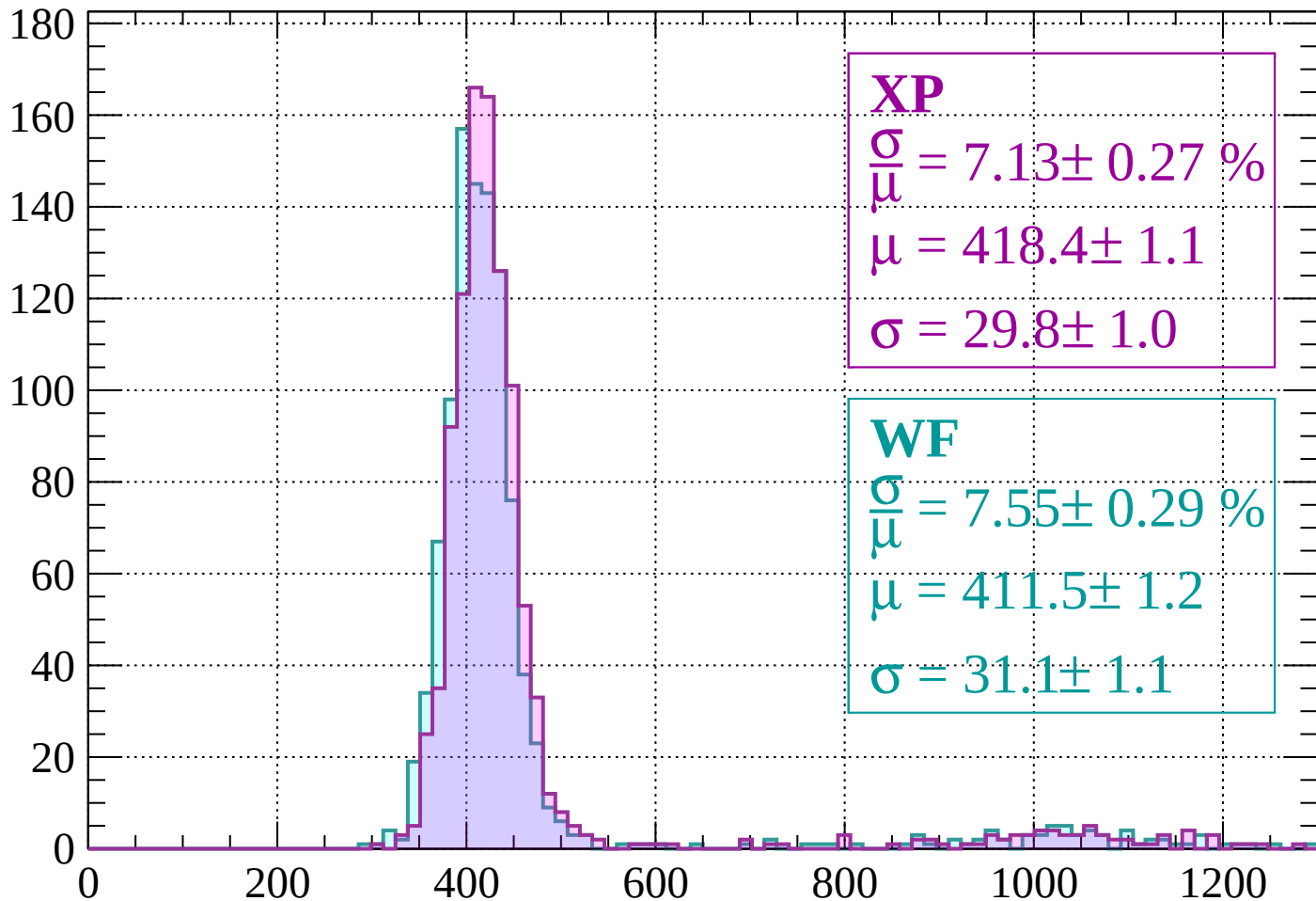
$$\mu = 407.2 \pm 1.1$$

$$\sigma = 26.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.27 \%$$

$$\mu = 418.4 \pm 1.1$$

$$\sigma = 29.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.55 \pm 0.29 \%$$

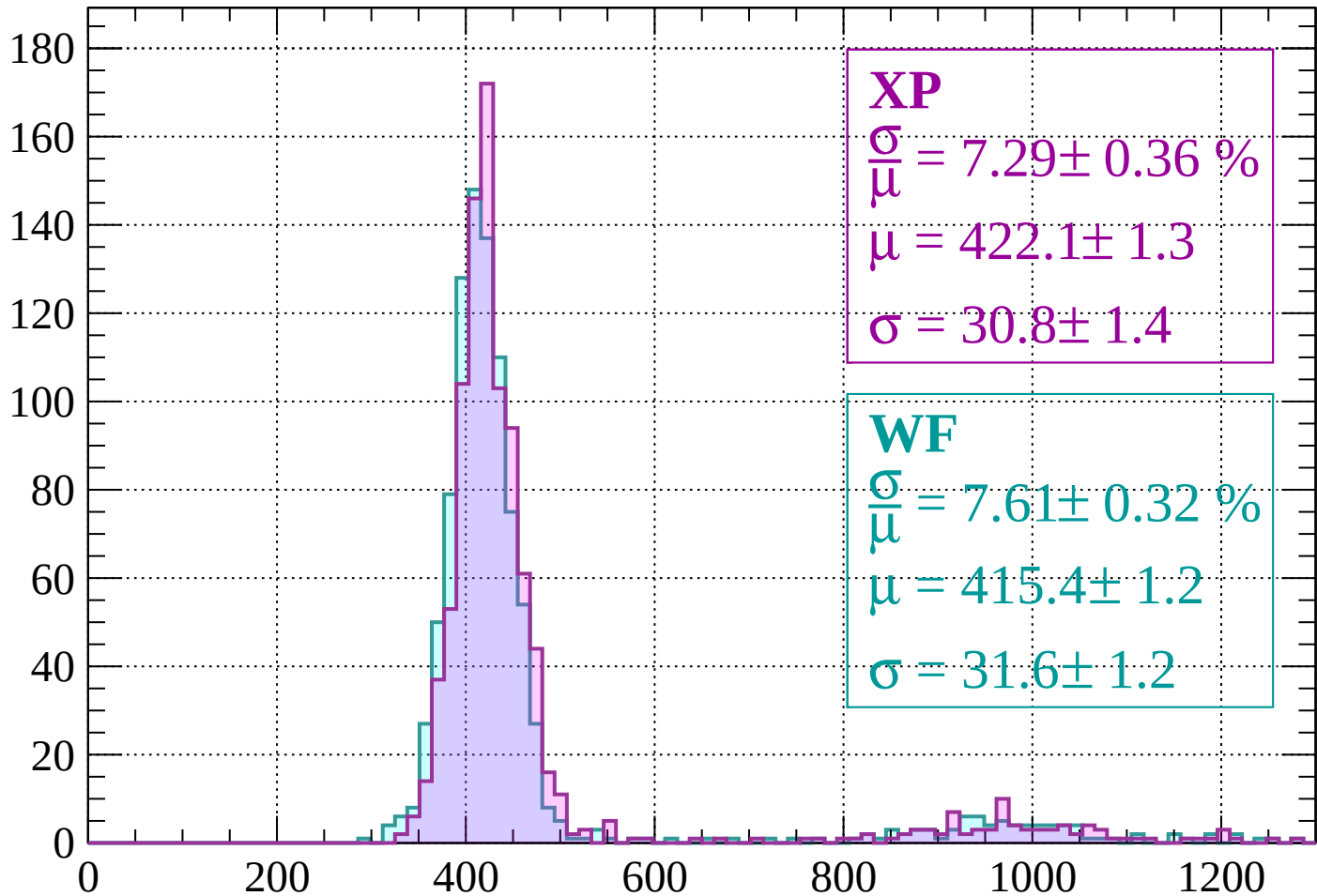
$$\mu = 411.5 \pm 1.2$$

$$\sigma = 31.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 7.29 \pm 0.36 \%$$

$$\mu = 422.1 \pm 1.3$$

$$\sigma = 30.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.32 \%$$

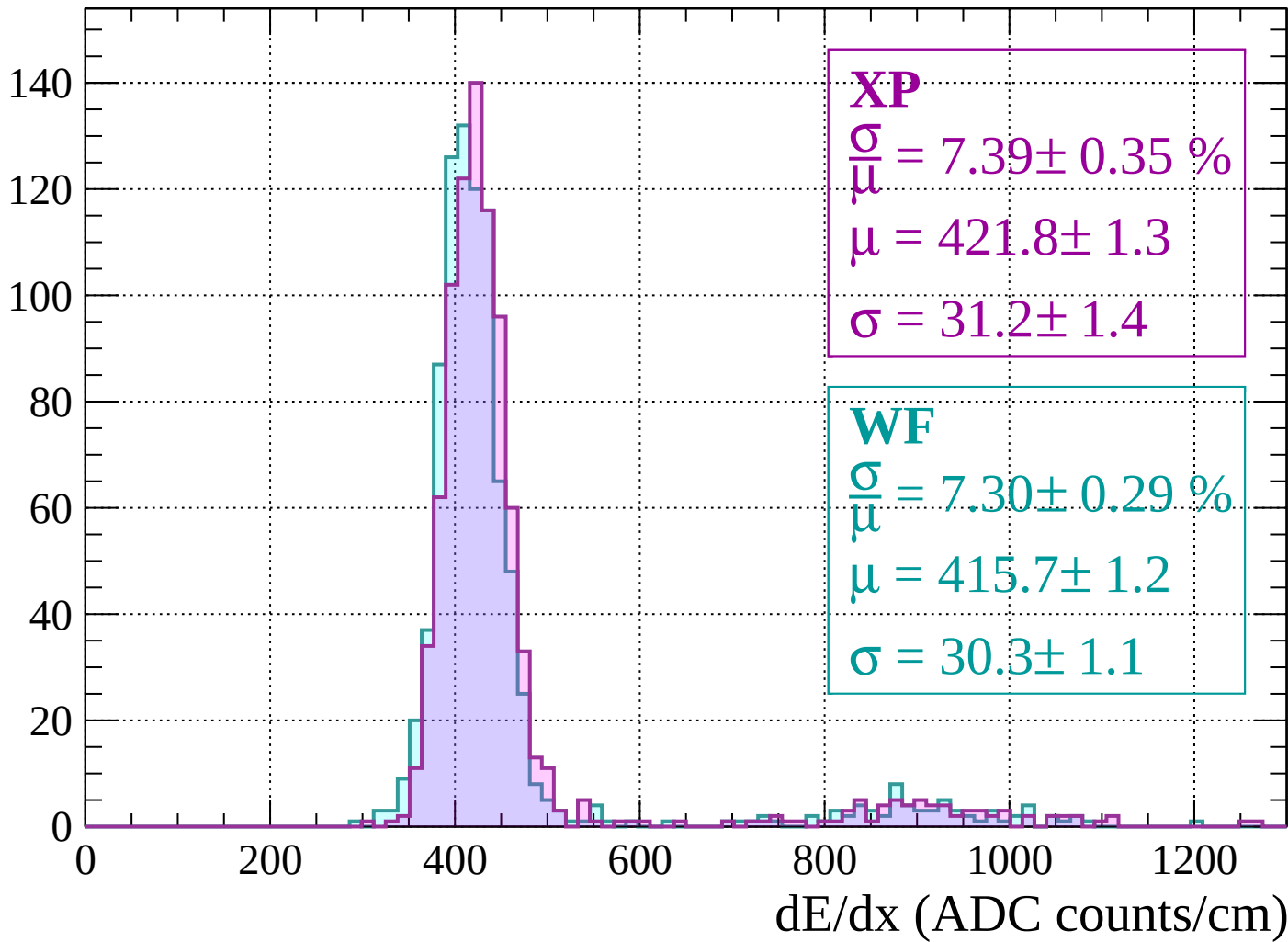
$$\mu = 415.4 \pm 1.2$$

$$\sigma = 31.6 \pm 1.2$$

dE/dx (ADC counts/cm)

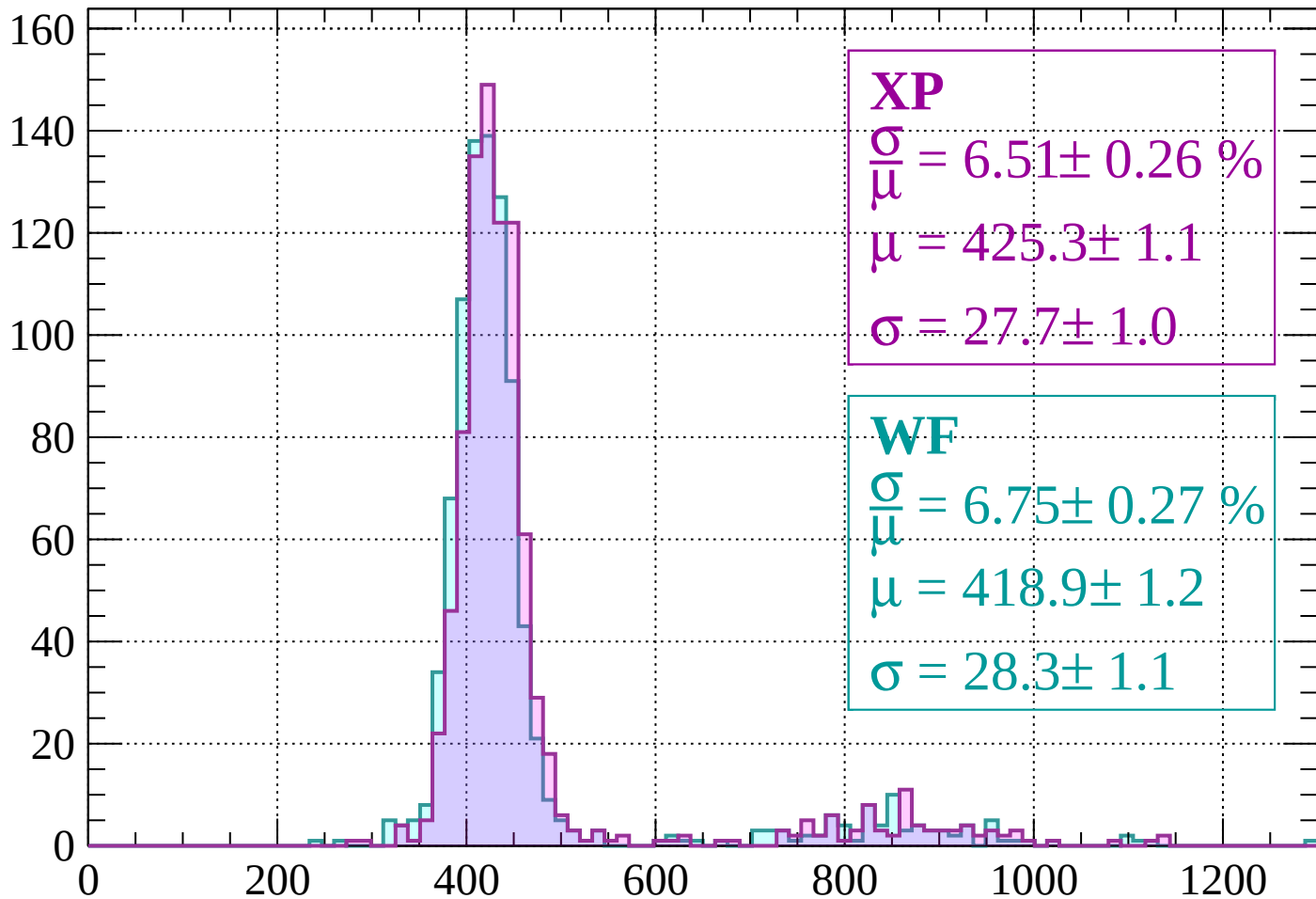
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 6.51 \pm 0.26 \%$$

$$\mu = 425.3 \pm 1.1$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.75 \pm 0.27 \%$$

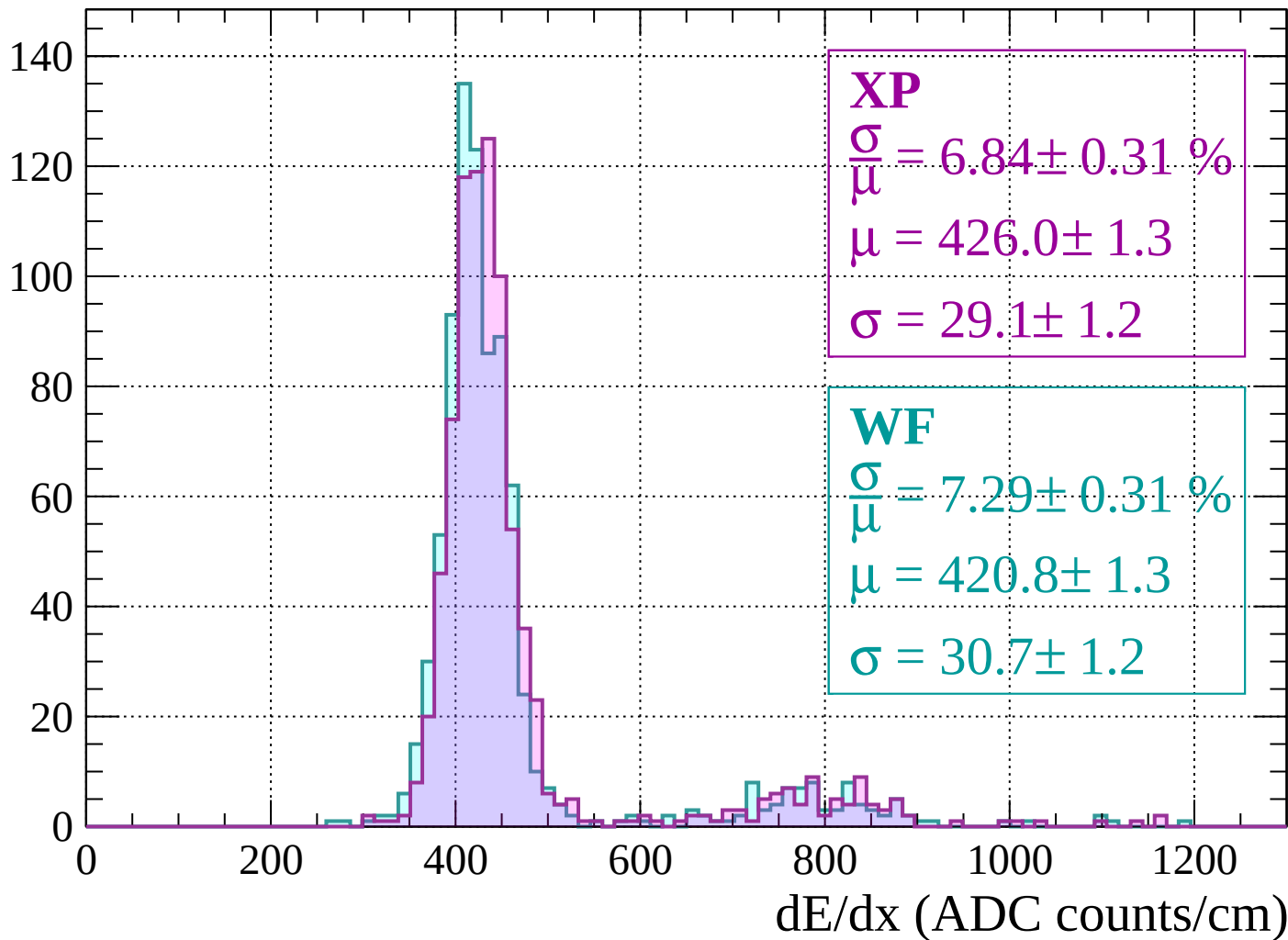
$$\mu = 418.9 \pm 1.2$$

$$\sigma = 28.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



Energy loss | $800 < p < 840$

Count

120

100

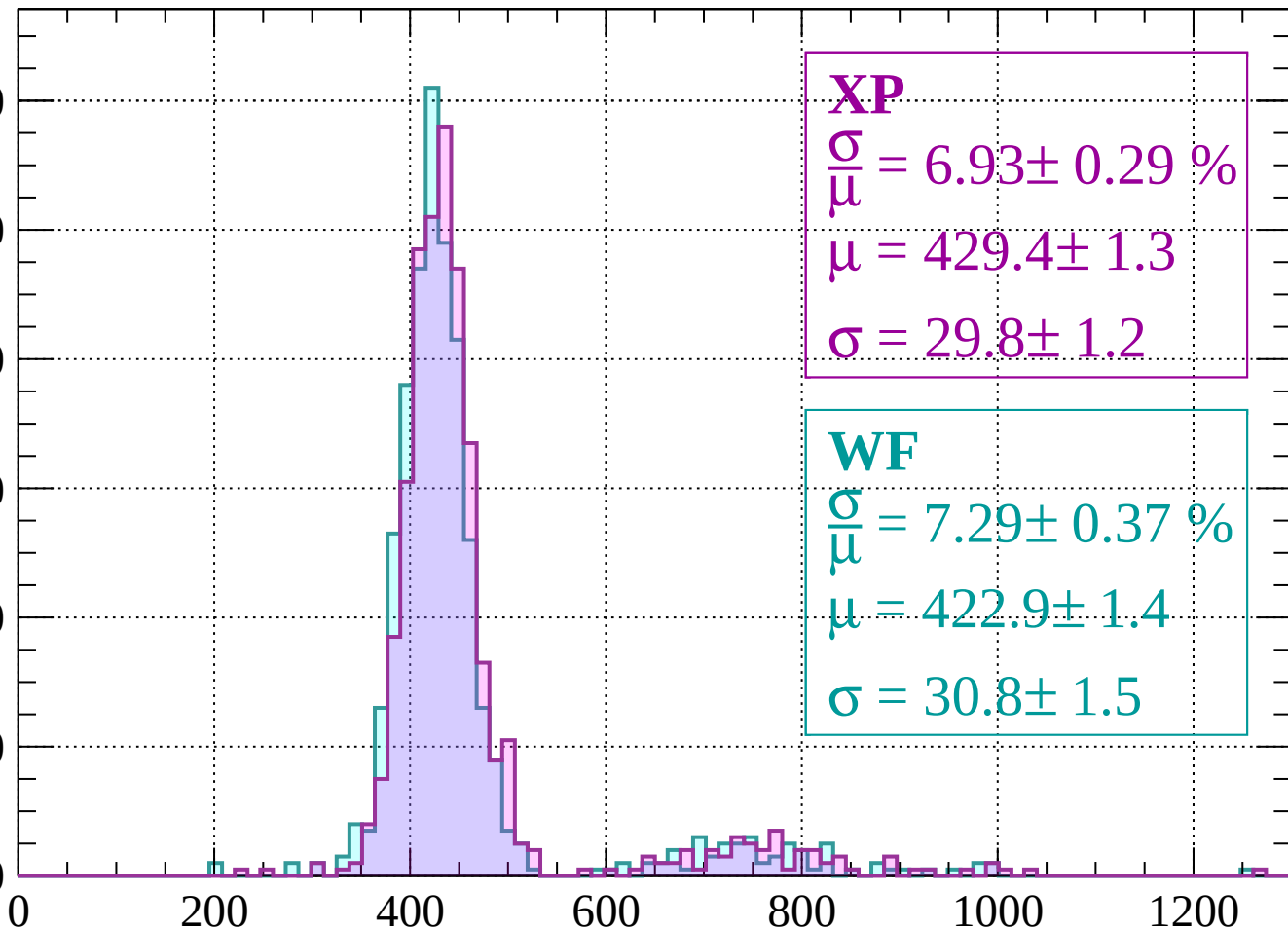
80

60

40

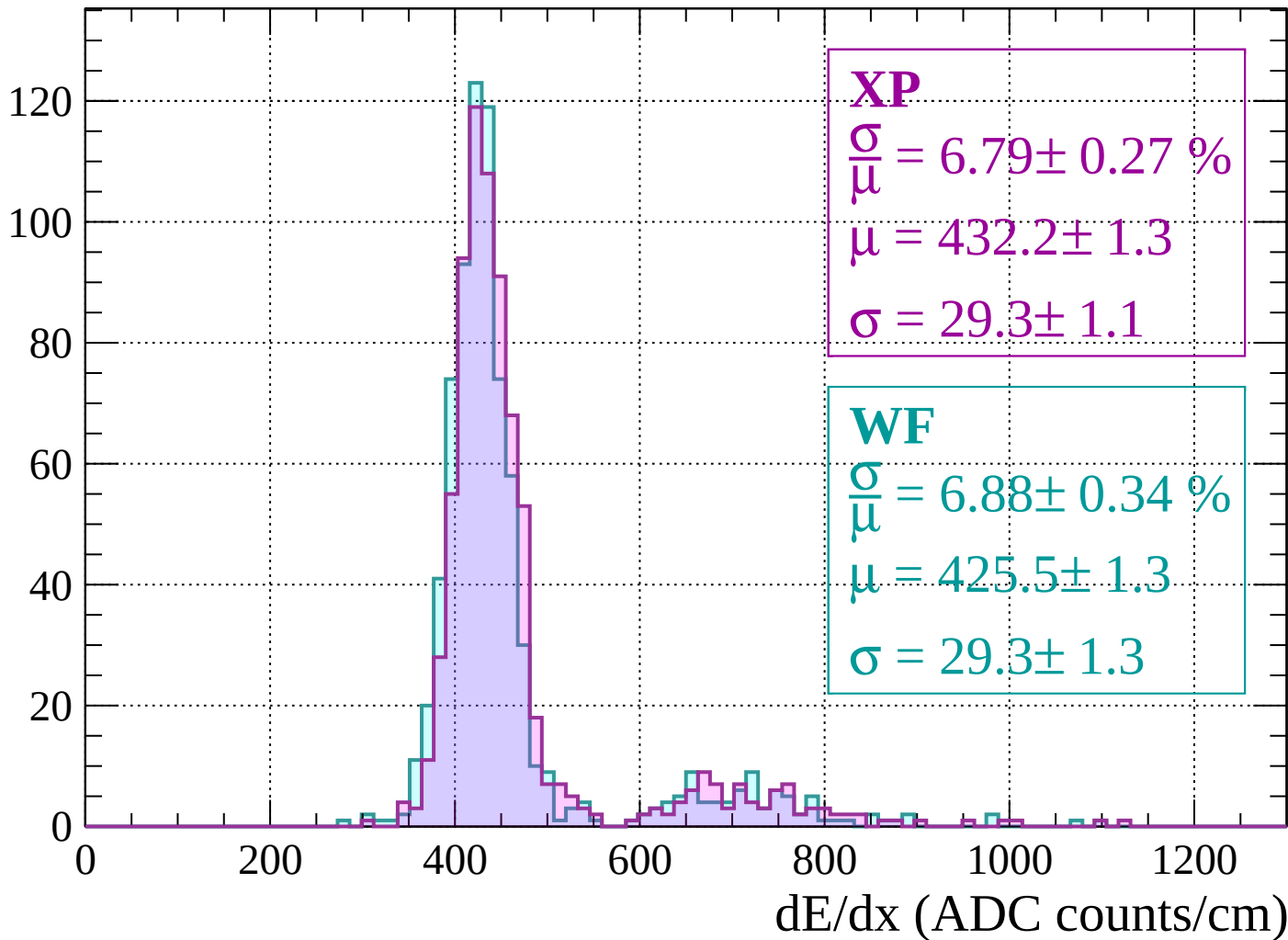
20

0



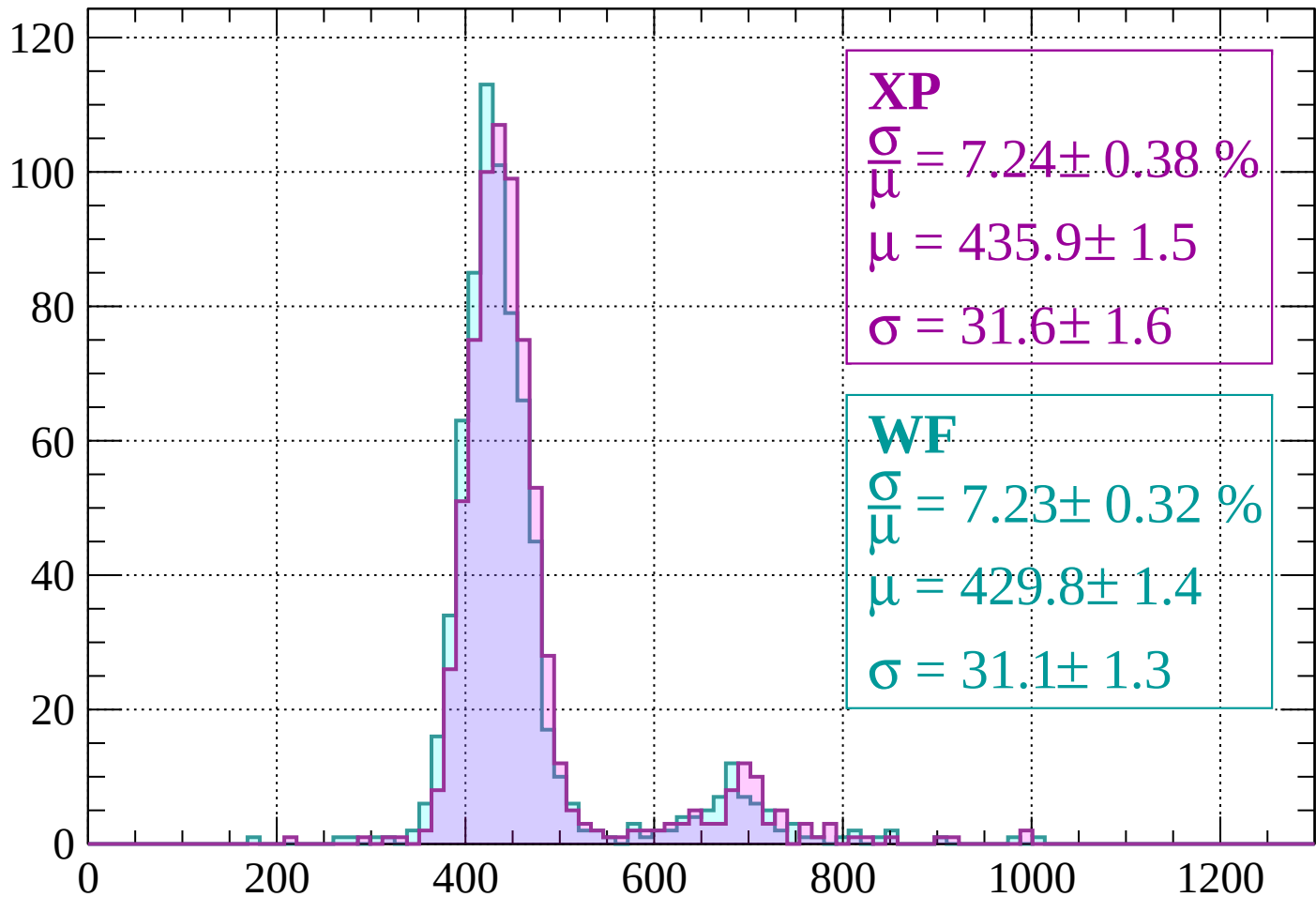
Energy loss | $840 < p < 880$

Count



Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 7.24 \pm 0.38 \%$$

$$\mu = 435.9 \pm 1.5$$

$$\sigma = 31.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.23 \pm 0.32 \%$$

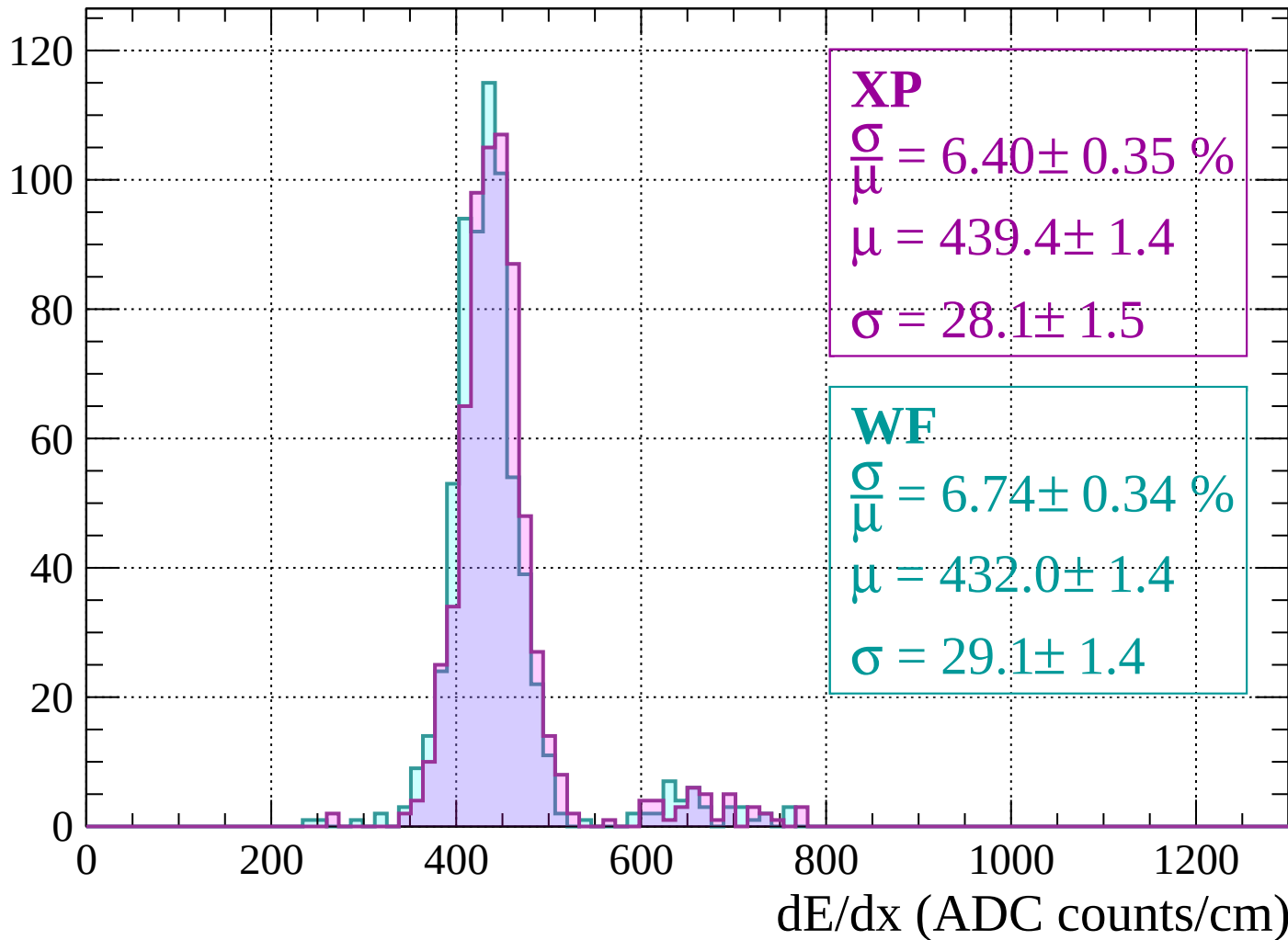
$$\mu = 429.8 \pm 1.4$$

$$\sigma = 31.1 \pm 1.3$$

dE/dx (ADC counts/cm)

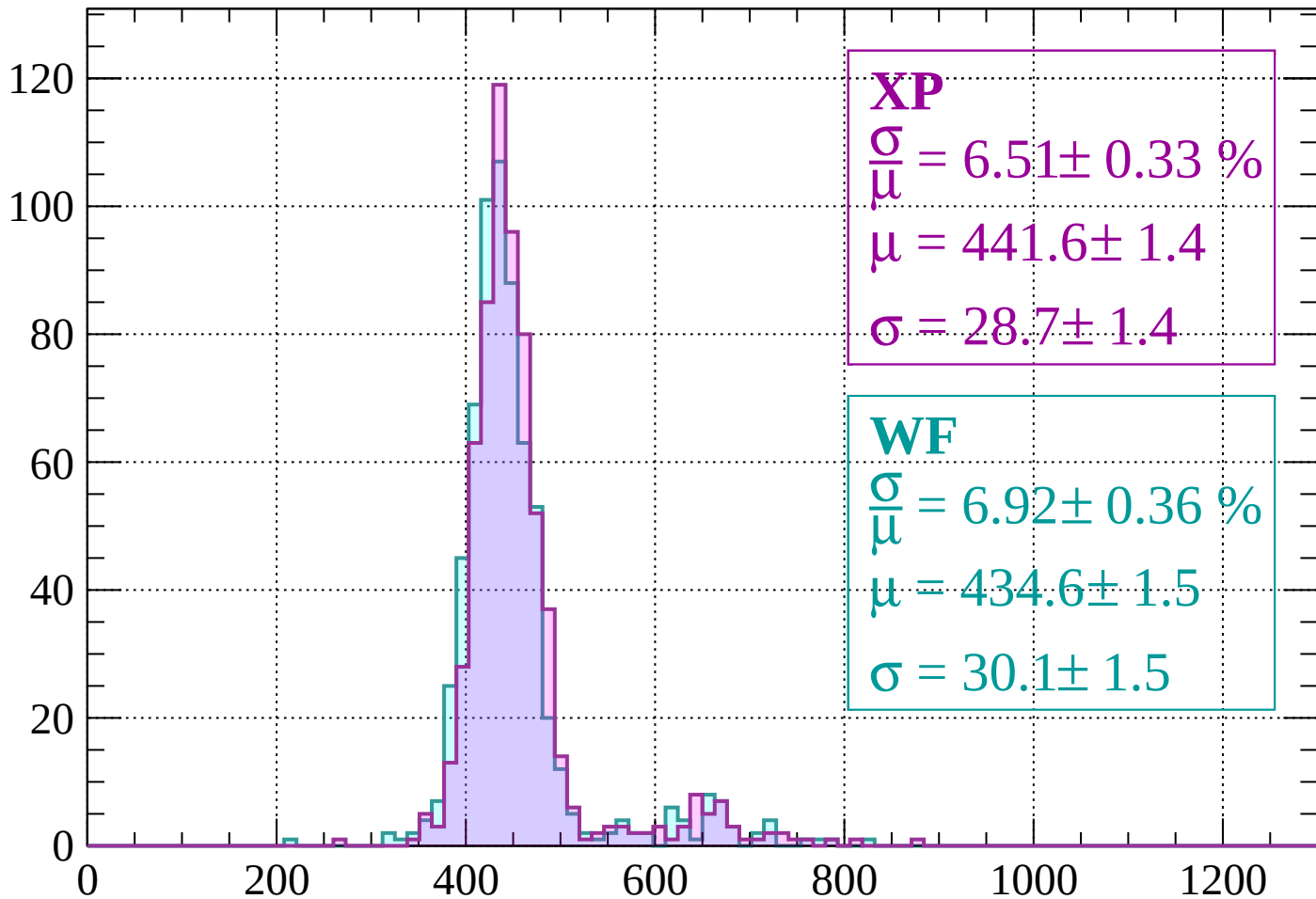
Energy loss | $920 < p < 960$

Count



Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 6.51 \pm 0.33 \%$$

$$\mu = 441.6 \pm 1.4$$

$$\sigma = 28.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.92 \pm 0.36 \%$$

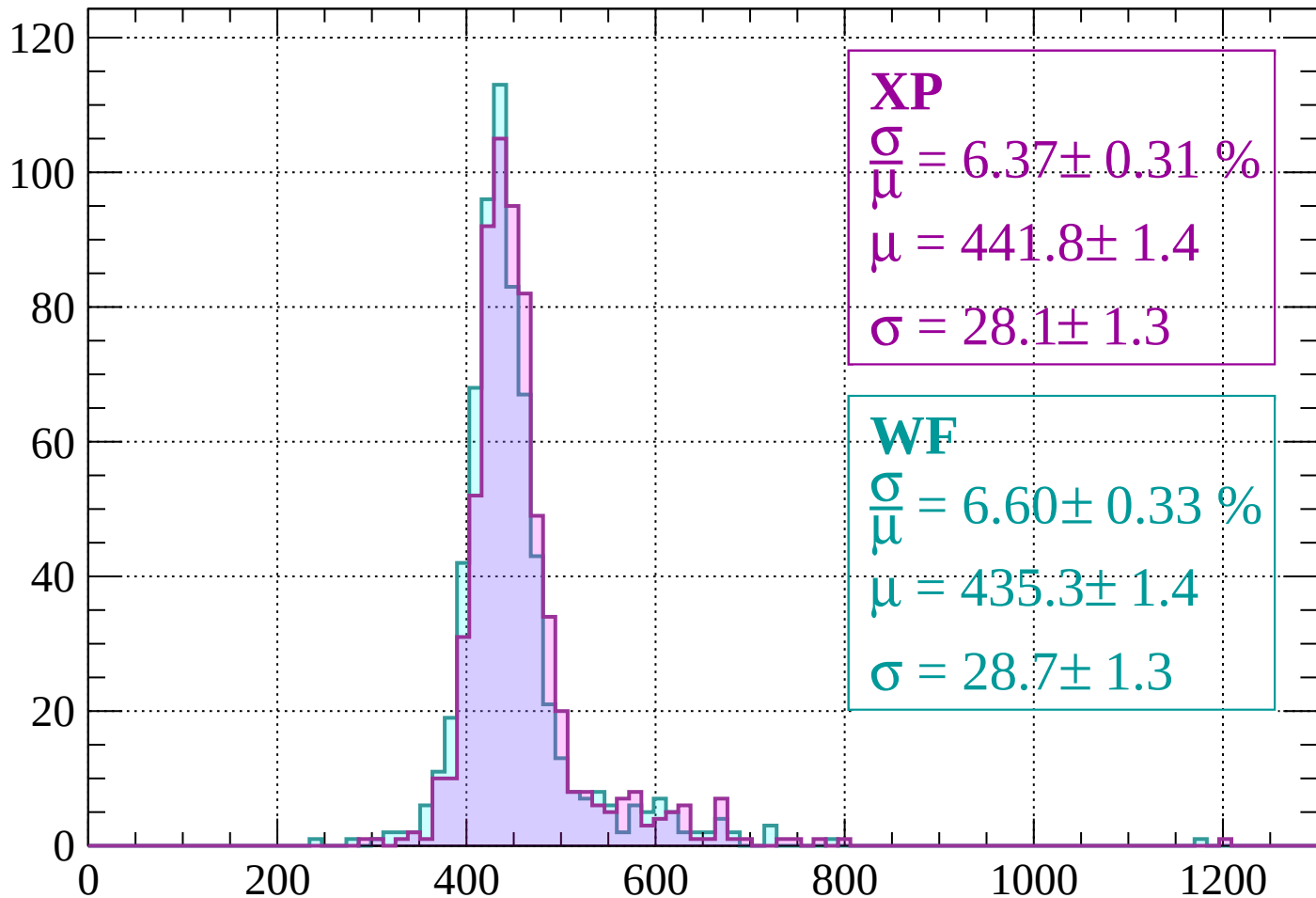
$$\mu = 434.6 \pm 1.5$$

$$\sigma = 30.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 6.37 \pm 0.31 \%$$

$$\mu = 441.8 \pm 1.4$$

$$\sigma = 28.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.60 \pm 0.33 \%$$

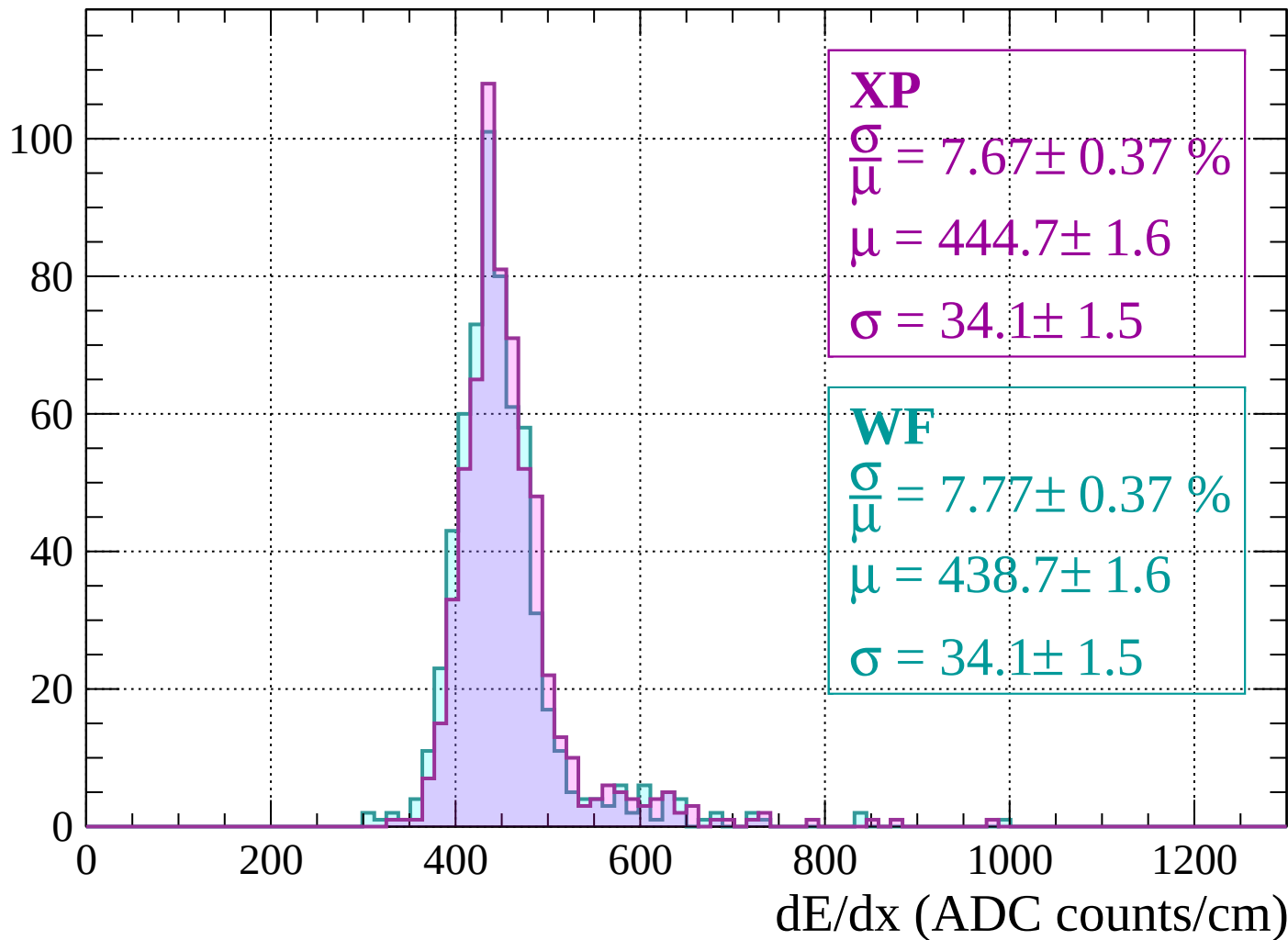
$$\mu = 435.3 \pm 1.4$$

$$\sigma = 28.7 \pm 1.3$$

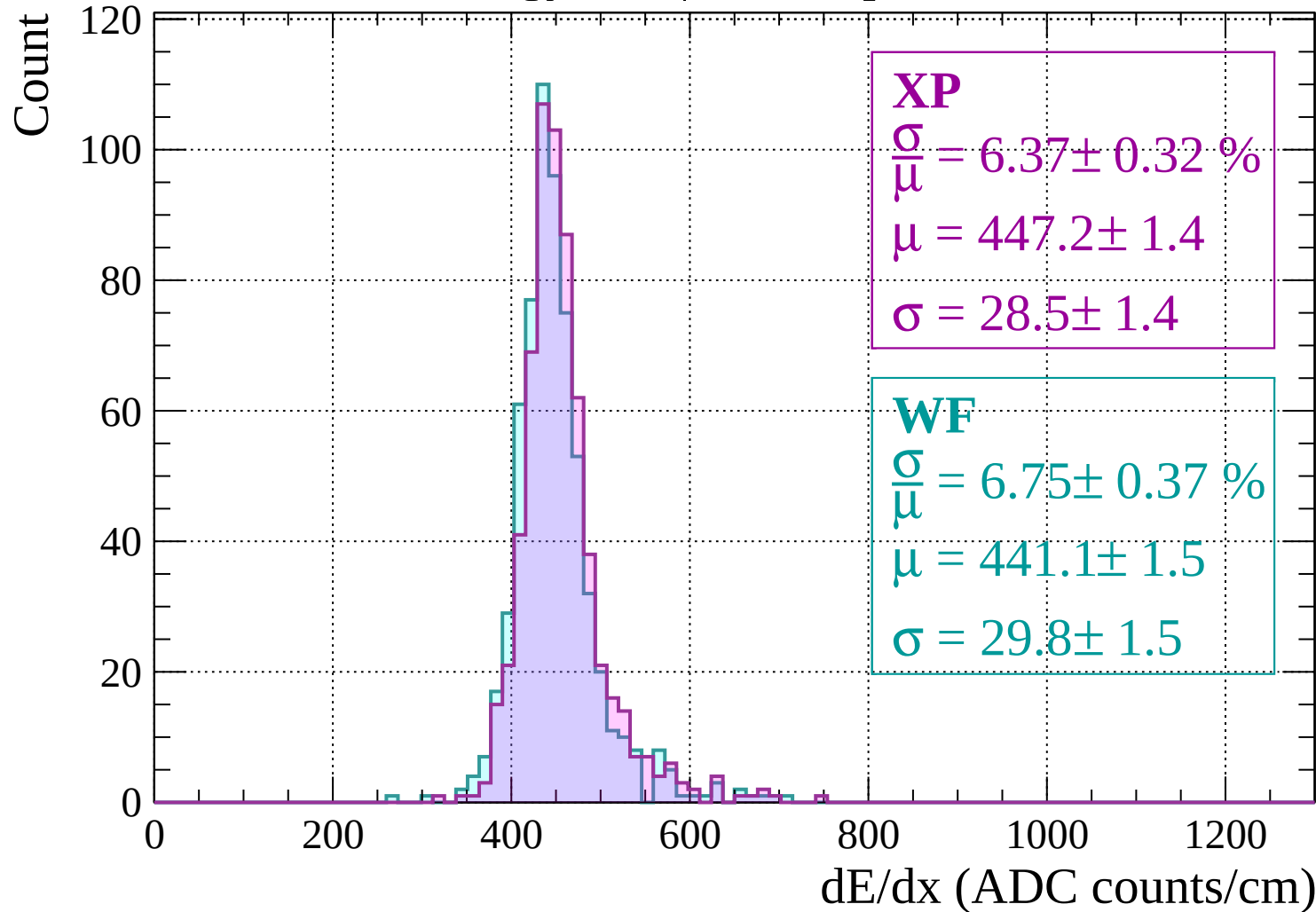
dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count

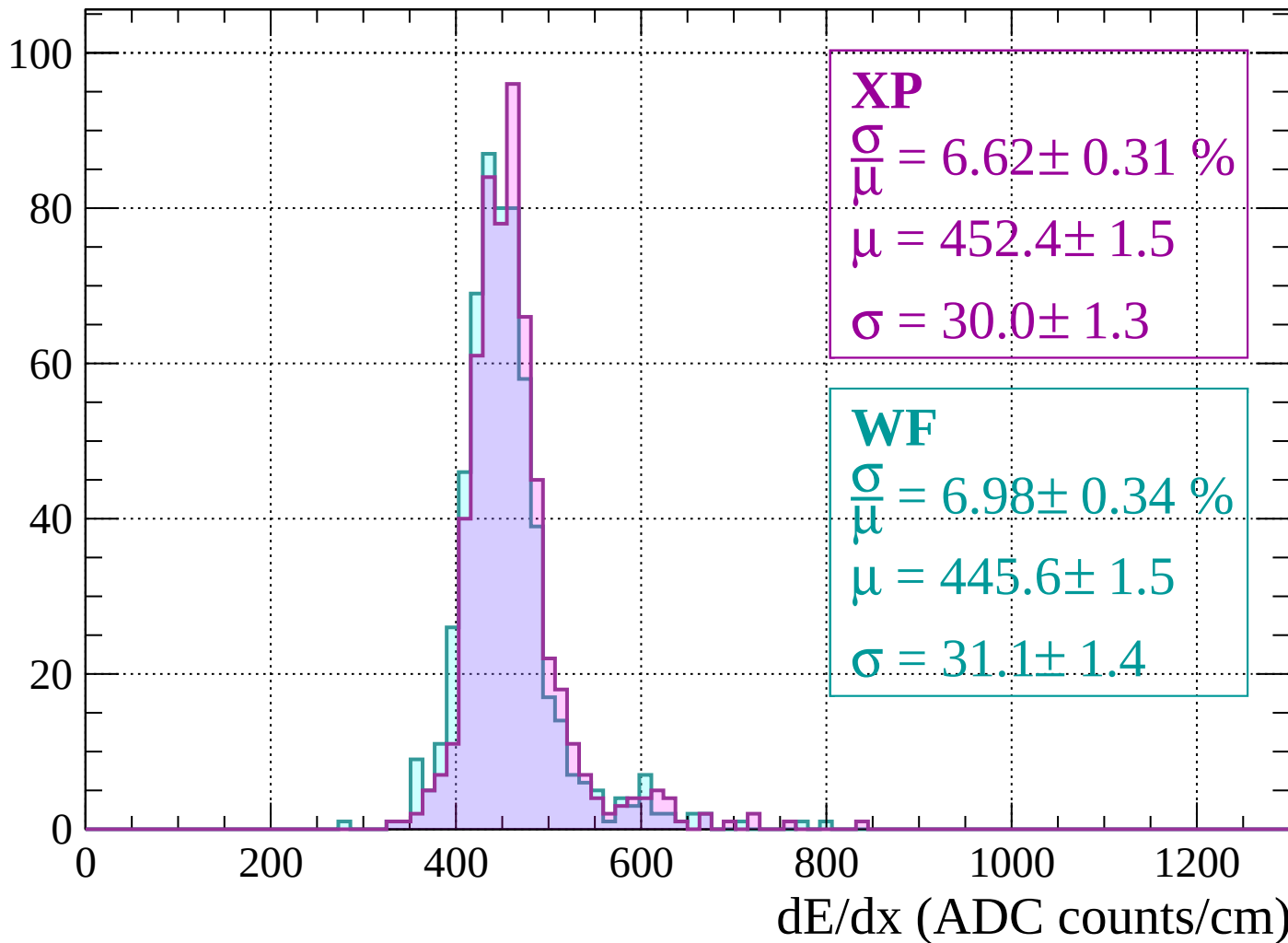


Energy loss | $1080 < p < 1120$



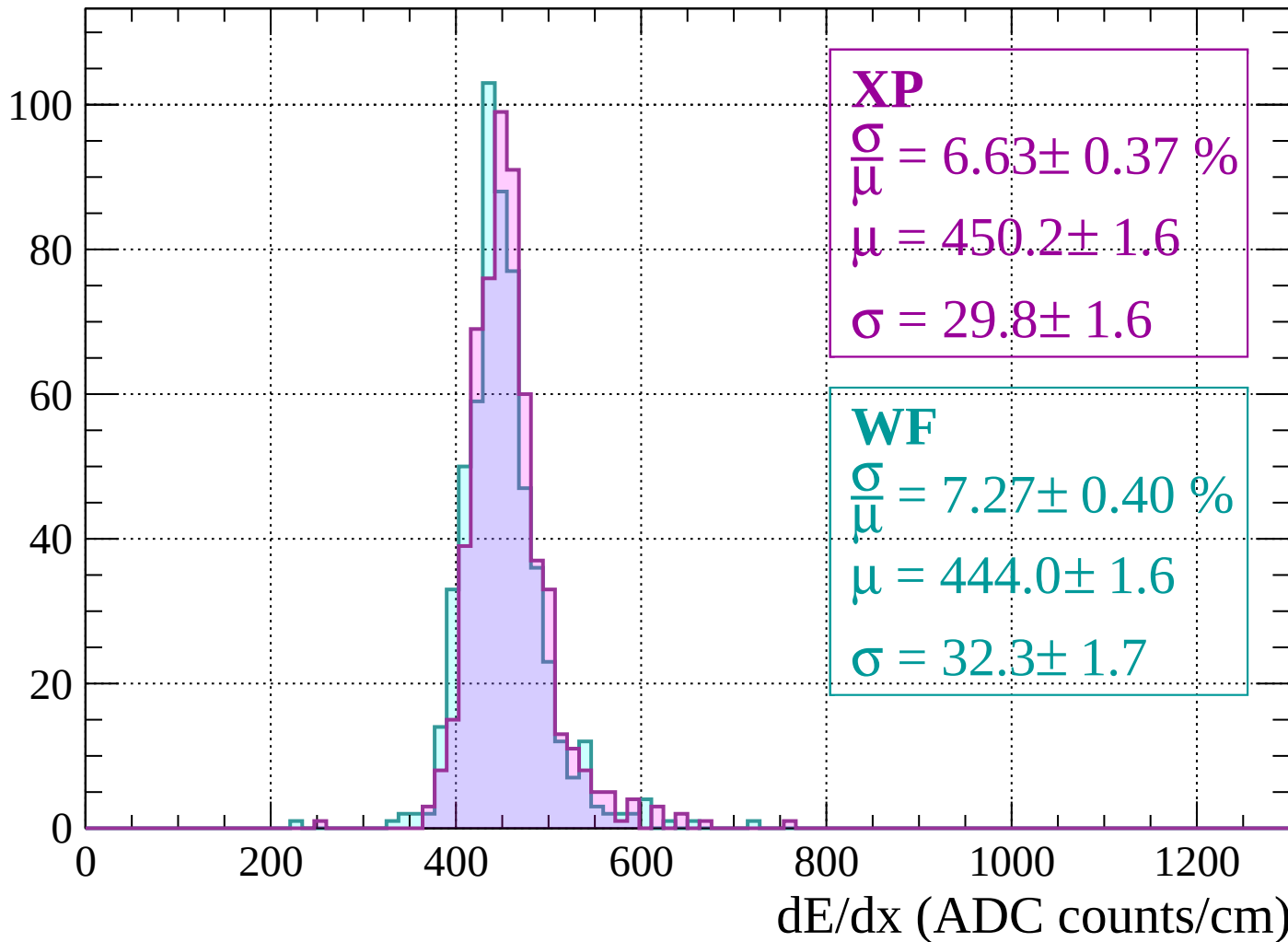
Energy loss | $1120 < p < 1160$

Count



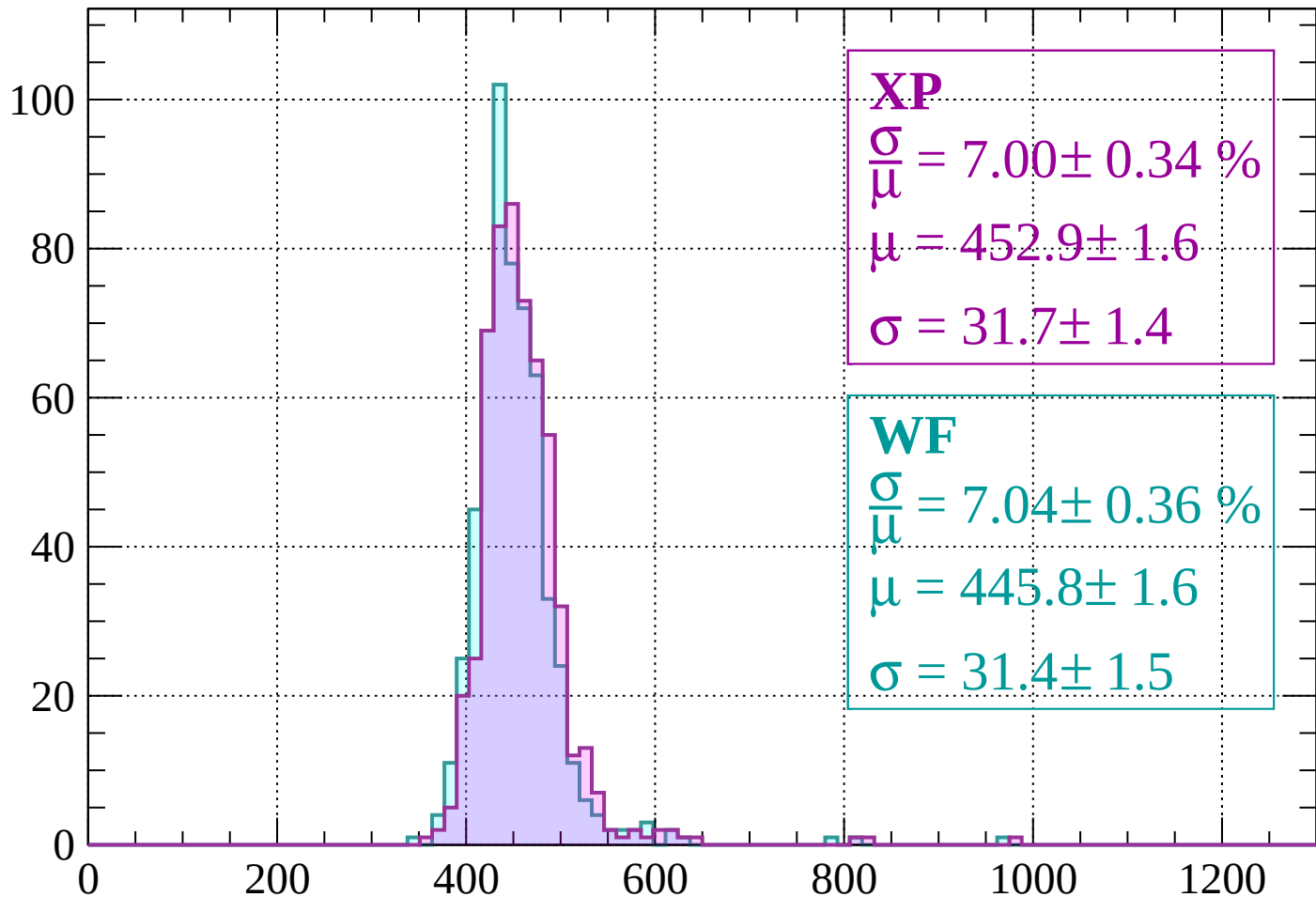
Energy loss | $1160 < p < 1200$

Count



Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 7.00 \pm 0.34 \%$$

$$\mu = 452.9 \pm 1.6$$

$$\sigma = 31.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.04 \pm 0.36 \%$$

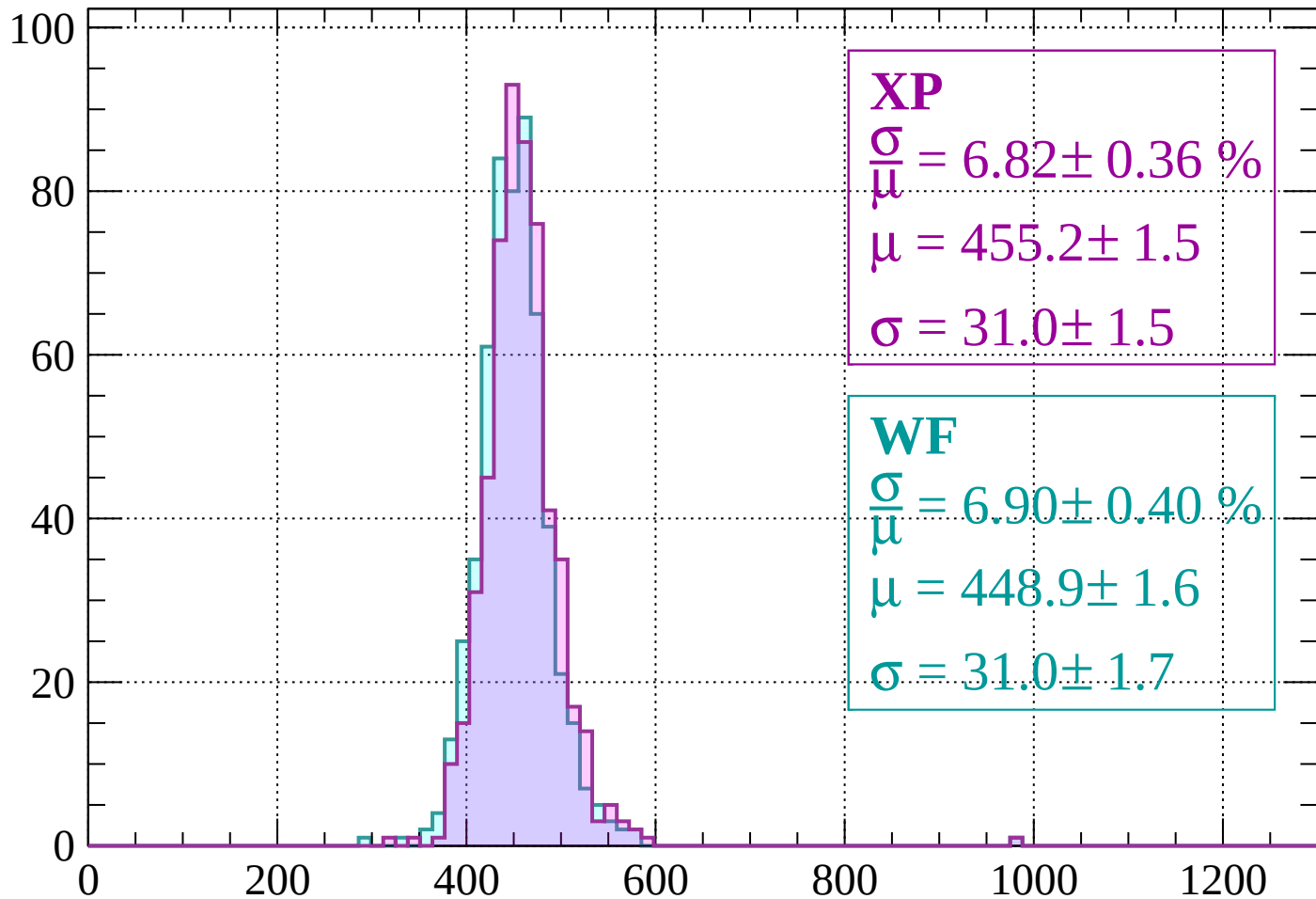
$$\mu = 445.8 \pm 1.6$$

$$\sigma = 31.4 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 6.82 \pm 0.36 \%$$

$$\mu = 455.2 \pm 1.5$$

$$\sigma = 31.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.40 \%$$

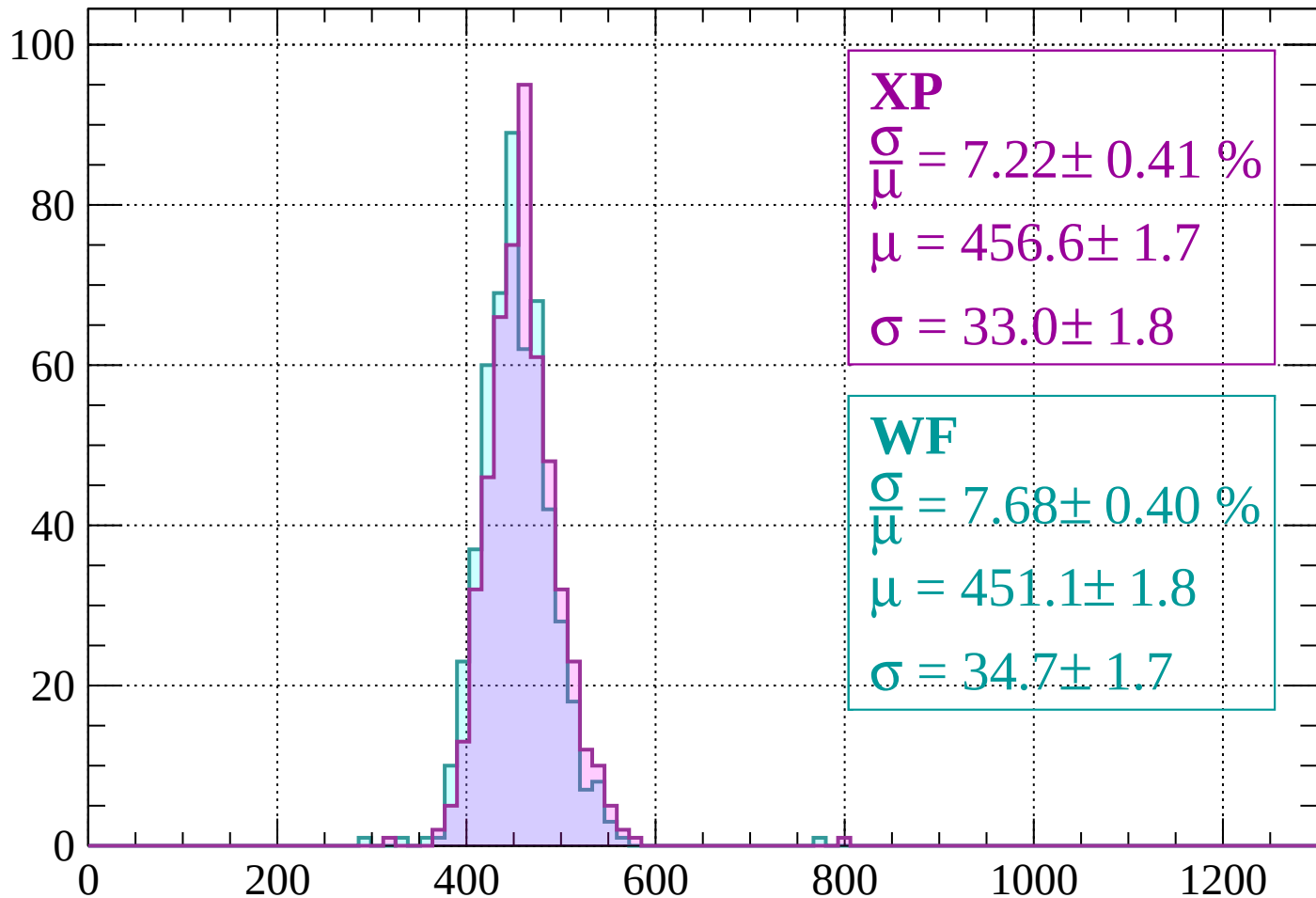
$$\mu = 448.9 \pm 1.6$$

$$\sigma = 31.0 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 7.22 \pm 0.41 \%$$

$$\mu = 456.6 \pm 1.7$$

$$\sigma = 33.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.68 \pm 0.40 \%$$

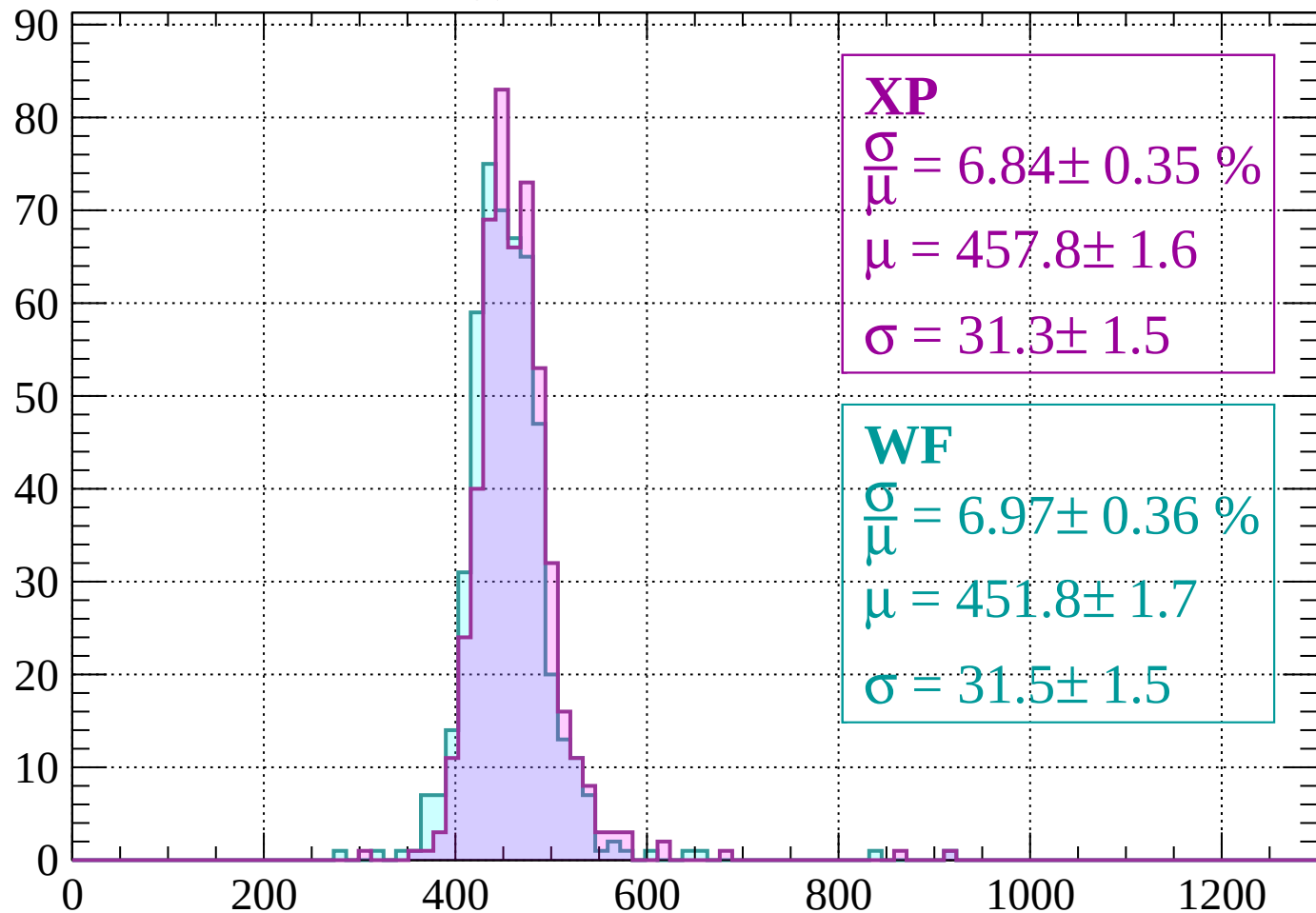
$$\mu = 451.1 \pm 1.8$$

$$\sigma = 34.7 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 6.84 \pm 0.35 \%$$

$$\mu = 457.8 \pm 1.6$$

$$\sigma = 31.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 6.97 \pm 0.36 \%$$

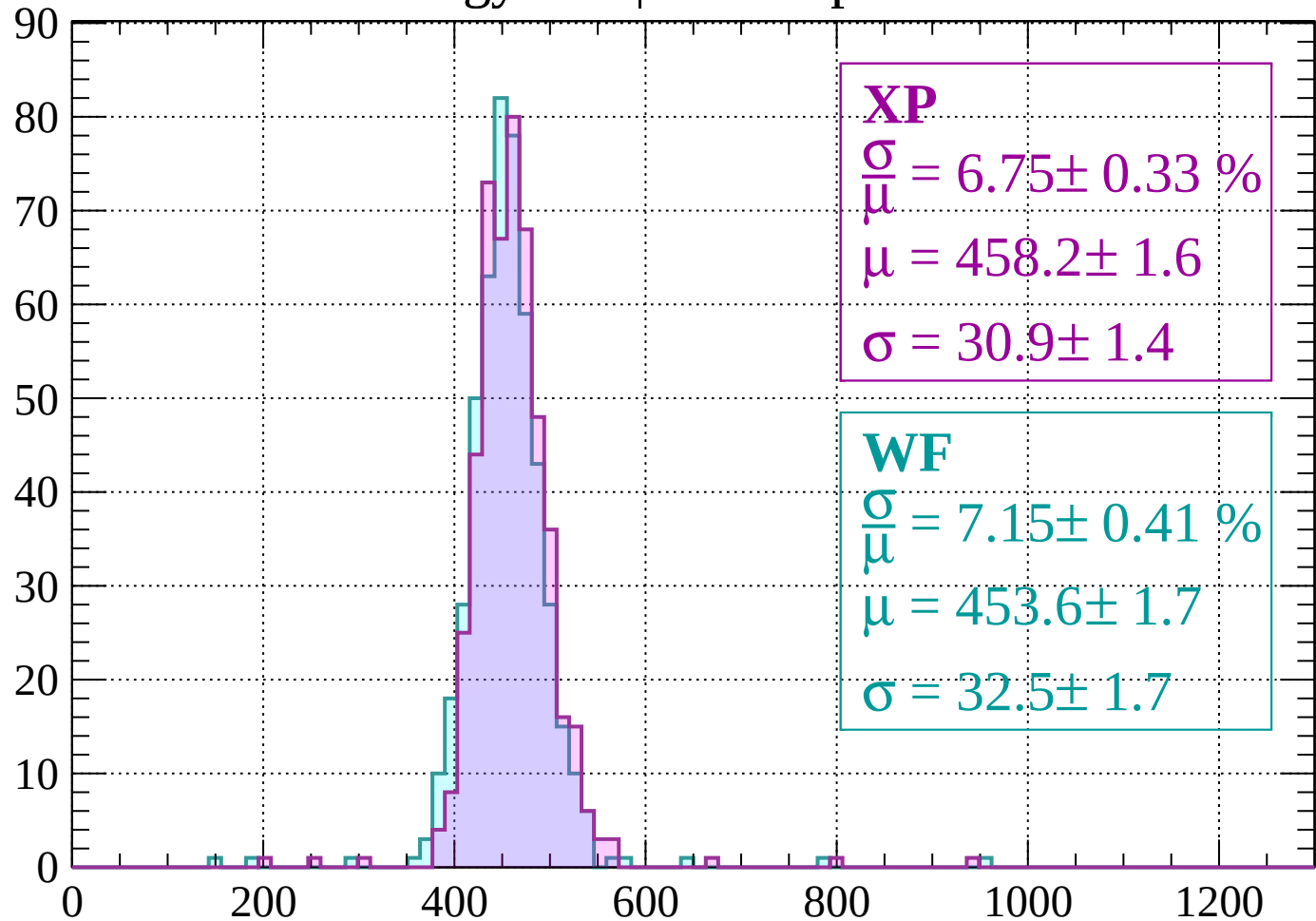
$$\mu = 451.8 \pm 1.7$$

$$\sigma = 31.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.33 \%$$

$$\mu = 458.2 \pm 1.6$$

$$\sigma = 30.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.15 \pm 0.41 \%$$

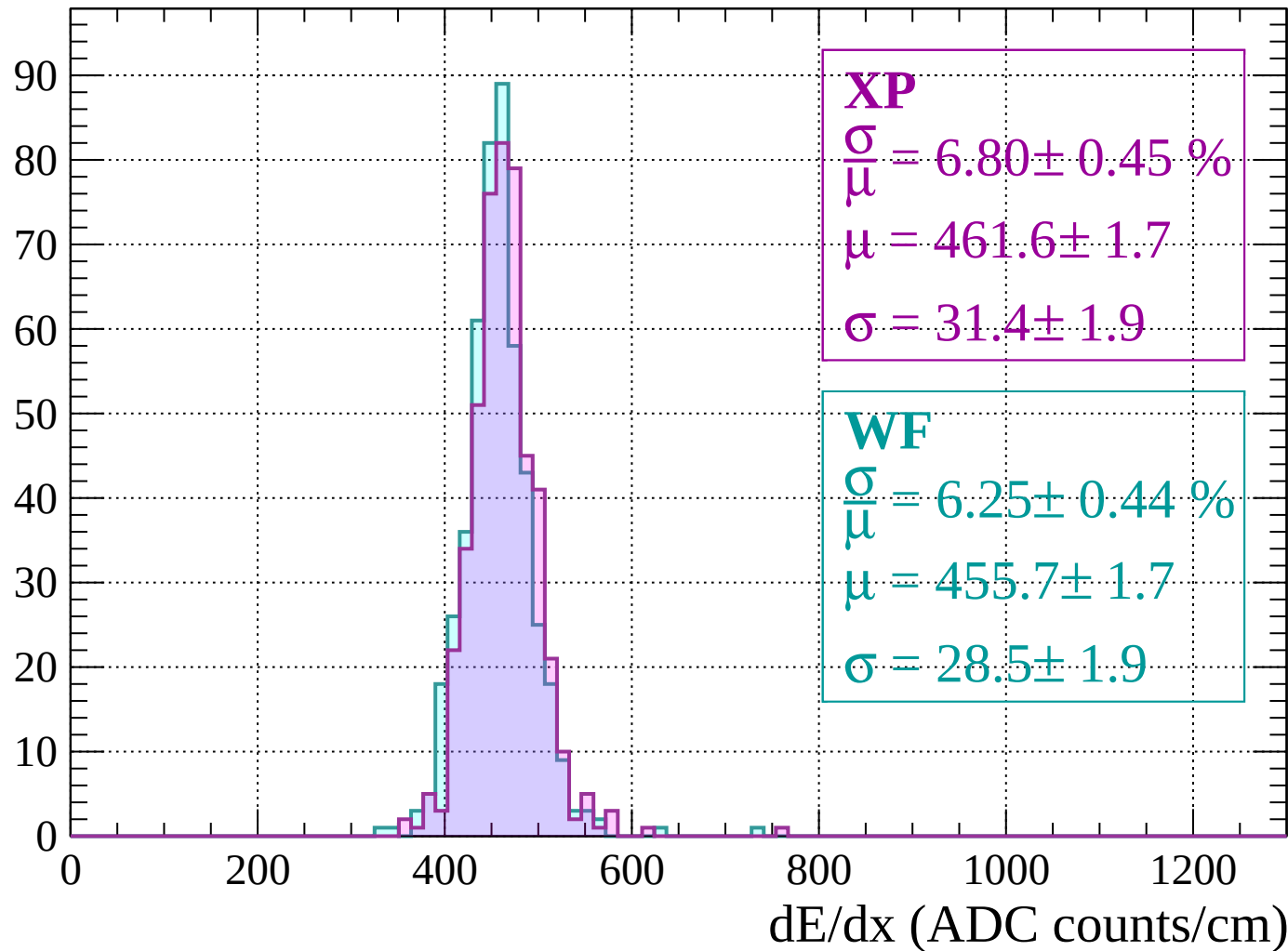
$$\mu = 453.6 \pm 1.7$$

$$\sigma = 32.5 \pm 1.7$$

dE/dx (ADC counts/cm)

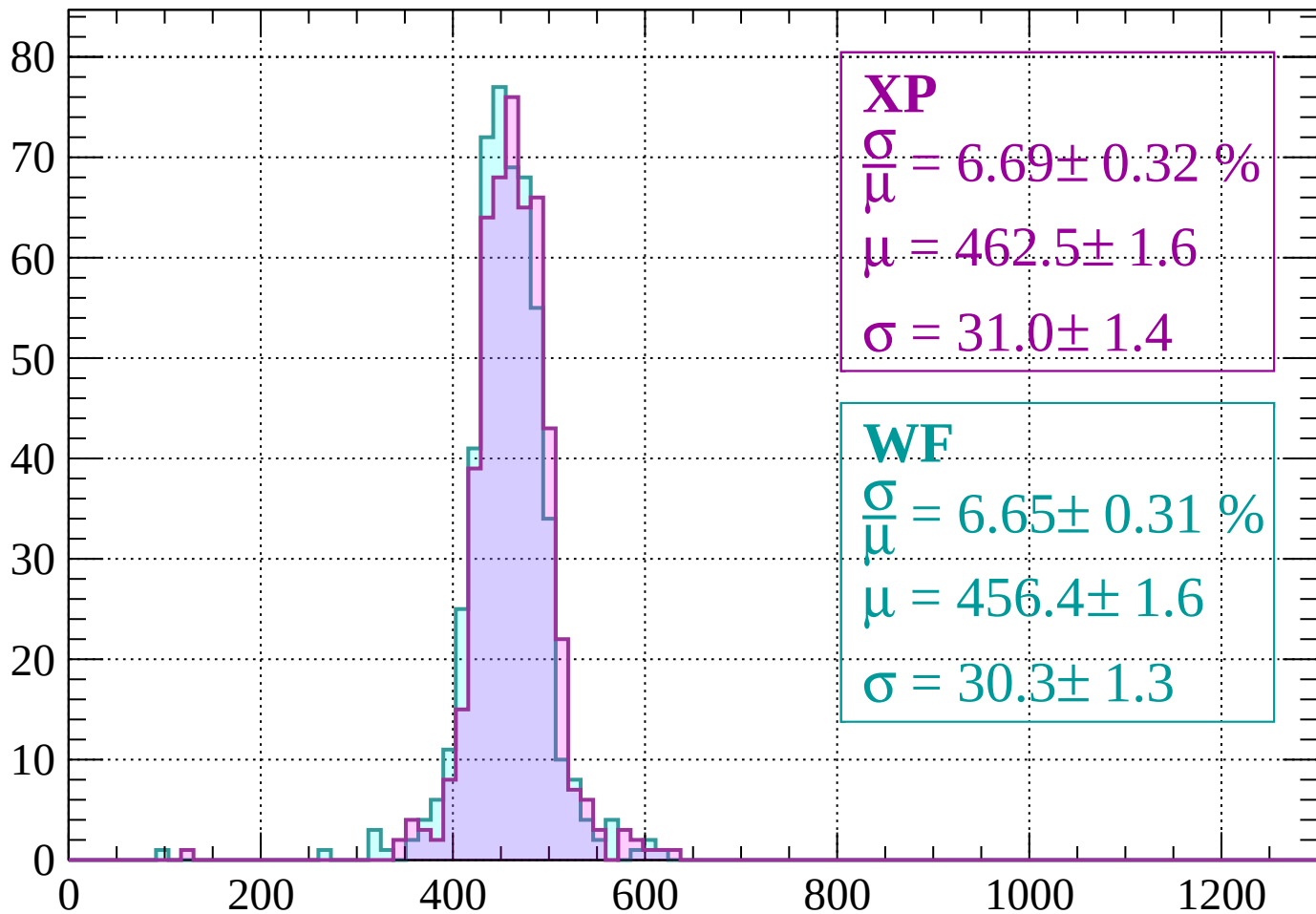
Energy loss | $1400 < p < 1440$

Count



Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 6.69 \pm 0.32 \%$$

$$\mu = 462.5 \pm 1.6$$

$$\sigma = 31.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.65 \pm 0.31 \%$$

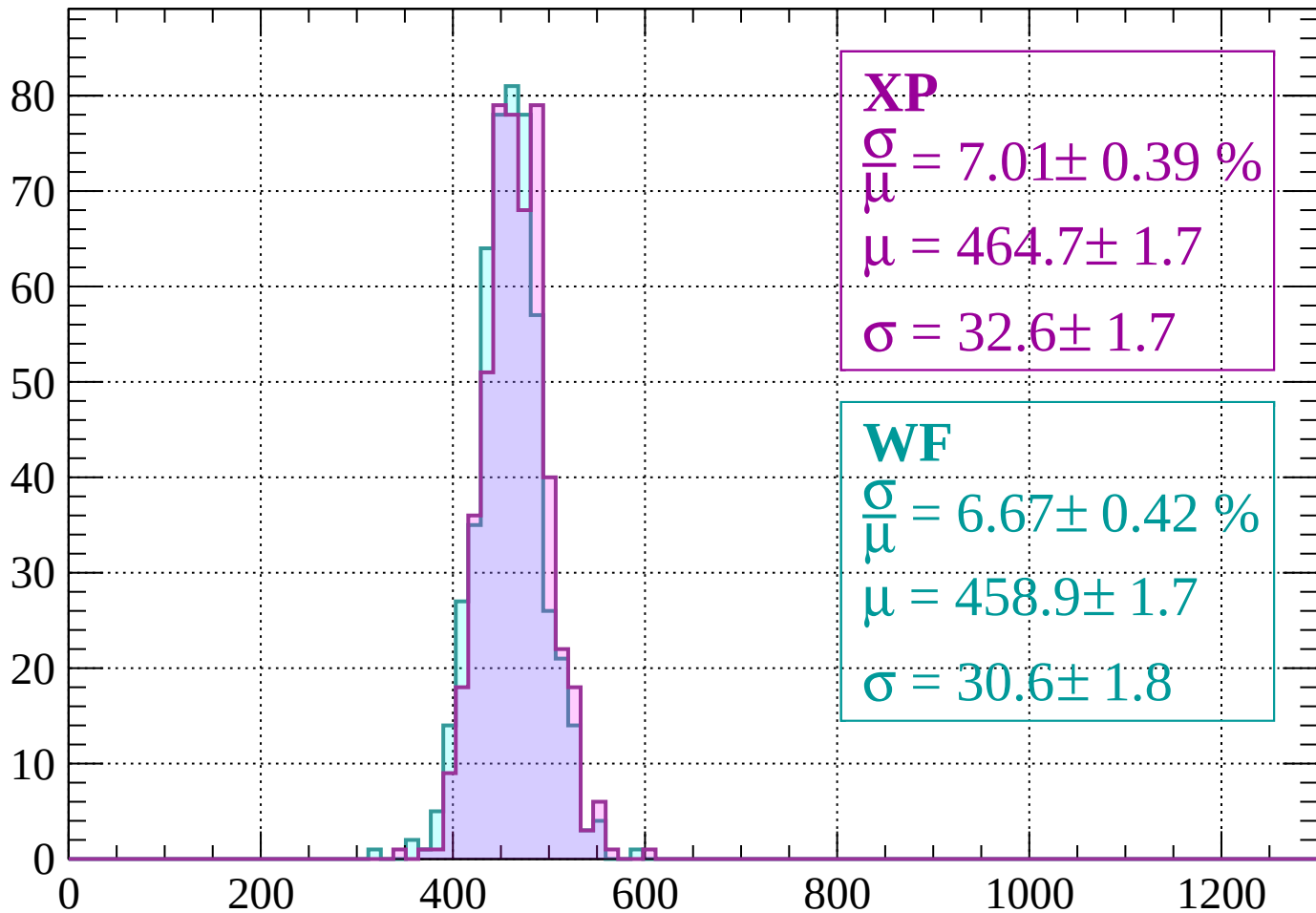
$$\mu = 456.4 \pm 1.6$$

$$\sigma = 30.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.01 \pm 0.39 \%$$

$$\mu = 464.7 \pm 1.7$$

$$\sigma = 32.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.67 \pm 0.42 \%$$

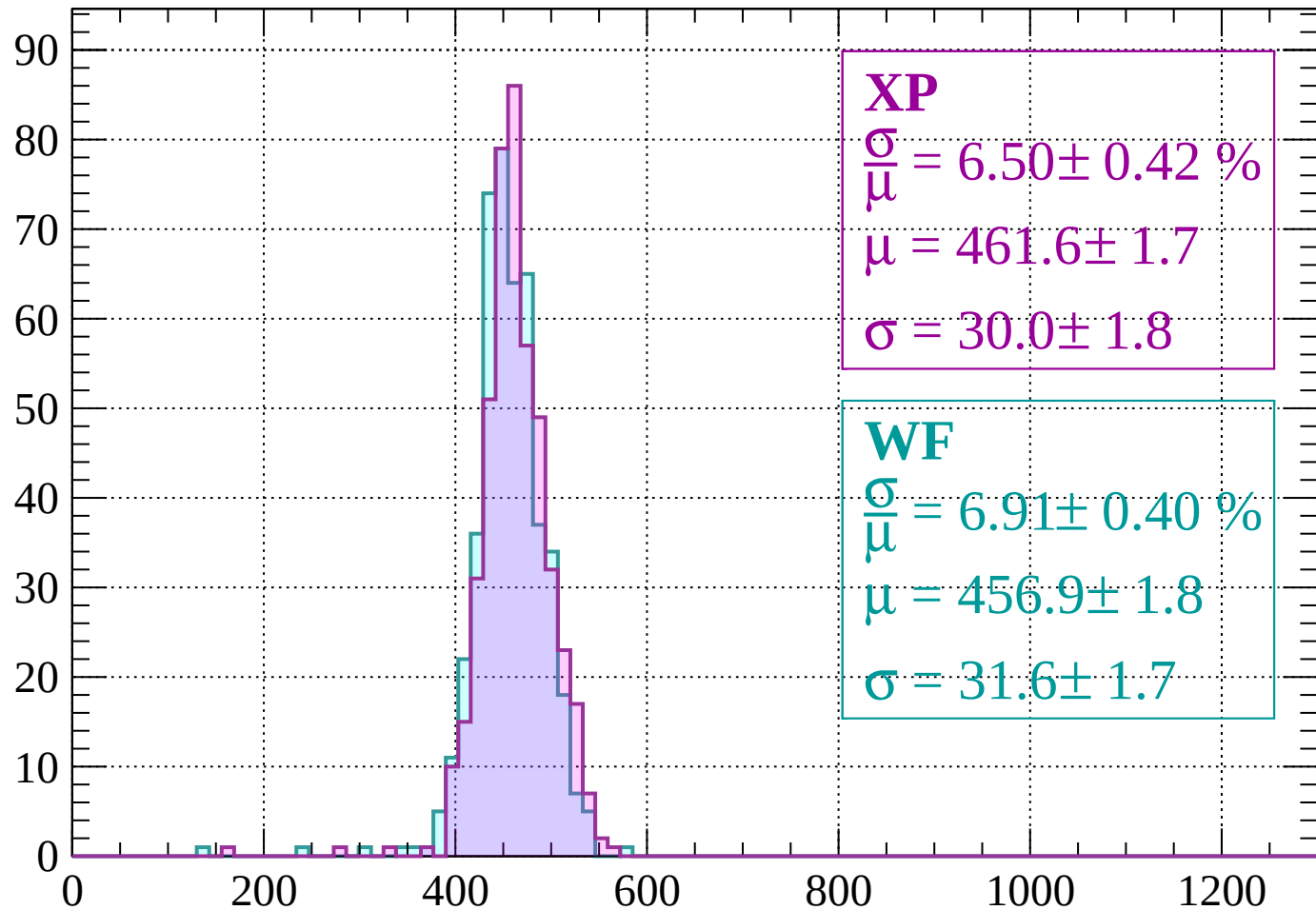
$$\mu = 458.9 \pm 1.7$$

$$\sigma = 30.6 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 6.50 \pm 0.42 \%$$

$$\mu = 461.6 \pm 1.7$$

$$\sigma = 30.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 6.91 \pm 0.40 \%$$

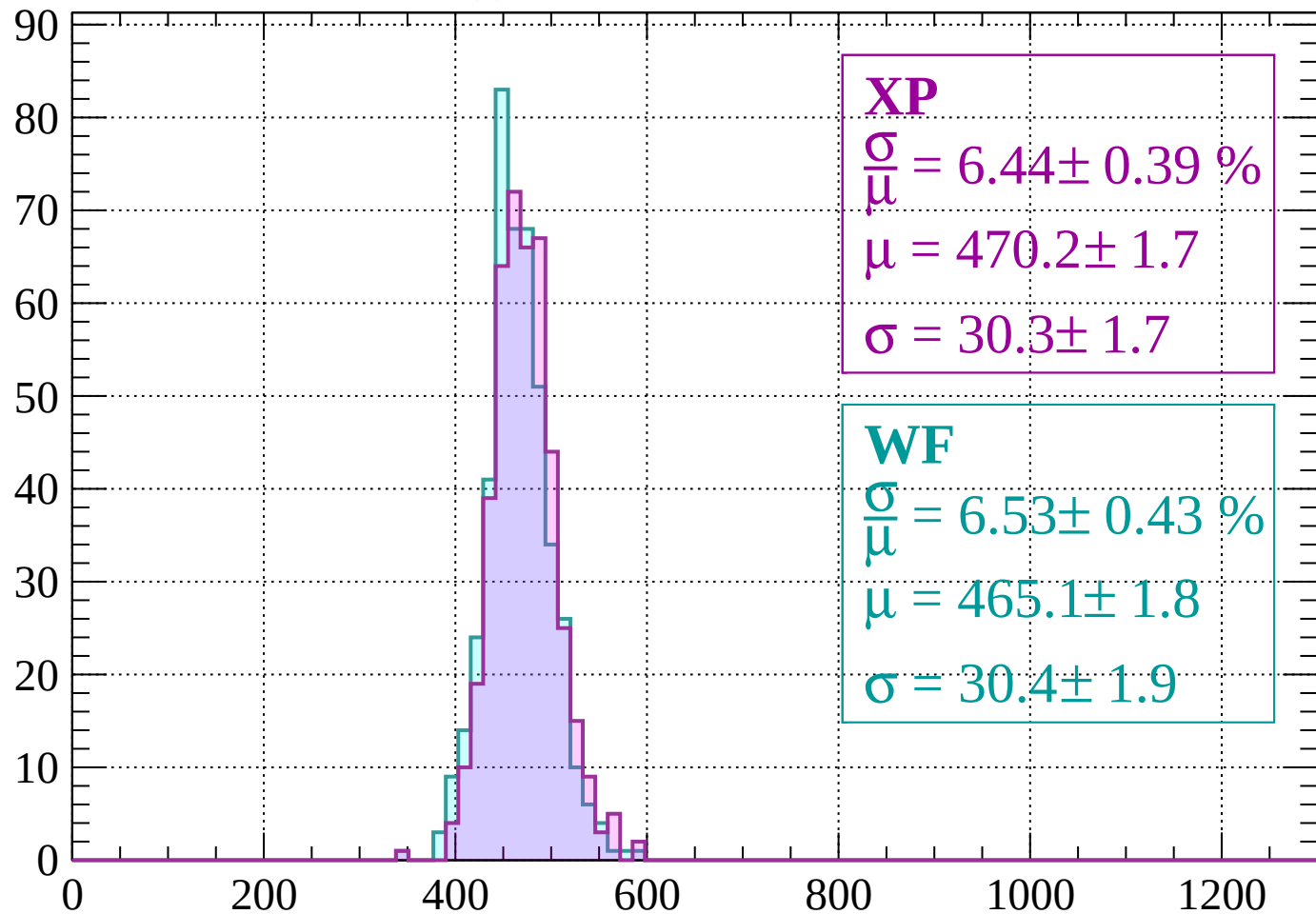
$$\mu = 456.9 \pm 1.8$$

$$\sigma = 31.6 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 6.44 \pm 0.39 \%$$

$$\mu = 470.2 \pm 1.7$$

$$\sigma = 30.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.53 \pm 0.43 \%$$

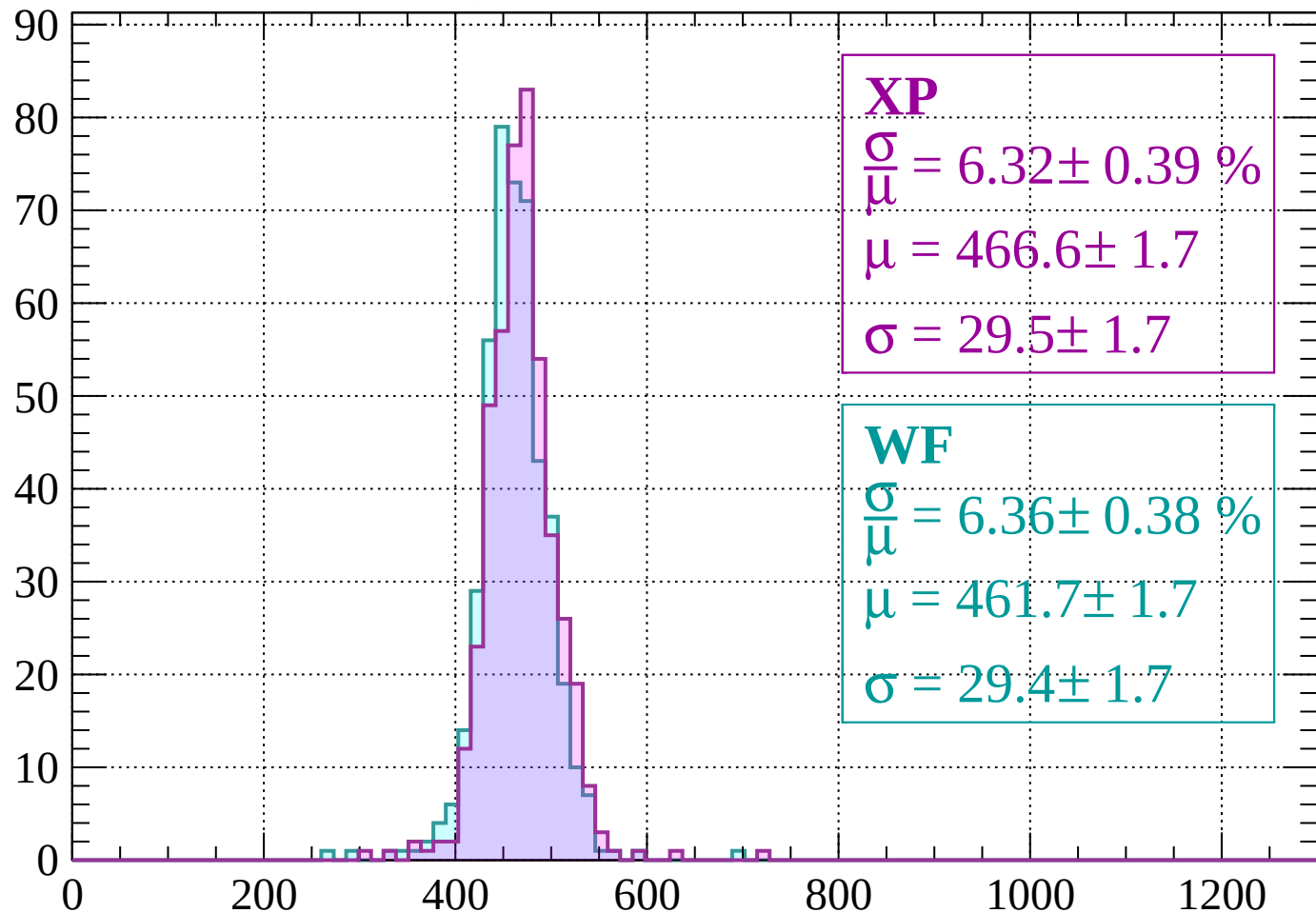
$$\mu = 465.1 \pm 1.8$$

$$\sigma = 30.4 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count



XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.39 \%$$

$$\mu = 466.6 \pm 1.7$$

$$\sigma = 29.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.36 \pm 0.38 \%$$

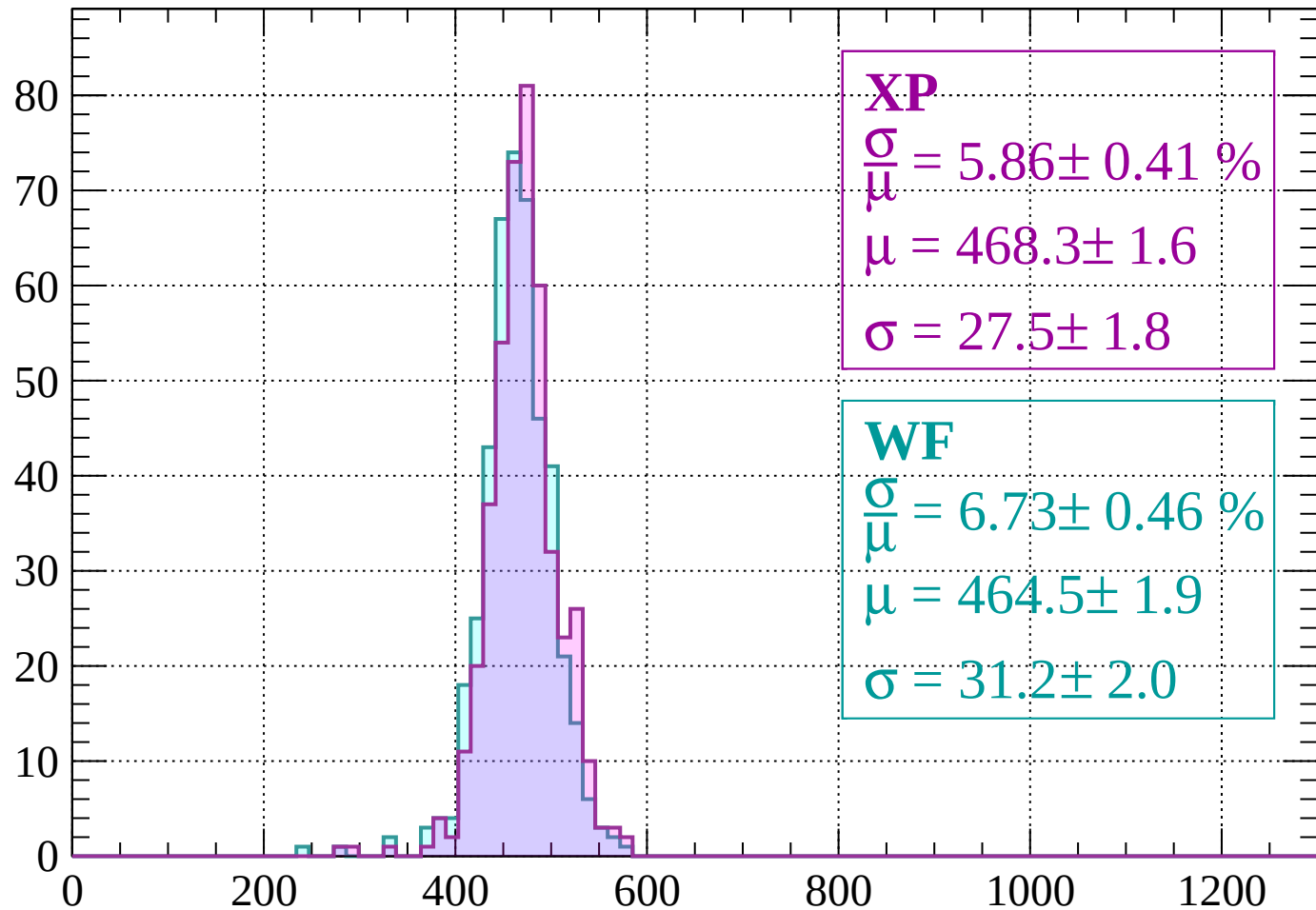
$$\mu = 461.7 \pm 1.7$$

$$\sigma = 29.4 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 5.86 \pm 0.41 \%$$

$$\mu = 468.3 \pm 1.6$$

$$\sigma = 27.5 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 6.73 \pm 0.46 \%$$

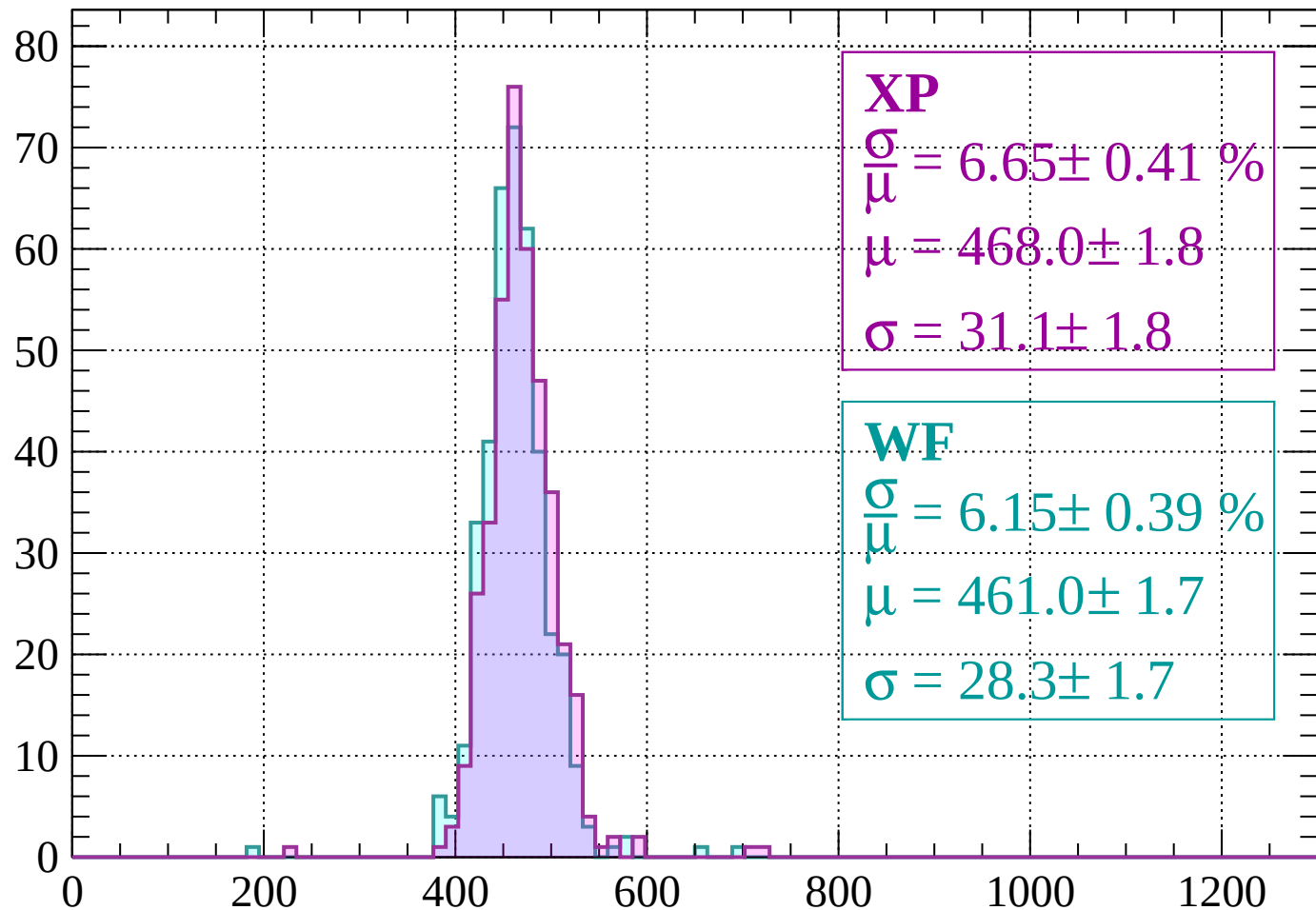
$$\mu = 464.5 \pm 1.9$$

$$\sigma = 31.2 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 6.65 \pm 0.41 \%$$

$$\mu = 468.0 \pm 1.8$$

$$\sigma = 31.1 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 6.15 \pm 0.39 \%$$

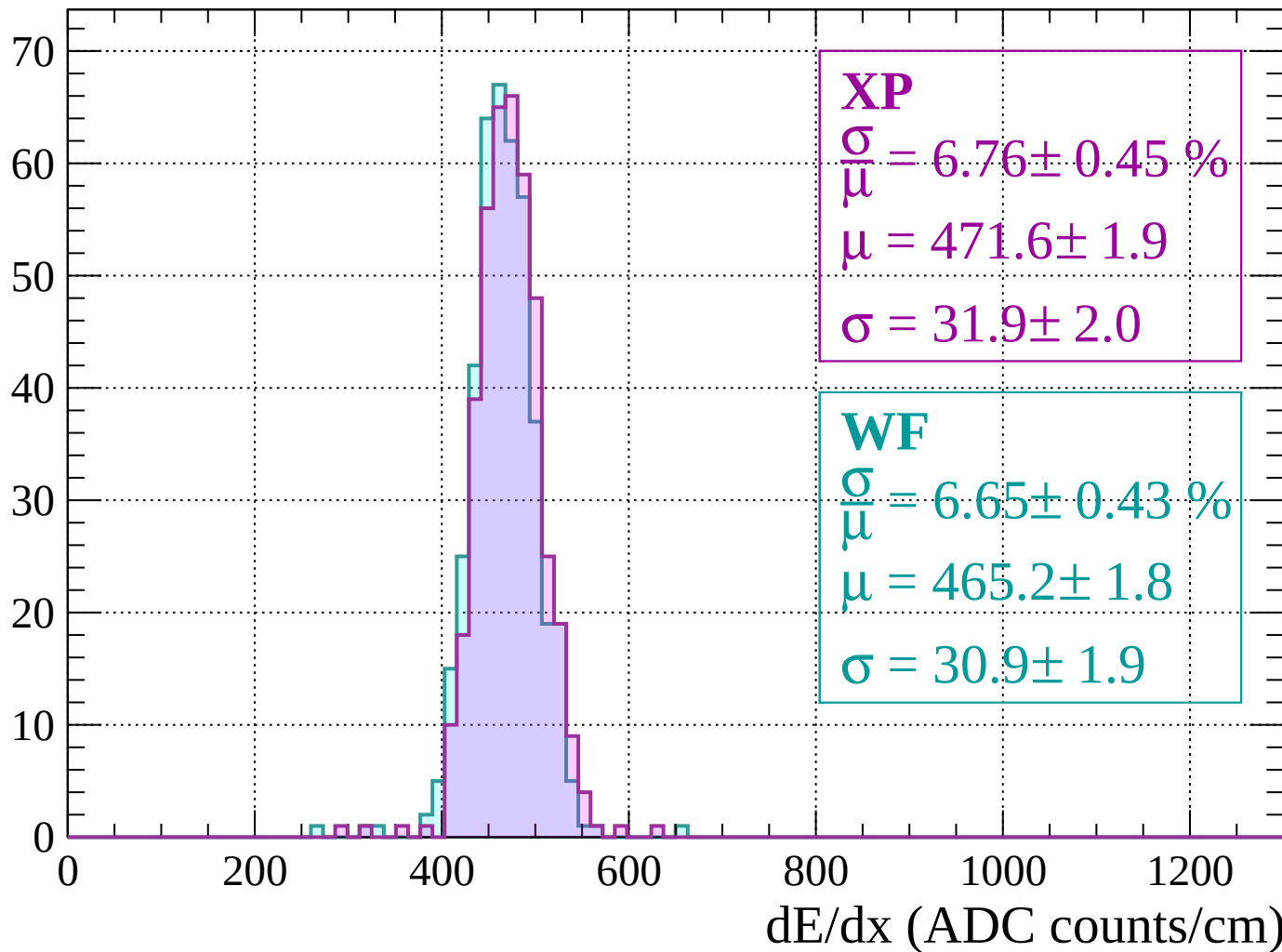
$$\mu = 461.0 \pm 1.7$$

$$\sigma = 28.3 \pm 1.7$$

dE/dx (ADC counts/cm)

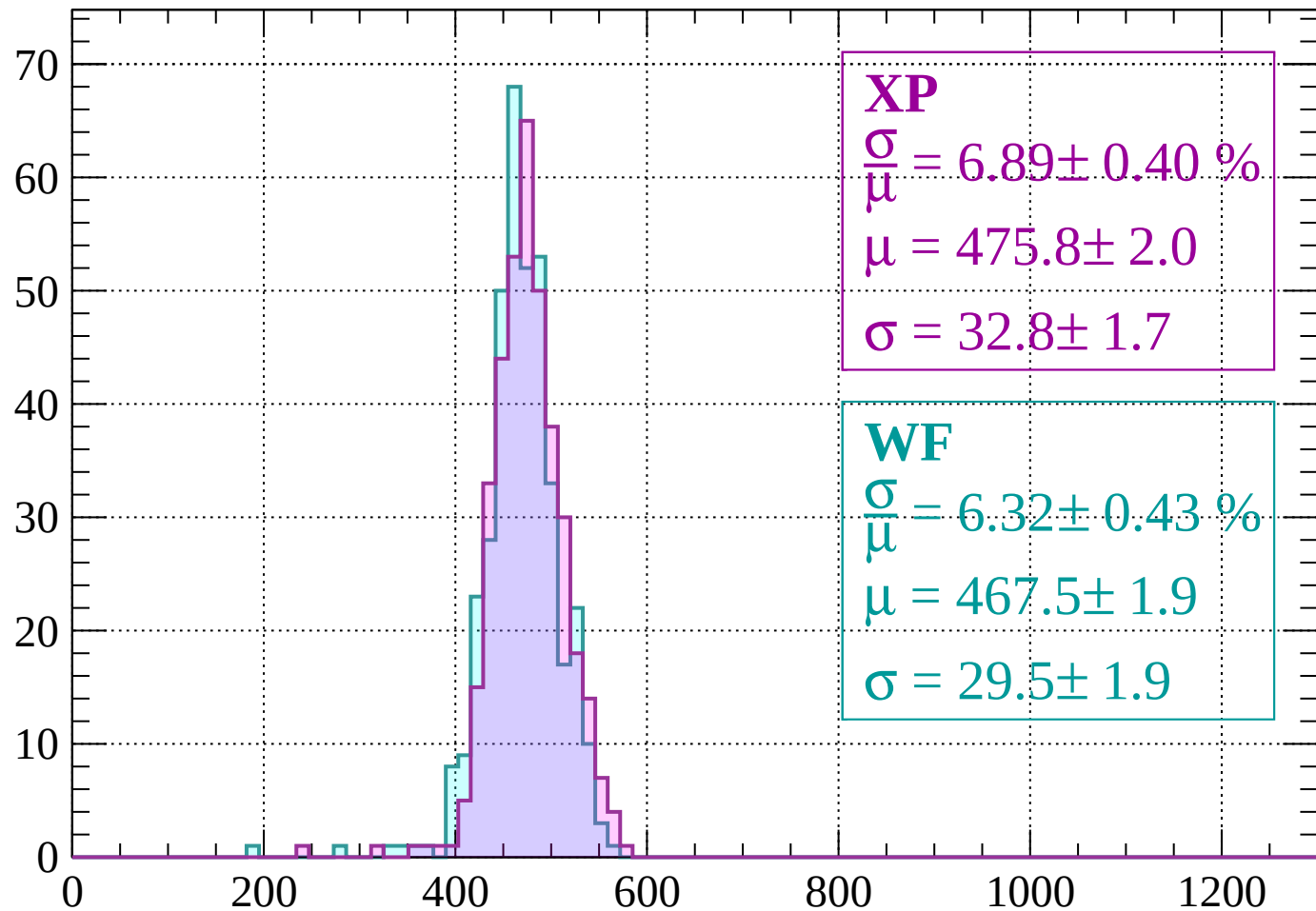
Energy loss | $1720 < p < 1760$

Count



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 6.89 \pm 0.40 \%$$

$$\mu = 475.8 \pm 2.0$$

$$\sigma = 32.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.32 \pm 0.43 \%$$

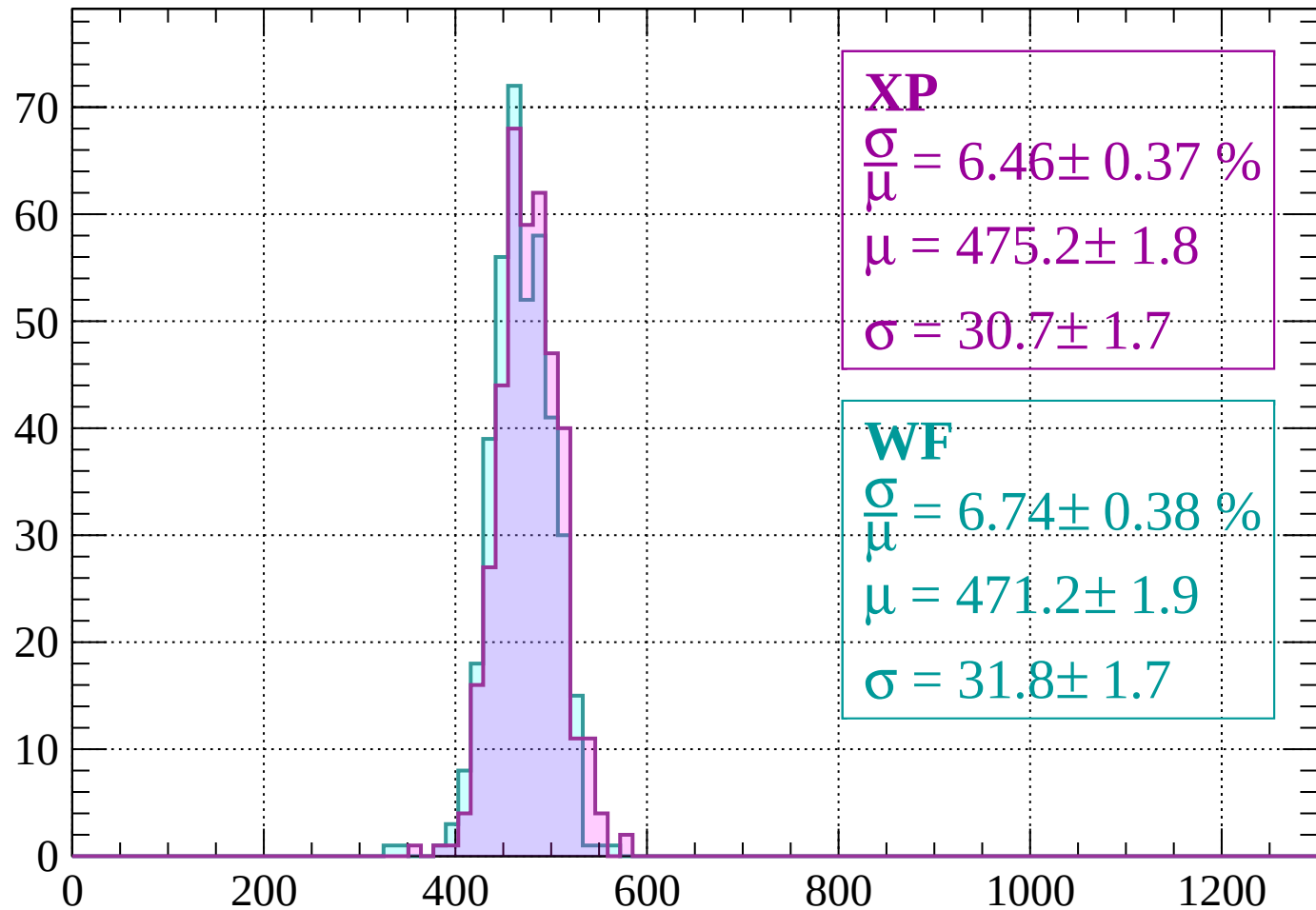
$$\mu = 467.5 \pm 1.9$$

$$\sigma = 29.5 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.46 \pm 0.37 \%$$

$$\mu = 475.2 \pm 1.8$$

$$\sigma = 30.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.38 \%$$

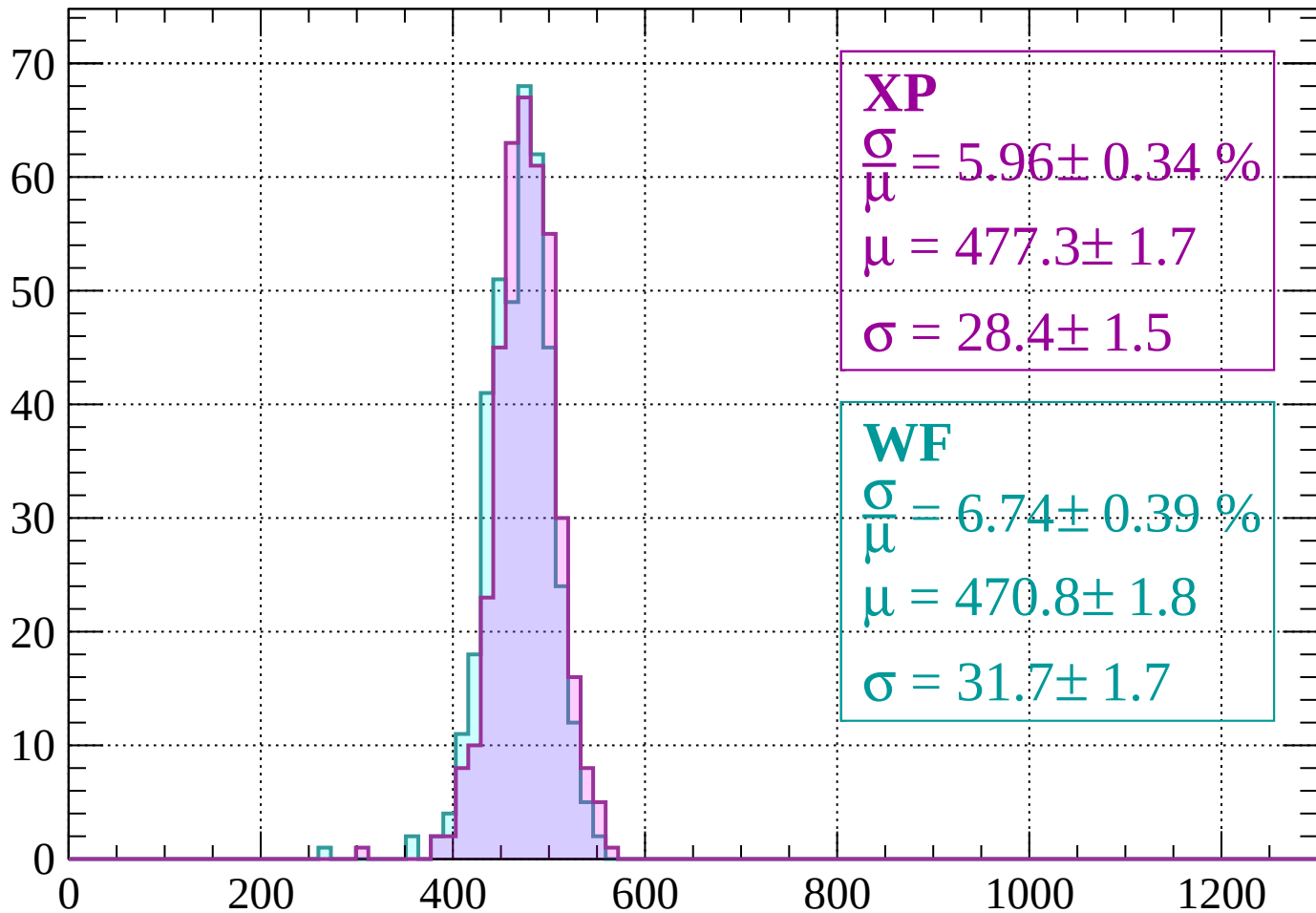
$$\mu = 471.2 \pm 1.9$$

$$\sigma = 31.8 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 5.96 \pm 0.34 \%$$

$$\mu = 477.3 \pm 1.7$$

$$\sigma = 28.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.39 \%$$

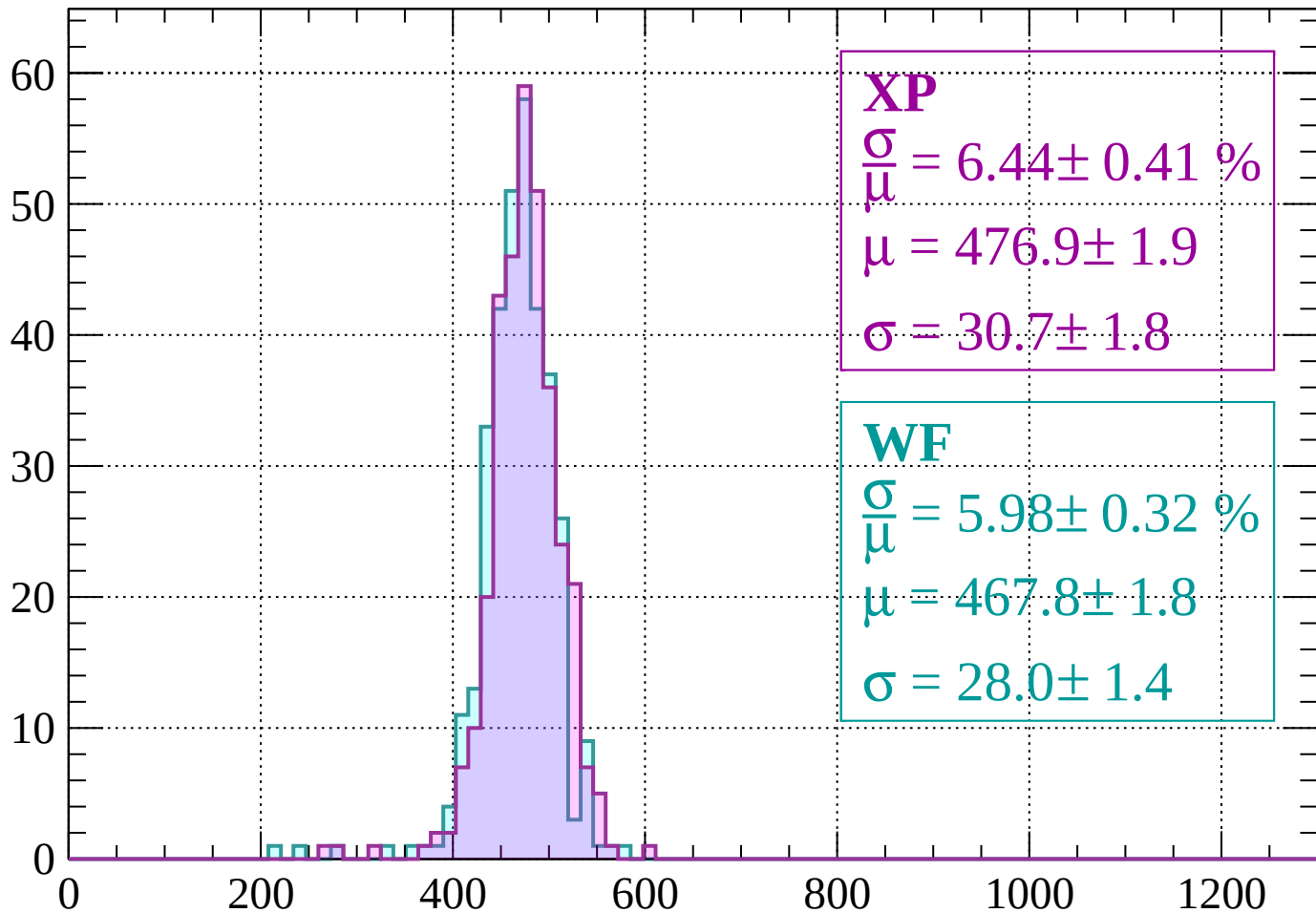
$$\mu = 470.8 \pm 1.8$$

$$\sigma = 31.7 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 6.44 \pm 0.41 \%$$

$$\mu = 476.9 \pm 1.9$$

$$\sigma = 30.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 5.98 \pm 0.32 \%$$

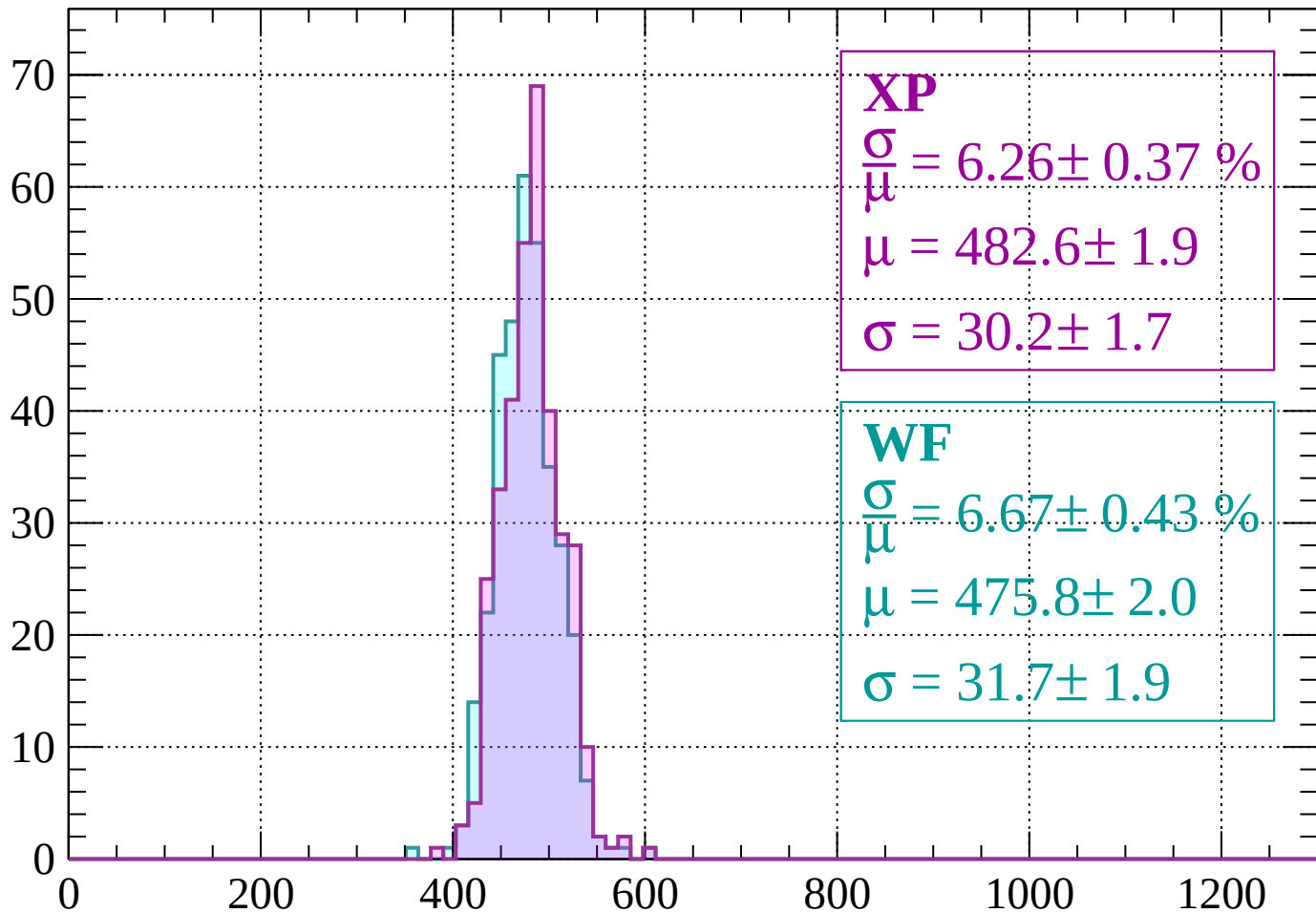
$$\mu = 467.8 \pm 1.8$$

$$\sigma = 28.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.37 \%$$

$$\mu = 482.6 \pm 1.9$$

$$\sigma = 30.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.67 \pm 0.43 \%$$

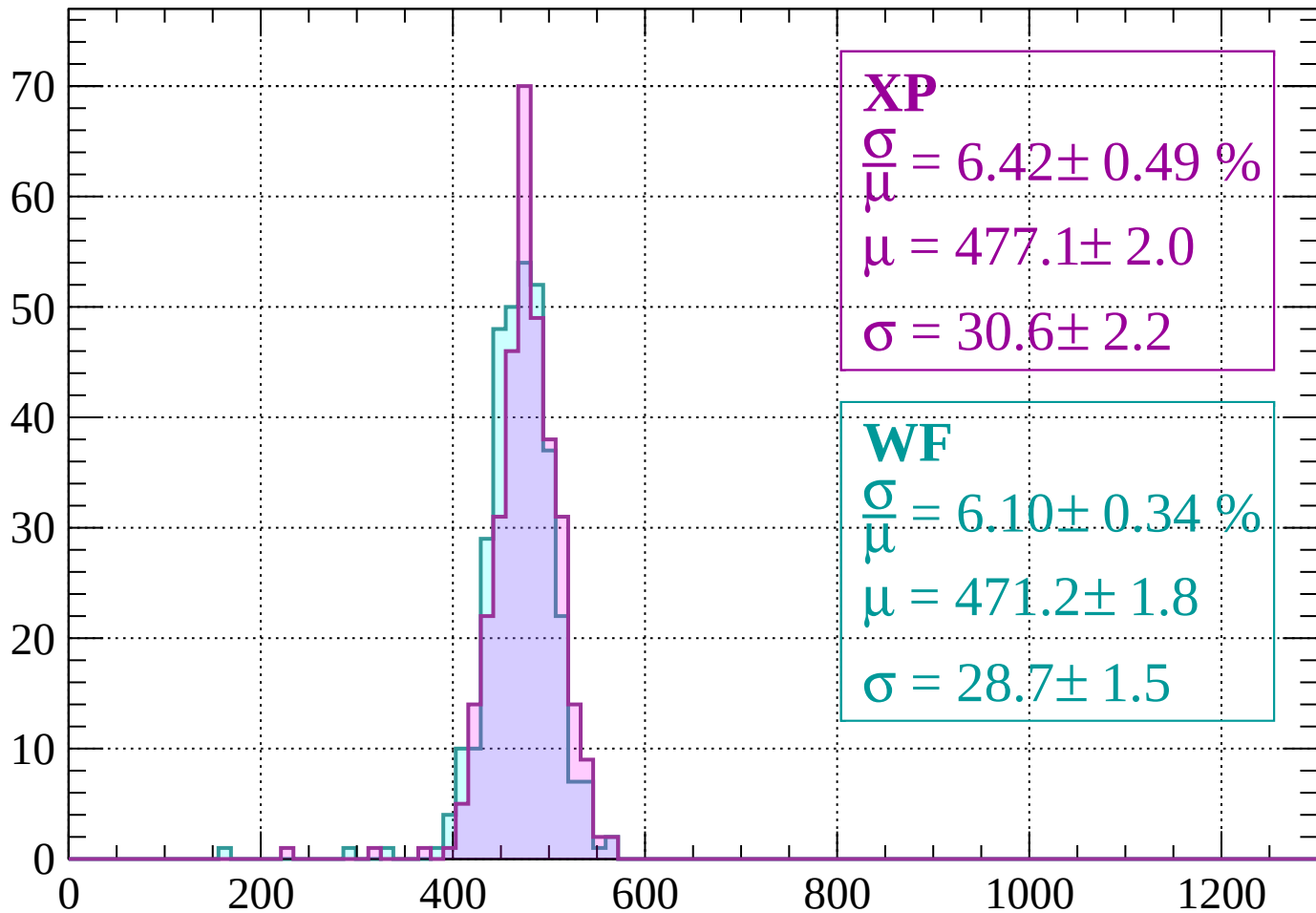
$$\mu = 475.8 \pm 2.0$$

$$\sigma = 31.7 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1960 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 6.42 \pm 0.49 \%$$

$$\mu = 477.1 \pm 2.0$$

$$\sigma = 30.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 6.10 \pm 0.34 \%$$

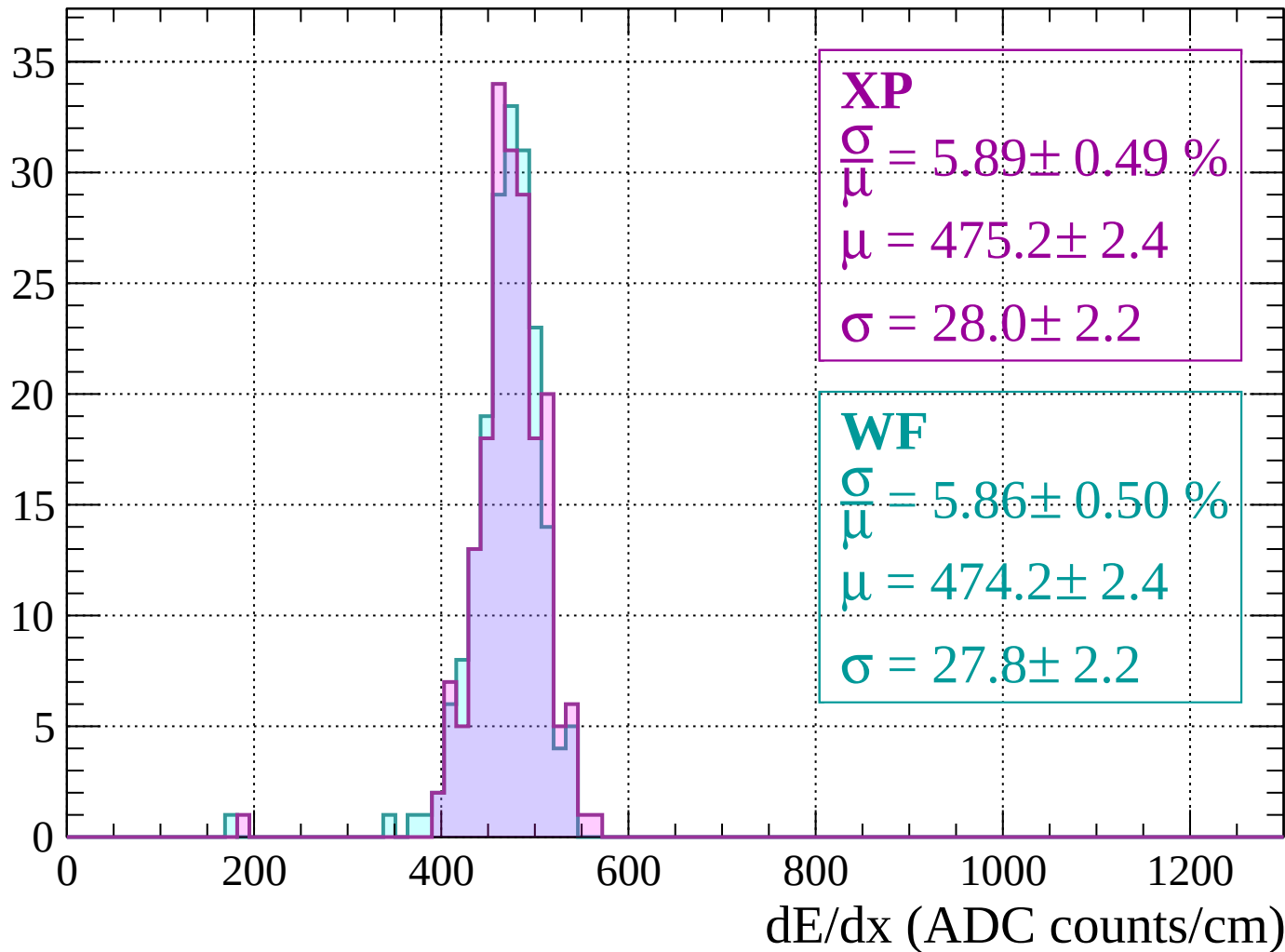
$$\mu = 471.2 \pm 1.8$$

$$\sigma = 28.7 \pm 1.5$$

dE/dx (ADC counts/cm)

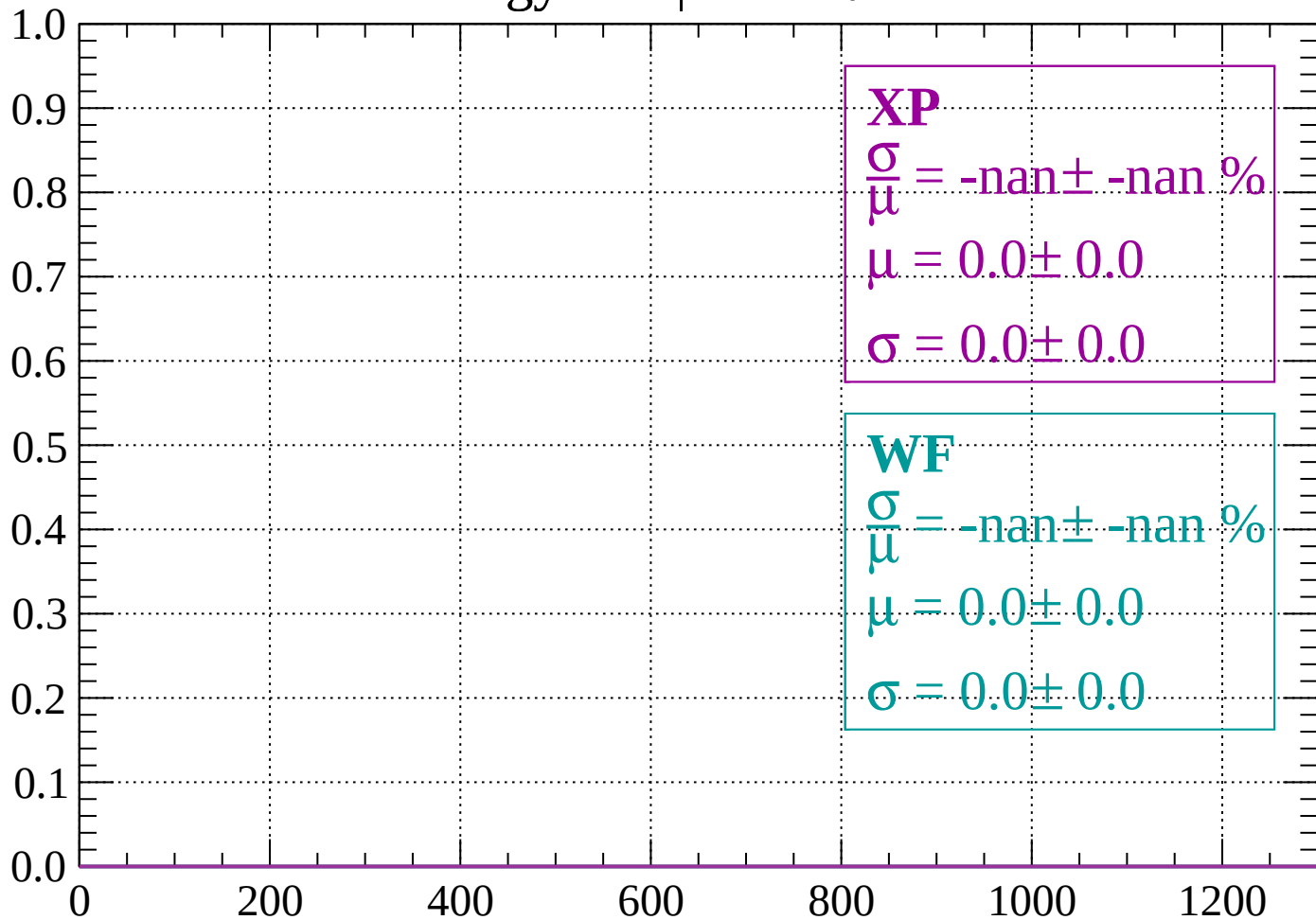
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

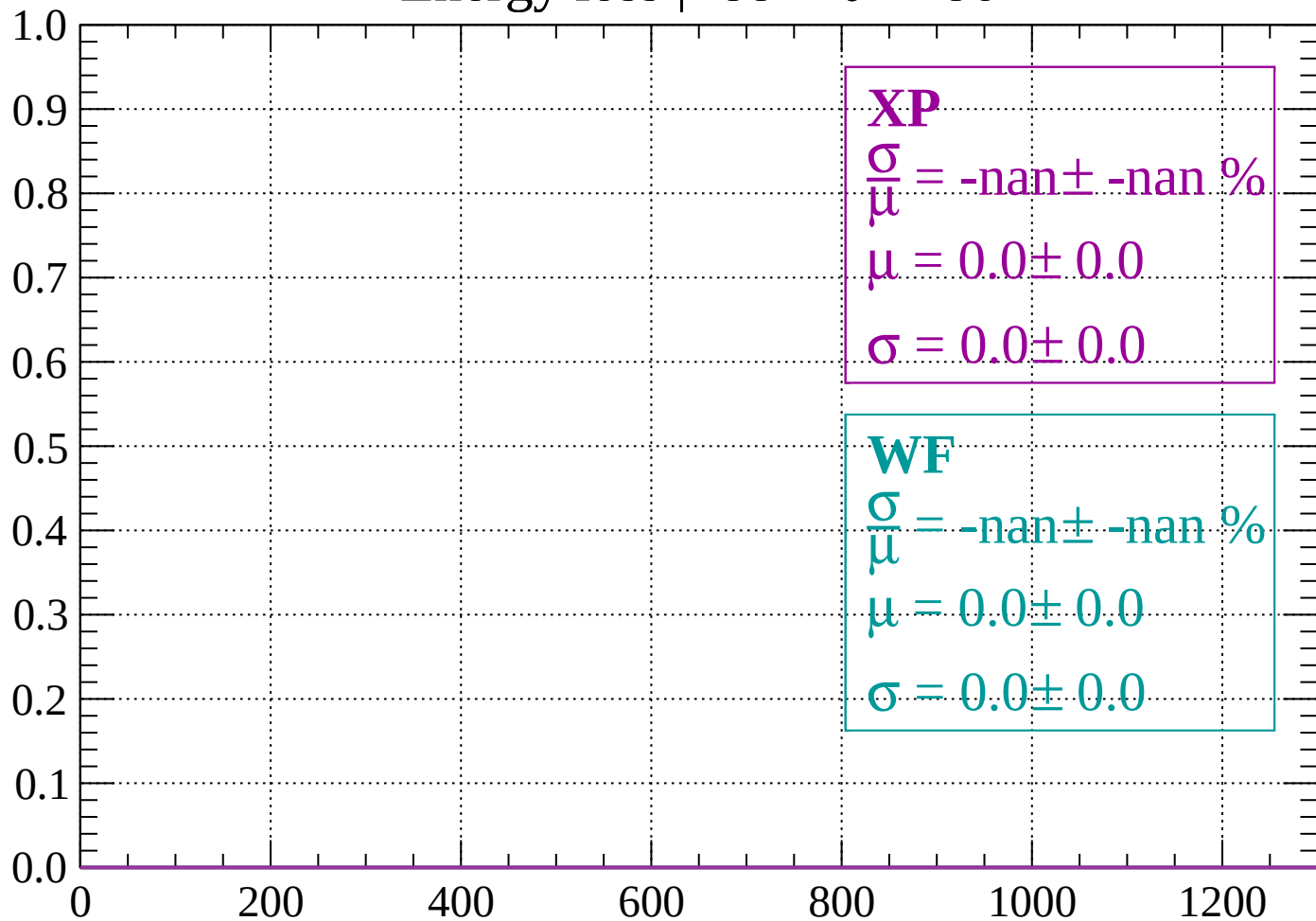
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

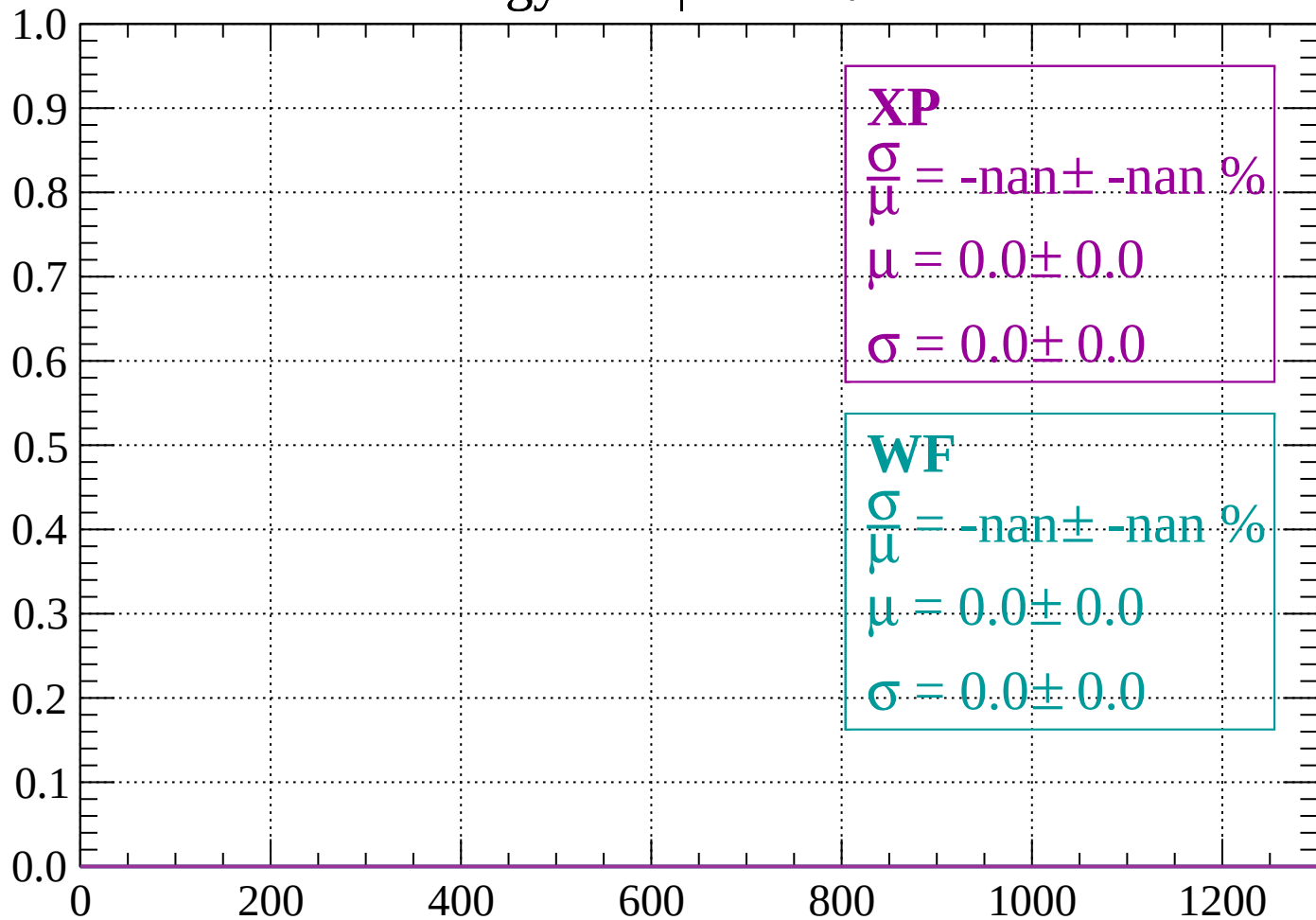
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

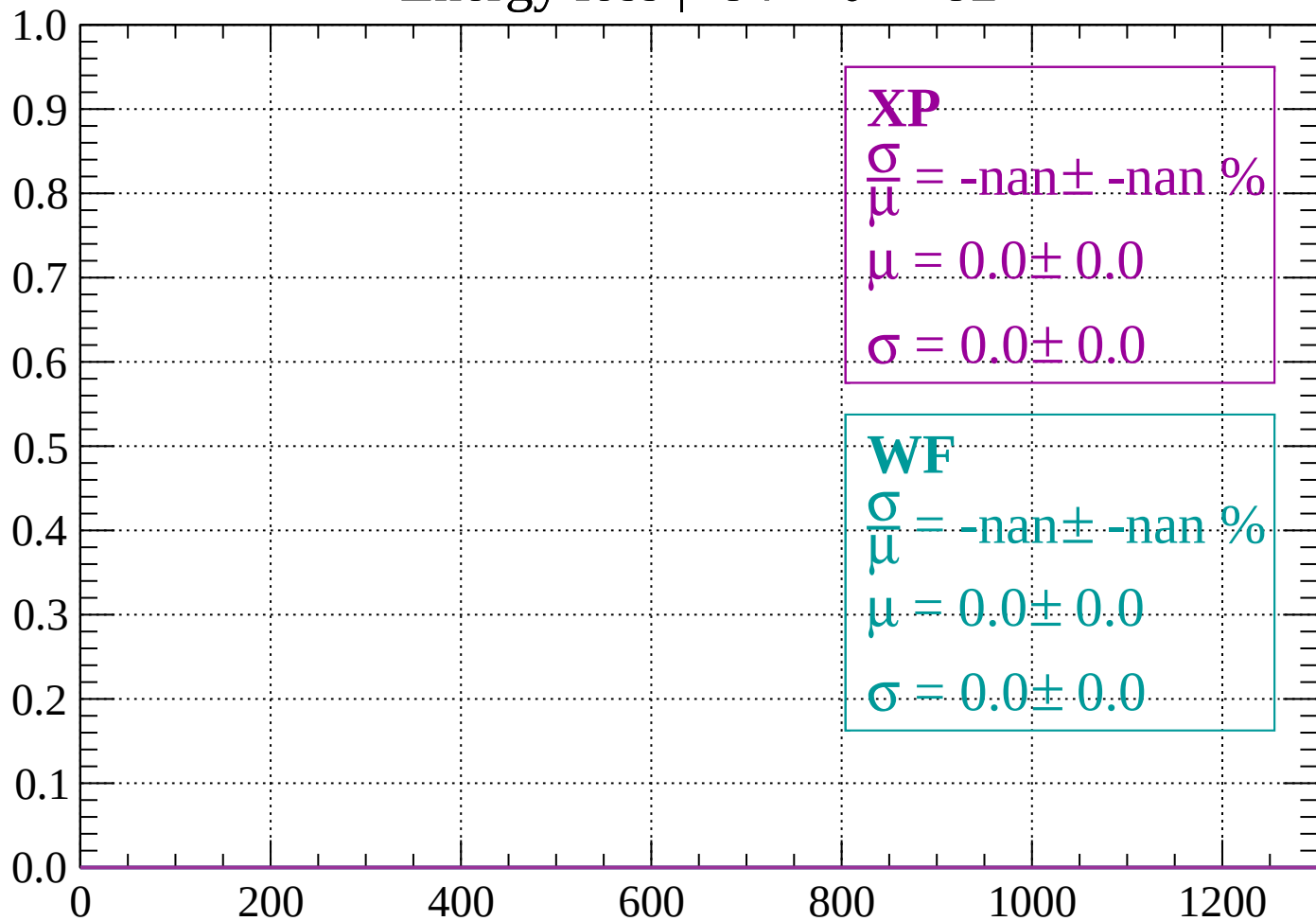
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

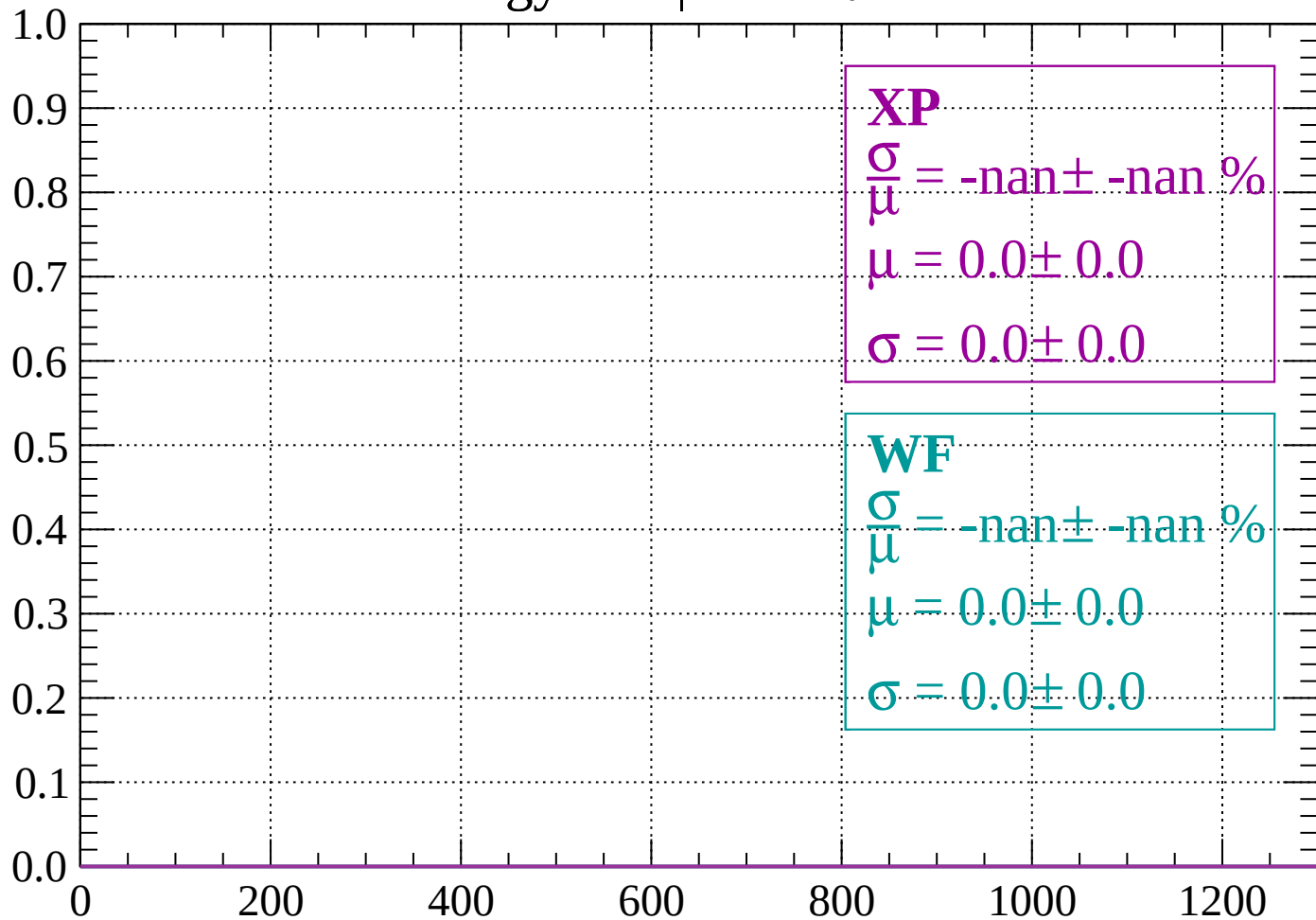
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

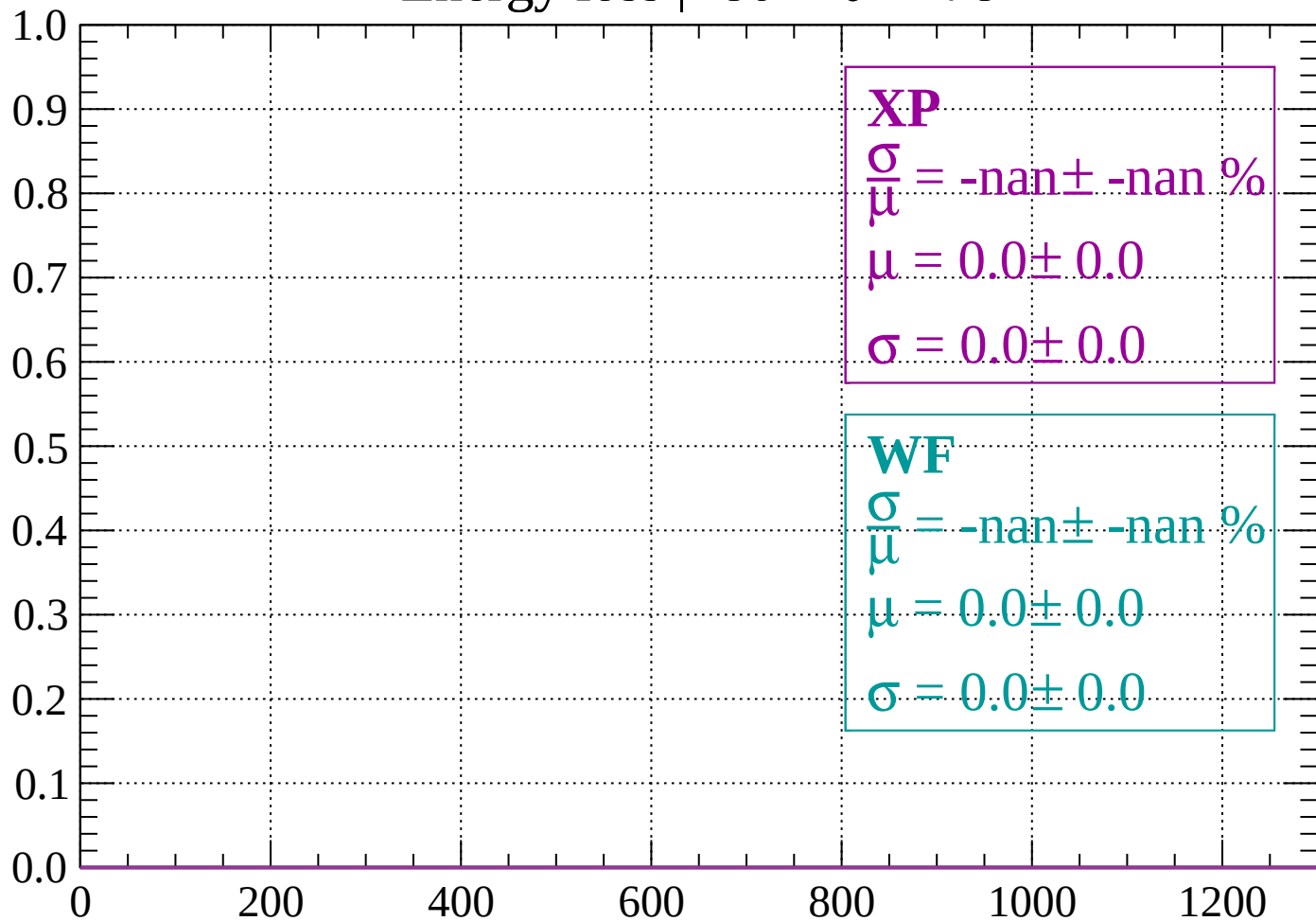
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

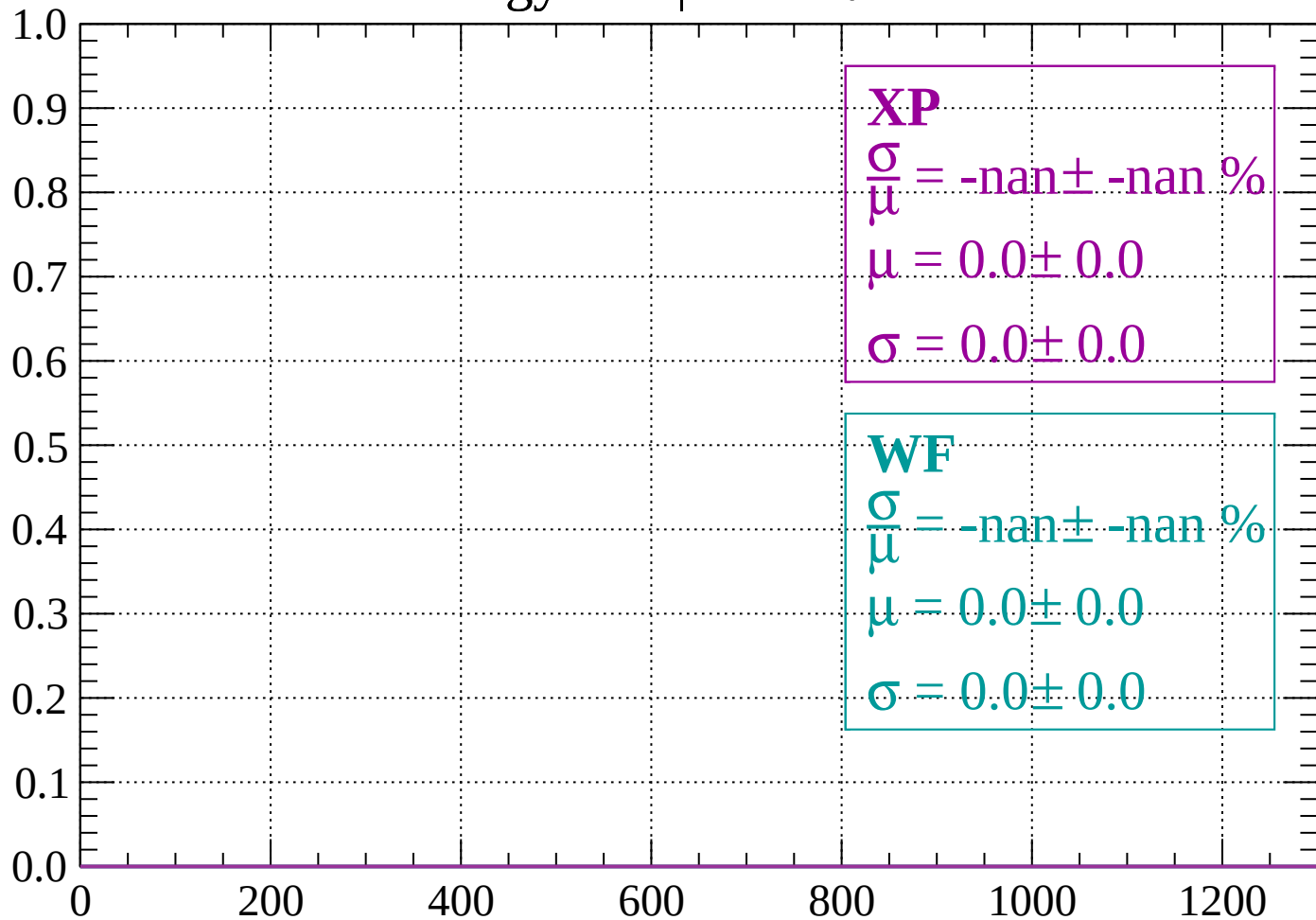
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

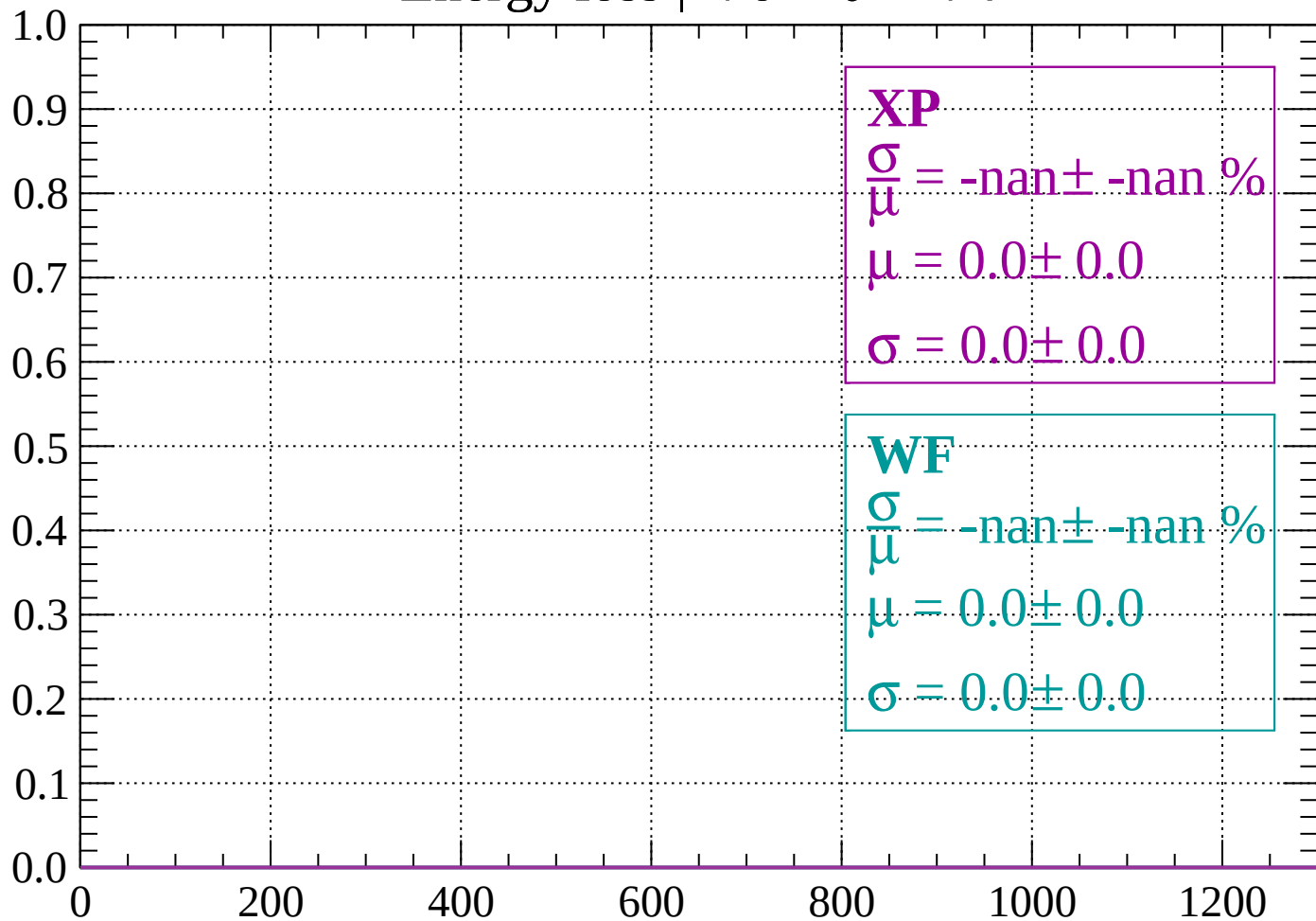
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

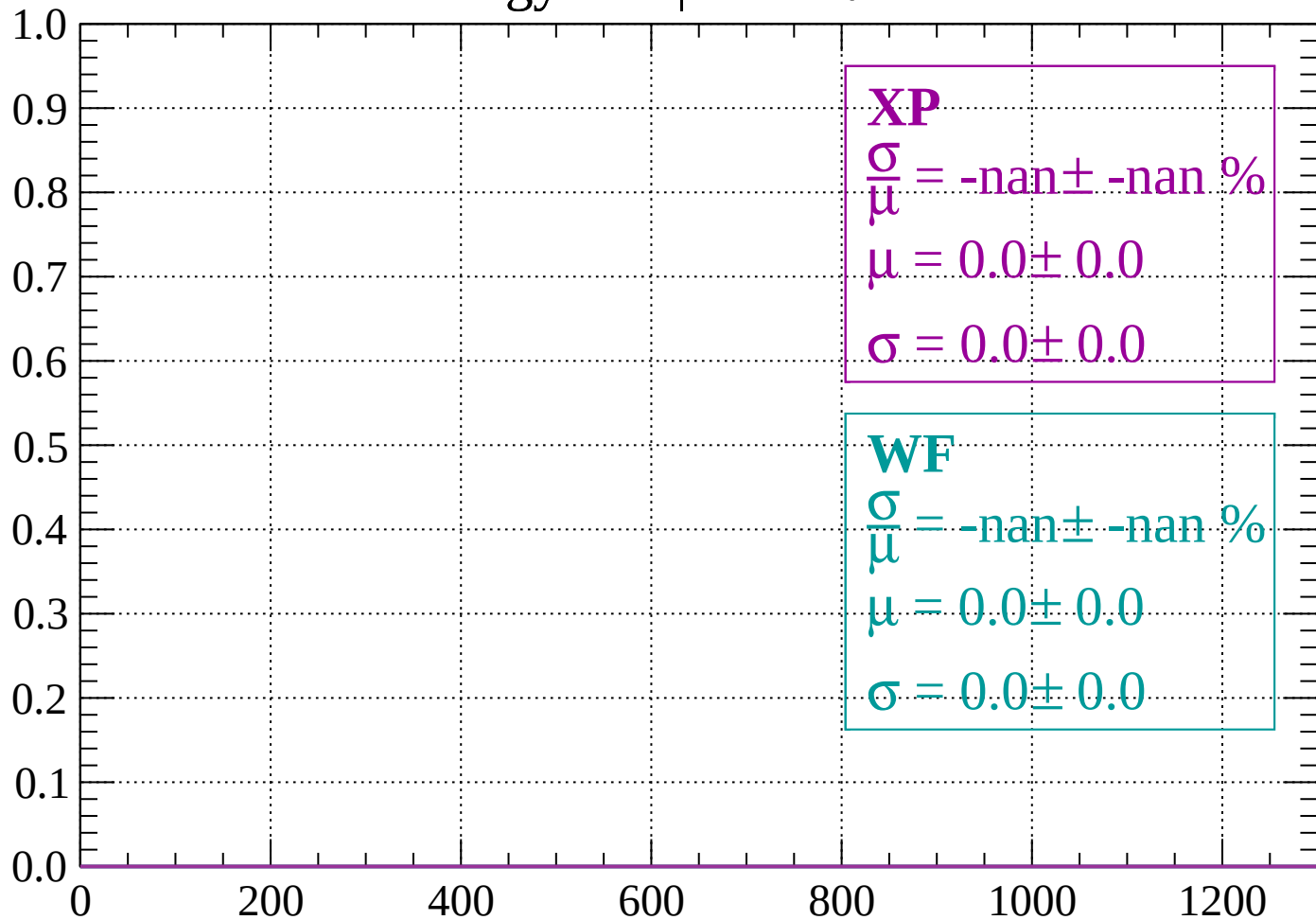
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

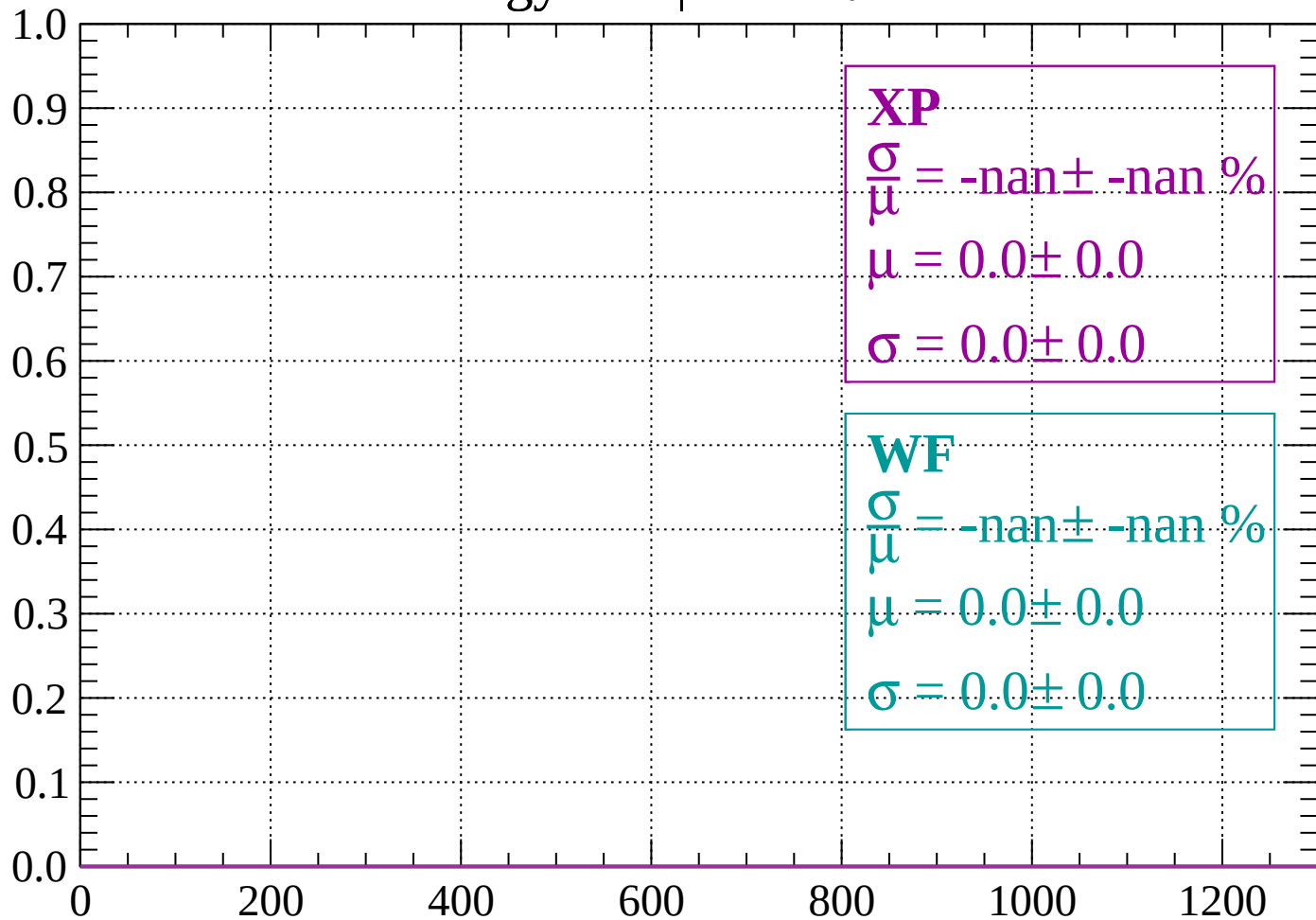
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

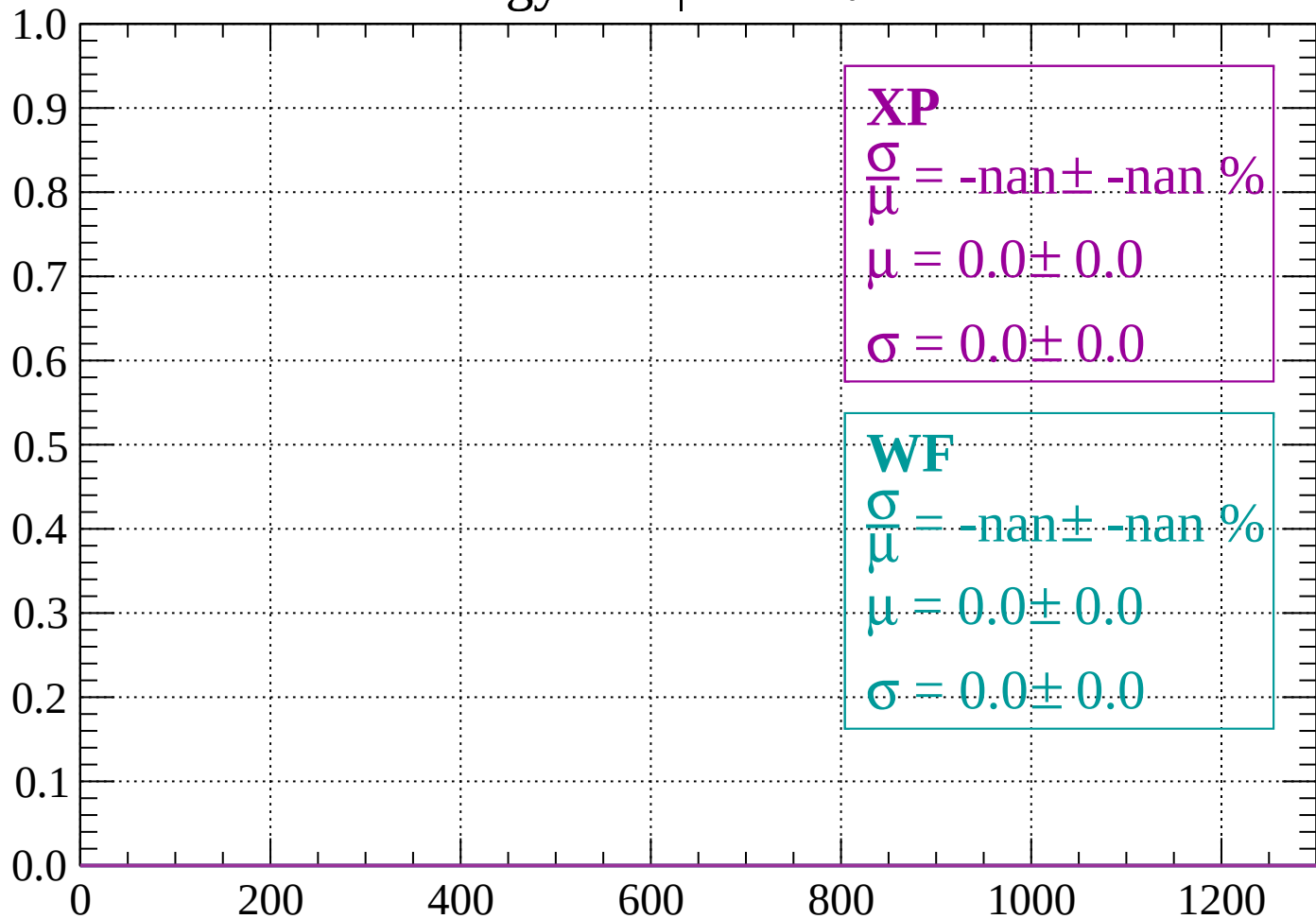
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

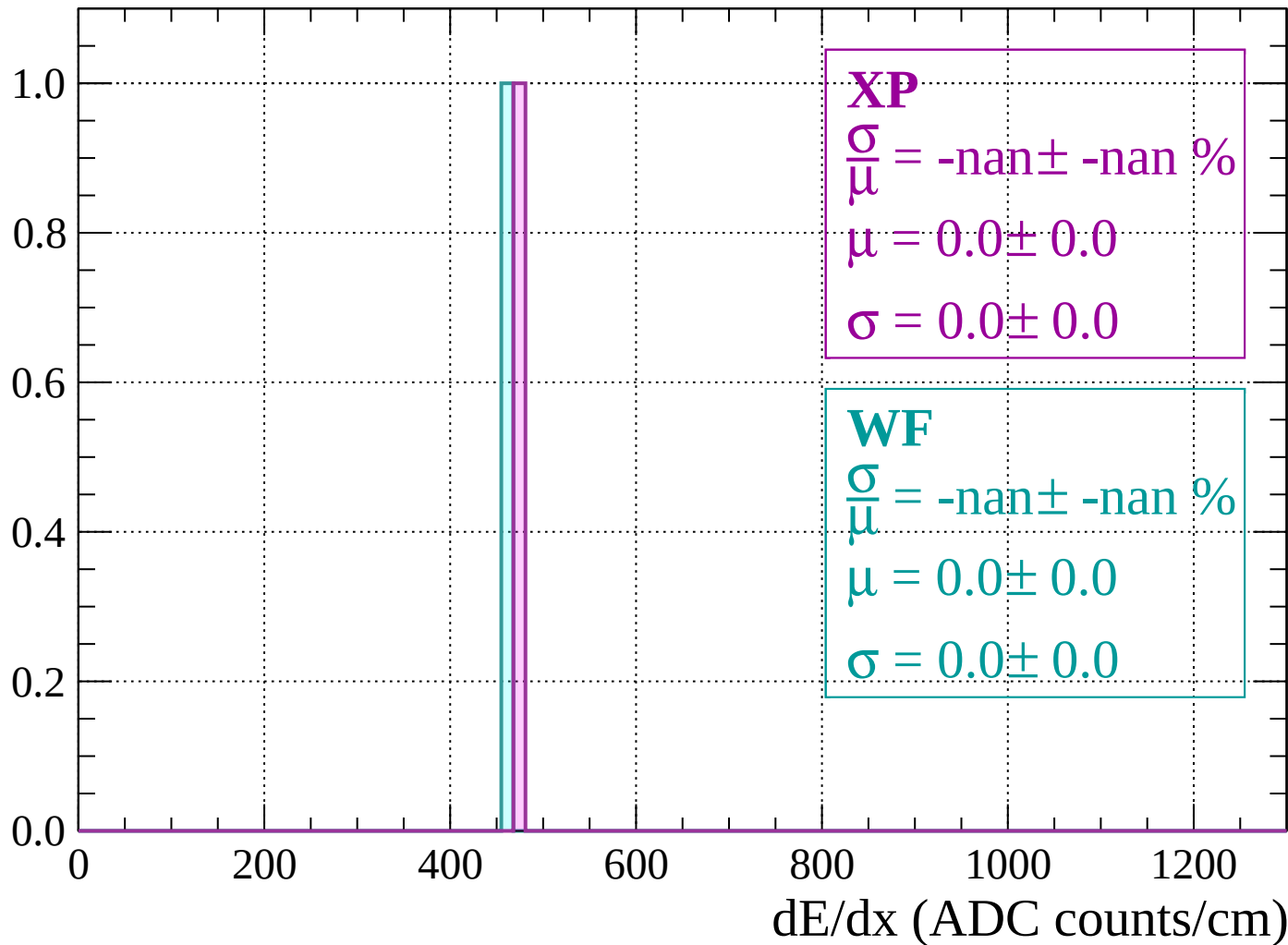
Count



dE/dx (ADC counts/cm)

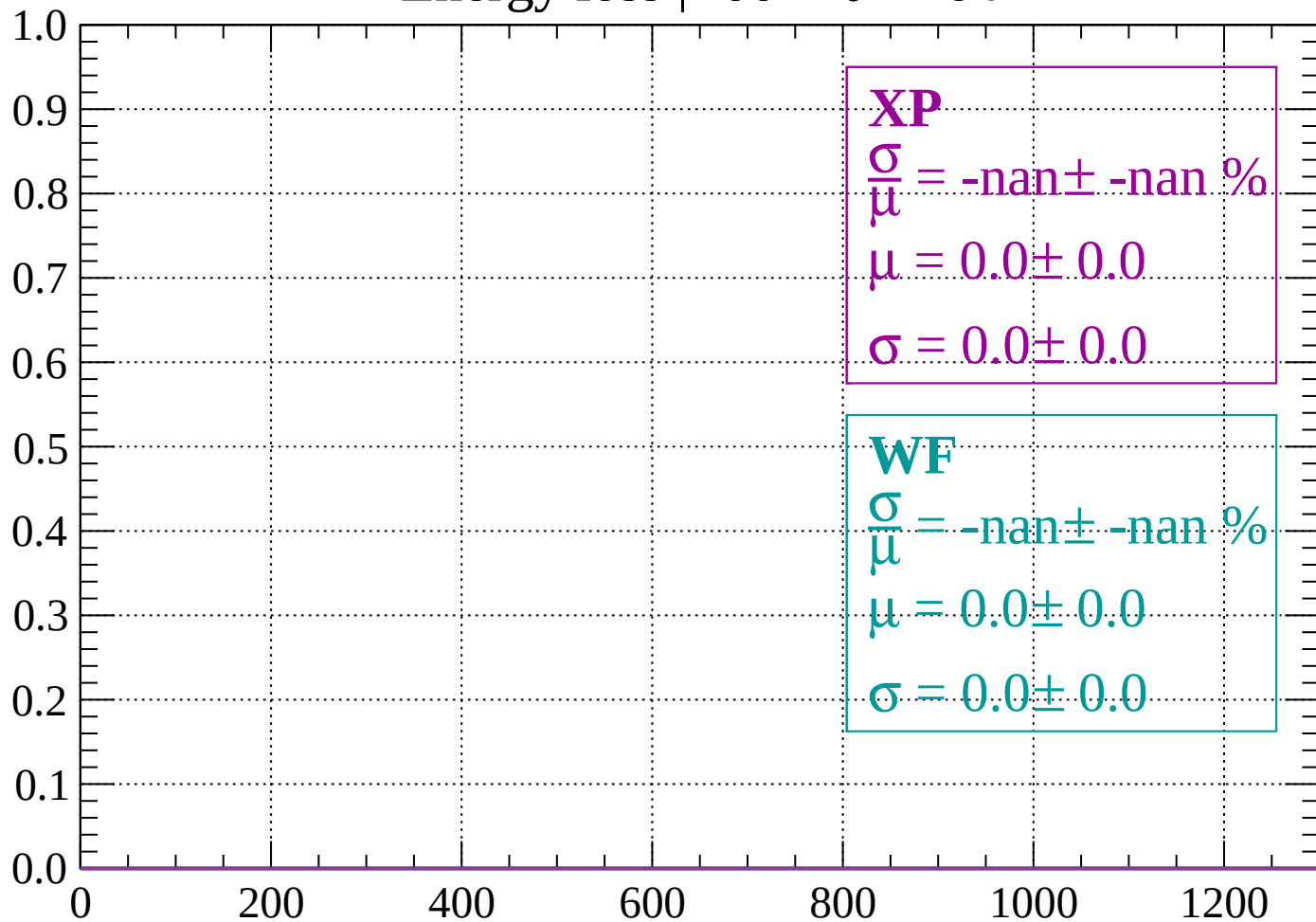
Energy loss | $-68 < \theta < -66$

Count



Energy loss | $-66 < \theta < -64$

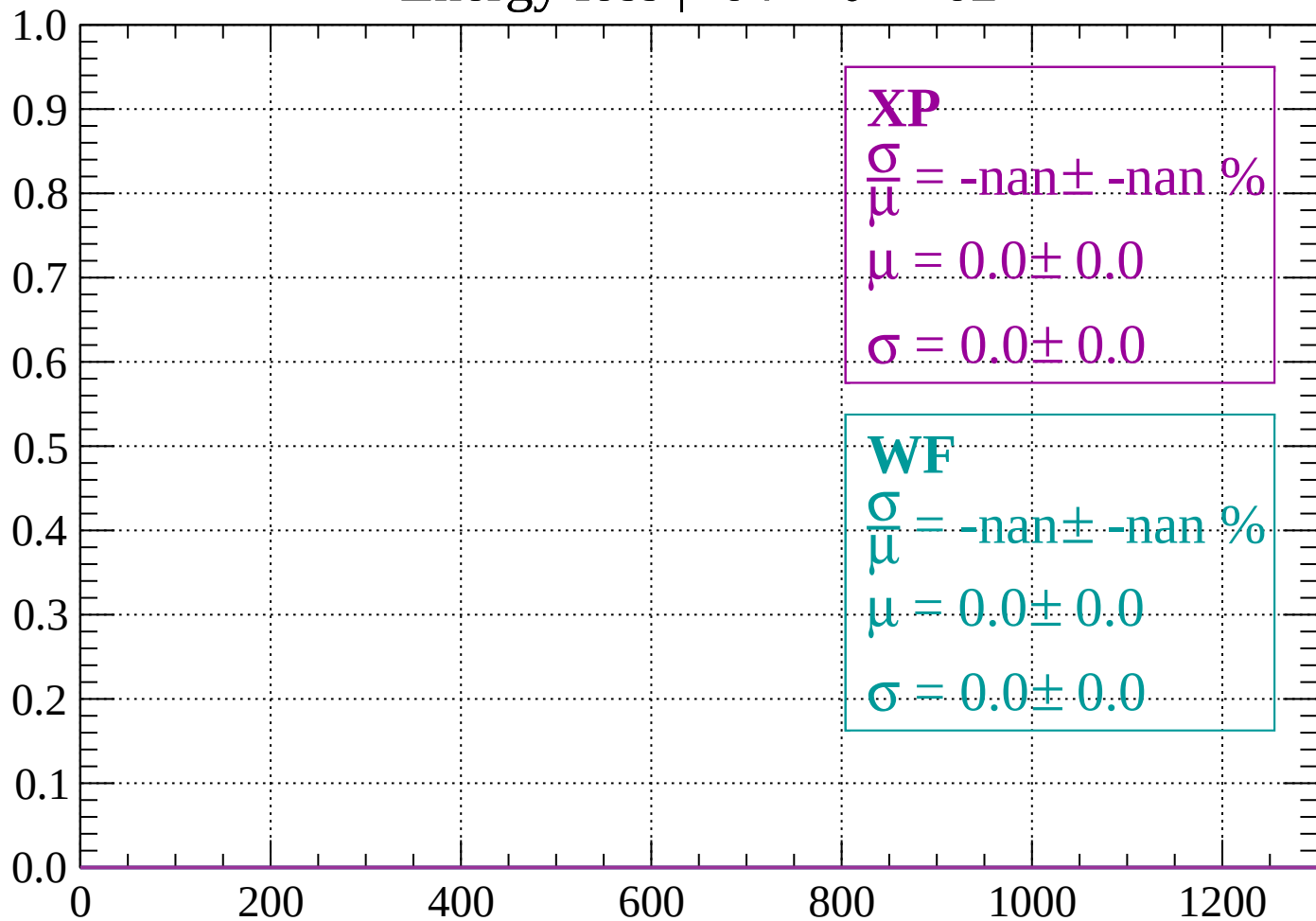
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

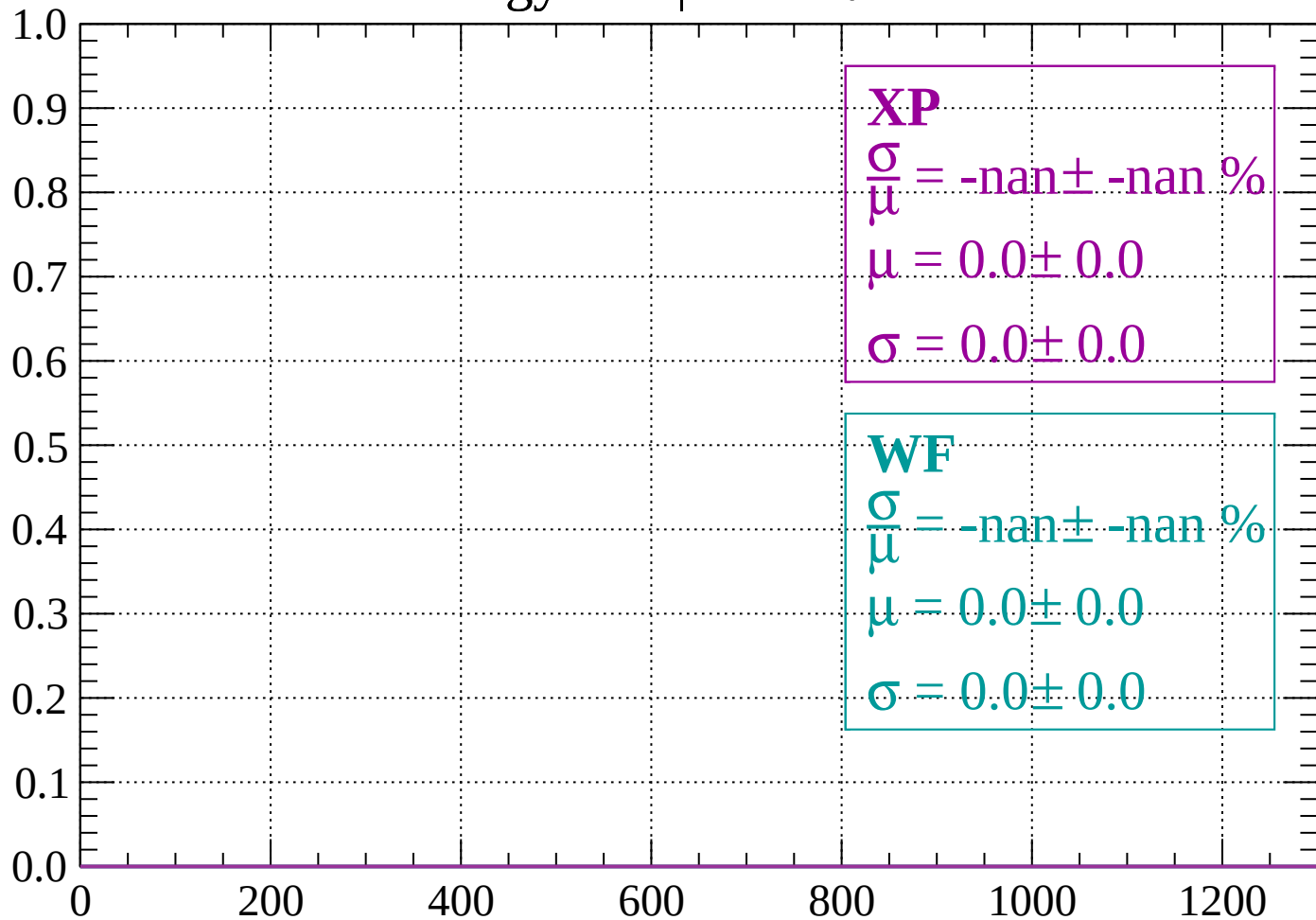
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

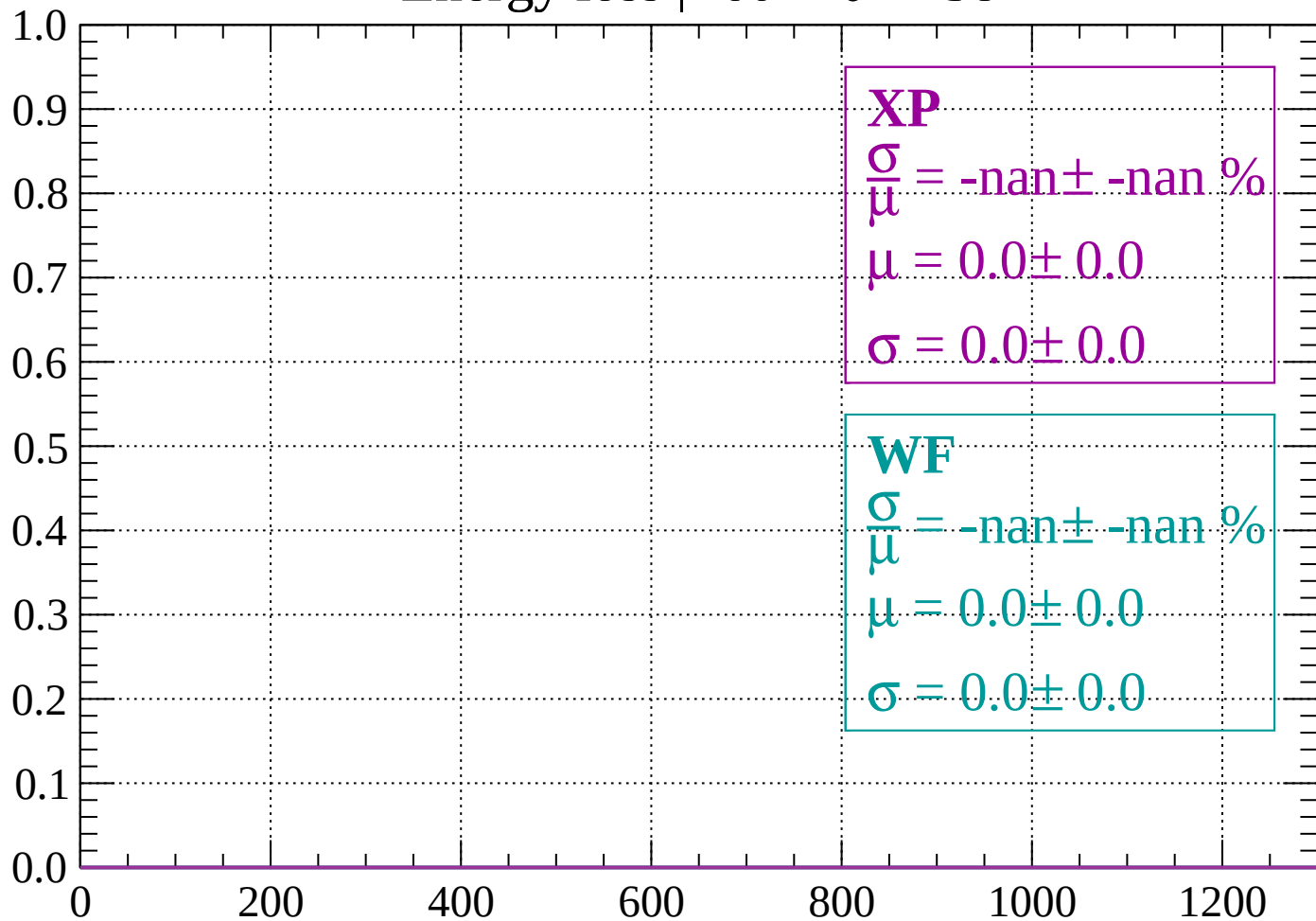
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

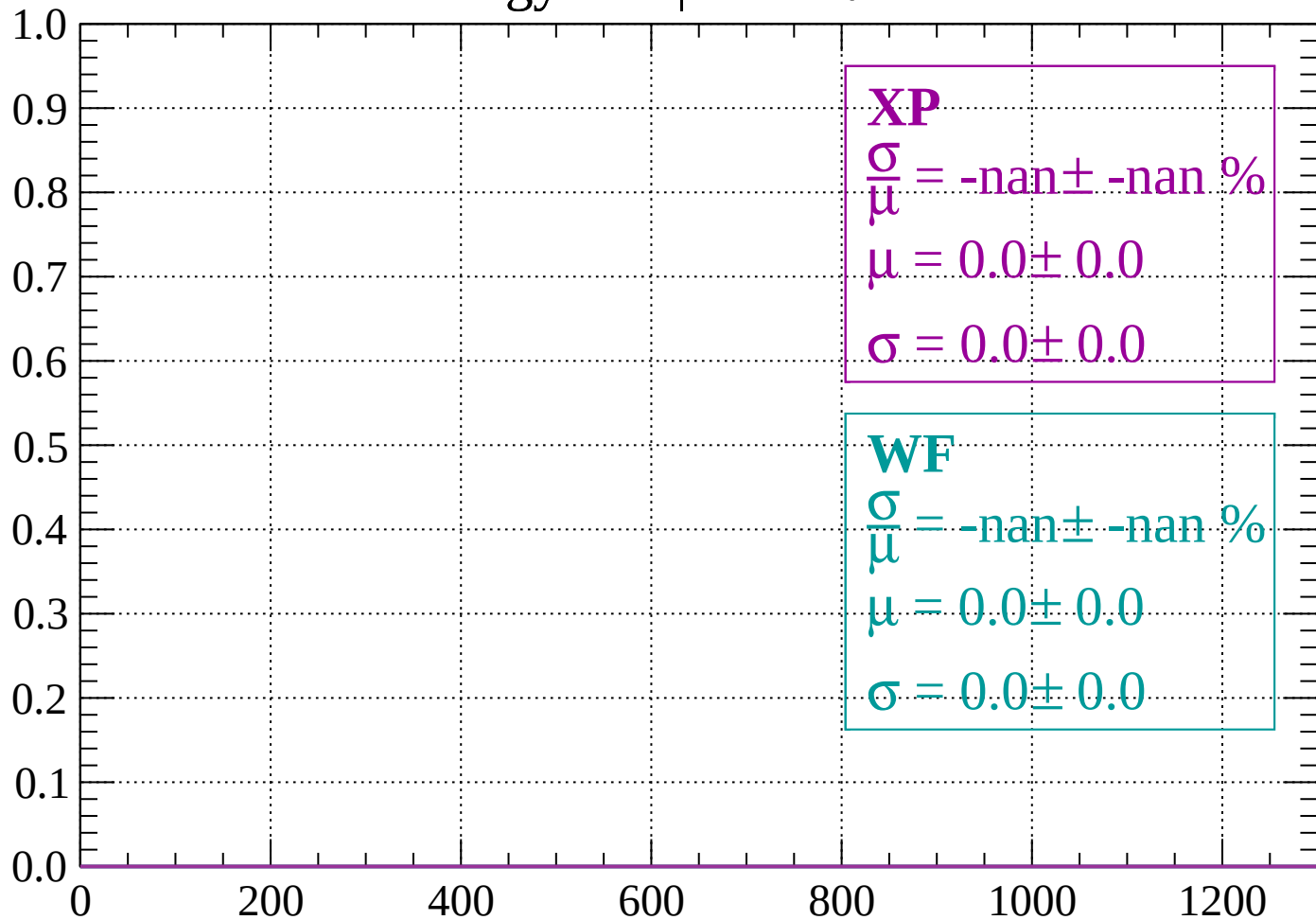
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

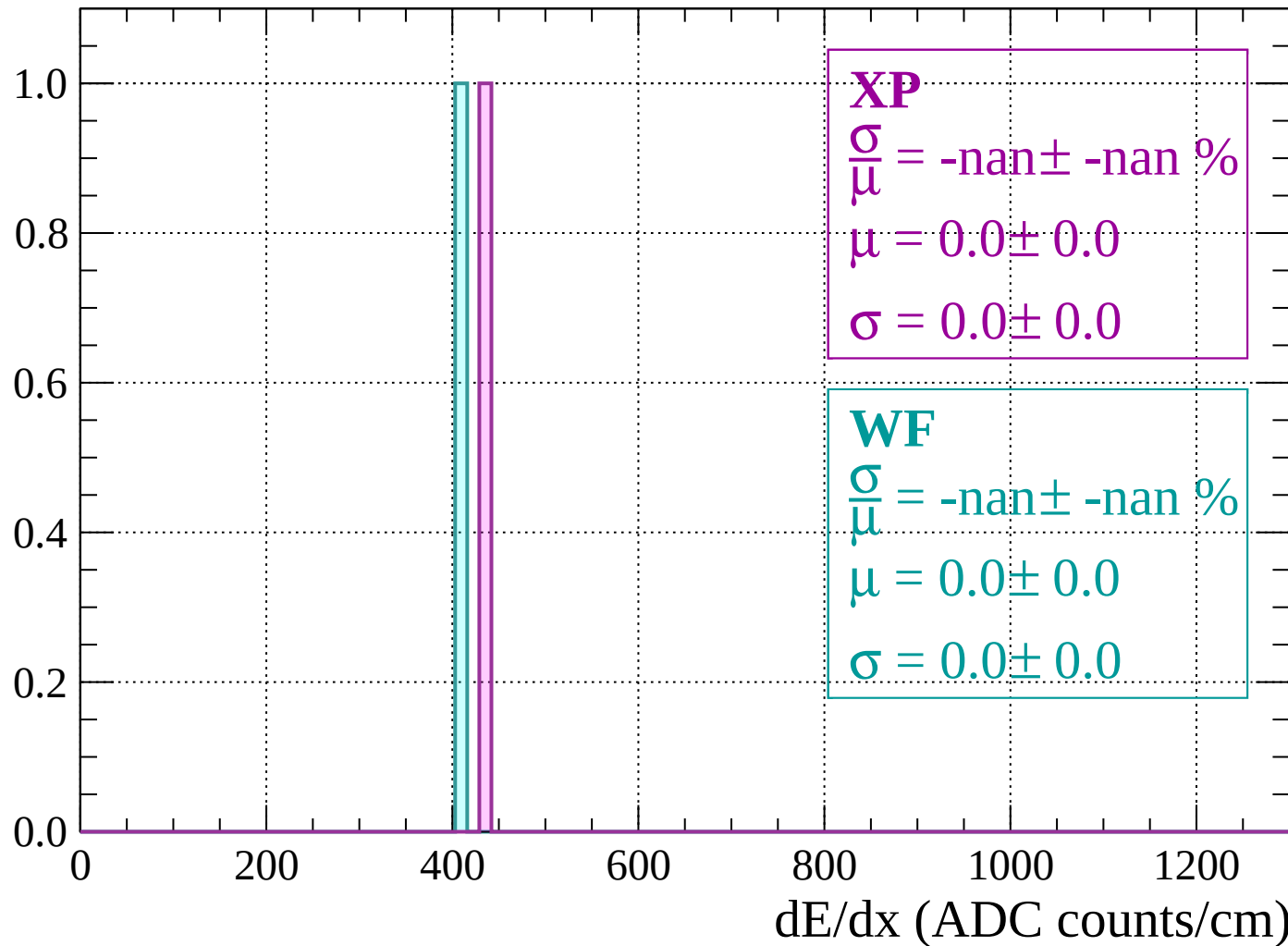
Count



dE/dx (ADC counts/cm)

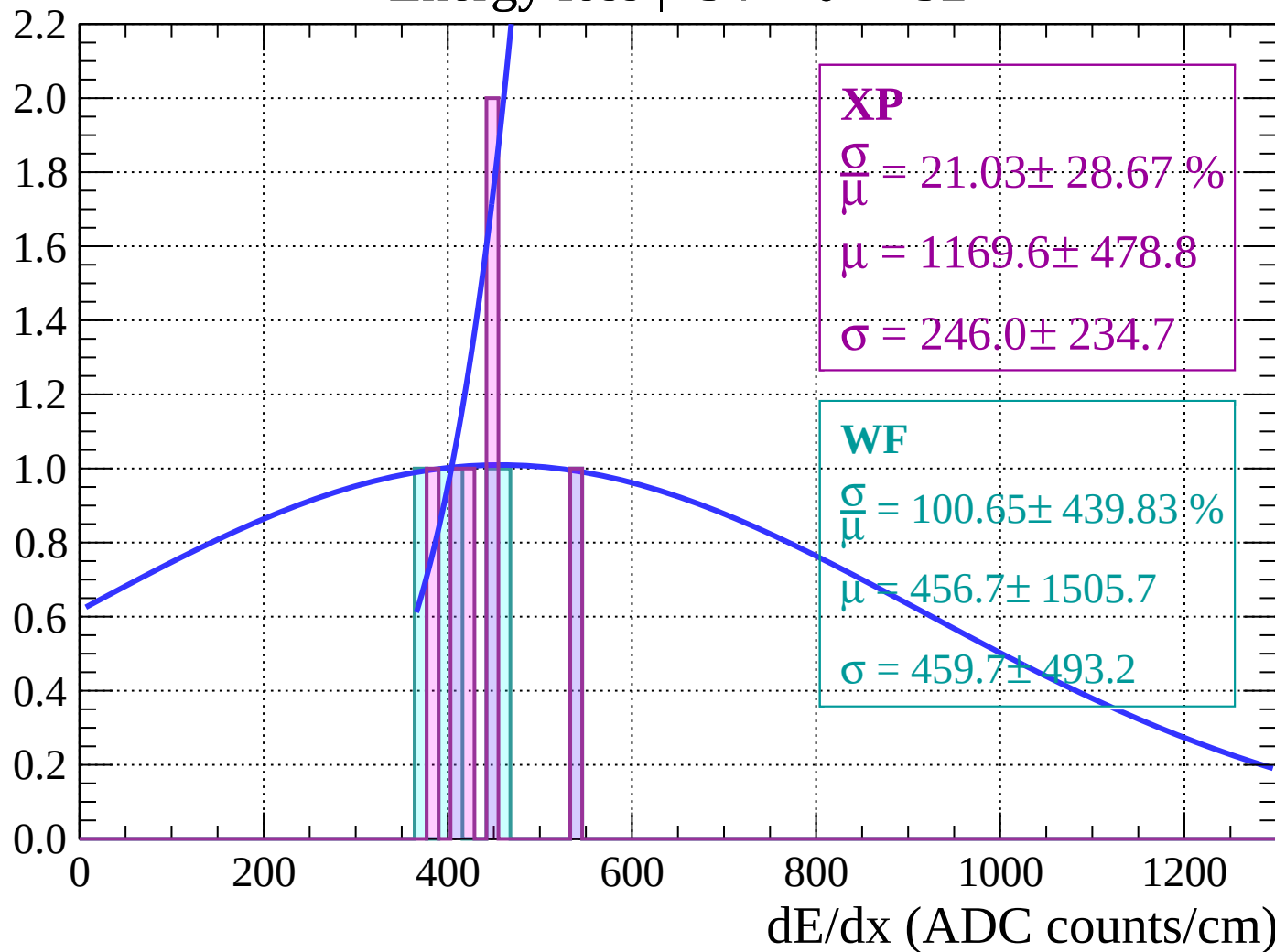
Energy loss | $-56 < \theta < -54$

Count



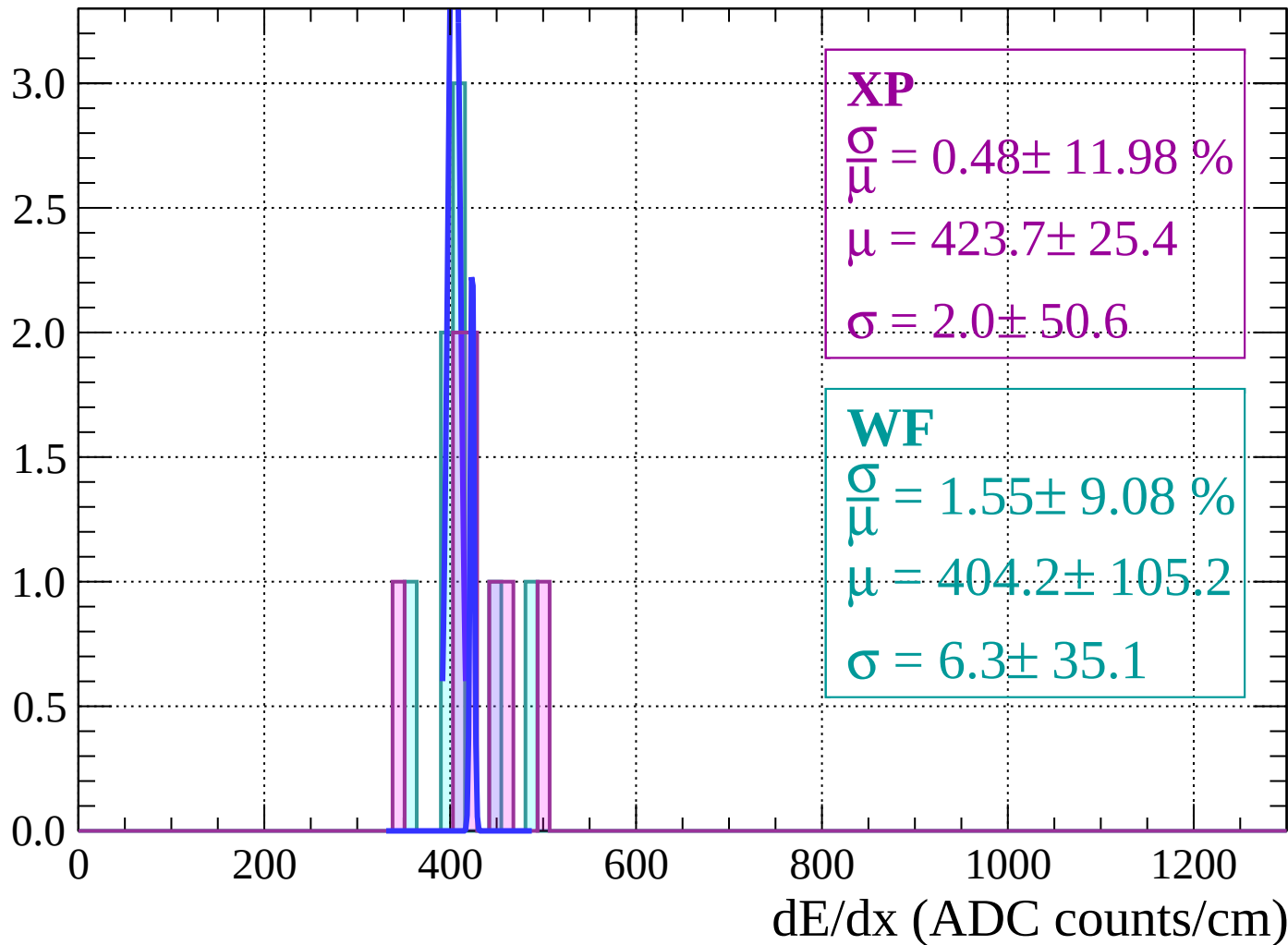
Energy loss $|-54 < \theta < -52$

Count



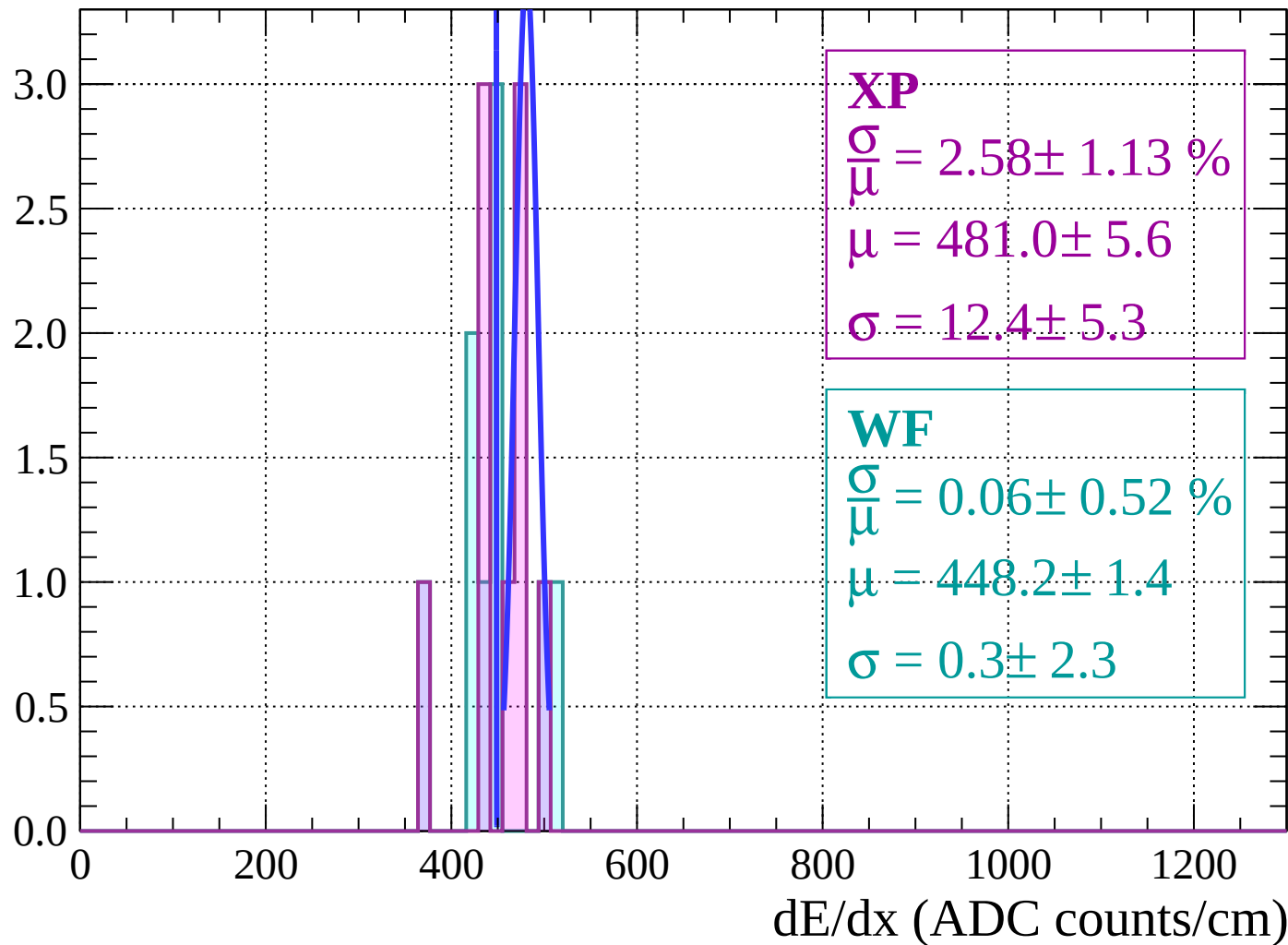
Energy loss | $-52 < \theta < -50$

Count



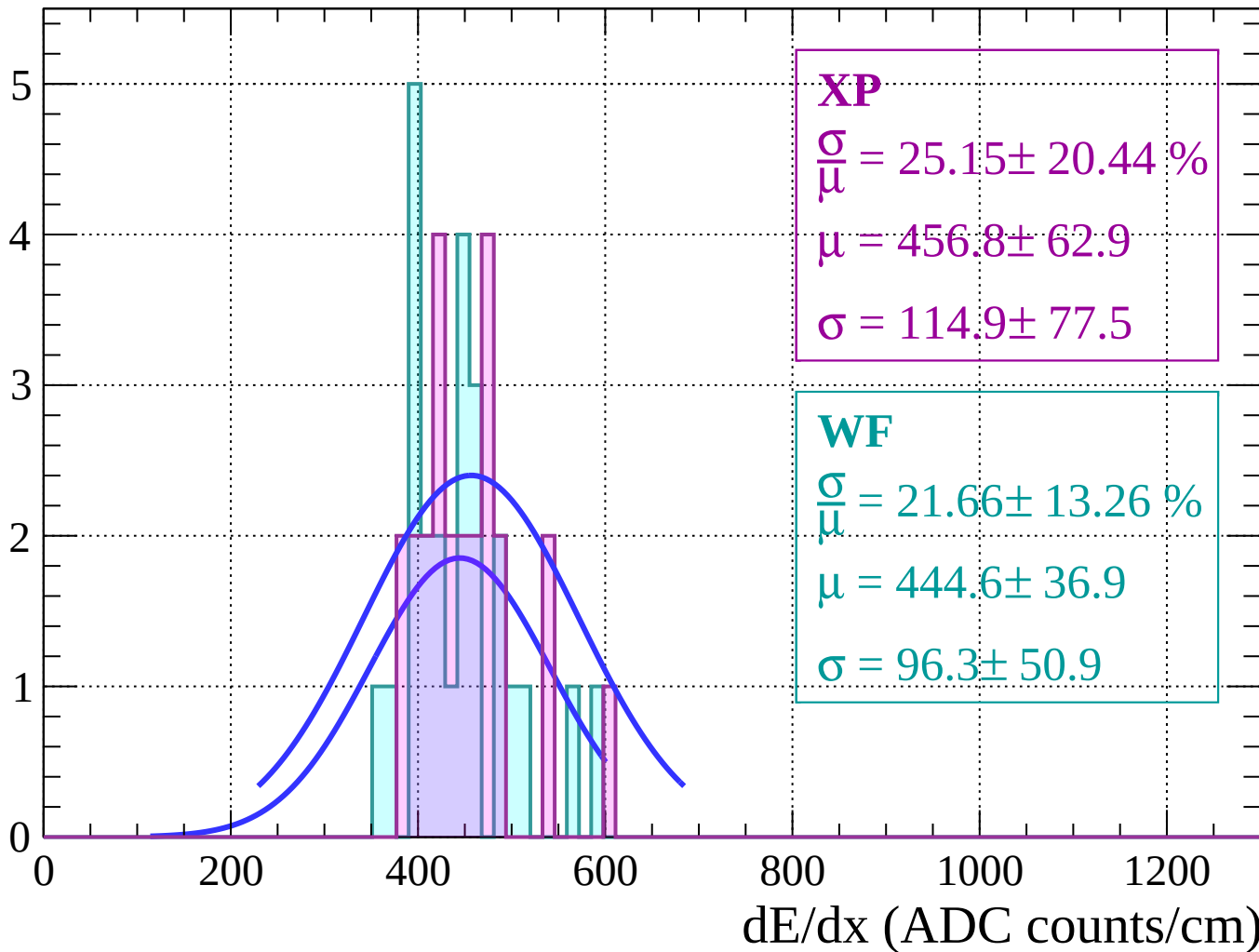
Energy loss | $-50 < \theta < -48$

Count



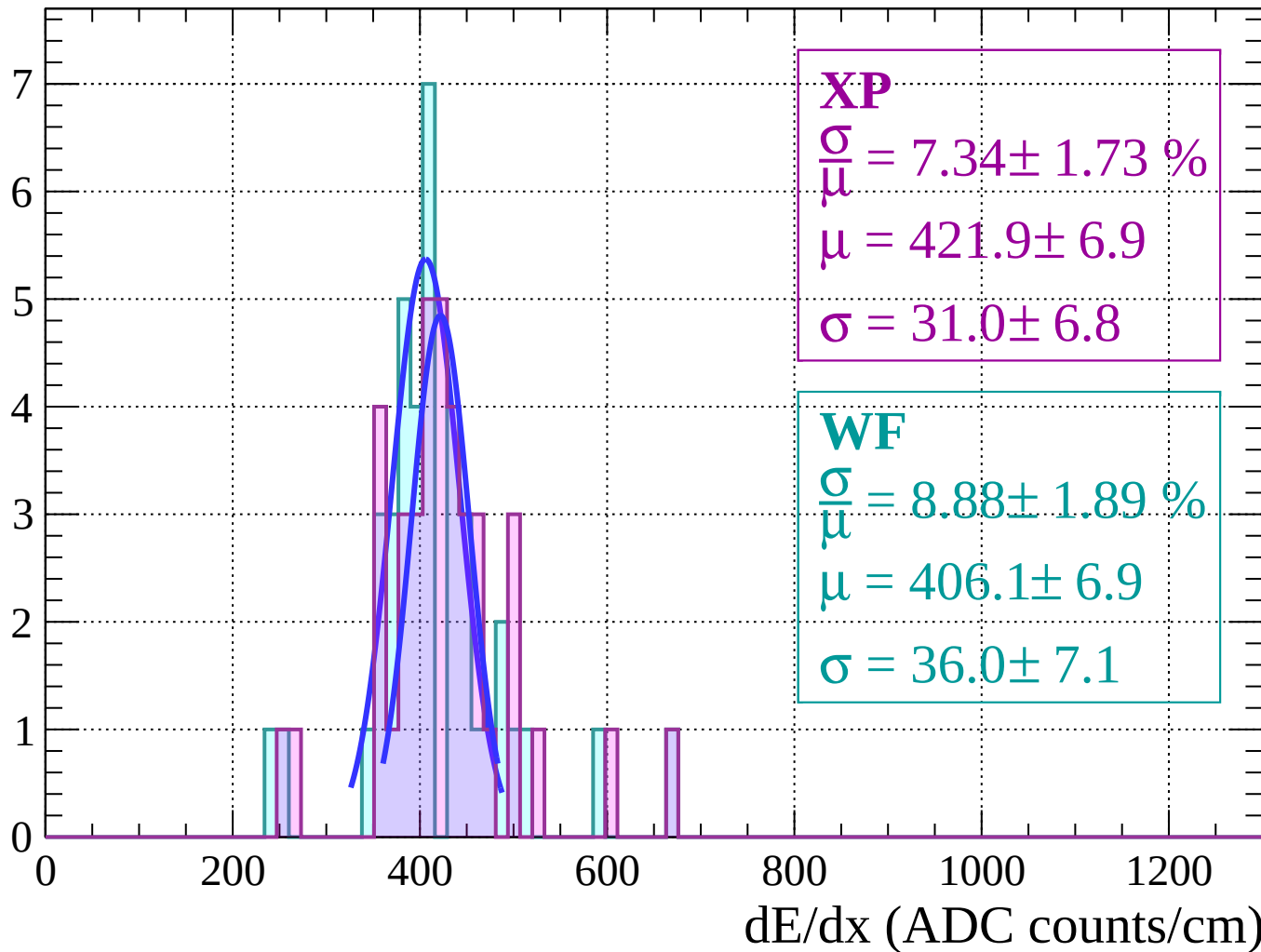
Energy loss | $-48 < \theta < -46$

Count



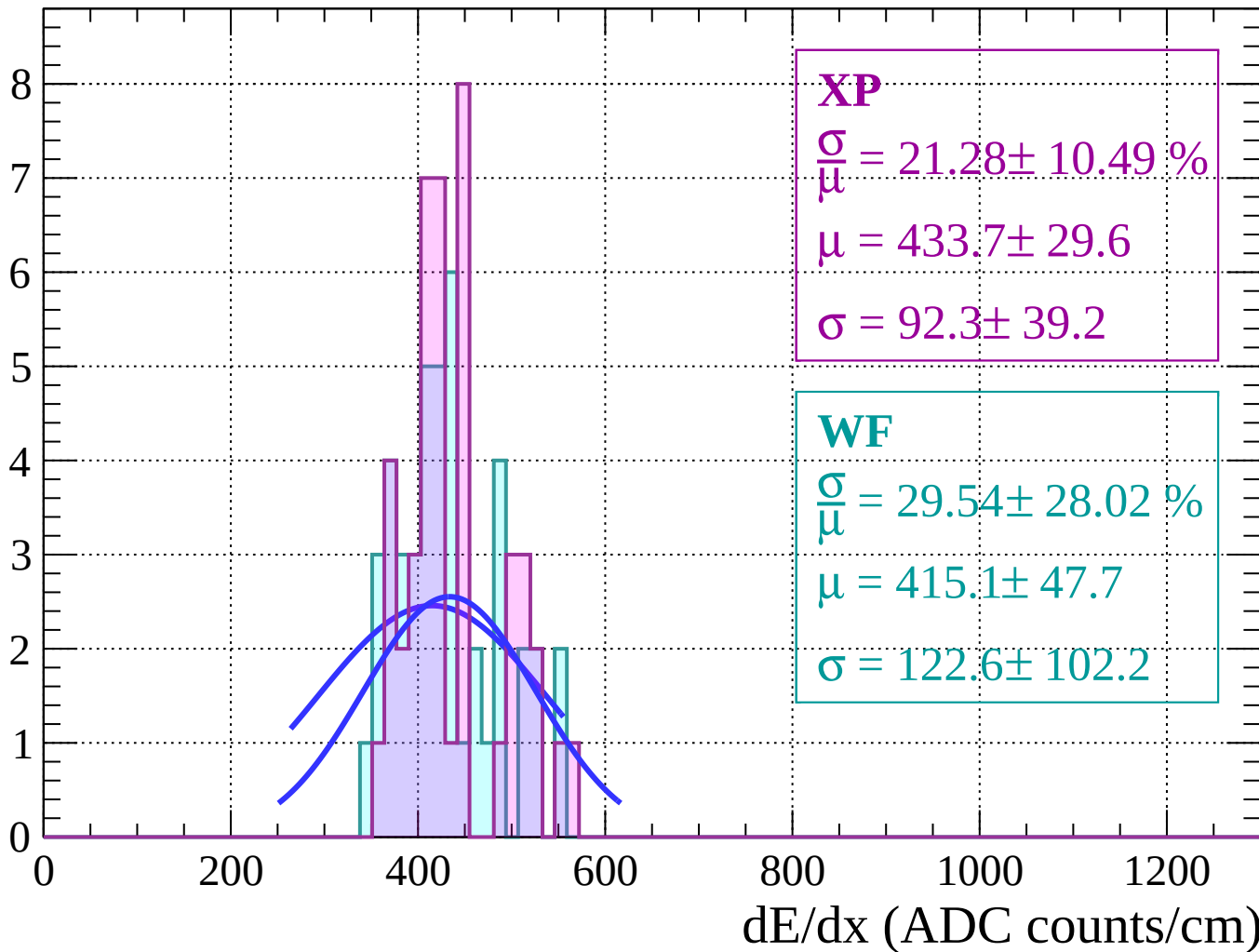
Energy loss | $-46 < \theta < -44$

Count



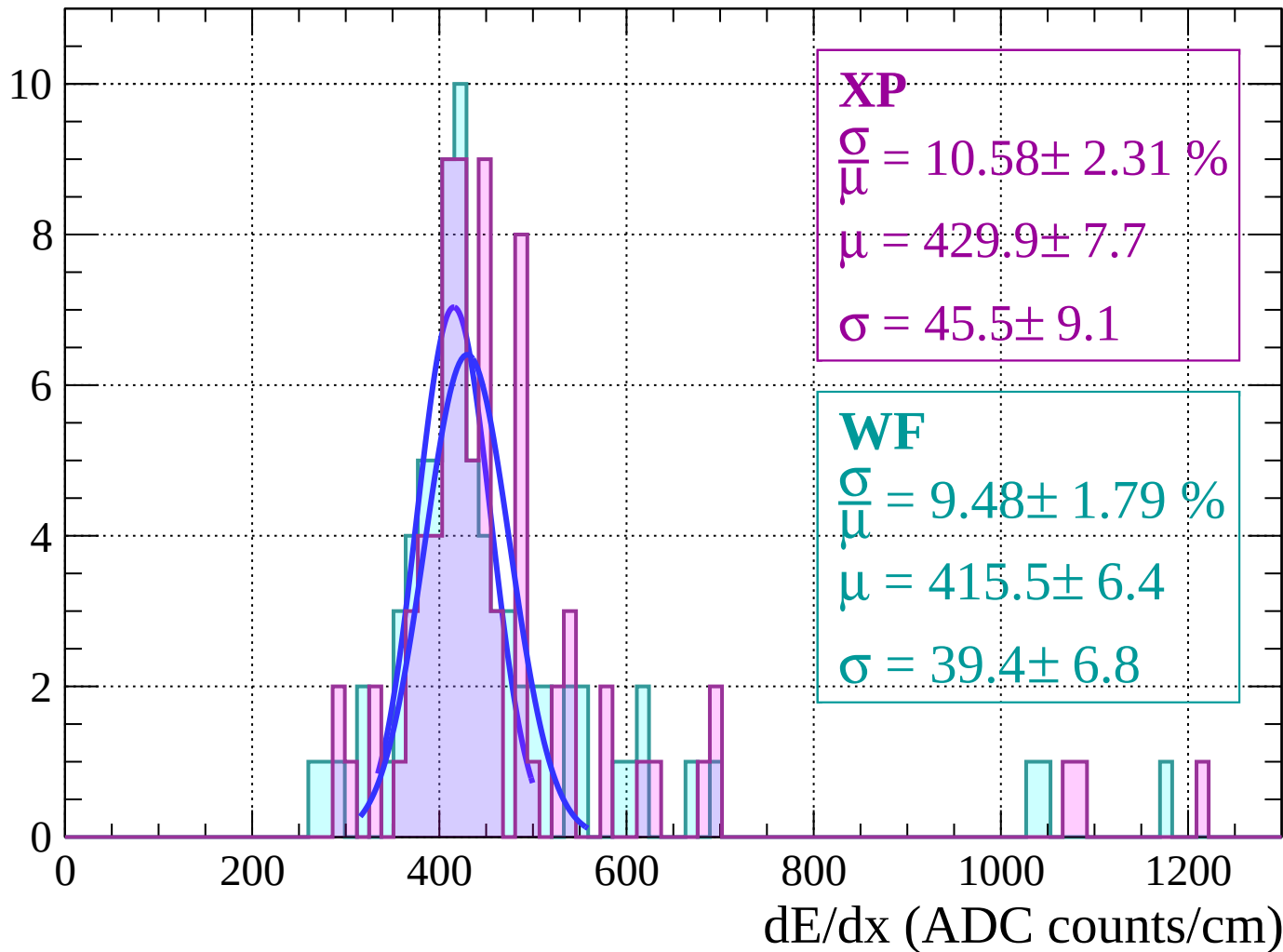
Energy loss | $-44 < \theta < -42$

Count



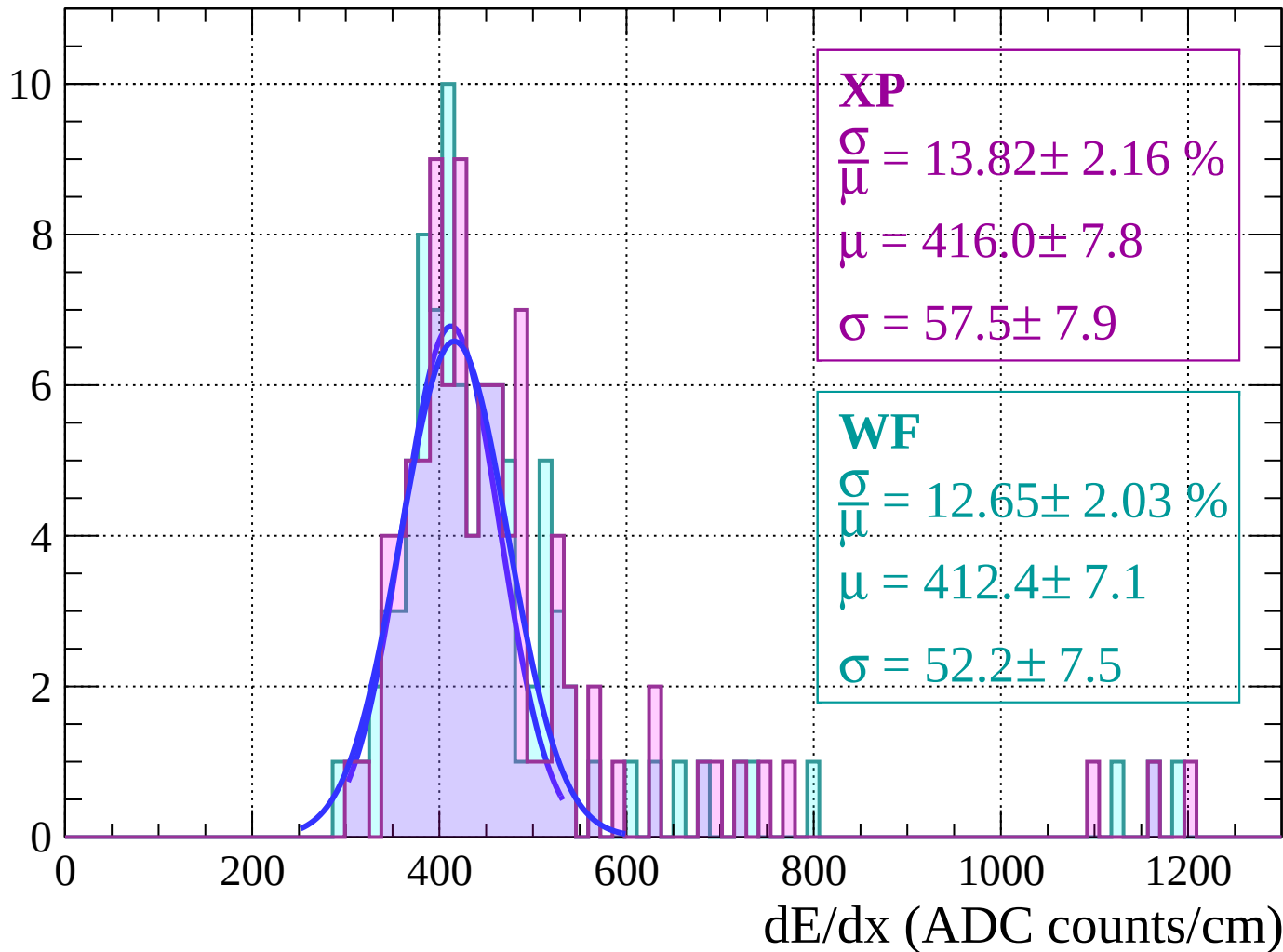
Energy loss | $-42 < \theta < -40$

Count



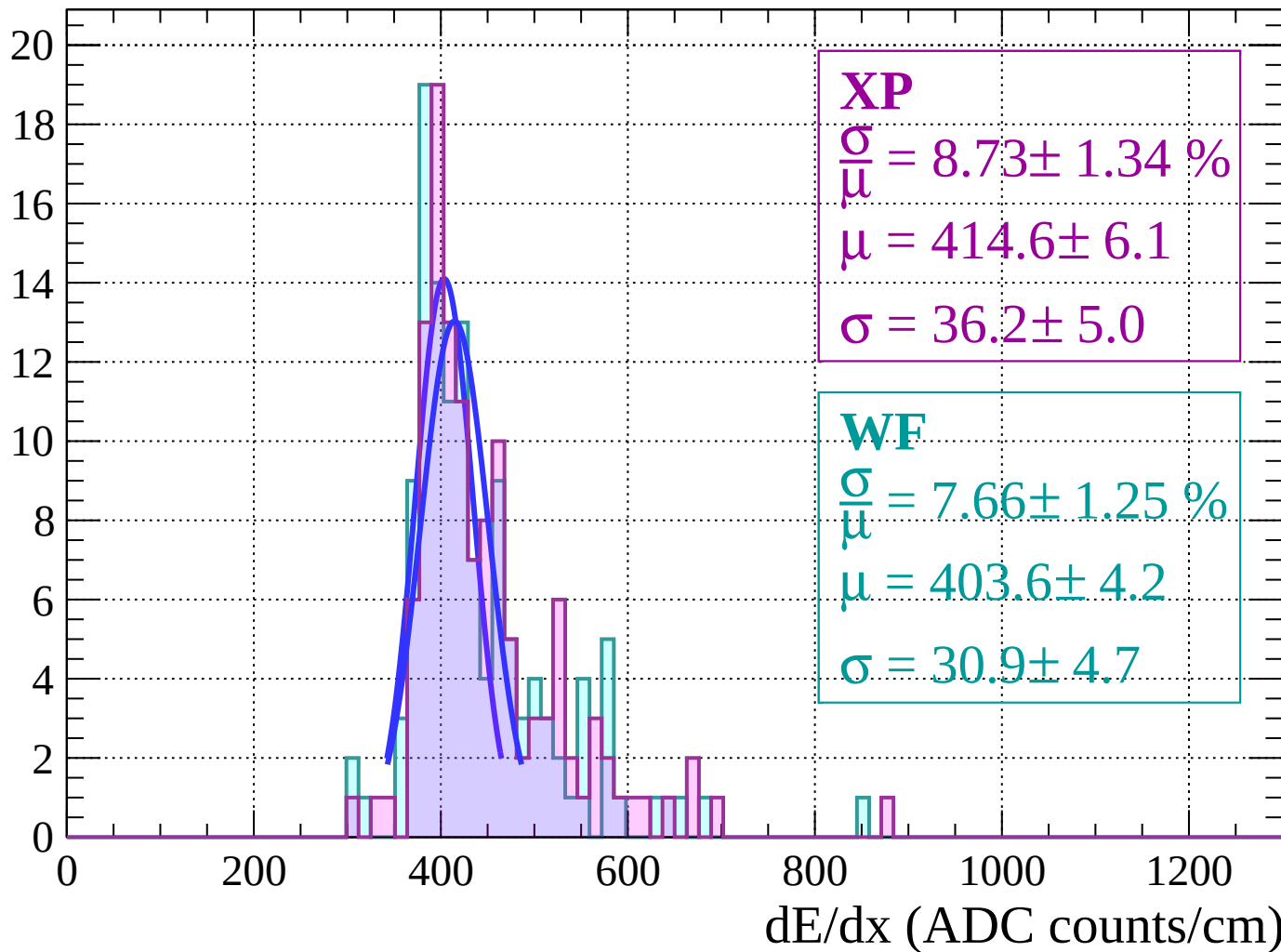
Energy loss | $-40 < \theta < -38$

Count



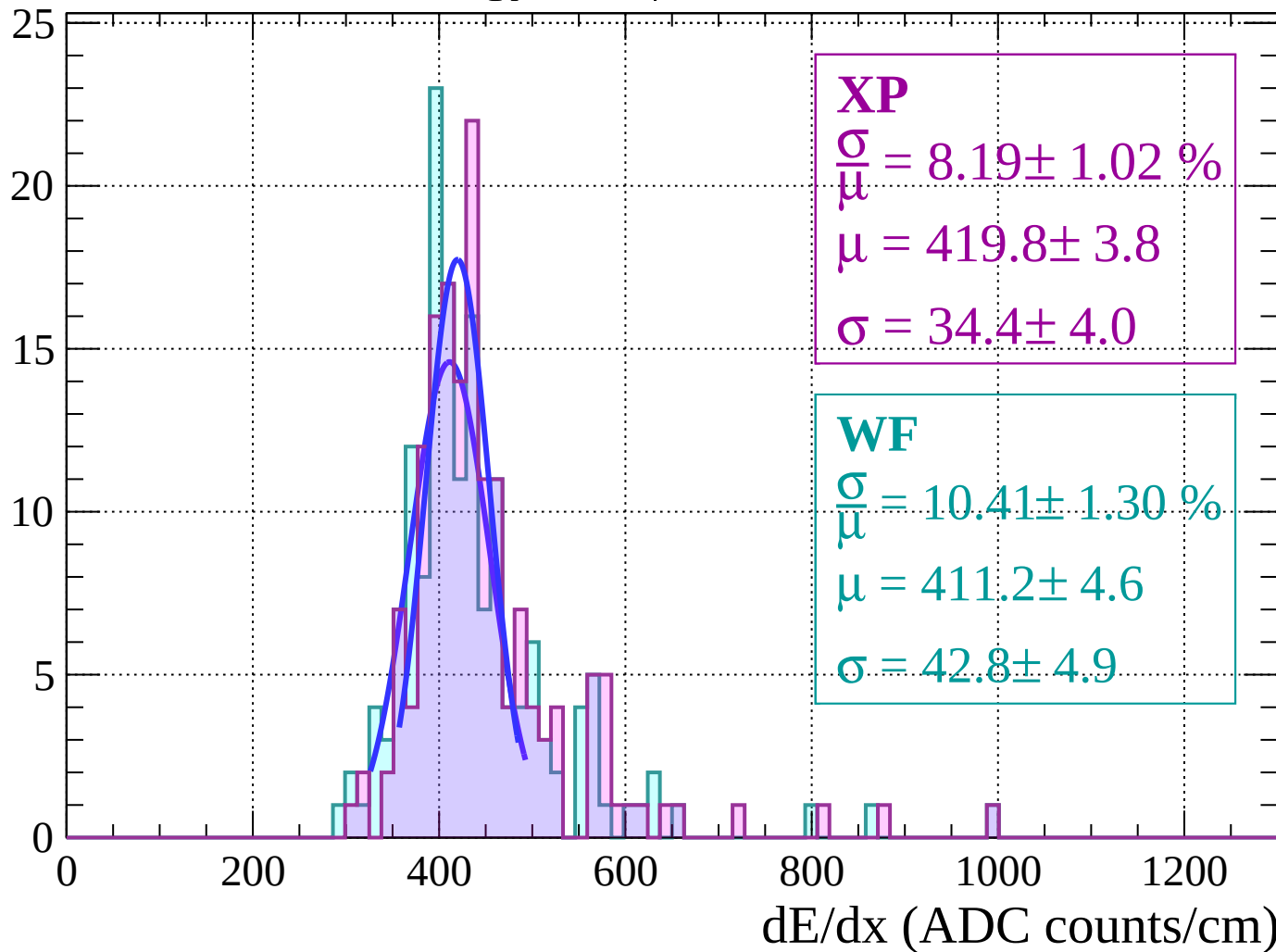
Energy loss | $-38 < \theta < -36$

Count



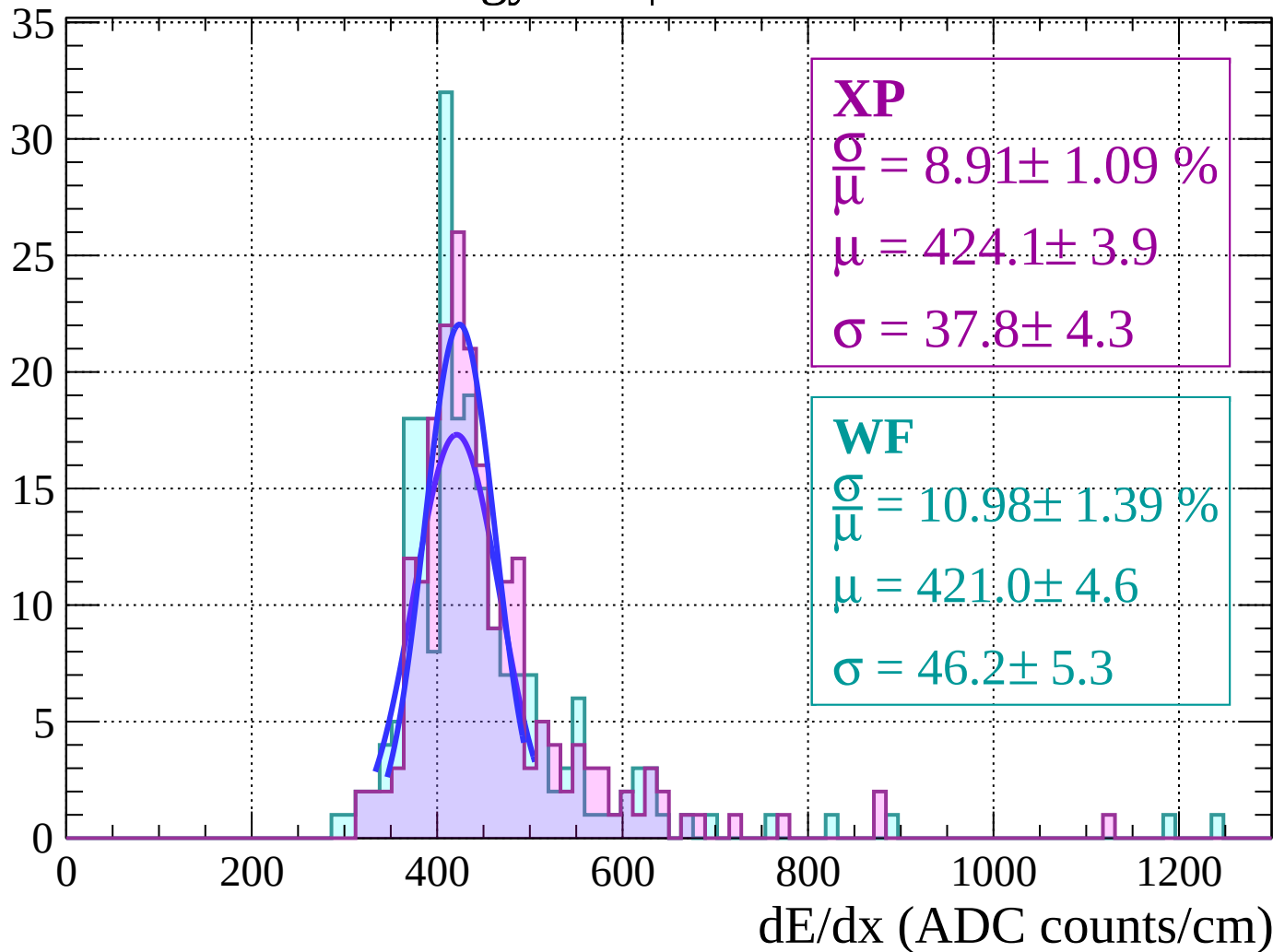
Energy loss | $-36 < \theta < -34$

Count



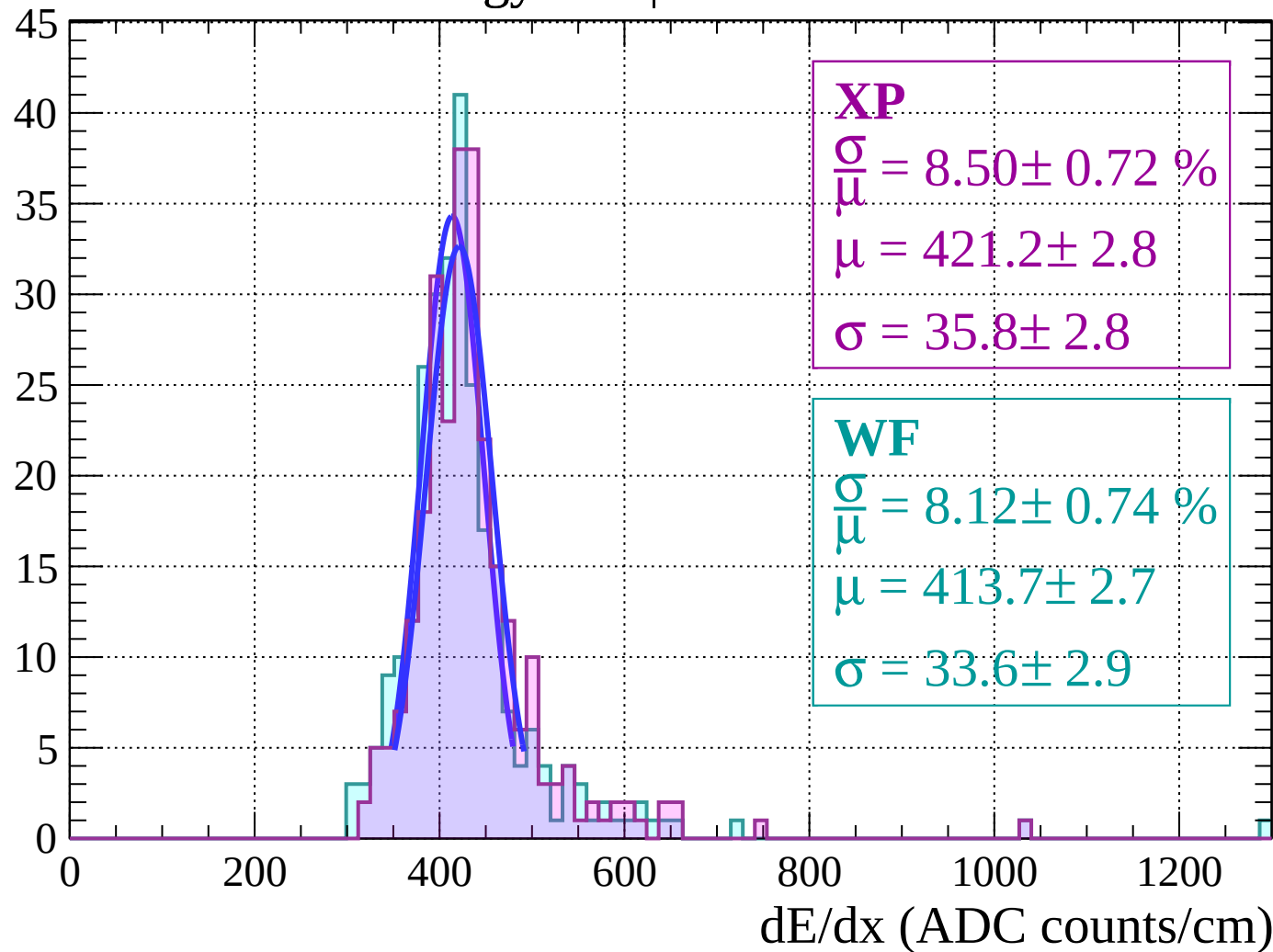
Energy loss | $-34 < \theta < -32$

Count



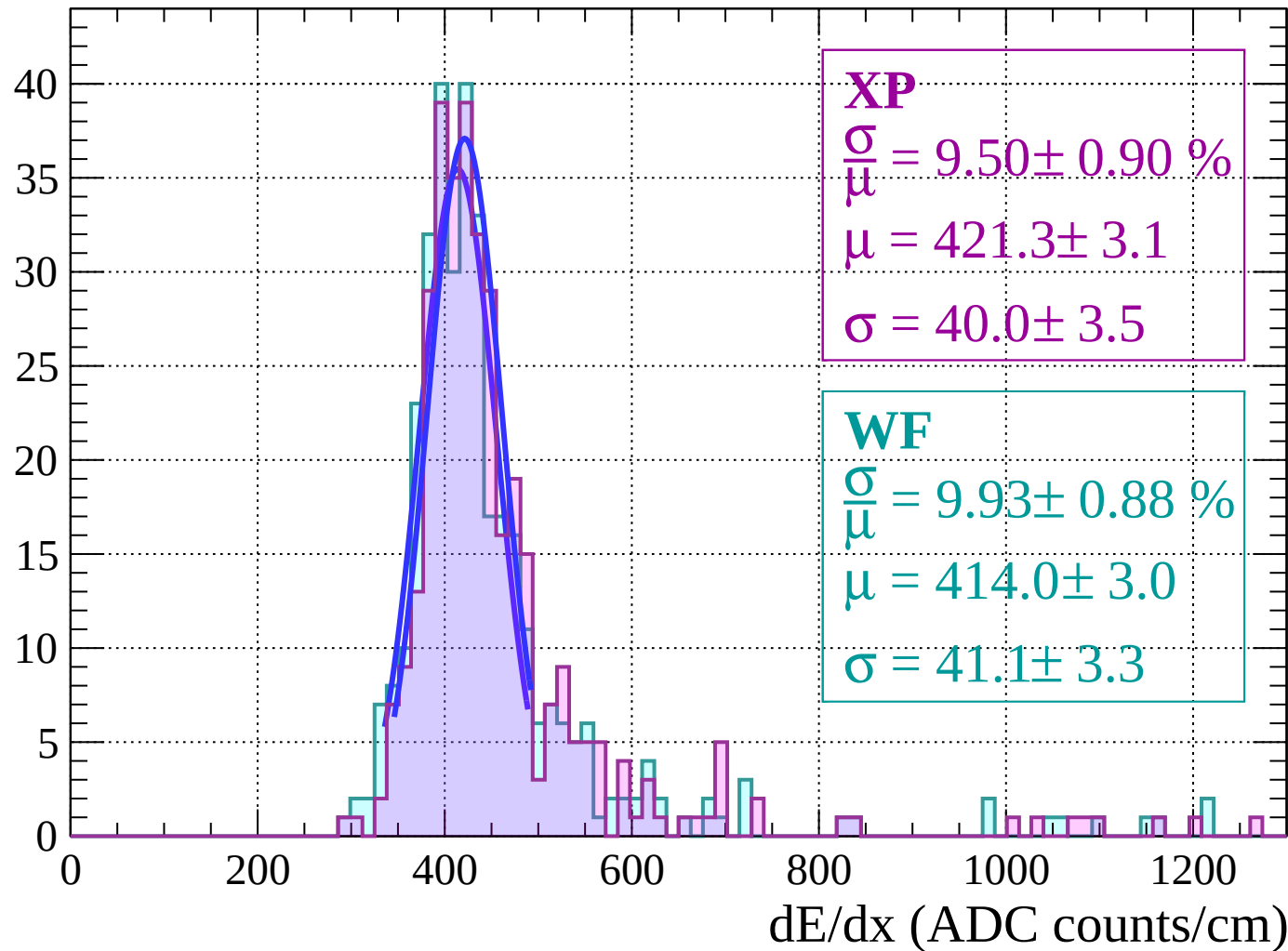
Energy loss | $-32 < \theta < -30$

Count



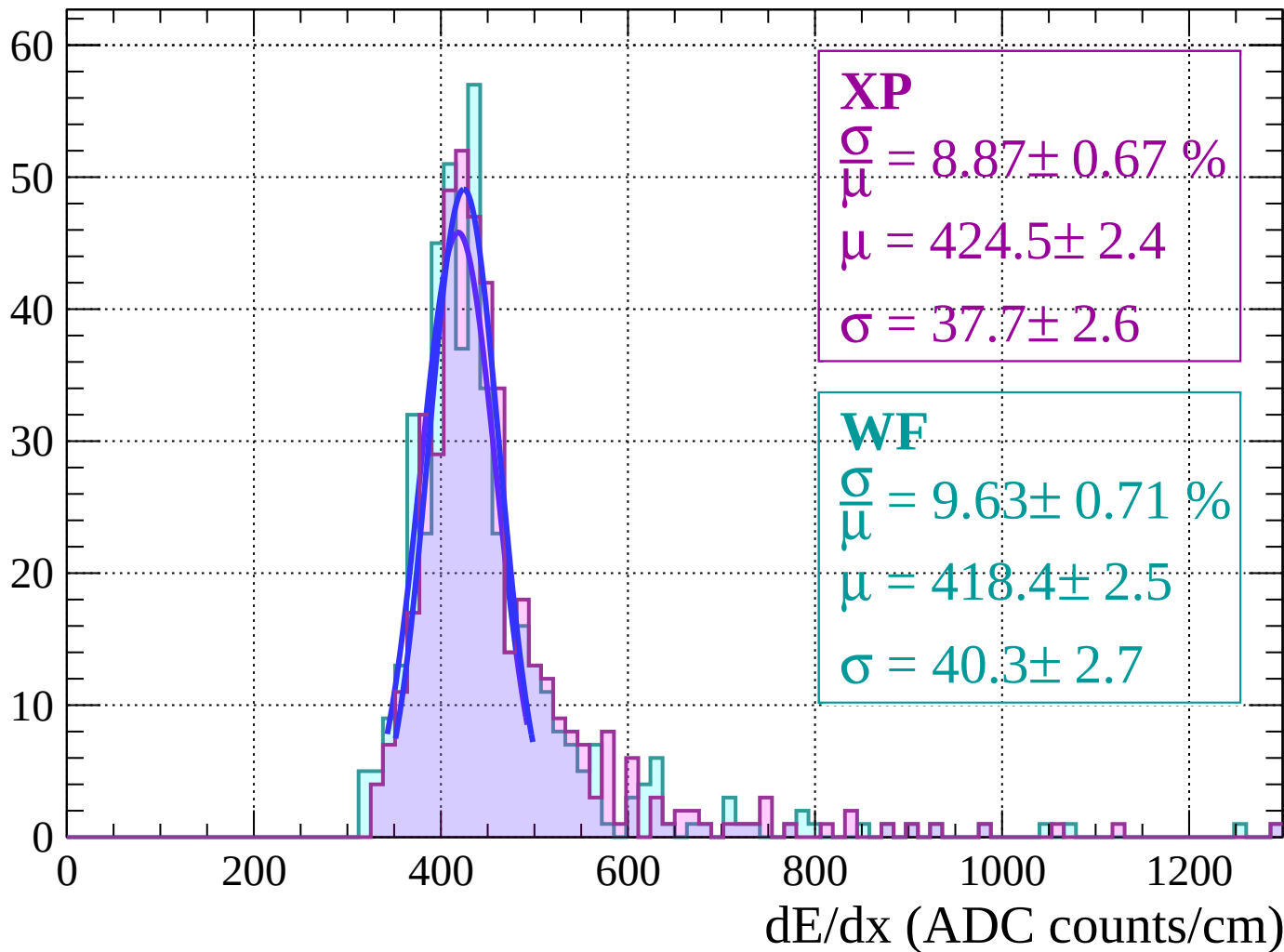
Energy loss | $-30 < \theta < -28$

Count



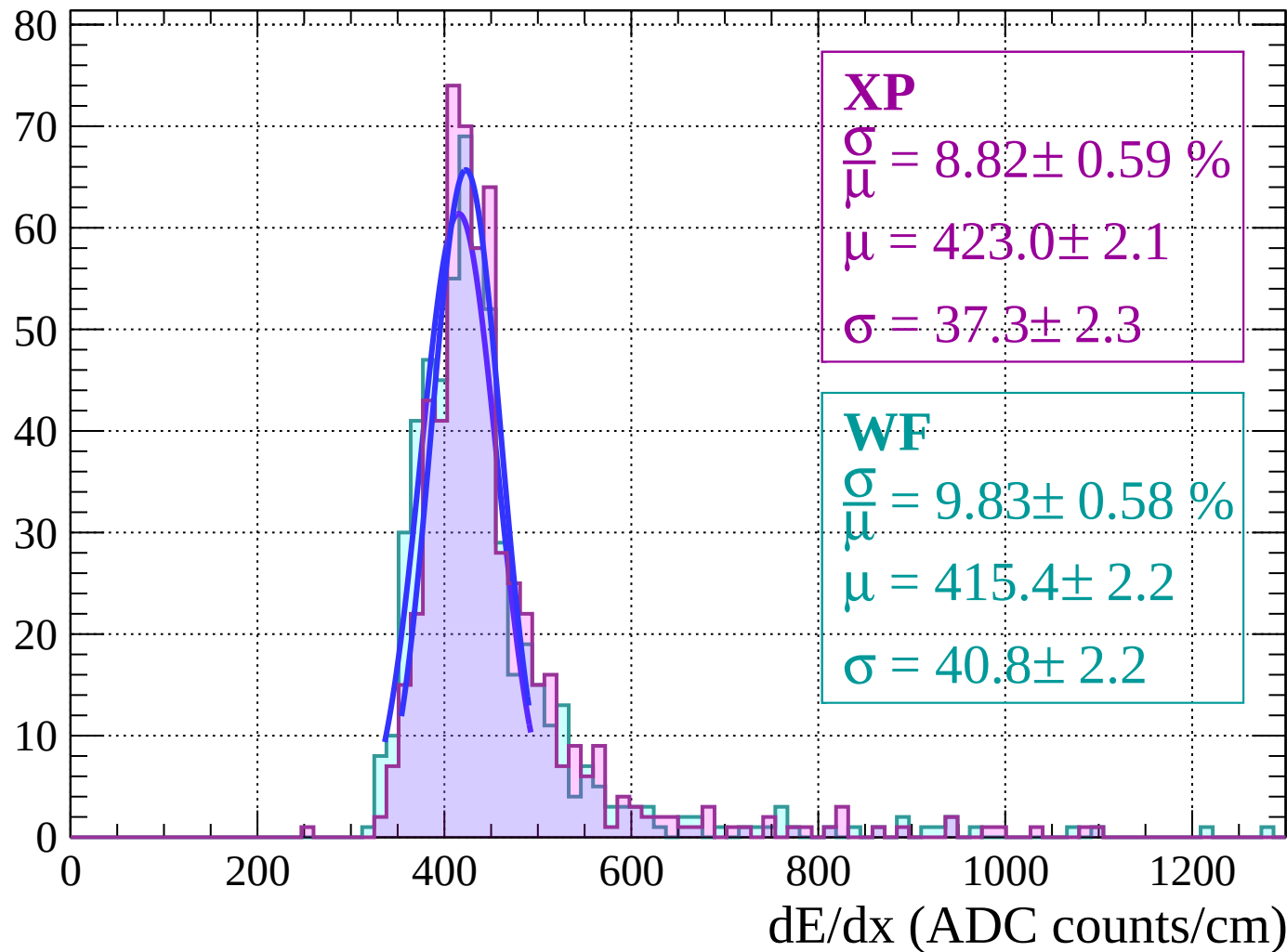
Energy loss | $-28 < \theta < -26$

Count



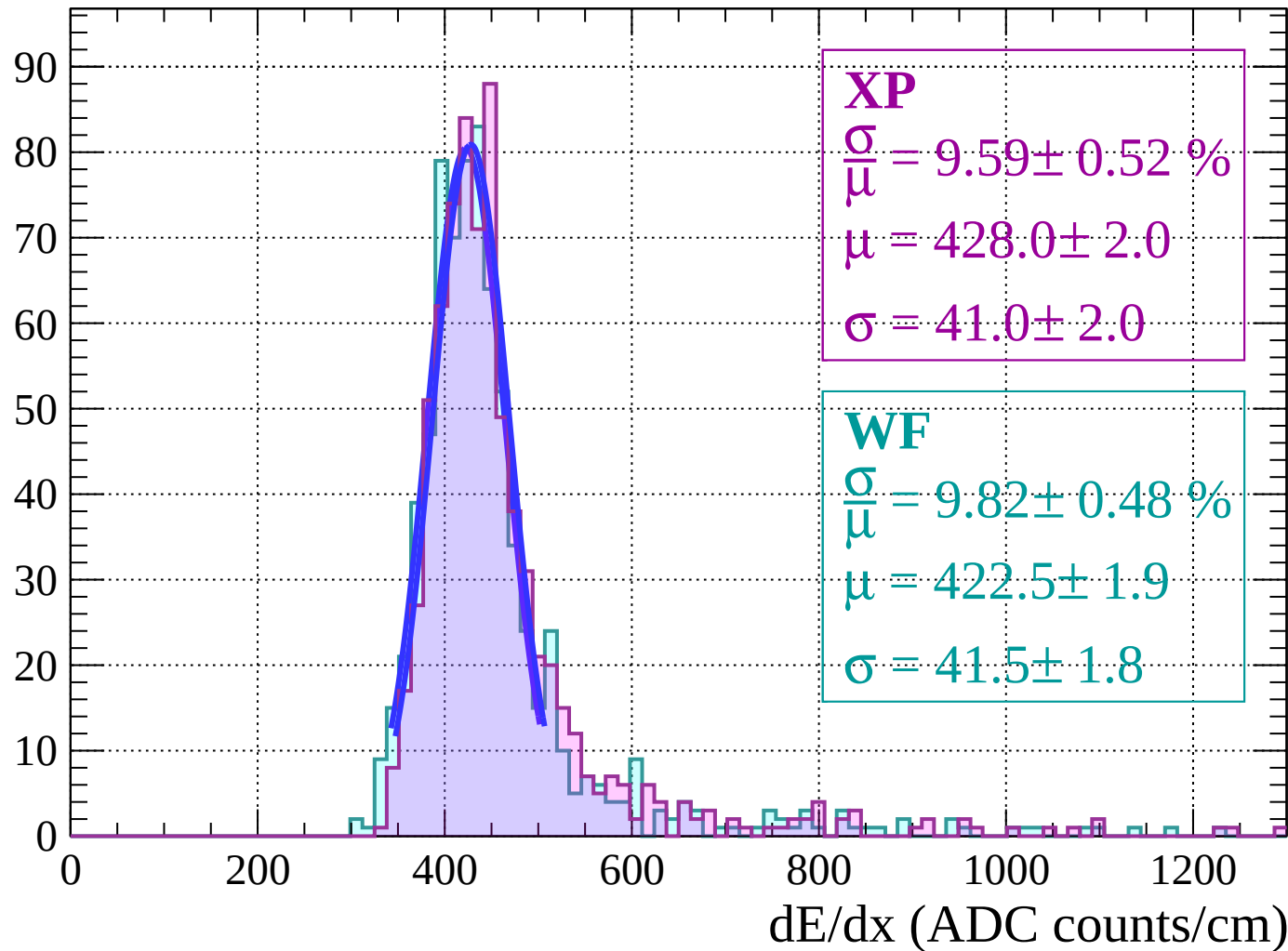
Energy loss | $-26 < \theta < -24$

Count



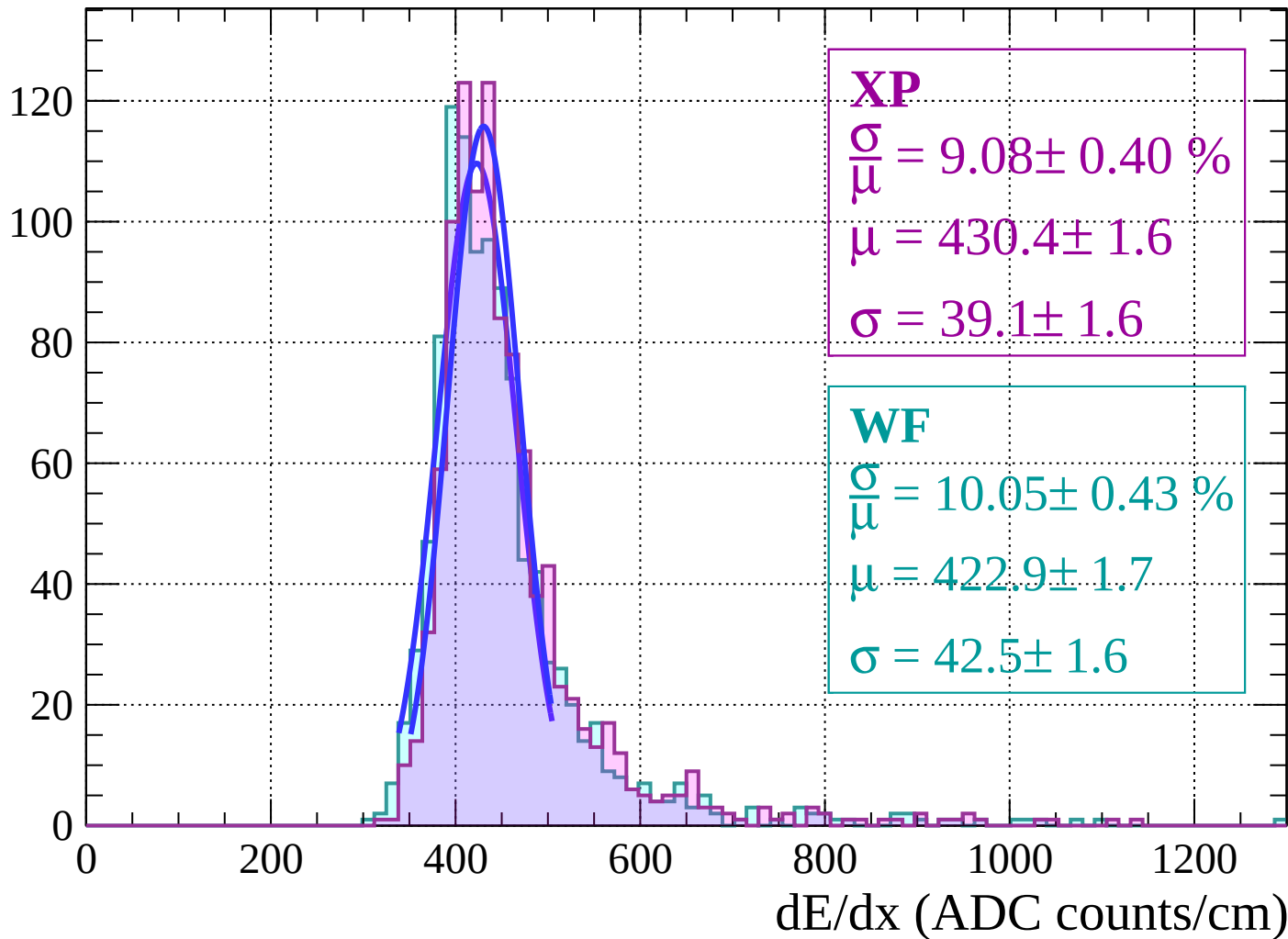
Energy loss | $-24 < \theta < -22$

Count



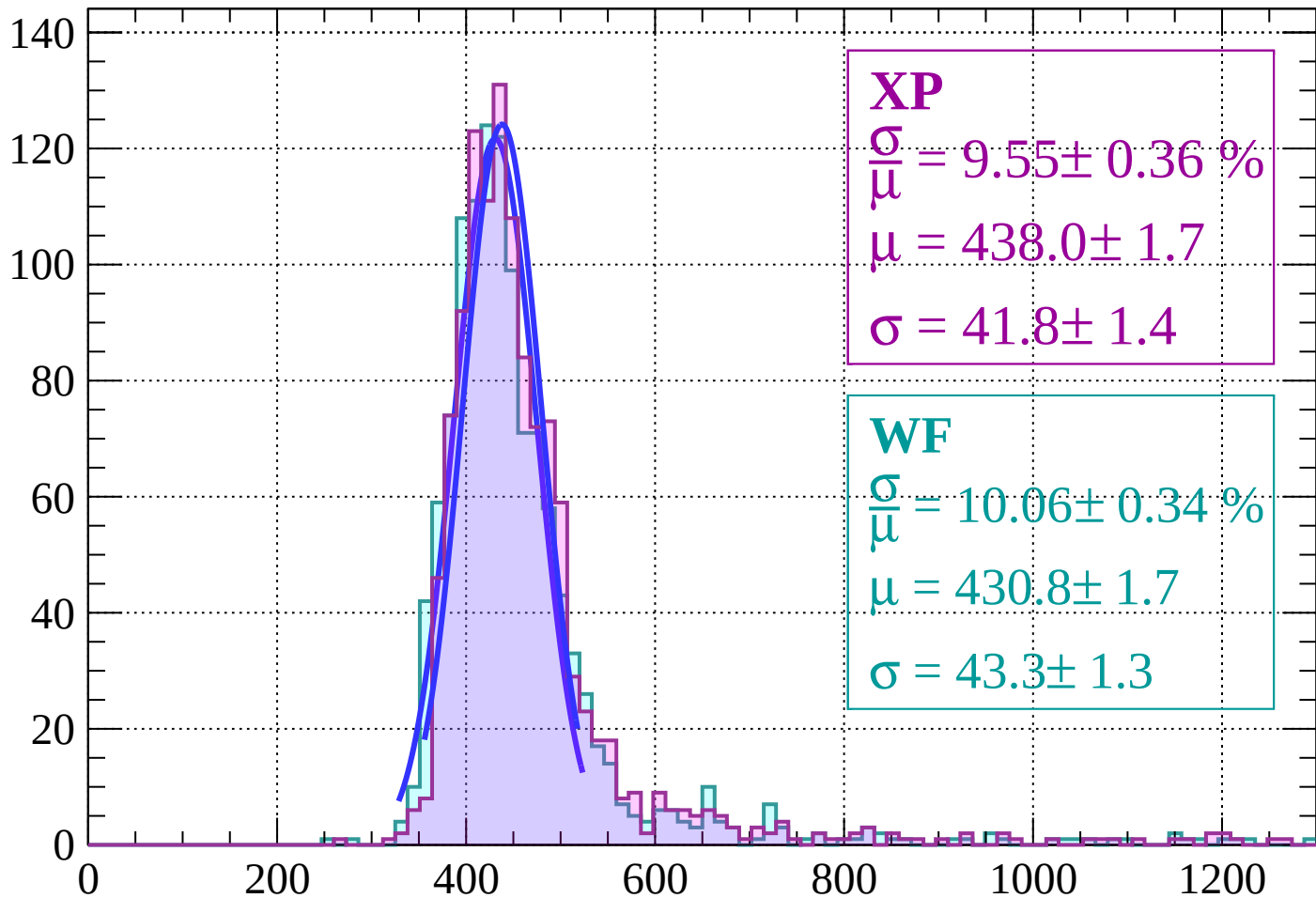
Energy loss $|-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.36 \%$$

$$\mu = 438.0 \pm 1.7$$

$$\sigma = 41.8 \pm 1.4$$

WF

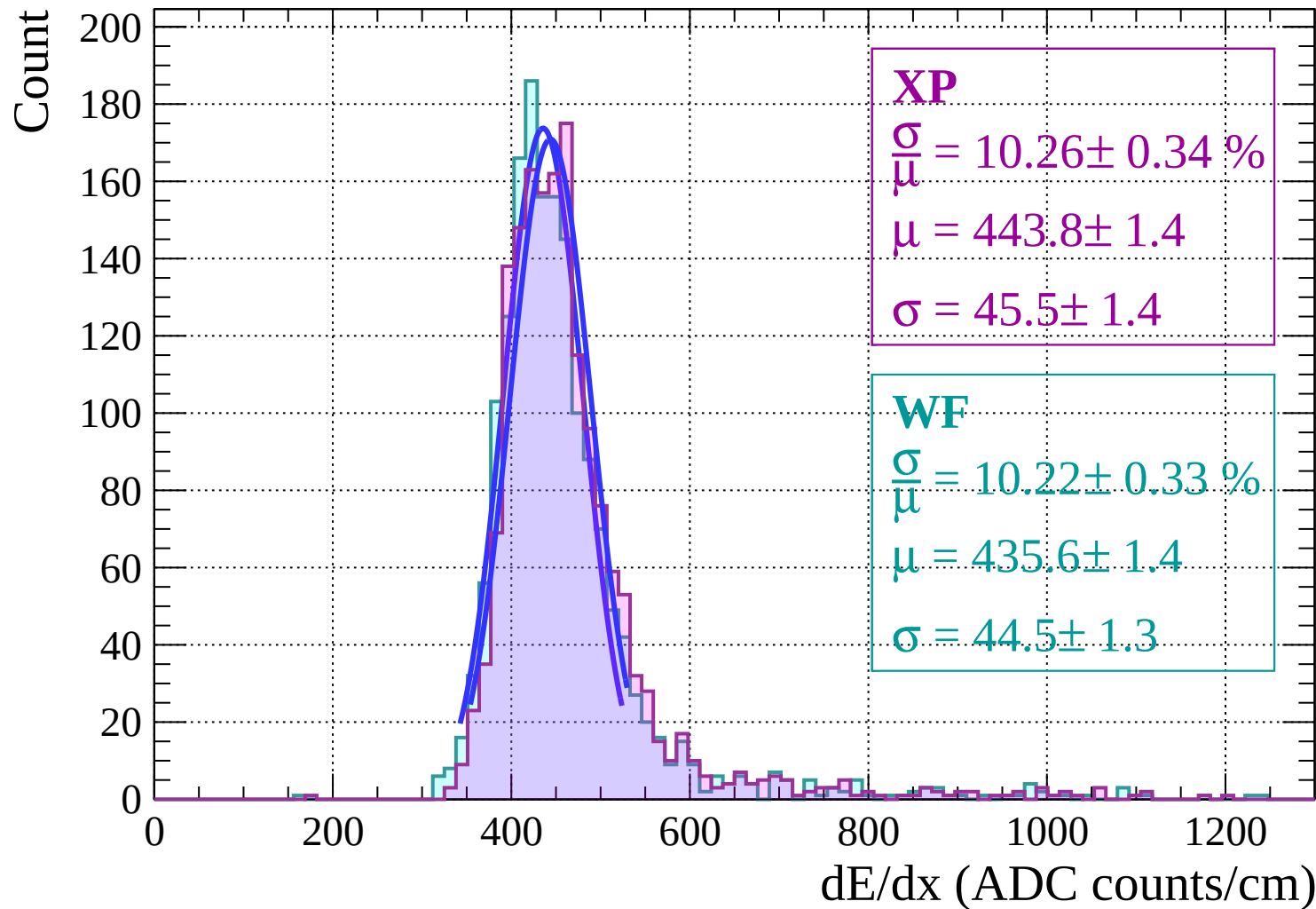
$$\frac{\sigma}{\mu} = 10.06 \pm 0.34 \%$$

$$\mu = 430.8 \pm 1.7$$

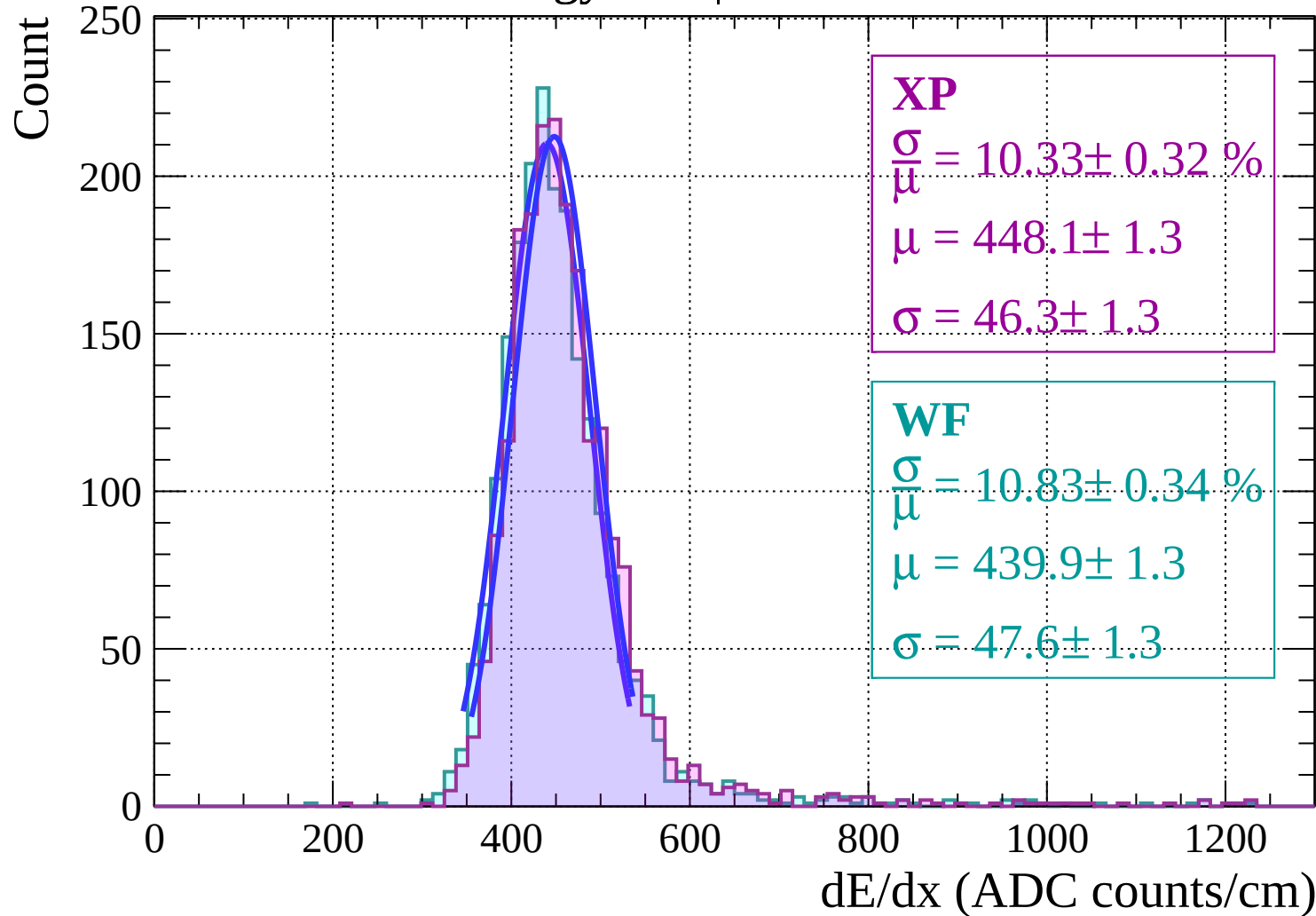
$$\sigma = 43.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

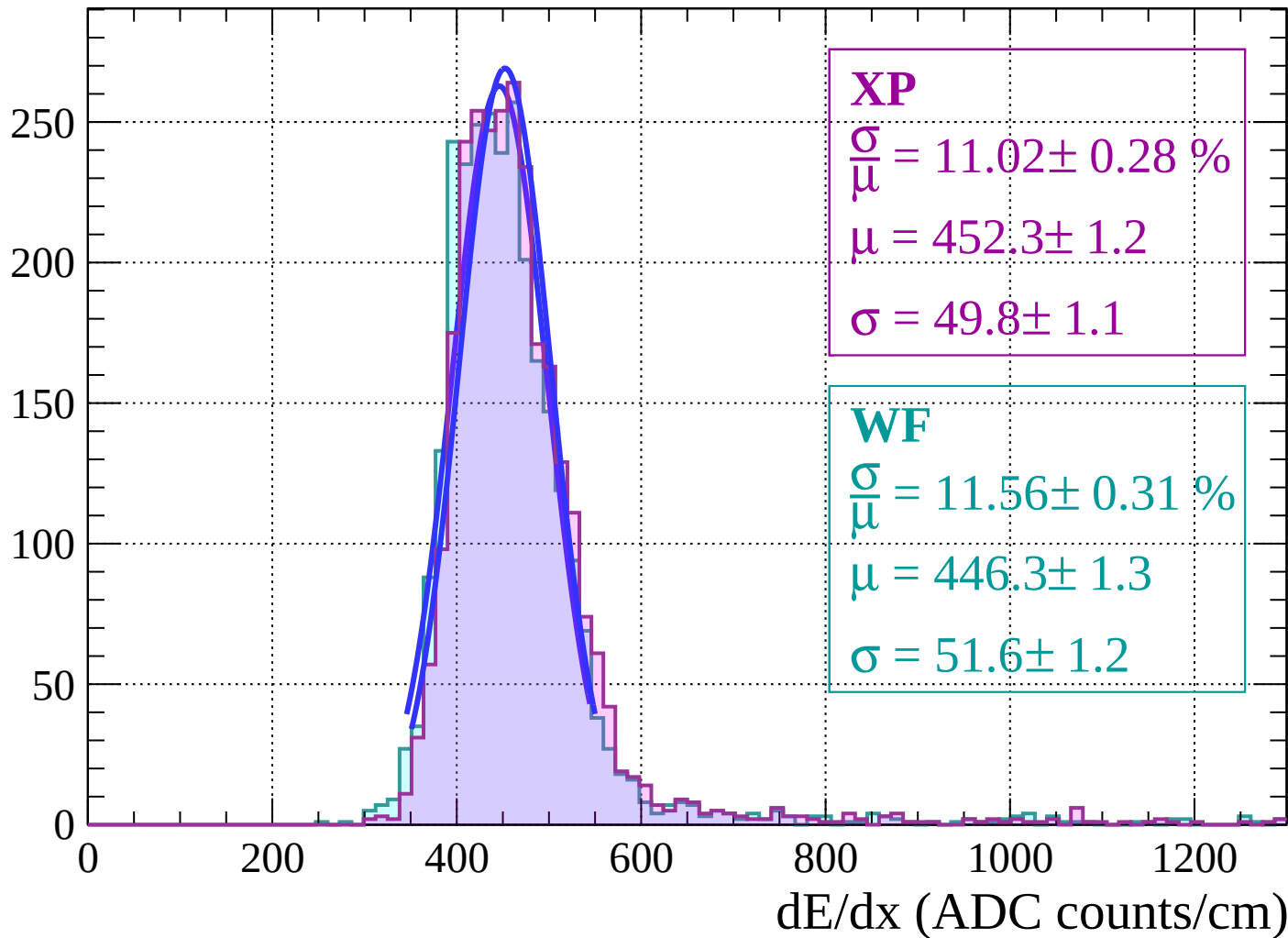


Energy loss | $-16 < \theta < -14$

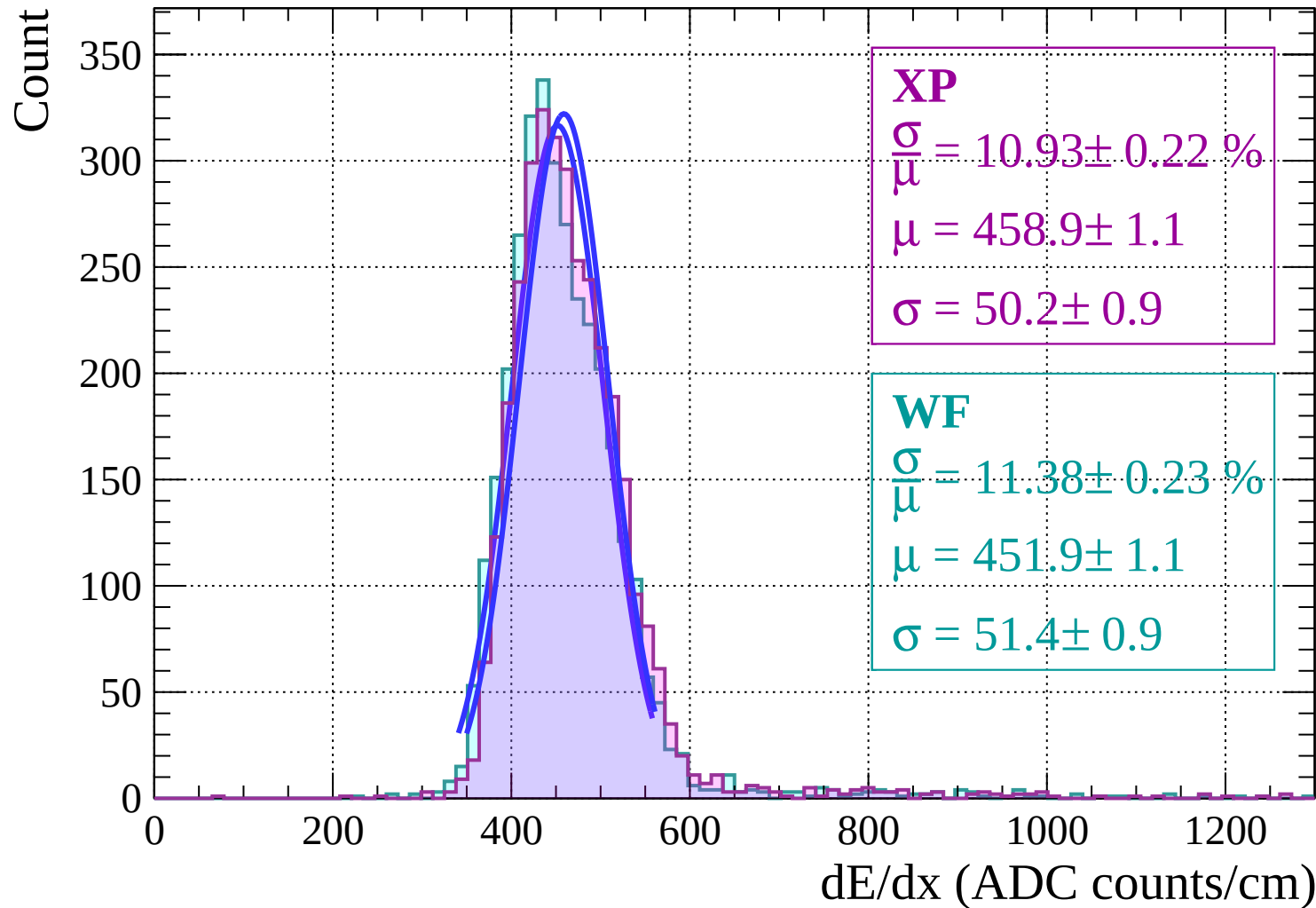


Energy loss $|-14 < \theta < -12$

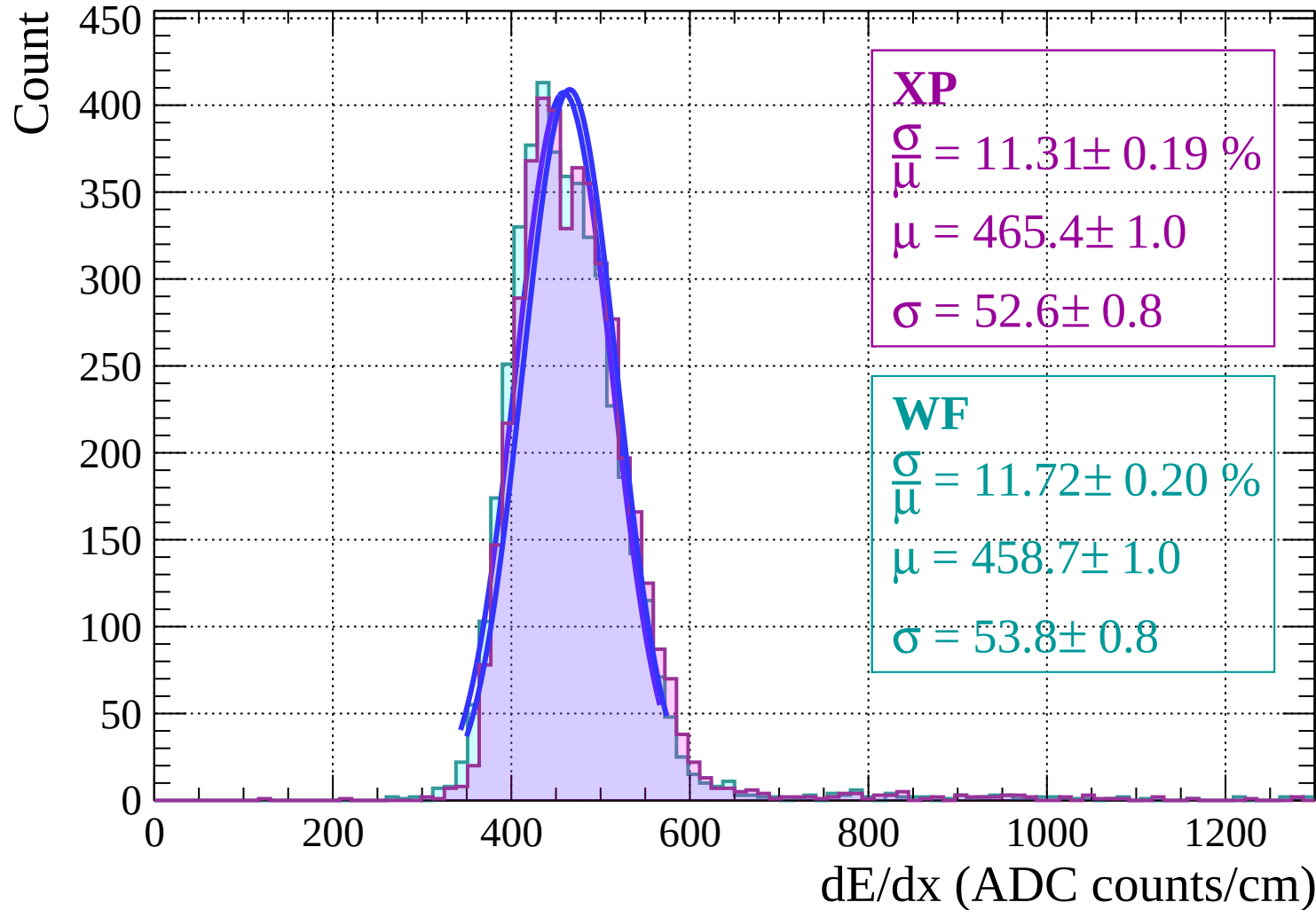
Count



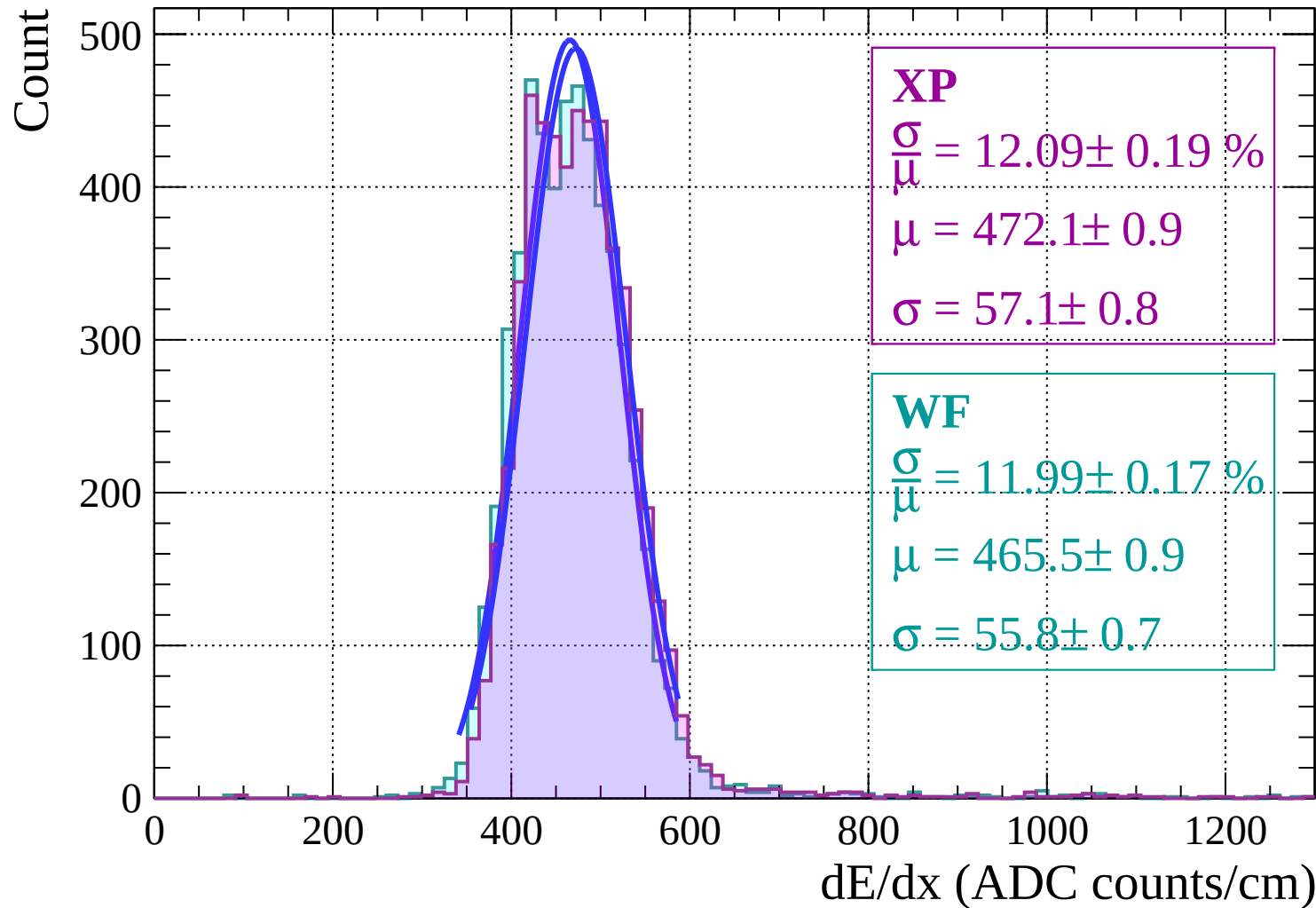
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

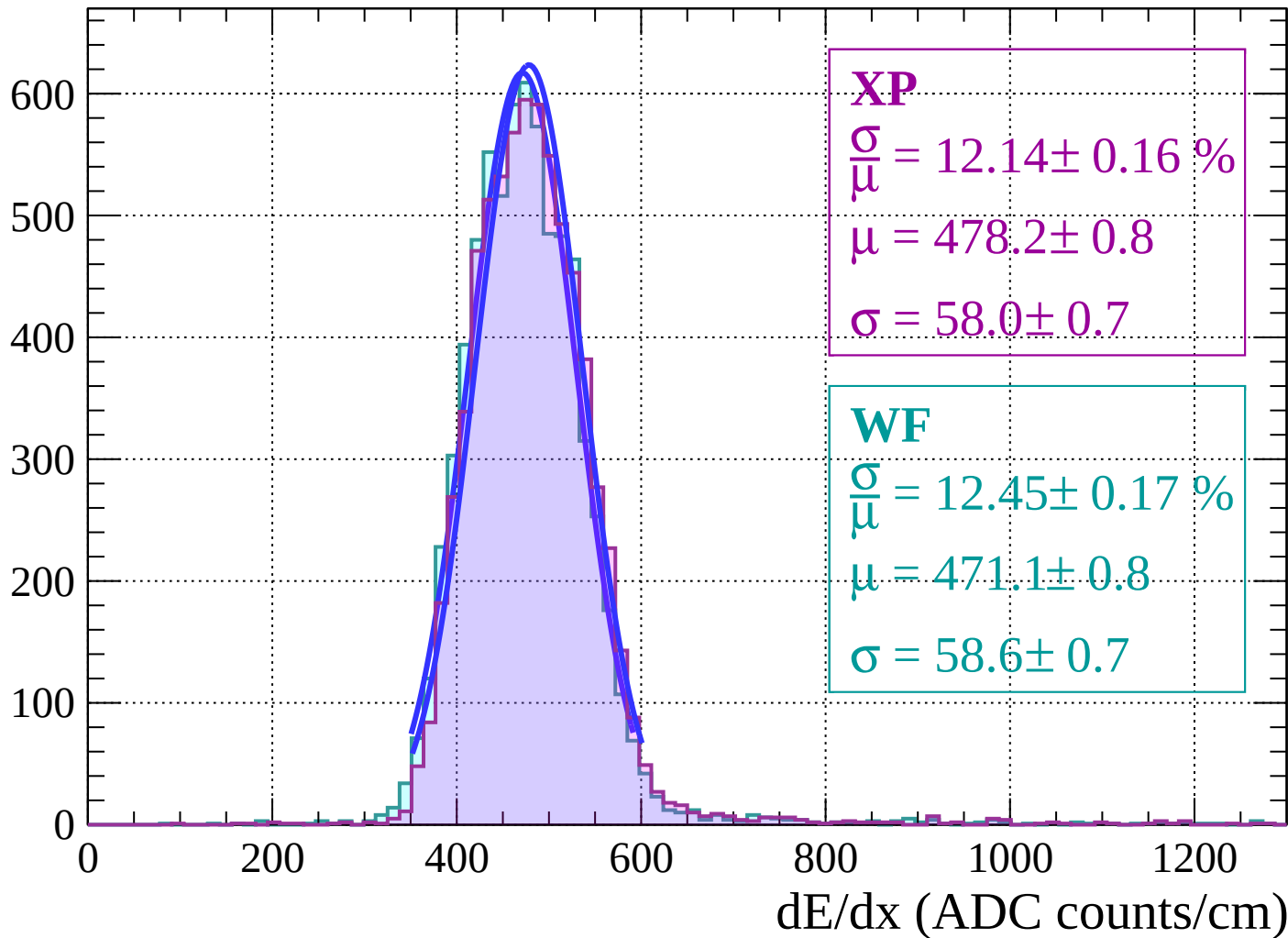


Energy loss | $-8 < \theta < -6$



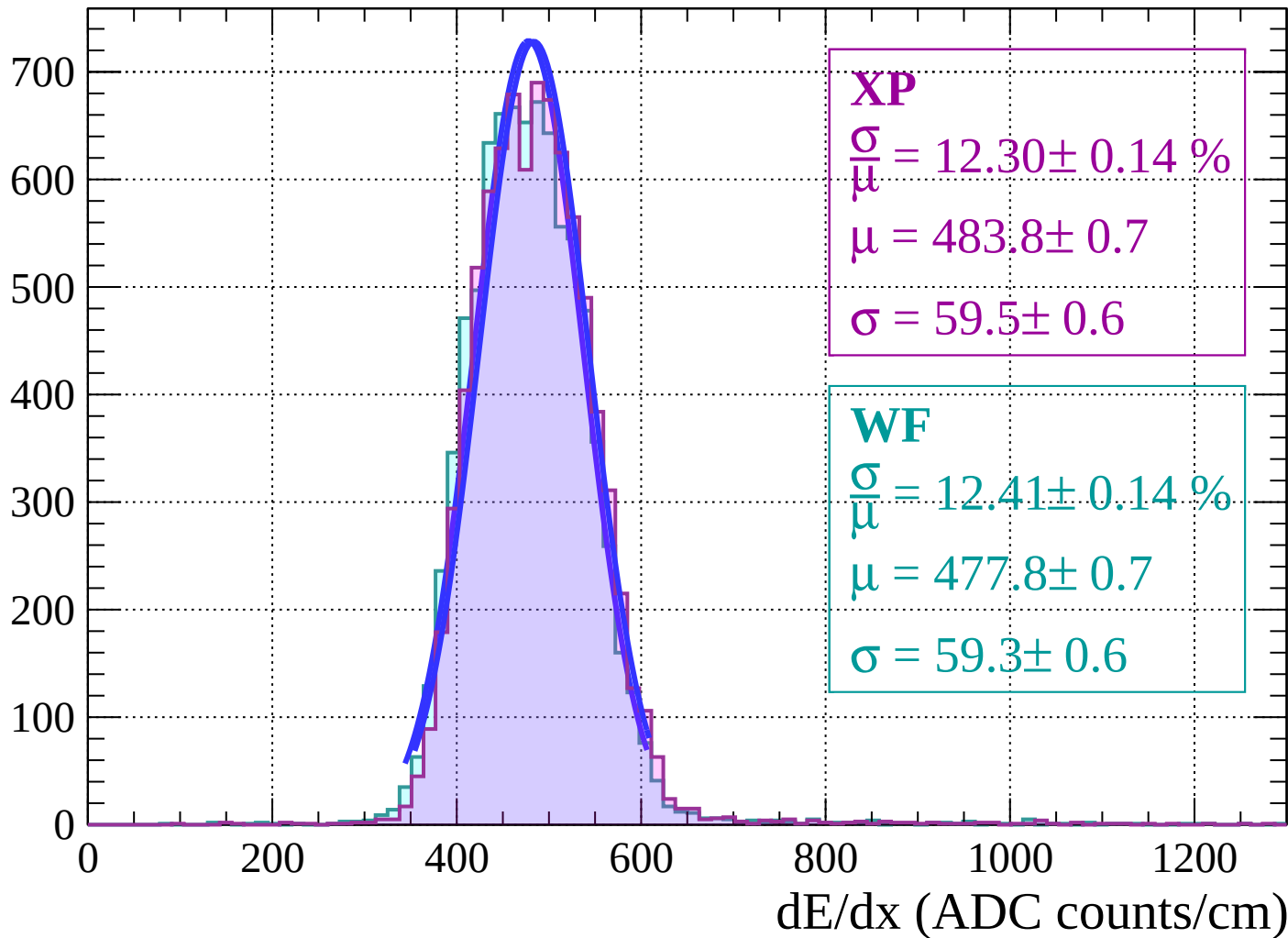
Energy loss | $-6 < \theta < -4$

Count

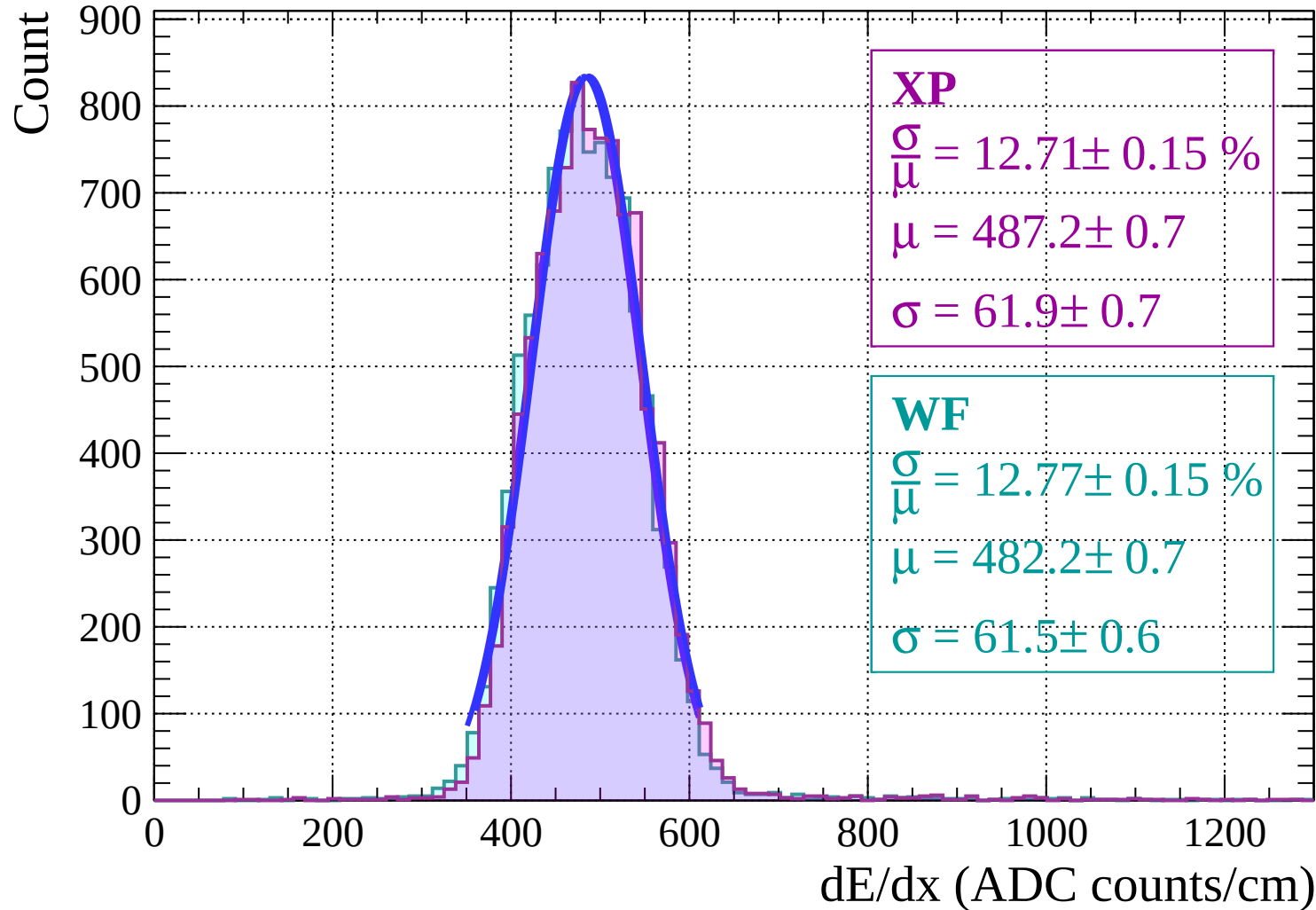


Energy loss $|-4 < \theta < -2$

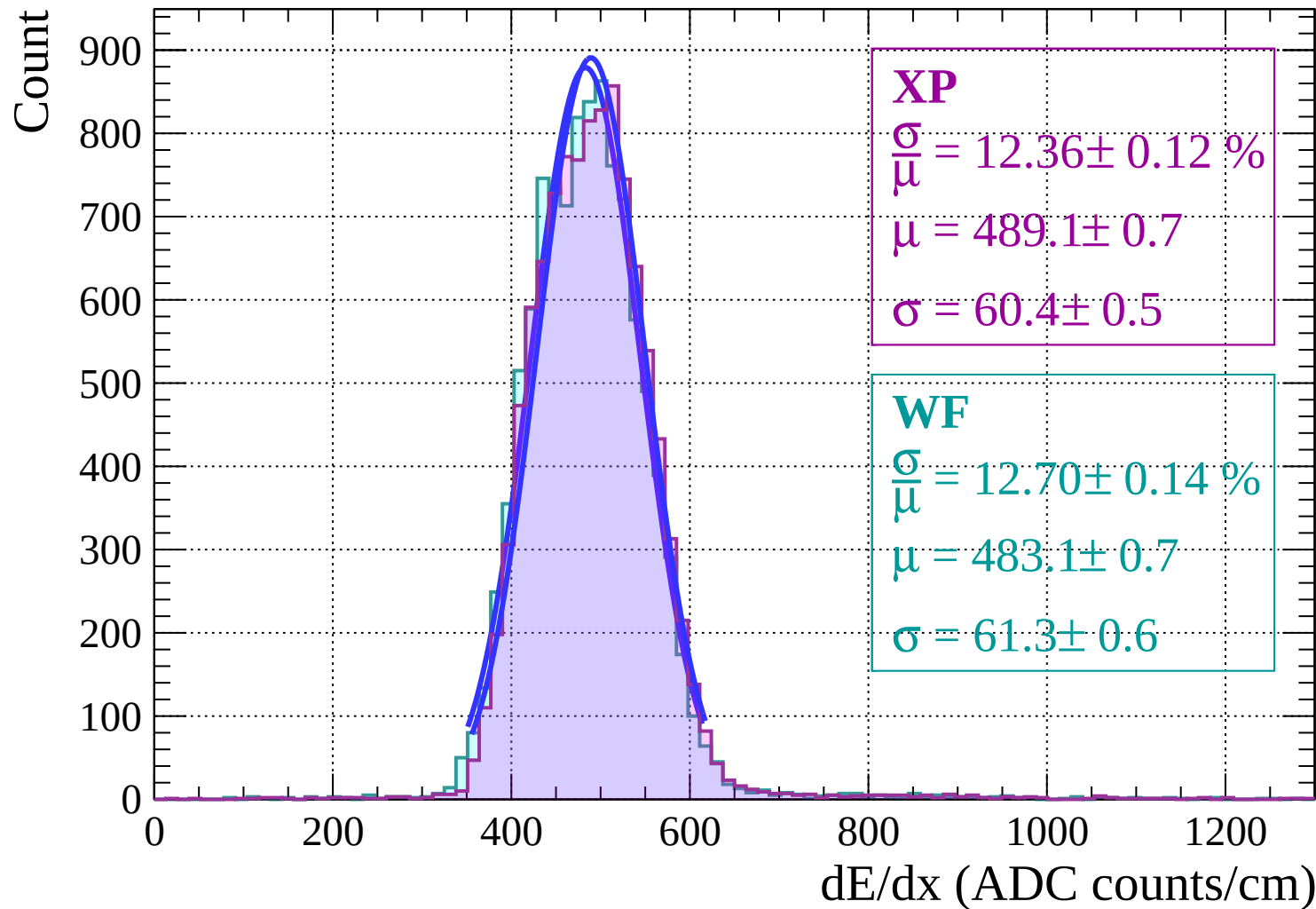
Count



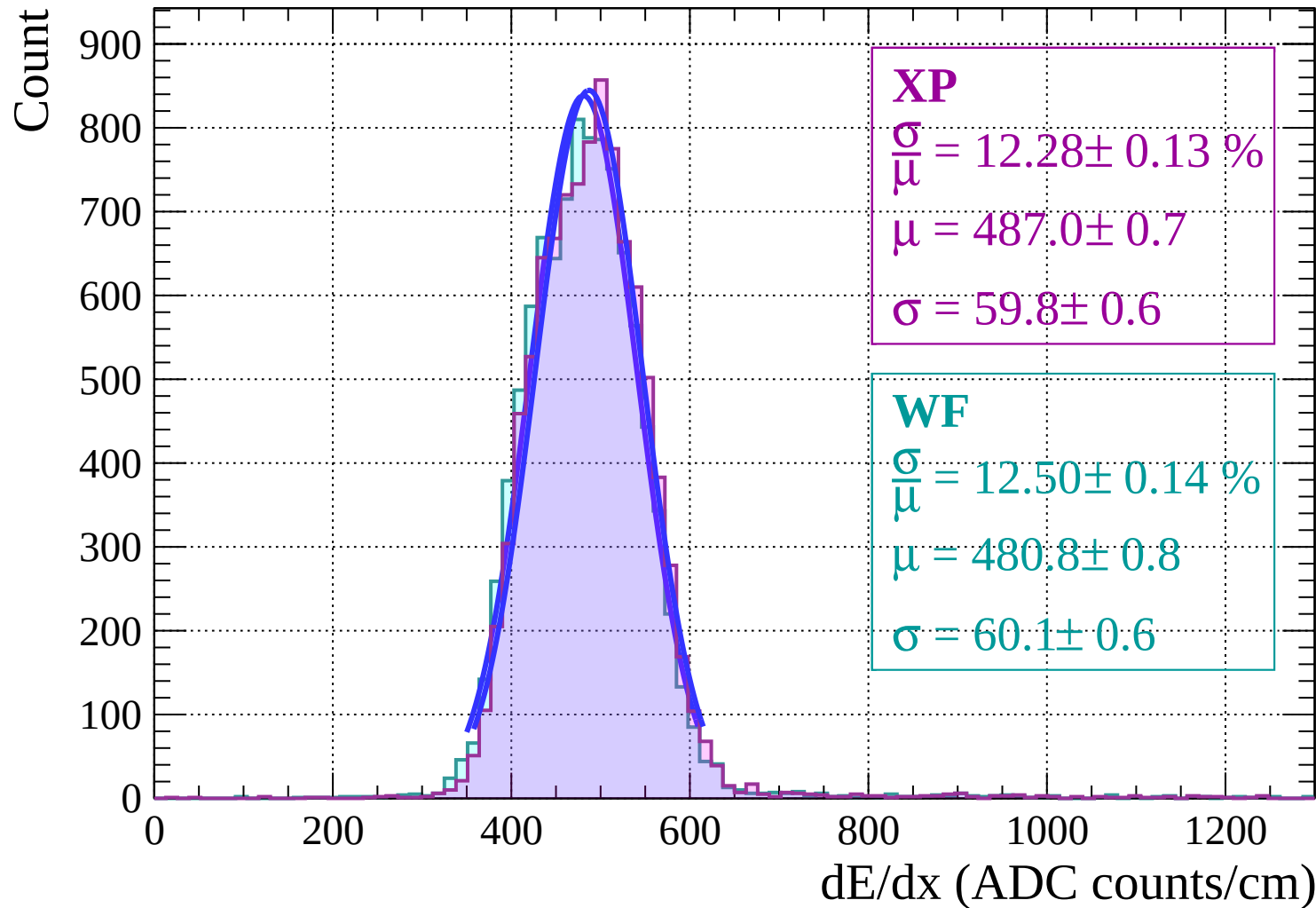
Energy loss | $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

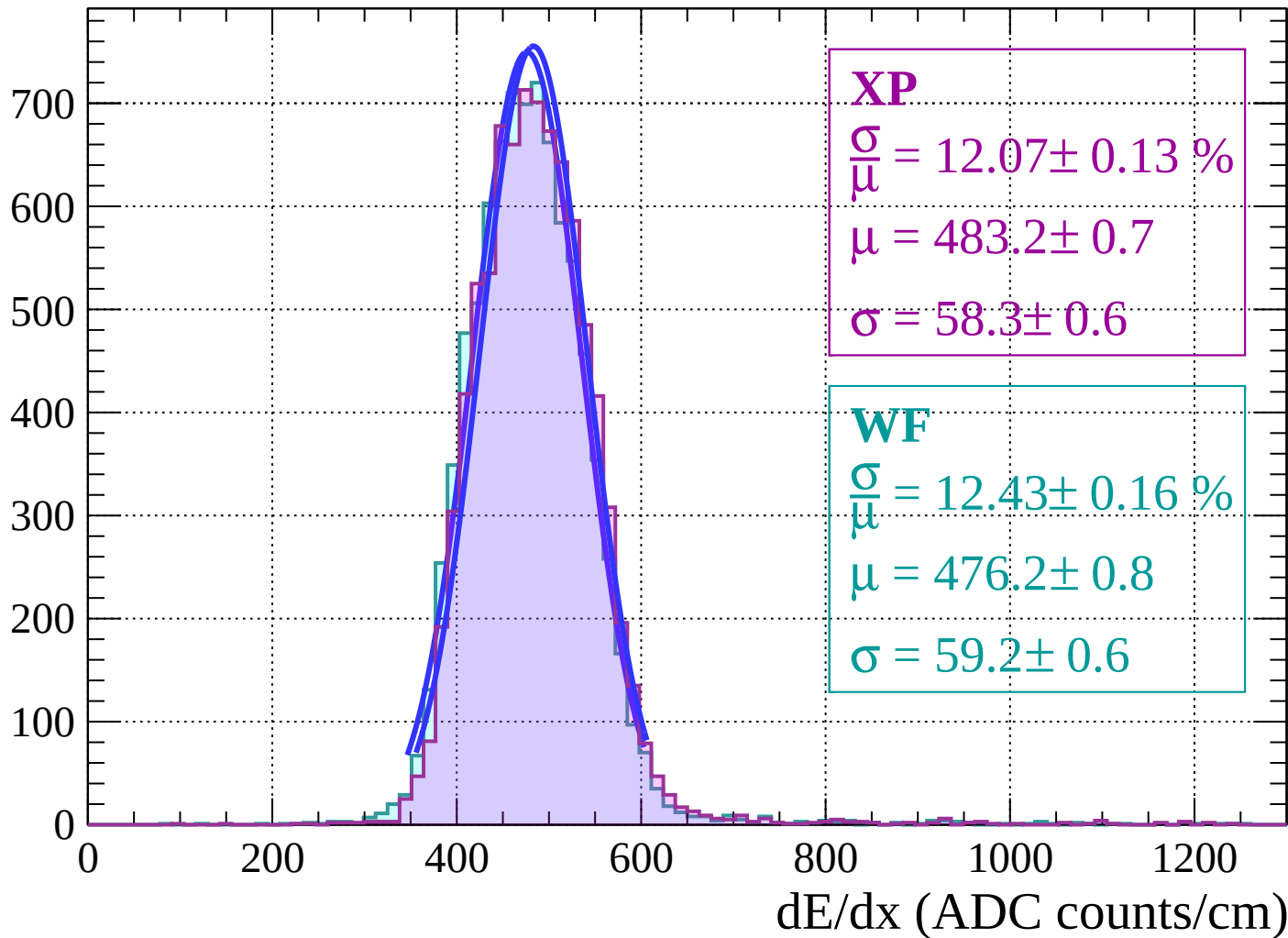


Energy loss | $2 < \theta < 4$



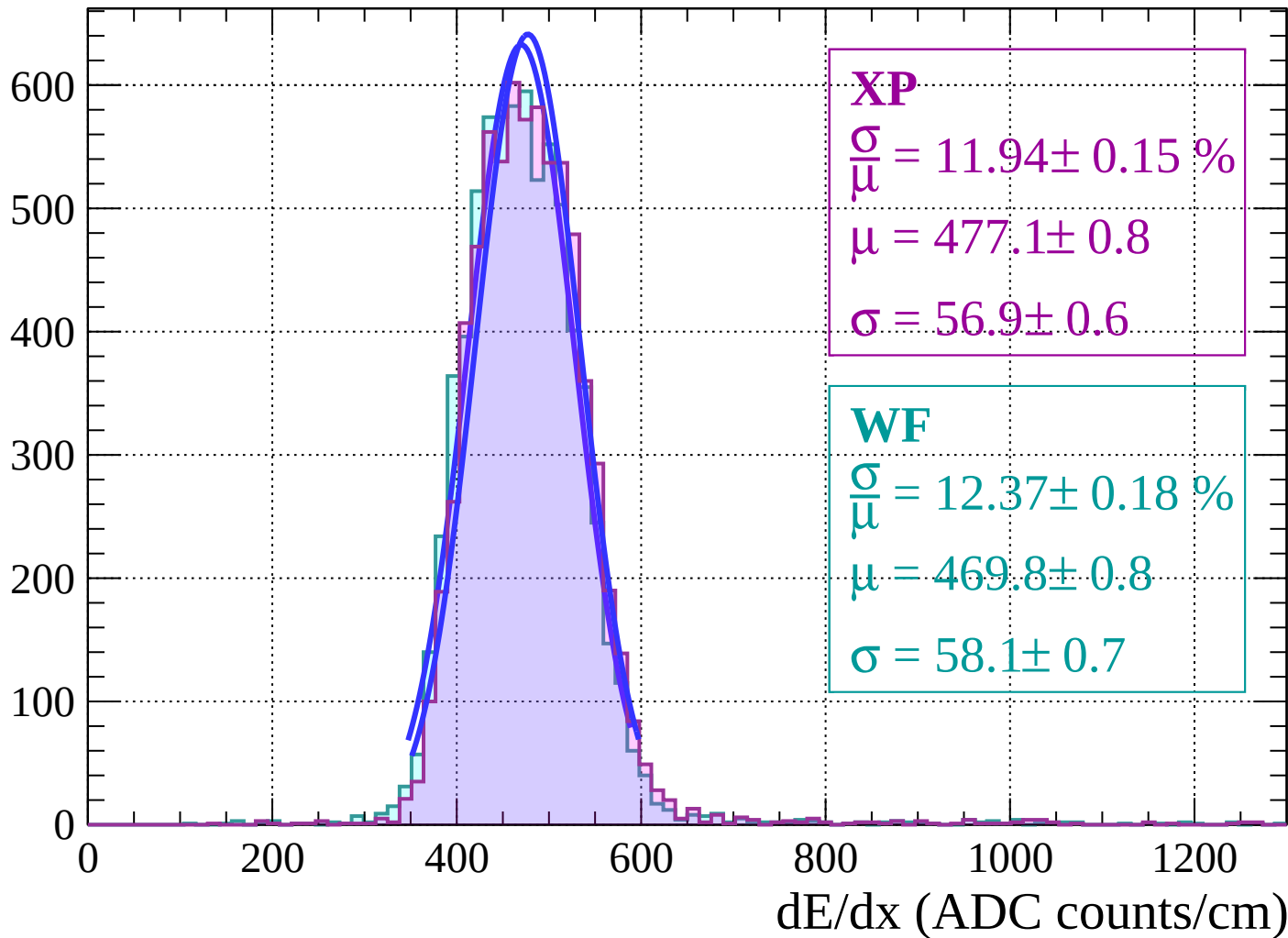
Energy loss | $4 < \theta < 6$

Count



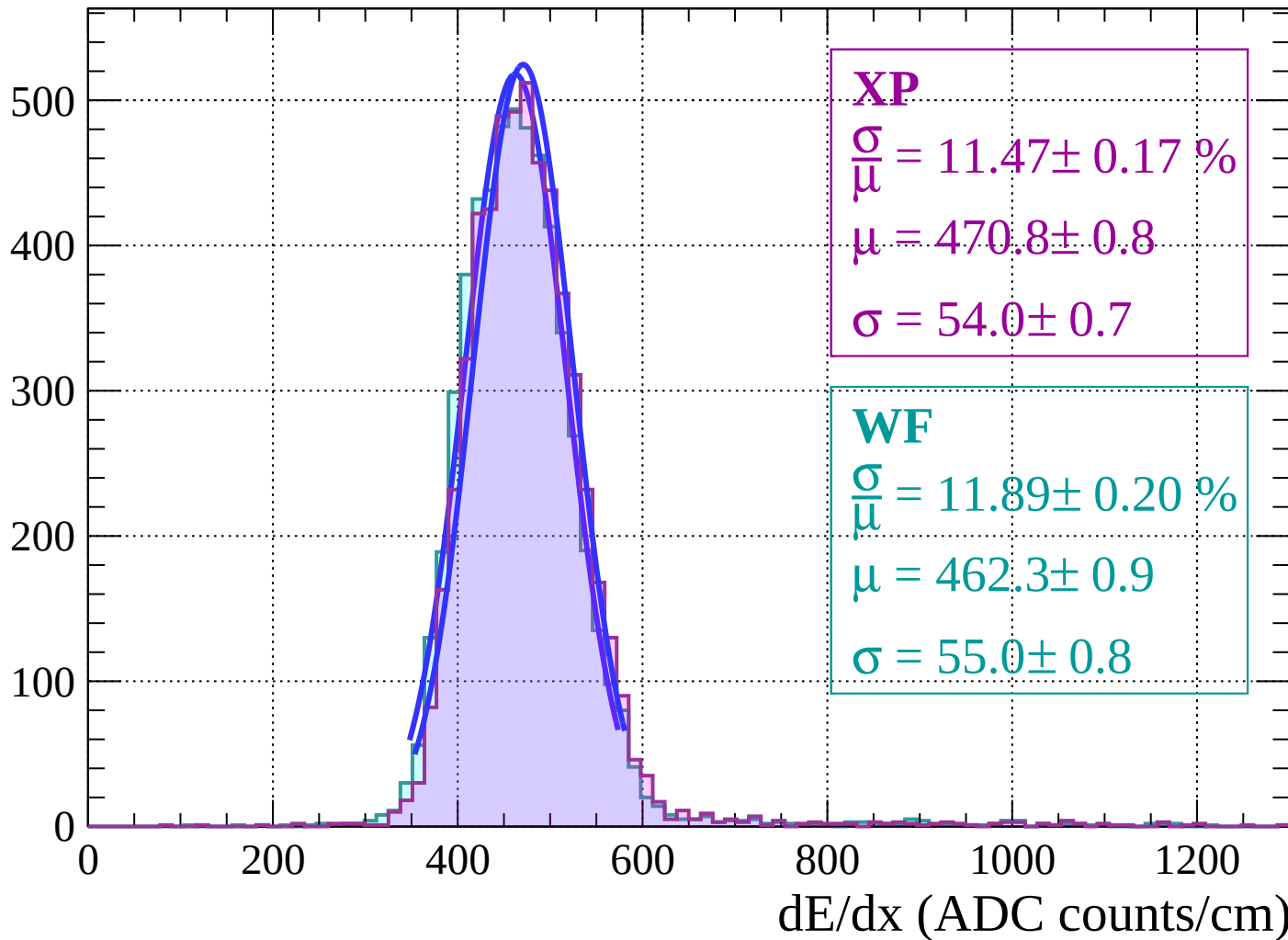
Energy loss | $6 < \theta < 8$

Count

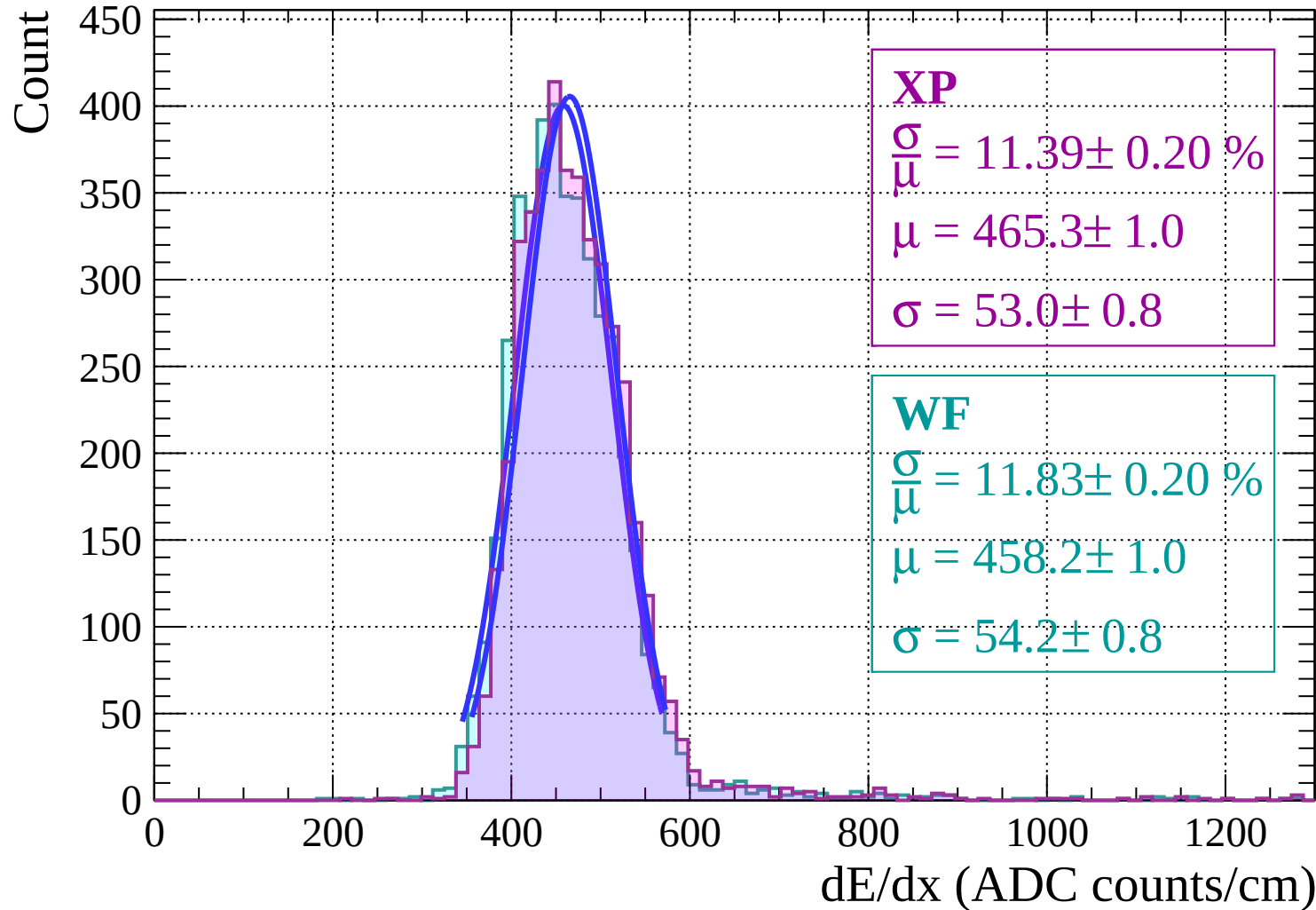


Energy loss | $8 < \theta < 10$

Count

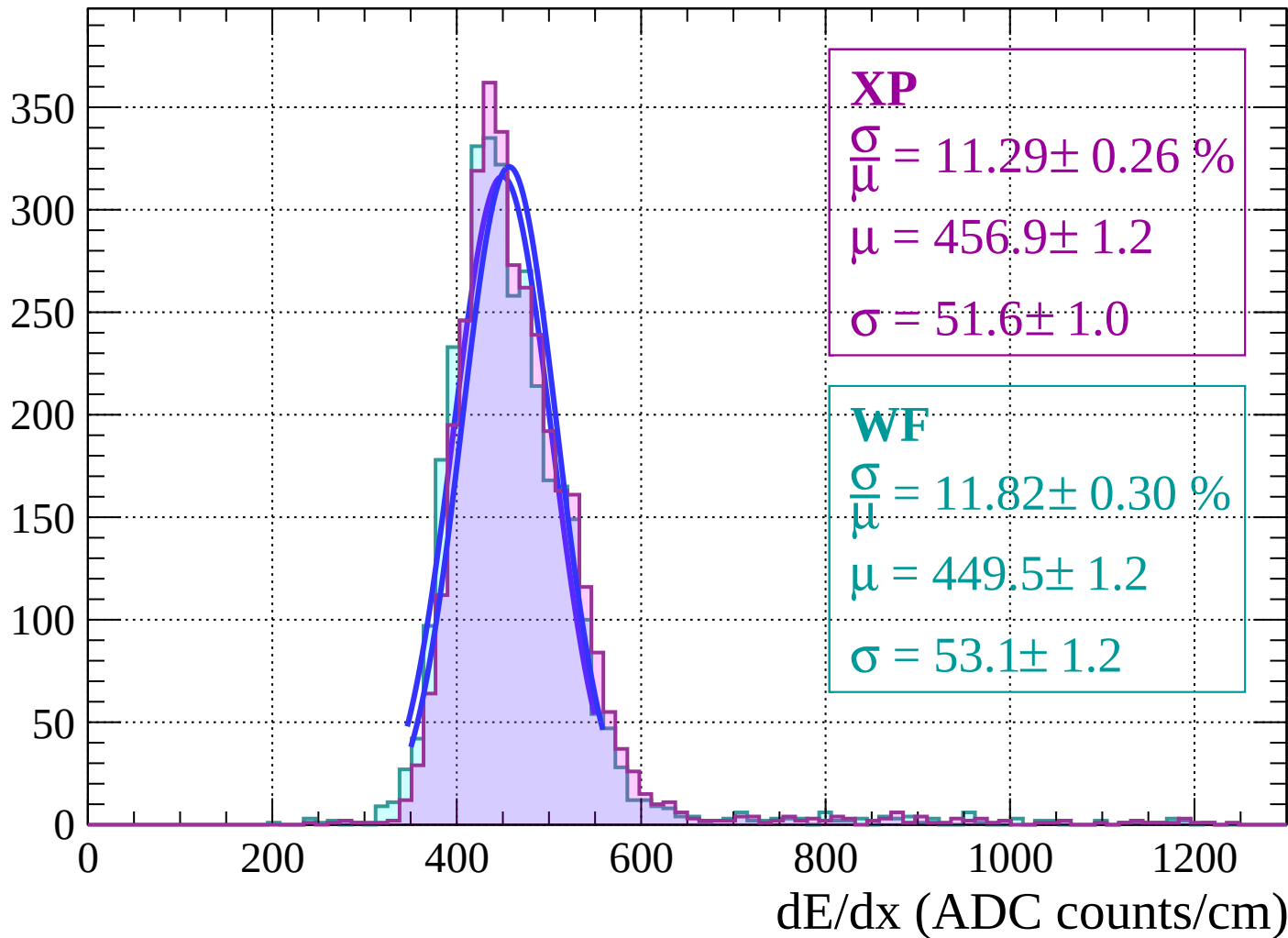


Energy loss | $10 < \theta < 12$



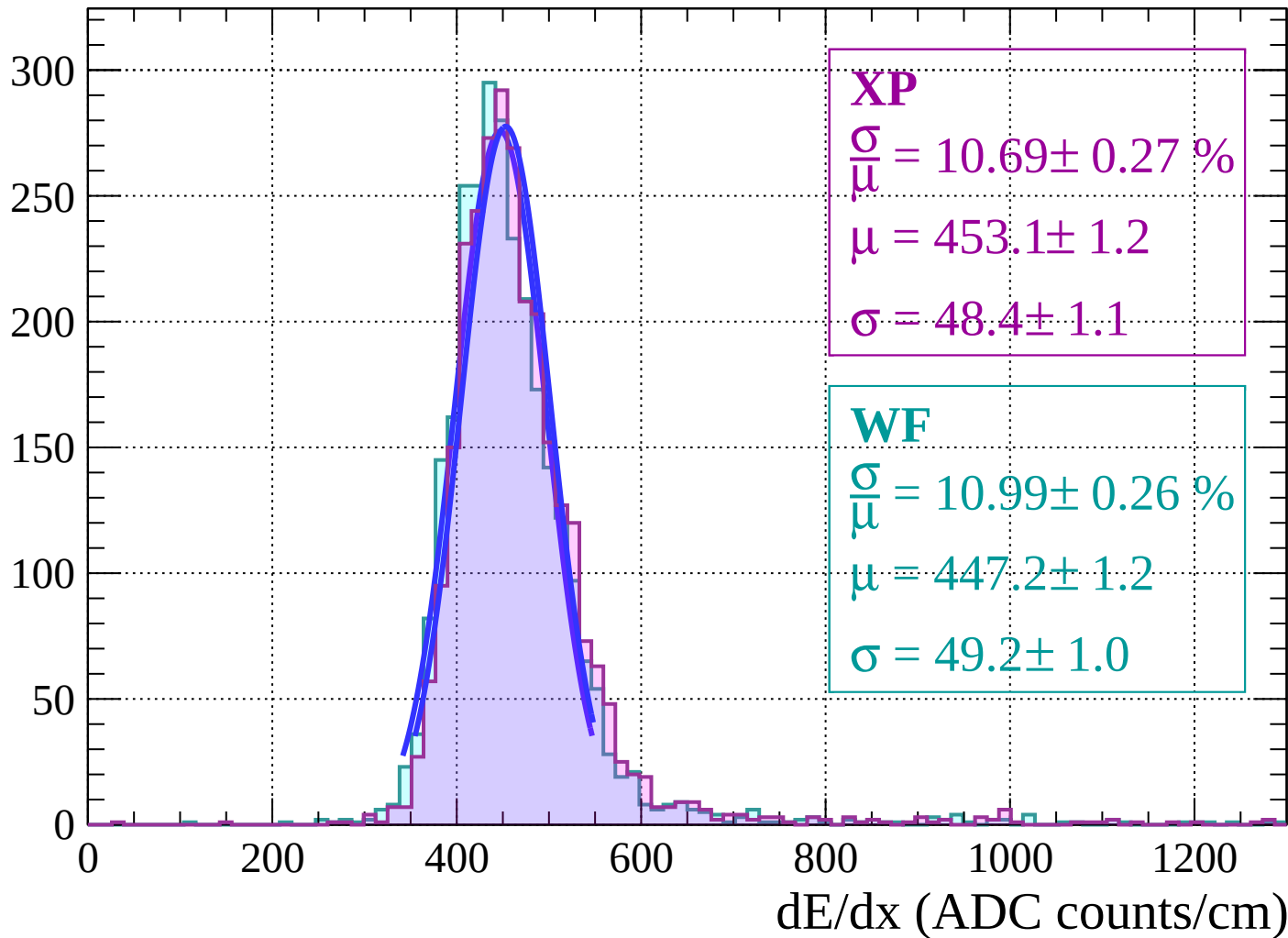
Energy loss | $12 < \theta < 14$

Count



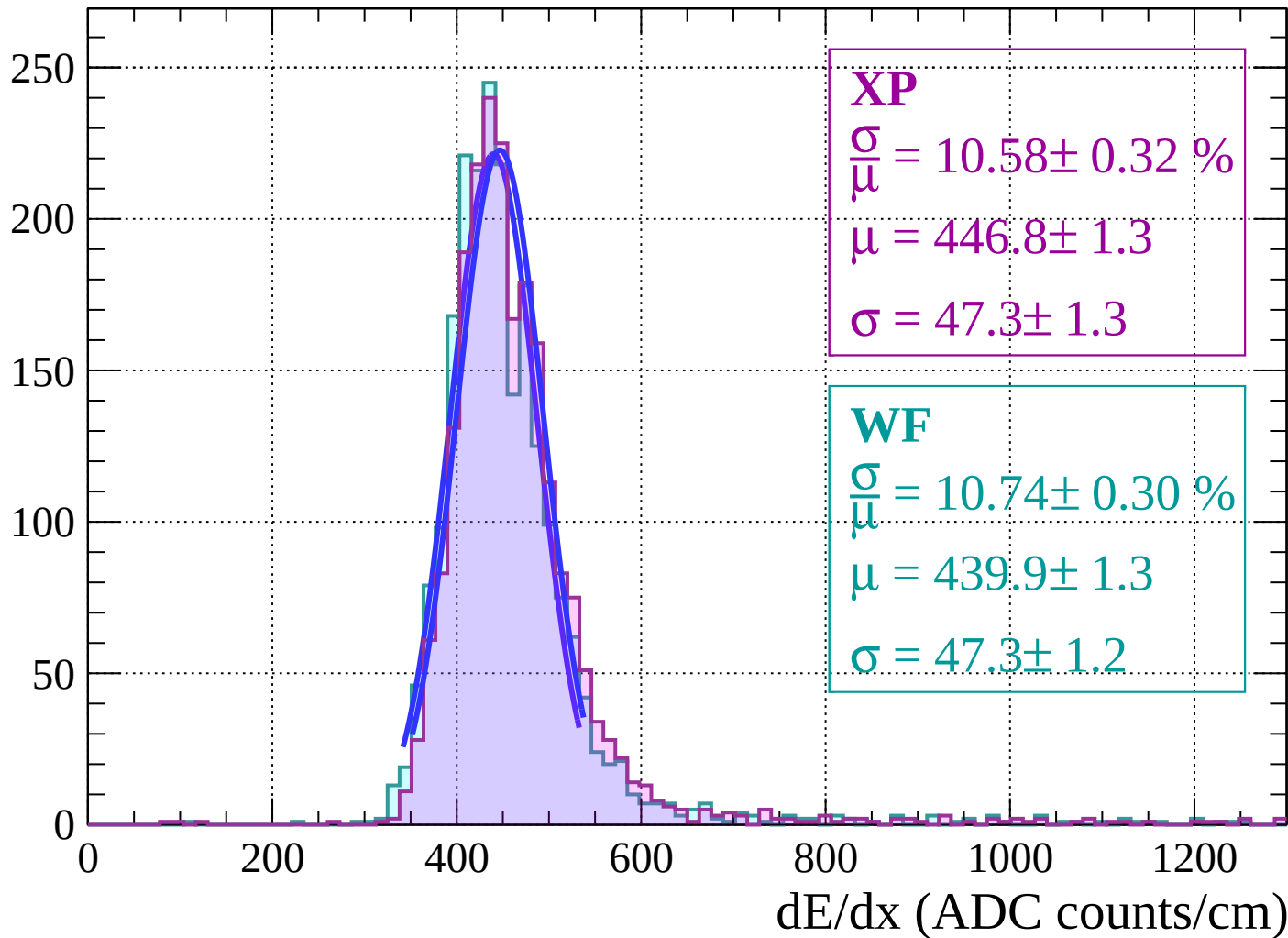
Energy loss | $14 < \theta < 16$

Count

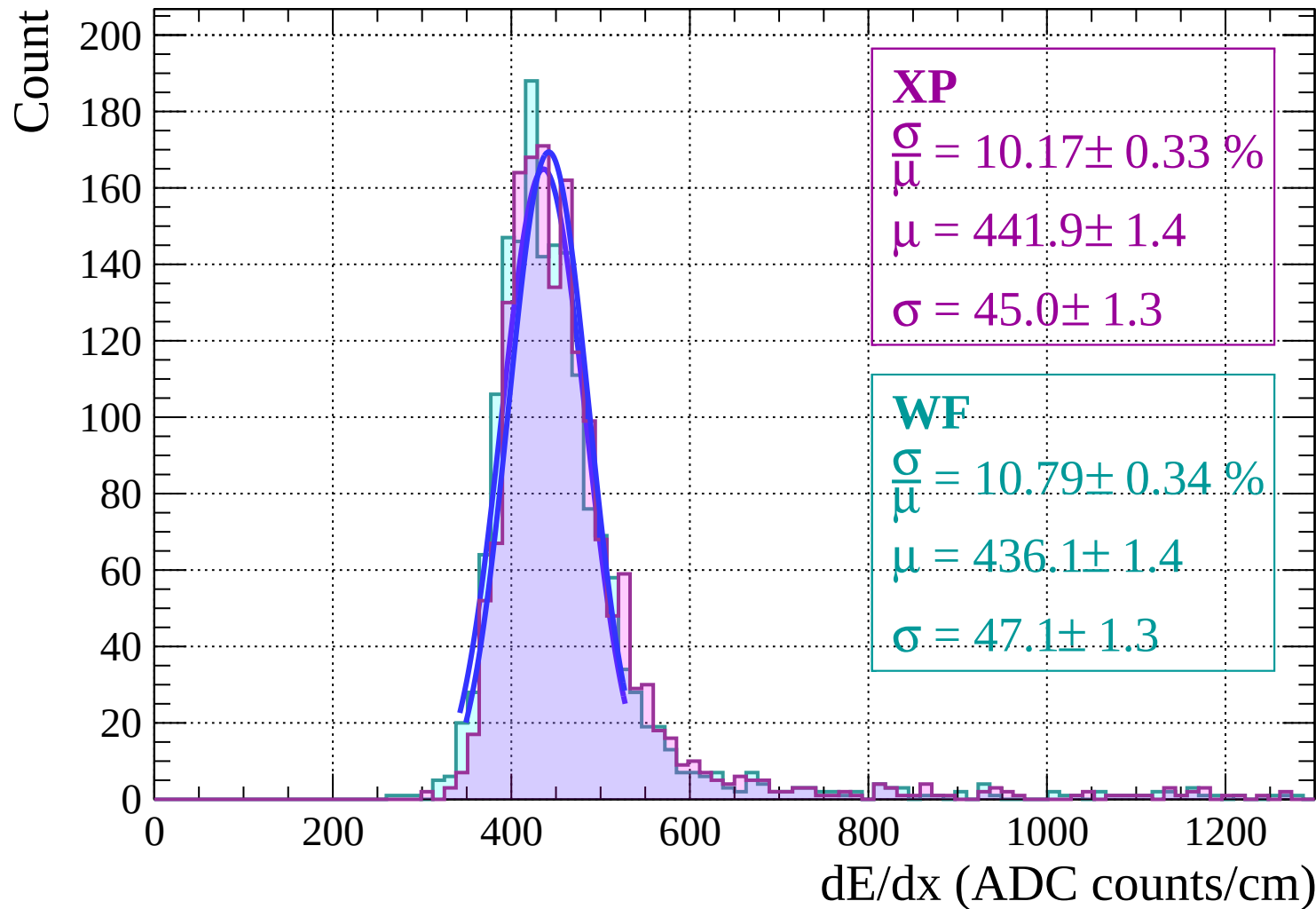


Energy loss | $16 < \theta < 18$

Count

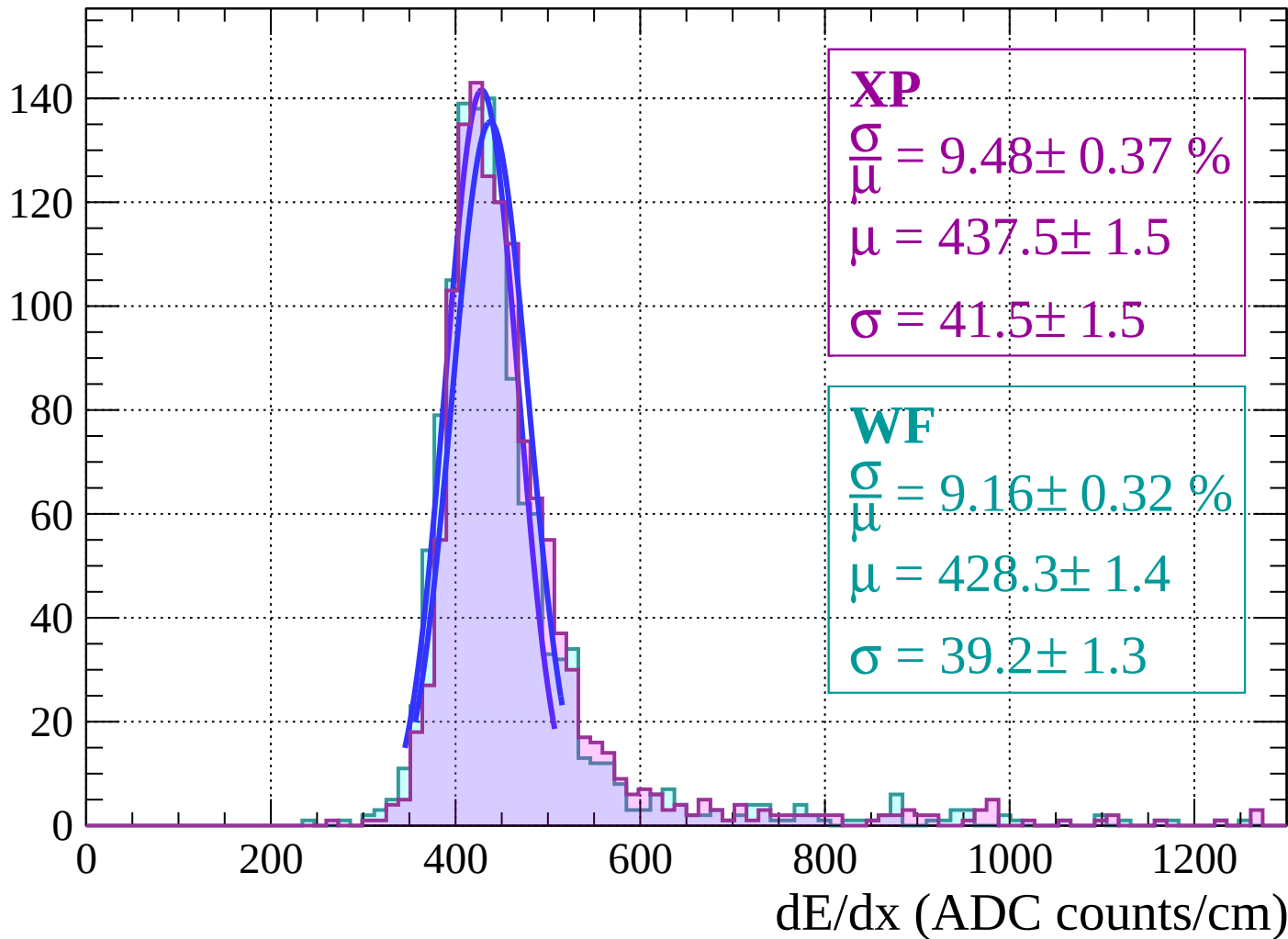


Energy loss | $18 < \theta < 20$



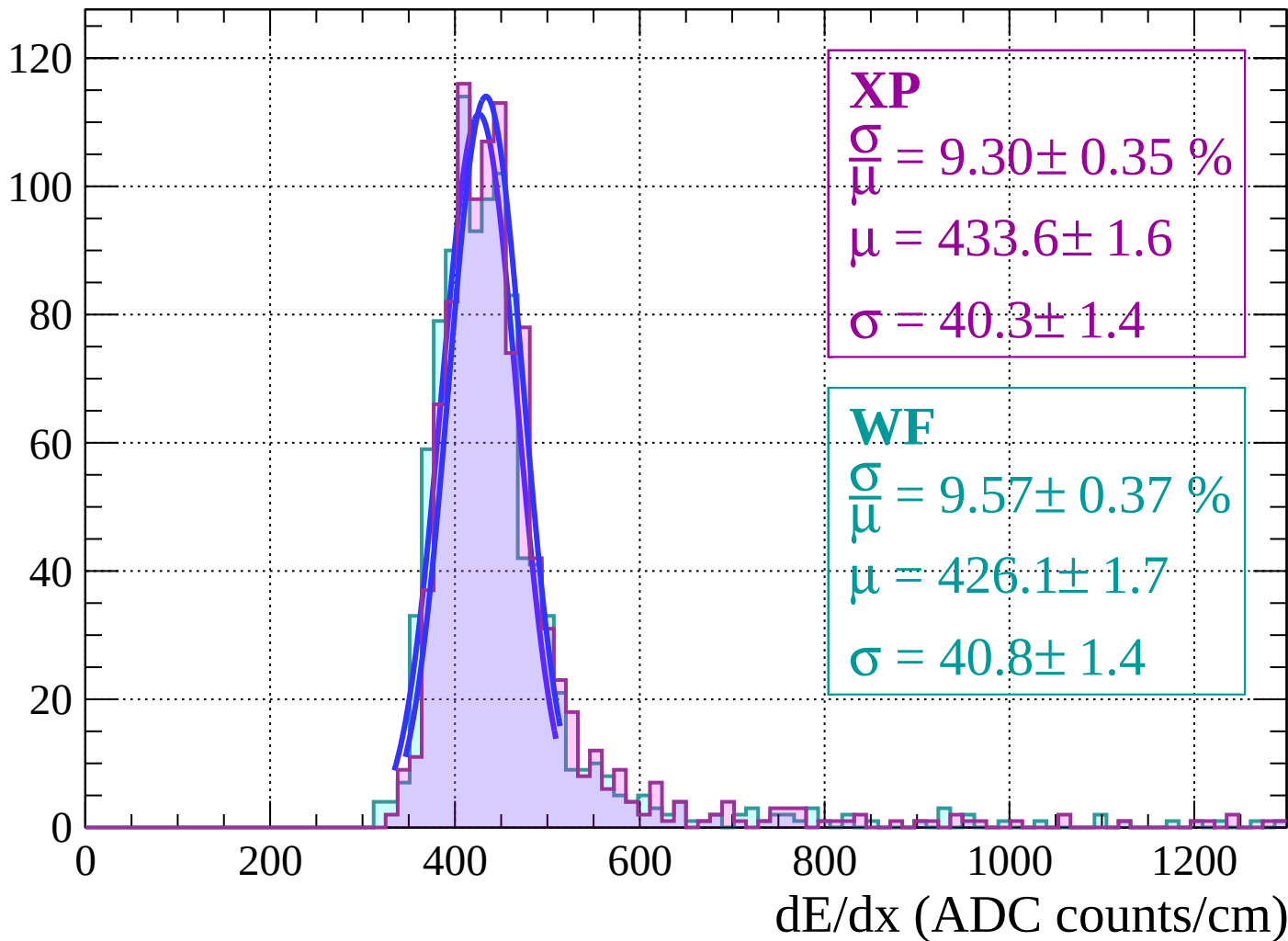
Energy loss | $20 < \theta < 22$

Count



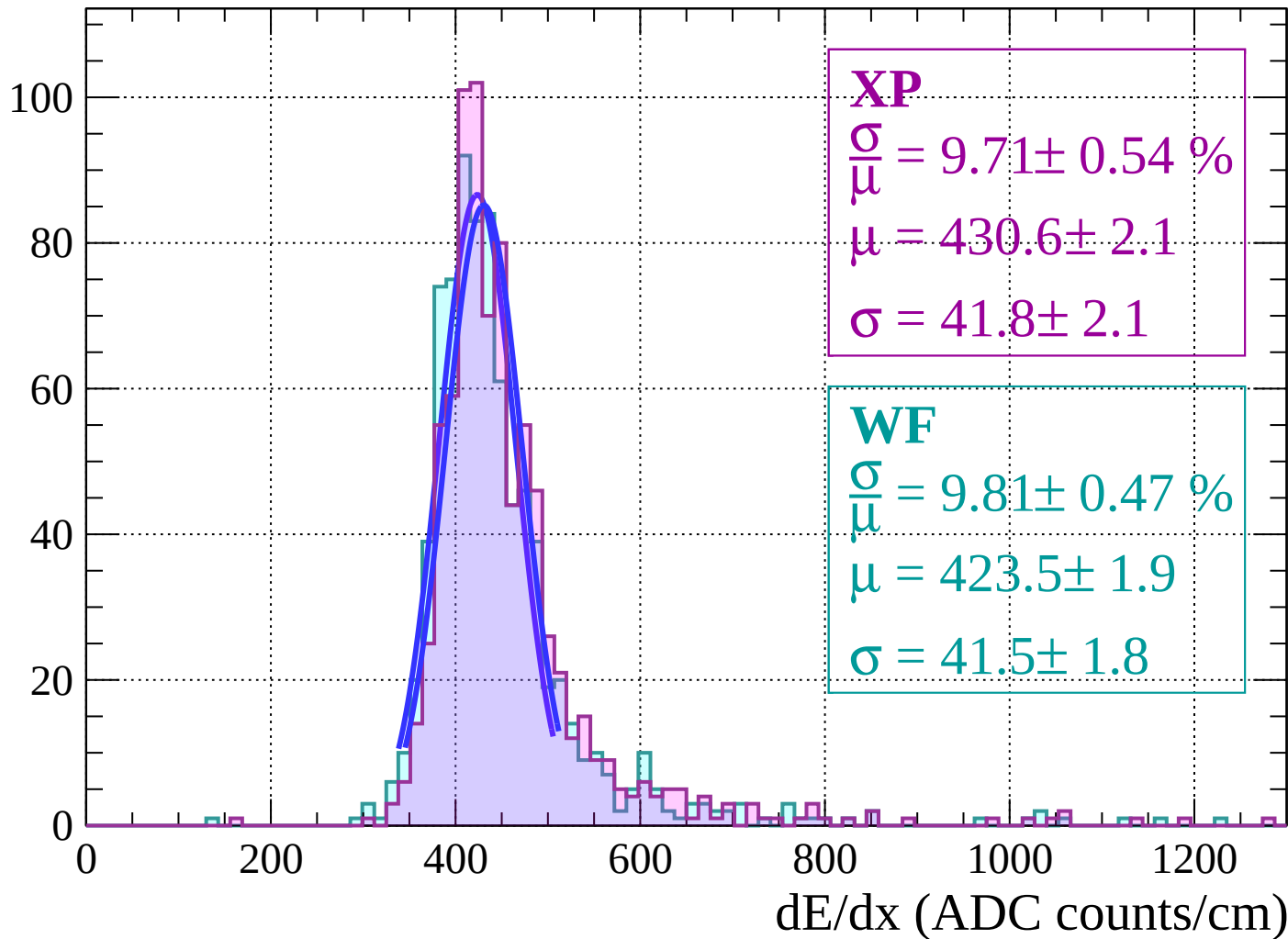
Energy loss | $22 < \theta < 24$

Count



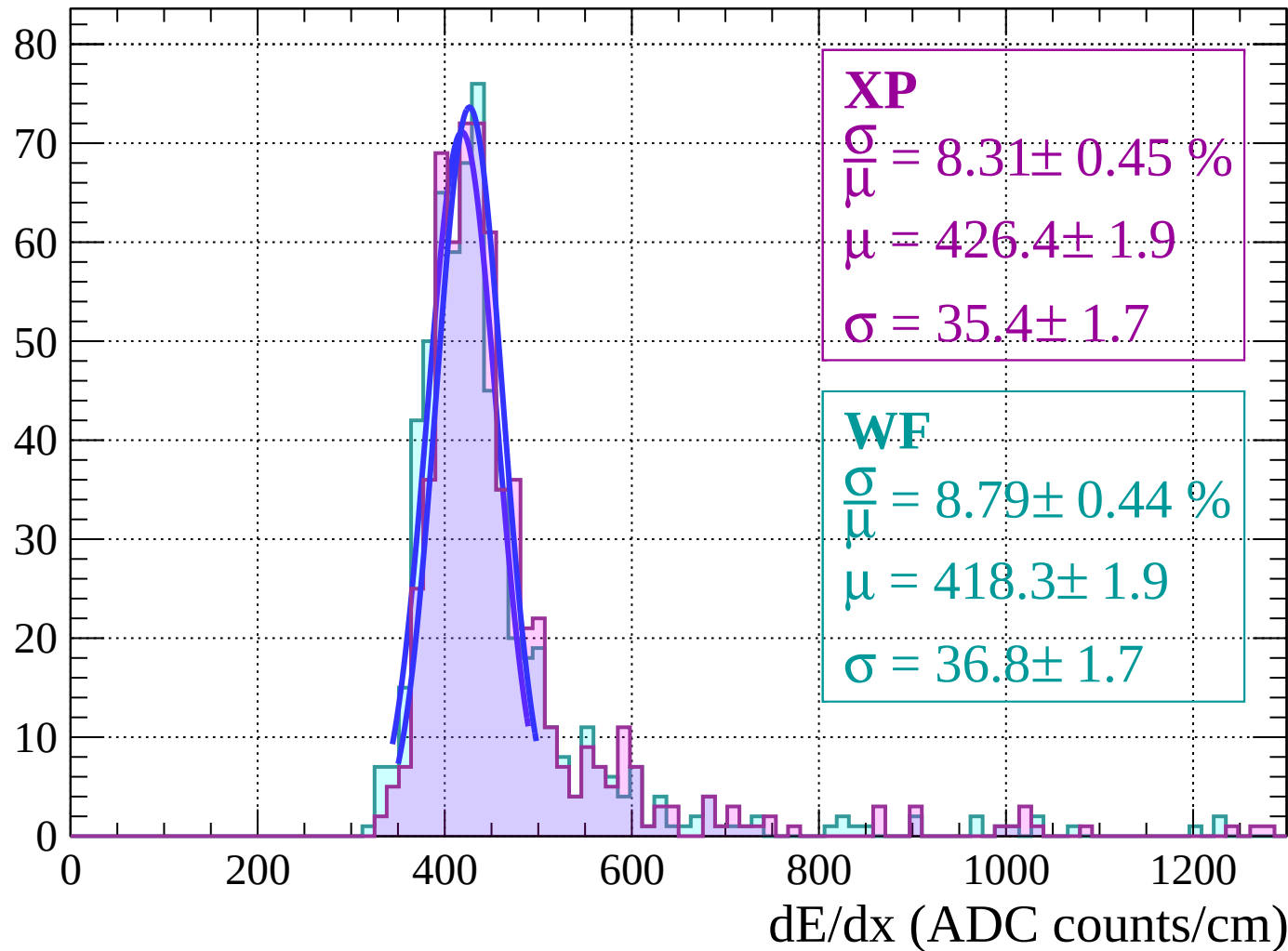
Energy loss | $24 < \theta < 26$

Count



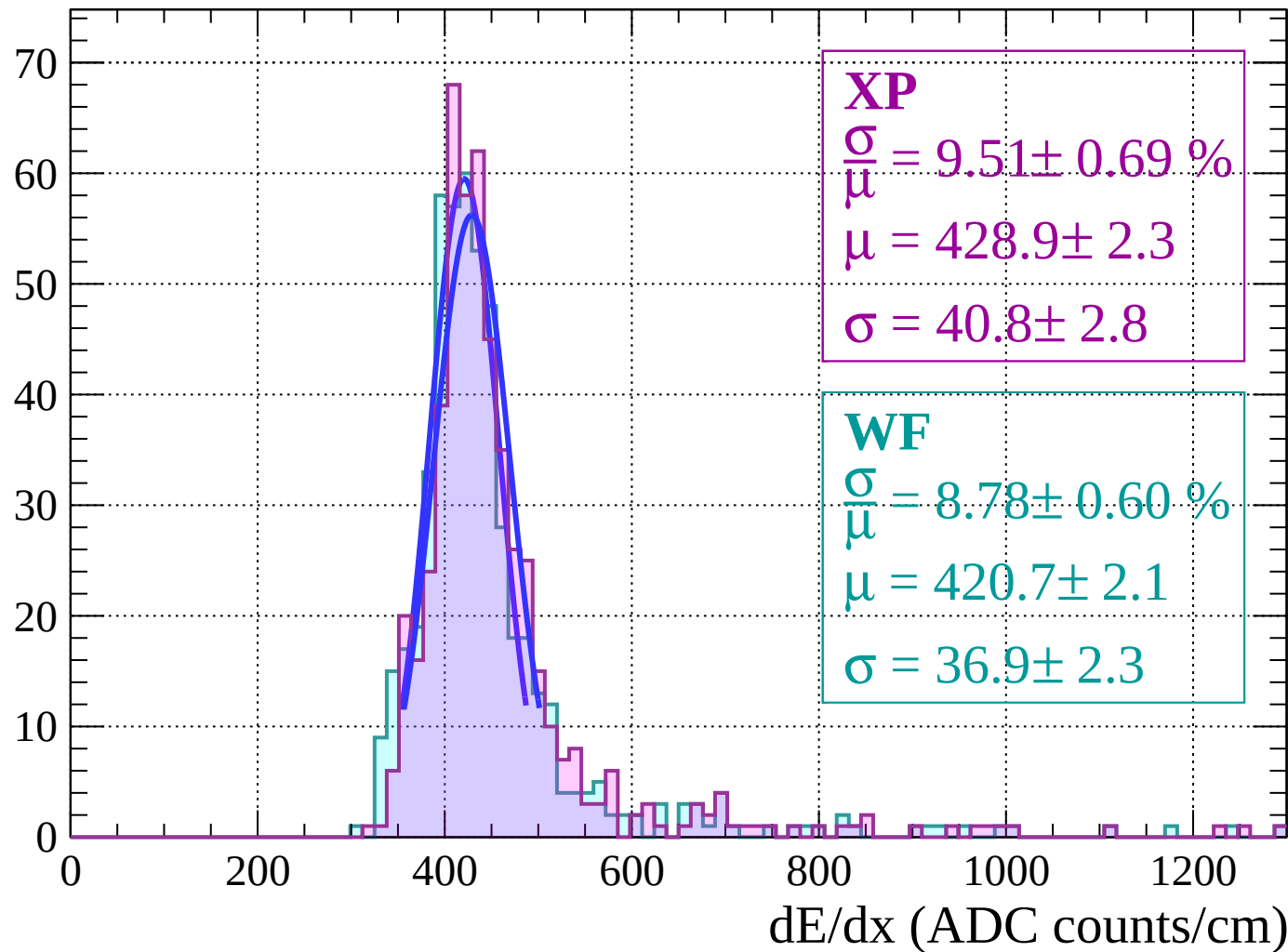
Energy loss | $26 < \theta < 28$

Count



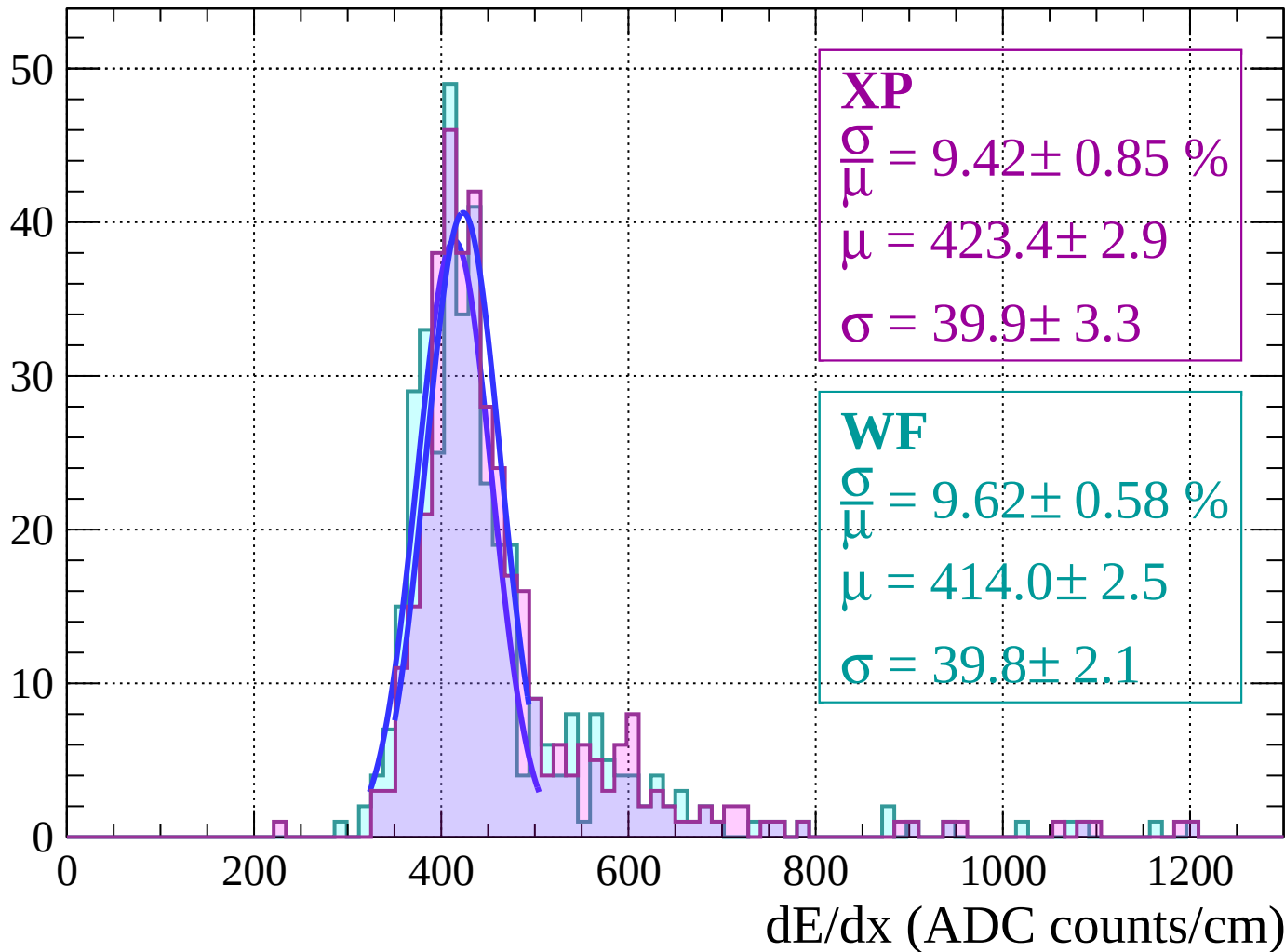
Energy loss | $28 < \theta < 30$

Count



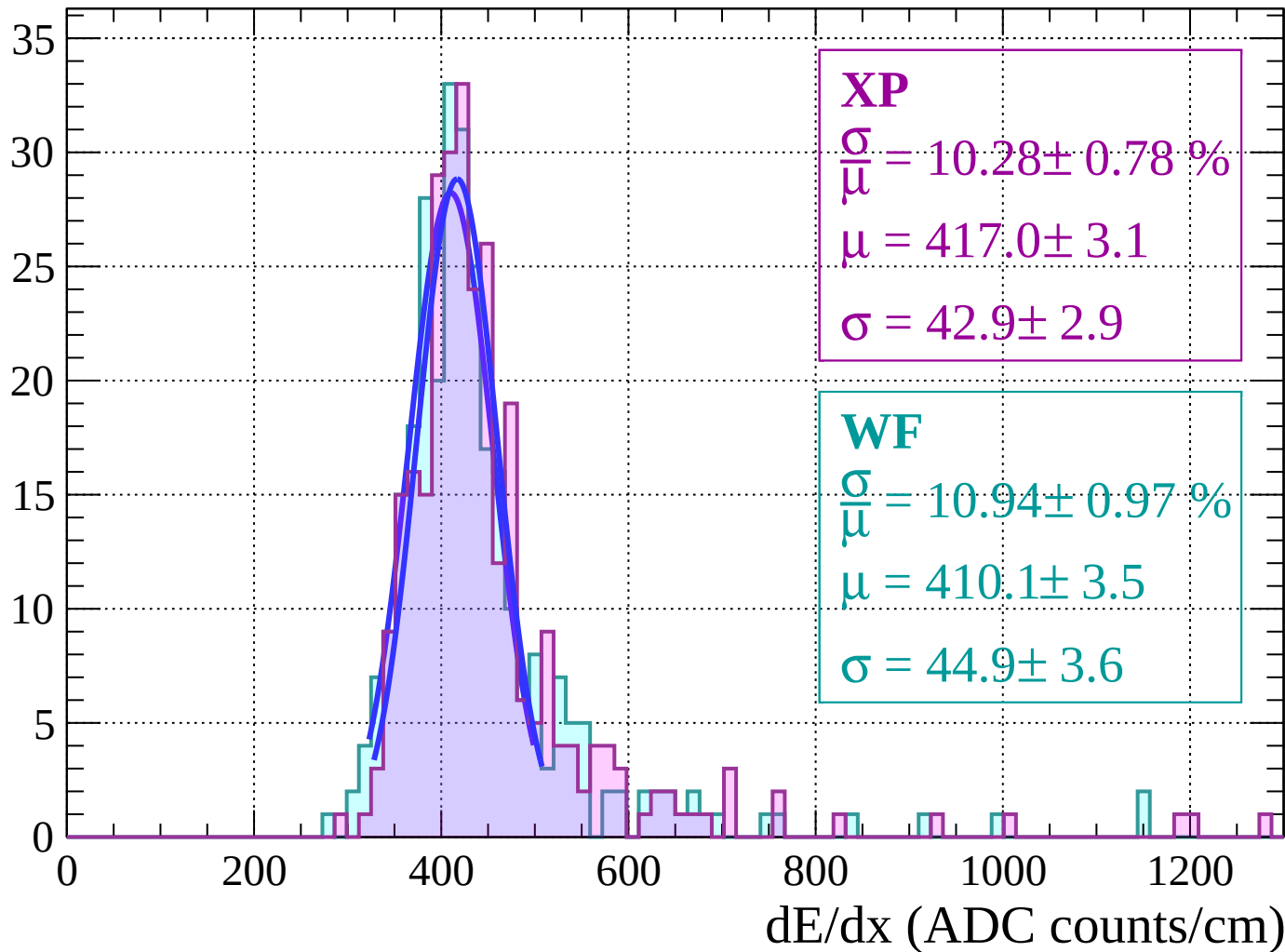
Energy loss | $30 < \theta < 32$

Count



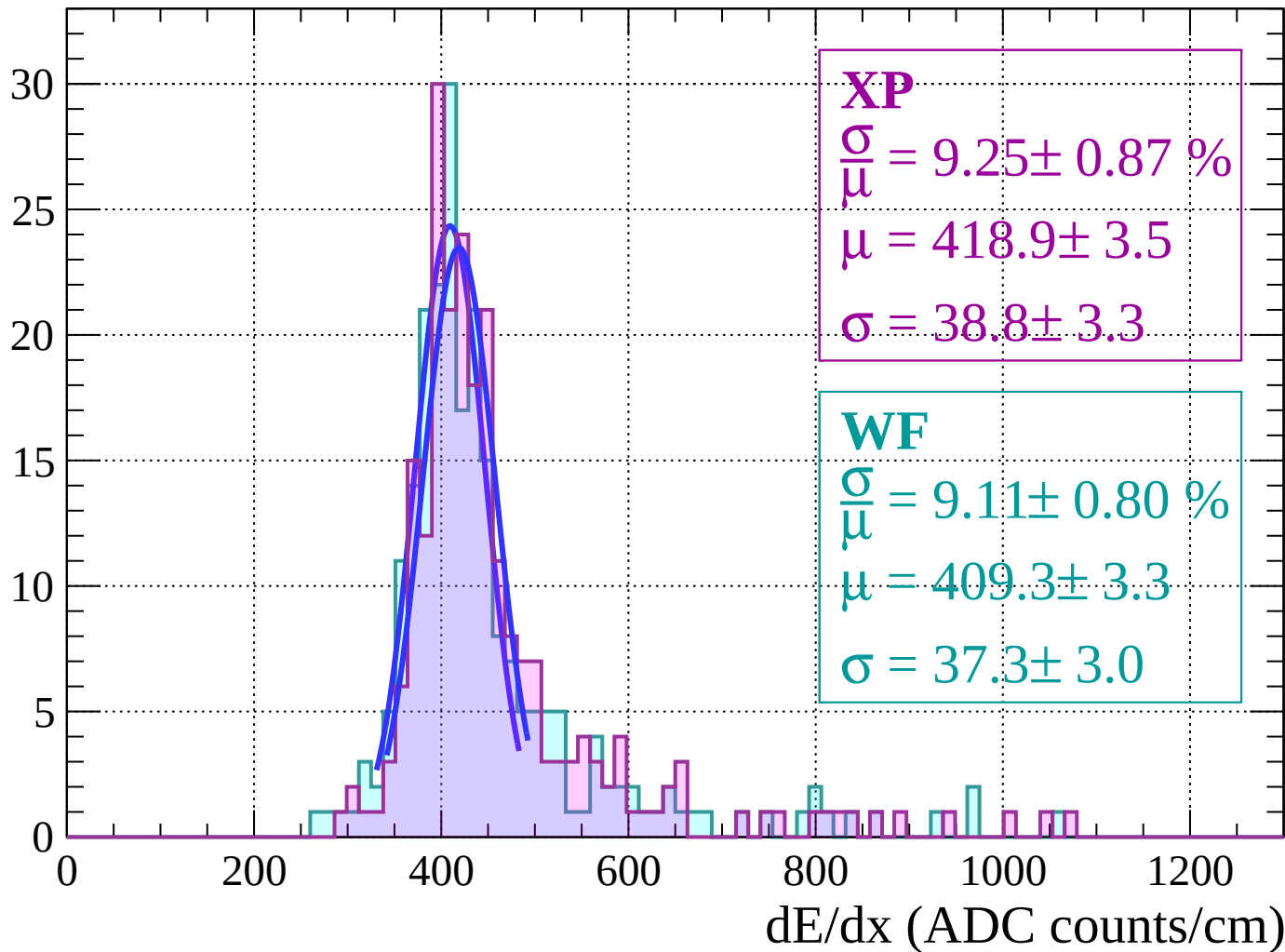
Energy loss | $32 < \theta < 34$

Count



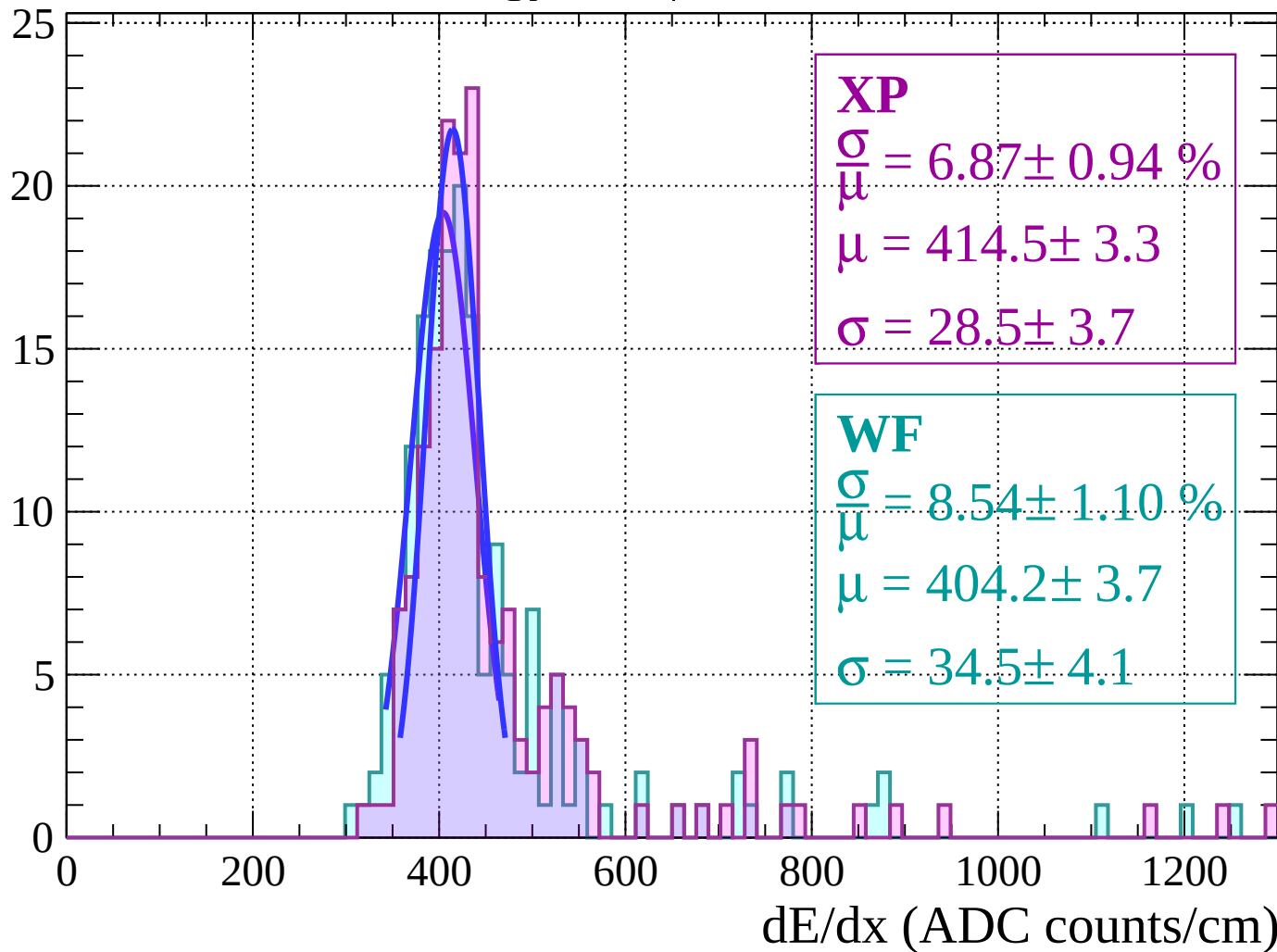
Energy loss | $34 < \theta < 36$

Count



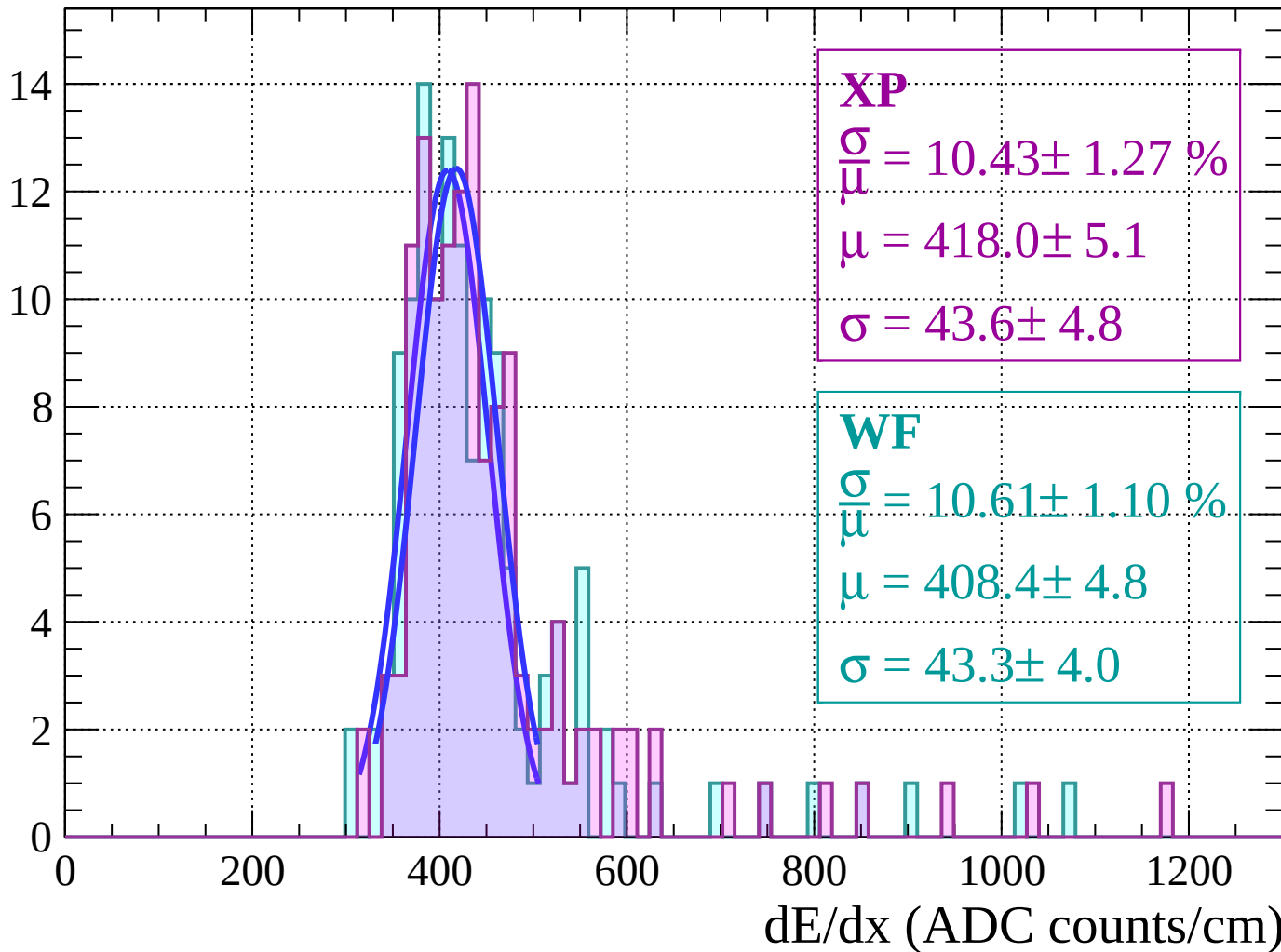
Energy loss | $36 < \theta < 38$

Count



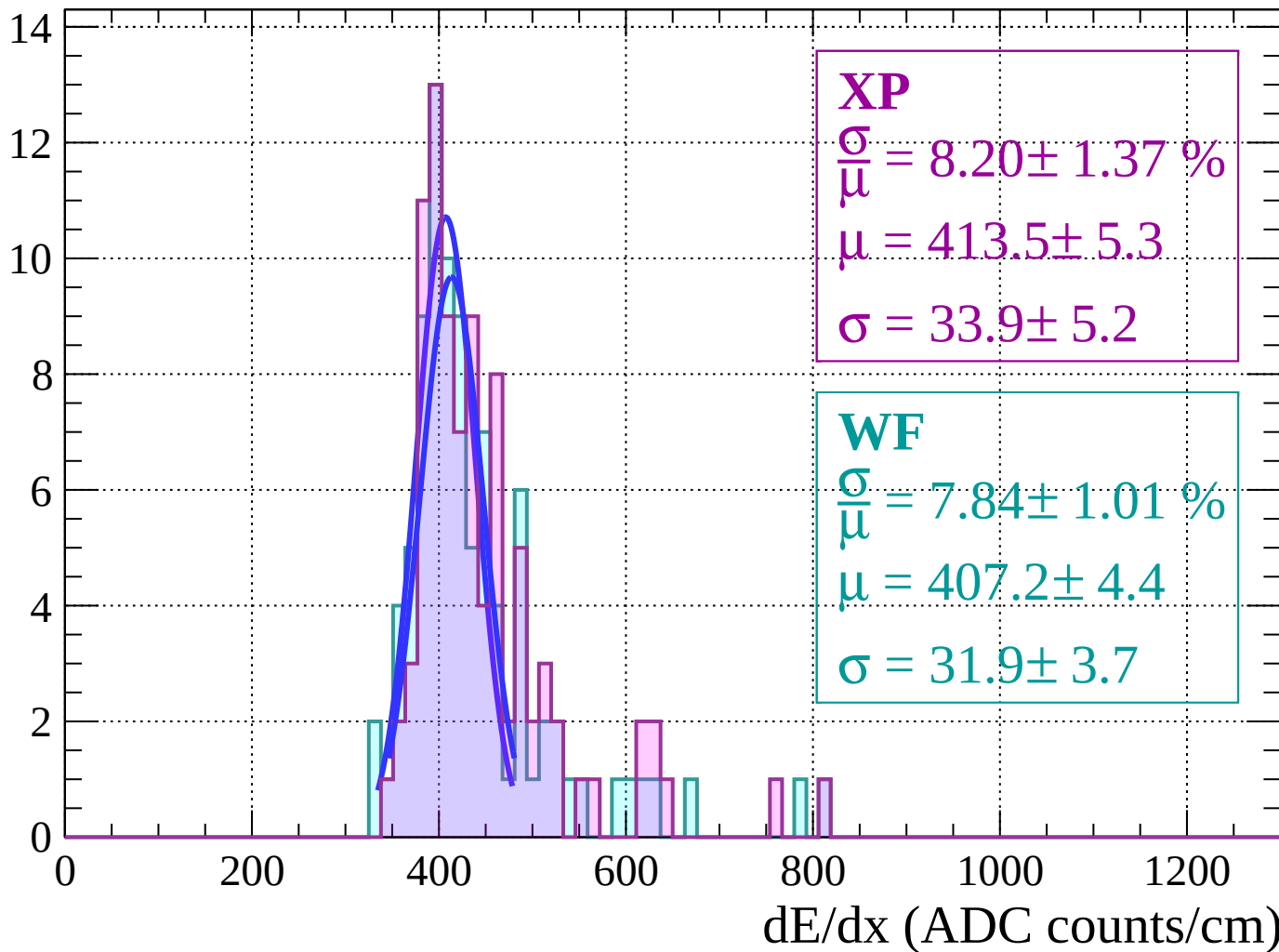
Energy loss | $38 < \theta < 40$

Count



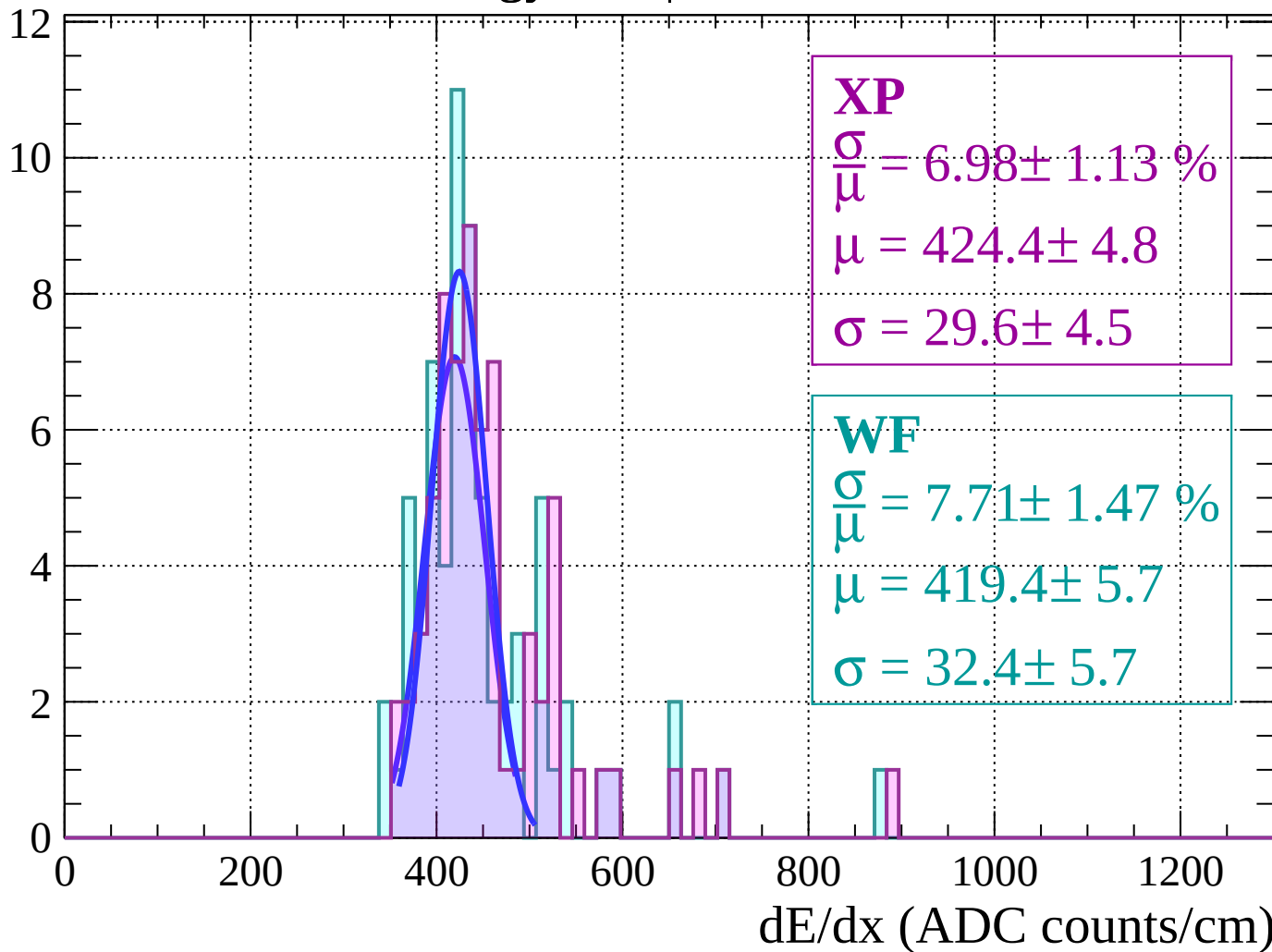
Energy loss | $40 < \theta < 42$

Count



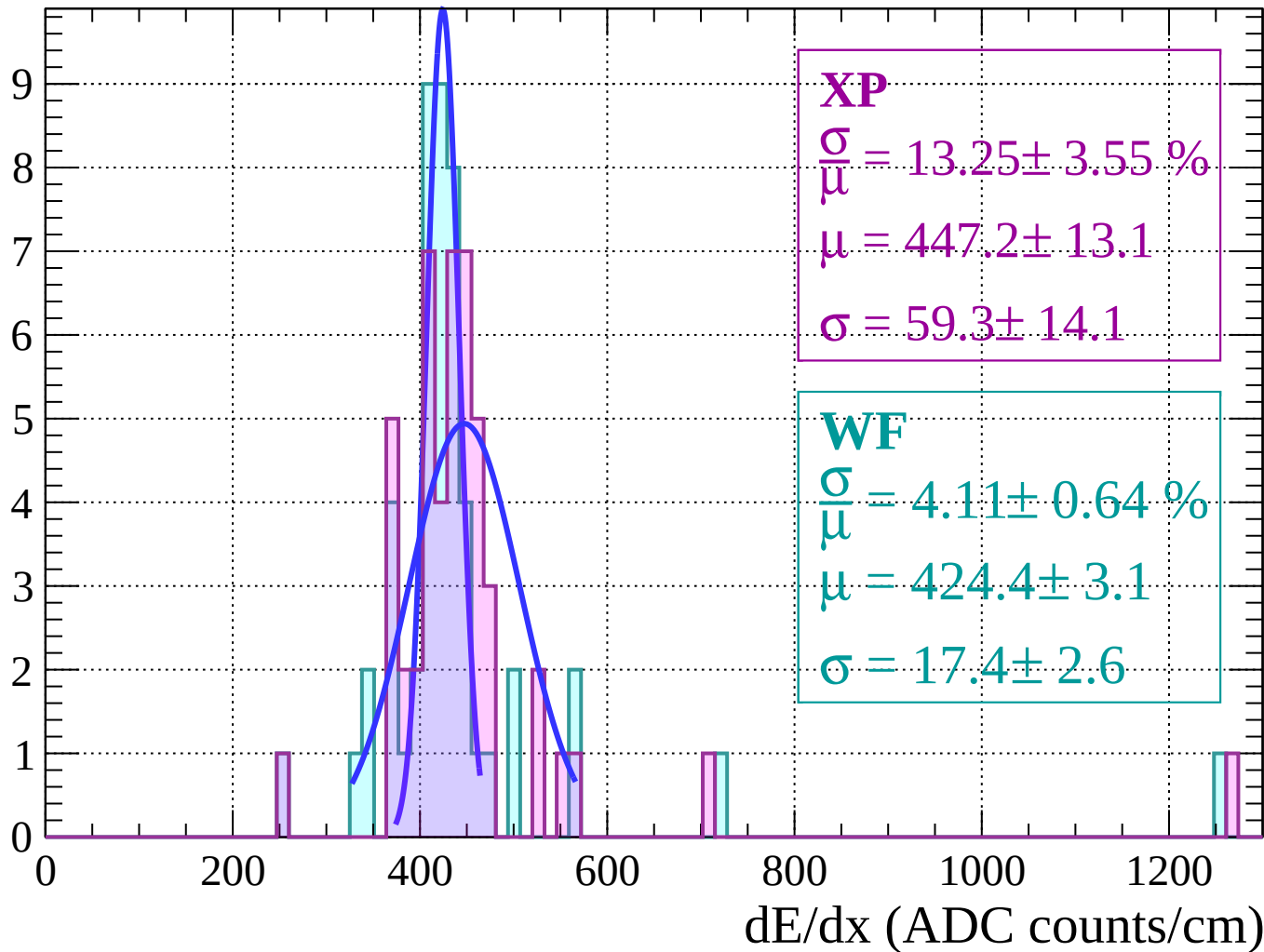
Energy loss | $42 < \theta < 44$

Count



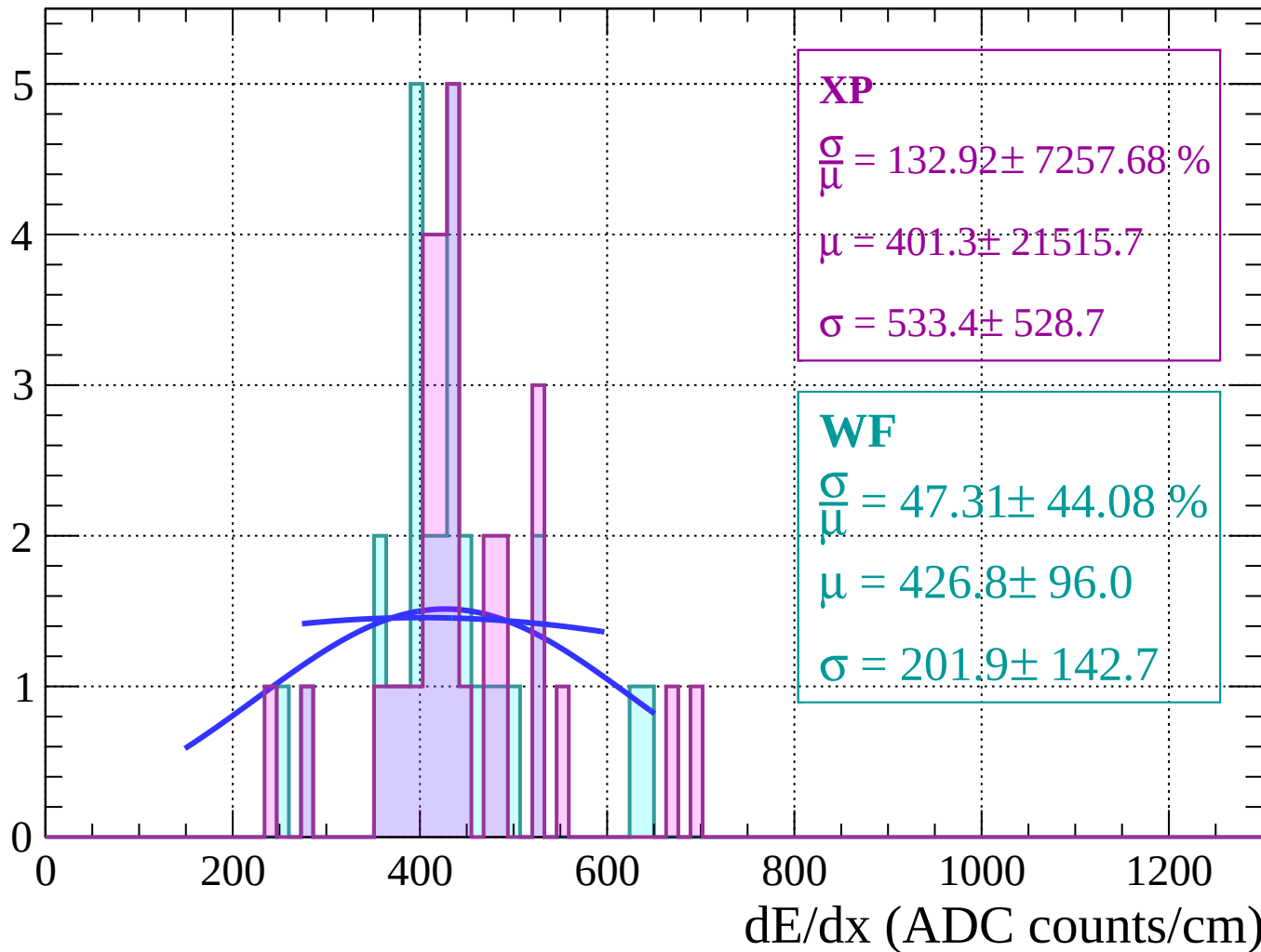
Energy loss | $44 < \theta < 46$

Count



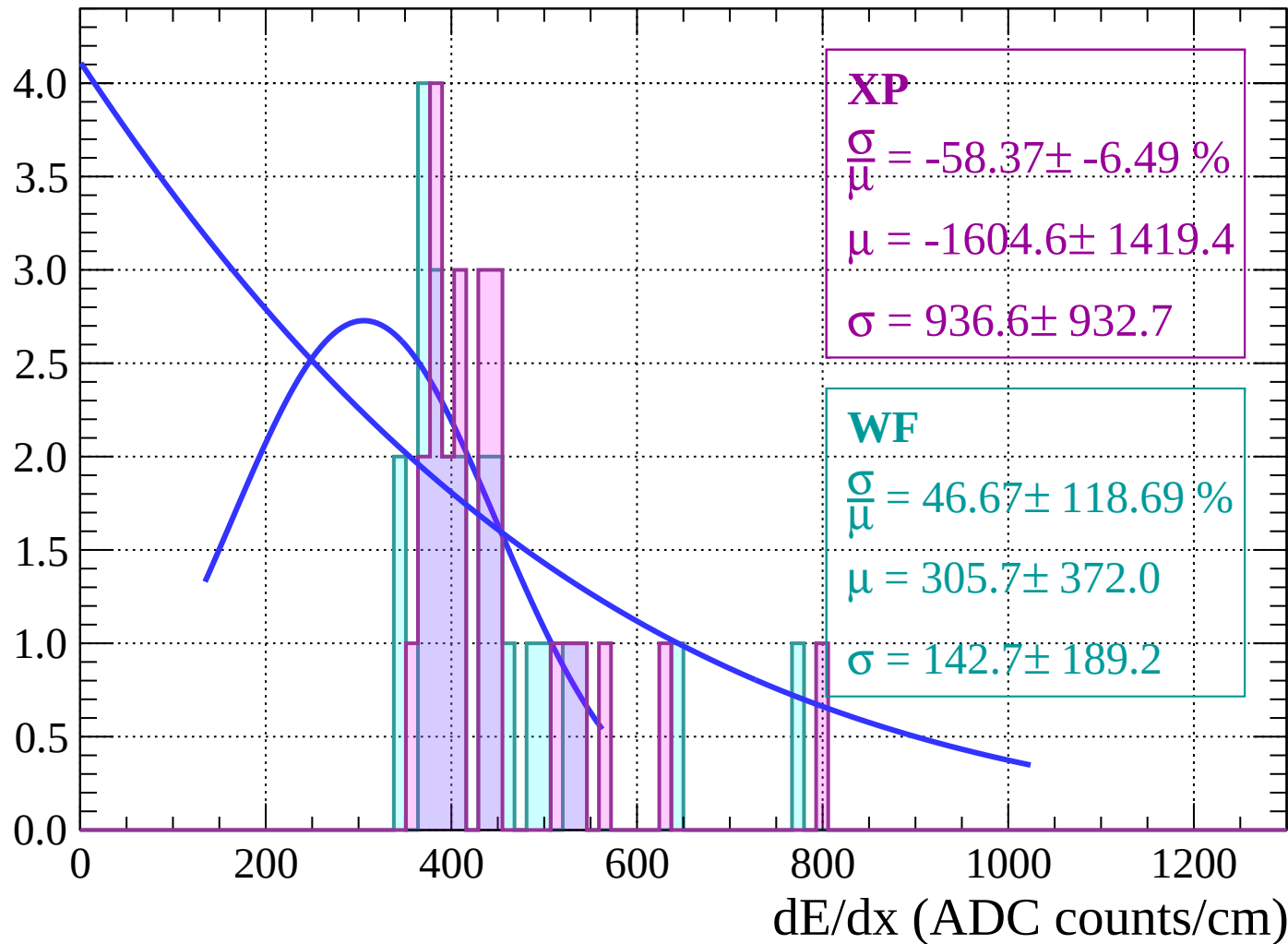
Energy loss | $46 < \theta < 48$

Count



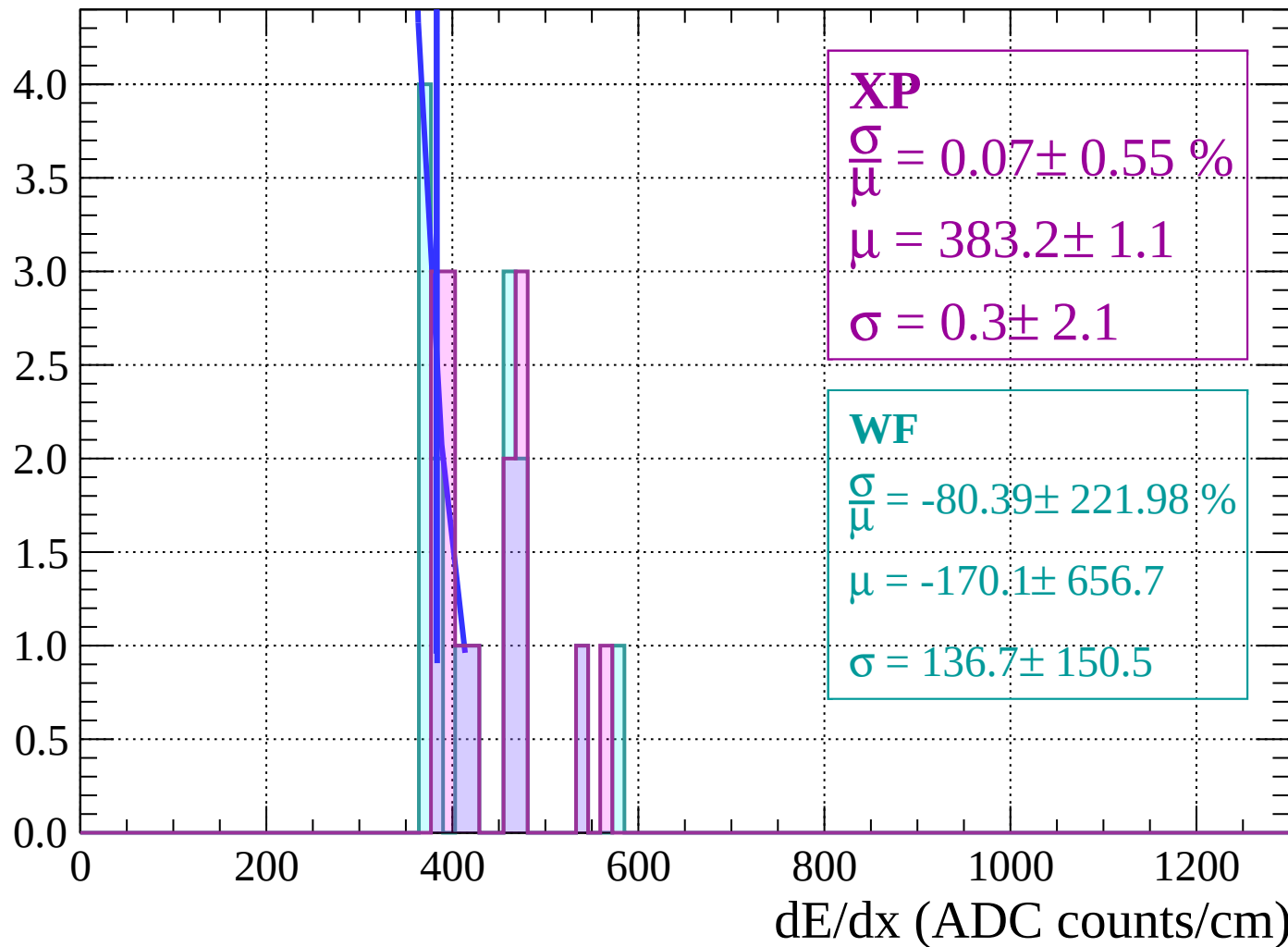
Energy loss | $48 < \theta < 50$

Count



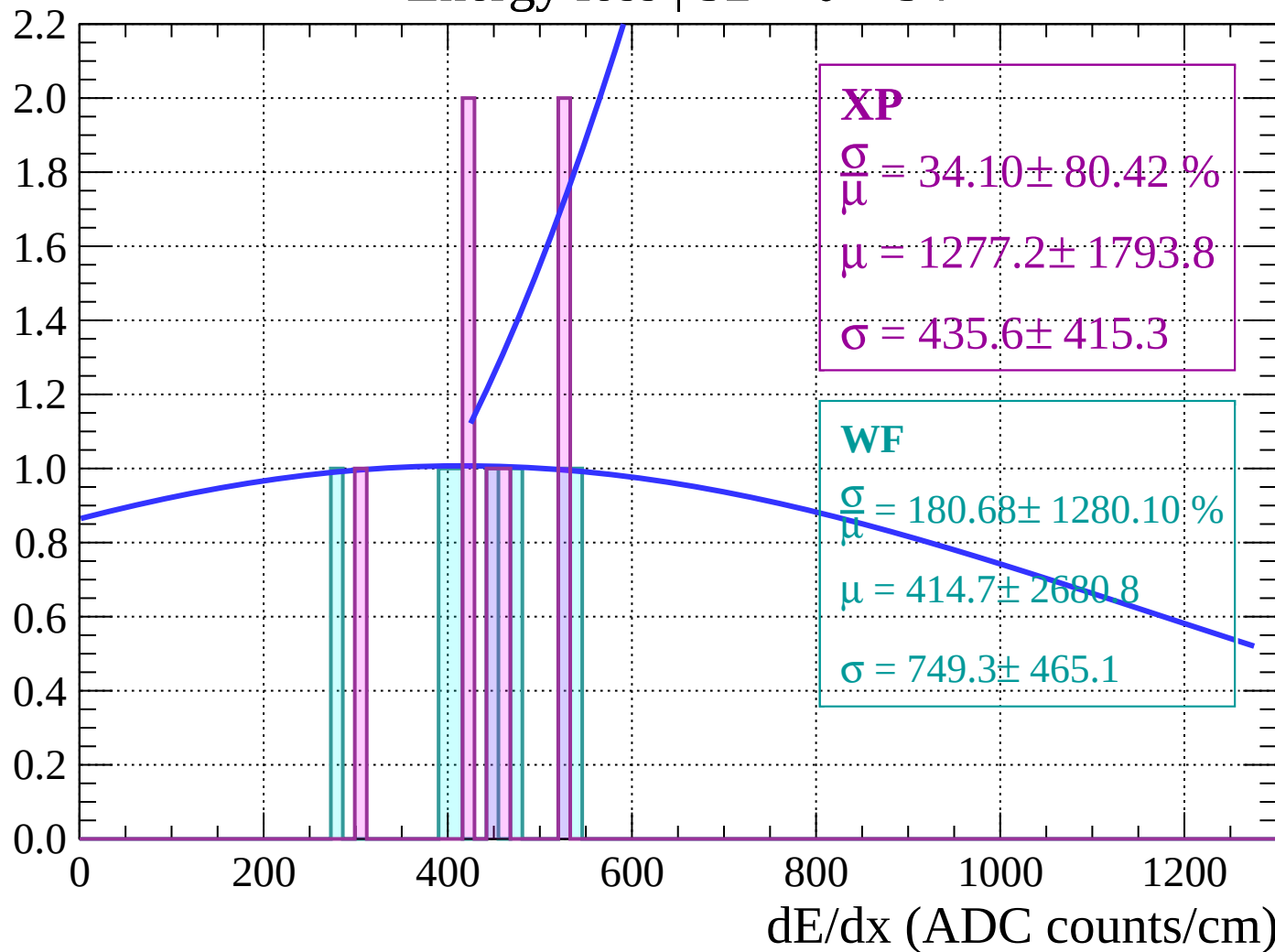
Energy loss | $50 < \theta < 52$

Count



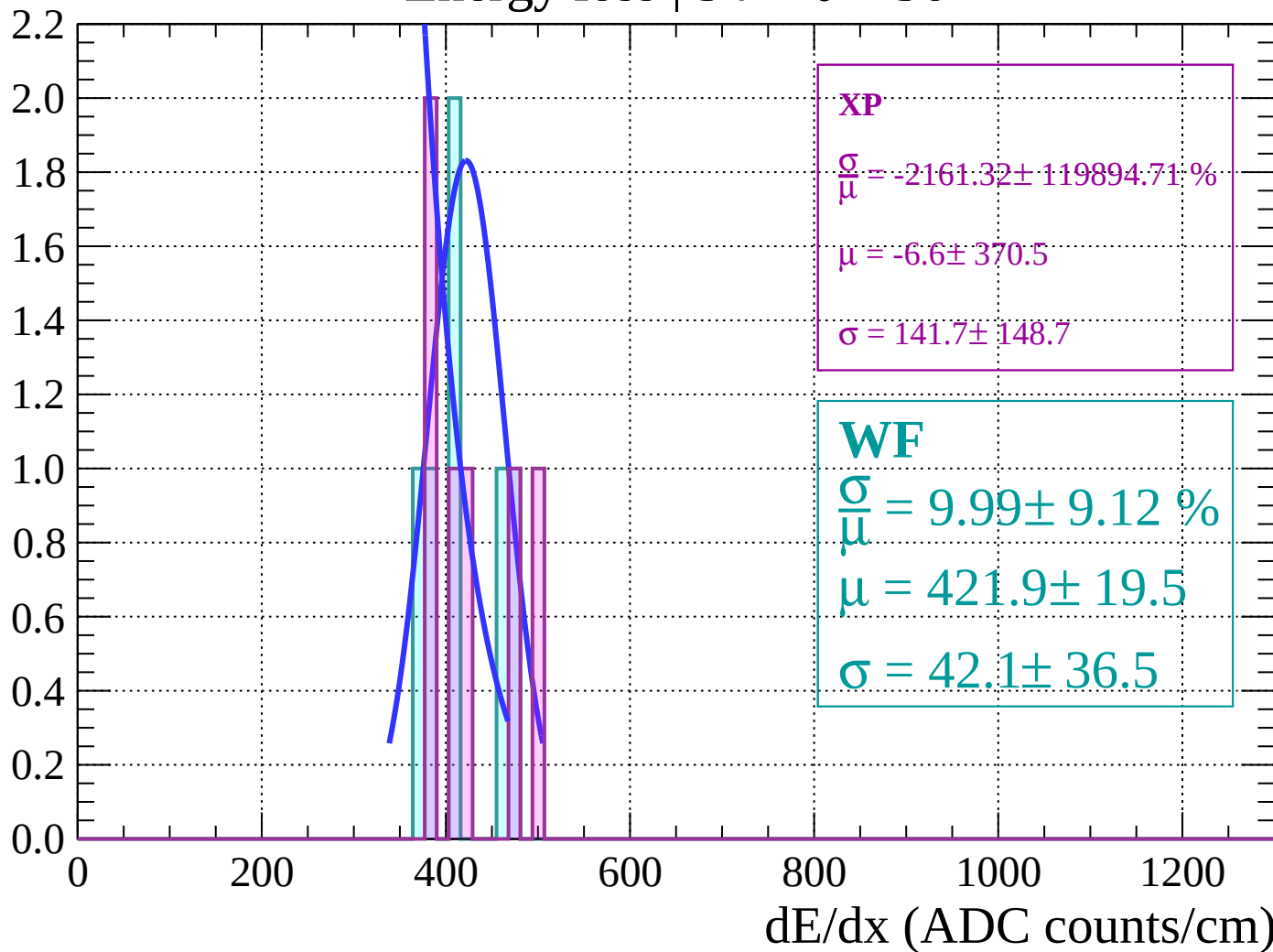
Energy loss | $52 < \theta < 54$

Count



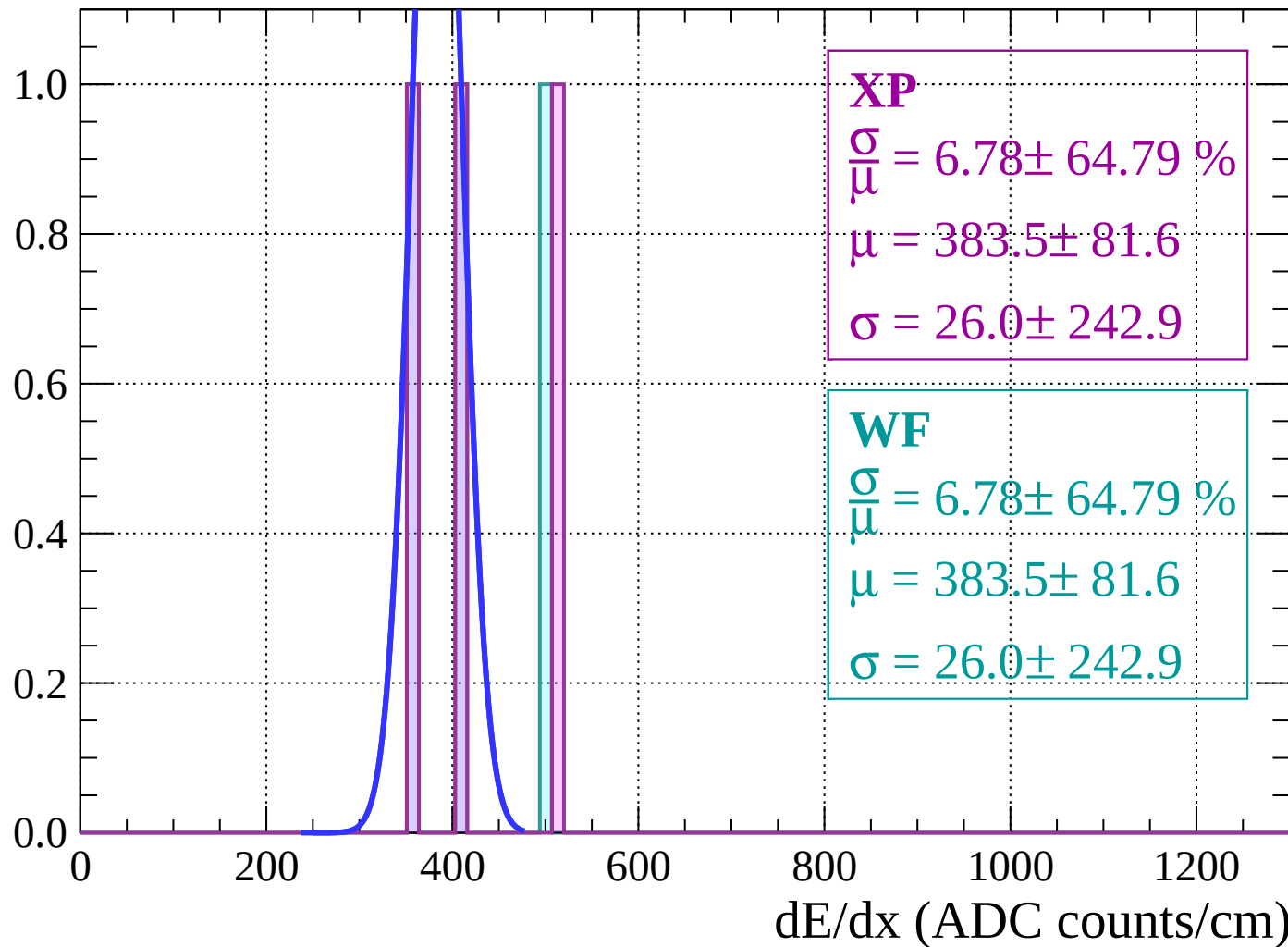
Energy loss | $54 < \theta < 56$

Count



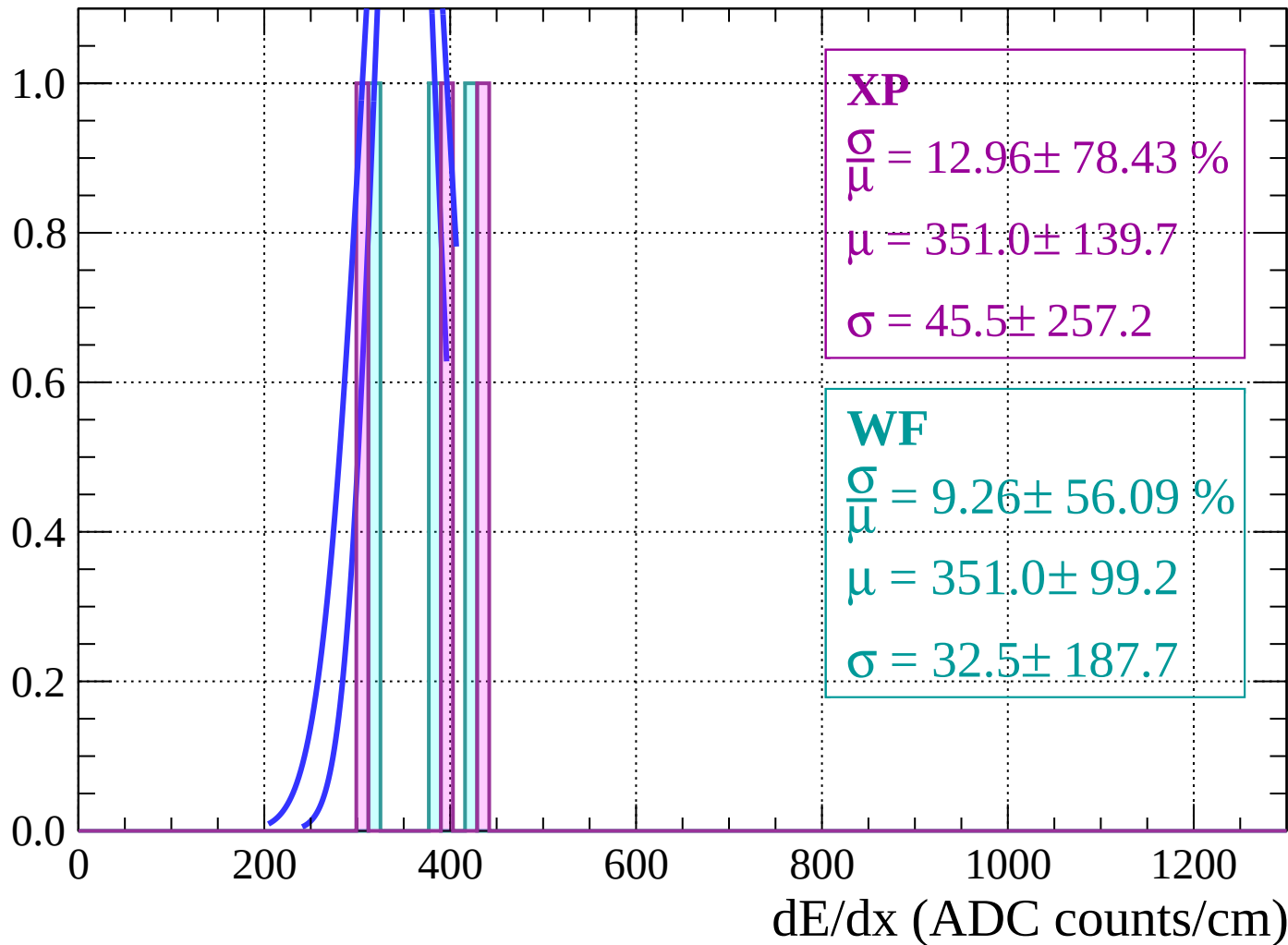
Energy loss | $56 < \theta < 58$

Count



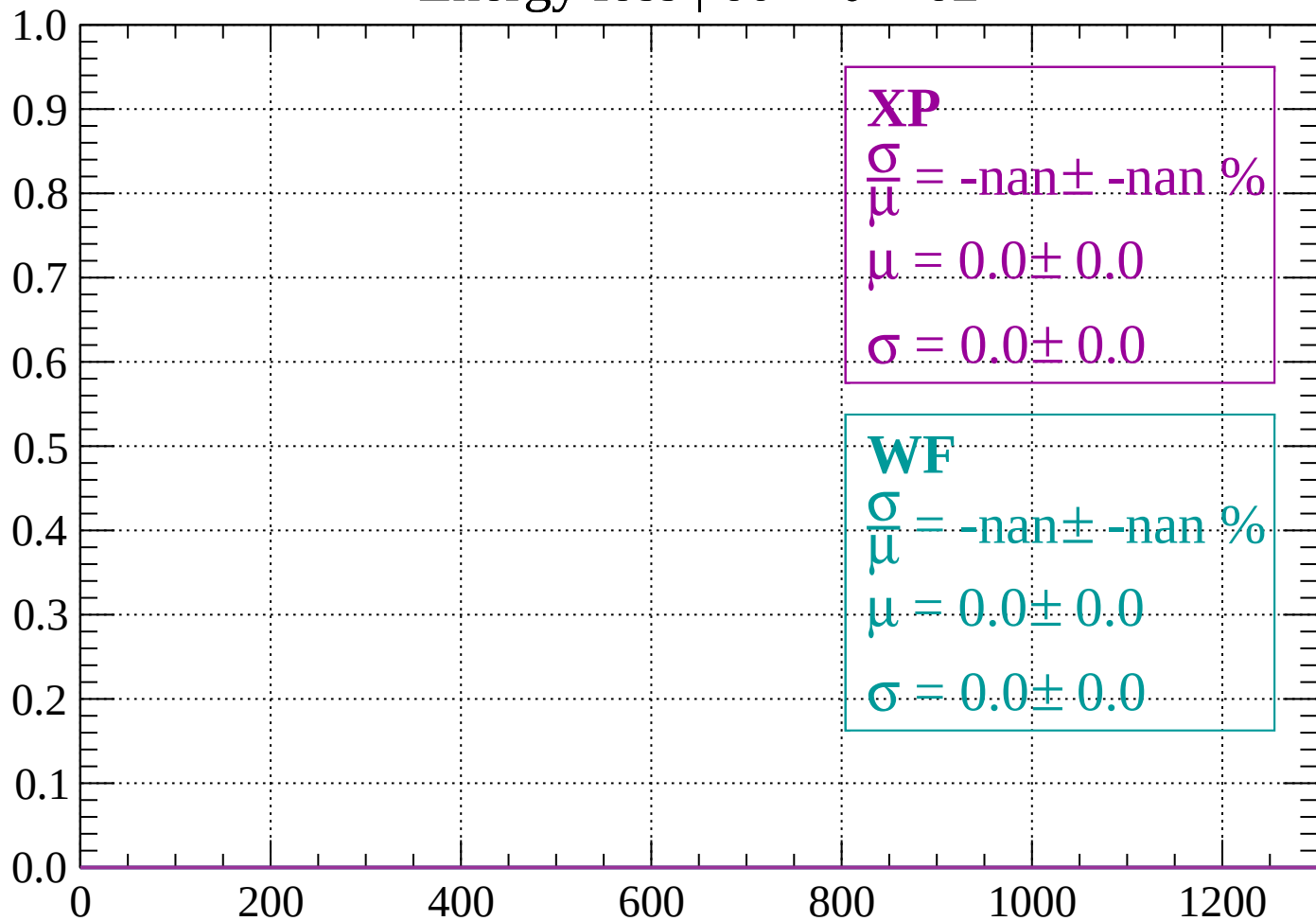
Energy loss | $58 < \theta < 60$

Count



Energy loss | $60 < \theta < 62$

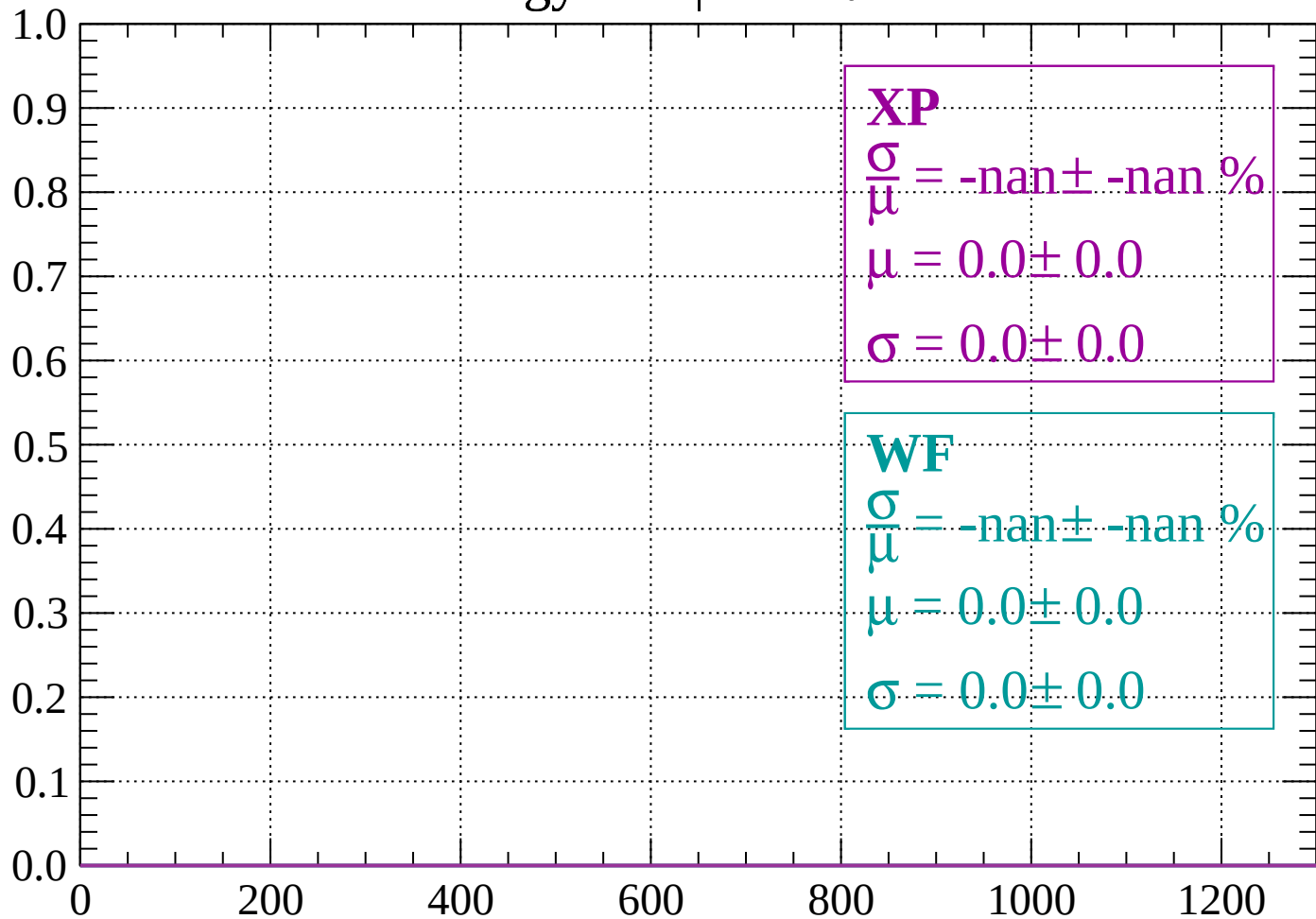
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

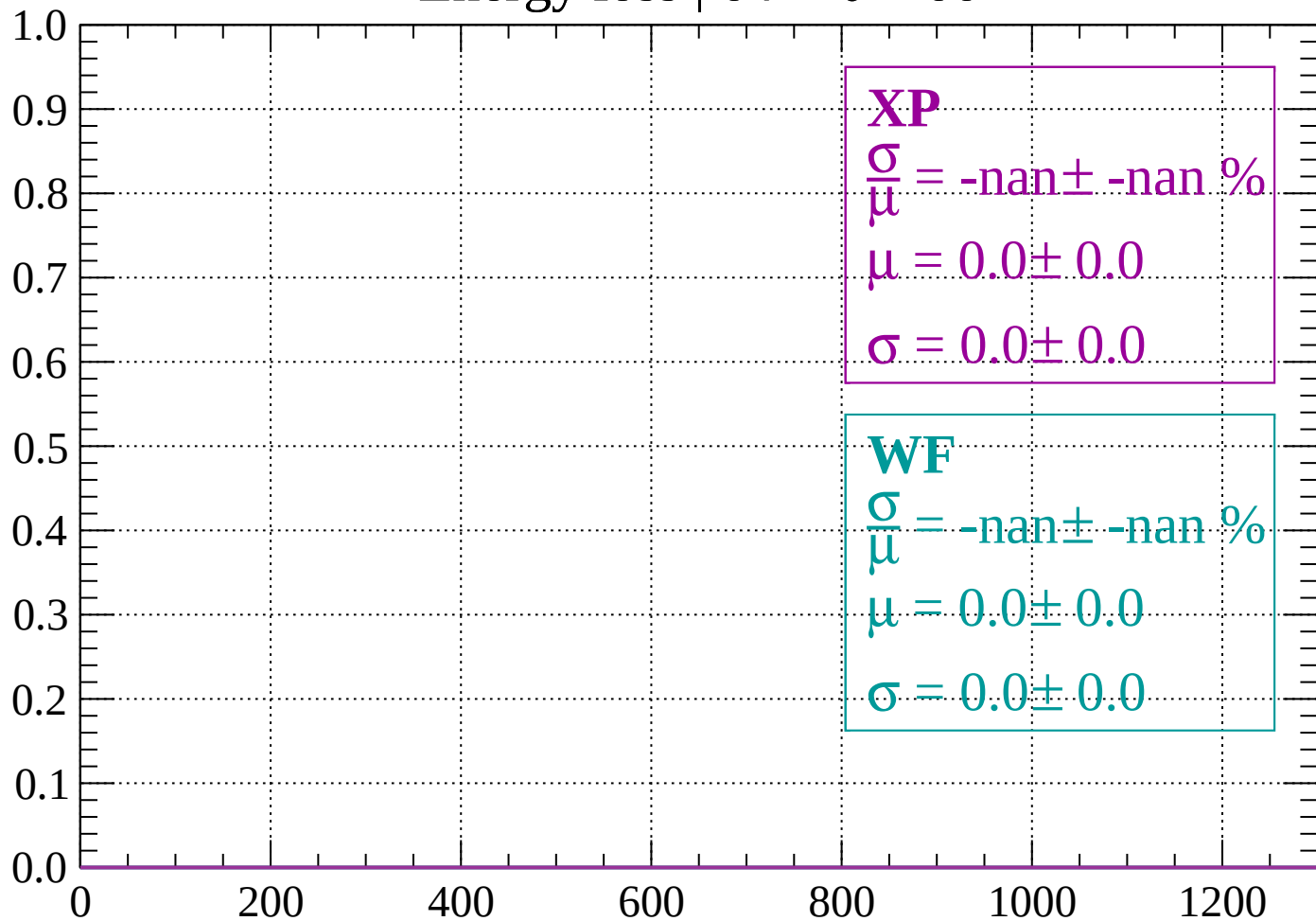
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

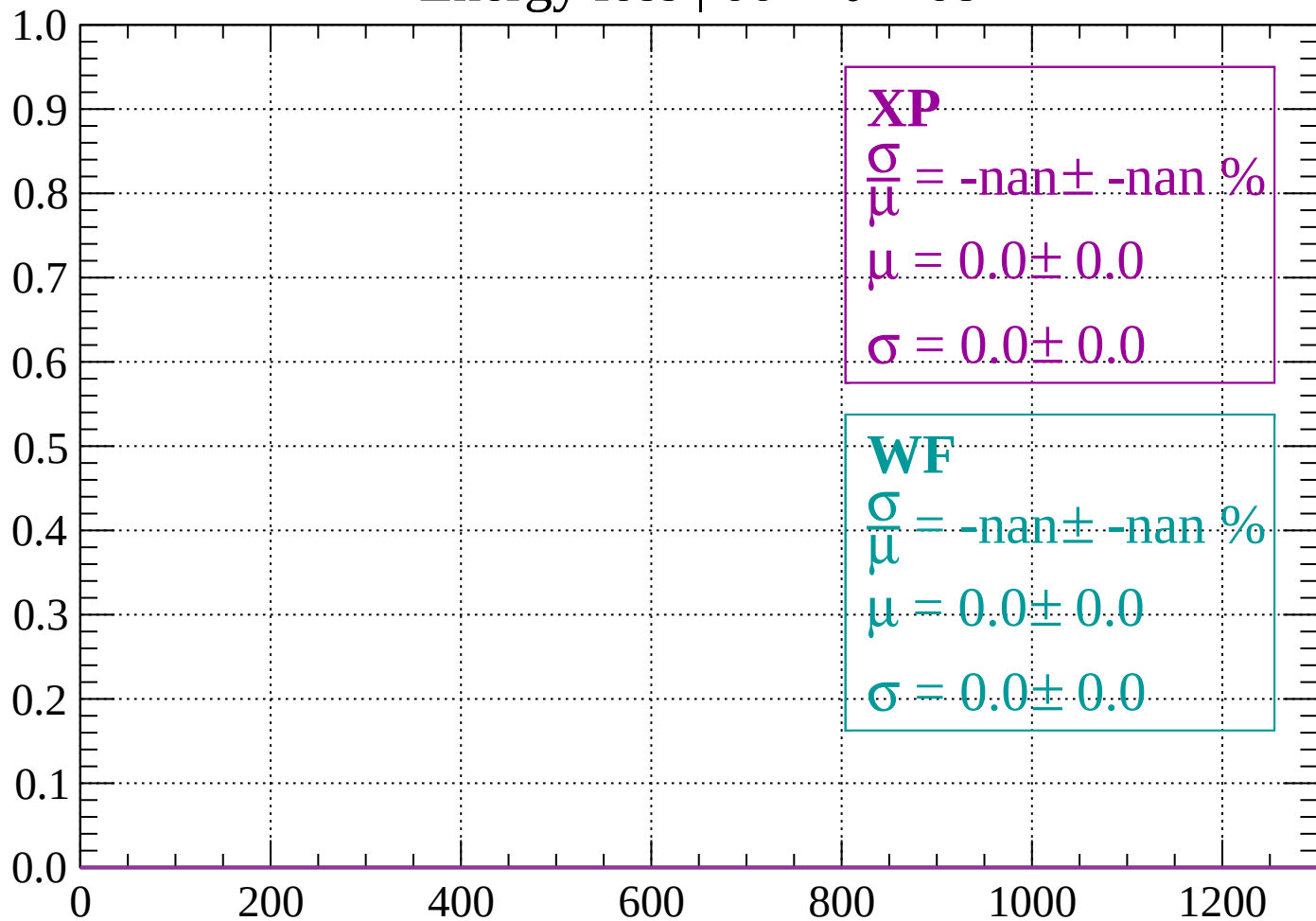
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

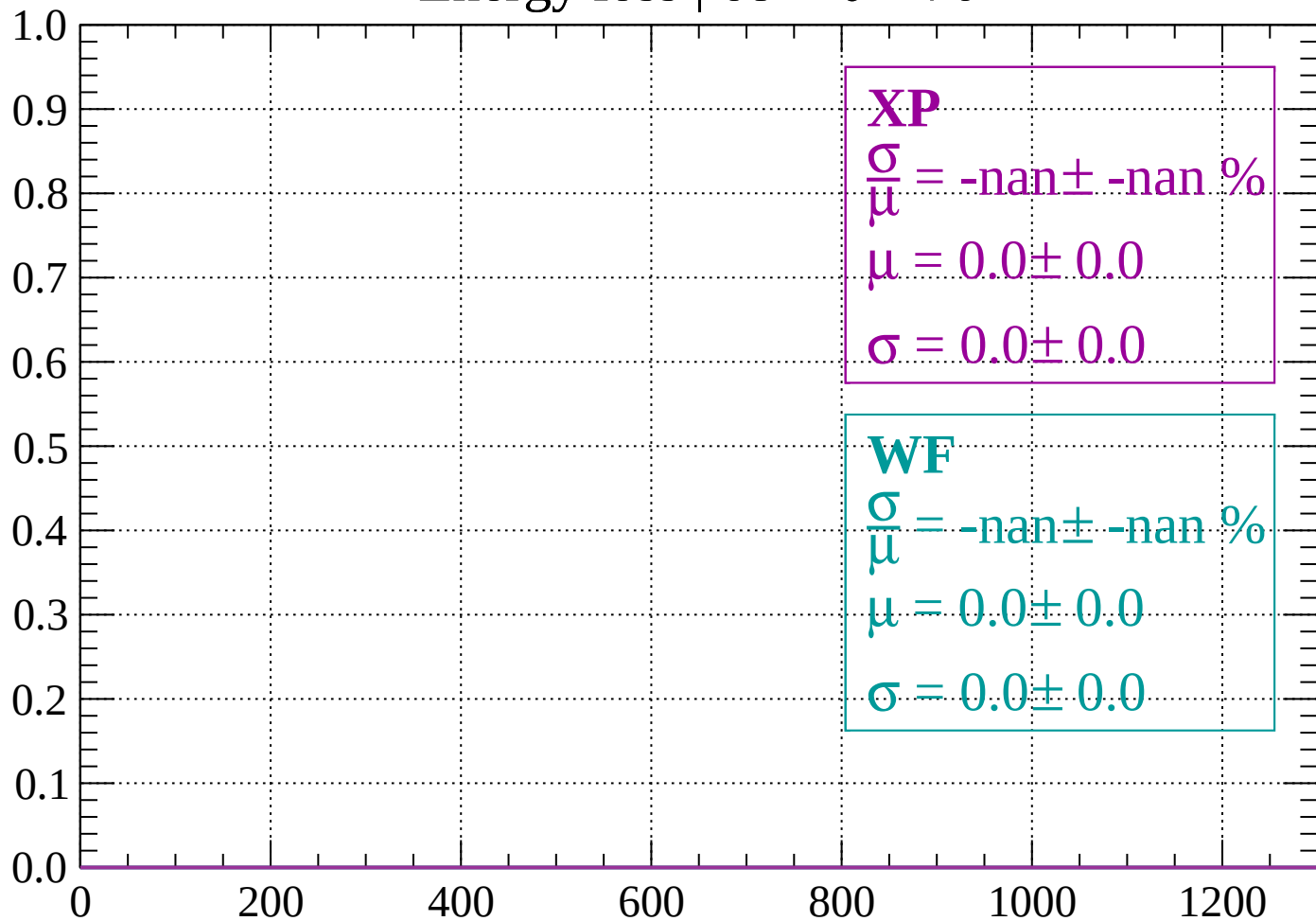
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

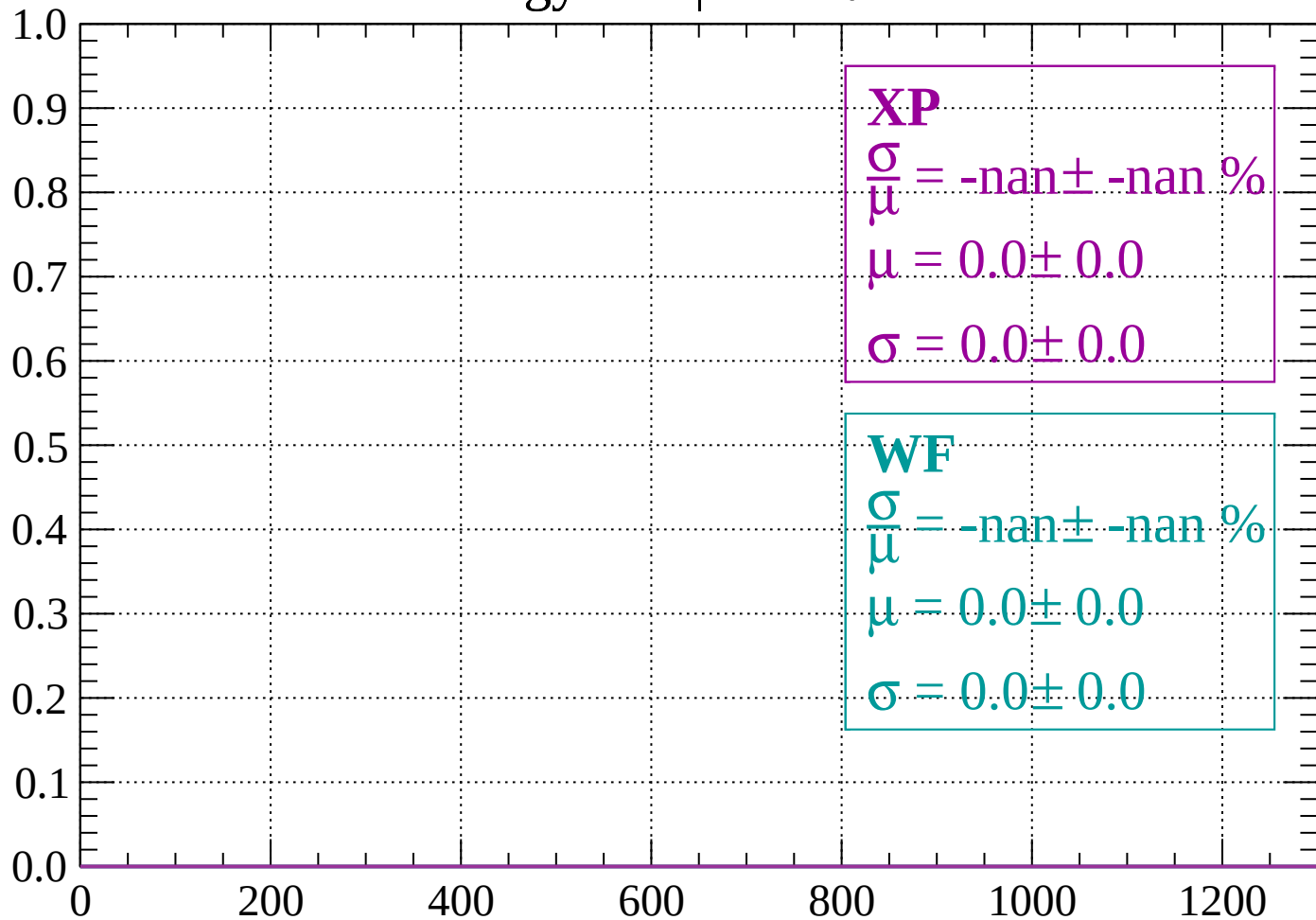
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

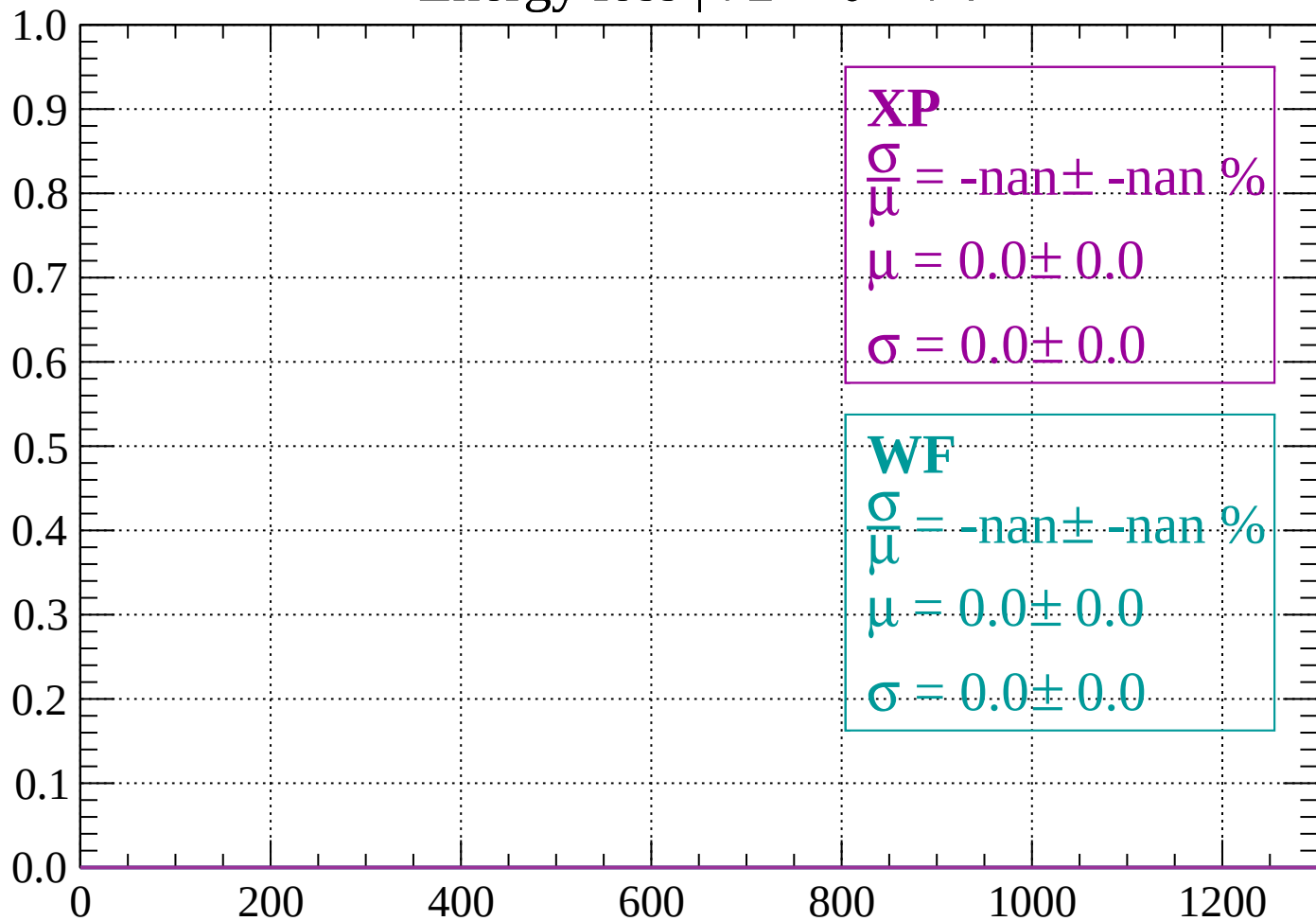
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

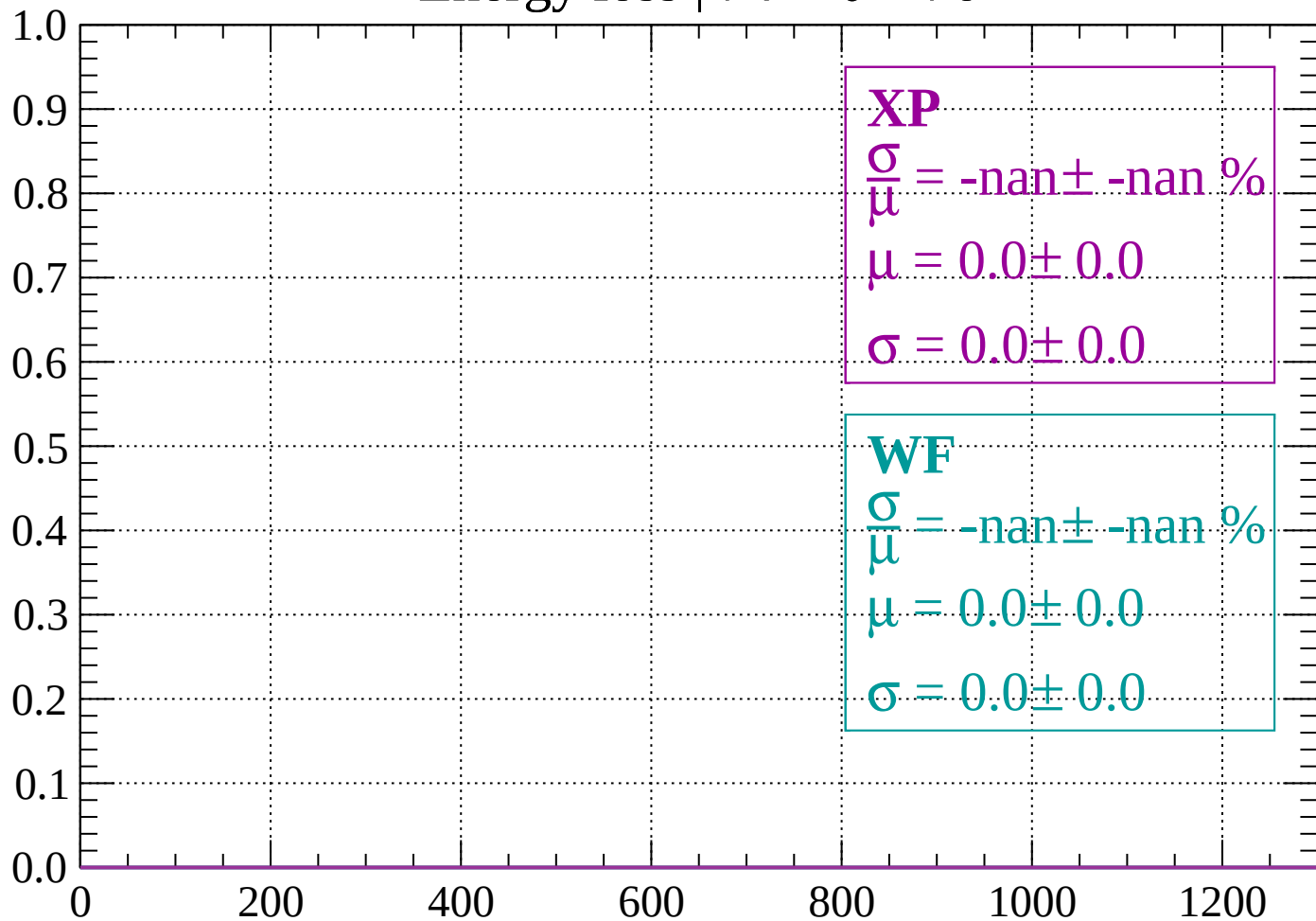
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

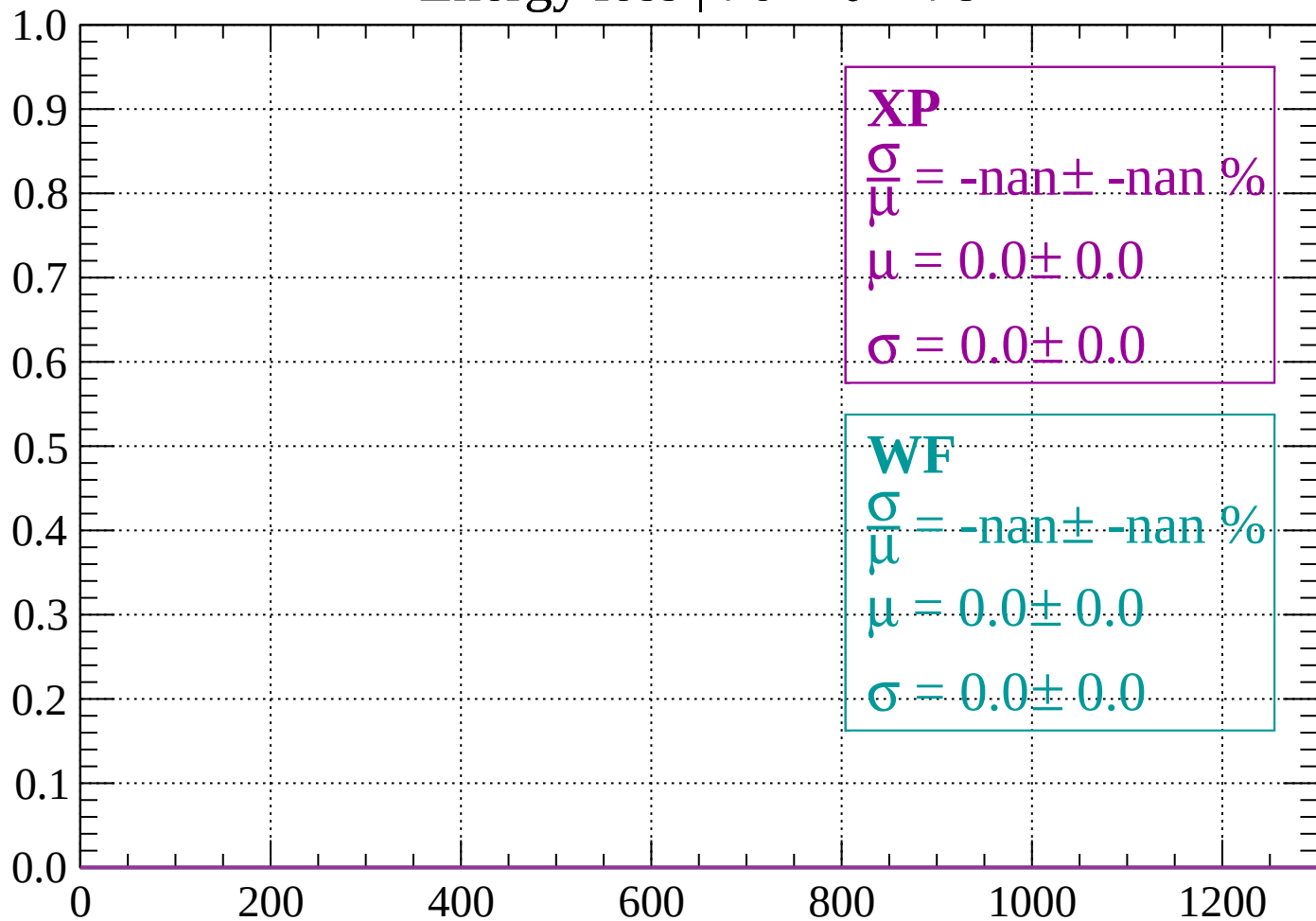
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

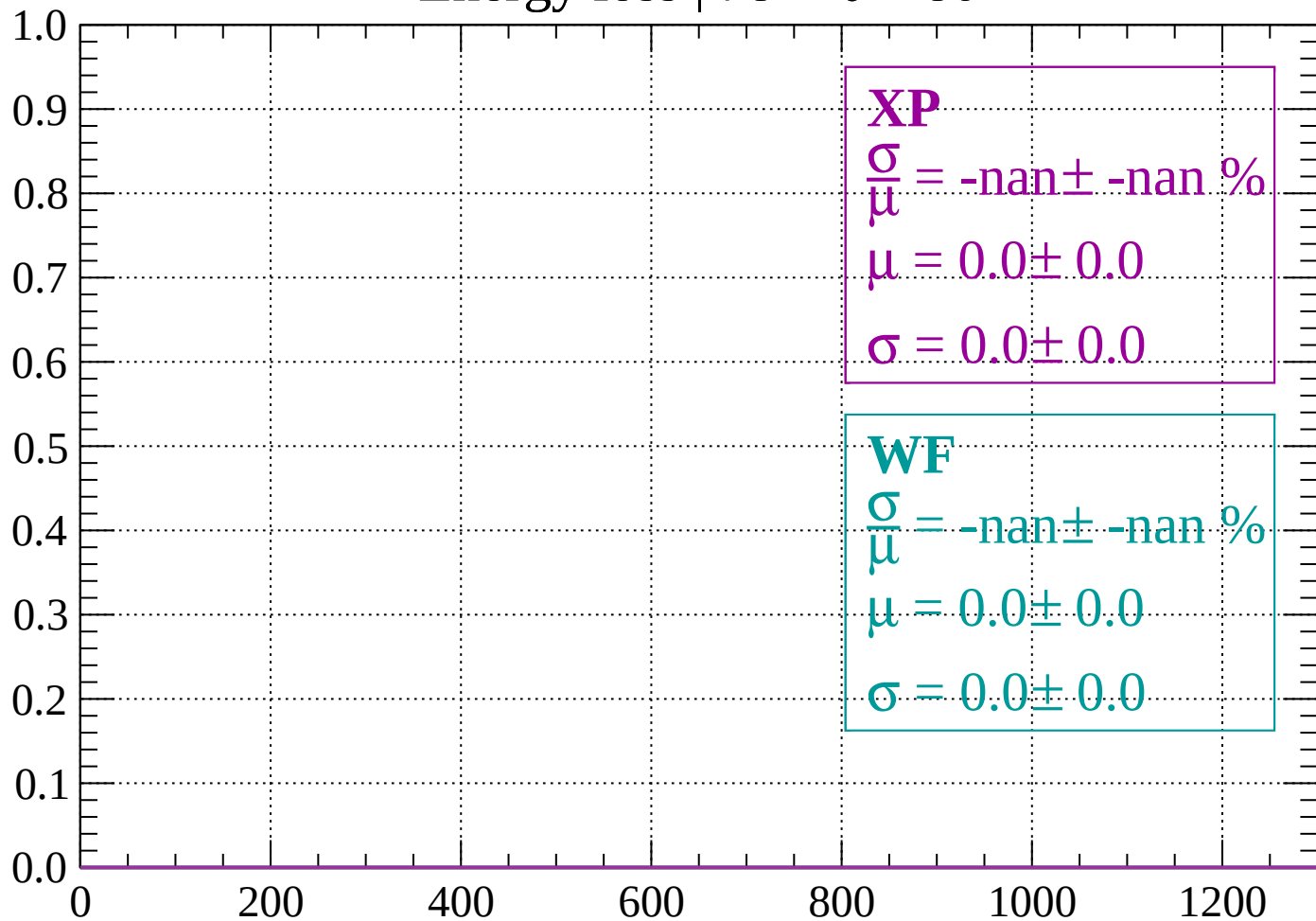
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

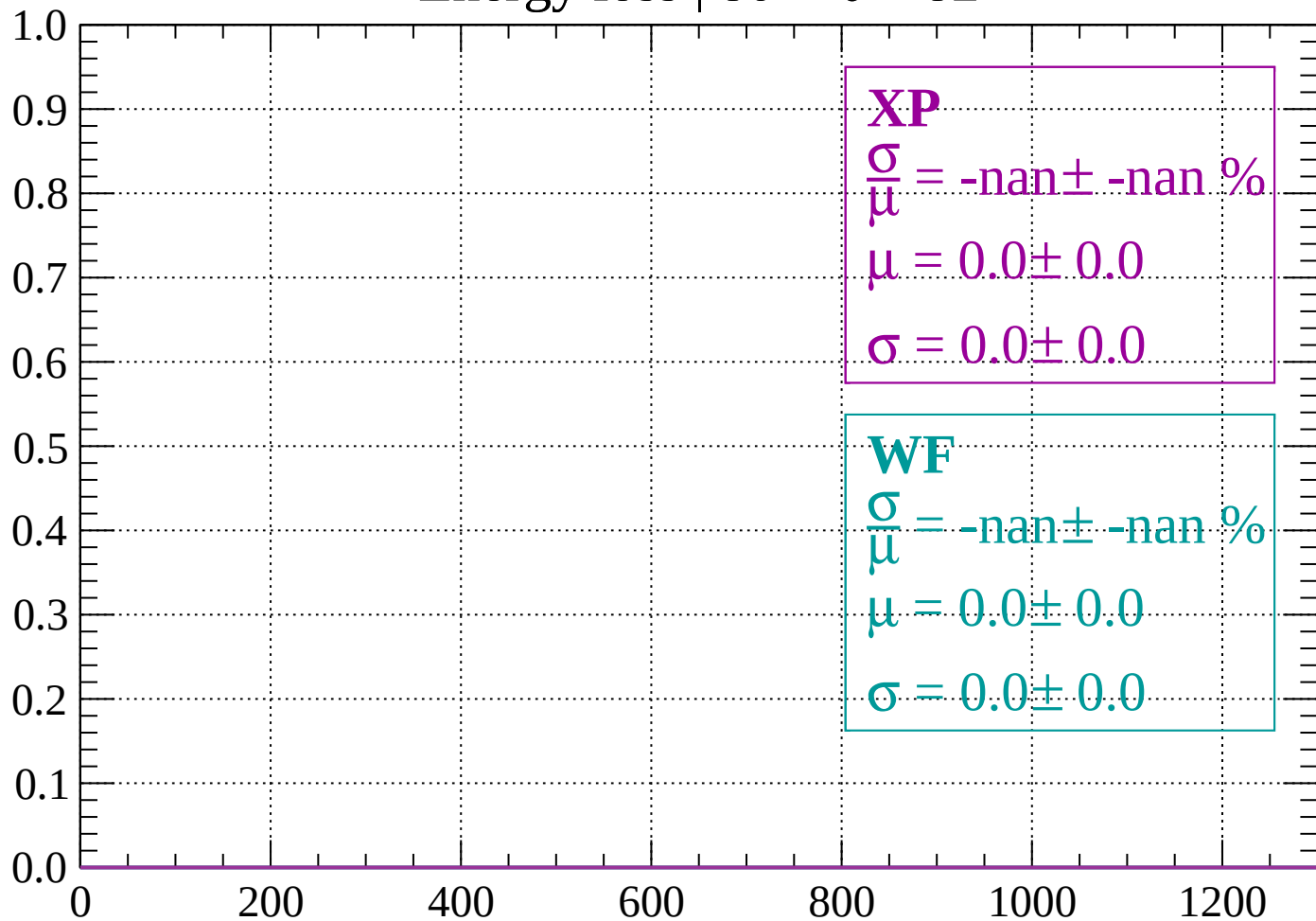
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

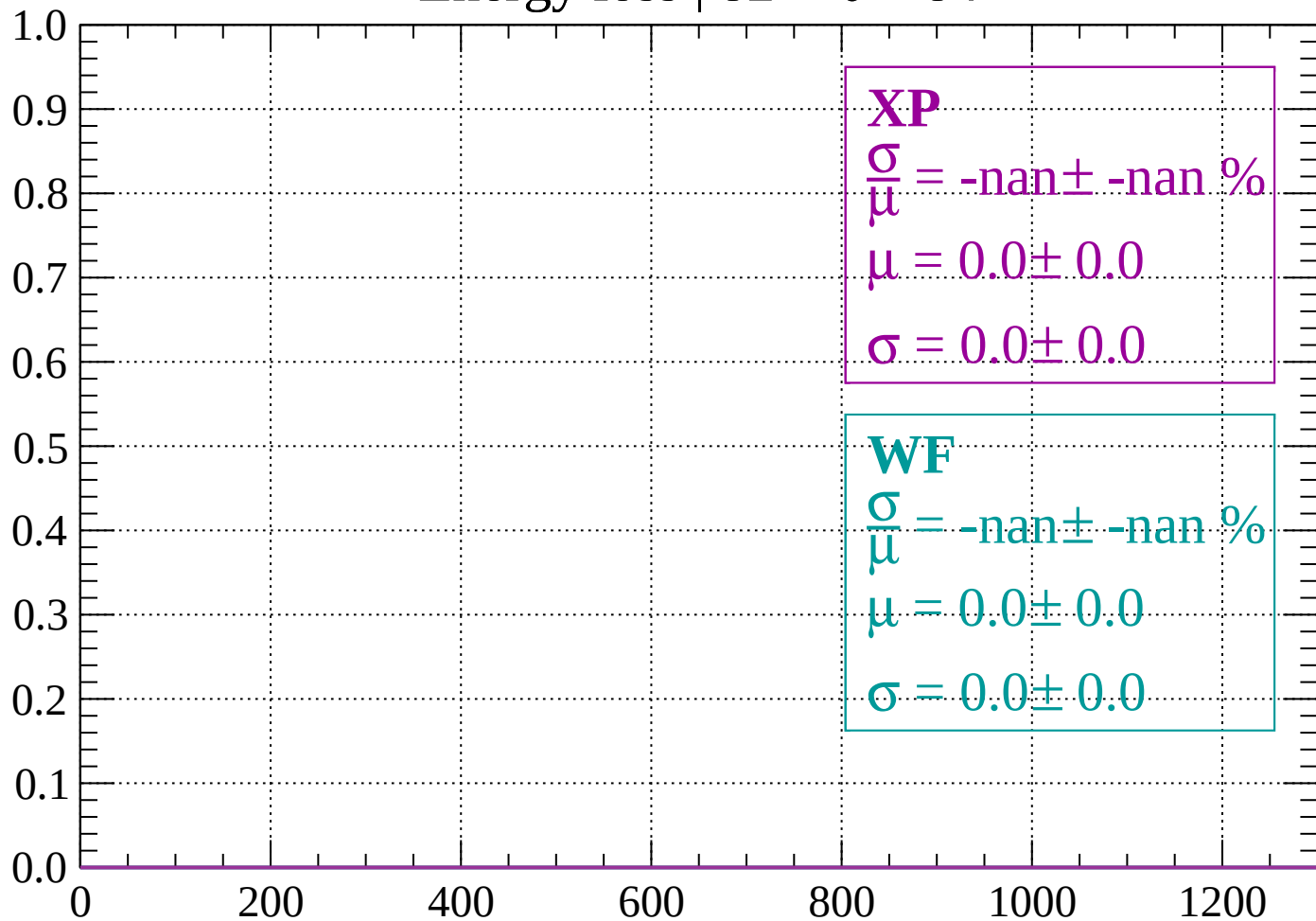
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

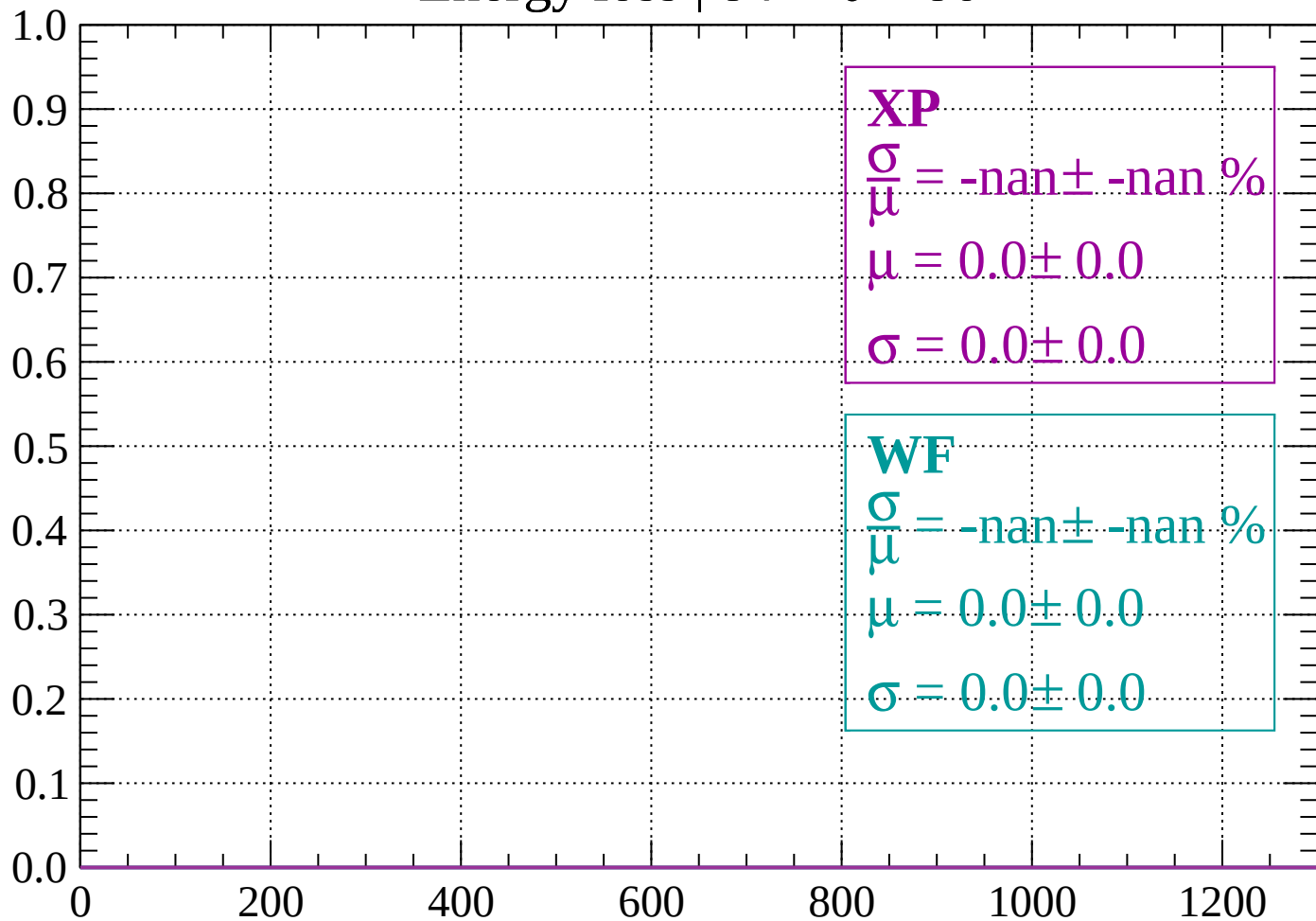
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

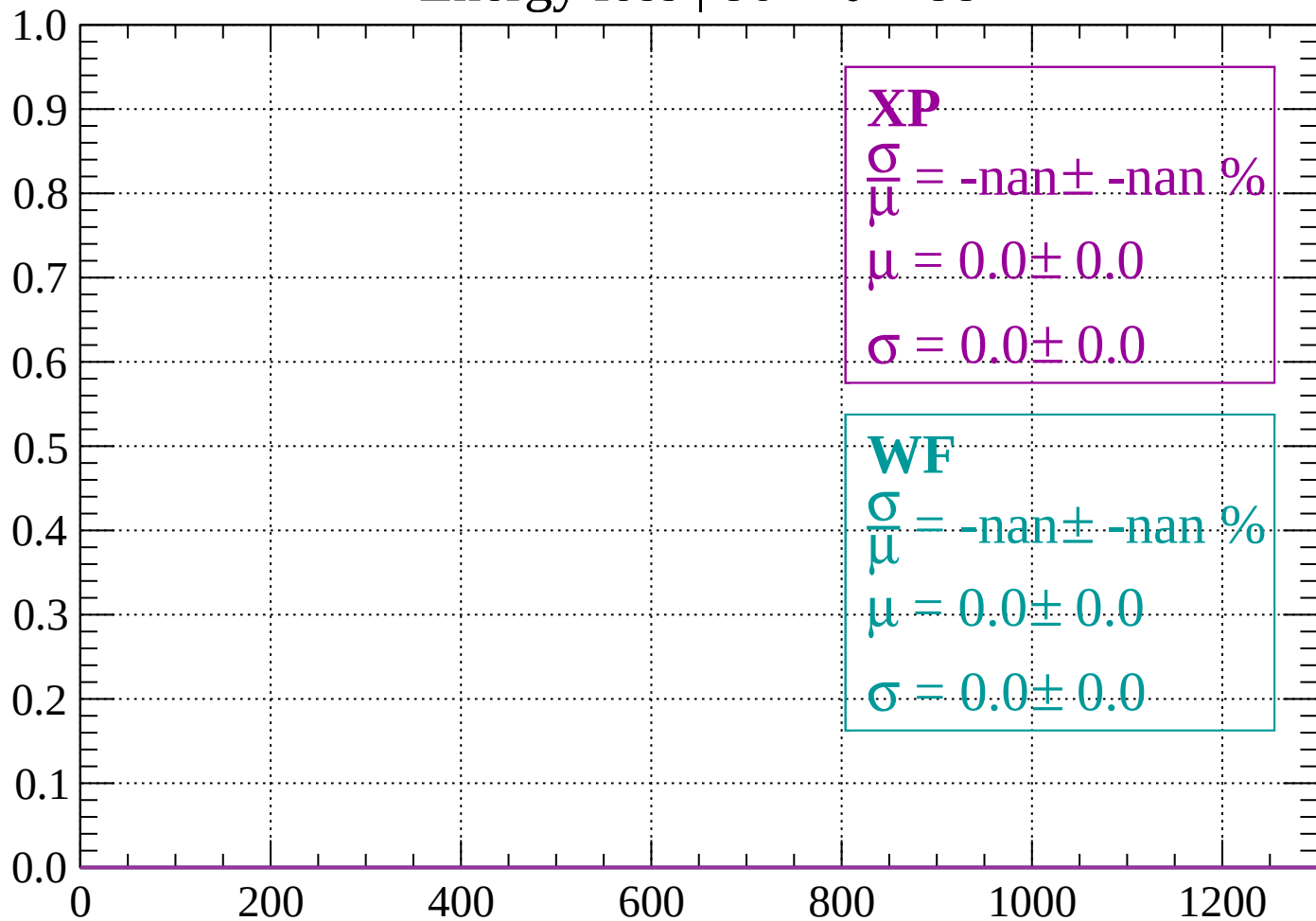
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

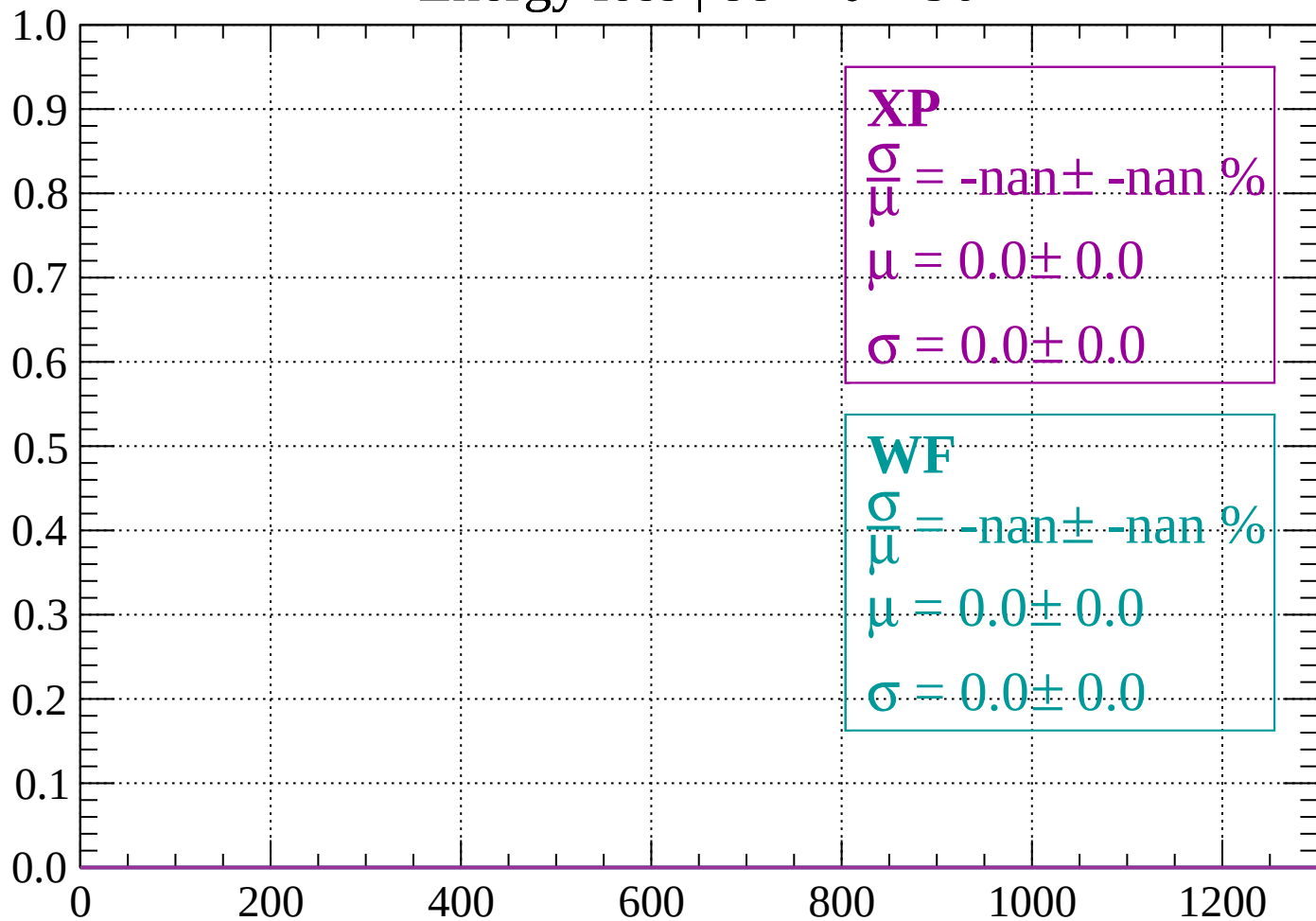
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

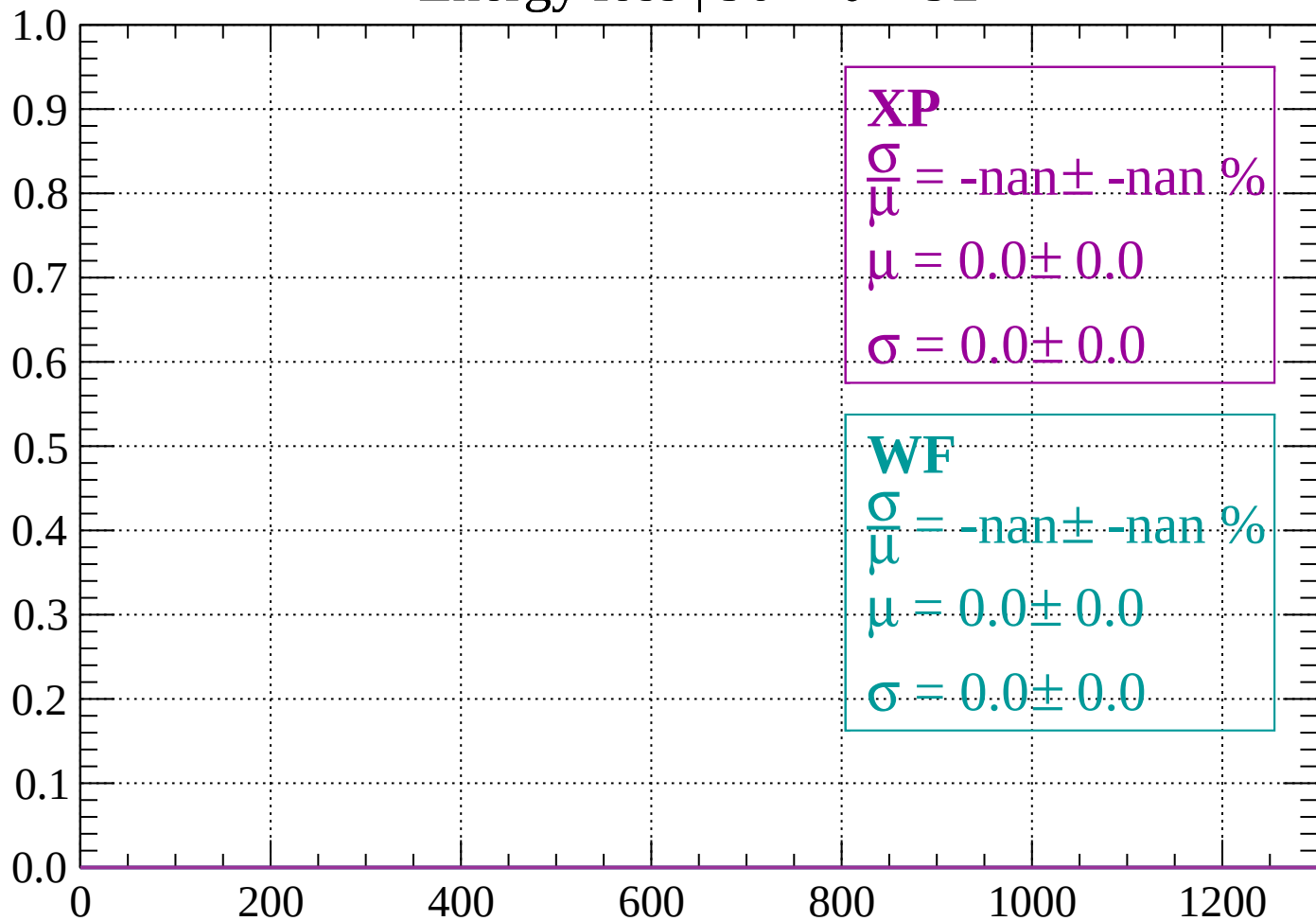
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

